MedMasters Collaborative

Cardiovascular System

Answer key and rationales

NPLEX Part I · by Dr. Sharilyn Rennie

Keep this volume separate from the workbook. Brain dumps can be right in more than one way, so grade the ideas and the relationships rather than the wording.

Cardiovascular System (CARD) · NPLEX-P1-CARD-01

CV-1 Cardiovascular Embryology

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-01-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the heart tube and name what each segment becomes

A competent response contains

· Sequence: cardiac progenitors pass through the cranial primitive streak into splanchnic lateral plate mesoderm and gather cranial to the neural plate as a horseshoe shaped cardiogenic field on days 16 to 18; endoderm signals of BMP and FGF with inhibition of Wnt induce NKX2-5; angioblastic cords canalize into paired endocardial tubes; lateral folding brings them to the ventral midline and they fuse into one tube by the end of week 3; cephalocaudal folding carries the tube into the future thorax; the heart beats at about day 22; looping runs from about day 23 to day 28.

· Segments from caudal to cranial: sinus venosus, primitive atrium, primitive ventricle, bulbus cordis, truncus arteriosus opening into the aortic sac.

· Derivatives: truncus arteriosus gives the proximal ascending aorta and pulmonary trunk; conus cordis gives the smooth outflow tracts of both ventricles; proximal bulbus cordis gives the trabeculated right ventricle; primitive ventricle gives most of the trabeculated left ventricle; primitive atrium gives the trabeculated atria including the auricles; right horn of the sinus venosus gives the smooth right atrium, the sinus venarum; left horn gives the coronary sinus and the oblique vein of the left atrium; the primitive pulmonary vein gives the smooth part of the left atrium.

· The primary heart field forms the initial tube, mainly the left ventricle and parts of the atria; the secondary heart field in pharyngeal mesoderm adds most of the right ventricle and the outflow tract, meaning the conus cordis and truncus arteriosus, plus part of the atria and the sinus venosus.

· Too few secondary field cells leave the outflow tract too short to rotate and align, which is one cause of outflow defects.

· Wall layers: inner endocardium, cardiac jelly that becomes the cushions, and outer myocardium; the proepicardium later supplies the epicardium, the coronary vessels and many cardiac fibroblasts.

· Loss of the dorsal mesocardium leaves the transverse pericardial sinus.

· Rough or trabeculated walls come from the original atrial and ventricular segments; smooth walls come from incorporated veins or from the outflow tract.

Distinctions that matter

· The two heart fields contribute different chambers, so a right ventricular or outflow problem points at the secondary field while the left ventricle comes mainly from the primary field.

· Smooth against trabeculated wall is a statement about embryonic origin, not about function.

Common misconceptions

· Writing that the whole heart arises from one field. The outflow tract and most of the right ventricle are added later by the secondary heart field.

· Treating the cardiac jelly as packing material. It is the tissue the endocardial cushions form in, so the valves and central septa come from it.

Where it is taught: Section C01, opening paragraphs and the table of segments and derivatives of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C01-BD2

Brain dump 2 · integration and application: A child with chronic sinusitis and an unusual chest film

A competent response contains

· The defect is in motile cilia, usually in the dynein arms, and the condition is primary ciliary dyskinesia, usually inherited in an autosomal recessive pattern.

· Nonfunctional dynein stops the rotary beat of the tilted monocilia at the primitive node, so there is no leftward nodal flow; NODAL, LEFTY2 and PITX2 are still expressed but on a side chosen at random, so sidedness is unpredictable rather than reversed.

· About half of patients have situs inversus totalis with dextrocardia, most of the remainder have normal situs, and a minority have heterotaxy, also called situs ambiguus.

· Kartagener syndrome is situs inversus with chronic sinusitis and bronchiectasis.

· Immotile airway cilia cannot clear mucus, giving neonatal respiratory distress, a wet cough, otitis media and bronchiectasis; immotile sperm flagella cause male infertility, and poor tubal transport raises the risk of ectopic pregnancy.

· Risk of structural heart disease depends on whether the heart and the body agree about sidedness: mirror image dextrocardia keeps the connections concordant so the heart is usually normal, while dextrocardia with normal abdominal situs, or heterotaxy with asplenia or polysplenia, carries a high risk of complex lesions.

· Mirror image dextrocardia inverts the P wave, the QRS complex and the T wave in lead I and gives diminishing R waves across the precordial leads.

Distinctions that matter

· Randomized sidedness is not the same as reversed sidedness, which is why only about half of these patients have mirrored organs.

· Concordance between heart and viscera, not the position of the heart by itself, is what predicts complex defects.

Common misconceptions

· Saying the cilia problem causes the heart defect directly. The cilia set the left and right signal, and the defects follow from discordant sidedness.

· Expecting every patient with this ciliary defect to have situs inversus. Most of those without it have normal situs.

Where it is taught: Section C01, paragraphs on primary ciliary dyskinesia and on concordance of situs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C01-RT01

Retrieval exercise: Rebuild the left and right signalling sequence

A competent response contains

· Step 1: dynein drives rotation of the tilted monocilia at the primitive node.

· Step 2: that rotation moves fluid leftward across the node.

· Step 3: NODAL and LEFTY2 are induced in left lateral plate mesoderm, and midline LEFTY1 forms a barrier that stops the signal spreading to the right.

· Step 4: NODAL induces PITX2 on the left.

· Step 5: PITX2 specifies left sided identity of the heart, lungs and gut, and the heart tube loops to the right.

· The cilia are tilted monocilia on the cells of the primitive node and the motor protein is dynein.

· If nodal flow fails, steps 3 to 5 still occur but on a randomly chosen side.

· In the loop the bulboventricular part bends ventrally, caudally and to the right while the atrial part moves dorsally, cranially and to the left, so the atria end up above and behind the ventricles, with the primitive ventricle, the future left ventricle, on the left and the bulbus cordis, the future right ventricle, on the right.

Distinctions that matter

· The midline LEFTY1 barrier is what keeps the pathway one sided; without it the left signal would cross over.

· Looping is a change in the position of segments already present, not the addition of new chambers.

Common misconceptions

· Writing that loss of nodal flow switches the signal to the right side. It removes the instruction, so the side is chosen at random.

· Placing the future left ventricle on the right after looping. The primitive ventricle ends up on the left.

Where it is taught: Section C01, flow diagram on left and right signalling and the looping paragraph of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw atrial septation in four panels

A competent response contains

· Septum primum grows from the atrial roof toward the endocardial cushions, leaving the ostium primum as a gap below its free edge.

· Apoptosis in the upper part of the septum primum opens the ostium secundum, a hole inside that septum.

· The septum primum then fuses with the cushions and the ostium primum closes at the end of week 5.

· The thick septum secundum grows down on the right side of the septum primum, and its free edge bounds the foramen ovale, which is the gap left by that second septum rather than a hole within it.

· The thin septum primum lies on the left and the rigid septum secundum on the right, and their openings are not aligned, so the lower part of the septum primum works as a flap valve.

· Before birth, higher right atrial pressure pushes the flap open and blood crosses right to left only; after birth higher left atrial pressure presses the flap shut.

· The adult remnant is the fossa ovalis: its floor is septum primum and its rim, the limbus, is the edge of the septum secundum.

· Anatomic fusion takes months.

Distinctions that matter

· The ostium secundum is a hole in the septum primum, while the foramen ovale is the gap left by the septum secundum; the names are easy to swap and the structures are not the same.

· Thin and left is septum primum, thick and right is septum secundum, and the offset is what makes the valve one way.

Common misconceptions

· Describing the foramen ovale as a hole in a single septum. It is an opening between two overlapping septa.

· Saying the flap closes because the hole shrinks. It closes because the pressure gradient reverses and presses the thin septum against the thick one.

Where it is taught: Section C02, flow diagram on atrial septation and the following paragraphs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C02-BD2

Brain dump 2 · integration and application: An unsealed flap against a true hole

A competent response contains

· Patent foramen ovale: no tissue is missing, the septa simply never fused, and this persists in about a quarter of adults; blood crosses only when right atrial pressure transiently exceeds left, as on release of a Valsalva strain, with coughing, or with acute pulmonary hypertension from pulmonary embolism, and the consequence is paradoxical embolism with stroke.

· Secundum atrial septal defect: a true tissue deficiency at the fossa ovalis from excess resorption of the septum primum or a deficient septum secundum; because the right ventricle is more compliant the shunt runs left to right.

· The extra volume loads the right atrium, right ventricle and pulmonary circulation; the left ventricle is spared because the shunt leaves before it.

· The murmur is a systolic ejection murmur at the left upper sternal border, produced by increased flow across a normal pulmonary valve rather than by flow through the defect; the interatrial pressure difference is small, so flow across the defect itself is nearly silent.

· A mid diastolic rumble across the tricuspid valve indicates a large shunt.

· The tracing shows right axis deviation with an rsR prime pattern in V1, which follows from right ventricular volume loading.

· Reversal sequence: high pulmonary flow over decades remodels the pulmonary arterioles, pulmonary vascular resistance rises, pulmonary pressure exceeds left sided pressure, the shunt reverses to right to left, and cyanosis appears; this is Eisenmenger syndrome and can end any large left to right shunt.

· The second sound is widely split because the right ventricle takes systemic return plus the shunt on every beat, so pulmonic closure is always late; it is fixed because inspiration raises right atrial pressure and shrinks the shunt by about the amount the systemic return rises, the atria behaving as one chamber, so right ventricular filling and the timing of pulmonic closure barely change with respiration.

· Right bundle branch block delays pulmonic closure electrically, so the wide split still varies with breathing.

· Known genetic causes of secundum defects are TBX5 mutation in Holt-Oram syndrome with upper limb defects and NKX2-5 mutation with atrioventricular conduction disease.

Distinctions that matter

· An unsealed flap needs a transient pressure reversal before anything crosses; a true deficiency shunts on every beat.

· Fixed splitting comes from a shunt that cancels the respiratory change, while a wide but moving split comes from delayed electrical activation of the right ventricle.

Common misconceptions

· Attributing the murmur to flow through the atrial defect. The gradient there is too small, and the sound comes from the extra flow across the pulmonary valve.

· Expecting left ventricular volume overload in an atrial septal defect. The shunt leaves the circuit before the left ventricle, so the right heart and lungs carry the load.

Where it is taught: Section C02, paragraphs on the patent foramen ovale and the secundum defect, and the worked example on the second heart sound of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C03-BD1

Brain dump 1 · foundational retrieval: What the endocardial cushions build

A competent response contains

· A single atrioventricular canal opens mainly over the future left ventricle and must shift to the right before dividing.

· At the end of week 4 the atrioventricular canal and outflow myocardium signal the endocardium through the TGF beta and BMP families; endocardial cells lose their junctions and invade the cardiac jelly, a change called epithelial to mesenchymal transformation, becoming the cushion mesenchyme.

· Superior and inferior atrioventricular cushions form with smaller lateral cushions; the outflow ridges form separately and receive neural crest cells.

· In week 5 the superior and inferior cushions fuse and divide the canal into right and left orifices.

· The fused mass closes the ostium primum, the lowest part of the atrial septum, at the end of week 5.

· Inferior cushion tissue joins the conal ridges and the muscular septum to close the interventricular foramen as the membranous septum by about the end of week 7.

· Cushion mesenchyme around each orifice is undermined on its ventricular side; the muscular strands left behind become the chordae tendineae, with muscle replaced by dense connective tissue, and the papillary muscles. The left orifice ends with two leaflets and the right with three.

· The muscular ventricular septum forms as the expanding ventricles press their medial walls together.

· The membranous septum closes last and needs three tissue sources to meet, the conal ridges, the muscular septum and the inferior cushion, which is why most ventricular septal defects are perimembranous; muscular defects, sometimes multiple, are less common and small ones often close.

Distinctions that matter

· The cushions build the valves, the lowest atrial septum and part of the ventricular septum, so they are central structures rather than valve tissue alone.

· The muscular septum forms by chamber growth, while the membranous part forms by three tissues meeting, and that difference explains where defects sit.

Common misconceptions

· Saying the valve leaflets grow out of the ventricular wall. They are carved out of cushion tissue by undermining from the ventricular side.

· Treating the chordae as tendon from the start. They are muscular strands in which the muscle is replaced by dense connective tissue.

Where it is taught: Section C03, first three paragraphs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C03-BD2

Brain dump 2 · integration and application: Why one cushion failure damages several structures

A competent response contains

· The failed step is fusion of the superior and inferior atrioventricular cushions, giving an atrioventricular septal defect, also called an atrioventricular canal defect.

· Because the cushions build several structures, one failure leaves a primum atrial septal defect, an inlet ventricular septal defect and a single common atrioventricular valve with five leaflets in the complete form.

· Failure at three months follows from pressure as well as flow: the ventricular defect transmits left ventricular pressure to the pulmonary arteries, so the lungs see high flow at high pressure, and pulmonary vascular disease arrives earlier than with an isolated atrial defect, where the shunt is at low atrial pressure.

· The atrioventricular node and His bundle are displaced inferiorly and posteriorly and the left anterior fascicle is short, giving left or superior axis deviation, often with first degree atrioventricular block.

· A secundum defect at the fossa ovalis gives right axis deviation instead, because the right ventricle is volume loaded and the conduction tissue sits normally.

· Partial form: two separate orifices with a primum atrial septal defect and a cleft anterior mitral leaflet producing mitral regurgitation.

· About 40 to 50 percent of children with trisomy 21 have congenital heart disease, and the complete atrioventricular septal defect is the characteristic lesion; the extra chromosome is thought to disturb cushion cell adhesion, migration and matrix.

· Ebstein anomaly: the septal and posterior tricuspid leaflets are poorly undermined, so their attachment is displaced toward the apex, part of the right ventricle is atrialized and the valve leaks.

· Tricuspid atresia: the right atrioventricular orifice is obliterated, so right atrial blood must cross the atrial septum, the hypoplastic right ventricle fills through a ventricular septal defect, and the lungs are supplied through that defect or through the ductus.

Distinctions that matter

· Left or superior axis deviation points at cushion failure with displaced conduction tissue; right axis deviation points at a volume loaded right ventricle with a normally placed node.

· A defect that carries ventricular pressure into the lungs behaves differently from one that carries only atrial pressure, even at the same shunt volume.

Common misconceptions

· Grouping all holes between the atria together. Only cushion defects involve the atrioventricular valves and give left axis deviation.

· Expecting a cushion defect to present like an atrial defect. The added ventricular communication brings heart failure and pulmonary vascular disease in months.

Where it is taught: Section C03, paragraph on atrioventricular septal defect, the comparison table, the trisomy 21 paragraph and the final paragraph of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C03-CP01

Comparison table: Three openings between the atria compared

A competent response contains

· Developmental step that failed: patent foramen ovale, the two septa do not fuse after birth; secundum defect, excess resorption of the septum primum or a deficient septum secundum; complete atrioventricular septal defect, the atrioventricular cushions fail to fuse.

· Tissue missing: no tissue loss in the patent foramen ovale; a true deficiency in the secundum defect; tissue missing in the lower septum and the inlet ventricular septum in the cushion defect.

· Location: patent foramen ovale between the overlapping septa; secundum defect at the fossa ovalis; cushion defect low in the septum above the atrioventricular valves, with a primum atrial defect and an inlet ventricular defect.

· Atrioventricular valves: normal in the patent foramen ovale and the secundum defect; a single common five leaflet valve in the complete cushion defect, and a cleft anterior mitral leaflet with regurgitation in the partial form.

· Flow: patent foramen ovale crosses right to left only when right atrial pressure transiently exceeds left; secundum defect shunts left to right every beat because the right ventricle is more compliant; complete cushion defect shunts left to right at both atrial and ventricular level, carrying left ventricular pressure to the lungs.

· Axis: normal in the patent foramen ovale; right axis deviation with rsR prime in V1 in the secundum defect; left or superior axis deviation, often with first degree block, in the cushion defect.

· Timing: the patent foramen ovale is silent until a transient pressure reversal allows paradoxical embolism in adult life; the secundum defect is often silent into adulthood and can end in Eisenmenger syndrome after decades; the complete cushion defect causes heart failure within months with early pulmonary vascular disease.

· Associations: Holt-Oram syndrome with TBX5 and conduction disease with NKX2-5 in the secundum defect; trisomy 21 in the complete cushion defect; none specific for the patent foramen ovale.

Distinctions that matter

· Only the patent foramen ovale lacks tissue loss, and only the cushion defects involve the atrioventricular valves and give left axis deviation.

· Direction of flow in the patent foramen ovale is right to left and intermittent, while the two true defects shunt left to right continuously until resistance changes.

Common misconceptions

· Calling a patent foramen ovale a small atrial septal defect. No septal tissue is missing and the physiology is different.

· Using murmur loudness to judge shunt size at the atrial level. The gradient is small, and the diastolic tricuspid rumble is what signals a large shunt.

Where it is taught: Section C03, table of interatrial and atrioventricular septal communications, with Section C02 for the patent foramen ovale and the secundum defect of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the spiral outflow septum and the three defects

A competent response contains

· The outflow tract must become a pulmonary trunk leaving the right ventricle anteriorly and to the left and an aorta leaving the left ventricle posteriorly and to the right.

· In week 5 the right superior and left inferior truncal ridges grow distally while swinging around the truncal wall, so the aorticopulmonary septum they form twists about 180 degrees.

· The right dorsal and left ventral conal ridges fuse into the conal or outlet septum, which separates the right ventricular infundibulum from the aortic vestibule and meets the muscular septum and the inferior cushion to complete the membranous septum.

· Cardiac neural crest cells come from the hindbrain and migrate through pharyngeal arches 3, 4 and 6; some support the arch arteries and help form the thymus and parathyroids, and others populate the truncal and conal ridges and the semilunar valves. Removing this crest in animal embryos produces persistent truncus arteriosus.

· Ridges that never form or fuse give persistent truncus arteriosus: one trunk and one valve sitting over a ventricular septal defect, supplying the coronary, pulmonary and systemic arteries, with mixing and mild cyanosis and then excess pulmonary flow and heart failure as resistance falls.

· A septum that fails to spiral gives D transposition of the great arteries: the aorta leaves the right ventricle and the pulmonary trunk leaves the left ventricle, the circuits run in parallel, survival requires mixing through the foramen ovale, the ductus or a ventricular septal defect, cyanosis is severe and early, the vascular pedicle is narrow, and maternal diabetes is a risk factor.

· A septum displaced anterosuperiorly gives tetralogy of Fallot: pulmonary stenosis, a malalignment ventricular septal defect, an overriding aorta and right ventricular hypertrophy.

· Because the outlet septum and the ventricular septum must align, displacing the outlet septum leaves a malalignment defect in the ventricular septum.

Distinctions that matter

· No septum, a straight septum and a displaced septum produce the three classic conotruncal defects, so naming the step tells you the anatomy.

· Truncal ridges divide the great vessels while conal ridges divide the outflow tracts of the ventricles, and they fail with different consequences.

Common misconceptions

· Treating transposition as a problem of vessel position alone. The septum formed but did not spiral, which is why each vessel sits over the wrong ventricle.

· Listing the four features of tetralogy as four separate faults. One displaced septum produces all four.

Where it is taught: Section C04, first two paragraphs and the table of conotruncal defects of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C04-BD2

Brain dump 2 · integration and application: Two blue newborns from two different failures

A competent response contains

· First newborn: D transposition of the great arteries, from a septum that failed to spiral, so the aorta arises from the right ventricle and the pulmonary trunk from the left ventricle.

· The circuits run in parallel, so systemic venous blood returns to the body and pulmonary venous blood returns to the lungs, and the two can only supply each other through a shunt: the foramen ovale, the ductus arteriosus or a ventricular septal defect.

· What changed on day two is that the ductus arteriosus closed as arterial oxygen rose and prostaglandin E2 fell, so mixing depended on the interatrial opening alone and systemic oxygen delivery fell.

· The great vessels lie one in front of the other, which narrows the vascular pedicle on the film.

· Second child: tetralogy of Fallot, from anterosuperior displacement of the infundibular septum, giving pulmonary stenosis, a malalignment ventricular septal defect, an overriding aorta and right ventricular hypertrophy; the raised apex with a concave upper left border is the boot shaped heart.

· Because the ventricular defect is large, both ventricles share systolic pressure, so shunt direction depends on right ventricular outflow resistance against systemic resistance; mild obstruction gives a mostly left to right shunt and a pink child.

· The murmur arises from the narrowed right ventricular outflow tract, so it softens during a spell because less blood is crossing that outflow.

· Chronic hypoxaemia causes secondary polycythaemia and clubbing.

· The 22q11.2 deletion, called DiGeorge or velocardiofacial syndrome, removes one copy of TBX1 and neighbouring genes and disrupts the pharyngeal tissue through which crest cells migrate, so outflow defects such as truncus arteriosus, tetralogy and interrupted aortic arch arrive with thymic hypoplasia and T cell deficiency, parathyroid hypoplasia with hypocalcaemia, and facial and palatal anomalies.

· Excess retinoic acid, as with maternal isotretinoin, disturbs crest migration and causes conotruncal and craniofacial defects.

Distinctions that matter

· Cyanosis in transposition is fixed by the anatomy of parallel circuits, while cyanosis in tetralogy varies with the balance of two resistances.

· A quieter murmur means less flow across the obstruction, which in tetralogy means a worse spell rather than a better heart.

Common misconceptions

· Reading a louder murmur as more severe disease in tetralogy. The murmur fades as the spell deepens.

· Expecting every child with tetralogy to be blue from birth. With mild outflow obstruction the shunt runs mostly left to right.

Where it is taught: Section C04, table of conotruncal defects, the paragraphs on transposition and tetralogy, and the paragraph on the 22q11.2 deletion of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C04-PR01

Prediction exercise: Predict what happens during a spell and during squatting

A competent response contains

· Systemic vascular resistance falls while the child cries or exerts.

· Right ventricular outflow resistance therefore rises relative to systemic resistance, because the outflow obstruction is fixed while the systemic bed dilates.

· Blood crosses the ventricular septal defect right to left, and more of it crosses, because the large defect lets both ventricles share pressure and blood takes the lower resistance route.

· Pulmonary blood flow falls and arterial oxygen saturation falls with it.

· The murmur softens, because it is generated by flow through the narrowed right ventricular outflow tract and less blood is crossing it.

· Hypoxaemia and acidosis lower systemic resistance further, which drives more right to left shunting, so the spell feeds itself.

· Squatting kinks the femoral arteries and raises systemic resistance, so the shunt swings back toward left to right, pulmonary flow and saturation rise, the murmur becomes louder again and the cycle is broken.

Distinctions that matter

· Shunt direction follows the comparison between two resistances, not the size of the defect, because the defect is large enough to equalize pressure.

· Murmur loudness tracks flow across the obstruction, so it moves in the opposite direction to the severity of the spell.

Common misconceptions

· Expecting the obstruction itself to worsen during a spell. The fixed obstruction stays the same while the systemic resistance falls around it.

· Reading squatting as an effect on the lungs. It acts on the systemic circulation by raising its resistance.

Where it is taught: Section C04, cycle diagram on the hypercyanotic spell and the preceding paragraph on shunt direction of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the fetal circulation with its saturations

A competent response contains

· Route: umbilical vein to the liver, part of the flow through the ductus venosus into the inferior vena cava, right atrium, across the foramen ovale to the left atrium, left ventricle, ascending aorta with the coronary, carotid and subclavian branches first; superior vena cava blood to the right atrium, right ventricle and pulmonary trunk, most of it across the ductus arteriosus into the descending aorta, then back through the umbilical arteries to the placenta.

· The umbilical vein is the most saturated fetal vessel at about 80 percent, with a partial pressure of oxygen of only about 30 to 35 mm Hg; the umbilical arteries carry blood at about 60 percent saturation.

· The crista dividens, the lower edge of the septum secundum, with the valve of the inferior vena cava, streams the better oxygenated inferior caval blood across the foramen ovale.

· Pulmonary vascular resistance is high because the lungs are fluid filled, their vessels are compressed, low alveolar oxygen tension causes hypoxic vasoconstriction, and mediators favour constriction with high endothelin and low nitric oxide.

· Systemic resistance is low because the placenta is a large low resistance circuit.

· Pulmonary trunk pressure therefore exceeds descending aortic pressure, which is why most right ventricular output crosses the ductus arteriosus and only a small fraction of combined output passes through the lungs.

· The right ventricle supplies slightly more than half of the combined output, which is why both ventricles have similar wall thickness at birth.

· Three shunts: ductus venosus past the liver, foramen ovale between the atria, ductus arteriosus from pulmonary trunk to descending aorta.

Distinctions that matter

· Saturation and partial pressure are separate numbers here: the umbilical vein is about 80 percent saturated at a partial pressure an adult would not tolerate, because fetal haemoglobin binds oxygen more tightly.

· Both ventricles pump into the systemic circuit in the fetus, which is what parallel circulation means.

Common misconceptions

· Describing the umbilical vein as fully saturated. It is the best fetal blood at about 80 percent, not arterial adult blood.

· Expecting a thin right ventricle before birth. It carries slightly more than half of the combined output, so the walls are of similar thickness.

Where it is taught: Section C05, opening paragraphs, flow diagram on the fetal circulation and the following paragraphs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C05-BD2

Brain dump 2 · integration and application: Why the fetal brain is well supplied at a low oxygen tension

A competent response contains

· Four reasons oxygenation works: fetal haemoglobin has a higher oxygen affinity, haemoglobin concentration is higher, cardiac output is high for body mass, and streaming delivers the best blood to the heart and brain.

· Fetal haemoglobin has two alpha and two gamma chains; the gamma chain carries serine where the beta chain has the 2,3 bisphosphoglycerate binding histidine at position 143, so 2,3 bisphosphoglycerate binds weakly and the half saturation pressure is about 19 mm Hg against about 27 mm Hg for adult haemoglobin.

· The double Bohr effect at the placenta: fetal blood loses carbon dioxide and gains affinity while maternal blood gains carbon dioxide and loses affinity, so oxygen moves from mother to fetus.

· Best oxygenated route: umbilical vein, ductus venosus, inferior vena cava, streamed across the foramen ovale by the crista dividens, left atrium with only slight dilution by the small pulmonary venous return, left ventricle, ascending aorta, then the coronary arteries and the carotid and subclavian arteries before the ductus joins the aorta distal to the left subclavian artery.

· The lower body receives the mixture of left ventricular blood and the desaturated ductal blood from the right ventricle, which carries superior caval and coronary sinus blood with part of the inferior caval stream.

· If the ductus arteriosus narrows in mid gestation the right ventricle has to eject into the high resistance pulmonary bed, so right ventricular pressure load rises, while flow reaching the descending aorta falls, so the lower body depends more on left ventricular output.

· If the foramen ovale narrows instead, the left atrium and left ventricle lose most of their filling and grow poorly while the right heart enlarges.

· Embryonic chambers grow in proportion to the flow that passes through them.

Distinctions that matter

· The shift in the dissociation curve is a property of the gamma chain and its weak 2,3 bisphosphoglycerate binding, not of a different oxygen binding site.

· Streaming is an anatomical effect inside the right atrium, separate from the affinity effect in the blood.

Common misconceptions

· Saying the fetal brain receives placental blood directly. It receives left ventricular blood that was streamed across the foramen ovale and only slightly diluted.

· Expecting a narrowed atrial opening to enlarge the left heart because pressure rises. Filling falls, so the chamber grows poorly.

Where it is taught: Section C05, paragraphs on streaming, on the lower body supply and on fetal haemoglobin, with the stop and predict box in Section C02 of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C05-RT02

Retrieval exercise: Rebuild the path of blood through the fetus

A competent response contains

· Step 1: the umbilical vein brings the best oxygenated blood, about 80 percent saturated, from the placenta.

· Step 2: the ductus venosus shunts part of that flow past the hepatic sinusoids into the inferior vena cava; the rest enters the sinusoids through the left portal vein.

· Step 3: the crista dividens streams inferior caval blood through the foramen ovale into the left atrium, which is possible because right atrial pressure exceeds left atrial pressure in the fetus.

· Step 4: the left ventricle ejects into the ascending aorta, and the coronary, carotid and subclavian arteries branch before any mixing point.

· Step 5: superior caval blood passes right atrium to right ventricle to pulmonary trunk.

· Step 6: most pulmonary trunk flow crosses the ductus arteriosus to the descending aorta, because pulmonary vascular resistance is high and pulmonary trunk pressure exceeds descending aortic pressure.

· Step 7: the umbilical arteries, branches of the internal iliac arteries, return blood at about 60 percent saturation to the placenta.

· The two shunt steps are step 2, bypassing the liver, and step 6, bypassing the lungs; the foramen ovale in step 3 bypasses the right ventricle and the lungs as well.

· The inlet of the ductus venosus regulates how much umbilical flow bypasses the liver and protects the heart from sudden volume loads such as those during uterine contractions.

· Inferior caval blood goes to the left heart and the brain; superior caval blood goes to the right heart, the ductus and the lower body.

Distinctions that matter

· Each step follows a pressure or resistance gradient, so the path is not a set of leaks but the route of least resistance at that moment.

· The ductus venosus bypasses the liver while the ductus arteriosus bypasses the lungs, and the similar names are easy to swap.

Common misconceptions

· Writing that all umbilical venous blood bypasses the liver. Part of it perfuses the hepatic sinusoids through the left portal vein.

· Treating the foramen ovale as a passive hole. Streaming by the crista dividens is what sends the better oxygenated blood through it.

Where it is taught: Section C05, flow diagram on the path of blood through the fetal circulation and the two paragraphs that follow it of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild the changes at birth and the remnants table

A competent response contains

· Step 1: the lungs expand with the first breaths and alveolar oxygen tension rises.

· Step 2: pulmonary vascular resistance falls and pulmonary flow rises several fold, because hypoxic vasoconstriction is relieved and because oxygen and shear stress release nitric oxide and prostacyclin.

· Step 3: the cord is clamped, systemic resistance rises and inferior caval return falls.

· Step 4: left atrial pressure exceeds right atrial pressure, so the septum primum is pressed against the septum secundum and the foramen ovale closes functionally within minutes to hours; anatomic fusion takes months.

· Step 5: rising arterial oxygen tension and falling prostaglandin E2 constrict the ductus arteriosus.

· Step 6: the ductus venosus and the umbilical vessels close and fibrose into ligaments.

· Ductal oxygen mechanism: oxygen sensitive potassium channels close, the membrane depolarizes and L type calcium channels open, so the ductal smooth muscle contracts.

· Prostaglandin mechanism: prostaglandin E2 falls because the placenta is gone and the perfused lungs metabolize it; in the fetus prostaglandin E2 acting on EP receptors raised cyclic AMP and kept the muscle relaxed. Functional ductal closure takes 1 to 3 days in term infants and fibrosis over weeks leaves the ligamentum arteriosum.

· Across the ductus the gradient reverses first, so flow briefly runs aorta to pulmonary trunk; the umbilical arteries constrict within minutes to mechanical, thermal and oxygen stimuli, and the umbilical vein and ductus venosus close slightly later, after which the liver is perfused by the portal vein and hepatic artery.

· Remnants: umbilical vein, placenta to liver, becomes the ligamentum teres hepatis in the free edge of the falciform ligament; ductus venosus, umbilical vein to inferior vena cava past the sinusoids with a regulated inlet, becomes the ligamentum venosum in the fissure between the caudate and left lobes; foramen ovale, right to left atrium by streaming and a right atrial pressure above left, becomes the fossa ovalis on the right side of the interatrial septum; ductus arteriosus, pulmonary trunk to descending aorta driven by high pulmonary resistance, low oxygen tension and prostaglandin E2, becomes the ligamentum arteriosum from

the left pulmonary artery to the underside of the aortic arch; umbilical arteries, internal iliac arteries to placenta, become the medial umbilical ligaments distally and the superior vesical arteries proximally; the allantois and urachus, never a vessel, become the median umbilical ligament from the bladder apex to the umbilicus.

· Pressure closes the foramen ovale, while chemical signals of oxygen and prostaglandin E2 close the ductus arteriosus.

Distinctions that matter

· The foramen ovale closes mechanically and the ductus arteriosus chemically, which is why one shuts in minutes and the other over days.

· The ligamentum venosum and the ligamentum arteriosum come from different shunts, and the paired medial umbilical ligaments are not the single median umbilical ligament.

Common misconceptions

· Saying ductal flow stops the moment the cord is clamped. The gradient reverses first and flow briefly runs aorta to pulmonary trunk before the muscle constricts.

· Treating functional closure as permanent. Anatomic closure takes weeks for the ductus and months for the foramen ovale.

Where it is taught: Section C06, opening paragraphs, flow diagram on closure of the shunts and the table of fetal shunts and remnants of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C06-BD2

Brain dump 2 · integration and application: A vessel that stays open and the lesions that depend on it

A competent response contains

· A patent ductus arteriosus is most common in prematurity because immature ductal muscle responds less to oxygen and more to prostaglandin E2; congenital rubella and high altitude also predispose.

· Aortic pressure exceeds pulmonary pressure throughout the cardiac cycle, so flow never stops and the murmur is continuous, a machinery murmur heard below the left clavicle.

· Diastolic runoff into the pulmonary artery widens the pulse pressure, and the shunt volume loads the left heart.

· Ductus dependent lesions: transposition of the great arteries depends on the ductus for mixing, pulmonary atresia depends on it for lung blood flow, and hypoplastic left heart or critical coarctation depends on it for body blood flow.

· Differential cyanosis in a long standing large ductus: years of high flow and pressure remodel the pulmonary arterioles until pulmonary pressure exceeds aortic pressure and the shunt reverses, which is Eisenmenger physiology; because the ductus enters the aorta distal to the left subclavian artery, the head and arms still receive unmixed left ventricular blood while the descending aorta carries desaturated blood to the lower body, so the toes are cyanotic and clubbed while the fingers stay pink.

· A reversed shunt at ventricular or atrial level mixes blood before the aorta, so fingers and toes are equally cyanotic.

· In persistent pulmonary hypertension of the newborn the pulmonary resistance stays high despite normal anatomy, so blood shunts right to left through the foramen ovale and the ductus arteriosus.

Distinctions that matter

· A continuous murmur reports a gradient that lasts all cycle, which is what separates ductal flow from a systolic shunt between the ventricles.

· Where the shunt sits relative to the arch branches decides whether cyanosis is differential or uniform.

Common misconceptions

· Expecting a ductus to load the right heart. The shunt leaves the aorta and returns through the lungs to the left atrium, so the left heart carries the volume.

· Reading cyanosis with normal heart anatomy as a structural defect. Persistent pulmonary hypertension of the newborn shunts right to left through normal fetal channels.

Where it is taught: Section C06, paragraph on patent ductus arteriosus and the worked example on differential cyanosis of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C06-CA01

Clinical application: A newborn who collapses on the third day

A competent response contains

· Recognize: the ductus arteriosus has closed. Until the third day it was carrying right ventricular blood into the descending aorta and supplying the lower body past an obstruction in the aorta, which is the pattern of a critical coarctation or an otherwise ductus dependent systemic circulation.

· Explain: in the fetus and in the first days the lower body is supplied through the ductus, so femoral pulses and perfusion were normal on day one. As the ductus constricted over 1 to 3 days the only route to the lower body became the obstructed aorta, so the femoral pulses disappeared, systemic perfusion fell with pallor and mottling, the left ventricle faced a sudden pressure load, left atrial pressure and pulmonary venous pressure rose to give fast breathing and congested lung fields, and systemic venous congestion enlarged the liver.

· Differentiate: a newborn with parallel circuits in transposition is deeply cyanotic within hours with poor saturation everywhere and a narrow vascular pedicle, not a pressure difference between upper and lower body; a newborn whose pulmonary resistance never fell is cyanotic with a structurally normal heart and shunts right to left at both the foramen ovale and the ductus, again without absent femoral pulses; this baby has preserved upper body saturation and pressure with absent femoral pulses and a saturation gradient, which points to an obstruction in the aorta distal to the arch branches.

· Investigate: compare pulses and blood pressure in the right arm with the legs, expecting a strong upper limb pulse with raised or normal pressure and absent or weak femoral pulses with a pressure gradient between arm and leg; compare pre and post ductal saturations, which here are 96 percent at the right hand and 82 percent at the right foot; palpate the liver edge, enlarged at 3 cm below the costal margin; and note the large heart with congested lung fields on the film.

· Decide: the ductus is normally closed by rising arterial oxygen tension, which shuts oxygen sensitive potassium channels and opens L type calcium channels, and by falling prostaglandin E2 once the placenta is gone and the perfused lungs metabolize it. The prostaglandin E2 supply was lost first, at delivery with the placenta, while the oxygen driven constriction then acted over the next days, and the lower body circulation now depends entirely on flow through the obstructed aorta.

· Justify: the ductus enters the aorta distal to the left subclavian artery, so the coronary, carotid and subclavian branches leave the arch before that point and carry fully saturated left ventricular blood, giving 96 percent at the right hand, while the descending aorta beyond the ductal insertion receives desaturated right ventricular blood through the ductus, giving 82 percent at the foot.

Distinctions that matter

· A saturation and pulse difference between upper and lower body localizes the problem distal to the arch branches, while uniform cyanosis points to mixing before the aorta.

· Loss of prostaglandin E2 happens at delivery, whereas the rise in oxygen tension acts over the following days, which is why collapse comes on day two or three rather than at birth.

Common misconceptions

· Reading a normal day one examination as evidence against a structural lesion. The ductus was supplying the lower body and hiding the obstruction.

· Expecting cyanosis to be the first sign in a ductus dependent systemic circulation. Pallor, absent femoral pulses and failure come first, with only a modest saturation gradient.

Where it is taught: Section C06, paragraphs on circulatory changes at birth, ductal closure and ductus dependent lesions, with the worked example on differential cyanosis of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C07-BD1

Brain dump 1 · foundational retrieval: Build the arch artery table and draw both nerve loops

A competent response contains

· Each pharyngeal arch artery runs from the aortic sac to the paired dorsal aortae, which fuse caudally into one descending aorta; the arteries appear in craniocaudal order during weeks 4 and 5 and are never all present at the same time.

· Truncus arteriosus gives the ascending aorta and the pulmonary trunk.

· Aortic sac: the right horn gives the brachiocephalic trunk and the left horn gives the arch as far as the left common carotid artery.

· First and second arch arteries: part of the maxillary artery, and the hyoid and stapedial arteries, with most of these vessels regressing.

· Third arch artery: the common carotid and the first part of the internal carotid, the rest of the internal carotid coming from the cranial dorsal aorta, and the external carotid sprouts from it.

· Fourth arch artery on the left: the arch between the left common carotid and the left subclavian arteries. On the right: the proximal right subclavian artery, with its distal part from the right dorsal aorta and the seventh intersegmental artery.

· Fifth arch artery: no derivative.

· Sixth arch arteries: the proximal right and left pulmonary arteries; the distal left becomes the ductus arteriosus and the distal right regresses.

· Left seventh intersegmental artery becomes the left subclavian artery, and the left dorsal aorta becomes the descending aorta beyond the left subclavian.

· Segments that disappear: the dorsal aorta between the third and fourth arches on both sides, and the right dorsal aorta below the right seventh intersegmental artery.

· The vagus supplies the fourth arch through the superior laryngeal nerve, which serves the cricothyroid muscle, and the sixth arch through the recurrent laryngeal nerve, which serves all other intrinsic laryngeal muscles.

· Each recurrent nerve first hooks under its sixth arch artery and is dragged caudally as the heart descends. On the right the distal sixth and the fifth arch arteries regress, so the nerve slides up to the right subclavian artery, a fourth arch derivative, at the root of the neck. On the left the distal sixth arch persists as the ductus, so the nerve stays under the aortic arch just behind the ligamentum arteriosum in the superior mediastinum. Both ascend in the tracheo-oesophageal groove.

Distinctions that matter

· The asymmetry of the two nerves comes from the asymmetry of the arch arteries, so the nerve course is a record of which vessel persisted.

· The internal carotid has two sources, the third arch artery proximally and the cranial dorsal aorta distally, so it is not a pure third arch derivative.

Common misconceptions

· Looking for a fifth arch derivative. There is none.

· Placing the right recurrent laryngeal nerve under the brachiocephalic trunk. It hooks under the right subclavian artery.

Where it is taught: Section C07, first paragraph, the table of origins of the great arteries and the paragraph on the laryngeal nerves of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C07-BD2

Brain dump 2 · integration and application: Four remodelling errors and a hoarse voice

A competent response contains

· Aberrant right subclavian artery: the right fourth arch artery and the cranial right dorsal aorta regress while the caudal right dorsal aorta persists, so the right subclavian becomes the last branch of the arch and crosses behind the oesophagus, which can cause dysphagia.

· Double aortic arch: the right dorsal aorta persists, leaving a vascular ring that can compress the trachea and the oesophagus.

· Interrupted aortic arch between the left common carotid and the left subclavian: abnormal regression of the left fourth arch artery, so the lower body depends on a patent ductus arteriosus; because neural crest supports the fourth arch artery, this type has the strongest link with the 22q11.2 deletion.

· Nonrecurrent laryngeal nerve: with an aberrant right subclavian artery there is no right fourth arch vessel to hold the nerve down, so the right laryngeal nerve does not recur.

· The recurrent nerve supplies the posterior cricoarytenoid, the only abductor of the vocal fold, so injury leaves the fold near the midline and the voice is hoarse.

· Left nerve course in the chest: near the aortic arch, the left pulmonary artery, the left main bronchus and the hilar nodes, so hoarseness can be caused by an aortic arch aneurysm, a left hilar or mediastinal tumour or nodes, a dilated left atrium or pulmonary artery, which is cardiovocal or Ortner syndrome, or ligation of a patent ductus arteriosus.

· The right nerve is at risk at the root of the neck from an apical lung tumour or a subclavian aneurysm.

· Thyroid surgery puts both nerves at risk.

Distinctions that matter

· An aberrant right subclavian artery and a nonrecurrent right laryngeal nerve are two consequences of the same remodelling error, so they travel together.

· Only the left nerve has a long thoracic course, which is why thoracic and cardiac disease causes left sided vocal fold paralysis.

Common misconceptions

· Expecting a vascular ring to come from a persisting fourth arch artery. It comes from persistence of the right dorsal aorta.

· Attributing hoarseness in mitral disease to the voice box itself. A dilated left atrium and pulmonary artery compress the left recurrent nerve against the arch and the ligamentum arteriosum.

Where it is taught: Section C07, final paragraph on abnormal remodelling, the paragraph on nerve vulnerability and the stop and predict box of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C02-Q01

Board-style question: An atrial opening that seals too early

Correct answer: C. Most left sided filling normally arrives across that opening, and chambers grow in proportion to the volume that passes through them

Why it is right: In the fetus pulmonary venous return is small, so the foramen ovale provides most left sided filling; when it seals early the left atrium and left ventricle receive little flow and grow poorly, and the right heart takes the volume and enlarges.

A is wrong: Oxygenated blood is not diverted into the pulmonary arteries, because high fetal pulmonary vascular resistance keeps pulmonary flow to a small fraction of combined output and the right ventricle still sends most of its output across the ductus arteriosus.

B is wrong: Left atrial pressure does not rise above right in this situation; with the opening sealed the left atrium is underfilled, and a small thick walled ventricle here reflects lost filling rather than an increased ejection load.

D is wrong: The endocardial cushions fuse and divide the atrioventricular canal in week 5, long before 24 weeks, and the mitral orifice is already formed, so cushion fusion is not what fails here.

A competent response contains

· Most left atrial and left ventricular filling in the fetus arrives across the foramen ovale, because pulmonary venous return is small while pulmonary vascular resistance is high.

· Embryonic and fetal chambers grow in proportion to the volume of blood that passes through them, so losing that filling leaves a small left atrium and a small left ventricle while the right heart enlarges.

Distinctions that matter

· Growth of a fetal chamber follows the flow through it, not the pressure it generates.

Common misconceptions

· Expecting an obstructed inflow to enlarge a chamber. Reduced filling leaves it underdeveloped.

Where it is taught: Section C02, stop and predict box, with the fetal circulation in Section C05 of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C07-Q02

Board-style question: Hoarseness with a dilated aortic arch

Correct answer: A. The distal part of the left sixth pharyngeal arch artery

Why it is right: The distal left sixth pharyngeal arch artery persists as the ductus arteriosus and becomes the ligamentum arteriosum, which keeps the left recurrent laryngeal nerve in the chest beneath the aortic arch where an arch aneurysm can compress it.

B is wrong: The left fourth pharyngeal arch artery becomes the aortic arch between the left common carotid and left subclavian arteries; the nerve passes under the arch but it is held there by the ligament behind it rather than by the arch itself, and on the right it is the fourth arch derivative that holds the nerve at the neck.

C is wrong: The left seventh intersegmental artery becomes the left subclavian artery, which is the vessel the right nerve hooks under, not the structure that anchors the left nerve in the chest.

D is wrong: The left horn of the aortic sac becomes the arch up to the left common carotid artery and has no relationship that traps the recurrent laryngeal nerve.

A competent response contains

· The left recurrent laryngeal nerve first hooks under the left sixth arch artery; the distal part of that artery persists as the ductus arteriosus and leaves the ligamentum arteriosum behind it in the adult.

· That remnant holds the nerve in the superior mediastinum under the aortic arch, so an arch aneurysm compresses it and the left vocal fold is paralysed near the midline because the posterior cricoarytenoid, the only abductor, is denervated.

Distinctions that matter

· The right nerve has no equivalent thoracic anchor, because the distal right sixth arch artery regresses and the nerve rises to the right subclavian artery.

Common misconceptions

· Assuming both recurrent nerves loop around the same structure. The vessel that persisted on each side decides where the nerve ends up.

Where it is taught: Section C07, paragraph on the laryngeal nerves and the paragraph on nerve vulnerability of the CV-1 Cardiovascular Embryology notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-02

CV-2 Histology of the Heart and Blood Vessels

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-02-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the heart wall and the fibrous skeleton

A competent response contains

· From the cavity outward: endocardium with its subendocardial layer, myocardium, epicardium.

· The endocardium is simple squamous endothelium on thin connective tissue, continuous with the endothelium of the great vessels.

· The subendocardial layer carries small vessels, nerves, and the Purkinje fibers.

· The myocardium is cardiac muscle, and the epicardium is the visceral serous pericardium: mesothelium over fat carrying the coronary vessels and nerves.

· The heart layers match the vessel layers: endocardium with intima, myocardium with media, epicardium with adventitia, and the fibrous skeleton has no vascular counterpart.

· Thickness follows chamber pressure: atria a few millimetres, right ventricle about 0.3 to 0.5 cm, left ventricle about 1.3 to 1.5 cm.

· The fibrous skeleton is dense irregular connective tissue, mainly type I collagen: four fibrous rings around the valve orifices, the right and left fibrous trigones, and the membranous interventricular septum.

· The right fibrous trigone is the thickest part and is called the central fibrous body.

· The bundle of His pierces the central fibrous body and then runs along the membranous septum, and it is the only normal electrical link between atria and ventricles.

· The skeleton anchors the valves and the myocardial sheets and completely separates atrial from ventricular muscle.

· A valve leaflet is a fold of endocardium around an essentially avascular core: a collagenous fibrosa continuous with the fibrous ring bears the load, a proteoglycan rich spongiosa cushions flexion, and an elastic layer lies on the inflow side.

· The spongiosa expands in myxomatous mitral valve disease, which is mitral valve prolapse, and vessels growing into scarred leaflets mark chronic rheumatic disease.

Distinctions that matter

· The endocardium is the layer; the subendocardium is only its deepest part, and that is where the Purkinje fibers and the ischemic vulnerability sit.

· The fibrous rings are mechanical anchors and electrical insulators at the same time, and students usually remember only the first job.

Common misconceptions

· Describing valve leaflets as vascular tissue. The normal leaflet core is essentially avascular, and vessels within it are a sign of old rheumatic scarring.

· Giving the right and left ventricle the same wall thickness. The left is about three to four times thicker because it generates systemic pressure.

Where it is taught: Section C01, paragraphs on endocardium and valves, myocardium, epicardium, and the fibrous cardiac skeleton of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C01-BD2

Brain dump 2 · integration and application: Wall stress, hypertrophy, and two failures of the skeleton

A competent response contains

· By the law of Laplace, wall stress is proportional to pressure times radius divided by twice the wall thickness.

· Case 1: (120 x 2.5) divided by (2 x 1.0) is 150 units, the baseline load on each myocyte.

· Case 2: (200 x 2.5) divided by (2 x 1.0) is 250 units, so raising pressure alone raises the load on every myocyte sharply.

· Thickening the wall to about 1.7 cm returns stress to about 147 units, which is the purpose of hypertrophy.

· Case 3: (120 x 3.5) divided by (2 x 0.8) is about 263 units, so dilation without matching thickening raises the energy cost of every beat even at a normal pressure.

· Pressure overload from hypertension or aortic stenosis adds sarcomeres in parallel, giving concentric hypertrophy with a thick wall and an unchanged or smaller cavity.

· Volume overload from regurgitation adds sarcomeres in series, giving eccentric hypertrophy with a larger cavity.

· Hypertrophied myocytes have large, dark, rectangular boxcar nuclei.

· The cost of thickening is a stiffer wall that outgrows its capillary supply, which predisposes to subendocardial ischemia.

· A strand of myocardium crossing a fibrous ring is an accessory pathway, the bundle of Kent, which bypasses the AV node and its delay.

· The three tracing findings that follow are a short PR interval, a delta wave, and reentrant tachycardia, the Wolff-Parkinson-White pattern.

· The His bundle lies beside the aortic ring and the membranous septum, so an aortic ring abscess in infective endocarditis, ring calcification, or surgery in that area can cause new AV block.

Distinctions that matter

· Concentric and eccentric hypertrophy differ in where the sarcomeres are added, which is why one thickens the wall and the other enlarges the cavity.

· Radius and thickness pull in opposite directions in the same relationship, so a dilated thin ventricle can carry a higher myocyte load than a hypertensive thick one.

Common misconceptions

· Treating hypertrophy as damage in itself. It is a response that returns wall stress toward normal, and its costs are stiffness and an outgrown capillary supply.

· Assuming an accessory pathway is an extra piece of conduction tissue. It is ordinary myocardium left crossing the ring, which is why it has no nodal delay.

Where it is taught: Section C01, myocardium paragraph, the worked example on wall stress, and the final paragraph of the section of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C01-RT01

Retrieval exercise: Rebuild why the subendocardium is injured first

A competent response contains

· Step 1: the coronary arteries run in epicardial fat and penetrate inward, so the inner wall is supplied last.

· Step 2: systole raises intramural pressure most near the cavity.

· Step 3: the inner wall vessels are compressed, so flow to the inner wall is mainly diastolic.

· Step 4: wall stress and oxygen demand are highest in the inner layers.

· Step 5: hypotension, tachycardia, or hypertrophy drops supply below demand.

· Step 6: necrosis appears first in the inner third of the wall.

· Tachycardia matters because it shortens diastole, which is when the inner wall is perfused.

· Wall stress is highest at the inner layers because stress rises with radius and the inner fibres sit at the largest effective radius for a given thickness of overlying muscle.

· The result is a subendocardial infarct, which can occur without complete occlusion of an artery because the inner third is already at the edge of its supply.

· The subendocardium is the watershed of the ventricular wall.

Distinctions that matter

· Epicardial coronary disease is about the vessel; subendocardial vulnerability is about the layer, and the two can produce ischemia independently.

· A subendocardial infarct reflects supply falling below demand across a whole layer, while a transmural infarct follows occlusion of a single artery.

Common misconceptions

· Expecting the layer nearest the blood in the cavity to be the best supplied. Cavity blood does not perfuse the wall; the coronary arteries do, from outside inward.

· Leaving systolic compression out of the explanation. Without it there is no reason the inner wall depends on diastole.

Where it is taught: Section C01, flow diagram on why the subendocardium is injured first of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C01-CA01

Clinical application: Chest tightness on exertion in a man with a thick left ventricle

A competent response contains

· Recognize: concentric hypertrophy from pressure overload, here aortic stenosis with a slow low volume carotid upstroke and a systolic murmur, with sarcomeres added in parallel.

· Explain: at a radius of 2.4 cm and 210 mm Hg, a 1.0 cm wall gives (210 x 2.4) divided by 2, which is 252 units, and a 1.8 cm wall gives (210 x 2.4) divided by 3.6, which is about 140 units, so thickening brought the load on each myocyte back toward normal.

· Differentiate: a plaque limits flow in a named epicardial artery, while this heart has smooth arteries and an oxygen demand that the supply cannot meet because the wall is thick, stiff, and has outgrown its capillary bed; the angiogram showing no narrowing is what separates them.

· Investigate: the subendocardium is at greatest risk, and on a stained section look for enlarged myocytes with large dark rectangular boxcar nuclei, the ratio of capillaries to enlarged myocytes, and interstitial fibrosis or myocyte loss concentrated in the inner third.

· Decide: necrosis would appear first in the inner third of the left ventricular wall, giving a subendocardial infarct.

· Justify: the coronary arteries run in epicardial fat and penetrate inward, systole compresses the inner vessels so inner flow is mainly diastolic, and wall stress and oxygen demand are highest in the inner layers, so the inner third is where supply fails first even with patent arteries.

Distinctions that matter

· Ischemia from demand outstripping supply across a layer is not the same as ischemia from a narrowed artery, even though the symptom is the same.

· Hypertrophy lowers wall stress while raising total oxygen demand, so the adaptation and the vulnerability come from the same change.

Common misconceptions

· Assuming exertional chest tightness always means an obstructive plaque. A thick ventricle with smooth arteries can be ischemic on demand alone.

· Expecting the outer wall to be injured first because it is furthest from the cavity. Supply arrives from the outside, so the inner third is the watershed.

Where it is taught: Section C01, myocardium paragraph, the worked example on wall stress, and the subendocardial flow diagram of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw a cardiac myocyte and its intercalated disc

A competent response contains

· A ventricular myocyte is a short branched cylinder, about 10 to 20 micrometres wide and 80 to 100 micrometres long, with one or two central nuclei.

· The clear perinuclear zone holds the Golgi apparatus, glycogen, and lipofuscin, a wear and tear pigment that accumulates with age.

· Mitochondria fill about a third or more of the cell volume, with close to one capillary per myocyte.

· Nearly all ATP comes from oxidative phosphorylation, mainly of fatty acids.

· Contraction stops within about a minute of complete ischemia, and irreversible injury begins after roughly 20 to 40 minutes.

· Myocytes join end to end at intercalated discs, dark stepped lines across the fibre; the transverse portion sits at the level of a Z line and holds the mechanical junctions, and the lateral portion holds most of the gap junctions.

· Fascia adherens: cadherins linked to the actin of the last sarcomere; it transmits contractile pull to the next cell and replaces the Z line at the cell end.

· Desmosome: desmosomal cadherins linked to desmin through plakoglobin, plakophilin, and desmoplakin; it keeps cells from pulling apart under repeated stress.

· Gap junction: connexons of connexin 43 in ventricular muscle; it passes ions between cells and makes the myocardium a functional syncytium.

· The T tubule pairs with one terminal cistern of sarcoplasmic reticulum, which is a diad, and it sits at the Z line.

· Cardiac T tubules are wide, and atrial myocytes have few of them.

Distinctions that matter

· Two junctions of the disc couple cells mechanically and one couples them electrically, and they sit in different portions of the disc.

· A diad has one cistern at the Z line, while a skeletal triad has two cisternae at the A and I band junction.

Common misconceptions

· Writing that the intercalated disc is a single junction. It is three junction types with separate proteins and separate jobs.

· Placing the nuclei at the periphery of the cell. Cardiac nuclei are central and few; peripheral and numerous nuclei belong to skeletal fibres.

Where it is taught: Section C02, paragraphs on the myocyte and intercalated discs, and the table of disc junctions of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-BD2

Brain dump 2 · integration and application: When the junctions fail

A competent response contains

· Heart 1: the cells stay attached because adhesion depends on the desmosomes and fascia adherens, but excitation spreads slowly and irregularly because current passes between myocytes mainly through gap junctions, which is a substrate for reentrant arrhythmia.

· Heart 2: the cells pull apart under stretch because the desmosomes hold them together; the lost myocytes are replaced by fibrofatty tissue, mostly in the right ventricle, and the result is ventricular arrhythmia and sudden death in young people.

· Heart 2 is arrhythmogenic cardiomyopathy, caused by mutations in desmosomal proteins such as plakophilin 2 or desmoplakin.

· The recessive form is Naxos disease, from a plakoglobin defect, which adds woolly hair and thickened palm and sole skin because the same desmosomal protein holds the epidermis together.

· Heart 3: adhesion is normal, but gap junctions shifted to the lateral borders slow conduction and change its direction, since conduction is normally faster along the fibre axis than across it; this occurs at the border of an old infarct and in a failing ventricle and creates a substrate for reentry.

· Adult cardiomyocytes are essentially postmitotic and have no satellite cell reserve.

· The heart therefore adapts to overload by hypertrophy and heals an infarct with collagen scar rather than new muscle.

Distinctions that matter

· Mechanical failure and electrical failure follow different junctions: desmosomes and fascia adherens for adhesion, gap junctions for current.

· Losing gap junctions and relocating them produce the same conduction problem by different routes, which is why both are arrhythmogenic.

Common misconceptions

· Expecting cells to separate when connexin 43 falls. Gap junctions pass current and contribute little to holding cells together.

· Explaining the hair and skin findings in Naxos disease as unrelated. The same desmosomal protein is used in the epidermis.

Where it is taught: Section C02, paragraph on gap junctions and desmosomal disease, the calcium handling paragraph, and the stop and predict box of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-CP01

Comparison table: Cardiac muscle against skeletal muscle

A competent response contains

· Cell shape and nuclei: cardiac cells are short and branched with one or two central nuclei; skeletal fibres are long and unbranched with many peripheral nuclei.

· Coupling: cardiac cells join at intercalated discs and behave as a functional syncytium; skeletal fibres have no junctions between them and each has its own motor ending.

· T tubules: cardiac tubules are wide, sit at the Z line, and pair with one cistern as a diad; skeletal tubules are narrow, sit at the A and I band junction, and pair with two cisternae as a triad.

· Trigger for sarcoplasmic reticulum release: in cardiac muscle calcium entering through L-type channels opens ryanodine receptor 2, which is calcium induced calcium release; in skeletal muscle the voltage sensor is mechanically linked to ryanodine receptor 1.

· Extracellular calcium: required in cardiac muscle and not required in skeletal muscle.

· Tetanus: not possible in cardiac muscle because of the long refractory period, and possible in skeletal muscle.

· Grading of force: cardiac force is graded by how much calcium enters and how loaded the sarcoplasmic reticulum is, since more cells cannot be recruited; skeletal force is graded by recruiting more motor units and by firing rate.

· Regeneration: minimal in cardiac muscle, which heals by scar; skeletal muscle has satellite cells that allow repair.

· What the cardiac version achieves: contraction as one unit, a chamber that cannot fuse into sustained contraction and so always refills, graded force without recruitment, and a cell built to work continuously for a lifetime.

Distinctions that matter

· Both muscle types are striated with the same sarcomere, so every difference in the table is about coordination, calcium supply, or repair, not about the contractile machinery.

· The diad against the triad is a structural difference; needing extracellular calcium is the functional consequence that follows from it.

Common misconceptions

· Saying the heart recruits more fibres to contract harder. There is no recruitment in a syncytium; force is graded by calcium.

· Treating the inability to tetanize as a weakness. It is what guarantees the ventricle relaxes and refills between beats.

Where it is taught: Section C02, calcium handling paragraph and the table comparing cardiac with skeletal muscle of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C02-Q01

Board-style question: Sudden death in a young athlete with a fibrofatty right ventricle

Correct answer: C. Plakoglobin of the desmosome

Why it is right: Plakoglobin is a desmosomal protein that links desmosomal cadherins to desmin, and the recessive defect, Naxos disease, lets myocytes detach under stretch so they are replaced by fibrofatty tissue in the right ventricle, while the same protein in epidermal desmosomes accounts for the woolly hair and the thickened palm and sole skin.

A is wrong: Loss of gap junction connexons slows and disorganizes conduction but leaves adhesion intact, so myocytes would not be pulled apart and there would be no fibrofatty replacement or skin and hair findings.

B is wrong: A defective cadherin to actin link in the fascia adherens would impair transmission of contractile force between cells, and it is not the junction shared with the epidermis that explains the hair and skin.

D is wrong: A ryanodine receptor 2 defect causes catecholamine triggered arrhythmia through disordered calcium release, with a structurally normal ventricle and no fibrofatty replacement or skin findings.

A competent response contains

· Myocytes pulled apart with fibrofatty replacement, mostly of the right ventricle, with ventricular arrhythmia and sudden death in a young person, is arrhythmogenic cardiomyopathy from a desmosomal protein defect.

· Woolly hair with thickened palm and sole skin identifies the recessive plakoglobin form, Naxos disease, because plakoglobin is also used by epidermal desmosomes.

Distinctions that matter

· Desmosomal failure separates cells mechanically; gap junction failure slows conduction without separating them.

Common misconceptions

· Reading any inherited ventricular arrhythmia as an ion channel defect. Here the histology shows cells pulled apart, which points to adhesion.

Where it is taught: Section C02 of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C03-BD1

Brain dump 1 · foundational retrieval: Purkinje fibers, working myocytes, and nodal cells

A competent response contains

· Purkinje fibers form a network in the subendocardium of both ventricles and are continuous with the bundle branches.

· They are larger than working myocytes, with pale glycogen rich cytoplasm that is periodic acid Schiff positive, a few myofibrils pushed to the periphery, and one or two central nuclei often ringed by a clear halo.

· Three features make them fast: a large diameter that lowers internal resistance to longitudinal current, abundant gap junctions that lower resistance between cells, and a large fast sodium current driving depolarization.

· Their conduction velocity is about 1.5 to 4 m/s, the fastest in the heart, so the ventricles depolarize almost together and the normal QRS is narrow.

· Table rows: size and staining, largest and pale and PAS positive for Purkinje, intermediate and eosinophilic and densely striated for the working myocyte, smallest and pale in dense connective tissue for the nodal cell.

· Myofibrils: few and peripheral in Purkinje fibers, abundant in working myocytes, few in nodal cells.

· Gap junctions: abundant in Purkinje fibers and working myocytes, sparse in nodal cells.

· Conduction velocity: about 1.5 to 4 m/s in Purkinje fibers, about 0.3 to 0.5 m/s in working ventricular muscle, and about 0.05 m/s in the AV node.

· Main role: rapid distribution of the impulse, force, and pacemaking with AV delay respectively.

· Purkinje fibers pace at about 20 to 40 beats per minute, which appears as a wide complex ventricular escape rhythm.

· The cerebellar Purkinje neuron is an unrelated cell that shares the name and is a common matching trap.

Distinctions that matter

· Conduction velocity and pacing rate are separate properties: Purkinje fibers conduct fastest and pace slowest.

· Pale cytoplasm has two different causes here: glycogen in the Purkinje fiber and sparse contractile material in the nodal cell.

Common misconceptions

· Expecting the fastest conducting cell to be packed with myofibrils. Purkinje fibers trade contractile machinery for speed.

· Placing Purkinje fibers in the mid myocardium. They lie in the subendocardium, which is also why surviving ones sit at the inner edge of an infarct.

Where it is taught: Section C03, Purkinje fiber paragraphs and the comparison table of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C03-BD2

Brain dump 2 · integration and application: Read two sections and follow the consequences

A competent response contains

· The first section shows Purkinje fibers, built for rapid distribution of the impulse through the ventricles.

· Their glycogen supports anaerobic glycolysis, which is thought to help them tolerate ischemia; together with diffusion from cavity blood this lets fibers at the inner edge of an infarct survive.

· Those surviving fibers can become a source of ectopic beats.

· The second section shows atrial myocytes with granules storing atrial natriuretic peptide, and the granules are most numerous in the right atrium.

· ANP is released when increased volume or pressure stretches the atrium, which makes the heart an endocrine organ.

· ANP binds natriuretic peptide receptor A, which carries its own guanylyl cyclase domain, so the second messenger is cGMP.

· Its actions: it dilates the afferent and constricts the efferent arteriole to raise GFR, inhibits collecting duct sodium reabsorption, suppresses renin and aldosterone, and relaxes vascular smooth muscle.

· Ventricular myocytes make B type natriuretic peptide, mainly on demand in response to wall stretch rather than storing it in prominent granules.

· Plasma BNP or NT-proBNP rises in heart failure because the ventricular wall is stretched, which is why it is the one measured in blood.

Distinctions that matter

· ANP is stored in atrial granules and released by stretch; BNP is made by the ventricle largely on demand, which is why only one of them shows as prominent granules on electron microscopy.

· The same cell type underlies both sections: both Purkinje fibers and ANP storing atrial cells are modified cardiomyocytes that traded contractile machinery for another job.

Common misconceptions

· Reading glycogen rich pale cells beneath the endothelium as fat or as degenerating muscle. The PAS reaction and the peripheral myofibrils identify Purkinje fibers.

· Assuming all natriuretic peptide comes from the atrium. Ventricular BNP is the one that rises in heart failure.

Where it is taught: Section C03, Purkinje glycogen paragraph, atrial granule paragraph, and the ANP receptor paragraph of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C03-RT01

Retrieval exercise: Rebuild the atrial stretch feedback loop

A competent response contains

· Step 1: blood volume or atrial pressure rises.

· Step 2: atrial myocytes stretch and release ANP from granules.

· Step 3: ANP binds natriuretic peptide receptor A and raises cGMP.

· Step 4: GFR rises while sodium reabsorption, renin, and aldosterone fall.

· Step 5: sodium and water loss lowers volume and atrial stretch.

· The releasing cells are atrial myocytes, the hormone was stored in membrane bound dense granules near the nucleus and Golgi apparatus, and the granules are most numerous in the right atrium.

· Natriuretic peptide receptor A carries its own guanylyl cyclase domain, so the second messenger is cGMP.

· At the kidney the afferent arteriole dilates and the efferent arteriole constricts, which raises glomerular filtration.

· This is negative feedback: the last step removes the stretch that released the hormone.

· ANP also relaxes vascular smooth muscle, which lowers the load against which the heart works.

Distinctions that matter

· ANP and nitric oxide both act through cGMP, but ANP uses a receptor with its own cyclase domain while nitric oxide uses soluble guanylyl cyclase inside the target cell.

· ANP raises GFR by acting on both arterioles in opposite directions, not by raising systemic pressure.

Common misconceptions

· Calling this positive feedback because the hormone rises with the stimulus. The output of the loop cancels the stimulus, which is negative feedback.

· Expecting ANP to act through cAMP by analogy with other peptide hormones. Its receptor makes cGMP directly.

Where it is taught: Section C03, cycle diagram on atrial stretch and natriuretic peptide feedback of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the wall of every vessel class

A competent response contains

· All arteries and veins have an intima of endothelium and thin subendothelial tissue, bounded in arteries by the internal elastic lamina, a media of smooth muscle and the matrix it secretes, and a mainly collagenous adventitia carrying nerves and, in large vessels, the vasa vasorum.

· Elastic artery, such as the aorta, pulmonary trunk, and major branches: thick intima and a media of about 40 to 70 fenestrated elastic lamellae; it stores energy in systole and recoils in diastole.

· Muscular artery, such as the coronary, radial, and renal: a sharp wavy internal elastic lamina and a thick smooth muscle media of up to about 40 layers; it distributes flow among organs.

· Arteriole: one or two smooth muscle layers, with an elastic lamina only in the larger ones; it is the main resistance vessel and controls capillary inflow, and the cutoff is generally under about 0.1 mm.

· Capillary: endothelium and basal lamina with pericytes only, about 5 to 10 micrometres wide; it is the exchange vessel, and its huge total cross section slows flow for exchange.

· Postcapillary venule: endothelium, pericytes, and loose junctions; it is the site of leukocyte emigration and inflammatory leak.

· Medium vein: thin media with intimal folds forming valves, for low pressure return.

· Large vein: thin media with the adventitia thickest and carrying longitudinal muscle, acting as a capacitance reservoir.

· The internal elastic lamina is most distinct in muscular arteries.

· The vasa vasorum run in the adventitia and outer media and supply the part of the wall too far from the lumen to be fed by luminal diffusion; they are more plentiful in veins.

· Valves are intimal folds in limb veins and work with the skeletal muscle pump; they are absent from the venae cavae, the portal system, and the dural sinuses.

· Veins are thin walled and compliant and hold about 60% to 70% of blood volume.

Distinctions that matter

· Arteries are built around the media and large veins around the adventitia, which is the quickest way to tell a large vein from an artery on a section.

· The elastic lamellae of an elastic artery and the smooth muscle layers of a muscular artery are both in the media, but one stores energy and the other sets calibre.

Common misconceptions

· Expecting the thickest wall layer to be the media in every vessel. In a large vein the adventitia is the thickest layer.

· Treating the vasa vasorum as a minor detail. They are why syphilitic aortitis damages the outer media and why a thick wall needs its own supply.

Where it is taught: Section C04, opening paragraph, the table of vessel classes, and the paragraphs on capillaries, venules, and veins of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C04-BD2

Brain dump 2 · integration and application: Which layer is damaged, and what the pressures do

A competent response contains

· High systolic with low diastolic pressure in an older patient: the elastin of the elastic artery media has fragmented and been replaced by collagen, so the aorta cannot store and return ejection energy, giving isolated systolic hypertension.

· Fibrillin 1 defect with a dilated root and tearing pain: Marfan syndrome weakens elastin in the media, producing cystic medial degeneration, aortic root dilation, and dissection, which splits the media.

· Tertiary syphilis with a tree bark intima: obliterative endarteritis of the vasa vasorum in the adventitia makes the outer media ischemic, so the ascending aorta becomes aneurysmal.

· Pipestem calcification with no luminal narrowing: Monckeberg medial calcific sclerosis calcifies the internal elastic lamina and the media of muscular arteries.

· Thickened renal arterioles with pink material in hypertension and diabetes: hyaline arteriolosclerosis deposits leaked plasma protein in the wall, and in severe hypertension hyperplastic arteriolosclerosis adds concentric onion skin smooth muscle; both narrow the lumen and reduce renal perfusion.

· Atherosclerosis, by contrast, affects the intima of elastic and muscular arteries.

· Resistance varies with the inverse fourth power of the radius, so a 20% narrowing raises resistance about 2.4 times, since 1 divided by 0.8 to the fourth power is about 2.44.

· Pressure falls across the arterioles from about 85 mmHg to about 30 to 35 mmHg, the largest drop anywhere in the circulation, because they combine a small radius with smooth muscle that can change it.

· In systole the aorta stretches and stores ejection energy; in diastole it recoils, maintaining diastolic pressure and nearly continuous distal flow, the windkessel effect.

· Loss of elastin raises systolic pressure, lowers diastolic pressure, and widens pulse pressure.

Distinctions that matter

· Intimal disease narrows the lumen, medial disease weakens or stiffens the wall, and adventitial disease starves the outer wall, so naming the layer predicts the clinical picture.

· Monckeberg calcification and atherosclerosis both calcify, but one sits in the media without narrowing the lumen and the other sits in the intima and does narrow it.

Common misconceptions

· Reading a wide pulse pressure in an older patient as a strong heart. It reflects a stiff aorta that no longer absorbs the stroke volume.

· Assuming a small narrowing has a small effect on resistance. The fourth power relationship means a 20% narrowing more than doubles it.

Where it is taught: Section C04, paragraphs on arteries and arterioles, the flow diagram on elastic recoil, and the arteriolar paragraph of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C04-Q02

Board-style question: Capillary type and what can leave the vessel

Correct answer: A. Whole red and white blood cells

Why it is right: Large intercellular gaps together with a discontinuous basal lamina leave no barrier to cells, which is why sinusoidal capillaries in the liver, spleen, and bone marrow allow whole blood cells and large proteins to cross the wall.

B is wrong: Water and dissolved gases alone describe a continuous capillary with unbroken endothelium and an intact basal lamina, as in muscle, lung, and the nervous system.

C is wrong: Passing small hormones while holding back albumin describes a fenestrated capillary, whose pores of about 60 to 80 nm sit in an otherwise continuous wall over a continuous basal lamina.

D is wrong: A wall sealed by tight junctions describes the continuous capillaries of the blood brain barrier, which is the opposite of the discontinuous wall in this micrograph.

A competent response contains

· A wide lumen with large gaps between endothelial cells and a discontinuous basal lamina describes a sinusoidal capillary.

· Sinusoidal capillaries are found in liver, spleen, and bone marrow, and they let large proteins and whole cells pass.

· Permeability rises from continuous to fenestrated to sinusoidal.

Distinctions that matter

· Fenestrated capillaries have pores of about 60 to 80 nm within an otherwise continuous wall and basal lamina, so they pass fluid and hormones but not cells.

Common misconceptions

· Treating any gap in a capillary wall as a fenestration. Fenestrations are pores in the cell; sinusoidal gaps are spaces between cells with a broken basal lamina.

Where it is taught: Section C04 of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the endothelial product table and the nitric oxide pathway

A competent response contains

· Nitric oxide: soluble guanylyl cyclase and cGMP; relaxation, with inhibition of platelets and leukocyte adhesion.

· Prostacyclin: IP receptor, Gs, cAMP; relaxation and inhibition of platelet aggregation.

· Endothelin-1: ETA receptor, Gq, IP3 and calcium; strong sustained constriction and smooth muscle growth.

· von Willebrand factor: bridges collagen to platelet GPIb and carries factor VIII; platelet adhesion.

· P-selectin: binds sialyl Lewis X on leukocytes; leukocyte rolling.

· Thrombomodulin: binds thrombin, which then activates protein C; anticoagulant.

· Heparan sulfate: enhances antithrombin; anticoagulant.

· Tissue plasminogen activator: converts plasminogen to plasmin; fibrinolytic.

· Endothelial nitric oxide synthase makes nitric oxide and L-citrulline from L-arginine and oxygen, using NADPH and tetrahydrobiopterin.

· Besides relaxing smooth muscle, nitric oxide inhibits platelet adhesion, platelet aggregation, leukocyte adhesion, and smooth muscle proliferation.

· Superoxide and hemoglobin destroy nitric oxide within seconds, so its action is local.

· Weibel-Palade bodies are rod shaped endothelial granules holding von Willebrand factor with P-selectin in their membrane, released within minutes by thrombin, histamine, epinephrine, or vasopressin acting on V2 receptors.

· von Willebrand factor bridges exposed collagen to platelet glycoprotein Ib, which matters most at high shear, and carries factor VIII, so severe deficiency lowers factor VIII.

· ADAMTS13 trims ultralarge von Willebrand factor multimers, and its deficiency lets them aggregate platelets in small vessels, causing thrombotic thrombocytopenic purpura.

· Preformed P-selectin starts leukocyte rolling within minutes, whereas E-selectin is synthesized over hours after interleukin 1 or tumor necrosis factor.

Distinctions that matter

· Nitric oxide and prostacyclin both relax and both inhibit platelets, but one works through cGMP and the other through cAMP.

· P-selectin is preformed and released in minutes, while E-selectin has to be made over hours, which is why the two act at different points of an inflammatory response.

Common misconceptions

· Treating nitric oxide as a circulating signal. It survives seconds and acts on the smooth muscle immediately beneath the endothelium.

· Thinking von Willebrand factor only sticks platelets down. It also carries factor VIII, which is why a severe deficiency affects the coagulation cascade too.

Where it is taught: Section C05, nitric oxide paragraph, the paragraph on Weibel-Palade bodies, and the table of endothelial products of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-BD2

Brain dump 2 · integration and application: Two vessels, one drug, and the same receptor

A competent response contains

· In the intact ring, acetylcholine binds endothelial M3 receptors, raises endothelial calcium, activates eNOS, and the nitric oxide produced relaxes the underlying smooth muscle, outweighing the direct effect on muscle.

· In the denuded ring, acetylcholine reaches only smooth muscle M3 receptors, which act through Gq, IP3, and calcium to cause contraction.

· The same receptor on two different cells therefore moves the vessel in opposite directions, which is why the response is a test of the endothelium rather than of the muscle.

· Infused into a human coronary artery, acetylcholine dilates a healthy vessel but constricts one with atherosclerosis or risk factor driven dysfunction, revealing lost bioavailable nitric oxide.

· This matters because light microscopy shows structure, not function, so a normal looking endothelium can be profoundly dysfunctional and is detected by its response to flow or acetylcholine.

· Rising flow raises shear stress and releases nitric oxide, so the upstream artery widens, which is flow mediated dilation.

· Aspirin irreversibly acetylates cyclooxygenase.

· Anucleate platelets cannot replace the enzyme, while endothelial cells can resynthesize it, so a low dose shifts the balance away from platelet thromboxane A2 and toward endothelial prostacyclin.

Distinctions that matter

· The acetylcholine response reports endothelial function; the drug itself is not vasodilator or vasoconstrictor in isolation.

· Thromboxane A2 and prostacyclin come from the same enzyme in two different cells, which is what makes a dose difference possible.

Common misconceptions

· Describing acetylcholine as a vasodilator without qualification. It dilates only when a functioning endothelium is there to make nitric oxide.

· Expecting a normal looking endothelium on a section to be working normally. Function has to be tested with flow or acetylcholine.

Where it is taught: Section C05, nitric oxide paragraph, the worked example on the acetylcholine paradox, the prostacyclin paragraph, and the integration note on structure against function of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-RT01

Retrieval exercise: Rebuild the nitric oxide relaxation pathway

A competent response contains

· Step 1: shear stress, acetylcholine, or bradykinin act on the endothelium.

· Step 2: calcium with calmodulin, together with phosphorylation, activate eNOS.

· Step 3: nitric oxide diffuses into the underlying smooth muscle.

· Step 4: soluble guanylyl cyclase raises cGMP and protein kinase G is activated.

· Step 5: calcium falls and myosin light chain phosphatase is activated.

· Step 6: the smooth muscle relaxes and the vessel dilates.

· The same nitric oxide and cGMP signal inhibits platelets, which is why the healthy endothelium keeps blood fluid and vessels relaxed at the same time.

· Atrial natriuretic peptide raises the same second messenger, cGMP, through natriuretic peptide receptor A, which carries its own guanylyl cyclase domain.

Distinctions that matter

· Nitric oxide is made in the endothelium and acts in the smooth muscle, so the signalling steps are split between two cells.

· Relaxation here comes from both less calcium and more phosphatase activity on myosin, not from calcium alone.

Common misconceptions

· Putting the cyclase in the endothelial cell. Soluble guanylyl cyclase is in the smooth muscle cell that nitric oxide diffuses into.

· Expecting nitric oxide to use a surface receptor. It is a small gas that crosses the membrane and binds its target enzyme directly.

Where it is taught: Section C05, flow diagram on nitric oxide and vascular relaxation, with the ANP receptor in Section C03 of the CV-2 Histology of the Heart and Blood Vessels notes.

NPLEX-P1-CARD-02-01-C05-PR01

Prediction exercise: Predict what happens to an endothelium under oxidative stress

A competent response contains

· Superoxide rises, because smoking, diabetes, hypertension, and oxidized LDL all raise endothelial oxidative stress.

· Available nitric oxide falls, because superoxide destroys it within seconds, and reduced bioavailable nitric oxide is the hallmark of endothelial dysfunction.

· Once tetrahydrobiopterin runs low, eNOS uncouples and produces superoxide instead of nitric oxide, so the enzyme adds to the problem it normally opposes.

· Endothelin-1 rises and the vessel constricts, since nitric oxide, prostacyclin, and laminar shear normally suppress it.

· VCAM-1, ICAM-1, and the selectins rise, so monocytes adhere and enter the intima along with LDL.

· Tissue factor and plasminogen activator inhibitor 1 rise while thrombomodulin falls, so the surface becomes prothrombotic, which is the endothelial injury of Virchow's triad.

· Infused acetylcholine constricts this vessel instead of dilating it, because endothelial nitric oxide no longer outweighs the direct effect on smooth muscle.

· Lesions appear first at branch points and curves, where shear is low and disturbed rather than laminar.

· The cycle is self reinforcing: the intimal lesion generates more reactive oxygen species, which drives progression toward atherosclerosis.

Distinctions that matter

· Endothelial dysfunction is a shift in the balance of products, not a loss of the endothelium, which is why the vessel can look normal on a section.

· Low laminar shear promotes dysfunction while high laminar shear protects, so the location of plaques follows flow rather than the burden of risk factors.

Common misconceptions

· Expecting an enzyme that makes nitric oxide to simply make less of it. Uncoupled eNOS makes superoxide, so it switches sides.

· Assuming plaques form where flow is fastest. They form where shear is low and disturbed, at branch points and curves.

Where it is taught: Section C05, endothelial dysfunction paragraph and the cycle diagram, with the acetylcholine example earlier in the section of the CV-2 Histology of the Heart and Blood Vessels notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-03

CV-3 Heart, Great Vessels, and Pericardium

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-03-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the heart in the chest and label what forms each edge

A competent response contains

· The heart lies in the middle mediastinum with the pericardium and the roots of the great vessels; the transverse thoracic plane through the sternal angle and the T4 to T5 disc separates the superior from the inferior mediastinum.

· Outline on the chest wall: superior border from the left second costal cartilage to the right third cartilage, right border down to the right sixth cartilage just beside the sternum, inferior border across to the left fifth intercostal space, left border back up to the left second cartilage.

· The apex lies behind the left fifth intercostal space at or just medial to the midclavicular line, about 9 cm from the midline, and the apex beat is normally no larger than one intercostal space.

· Anterior or sternocostal surface: mainly right ventricle, with the right atrium on the right and the left ventricle along the left edge.

· Inferior or diaphragmatic surface: mainly left ventricle with some right ventricle.

· Base, which is the posterior surface: mainly left atrium with some right atrium.

· Apex: left ventricle.

· Right border: right atrium, between the two venae cavae.

· Left border: left ventricle, with the left auricle near the top.

· Inferior border: right ventricle, with the left ventricle near the apex.

· The heart is rotated so that the base faces backward and to the right and the apex points forward, downward and to the left, with about two thirds of the heart left of the median plane, so the right side faces forward and the right ventricle forms most of the front.

Distinctions that matter

· The base is the posterior surface opposite the apex, not the upper border where the great vessels leave.

· The left ventricle forms the apex and the left border but only a strip of the front; the right ventricle forms most of the anterior surface.

Common misconceptions

· Placing the left ventricle at the front because it is the pumping chamber for the body. Looping of the heart tube rotated the right ventricle forward.

· Putting the apex beat in the midclavicular line in the fourth space. It lies behind the fifth intercostal space at or just medial to that line.

Where it is taught: Section C01 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C01-BD2

Brain dump 2 · integration and application: Read the chest film and the apex beat

A competent response contains

· Right contour from top to bottom: superior vena cava, then right atrium.

· Left contour from top to bottom: aortic knob, pulmonary trunk, left auricle, left ventricle.

· The right ventricle and left atrium form no contour on the posteroanterior film because they are the most anterior and most posterior chambers; both are judged on the lateral view, where the right ventricle lies behind the lower sternum and the left atrium above the left ventricle posteriorly.

· The cardiothoracic ratio is the widest transverse cardiac diameter divided by the widest internal thoracic diameter on a posteroanterior film in full inspiration, and it is normally below 0.5.

· In the woman the left ventricle has dilated under the volume load of chronic mitral regurgitation; it pushes the left border down and out, alters the lower left contour and raises the ratio above 0.5.

· The apex beat has moved because the enlarged left ventricle still forms the apex, so the apex now sits lateral to the midclavicular line and often in the sixth intercostal space.

· The portable film overstates heart size because an anteroposterior projection puts the heart farther from the detector and magnifies it, and a shallow breath raises the diaphragm and widens the cardiac shadow.

· Before calling his heart enlarged, repeat the measurement on a posteroanterior film taken in full inspiration.

Distinctions that matter

· An edge is visible on a radiograph only where the heart touches air filled lung, which is why chamber position and not chamber size decides which chamber forms a contour.

· A raised ratio reports a wide shadow, which is not the same as a large heart until projection and depth of inspiration are accounted for.

Common misconceptions

· Reading any ratio above 0.5 as cardiomegaly regardless of the film. Projection and depth of breath change the number.

· Expecting left atrial enlargement to bulge a contour in the way the left ventricle does. It shows indirectly, through a straight left border and a double density.

Where it is taught: Section C01, radiograph paragraphs and the table of surfaces, borders and contours of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-BD1

Brain dump 1 · foundational retrieval: Lay out the chambers from front to back with their neighbors

A competent response contains

· Front to back order: right ventricle, then right atrium and left ventricle together, then left atrium.

· The right ventricle sits directly behind the sternum and the left fourth and fifth costal cartilages.

· The left atrium is the most posterior chamber; the esophagus lies immediately behind it, the descending aorta behind and to its left, and the tracheal bifurcation with the left main bronchus just above it.

· Only the pericardium and the oblique pericardial sinus separate the left atrium from the esophagus.

· Looping of the heart tube rotated the right ventricle to the front and left the left atrium at the back.

Distinctions that matter

· The right atrium and left ventricle lie at the same depth in the middle of the order, so neither is the most anterior or the most posterior chamber.

· The descending aorta is a posterior mediastinal neighbor, while the esophagus passes immediately behind the pericardium.

Common misconceptions

· Naming the right atrium as the most anterior chamber because it forms the right border. The right ventricle is the front chamber.

· Forgetting that the oblique sinus lies between the left atrium and the esophagus, which is why a probe in the esophagus is still outside the heart.

Where it is taught: Section C02, opening paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-BD2

Brain dump 2 · integration and application: Trace each finding of left atrial enlargement to a neighbor

A competent response contains

· Breathlessness: raised left atrial pressure passes back through the pulmonary veins to the pulmonary capillaries and increases filtration into the lung.

· Dysphagia for solids: the enlarging left atrium compresses the esophagus from in front, because the esophagus lies immediately behind it.

· Hoarseness: the enlarged atrium with a dilated pulmonary artery stretches the left recurrent laryngeal nerve under the aortic arch.

· Straight left border: the enlarged left auricle straightens the left border below the pulmonary trunk.

· Double density inside the right border: the atrium projects to the right behind the right atrium.

· Raised left main bronchus with a widened carinal angle: the bronchus sits just above the atrium and is lifted as the atrium expands upward.

· The atrium can expand only backward and upward, because the ventricles lie in front of it and block forward expansion.

· Her history of rheumatic fever points to mitral stenosis: the narrowed valve raises left atrial pressure, the atrium dilates over months to years, and the pressure is transmitted back to the pulmonary capillaries.

· A probe in the esophagus lies millimeters from the left atrium with no lung or chest wall in between, so transesophageal echocardiography images the atrium and auricle far better than a chest wall probe and is the preferred way to look for thrombus in the auricle.

Distinctions that matter

· Every finding outside the heart comes from a neighbor of the left atrium, while the breathlessness comes from pressure transmitted upstream through the pulmonary veins.

· The straight left border is the auricle and the double density is the body of the atrium, so the two signs report different parts of the same chamber.

Common misconceptions

· Explaining the dysphagia by an aortic cause. The indentation is on the front of the esophagus, which places the compressing structure in front of it.

· Expecting the enlarged left atrium to push the sternum forward. The ventricles sit in front of it, so it enlarges backward and upward.

Where it is taught: Section C02, left atrium paragraph, the flow figure and the worked example of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-RT01

Retrieval exercise: Rebuild the chain from a narrowed mitral valve to hoarseness

A competent response contains

· Step 1: a narrowed mitral valve raises left atrial pressure.

· Step 2: the left atrium dilates over months to years.

· Step 3: the atrium can expand only backward and upward, because the ventricles block the way forward.

· Step 4: the compressed esophagus causes dysphagia, with an indentation on the front of the barium filled esophagus.

· Step 5: the left main bronchus is lifted and the carinal angle widens.

· Step 6: the stretched left recurrent laryngeal nerve causes hoarseness; the nerve hooks under the aortic arch at the ligamentum arteriosum, and a dilated pulmonary artery from secondary pulmonary hypertension adds to the stretch.

· On the posteroanterior film the same enlargement straightens the left border below the pulmonary trunk and produces a double density on the right.

· Slow flow in the dilated atrium favors atrial fibrillation and thrombus in the left auricle.

Distinctions that matter

· The symptoms belong to the neighbors of the atrium, while the rhythm and the thrombus belong to flow inside it.

· Pulmonary hypertension here is secondary to the raised left atrial pressure, and the dilated pulmonary artery is what brings the nerve into the picture.

Common misconceptions

· Treating the hoarseness as a separate disease. It is the same enlargement reaching the nerve in the aortopulmonary window.

· Expecting dysphagia for liquids first. The compression affects solids.

Where it is taught: Section C02, flow figure on how mitral stenosis produces symptoms outside the heart of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-BD1

Brain dump 1 · foundational retrieval: Build the pericardium layer by layer

A competent response contains

· The fibrous pericardium is dense connective tissue, fused below to the central tendon of the diaphragm, blended above with the adventitia of the great vessels, and tied to the sternum by sternopericardial ligaments.

· Because it is nearly inelastic over short periods it anchors the heart, moves it with the diaphragm and prevents sudden overdistension, but it also lets rapidly entering fluid compress the heart.

· The serous pericardium is a closed mesothelial sac: the parietal layer lines the fibrous sac and reflects around the great vessels to become the visceral layer, the epicardium.

· The pericardial cavity is a potential space between the two serous layers holding about 15 to 50 mL of serous fluid that lets the surfaces glide.

· The heart lies outside the cavity in the way a fist pushed into an underinflated balloon lies outside the balloon cavity, covered by one wall of it.

· Two reflection sleeves: one around the ascending aorta and pulmonary trunk, the other around the venae cavae and pulmonary veins.

· The transverse sinus lies behind the ascending aorta and pulmonary trunk and in front of the superior vena cava; the oblique sinus is a blind pouch behind the left atrium.

· The pericardium is supplied mainly by the pericardiacophrenic arteries from the internal thoracic arteries, and the epicardium by the coronary arteries.

Distinctions that matter

· The fibrous layer is one sac of connective tissue, while the serous pericardium is two layers of mesothelium with the cavity between them.

· The pericardial cavity is a potential space with a few tens of milliliters in it, not an empty chamber.

Common misconceptions

· Drawing the heart inside the serous cavity. The heart is covered by the visceral layer and lies outside the cavity itself.

· Calling the fibrous pericardium rigid in every circumstance. It stretches over weeks, which is why a slow effusion can become large.

Where it is taught: Section C03, fibrous and serous layer paragraphs and the sinus paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-BD2

Brain dump 2 · integration and application: Separate pericardial pain from ischemic pain

A competent response contains

· Pericardial pain travels in the phrenic nerves, which enter the cord at C3 to C5 and descend between the fibrous pericardium and the mediastinal pleura in front of the lung roots.

· The fibrous pericardium and the parietal serous layer are pain sensitive; the visceral layer is largely insensitive. Vagal and sympathetic fibers also reach the pericardium but do not explain the characteristic pain.

· The supraclavicular nerves, C3 and C4, bring skin input from the shoulder into the same segments, so the brain attributes the signal to the skin and the pain is felt at the trapezius ridge.

· Ischemic myocardial pain travels with sympathetic afferents to about T1 to T5 and is felt as central chest pressure spreading to the medial left arm or jaw.

· Comparison: pericardial pain is sharp, felt over the chest and at the trapezius ridge, worse on deep inspiration and on lying flat, and often eased by sitting forward. Ischemic pain is pressure like, central, spreads to the medial left arm or jaw, and changes little with breathing or posture.

· The inspiratory worsening reflects movement against the neighboring pleura; the postural effect is less certain.

· Inflamed serous surfaces rubbing together give a scratchy friction rub with up to three components, matching atrial systole, ventricular systole and rapid filling.

Distinctions that matter

· Referral depends on shared spinal segments, not on any direct connection between pericardium and shoulder.

· Segment level separates the two pains: C3 to C5 by the phrenic nerve sends pain to the shoulder, T1 to T5 by sympathetic afferents sends it to the chest and arm.

Common misconceptions

· Attributing shoulder pain to spread of inflammation up to the shoulder. Nothing travels there; the signal is misattributed in the dorsal horn.

· Expecting the visceral layer to be the source of the pain. It is largely insensitive, and the fibrous and parietal layers carry it.

Where it is taught: Section C03, blood supply and nerves paragraph, the referral flow figure and the final paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-RT01

Retrieval exercise: Rebuild the referral of pericardial pain to the shoulder

A competent response contains

· Step 1: inflammation irritates the parietal pericardium.

· Step 2: phrenic nerve afferents carry the signal; the phrenic nerve comes from C3 to C5 and descends between the fibrous pericardium and the mediastinal pleura in front of the root of the lung.

· Step 3: the fibers enter the cord at C3 to C5.

· Step 4: the same dorsal horn neurons receive shoulder skin input from the supraclavicular nerves, C3 and C4.

· Step 5: the brain attributes the signal to the skin.

· Step 6: pain is felt at the trapezius ridge and shoulder.

· Referral depends on shared spinal segments rather than on any direct connection.

· Irritation of the diaphragm is carried by the same phrenic nerve, so it is felt in the same place.

Distinctions that matter

· The phrenic nerve is the common route for both pericardial and diaphragmatic pain, which is why the two refer alike.

· The convergence happens in the dorsal horn, not in the periphery.

Common misconceptions

· Writing that the pain is referred because the pericardium touches the diaphragm. Contact is not the mechanism; shared segments are.

Where it is taught: Section C03, flow figure on referral of pericardial pain of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C03-CP01

Comparison table: Transverse against oblique pericardial sinus

A competent response contains

· Location: the transverse sinus lies behind the ascending aorta and pulmonary trunk and in front of the superior vena cava; the oblique sinus lies behind the left atrium.

· Boundaries: the transverse sinus has the arterial reflection in front and the venous reflection behind; the oblique sinus is bounded by the reflections around the pulmonary veins and the inferior vena cava.

· Shape: the transverse sinus is a passage open at both ends; the oblique sinus is a blind pouch open only below.

· Origin: the transverse sinus forms when the dorsal mesocardium breaks down during looping; the oblique sinus forms as the venous reflection folds while the pulmonary veins are incorporated into the left atrium.

· Practical importance: a finger or tape passed through the transverse sinus isolates the great arteries from the venous side during surgery; the oblique sinus lies between the left atrium and the esophagus.

· Closest neighbor: the transverse sinus has the great arteries and the superior vena cava around it inside the sac, while the back wall of the oblique sinus lies against the esophagus.

Distinctions that matter

· The transverse sinus is behind the arteries and the oblique sinus is behind the left atrium.

· One is a through passage and the other a blind pouch, which follows from how each reflection formed.

Common misconceptions

· Describing the oblique sinus as a passage behind the aorta. That is the transverse sinus.

· Treating the sinuses as separate cavities. Both are recesses of the one pericardial cavity.

Where it is taught: Section C03, reflections and sinuses paragraph and the sinus table of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-BD1

Brain dump 1 · foundational retrieval: Build the great vessel table from memory

A competent response contains

· Superior vena cava: begins from the brachiocephalic veins behind the right first costal cartilage, ends in the right atrium at the right third costal cartilage, lies to the right of the ascending aorta and receives the azygos vein at about T4.

· Inferior vena cava: begins from the common iliac veins, ends in the lower right atrium, and pierces the central tendon of the diaphragm at about T8.

· Pulmonary veins, usually four: begin in each lung, end in the posterior wall of the left atrium, vary in number and bound the oblique sinus.

· Pulmonary trunk: begins at the conus arteriosus, ends under the arch at the level of the sternal angle, and its right branch passes behind the ascending aorta and superior vena cava.

· Ascending aorta: begins at the aortic valve, becomes the arch at the sternal angle, and its only branches are the coronary arteries.

· Aortic arch: begins behind the right second sternocostal joint, ends on the left side of the T4 to T5 disc, and gives the brachiocephalic trunk, left common carotid and left subclavian arteries from right to left; the left brachiocephalic vein crosses in front of these branches.

· The ascending aorta and pulmonary trunk lie inside the pericardium, which is why a tear in the ascending aorta causes tamponade; the arch and descending aorta lie outside it.

· The ligamentum arteriosum runs from the origin of the left pulmonary artery to the underside of the arch just beyond the origin of the left subclavian artery.

· The aortic isthmus is the segment of aorta between the left subclavian origin and the attachment of the ligamentum arteriosum.

· The aortopulmonary window is the space between the underside of the arch and the left pulmonary artery, and the left recurrent laryngeal nerve passes through it.

· The ligamentum arteriosum is the fibrosed ductus arteriosus, from the distal left sixth arch artery, which in fetal life carried right ventricular output from the left pulmonary artery to the aorta and closed within a day or two of birth.

Distinctions that matter

· The coronary arteries are the only branches of the ascending aorta, so any other branch described as arising there is wrong.

· The isthmus is a segment of aorta while the ligamentum arteriosum is a cord joining two vessels, and the ligamentum marks the lower end of the isthmus.

Common misconceptions

· Placing the whole aorta inside the pericardium. Only the ascending part is intrapericardial.

· Listing the arch branches from left to right. Their order from right to left is brachiocephalic trunk, left common carotid, left subclavian.

Where it is taught: Section C04, great vessel table and the ligamentum arteriosum paragraph of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-BD2

Brain dump 2 · integration and application: Three patients with vessels pressing on neighbors

A competent response contains

· Patient one: a mass in the aortopulmonary window, between the underside of the arch and the left pulmonary artery, stretches or invades the left recurrent laryngeal nerve, which passes through that window after hooking under the arch behind the ligamentum arteriosum.

· The nerve supplies all intrinsic laryngeal muscles on its side except the cricothyroid, so injury paralyzes that vocal fold and gives a hoarse, breathy voice.

· A mass at the same level on the right would be much less likely to change the voice, because the right nerve hooks under the right subclavian artery in the root of the neck well above the arch, the right distal sixth and fifth arch arteries having regressed.

· Patient two: superior vena cava obstruction by a mediastinal tumor or nodes, so the face, neck and arms swell and chest wall veins distend as collaterals carry blood to the inferior vena cava.

· Patient three: an aberrant right subclavian artery arising last from the arch and passing behind the esophagus, which can cause dysphagia; on that side the laryngeal nerve does not recur.

· Position decides which structure suffers in each case: the nerve is in the window, the superior vena cava is the compressible low pressure vessel of the superior mediastinum, and the aberrant artery crosses behind the esophagus.

· The commonest arch variation is a shared origin of the brachiocephalic trunk and left common carotid artery.

Distinctions that matter

· Hoarseness from a lesion in the chest points to the left nerve, because only the left nerve has a long intrathoracic course.

· A vessel compressing the esophagus from behind points to an arch variant, while a chamber compressing it from in front points to the left atrium.

Common misconceptions

· Expecting cricothyroid weakness with a recurrent laryngeal injury. That muscle has a different nerve.

· Assuming swelling of the face and arms means heart failure. The distended chest wall collaterals point to obstruction of the superior vena cava.

Where it is taught: Section C04, first paragraph, arch variation paragraph, aortopulmonary window paragraph and the stop and predict box of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-RT01

Retrieval exercise: Rebuild why the two recurrent laryngeal nerves loop at different levels

A competent response contains

· Step 1: both nerves hook under the sixth arch arteries in the embryo.

· Step 2: the heart and arteries descend, pulling the nerves down with them.

· Step 3: the right distal sixth and fifth arch arteries regress.

· Step 4: the right nerve rises to hook under the right subclavian artery, a fourth arch derivative, in the root of the neck.

· Step 5: the left distal sixth arch artery persists as the ductus arteriosus.

· Step 6: the left nerve stays hooked under the arch at the ligamentum arteriosum.

· Both nerves ascend in the tracheoesophageal groove to the larynx, and the left vagus crosses the arch behind the left phrenic nerve before giving off the left recurrent laryngeal nerve.

· Injuries to the left nerve in the chest: an aneurysm of the arch, lung cancer or nodes in the aortopulmonary window, an enlarged left atrium with a dilated pulmonary artery, and surgery on a patent ductus arteriosus.

· The long intrathoracic course is what exposes the left nerve to disease of the arch, left atrium, pulmonary artery and left lung.

Distinctions that matter

· Both nerves follow the same embryological rule, and only the fate of the sixth arch artery differs between the sides.

· The right subclavian artery is a fourth arch derivative, which is why the right nerve ends up higher.

Common misconceptions

· Explaining the difference by the length of the nerves rather than by which arch artery persisted on each side.

· Expecting the right nerve to be at risk from an arch aneurysm. It never reaches the arch.

Where it is taught: Section C04, flow figure on why the two nerves loop at different levels of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the four chambers from the inside

A competent response contains

· Right atrium: smooth part is the sinus venarum, rough part carries pectinate muscles and the right auricle, and its features are the crista terminalis, the fossa ovalis and the openings of the two venae cavae and the coronary sinus.

· Right ventricle: smooth part is the conus arteriosus, rough part has coarse trabeculae carneae, and its features are the moderator band, three papillary muscles and the supraventricular crest.

· Left atrium: smooth over most of the wall where the pulmonary veins were incorporated, rough only in the left auricle, with four pulmonary vein openings.

· Left ventricle: smooth part is the aortic vestibule, rough part has fine trabeculae carneae, with two papillary muscles and the mitral and aortic valves in fibrous continuity.

· The crista terminalis, with the sulcus terminalis outside, separates the smooth sinus venarum from the pectinate wall.

· The floor of the fossa ovalis is septum primum and its rim or limbus is septum secundum.

· The coronary sinus opens between the opening of the inferior vena cava and the tricuspid orifice.

· The moderator band, the septomarginal trabecula, crosses from the interventricular septum to the anterior papillary muscle of the right ventricle.

· The sinoatrial node lies at the upper end of the crista terminalis near the superior vena cava, and the atrioventricular node lies in the septum just above the opening of the coronary sinus.

· The moderator band carries part of the right bundle branch to the anterior papillary muscle so it contracts on time.

· Right ventricular chordae reach adjacent edges of the tricuspid cusps, while each left ventricular papillary muscle sends chordae to both mitral cusps; the valves move passively and the papillary muscles tighten the chordae in systole so the cusps do not evert.

· Smooth atrial walls come from absorbed venous structures and rough walls from the primitive atrium; the left atrium is smooth because it incorporates the primitive pulmonary vein, which is also why four veins open into it.

· Smooth ventricular walls are the outflow tracts, the conus arteriosus and the aortic vestibule.

Distinctions that matter

· Pectinate muscles belong to the atria and trabeculae carneae to the ventricles.

· The posteromedial papillary muscle of the left ventricle usually has a single arterial supply and ruptures more often after inferior infarction, while the anterolateral has more than one source.

Common misconceptions

· Describing pectinate muscles throughout the left atrium. They are confined to the narrow, hooked left auricle, whose stagnant flow makes it the usual site of thrombus in atrial fibrillation.

· Placing the atrioventricular node at the top of the interatrial septum. It sits in the lower septum just above the coronary sinus opening.

Where it is taught: Section C05, chamber table and the atria and ventricles paragraphs of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C05-BD2

Brain dump 2 · integration and application: Wall thickness, wall stress and two loaded ventricles

A competent response contains

· The left ventricular wall is about 8 to 12 mm and the right about 3 to 5 mm, so the left is two to three times thicker, because peak systolic pressure is about 120 mm Hg on the left and about 25 mm Hg on the right.

· Law of Laplace: wall stress equals pressure times radius divided by twice the wall thickness, and wall stress is a main determinant of myocardial oxygen demand.

· The patient with years of high aortic pressure has a pressure load, adds sarcomeres in parallel, thickens the wall without enlarging the cavity, and returns wall stress toward normal through the thickness term in the denominator.

· The patient with a large valve leak has a volume load, adds sarcomeres in series, enlarges the cavity, and the larger radius raises wall stress through the numerator.

· Both ventricles are equally thick at birth because the ductus arteriosus exposes them to systemic pressure; after birth pulmonary resistance falls and the right wall stays thin.

· The thin right wall handles a volume load easily but tolerates an acute pressure rise poorly, so a large pulmonary embolism can cause right ventricular failure.

· Paradoxical embolism: the embolism raises right heart pressures, right atrial pressure comes to exceed left atrial pressure, the flap of septum primum over the fossa ovalis opens right to left, and a further fragment of thrombus passes into the left heart and aorta, bypassing the lung, to lodge in a cerebral artery.

· In normal life the higher left atrial pressure holds the flap shut; the septa fail to fuse in about one quarter of adults, leaving a probe patent foramen ovale that can open with coughing, straining or pulmonary hypertension.

Distinctions that matter

· Concentric hypertrophy lowers wall stress by thickening the wall, while eccentric hypertrophy raises it by enlarging the radius, and the equation shows why.

· A probe patent foramen ovale is an unfused flap valve rather than a fixed hole, which is why it opens only when the gradient reverses.

Common misconceptions

· Reading a thick wall as a sick wall in itself. Thickening is what keeps wall stress near normal under a pressure load.

· Expecting the right ventricle to fail from a volume load as readily as from a pressure load. The thin wall accommodates volume and struggles with pressure.

Where it is taught: Section C05, wall thickness and law of Laplace paragraphs and the stop and predict box of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C05-PR01

Prediction exercise: Predict what happens when right heart pressure rises above left

A competent response contains

· Pulmonary vascular resistance rises and right ventricular systolic pressure rises, because the embolus obstructs the pulmonary arterial bed.

· Right ventricular wall stress rises through the pressure term of the law of Laplace, and the thin right wall tolerates an acute pressure rise poorly.

· Right atrial pressure rises until it exceeds left atrial pressure, so the gradient across the fossa ovalis reverses and the flap of septum primum, normally held shut by higher left atrial pressure, opens.

· Flow through the fossa ovalis is now right to left rather than absent.

· A fragment of thrombus arriving in the right atrium can cross to the left atrium and out through the aorta, escaping the pulmonary capillary filter, which is paradoxical embolism and can lodge in a cerebral artery.

· The interventricular septum shifts toward the left ventricle as the right ventricle distends, and left ventricular filling falls because the two ventricles share the fixed space inside the pericardium.

Distinctions that matter

· The foramen ovale opens because the gradient reverses, not because the embolus enlarges it.

· Reduced left ventricular filling here comes from septal shift and shared pericardial space, not from a problem in the left ventricle itself.

Common misconceptions

· Expecting a right to left shunt at rest in anyone with a probe patent foramen ovale. The flap stays shut until right atrial pressure exceeds left.

· Assuming a venous clot cannot reach the brain. With a reversed gradient it bypasses the lung entirely.

Where it is taught: Section C05, fossa ovalis paragraph, the stop and predict box, and the wall thickness paragraphs of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-BD1

Brain dump 1 · foundational retrieval: Blunt and penetrating injury, structure by structure

A competent response contains

· Blunt trauma injures by deceleration and compression, so it damages tethered or trapped structures; penetrating trauma injures whatever lies on the track, so anterior structures suffer most.

· Deceleration tears the aorta at the isthmus, between the origin of the left subclavian artery and the attachment of the ligamentum arteriosum.

· The tear happens there because intercostal arteries and pleura fix the descending aorta to the spine while the heart and arch hang mobile from the great vessels, so shear concentrates at the junction.

· Most full thickness tears are fatal at the scene; survivors usually have a pseudoaneurysm contained by the collagen rich adventitia, which can rupture without warning.

· Chest film findings: a widened mediastinum, a blurred aortic knob, the trachea and nasogastric tube pushed to the right, a depressed left main bronchus and sometimes a left apical cap; fractures of the first and second ribs signal a large force.

· Because much of the mediastinal blood is venous, these signs only suggest the diagnosis, which rests on computed tomographic angiography.

· A sternal blow bruises the right ventricle most often because it is the anterior chamber; it releases troponin, sinus tachycardia is the commonest rhythm disturbance, and right bundle branch block fits the anterior course of the right bundle and moderator band.

· A precordial blow landing on the upstroke of the T wave, the vulnerable period, can trigger ventricular fibrillation with no structural injury, which is commotio cordis.

· Stab wounds injure the right ventricle most often, then the left ventricle, the right atrium and the left atrium.

· If the pericardial wound seals, blood collects in the sac as hemopericardium; if it stays open, blood drains into the pleural cavity as hemothorax.

Distinctions that matter

· Blunt injury is predicted by what is tethered and penetrating injury by what is in front, which is why the isthmus and the right ventricle head the two lists.

· A widened mediastinum is a clue rather than a diagnosis, because venous bleeding widens it too.

Common misconceptions

· Expecting a normal chest film to exclude an aortic tear. The signs are insensitive and the diagnosis rests on computed tomographic angiography.

· Explaining commotio cordis by a bruised myocardium. There is no structural injury; the blow lands in the vulnerable period.

Where it is taught: Section C06, blunt trauma paragraphs, the deceleration flow figure, and the contusion and penetrating injury paragraphs of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-BD2

Brain dump 2 · integration and application: Explain every sign of tamponade from a stiff sac

A competent response contains

· The blade most likely entered the right ventricle, because it is the most anterior chamber and lies directly behind the sternum and the left fourth and fifth costal cartilages.

· About 150 to 200 mL entering over minutes can cause tamponade, because the fibrous sac is stiff over short periods and its pressure and volume curve turns steeply upward; a slow effusion can reach a liter with little pressure rise because the sac stretches over weeks.

· The right chambers are affected first because they are low pressure chambers, so pericardial pressure reaches their diastolic pressure soonest and compresses them in diastole.

· Filling falls because the chambers cannot expand against the pericardial pressure, so stroke volume and arterial pressure fall.

· Venous pressure rises because blood cannot enter the compressed right chambers, and diastolic pressures in all four chambers equalize with the pericardial pressure.

· Fluid between the heart and the chest wall muffles the heart sounds, and the fall in stroke volume drives a reflex tachycardia; muffled sounds, hypotension and raised venous pressure are the Beck triad.

· Pulsus paradoxus: on inspiration the extra right ventricular filling can only push the interventricular septum toward the left ventricle, because the sac is fixed, so left ventricular filling and stroke volume fall and systolic pressure falls by more than 10 mm Hg.

· The jugular y descent is blunted because the ventricle cannot accept rapid filling when the pericardium is already at its limit.

· In a patient who has also lost a lot of blood the venous pressure may be too low to distend the neck veins, so they can be flat.

· Tension pneumothorax instead would give absent breath sounds with a hyperresonant percussion note on the affected side, and the trachea shifted away as a late sign.

Distinctions that matter

· The volume needed for tamponade depends on the rate of filling rather than on the amount alone, because the sac is stiff over minutes and compliant over weeks.

· Electrical alternans and low voltage belong to large effusions rather than to a small acute hemopericardium.

Common misconceptions

· Expecting a large volume before tamponade appears. A rapidly entering 150 to 200 mL is enough.

· Ruling out tamponade because the neck veins are flat. Blood loss can keep them flat.

Where it is taught: Section C06, penetrating injury and tamponade paragraphs and the comparison table of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-RT01

Retrieval exercise: Rebuild the downward spiral of acute tamponade

A competent response contains

· Step 1: blood fills the stiff pericardial sac rapidly, and about 150 to 200 mL is enough because the fibrous pericardium is nearly inelastic over minutes.

· Step 2: the pressure compresses the right atrium and right ventricle in diastole, because they are the low pressure chambers.

· Step 3: ventricular filling and stroke volume fall.

· Step 4: arterial and aortic diastolic pressure fall.

· Step 5: coronary perfusion pressure falls, because it is the difference between aortic diastolic pressure, which is falling, and ventricular diastolic pressure, which is rising.

· Step 6: ischemia weakens contraction and output falls further, feeding back into step 3.

· Unless the blood is removed each turn of the loop lowers output further, and the spiral ends in pulseless electrical activity.

· Removing the blood from the sac lowers pericardial pressure and breaks the loop.

Distinctions that matter

· The loop is self sustaining because falling output starves the muscle that generates the output.

· Compression happens in diastole, which is why this starts as a filling problem rather than a contraction problem.

Common misconceptions

· Treating the fall in blood pressure as the start of the problem. It is a consequence of blocked filling.

· Expecting a faster heart rate to compensate indefinitely. Short diastole reduces filling and coronary perfusion further.

Where it is taught: Section C06, cycle figure on the downward spiral of acute tamponade of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-CP01

Comparison table: Tamponade against tension pneumothorax after chest trauma

A competent response contains

· Why output falls: in tamponade pericardial pressure blocks ventricular filling; in tension pneumothorax raised intrathoracic pressure and mediastinal shift reduce venous return.

· Neck veins: distended in both, unless the patient is hypovolemic.

· Breath sounds and percussion: normal on both sides in tamponade; absent and hyperresonant on the affected side in tension pneumothorax.

· Trachea: midline in tamponade; shifted away in tension pneumothorax, and that shift is a late sign.

· Heart sounds: muffled in tamponade; usually audible in tension pneumothorax.

· Flat neck veins in either condition point to coexisting blood loss, because venous pressure is then too low to distend them.

· Both conditions cause shock with distended neck veins, and the lung findings are what separate them.

Distinctions that matter

· Shock with distended neck veins narrows the field to obstructed filling or obstructed venous return, and the chest examination then decides which.

· A midline trachea with normal breath sounds argues for a pericardial cause rather than a pleural one.

Common misconceptions

· Relying on tracheal deviation to identify a tension pneumothorax early. It appears late.

· Expecting muffled heart sounds in both. Muffling belongs to fluid around the heart.

Where it is taught: Section C06, table comparing tamponade with tension pneumothorax of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-CA01

Clinical application: Shock with distended neck veins after a stab wound

A competent response contains

· Recognize: the blade entered the right ventricle, the most anterior chamber, which lies behind the sternum and the left fourth and fifth costal cartilages, and blood has collected in the sealed pericardial sac as hemopericardium causing tamponade; the triad present here is hypotension, distended neck veins from raised venous pressure, and muffled heart sounds.

· Explain: the fibrous pericardium is nearly inelastic over minutes, so its pressure and volume curve turns steeply upward and about 150 to 200 mL is enough to raise pericardial pressure above the diastolic

pressure of the low pressure right chambers, which are therefore compressed in diastole before the left chambers.

· Differentiate: a tension pneumothorax would give absent breath sounds and a hyperresonant percussion note on one side with the trachea shifted away, while here breath sounds and percussion are normal and the trachea is midline; blood loss alone would give flat rather than distended neck veins, with no muffling of the heart sounds and no inspiratory fall of this size.

· Investigate: the inspiratory fall in systolic pressure is 18 mm Hg, above the 10 mm Hg threshold for pulsus paradoxus; the chest film would show little or no enlargement of the cardiac shadow because the sac has had no time to stretch; bedside echocardiography would show fluid in the pericardial sac with collapse of the right chambers in diastole; the electrocardiogram shows sinus tachycardia with normal voltages, because this is a small acute collection rather than a large effusion.

· Decide: diastolic pressures in all four chambers equalize with the pericardial pressure; the inspiratory fall in systolic pressure shows that extra right ventricular filling can only come at the expense of the left ventricle, and the blunted y descent shows the ventricle cannot accept rapid filling in early diastole.

· Justify: the fibrous pericardium is a stiff closed sac fused to the diaphragm and blended with the adventitia of the great vessels, so the total volume inside it is fixed over minutes and the two ventricles compete for it through the interventricular septum; the same volume accumulating slowly would let the sac stretch, so pressure would stay low and a liter could gather with little effect.

Distinctions that matter

· The rate of accumulation and not the volume alone decides whether a pericardial collection causes tamponade.

· Low voltage and electrical alternans belong to large chronic effusions, so their absence here does not argue against acute tamponade.

Common misconceptions

· Expecting a large cardiac shadow on the chest film in acute tamponade. The sac has not stretched.

· Reading a normal chest examination as evidence against a serious thoracic injury. Here it is what places the problem in the pericardium.

Where it is taught: Section C06, penetrating injury and tamponade paragraphs, the tamponade cycle figure, and the pericardium in Section C03 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C01-Q01

Board-style question: Which chamber forms the apex beat

Correct answer: D. Left ventricle

Why it is right: The left ventricle forms the apex and the left border, so when it dilates the left border moves down and out, the lower left contour is displaced, the apex beat shifts lateral to the midclavicular line and often into the sixth intercostal space, and the cardiothoracic ratio rises above 0.5.

A is wrong: Right atrial enlargement bulges the lower right border and does not move the apex beat, because the right atrium forms the right border rather than the apex.

B is wrong: Right ventricular enlargement produces a parasternal lift and fills the clear space behind the lower sternum on the lateral film, because the right ventricle forms no contour on the posteroanterior view.

C is wrong: Left atrial enlargement straightens the left border below the pulmonary trunk and gives a double density on the right without displacing the apex, because the left atrium is the most posterior chamber.

A competent response contains

· The left ventricle forms the apex and the left border, so its dilation moves the apex beat down and out and displaces the lower left contour.

· A cardiothoracic ratio of 0.59 on a posteroanterior film in full inspiration is above the normal limit of 0.5.

Distinctions that matter

· An apex beat lateral to the midclavicular line and below the fifth space reports enlargement of the chamber that forms the apex.

Common misconceptions

· Attributing any displaced apex beat to the right ventricle. The right ventricle lifts the parasternal area instead.

Where it is taught: Section C01 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C02-Q02

Board-style question: Dysphagia and a widened carina

Correct answer: A. An enlarged left atrium expanding backward and upward against the esophagus and left main bronchus

Why it is right: The left atrium lies immediately in front of the esophagus and just below the carina, and because the ventricles block forward expansion a dilating atrium presses backward on the esophagus and upward on the left main bronchus, giving an anterior esophageal indentation, an elevated bronchus and a widened carinal angle.

B is wrong: A dilated right ventricle occupies the retrosternal space in front of the heart, so it does not reach the esophagus or the carina.

C is wrong: An aneurysm of the descending thoracic aorta lies behind and to the left of the left atrium, so its indentation would be on the back of the esophagus and it would not lift the left main bronchus.

D is wrong: An aberrant right subclavian artery also passes behind the esophagus, so its indentation is posterior, and it arises from the arch rather than from a cardiac chamber.

A competent response contains

· The left atrium is the most posterior chamber, with the esophagus immediately behind it and the carina and left main bronchus just above.

· Rheumatic fever scars the mitral valve, and mitral stenosis raises left atrial pressure so the atrium dilates backward and upward.

Distinctions that matter

· An indentation on the anterior wall of the esophagus places the compressing structure in front of the esophagus, which is where the left atrium sits.

Common misconceptions

· Reading dysphagia in heart disease as unrelated to the heart. The position of the left atrium explains it.

Where it is taught: Section C02 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C04-Q03

Board-style question: Hoarseness from a mediastinal mass

Correct answer: C. The left recurrent laryngeal nerve where it hooks under the arch at the ligamentum arteriosum

Why it is right: The left recurrent laryngeal nerve leaves the vagus at the arch, hooks under it behind the ligamentum arteriosum and passes through the aortopulmonary window, which is the space described, so a mass there stretches or invades it and paralyzes the left vocal fold.

A is wrong: The right recurrent laryngeal nerve hooks under the right subclavian artery in the root of the neck well above the arch, because the right distal sixth arch artery regressed, so a mass at this level does not reach it.

B is wrong: The left phrenic nerve crosses the arch in front of the vagus and supplies the diaphragm and carries pericardial pain, so injuring it would weaken the diaphragm rather than change the voice.

D is wrong: The left vagus behind the root of the left lung has already given off the recurrent laryngeal nerve at the arch, so a lesion there would not account for isolated vocal fold paralysis.

A competent response contains

· The space between the underside of the arch and the left pulmonary artery is the aortopulmonary window, and the left recurrent laryngeal nerve passes through it after hooking under the arch at the ligamentum arteriosum.

· The nerve supplies all intrinsic laryngeal muscles on that side except the cricothyroid, so injury paralyzes the vocal fold and gives a hoarse, breathy voice.

Distinctions that matter

· Hoarseness from a lesion inside the chest points to the left nerve, because the right nerve never descends below the root of the neck.

Common misconceptions

· Expecting either recurrent laryngeal nerve to be at risk at this level. Only the left one is there.

Where it is taught: Section C04 of the CV-3 Heart, Great Vessels, and Pericardium notes.

NPLEX-P1-CARD-03-01-C06-Q04

Board-style question: Where deceleration tears the aorta

Correct answer: B. The aortic isthmus, because the mobile arch shears against the descending aorta held to the spine

Why it is right: Intercostal arteries and pleura fix the descending aorta to the spine while the heart and arch hang mobile from the great vessels, so sudden deceleration concentrates shear at the isthmus near the ligamentum arteriosum, where the intima and media tear and the adventitia may contain a pseudoaneurysm.

A is wrong: The ascending aorta lies inside the pericardium, so a tear there fills the pericardial sac and causes tamponade rather than the mediastinal widening and rib fractures described.

C is wrong: Fibrous continuity between the aortic and mitral valves is a feature of normal left ventricular anatomy and has nothing to do with where deceleration shear concentrates.

D is wrong: The aortic opening in the diaphragm transmits the aorta between the crura rather than tethering it at a junction between mobile and fixed segments, so it is not the usual site of a deceleration tear.

A competent response contains

· The aortic isthmus lies between the origin of the left subclavian artery and the attachment of the ligamentum arteriosum.

· A widened mediastinum with a blurred aortic knob, rightward deviation of the trachea and nasogastric tube and upper rib fractures fit a deceleration injury at that site.

Distinctions that matter

· Blunt injury concentrates where a mobile segment meets a fixed one, which is what makes the isthmus predictable.

Common misconceptions

· Expecting the chest film alone to establish the diagnosis. It suggests it, and computed tomographic angiography confirms it.

Where it is taught: Section C06 of the CV-3 Heart, Great Vessels, and Pericardium notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-04

CV-4 Heart Valves and the Cardiac Cycle

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-04-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the four valves and where each is heard

A competent response contains

· Tricuspid: right atrium to right ventricle, anterior, posterior, and septal cusps, held closed by chordae running to three papillary muscles, heard at the left lower sternal border in the 4th to 5th intercostal space.

· Mitral: left atrium to left ventricle, anterior and posterior cusps, held closed by chordae to the anterolateral and posteromedial papillary muscles, heard at the apex in the left 5th intercostal space at the midclavicular line.

· Pulmonary: right ventricle to pulmonary trunk, anterior, right, and left cusps, sealed by cusp shape with no chordae, heard at the left 2nd intercostal space at the sternal border.

· Aortic: left ventricle to aorta, right, left, and posterior cusps also called right coronary, left coronary, and noncoronary, sealed by cusp shape with no chordae, heard at the right 2nd intercostal space at the sternal border.

· The fibrous skeleton is dense connective tissue rings that hold the four valve openings in shape and electrically insulate the atria from the ventricles.

· Each cusp is a fold of endocardium with a connective tissue core and no muscle, so the valve has no power of its own and moves only as the pressure gradient across it changes: open when the pressure behind exceeds the pressure in front, closed when that reverses.

· Semilunar cusps seal because backflow at the end of systole fills the pockets and presses them together, with a central nodule and thin lateral lunules completing the seal.

· The coronary arteries arise from the right and left aortic sinuses behind the cusps, which is where the cusp names come from.

· Aortic stenosis radiates to the carotids because aortic sounds follow the ascending aorta up to the right 2nd intercostal space.

· Mitral regurgitation radiates to the left axilla along the direction of the backward jet.

· A bicuspid aortic valve occurs in about 1% to 2% of people and calcifies and becomes stenotic early.

Distinctions that matter

· AV valves need chordae because their large thin cusps face full systolic pressure; semilunar valves need none because their pocket shape does the sealing.

· The valves all sit close together behind the sternum, so an auscultation area is where the sound is carried downstream or over the nearest chamber rather than where the valve lies.

Common misconceptions

· Writing that papillary muscles pull the AV valves open in diastole. They contract with the ventricle and hold the cusps shut in systole.

· Treating an auscultation area as the surface marking of the valve itself. The areas follow the path of the sound, which is why aortic sounds are loudest to the right of the sternum.

Where it is taught: Section C01 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C01-BD2

Brain dump 2 · integration and application: Why an inferior infarct tears one papillary muscle and not the other

A competent response contains

· The posteromedial papillary muscle failed, because it usually depends on the posterior descending artery alone, which arises from the right coronary artery in most hearts.

· The anterolateral papillary muscle usually has a dual supply from left anterior descending and circumflex branches, so it is far less likely to infarct and rupture.

· Sequence: occlusion of the right coronary or posterior descending artery, necrosis of the posteromedial muscle, rupture of the softened muscle usually within the first week, mitral cusps flail into the left atrium during systole, acute severe mitral regurgitation into a small noncompliant atrium, large v waves with pulmonary edema and low forward stroke volume.

· The regurgitant volume goes backward into the left atrium from the onset of contraction, so left atrial and pulmonary venous pressures rise steeply.

· Acute regurgitation is tolerated badly because the atrium has had no time to dilate, so a given regurgitant volume produces a much higher pressure than in chronic disease, where a dilated compliant atrium absorbs it.

· In chronic mitral regurgitation the ventricle also has time to undergo eccentric hypertrophy and dilate, which protects forward output for years.

· Chordal rupture from myxomatous degeneration and from endocarditis produces the same loss of cusp support.

Distinctions that matter

· Which papillary muscle ruptures is decided by blood supply, not by size or position.

· Severity of the regurgitation and tolerance of it are separate questions: compliance of the receiving atrium decides the pressure, and time decides the compliance.

Common misconceptions

· Expecting a loud murmur to mean a large leak. The murmur may be short or soft when the atrium is small and stiff and pressures equalise quickly.

· Assuming total stroke volume must fall. The ventricle ejects a large total volume, but much of it goes backward, so forward output drops.

Where it is taught: Section C01, papillary muscle paragraphs and the rupture flow diagram of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C01-CA01

Clinical application: Sudden breathlessness five days after an inferior infarction

A competent response contains

· Recognize: acute severe mitral regurgitation from rupture of the posteromedial papillary muscle, following right coronary or posterior descending artery occlusion in an inferior infarction.

· Explain: the posteromedial muscle depends on the posterior descending artery alone, so it necroses and then ruptures in the first week; the cusps flail into the left atrium during systole, giving a holosystolic murmur at the apex radiating to the axilla, a large late systolic v wave on the wedge tracing, and a high mean wedge pressure that drives pulmonary edema and crackles.

· Differentiate: ventricular septal rupture gives a harsh holosystolic murmur at the left lower sternal border with a left to right shunt rather than a giant v wave, and free wall rupture with tamponade gives hypotension with a raised venous pressure, a prominent x descent, and no new loud murmur.

· Investigate: on the wedge tracing look for a small a wave with a very large late systolic v wave and a raised mean pressure; on the jugular tracing look at the height of the pulsation and the wave pattern, which here reflects the raised filling pressures rather than tricuspid regurgitation.

· Decide: blood pressure is low because a large share of each ejected volume goes backward into the left atrium, so forward stroke volume into the aorta falls even though total stroke volume has risen.

· Justify: the leak begins as soon as ventricular pressure exceeds atrial pressure, well before the aortic valve opens, and the atrium has not dilated, so it is stiff and its pressure rises steeply, which both limits further backward flow into a high pressure chamber and transmits pressure to the pulmonary veins; the tachycardia is the compensation for the small forward volume.

Distinctions that matter

· A new murmur after infarction is placed by its location and timing: apex radiating to the axilla points to the mitral valve, left lower sternal border points to the septum.

· A giant v wave reports atrial compliance and the volume arriving backward, not the size of the valve opening.

Common misconceptions

· Reading the normal electrocardiogram as evidence against a mechanical complication. Rupture is a structural event and does not change the tracing.

· Assuming low blood pressure means poor contractility. Contractility can be preserved while forward output falls because the ventricle has a second low pressure outlet.

Where it is taught: Section C01, papillary muscle paragraphs and rupture flow diagram, with the large v wave in section C04 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the Wiggers diagram and mark every valve event

A competent response contains

· Four valve events: mitral closure when left ventricular pressure exceeds left atrial pressure, aortic opening when ventricular pressure exceeds aortic diastolic pressure of about 80 mm Hg, aortic closure when flow reverses as ventricular pressure falls below aortic pressure, and mitral opening when ventricular pressure falls below left atrial pressure.

· Phases in order: atrial systole, isovolumetric contraction of about 0.05 s, rapid ejection, reduced ejection, isovolumetric relaxation, rapid filling, diastasis.

· Normal numbers: aortic diastolic pressure about 80 mm Hg, peak ventricular and aortic pressure about 120 mm Hg, end-diastolic volume about 120 mL, end-systolic volume about 50 mL, stroke volume about 70 mL, ejection fraction about 55% to 70%, pulmonary artery pressure about 25/8 mm Hg.

· At about 75 beats per minute the cycle lasts about 0.8 s, with systole about 0.3 s and diastole about 0.5 s.

· About two thirds of the stroke volume leaves during rapid ejection; in reduced ejection the T wave begins and ventricular pressure falls slightly below aortic pressure while momentum carries blood forward.

· S1 is mitral then tricuspid closure, best at the apex, from vibration of the tensed cusps, chordae, walls, and blood; aortic valve opening is silent unless a stiff domed valve gives an ejection click.

· S2 is aortic then pulmonic closure, best at the base, and the dicrotic notch appears on the aortic curve at that moment from the backflow and aortic rebound.

· Atrial contraction after the P wave supplies about the last 20% of end-diastolic volume at rest.

· Rapid filling delivers most of the filling and is aided by active relaxation; diastasis is the reduced filling period that follows.

Distinctions that matter

· Every valve event on the diagram is a crossing of two pressure curves, so each one is read by asking which pressure across that valve is higher.

· Only the two isovolumetric phases have all four valves closed, which is why volume is flat in both.

Common misconceptions

· Drawing aortic pressure falling to zero in diastole. It falls to about 80 mm Hg, which is the pressure the ventricle has to exceed to open the valve.

· Marking S1 and S2 as valve opening sounds. Both are closure sounds, and normal opening is silent.

Where it is taught: Section C02, systole and diastole paragraphs and the cardiac cycle diagram of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C02-BD2

Brain dump 2 · integration and application: Name the phase from four numbers, and what a fast rate costs

A competent response contains

· Tracing A: ventricular pressure above atrial pressure so the mitral valve is closed, aortic pressure above ventricular pressure so the aortic valve is closed, and a volume of 120 mL is end-diastolic, so this is isovolumetric contraction just after S1.

· Tracing B: the same pressure relationships with a volume of 50 mL, which is end-systolic, so this is isovolumetric relaxation just after S2.

· Volume, not pressure, separates the two isovolumetric phases, because both have all valves closed and the same pressures can occur on the way up and on the way down.

· Tracing C: atrial pressure above ventricular pressure so the mitral valve is open, aortic pressure well above ventricular pressure so the aortic valve is closed, and a volume between end-systolic and end-diastolic, so this is ventricular filling.

· Raising the rate to 160 beats per minute shortens the whole cycle, but diastole shortens far more than systole.

· Diastasis, the reduced filling period, is lost first, so filling depends more on rapid filling and on atrial contraction, whose share of end-diastolic volume rises.

· Left ventricular coronary flow occurs mainly in diastole, so a short diastole reduces perfusion at the same time as demand is higher.

· The pulmonary valve opens before the aortic valve because the right ventricle has to exceed only about 8 mm Hg of pulmonary diastolic pressure, and it closes after the aortic valve because the low impedance pulmonary circulation prolongs forward flow, so A2 precedes P2.

Distinctions that matter

· Comparing two pressures at a time settles the state of each valve; trying to read the phase from one pressure alone does not.

· Shorter diastole reduces both filling and coronary perfusion, which are separate consequences of the same change.

Common misconceptions

· Assuming systole and diastole shorten in proportion as the rate rises. Diastole takes almost all of the loss.

· Reading a high ventricular pressure as proof the aortic valve is open. It is open only while ventricular pressure exceeds aortic pressure.

Where it is taught: Section C02, opening paragraph, right heart timing paragraph, and the worked example on identifying the phase of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C02-RT01

Retrieval exercise: Rebuild the valve events of one left-sided cycle

A competent response contains

· Step 1: atrial systole completes filling to end-diastolic volume.

· Step 2: left ventricular pressure exceeds left atrial pressure and the mitral valve closes, producing S1.

· Step 3: isovolumetric contraction, all valves closed and pressure rising at constant volume.

· Step 4: ventricular pressure exceeds aortic pressure at about 80 mm Hg and the aortic valve opens.

· Step 5: rapid then reduced ejection, volume falling to end-systolic volume.

· Step 6: flow reverses and the aortic valve closes, producing S2 and the dicrotic notch.

· Step 7: isovolumetric relaxation, all valves closed and pressure falling at constant volume.

· Step 8: ventricular pressure falls below left atrial pressure and the mitral valve opens.

· Step 9: rapid filling, with an S3 if one is present, then diastasis.

· The dicrotic notch is the small downward deflection and rebound on the aortic curve produced by the brief backflow and the elastic recoil of the aorta at the moment the valve closes.

· S1 is loud with a short PR interval or mitral stenosis, because the cusps are still widely open when contraction begins and travel further to close, and soft with a long PR interval, because the cusps have already drifted together.

Distinctions that matter

· Steps 3 and 7 are the only all valves closed steps, and the volume is flat in both; what differs is whether the ventricle holds end-diastolic or end-systolic volume.

· Pressure thresholds belong to the opening steps and pressure crossings to the closing steps, so name the two pressures each time rather than a single number.

Common misconceptions

· Putting the aortic valve opening at the same moment as S1. Isovolumetric contraction sits between them.

· Explaining a loud S1 by stronger contraction alone. Cusp position at the start of contraction is what the notes make the deciding factor.

Where it is taught: Section C02, the cardiac cycle diagram and the paragraph on S1 intensity of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-BD1

Brain dump 1 · foundational retrieval: Draw the normal pressure-volume loop

A competent response contains

· The loop is traced counterclockwise.

· Bottom right corner: mitral valve closure, at about 120 mL and about 8 mm Hg.

· Top right corner: aortic valve opening, at about 80 mm Hg.

· Top left corner: aortic valve closure, at about 50 mL and about 100 mm Hg.

· Bottom left corner: mitral valve opening.

· Right side is vertical isovolumetric contraction, the top is ejection peaking near 120 mm Hg, the left side is vertical isovolumetric relaxation, and the bottom edge is filling with little rise in pressure in a compliant ventricle.

· The width of the loop, end-diastolic volume minus end-systolic volume, is the stroke volume, and the enclosed area is the stroke work.

· The bottom edge follows the end-diastolic pressure-volume relationship, which is steeper in a stiff ventricle.

· The top left corner lies on the end-systolic pressure-volume line, whose slope rises with contractility.

· Normal values: end-diastolic volume about 120 mL, end-systolic volume about 50 mL, stroke volume about 70 mL, ejection fraction about 55% to 70%.

Distinctions that matter

· Width measures stroke volume and area measures stroke work, so a tall narrow loop can do more work while moving less blood.

· The diastolic edge reports compliance and the end-systolic corner reports contractility; they are different properties on different parts of the loop.

Common misconceptions

· Tracing the loop clockwise. Filling runs left to right along the bottom and ejection runs right to left along the top, which makes it counterclockwise.

· Reading the bottom edge as passive and featureless. Its slope is the stiffness of the ventricle and it rises steeply in a hypertrophied one.

Where it is taught: Section C03, first two paragraphs of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-BD2

Brain dump 2 · integration and application: Name the lesion from the loop

A competent response contains

· Loop 1: stroke volume 125 mL, a volume load, aortic regurgitation; the giveaway is that the left side slopes to the right, so the ventricle gains volume while pressure falls after ejection, which means it is filling from the aorta.

· Loop 2: stroke volume 50 mL, a pressure load, aortic stenosis; the giveaway is a peak ventricular pressure of 215 mm Hg against a normal aortic pressure, which is a gradient across the valve.

· Loop 3: stroke volume 130 mL, a volume load, mitral regurgitation; the giveaway is that volume starts falling at the bottom right corner while pressure is only about 15 mm Hg, far too low to have opened the aortic valve.

· In aortic regurgitation relaxation is no longer isovolumetric because blood leaks back from the aorta as soon as ventricular pressure falls, so volume rises during it; chronic eccentric hypertrophy follows, the ventricle becomes more compliant and the loop shifts right, pulse pressure widens, and the large total stroke volume comes from filling by both atrium and aorta with the Frank-Starling mechanism.

· In mitral regurgitation contraction is no longer isovolumetric because blood escapes into the left atrium as soon as ventricular pressure exceeds atrial pressure, so volume falls during it; eccentric hypertrophy with left atrial dilation follows, total stroke volume rises but forward output falls, and end-systolic volume tends to fall early because the low pressure atrium is a second outlet.

· In aortic stenosis both isovolumetric phases are preserved; concentric hypertrophy follows, which normalises wall stress but stiffens the ventricle, steepening the diastolic edge and raising end-diastolic pressure, and the extra loop area means greater stroke work and oxygen demand.

· Peak pressures of about 180 to 250 mm Hg are quoted for severe aortic stenosis while aortic pressure stays normal or low, and incomplete ejection raises end-systolic volume and narrows the loop.

· Mitral stenosis underfills the ventricle, so the loop is small and shifted left.

Distinctions that matter

· A regurgitant lesion abolishes the isovolumetric phase in which the leaking valve ought to be closed, which is the quickest way to tell the two regurgitations apart on a loop.

· Total and forward stroke volume separate in mitral regurgitation, so loop width alone overstates what reaches the aorta.

Common misconceptions

· Taking a wide loop as a healthy one. In mitral regurgitation much of that width is volume going backward.

· Expecting raised peak pressure in every valve lesion. Peak pressure is raised in aortic stenosis and normal or low in mitral regurgitation, which is what rules stenosis out.

Where it is taught: Section C03, valve lesion paragraphs, the loop comparison table, and the worked example on reading a lesion of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-CP01

Comparison table: Pressure-volume loop in three valve lesions

A competent response contains

· Type of load: aortic stenosis is a pressure load on afterload, aortic regurgitation and mitral regurgitation are volume loads.

· Peak left ventricular pressure: greatly increased in aortic stenosis with a gradient from ventricle to aorta, increased in aortic regurgitation, normal or decreased in mitral regurgitation.

· End-diastolic volume: normal or mildly increased in aortic stenosis, greatly increased in aortic regurgitation, increased in mitral regurgitation.

· End-systolic volume: increased in aortic stenosis, increased in aortic regurgitation, normal or decreased early in mitral regurgitation.

· Total stroke volume: normal or decreased in aortic stenosis, greatly increased in aortic regurgitation, increased in mitral regurgitation with forward volume reduced.

· Phase that is not truly isovolumetric: none in aortic stenosis, relaxation in aortic regurgitation with volume rising, contraction in mitral regurgitation with volume falling.

· Chronic hypertrophy: concentric in aortic stenosis, eccentric in aortic regurgitation, eccentric with left atrial dilation in mitral regurgitation.

· Loop shape: aortic stenosis is tall and narrow with a steep diastolic edge, aortic regurgitation is wide and shifted right with a sloping left side, mitral regurgitation is wide and shifted right with a sloping right side.

Distinctions that matter

· The sloping side names the lesion: a sloping left side is aortic regurgitation and a sloping right side is mitral regurgitation.

· Pressure loading gives concentric hypertrophy with sarcomeres added in parallel, volume loading gives eccentric hypertrophy with sarcomeres added in series.

Common misconceptions

· Filling the end-systolic volume row for mitral regurgitation as increased. Early on it tends to fall, because the atrium is a second low pressure outlet.

· Writing a raised peak pressure for every lesion that enlarges the loop. Only aortic stenosis raises it greatly, and it does so against a normal aortic pressure.

Where it is taught: Section C03, valve lesion paragraphs and the loop comparison table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-Q01

Board-style question: A loop that loses volume before the aortic valve opens

Correct answer: C. Blood escaping into the left atrium from the onset of ventricular contraction

Why it is right: Mitral regurgitation lets blood escape into the low pressure left atrium as soon as ventricular pressure exceeds atrial pressure, so volume falls during what should be isovolumetric contraction, the right side of the loop slopes, total stroke volume rises, and forward output falls.

A is wrong: A fixed outflow obstruction would raise peak ventricular pressure well above aortic pressure and give a tall narrow loop with both isovolumetric phases intact, but the peak here is 108 mm Hg.

B is wrong: Reduced compliance steepens the bottom edge of the loop and raises end-diastolic pressure without letting volume leave before the aortic valve opens, and the end-diastolic volume here is large rather than small.

D is wrong: Blood returning from the aorta would add volume during relaxation and make the left side of the loop slope to the right, which is the opposite side and the opposite direction of volume change.

A competent response contains

· Stroke volume is 125 mL, a volume load, with a normal peak ventricular pressure.

· Volume falling at 16 mm Hg means the ventricle is emptying long before it could open the aortic valve at about 80 mm Hg, so the exit is backward into the left atrium.

· Total stroke volume is large while forward output is low, which is the pattern of mitral regurgitation.

Distinctions that matter

· Losing volume during what should be isovolumetric contraction localises the leak to the mitral valve; gaining volume during what should be isovolumetric relaxation localises it to the aortic valve.

Common misconceptions

· Taking a wide loop as evidence of good forward output. The width here is largely regurgitant volume.

Where it is taught: Section C03 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the jugular venous pulse against the heart sounds

A competent response contains

· a wave: right atrial contraction, after the P wave and just before S1.

· c wave: the closed tricuspid valve bulging into the atrium during right ventricular isovolumetric contraction, just after S1, with some transmitted carotid pulsation.

· x descent: the systolic fall in atrial pressure from atrial relaxation and the downward pull of the tricuspid ring as the ventricle ejects.

· v wave: atrial filling from the venae cavae against the closed tricuspid valve, peaking near S2.

· y descent: atrial emptying when the tricuspid valve opens, during rapid filling in early diastole.

· The right internal jugular vein runs nearly straight through the brachiocephalic vein and superior vena cava into the right atrium, so right atrial pressure changes appear in the neck.

· Differences from the carotid: the jugular pulse is not palpable, usually shows two peaks per cycle, falls with inspiration, and changes with position.

· The sternal angle lies about 5 cm above the right atrium, so with the patient at about 45 degrees a pulsation more than about 3 cm above the sternal angle suggests raised right atrial pressure.

· The left atrial or wedge tracing has the same waves.

Distinctions that matter

· The a wave is produced by the atrium contracting and the v wave by the atrium filling, so one precedes S1 and the other peaks near S2.

· The c wave comes from the valve bulging back into the atrium, not from blood crossing the valve.

Common misconceptions

· Calling the v wave right ventricular contraction. It is passive atrial filling against a closed valve.

· Timing the y descent in systole. The tricuspid valve opens in early diastole, which is when the atrium empties.

Where it is taught: Section C04, first two paragraphs and the jugular wave table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C04-BD2

Brain dump 2 · integration and application: Read four abnormal neck tracings

A competent response contains

· Patient 1: intermittent cannon a waves from complete heart block; the atria and ventricles beat independently, so on some beats the atrium contracts against a closed tricuspid valve and the contraction is forced backward into the neck, and because the rhythms are unrelated the coincidence happens only on some beats.

· Patient 2: regular cannon a waves, because atria and ventricles activate together on every beat, as in some junctional rhythms and in AV nodal reentrant tachycardia.

· Patient 3: absent a wave in atrial fibrillation, because there is no organised atrial contraction.

· Patient 4: tricuspid regurgitation; the atrium receives blood ejected backward from the right ventricle, so pressure rises throughout systole, the x descent is lost, the c and v waves merge into a large systolic c-v wave, the y descent is steep, and systolic hepatic pulsation may be felt.

· The wedge tracing in acute mitral regurgitation shows the same large v wave on the left side.

· A large a wave that is not a cannon wave means resistance to atrial emptying through an open valve, as in tricuspid stenosis or a stiff right ventricle.

· Ventricular tachycardia with AV dissociation and ventricular pacing also produce irregular cannon a waves.

Distinctions that matter

· Cannon a waves mean the atrium is contracting against a closed valve, while large a waves mean it is straining to empty through an open one.

· Regular against irregular cannon waves follows from whether the atria and ventricles are linked on every beat or beating independently.

Common misconceptions

· Reading any large neck pulsation as tricuspid regurgitation. Timing separates them: an a wave is presystolic and the c-v wave of regurgitation fills systole.

· Expecting cannon a waves in atrial fibrillation. Without organised atrial contraction there is no a wave at all.

Where it is taught: Section C04, abnormal wave paragraphs, the jugular wave table, and the stop and predict box of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C04-Q02

Board-style question: Intermittent giant neck pulsations

Correct answer: A. Right atrial contraction against a closed tricuspid valve on beats where the timing coincides

Why it is right: In complete heart block the atria and ventricles beat independently, so on the beats where atrial contraction lands during ventricular systole the atrium contracts against a closed tricuspid valve and the pressure is forced back up the jugular vein as a cannon a wave, which appears only on those beats.

B is wrong: Tricuspid regurgitation raises atrial pressure through the whole of systole on every beat, giving a large c-v wave with a lost x descent rather than occasional pulsations.

C is wrong: Tricuspid stenosis obstructs emptying through an open valve, producing a large a wave on every beat with a slow y descent, not an intermittent giant wave.

D is wrong: A transmitted carotid pulsation would be palpable and would occur with every ejection, and the jugular pulse is specifically not palpable.

A competent response contains

· P waves at 92/min and QRS complexes at 40/min with no fixed relationship are complete heart block with AV dissociation.

· When a P wave falls during ventricular systole the right atrium contracts against a closed tricuspid valve, which forces the contraction backward into the neck as a cannon a wave.

· The pulsations are intermittent because the atrial and ventricular rhythms are unrelated, so the timing coincides only on some beats.

Distinctions that matter

· Intermittent cannon waves point to AV dissociation, while cannon waves on every beat point to atria and ventricles activating together.

Common misconceptions

· Reading a large venous pulsation as a sign of valve disease without timing it against the cardiac cycle.

Where it is taught: Section C04 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the S2 splitting table from memory

A competent response contains

· A2 is the aortic component, loud and heard widely; P2 is the pulmonic component, soft and best heard at the left 2nd intercostal space, which is where splitting is assessed; A2 normally comes first and a split is audible when the components are more than about 0.03 s apart.

· Physiologic: A2 then P2, single or very narrow in expiration, split by about 0.04 s on inspiration, because inspiration delays P2 and slightly advances A2; normal, especially in the young.

· Wide: A2 then P2, split in expiration and wider on inspiration, because P2 is delayed throughout or A2 is early; right bundle branch block, pulmonic stenosis, and severe mitral regurgitation through an early A2.

· Fixed: A2 then P2, a wide split that does not change with breathing, because the right ventricular volume load stays constant as shunt changes offset changes in venous return; atrial septal defect.

· Paradoxical: P2 then A2, split in expiration and single or narrower on inspiration, because A2 is delayed past P2; left bundle branch block, severe aortic stenosis, right ventricular pacing, and obstructive hypertrophic cardiomyopathy.

· The two bedside questions: is a split heard in expiration, and does inspiration widen it, leave it unchanged, or narrow it.

· Physiologic splitting is easiest to hear in children and young adults.

Distinctions that matter

· Wide and fixed splitting both give a split in expiration; what separates them is whether inspiration widens it further.

· Paradoxical splitting is the only pattern in which the components are reversed, and it is also the only one that narrows on inspiration.

Common misconceptions

· Listening for splitting at the apex. P2 is soft and is assessed at the left 2nd intercostal space.

· Treating any wide split as pathological without timing it to breathing. A physiologic split widens on inspiration and closes in expiration.

Where it is taught: Section C05, opening paragraph and the S2 splitting table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C05-BD2

Brain dump 2 · integration and application: Four patients with a split S2

A competent response contains

· Patient 1: wide splitting from right bundle branch block; the right ventricle is activated late so P2 is delayed throughout, and the respiratory mechanism still works, which is why inspiration widens it further.

· Patient 2: fixed splitting from an atrial septal defect; the left to right shunt volume loads the right ventricle and distends the pulmonary bed, delaying P2, and because the atria act as one chamber the extra systemic return of inspiration is offset by a smaller shunt, with the reverse in expiration, so right ventricular volume and P2 timing stay nearly constant.

· Patient 3: paradoxical splitting from severe aortic stenosis; A2 is delayed mechanically past P2, so the split is heard in expiration and the normal inspiratory delay of P2 moves it toward the late A2 and the sound becomes single.

· Patient 4: wide splitting from severe mitral regurgitation; A2 is early rather than P2 late, because the regurgitant orifice gives the left ventricle a second outlet so ejection finishes sooner, and the respiratory mechanism still works.

· Electrical delay of a component comes from late activation of that ventricle, as in right bundle branch block for P2 and left bundle branch block or right ventricular pacing for A2.

· Mechanical delay comes from prolonged ejection, as in pulmonic stenosis for P2 and severe aortic stenosis or obstructive hypertrophic cardiomyopathy for A2.

Distinctions that matter

· A wide split can come from a late P2 or an early A2, so name which component has moved rather than only the width.

· Fixed splitting is not simply a very wide split: the respiratory variation is abolished, which is what the shunt explains.

Common misconceptions

· Expecting inspiration to widen every split. It narrows a paradoxical split and leaves a fixed one unchanged.

· Assuming severe mitral regurgitation delays P2. It moves A2 earlier, and the width comes from that.

Where it is taught: Section C05, abnormal splitting paragraphs and the S2 splitting table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C05-RT01

Retrieval exercise: Rebuild the sequence that widens the split on inspiration

A competent response contains

· Step 1: inspiration makes intrathoracic pressure more negative.

· Step 2: systemic venous return to the right heart rises.

· Step 3: right ventricular ejection lasts longer.

· Step 4: pulmonary vascular impedance falls, which prolongs forward flow.

· Step 5: P2 is delayed.

· Step 6: blood pools in the lungs, left ventricular filling dips, and A2 comes slightly earlier.

· Step 7: the split widens.

· Steps 1 to 5 act on the right heart; step 6 acts on the left.

· The expanded pulmonary vessels briefly hold more blood, so less reaches the left ventricle, its stroke volume dips, ejection is shorter, and A2 moves earlier.

· The split goes from single or very narrow in expiration to about 0.04 s on inspiration, with about 0.03 s needed for a split to be audible.

· It is heard most easily in children and young adults.

Distinctions that matter

· Breathing changes right heart filling more than left, which is why the split widens rather than simply shifting both components.

· Two separate changes contribute: P2 moves later and A2 moves slightly earlier.

Common misconceptions

· Explaining the widening by venous return alone. Falling pulmonary impedance also prolongs right ventricular ejection.

· Expecting left ventricular filling to rise on inspiration along with the right. It dips, because blood is briefly held in the expanded pulmonary vessels.

Where it is taught: Section C05, physiologic splitting paragraph and flow diagram of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C06-BD1

Brain dump 1 · foundational retrieval: S3 and S4 side by side

A competent response contains

· S3: early diastole, about 0.12 to 0.16 s after S2, during rapid filling; a large fast inflow is abruptly decelerated as the ventricle reaches its limit of expansion, setting walls and blood vibrating.

· S4: late diastole just before S1, with atrial contraction; atrial ejection into a stiff ventricle causes a sharp pressure rise and a low frequency vibration.

· Both are low pitched and made by the ventricle and its blood, not by valves.

· Left-sided sounds are heard best with the bell at the apex in the left lateral position; right-sided sounds at the left lower sternal border, and they are louder on inspiration because inspiration increases right heart filling.

· S3 is often normal in children, young adults up to about age 40, trained athletes, and pregnancy, where a compliant ventricle fills briskly.

· In older adults S3 signals volume overload: a dilated failing ventricle with high atrial pressure, regurgitant volume returning in mitral or aortic regurgitation, a left to right shunt, or a high output state.

· S4 indicates reduced compliance: hypertrophy from hypertension, aortic stenosis, or hypertrophic cardiomyopathy, and acute ischemia, which impairs relaxation; it may occur with age-related stiffening but is abnormal in the young.

· S4 is absent in atrial fibrillation because it requires organised atrial contraction; S3 can persist in atrial fibrillation because it depends on passive rapid filling.

· Significant mitral stenosis prevents the brisk early inflow that makes an S3, so no S3 occurs.

Distinctions that matter

· S3 is a volume sound in early diastole and S4 is a stiffness sound in late diastole, which is why atrial fibrillation abolishes one and not the other.

· An S3 can be a normal finding in the young, whereas an S4 in a young person is abnormal.

Common misconceptions

· Attributing S3 or S4 to a valve snapping. Both come from the ventricular wall and the blood inside it.

· Expecting an S3 in severe mitral stenosis because atrial pressure is high. The narrowed valve prevents the rapid inflow the sound depends on.

Where it is taught: Section C06, S3 and S4 paragraphs and the stop and predict box of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C06-BD2

Brain dump 2 · integration and application: Decide which murmur is just flow

A competent response contains

· A murmur is turbulent flow; turbulence appears when the Reynolds number exceeds a critical value, and that number rises with flow velocity and vessel diameter and falls with viscosity.

· Pregnancy: blood volume rises about 40% to 50% and cardiac output about 30% to 50%, so velocity rises; the murmur goes after delivery.

· Anemia: hypoxic vasodilation raises cardiac output and low red cell mass lowers viscosity, so both terms push toward turbulence; the murmur goes when the anemia is corrected.

· A six-year-old: high output relative to body size with a thin chest wall that transmits sound, so a normal valve is audible; the murmur goes as the child grows.

· Fever at 39.4 degrees C (103 degrees F): raised cardiac output and velocity; the murmur goes when the fever settles. Thyrotoxicosis and exercise act the same way.

· Functional murmurs are soft, grade 1 or 2 with no thrill, early to mid systolic ejection murmurs at the base or left sternal border, with normal S2 splitting and no extra sounds.

· Findings that argue for structural disease: a diastolic murmur, a holosystolic murmur, a thrill, or an abnormal S2.

· The venous hum is the innocent continuous murmur, heard above the clavicle, and it disappears when the child lies down or the jugular vein is compressed.

· Common innocent murmurs in children: a vibratory musical early systolic murmur at the left lower sternal border or apex known as the Still murmur, a soft pulmonary flow murmur at the left 2nd intercostal space, and the venous hum above the clavicle.

Distinctions that matter

· High flow and low viscosity both raise the Reynolds number, so a flow murmur can arise from more output, thinner blood, or both.

· Timing is the strongest structural clue: a diastolic or holosystolic murmur is never functional, while the venous hum is continuous and still innocent.

Common misconceptions

· Reading any murmur in a child or a pregnant woman as innocent. Grade, timing, S2, and extra sounds still have to be checked.

· Assuming a functional murmur means a mild structural lesion. The valve is normal, and removing the cause removes the murmur.

Where it is taught: Section C06, functional flow murmur paragraphs of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C07-BD1

Brain dump 1 · foundational retrieval: Build the murmur timing table from memory

A competent response contains

· Rule: stenosis is heard when the valve should be open and regurgitation when it should be closed, so aortic and pulmonic stenosis are systolic while their regurgitations are diastolic, and mitral and tricuspid regurgitation are systolic while their stenoses are diastolic.

· Grading runs 1 for faint to 6 for audible with the stethoscope off the chest, and a palpable thrill means grade 4 or higher.

· Ejection crescendo-decrescendo: aortic stenosis at the right 2nd intercostal space with carotid radiation and a soft A2; pulmonic stenosis at the left 2nd intercostal space with a wide split S2; hypertrophic cardiomyopathy at the left lower sternal border with no carotid radiation.

· Holosystolic: mitral regurgitation at the apex radiating to the left axilla; tricuspid regurgitation at the left lower sternal border and louder on inspiration; ventricular septal defect at the left lower sternal border and harsh.

· Click then late systolic: mitral valve prolapse at the apex, with the click earlier when the left ventricle is smaller.

· Early diastolic decrescendo: aortic regurgitation at the left sternal border in the 3rd to 4th space with a wide pulse pressure, best heard sitting forward in held expiration.

· Opening snap then mid-diastolic rumble: mitral stenosis at the apex in the left lateral position with the bell, with presystolic accentuation.

· Mid-diastolic rumble: tricuspid stenosis at the left lower sternal border, louder on inspiration.

· Continuous peaking at S2: patent ductus arteriosus at the left infraclavicular area, machinery-like, because aortic pressure exceeds pulmonary pressure throughout the cycle.

· An ejection murmur starts after S1 because flow across the outflow valve begins only after isovolumetric contraction, and it ends before S2.

· A holosystolic murmur runs from S1 to S2 because blood escapes into a lower pressure chamber from the onset of contraction; a small ventricular septal defect may be loud because its jet is fast.

· In mitral valve prolapse the cusps hold until the ventricle becomes small enough in mid systole, so a midsystolic click precedes a late systolic murmur.

· As aortic stenosis worsens the murmur peaks later and A2 softens; in mitral stenosis a shorter S2 to opening snap interval means higher atrial pressure and more severe disease.

Distinctions that matter

· Aortic stenosis and hypertrophic cardiomyopathy are both ejection murmurs, and carotid radiation separates them.

· Timing within systole carries the diagnosis: after S1 and ending before S2 means an outflow obstruction, S1 to S2 throughout means a leak into a lower pressure chamber, and late with a click means prolapse.

Common misconceptions

· Taking a louder murmur as a more severe lesion. Loudness depends on flow as well as severity, so a severely stenotic valve may soften as output falls and a small septal defect may be louder than a large one.

· Expecting A2 to be loud in severe aortic stenosis. It softens as the valve becomes immobile.

Where it is taught: Section C07, systolic, diastolic, and continuous murmur paragraphs and the murmur timing table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C07-BD2

Brain dump 2 · integration and application: Separate two systolic murmurs with the bedside maneuvers

A competent response contains

· Inspiration increases right heart return, so right-sided murmurs get louder while hypertrophic cardiomyopathy and mitral valve prolapse change little.

· Squatting raises both preload and afterload, so most murmurs get louder, hypertrophic cardiomyopathy gets softer, and the prolapse click moves later.

· Standing from a squat lowers preload, so most murmurs get softer, hypertrophic cardiomyopathy gets louder, and the prolapse click moves earlier.

· Handgrip raises afterload, so mitral regurgitation, aortic regurgitation, and ventricular septal defect get louder while the aortic stenosis gradient and murmur fall; hypertrophic cardiomyopathy gets softer and the prolapse click moves later.

· Valsalva strain lowers preload, so most murmurs get softer, hypertrophic cardiomyopathy gets louder, and the prolapse click moves earlier.

· More preload means a larger stroke volume across the lesion and a louder murmur; more afterload increases the backward gradient in a regurgitant lesion and makes those murmurs louder.

· The two conditions that grow louder on standing are obstructive hypertrophic cardiomyopathy and mitral valve prolapse, and both worsen as the left ventricular cavity shrinks.

· In hypertrophic cardiomyopathy a smaller cavity brings the hypertrophied septum and the anterior mitral cusp closer together and worsens outflow obstruction; in prolapse a smaller cavity lets the redundant cusps prolapse earlier in contraction.

· In aortic stenosis the obstruction is fixed, so a lower preload only reduces flow across it and softens the murmur, and handgrip reduces the gradient, which also softens it.

Distinctions that matter

· Most murmurs follow the volume crossing the lesion, while hypertrophic cardiomyopathy and mitral valve prolapse follow cavity size and therefore move the opposite way with preload.

· Preload and afterload are separate levers: squatting raises both, Valsalva and standing drop preload, and handgrip raises afterload alone.

Common misconceptions

· Expecting handgrip to make every murmur louder. It increases regurgitant murmurs and softens the aortic stenosis murmur.

· Treating the prolapse click and murmur as moving together in loudness alone. The click shifts in timing, and the murmur lengthens as the click moves earlier.

Where it is taught: Section C07, dynamic maneuvers paragraph and the maneuver table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C07-PR01

Prediction exercise: Predict what standing does to a click and a late systolic murmur

A competent response contains

· Venous return falls on standing, so left ventricular end-diastolic volume falls.

· The cavity is smaller at the start of systole.

· The redundant cusps reach the point of giving way earlier in contraction, because a smaller cavity lets them prolapse sooner.

· The click moves earlier in systole.

· The murmur becomes longer, since it runs from the click to S2, and it is a late systolic murmur that now begins sooner.

· Squatting raises preload and afterload, so the cavity is larger, the click moves later, and the murmur shortens; handgrip raises afterload, which also moves the click later.

Distinctions that matter

· The click reports the moment the cusps give way and the murmur reports the leak that follows, so the two findings move together in timing.

· Mitral valve prolapse moves opposite to most murmurs with preload, in the same direction as hypertrophic cardiomyopathy.

Common misconceptions

· Expecting a lower preload to soften this murmur as it does most murmurs. A smaller cavity makes the prolapse worse.

· Reading an earlier click as a shorter murmur. An earlier click leaves more of systole for the murmur to fill.

Where it is taught: Section C07, dynamic maneuvers paragraph, the maneuver table, and the stop and predict box of the CV-4 Heart Valves and the Cardiac Cycle notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-05

CV-5 Cardiac Muscle Contraction

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-05-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the ventricular action potential

A competent response contains

· Phase 4 rest is about -85 to -90 mV, held near EK (about -95 mV with K+ of 140 mmol/L inside and 4 mmol/L outside) by the inward rectifier IK1.

· Phase 0 begins at a threshold of about -70 mV and reaches about +20 mV within 1 to 2 ms through fast Na+ channels (Nav1.5, gene SCN5A), carrying positive charge inward; the steepness of this upstroke sets conduction velocity.

· Phase 1 is a brief notch from Na+ channel inactivation plus transient outward K+ current (Ito), and corresponds to the J point.

· Phase 2 holds the membrane near 0 mV for about 0.2 s, with sustained inward Ca2+ current through L-type channels (Cav1.2) balanced by small outward K+ currents, and corresponds to the ST segment.

· Phase 3 is repolarization: L-type channels inactivate, the delayed rectifiers IKr (KCNH2, hERG) and IKs (KCNQ1) grow, and below about -40 mV IK1 unblocks and finishes the job; this is the T wave.

· Phase 0 is the QRS complex, phase 4 is the interval from the T wave to the next QRS, and the whole action potential of about 0.25 to 0.3 s corresponds approximately to the QT interval.

· Inward rectification means IK1 conducts well near the resting potential but very little once the cell is depolarized, because intracellular Mg2+ and polyamines plug the pore.

· That property gives working myocytes a stable resting potential and limits K+ loss during the long plateau.

· Current arrives from an already depolarized neighbor through gap junctions in the intercalated discs, so the ventricle behaves as a functional syncytium and no myocyte needs its own nerve terminal.

Distinctions that matter

· The Ca2+ current of phase 2 is what holds the plateau; the Na+ current of phase 0 only sets the upstroke and is already inactivating by phase 1.

· IK1 and the delayed rectifiers are different K+ currents with different jobs: IK1 sets the resting potential and finishes phase 3, while IKr and IKs do most of the work of phase 3.

Common misconceptions

· Drawing the plateau as electrical quiet. Two large currents are flowing and nearly cancelling, which is why small changes in either one shift the duration of the action potential.

· Saying working myocytes drift to threshold on their own. They have abundant IK1 and sit still until a neighbor depolarizes them.

Where it is taught: Section C01, opening paragraphs, the ventricular-ap diagram, and the table of phases of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C01-BD2

Brain dump 2 · integration and application: Refractoriness and a change in plasma potassium

A competent response contains

· The effective refractory period lasts about 0.25 to 0.3 s and ends near -50 to -60 mV; during it no stimulus of any strength produces a propagated action potential because too few fast Na+ channels have recovered from inactivation.

· The relative refractory period lasts about 0.05 s; a stronger than normal stimulus fires the cell, and the upstroke is slower and smaller.

· Inactivated Na+ channels cannot reset until the membrane repolarizes, which is why refractoriness lasts nearly as long as contraction.

· Skeletal muscle has a brief refractory period, so twitches summate into tetanus; the cardiac refractory period covers most of the twitch, so the ventricle must relax and refill before it can be excited again.

· The long refractory period also resists reentry, and that protection is undermined where repolarization is uneven, as in ischemia or long QT syndromes.

· With K+ of 8.0 mmol/L, EK = 61.5 x log(8/140), about -76 mV instead of about -95 mV.

· Because IK1 dominates resting conductance, the resting potential follows EK up to about -70 mV.

· At that potential many fast Na+ channels remain inactivated between beats, so phase 0 is slower and smaller, conduction slows, and the QRS widens.

· High extracellular K+ also increases IKr conductance, so phase 3 speeds up and T waves become peaked, and this usually appears before the QRS widens.

· If most Na+ channels are inactivated, conduction fails and ventricular fibrillation or asystole can follow.

· A low plasma K+ of 2.6 mmol/L reduces IKr and IK1, so phase 3 lengthens and afterdepolarizations are favored.

Distinctions that matter

· High and low plasma K+ move phase 3 in opposite directions: high K+ speeds repolarization, low K+ slows it, even though both disturb rhythm.

· Peaked T waves are a repolarization change and a widened QRS is a depolarization change, which is why they appear in that order.

Common misconceptions

· Expecting hyperkalemia to make cells more excitable because the membrane is depolarized. The depolarization sits in the range that inactivates Na+ channels, so the cell becomes harder to excite and conducts poorly.

· Explaining the lack of tetany by saying cardiac contraction is slow. It is the length of the refractory period relative to the twitch that prevents summation.

Where it is taught: Section C01, refractory period heading and the worked example on hyperkalemia of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the dyad and the path from voltage to force

A competent response contains

· T tubules carry the action potential into the cell at the Z lines, and where a T tubule meets one sarcoplasmic reticulum terminal cisterna it forms a dyad with a cleft of about 12 to 15 nm.

· Across that cleft, L-type Ca2+ channels (dihydropyridine receptors, Cav1.2) face clusters of RyR2, and the store inside the sarcoplasmic reticulum is buffered by calsequestrin.

· Diastolic cytosolic free Ca2+ is about 0.1 µmol/L, peak cytosolic Ca2+ in a beat is about 1 µmol/L, and extracellular ionized Ca2+ is about 1.1 to 1.3 mmol/L.

· In human ventricle the sarcoplasmic reticulum supplies about 70 percent of activator Ca2+.

· Calcium-induced calcium release: trigger Ca2+ entering the cleft during the plateau binds the cytosolic side of RyR2 and opens it, releasing a much larger amount of Ca2+ from the store.

· One RyR2 cluster releasing Ca2+ gives a Ca2+ spark; thousands of synchronized sparks make up the Ca2+ transient.

· Release is graded because each L-type channel controls only the cluster it faces, so a larger trigger or a fuller store releases more Ca2+.

· Release ends within tens of milliseconds, mainly because falling local sarcoplasmic reticulum Ca2+ lowers RyR2 open probability.

· Cross-bridge cycle: Ca2+ binds the single regulatory site of cardiac troponin C, troponin C grips troponin I, troponin I lets go of actin, tropomyosin rolls deeper into the groove and uncovers myosin-binding sites, a myosin head carrying ADP and Pi binds actin, Pi leaves in the power stroke, ADP leaves into the rigor state, a new ATP must bind before the head detaches, and hydrolysis of that ATP recocks the head.

· Atrial myocytes have few T tubules, so their release spreads inward from the surface, and adult human ventricle expresses mainly the slower, economical beta myosin heavy chain.

Distinctions that matter

· Trigger Ca2+ and activator Ca2+ are not the same pool: a small entry through L-type channels releases the much larger store that actually binds troponin C.

· Cardiac troponin C has one regulatory Ca2+ site while fast skeletal troponin C has two.

Common misconceptions

· Saying ATP powers the power stroke itself. The power stroke comes from Pi release, and ATP binding is what detaches the head, with hydrolysis recocking it.

· Treating release as all or nothing. It is graded by the size of the trigger and the loading of the store, which is exactly what the heart regulates.

Where it is taught: Section C02, first three paragraphs and the excitation-contraction coupling flow of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-BD2

Brain dump 2 · integration and application: Remove the trigger, then let the store leak

A competent response contains

· In a Ca2+ free solution cardiac contractions weaken within seconds to a minute and stop, because no trigger Ca2+ enters to open RyR2 and the store runs down as the Na+/Ca2+ exchanger keeps extruding Ca2+.

· The skeletal muscle keeps twitching, because Cav1.1 opens RyR1 through a mechanical link and needs no extracellular Ca2+ for a twitch.

· The structural difference is the mechanical coupling of the skeletal triad against the diffusional, Ca2+ dependent coupling of the cardiac dyad.

· Blocking cardiac L-type Ca2+ channels removes the trigger, so force falls promptly, while skeletal twitches are spared for the same reason the Ca2+ free bath spares them.

· A leaky RyR2, or loss of calsequestrin, lets the store escape in diastole under adrenergic stress.

· The Na+/Ca2+ exchanger extrudes that leaked Ca2+ while bringing in 3 Na+ for each Ca2+, so it carries net inward positive charge and is electrogenic.

· That net inward current produces delayed afterdepolarizations, which can reach threshold and start ventricular arrhythmia during exercise.

· The condition is catecholaminergic polymorphic ventricular tachycardia.

· The notes describe the same mechanism in glycoside toxicity and in failing myocardium.

Distinctions that matter

· Loss of the trigger weakens contraction, while loss of diastolic RyR2 closure causes arrhythmia; both are RyR2 problems with opposite presentations.

· The exchanger is the arrhythmogenic step, not the leak itself, because it converts a Ca2+ movement into a depolarizing current.

Common misconceptions

· Expecting a Ca2+ free solution to stop skeletal twitches quickly. Skeletal release does not need extracellular Ca2+, so the twitch persists.

· Calling the exchanger electroneutral. It moves 3 Na+ in for 1 Ca2+ out, so each cycle carries net positive charge inward.

Where it is taught: Section C02, paragraph on coupling without a mechanical link and the stop and predict box of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-RT01

Retrieval exercise: Rebuild excitation-contraction coupling

A competent response contains

· Step 1: the action potential spreads through gap junctions and down the T tubules.

· Step 2: L-type Ca2+ channels in the T tubule membrane open and trigger Ca2+ enters the dyadic cleft.

· Step 3: Ca2+ in the cleft binds the cytosolic side of RyR2 and opens it, which is calcium-induced calcium release.

· Step 4: cytosolic Ca2+ rises from about 0.1 to about 1 µmol/L.

· Step 5: Ca2+ binds troponin C, troponin I releases actin, and tropomyosin moves to uncover myosin-binding sites.

· Step 6: cross-bridges cycle using ATP and the sarcomere shortens.

· Relaxation reverses the sequence: SERCA2a and the Na+/Ca2+ exchanger lower cytosolic Ca2+ to about 0.1 µmol/L, Ca2+ leaves troponin C, tropomyosin covers the binding sites again, and each attached myosin head binds a fresh ATP in order to release actin.

Distinctions that matter

· Ca2+ entry and Ca2+ release are separate steps in the sequence, and only the second one supplies most of the Ca2+ that binds troponin C.

· Relaxation is not simply the absence of excitation: it needs ATP for SERCA2a and ATP at every myosin head.

Common misconceptions

· Writing that the action potential opens RyR2. Voltage opens the L-type channel, and Ca2+ opens RyR2.

· Leaving ATP out of relaxation. A head without a fresh ATP stays attached, which is what produces rigor.

Where it is taught: Section C02, excitation-contraction coupling flow, with relaxation in C03 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-CP01

Comparison table: Skeletal against cardiac excitation-contraction coupling

A competent response contains

· Excitation: skeletal fibers are excited by acetylcholine at their own neuromuscular junction; cardiac myocytes are excited by current from neighbors through gap junctions.

· T tubule position and junction: skeletal at the A-I junction forming a triad; cardiac at the Z line forming a dyad.

· Coupling: Cav1.1 mechanically opens RyR1 in skeletal muscle; in cardiac muscle Ca2+ entering through Cav1.2 opens RyR2, which is calcium-induced calcium release.

· Extracellular Ca2+: not needed for a skeletal twitch, required for a cardiac contraction.

· Action potential duration: about 1 to 5 ms in skeletal muscle, about 250 to 300 ms in cardiac muscle.

· Ca2+ sensor: troponin C in both, with two regulatory sites in fast skeletal troponin C and one in cardiac troponin C.

· A Ca2+ free solution leaves skeletal twitches largely unchanged within a minute, while cardiac force weakens and stops.

· Smooth muscle differs from both: it uses calmodulin and myosin light chain kinase rather than troponin C.

Distinctions that matter

· Both striated muscles use troponin C, so the real difference is in how the release channel is opened, not in the Ca2+ sensor.

· The long cardiac action potential is what supplies Ca2+ throughout the plateau, which is why its duration belongs in this comparison.

Common misconceptions

· Pairing the triad with cardiac muscle. The triad is skeletal, at the A-I junction; the cardiac junction is a dyad at the Z line.

· Assuming a drug that blocks L-type channels weakens both muscles. Skeletal coupling is mechanical, so it is spared.

Where it is taught: Section C02, comparison table on excitation-contraction coupling of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-BD1

Brain dump 1 · foundational retrieval: Build the calcium removal table from memory

A competent response contains

· SERCA2a sits in the sarcoplasmic reticulum membrane, uses ATP at 2 Ca2+ per ATP, removes about 70 percent of activator Ca2+, and is regulated by phospholamban.

· The Na+/Ca2+ exchanger sits in the sarcolemma, is driven by the Na+ gradient at 3 Na+ in per Ca2+ out, removes about 25 to 30 percent, and is governed by intracellular Na+ and membrane potential.

· The plasma membrane Ca2+-ATPase and mitochondrial uptake together handle about 1 to 2 percent, the pump using ATP in the sarcolemma under calmodulin control and mitochondrial uptake across the inner mitochondrial membrane.

· Cytosolic Ca2+ falls from about 1 µmol/L back to about 0.1 µmol/L during relaxation.

· Relaxation is active on two counts: SERCA2a and the plasma membrane pump hydrolyse ATP to move Ca2+ against its gradient, and each attached myosin head needs a fresh ATP before it can release actin.

· SERCA2a speed largely sets the rate of early diastolic relaxation.

· Unphosphorylated phospholamban lowers the Ca2+ affinity of SERCA2a and so inhibits it.

· Phosphorylation of phospholamban releases that inhibition, so phosphorylation activates the pump.

· Protein kinase A and CaMKII are the two kinases that phosphorylate phospholamban.

· The exchanger is electrogenic and depends on the Na+ gradient maintained by the Na+/K+-ATPase, so it slows or reverses when cytosolic Na+ rises or the membrane stays depolarized.

Distinctions that matter

· SERCA2a returns Ca2+ to the store inside the cell while the exchanger removes Ca2+ from the cell altogether, which is why one sets relaxation speed and the other sets the Ca2+ balance across the sarcolemma.

· Phospholamban is an inhibitor that is switched off by phosphorylation, so more phosphorylation means faster pumping.

Common misconceptions

· Describing relaxation as passive recoil. It costs ATP at the pumps and at every myosin head.

· Reading phospholamban phosphorylation as inhibition of SERCA2a. It removes inhibition.

Where it is taught: Section C03, first paragraph and the table of Ca2+ removal pathways of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-BD2

Brain dump 2 · integration and application: Partial inhibition of the sodium pump, beat by beat

A competent response contains

· Step 1: partial inhibition of the Na+/K+-ATPase raises cytosolic Na+ by a few mmol/L.

· Step 2: the smaller Na+ gradient slows the Na+/Ca2+ exchanger, so less Ca2+ leaves the cell during each diastole.

· Step 3: SERCA2a pumps the retained Ca2+ into the sarcoplasmic reticulum.

· Step 4: over several beats the store builds until extrusion again matches entry.

· Step 5: each beat then releases more Ca2+ through RyR2, so the Ca2+ transient is larger and contraction is stronger.

· Step 6: the effect needs no receptor signaling at all, which is why it works on a denervated or beta blocked heart.

· The change takes several beats because it works by accumulating store content, not by altering a single beat's trigger.

· With excessive inhibition the store becomes overloaded, releases Ca2+ spontaneously in diastole, and the exchanger turns that release into net inward current and delayed afterdepolarizations.

· Ischemia raises cytosolic Na+ by the same arithmetic: the Na+/K+-ATPase slows as ATP falls and Na+/H+ exchange adds more Na+ while the cell is acidotic, so the exchanger fails and Ca2+ accumulates.

Distinctions that matter

· Loading the store raises force over several beats, while a larger trigger raises force on the next beat; both act on release but on different timescales.

· A stronger beat and an afterdepolarization come from the same overload, which is why the margin between the two is narrow.

Common misconceptions

· Saying the pump inhibitor acts on the contractile proteins. It changes Na+, then the exchanger, then store loading.

· Expecting a receptor or a second messenger to be involved. This route is entirely transport based.

Where it is taught: Section C03, worked example on inhibiting the Na+/K+-ATPase and the paragraph on ischemia of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-RT01

Retrieval exercise: Rebuild calcium cycling in one heartbeat

A competent response contains

· Step 1: L-type channels admit a small trigger Ca2+ influx across the sarcolemma.

· Step 2: RyR2 releases most activator Ca2+ from the sarcoplasmic reticulum, about 70 percent of the total in human ventricle.

· Step 3: Ca2+ binds troponin C and cross-bridges generate force.

· Step 4: SERCA2a returns most Ca2+ to the sarcoplasmic reticulum, about 70 percent of what was removed.

· Step 5: the Na+/Ca2+ exchanger exports an amount equal to what entered, about 25 to 30 percent of removal.

· Step 6: cytosolic Ca2+ returns to about 0.1 µmol/L and the muscle relaxes.

· Balance one: reuptake into the store balances release from it.

· Balance two: extrusion across the sarcolemma balances entry across it.

· If extrusion falls behind entry, store content rises and later beats are stronger, up to the point of overload and diastolic release.

· If reuptake falls behind release, the store empties, diastolic Ca2+ stays high, relaxation slows, and later beats are weaker.

Distinctions that matter

· The two balances are separate: one governs store content, the other governs total cell Ca2+, and a pump inhibitor acts on the second to change the first.

· Most Ca2+ moves in and out of the store each beat while only a small amount crosses the sarcolemma.

Common misconceptions

· Treating the trigger influx as the main source of activator Ca2+. It is small, and its job is to open RyR2.

· Assuming entry and extrusion must be equal at every instant. They balance over the cycle, which is what lets store content change.

Where it is taught: Section C03, Ca2+ cycling cycle and the table of removal pathways of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-PR01

Prediction exercise: Predict what happens when coronary flow stops

A competent response contains

· Creatine phosphate falls quickly and anaerobic glycolysis produces lactate and H+, with Pi accumulating as ATP and creatine phosphate are broken down.

· Ca2+ sensitivity of troponin C falls because of the H+, and force per cross-bridge falls because of the Pi, so force drops within about 60 s while ATP is still substantial.

· Cytosolic Na+ rises as the Na+/K+-ATPase slows and Na+/H+ exchange imports Na+ in exchange for H+, so the Na+ gradient that drives the Na+/Ca2+ exchanger is lost and extrusion fails.

· SERCA2a slows as ADP and Pi accumulate and ATP falls, so diastolic cytosolic Ca2+ stays high and cross-bridges stay attached.

· The ventricle becomes stiff and end-diastolic pressure rises.

· Relaxation is impaired first, before reduced wall motion, ECG changes, and angina, which is the order of the ischemic cascade.

· After about 20 to 40 minutes Ca2+ overload causes irreversible injury, and on reperfusion that overload drives hypercontraction and contraction band necrosis.

· Severe ATP depletion leaves heads attached without fresh ATP, giving rigor and ischemic contracture.

Distinctions that matter

· Force falls from acidosis and Pi rather than from running out of ATP, which is why it falls so early.

· Impaired relaxation and impaired contraction come from the same energy failure, but they appear in a fixed order.

Common misconceptions

· Assuming contractile failure means ATP has run out. At 60 s ATP is still substantial and the problem is H+ and Pi at the myofilaments.

· Expecting ECG changes to be the earliest sign. Impaired relaxation comes first in the cascade.

Where it is taught: Section C03, heading on why ischemia impairs contraction and relaxation and the flow from ATP loss to stiff, weak myocardium of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the beta-1 cascade and its targets

A competent response contains

· Norepinephrine comes from sympathetic nerve terminals and epinephrine from the adrenal medulla; both act on beta-1 receptors, the predominant beta receptor of ventricular myocytes.

· The beta-1 receptor couples to Gs; the alpha subunit stimulates adenylyl cyclase, mainly types 5 and 6, and cAMP rises.

· cAMP frees the catalytic subunits of protein kinase A, and A-kinase anchoring proteins hold the kinase close to its targets.

· L-type Ca2+ channel: phosphorylation raises open probability and lengthens openings, giving a larger trigger and more store release, so contraction is stronger.

· Phospholamban: phosphorylation removes its inhibition of SERCA2a, so reuptake is faster and store loading is higher, giving faster relaxation and stronger beats.

· Troponin I: phosphorylation lowers the Ca2+ affinity of troponin C, so Ca2+ dissociates faster and relaxation is faster.

· Myosin-binding protein C: phosphorylation speeds cross-bridge kinetics, so force rises and falls faster.

· IKs channel (KCNQ1): phosphorylation gives a larger repolarizing K+ current, so the action potential shortens at high rates.

· The net result within seconds is a stronger, faster, shorter twitch with faster relaxation: positive inotropy and positive lusitropy.

· The signal is ended by phosphodiesterases, mainly PDE3 and PDE4, by phosphatases, and by receptor desensitization through G protein-coupled receptor kinase 2 and beta-arrestin.

Distinctions that matter

· Inotropy is the force and speed of contraction at a given sarcomere length; lusitropy is the rate of relaxation, and phospholamban phosphorylation serves both.

· The first three targets are the core of the response; myosin-binding protein C and IKs shape the timing rather than the size of the transient.

Common misconceptions

· Treating faster relaxation as a side effect. A faster heart rate shortens diastole more than systole, so filling depends on faster relaxation.

· Leaving out the ways the signal ends. Phosphodiesterases, phosphatases and desensitization are what make the response brief.

Where it is taught: Section C04, first two paragraphs, the beta-1 signaling flow, and the table of protein kinase A targets of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C04-BD2

Brain dump 2 · integration and application: Take one target out of the cascade

A competent response contains

· Heart rate still rises and the L-type Ca2+ current still increases, so contractility increases somewhat in the mutant animal.

· SERCA2a stays inhibited, so reuptake does not speed up and the store grows less than it should.

· The normal phospholamban response does two jobs: within a beat it makes cytosolic Ca2+ fall faster, and across successive beats it loads the store so later releases are larger.

· Relaxation therefore accelerates far less, with mainly the troponin I effect left, and filling suffers at fast heart rates because diastole is short and the ventricle relaxes slowly.

· In a normal heart troponin I phosphorylation alone would slightly lower force at a given Ca2+, but the larger Ca2+ transient more than compensates.

· Enhanced IKs keeps repolarization in step with the larger plateau Ca2+ current, so losing IKs means the action potential fails to shorten when sympathetic drive raises the plateau current, which is why exercise provokes arrhythmias in long QT type 1.

· In chronic heart failure the notes give four changes: beta-1 receptors are downregulated, G protein-coupled receptor kinase 2 is increased, SERCA2a is reduced, and phospholamban is left underphosphorylated.

· Consequences: weaker contraction from a smaller trigger and a smaller store, slower relaxation from reduced and inhibited SERCA2a, a blunted response to exercise from fewer and desensitized receptors, and more arrhythmias from the resulting Ca2+ handling.

· RyR2 phosphorylation by protein kinase A is also reported, but its importance is debated.

Distinctions that matter

· Within a beat and across beats are two different effects of the same phosphorylation, which is why losing it costs both lusitropy and inotropy.

· Receptor level changes and downstream protein changes both blunt the response in failure, and they are separate faults in the same pathway.

Common misconceptions

· Expecting no inotropic response at all in the mutant. The L-type channel and the rate response are intact.

· Reading troponin I phosphorylation as a weakening of the heart. The bigger transient outweighs it, and the gain is faster relaxation.

Where it is taught: Section C04, paragraph on phospholamban, the final paragraph on heart failure, and the stop and predict box of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C04-CA01

Clinical application: Breathlessness on exertion in a man with a dilated ventricle

A competent response contains

· Recognize: the failing pathway is beta-1 adrenergic signaling through Gs, cAMP and protein kinase A, shown by reduced beta-1 receptor density, increased G protein-coupled receptor kinase 2, reduced SERCA2a, and largely unphosphorylated phospholamban.

· Explain: fewer receptors and more G protein-coupled receptor kinase 2 mean less cAMP and less protein kinase A activity, so the L-type Ca2+ channel is phosphorylated less and the trigger is smaller; unphosphorylated phospholamban keeps SERCA2a inhibited and reduced SERCA2a protein lowers pumping further, so reuptake is slow, the store is underfilled, the Ca2+ transient is smaller and slower to fall, and troponin I stays unphosphorylated so Ca2+ leaves troponin C slowly.

· Differentiate: in low filling the Ca2+ transient is normal and the heart simply sits lower on a normal ventricular function curve, so force at any given sarcomere length is normal; here the transient is smaller and the whole curve is shifted downward, so force is low at every sarcomere length.

· Investigate: on the exercise test the failure of heart rate to rise much above resting and the flat stroke volume show chronotropic and inotropic responses to catecholamines are blunted despite elevated plasma norepinephrine; on the echocardiogram the dilated ventricle with slowed early diastolic filling shows the relaxation defect, and filling pressures that are adequate rule out simple underfilling.

· Decide: slowed early diastolic filling comes mainly from SERCA2a, which is both reduced in amount and left inhibited by unphosphorylated phospholamban, and once heart rate rises the shortened diastole leaves too little time for a slow pump to lower cytosolic Ca2+, so filling falls further.

· Justify: SERCA2a normally removes about 70 percent of activator Ca2+ and its speed largely sets early diastolic relaxation, so inhibition plus reduced protein slows that removal directly; the Ca2+ it fails to pump back leaves a smaller store, and because release is graded by store content, later beats release less Ca2+ and contract more weakly, which is how one pump defect produces both the stiff filling and the weak ejection.

Distinctions that matter

· A preload problem and an inotropic problem both lower stroke volume, and the Ca2+ transient is what separates them.

· Reduced SERCA2a protein and unphosphorylated phospholamban are two independent reasons for the same slow reuptake.

Common misconceptions

· Reading the high plasma norepinephrine as evidence that beta-1 signaling is intact. The drive is high precisely because the response is blunted.

· Treating slow relaxation as a passive stiffness problem only. Ca2+ removal is active, and its failure is what keeps cross-bridges attached in diastole.

Where it is taught: Section C04, final paragraph on chronic heart failure, with Ca2+ removal in C03 and curve shifts in C05 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the length-tension curves

A competent response contains

· Thick filaments are about 1.6 µm and thin filaments about 1 µm long, so overlap is optimal at about 2.0 to 2.2 µm.

· Ventricular sarcomeres work from about 1.8 to 1.9 µm at low filling to about 2.2 µm at normal to high filling, which is the ascending limb.

· On the descending limb stretch reduces overlap until active force reaches zero near 3.6 µm in the classic skeletal muscle studies.

· On the ascending limb force is limited because thin filaments from opposite sides interfere with each other, and at very short lengths thick filaments butt against the Z lines.

· Length-dependent activation is the rise in myofilament Ca2+ sensitivity with stretch, which shifts the force versus Ca2+ curve to the left so the same Ca2+ transient produces more force.

· The three proposed mechanisms are titin strain positioning more myosin heads to bind actin, narrower filament spacing as the cell thins, and cooperative Ca2+ binding to troponin C.

· Over the working range force rises several times more than filament overlap alone predicts, which is why the cardiac curve is steep.

· Stiff cardiac titin isoforms and collagen make passive tension climb steeply above about 2.2 µm, so intact hearts rarely reach the descending limb.

· A slower additional rise in the Ca2+ transient develops over minutes after sustained stretch, but the beat-to-beat response comes from overlap and Ca2+ sensitivity.

Distinctions that matter

· Active and passive tension are separate curves: active tension comes from cross-bridges, passive tension from titin and collagen.

· Overlap and Ca2+ sensitivity are two separate contributions to the same steep ascending limb.

Common misconceptions

· Drawing the ventricle working near the peak of the curve. It works on the ascending limb, below the optimum.

· Explaining the steep cardiac curve by overlap alone. Overlap changes modestly over the working range and sensitivity supplies most of the rise.

Where it is taught: Section C05, first two paragraphs and the length-tension comparison table of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-BD2

Brain dump 2 · integration and application: Tell a preload change from an inotropic change

A competent response contains

· Skeletal muscle grades force by recruiting motor units according to the size principle and by summating twitches at higher firing rates.

· The heart cannot recruit because gap junctions make it a functional syncytium and every myocyte contracts on every beat, and it cannot summate because the refractory period lasts nearly as long as the twitch.

· Lever one is Ca2+ delivery: raised by beta-1 stimulation and by a higher rate, which loads the store in the positive force-frequency relationship, and lowered in failing hearts where that relationship is blunted.

· Lever two is myofilament Ca2+ sensitivity: raised by stretch, lowered by acidosis and by troponin I phosphorylation.

· Lever three is sarcomere length, set by filling.

· These levers work only because a normal Ca2+ transient does not saturate troponin C, so there is room for the same transient to produce more or less force.

· The first heart shows a preload effect: it moves along its existing ventricular function curve with little change in the Ca2+ transient, and force at a given sarcomere length is unchanged.

· The second shows an inotropic effect: the larger transient shifts the curve upward, so force rises at any sarcomere length, and this is what a beta-1 agonist produces.

· A dilated failing ventricle is weak from reduced contractility, higher wall stress, and mitral regurgitation, not from overstretch.

· Chronic volume overload adds sarcomeres in series, which keeps individual sarcomeres off the descending limb even as the chamber enlarges.

· A stiff ventricle from hypertrophy, fibrosis or ischemia needs a higher filling pressure to reach the same volume.

Distinctions that matter

· Moving along a curve and shifting a curve are different events, and the Ca2+ transient tells you which one happened.

· Chamber dilation and sarcomere overstretch are not the same thing, because sarcomeres are added in series.

Common misconceptions

· Explaining failure in a dilated heart as sarcomeres on the descending limb. Stiff titin and collagen, plus added sarcomeres, keep them off it.

· Saying the heart recruits more myocytes when it needs more force. Every myocyte already contracts on every beat.

Where it is taught: Section C05, third and fourth paragraphs and the stop and predict box of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-CP01

Comparison table: Length-tension and force grading in skeletal against cardiac muscle

A competent response contains

· Working sarcomere length: skeletal muscle operates near the optimum, set by its bone attachments; cardiac muscle operates on the ascending limb at about 1.8 to 2.2 µm.

· Basis of the active length-tension curve: filament overlap in skeletal muscle; overlap plus length-dependent Ca2+ sensitivity in cardiac muscle, which makes the curve steep.

· Passive tension at long lengths: low in skeletal muscle until well beyond the optimum; high in cardiac muscle above about 2.2 µm because of stiff titin isoforms and collagen.

· Recruitment and summation: available in skeletal muscle through motor units and tetanus; unavailable in cardiac muscle because all cells contract each beat and the refractory period is long.

· Ca2+ delivery and sensitivity: a minor role in skeletal force grading, the major levers in cardiac force grading.

· Starting length: not used physiologically in skeletal muscle; a major lever in the heart, which is the Frank-Starling mechanism.

· Skeletal muscle varies how many fibers are active and how often they fire, while cardiac muscle varies how strongly each myocyte contracts.

Distinctions that matter

· Both muscles have a length-tension relationship, but only the heart uses starting length as a physiological control.

· High passive tension is protective in the heart, because it is what keeps sarcomeres off the descending limb.

Common misconceptions

· Filling the recruitment row for the heart with a partial answer. The heart has no recruitment and no summation, and the reasons are different for each.

· Writing that cardiac muscle works at its optimal overlap. It works below the optimum, on the ascending limb.

Where it is taught: Section C05, length-tension and force grading comparison table of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C01-Q01

Board-style question: A widened QRS with a high plasma potassium

Correct answer: C. A less negative resting potential leaving fast Na+ channels inactivated

Why it is right: Hyperkalemia moves the resting potential to about -70 mV because IK1 holds the membrane near EK, and at that voltage most fast Na+ channels remain inactivated, so the phase 0 upstroke is slow and small, ventricular conduction velocity falls, and the QRS widens.

A is wrong: Increased conductance of the rapid delayed rectifier K+ current is a real effect of hyperkalemia, but it speeds phase 3 and produces the peaked T waves rather than widening the QRS.

B is wrong: Prolonged opening of L-type Ca2+ channels would lengthen the plateau and the ST segment and QT interval, not the duration of ventricular depolarization.

D is wrong: Reduced Ca2+ release through RyR2 would weaken contraction, and release from the sarcoplasmic reticulum generates no deflection on the surface tracing.

A competent response contains

· A plasma K+ of 8.1 mmol/L raises EK from about -95 mV to about -76 mV, and because IK1 dominates resting conductance the resting potential follows to about -70 mV.

· At that voltage many fast Na+ channels stay inactivated between beats, so phase 0 is slower and smaller, conduction slows, and the QRS widens.

· The peaked T waves come from the separate effect of high extracellular K+ on IKr, which speeds phase 3.

Distinctions that matter

· The T wave change is a repolarization effect and the QRS change is a depolarization effect, with different currents behind each.

Common misconceptions

· Expecting a depolarized resting potential to make the cell easier to excite. It sits in the range that inactivates Na+ channels.

Where it is taught: Section C01 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-Q02

Board-style question: Two muscles in a calcium free bath

Correct answer: A. Cardiac release channels open only in response to Ca2+ entering through L-type channels, while the skeletal voltage sensor opens its release channel mechanically

Why it is right: Cardiac excitation-contraction coupling depends on trigger Ca2+ crossing the sarcolemma to open RyR2, so a Ca2+ free bath abolishes release within seconds to a minute, while the skeletal voltage sensor Cav1.1 opens RyR1 mechanically and keeps releasing the internal store.

B is wrong: Calmodulin and myosin light chain kinase belong to smooth muscle; both skeletal and cardiac muscle use troponin C as the Ca2+ sensor.

C is wrong: The cardiac sarcoplasmic reticulum holds a large store buffered by calsequestrin, and it supplies about 70 percent of activator Ca2+ in human ventricle.

D is wrong: The A-I junction and the triad are the skeletal arrangement; the cardiac T tubule meets the sarcoplasmic reticulum at the Z line in a dyad with a cleft of about 12 to 15 nm.

A competent response contains

· Cardiac coupling is Ca2+ dependent: trigger Ca2+ entering the dyadic cleft through L-type channels opens RyR2, so removing extracellular Ca2+ removes the trigger and the store also runs down through the Na+/Ca2+ exchanger.

· Skeletal coupling is mechanical: Cav1.1 in the triad opens RyR1 through a physical link, so a twitch needs no extracellular Ca2+.

Distinctions that matter

· Both muscles have a sarcoplasmic reticulum store and both use troponin C; the difference is only in how the release channel is opened.

Common misconceptions

· Assuming any Ca2+ dependent muscle must fail without extracellular Ca2+. Skeletal muscle depends on its internal store and a mechanical trigger.

Where it is taught: Section C02 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-Q03

Board-style question: Force doubles with a small change in overlap

Correct answer: D. Greater myofilament Ca2+ sensitivity at the longer sarcomere length

Why it is right: Length-dependent activation raises myofilament Ca2+ sensitivity when the sarcomere is stretched, through titin strain, narrower filament spacing and cooperative Ca2+ binding to troponin C, so the same peak Ca2+ produces several times more force than the change in overlap alone would give.

A is wrong: Recruitment does not exist in the heart, because gap junctions make it a functional syncytium and every myocyte contracts on every beat.

B is wrong: Summation is prevented by a refractory period that lasts nearly as long as the twitch, and the stem fixes the stimulus interval in any case.

C is wrong: A larger trigger influx would raise the peak cytosolic Ca2+, and the stem states that the peak Ca2+ of each twitch is unchanged.

A competent response contains

· Ventricular sarcomeres stretched from 1.9 to 2.2 µm stay on the ascending limb, where overlap changes only modestly.

· Most of the extra force comes from length-dependent activation: stretch raises myofilament Ca2+ sensitivity and shifts the force versus Ca2+ curve to the left, so an unchanged Ca2+ transient recruits more cross-bridges.

Distinctions that matter

· Changing the Ca2+ transient and changing the sensitivity to it are separate levers, and the stem fixes the transient.

Common misconceptions

· Explaining the steep cardiac length-tension curve by overlap alone, which accounts for only part of the rise.

Where it is taught: Section C05 of the CV-5 Cardiac Muscle Contraction notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-06

CV-6 Cardiac Output and Its Regulation

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-06-01-C01-BD1

Brain dump 1 · foundational retrieval: Write out the output and pressure equations

A competent response contains

· CO = SV x HR; resting CO is about 5 L/min at an SV of about 70 mL and a rate of about 72 bpm.

· SV = EDV - ESV; a resting left ventricle holds an EDV of about 120 mL and an ESV of about 50 mL.

· EF = SV / EDV, normally about 55 to 70%.

· Cardiac index is CO divided by body surface area, about 2.5 to 4 L/min per square metre.

· CO = (MAP - RAP) / TPR, and with RAP near 0 mmHg this becomes MAP = CO x TPR; normal TPR is about 18 mmHg per L/min and normal MAP about 70 to 100 mmHg.

· From a cuff, MAP is about diastolic pressure plus one third of the pulse pressure, so 120/80 mmHg gives about 93 mmHg.

· The one third weighting reflects diastole filling about two thirds of the cycle at resting rates; at fast rates diastole shortens more than systole, so the rule underestimates MAP.

· Fick: CO = VO2 / (CaO2 - CvO2), with VO2 about 250 mL/min at rest from expired gas analysis.

· O2 content = (1.34 x hemoglobin in g/dL x fractional saturation) + (0.003 x PO2 in mmHg), in mL O2 per dL; the first term is hemoglobin-bound and the second is dissolved, and dissolved O2 adds only about 0.3 mL/dL.

· Multiply content in mL/dL by 10 to express it per litre.

· Arterial content can be sampled from any systemic artery, but mixed venous content must come from the pulmonary artery, because venous blood from different organs differs widely in saturation.

Distinctions that matter

· Stroke volume is a per-beat volume and cardiac output is a per-minute flow, so a change in rate alters output without altering stroke volume.

· Ejection fraction is a ratio and partly corrects for chamber size, while stroke volume is an absolute volume.

Common misconceptions

· Taking MAP as the average of systolic and diastolic pressure. Diastole occupies about two thirds of the resting cycle, so the weighting is toward diastolic pressure.

· Leaving the dissolved term out of O2 content as if it were zero. It is small at about 0.3 mL/dL but it is part of the formula.

Where it is taught: Section C01, opening paragraphs and the Fick principle paragraphs of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C01-BD2

Brain dump 2 · integration and application: Work one patient two ways and read the venous saturation

A competent response contains

· SV is 150 minus 90, which is 60 mL, below the normal 70 mL.

· CO is 60 mL times 60 bpm, which is 3.6 L/min, below the normal 5 L/min.

· EF is 60 divided by 150, which is 40%, below the normal 55 to 70%.

· CaO2 = (1.34 x 12 x 0.97) + (0.003 x 90) = 15.6 + 0.27, about 15.9 mL/dL.

· CvO2 = (1.34 x 12 x 0.55) + (0.003 x 30) = 8.8 + 0.09, about 8.9 mL/dL.

· The arteriovenous difference is about 7.0 mL/dL, which is 70 mL O2 per litre of blood.

· CO = 240 / 70, which is about 3.4 L/min, close to the volumetric value of 3.6 L/min.

· A mixed venous saturation of 55% is low, so tissues are extracting more O2 from each litre and delivery is inadequate for demand.

· The arteriovenous difference widened because VO2 stayed the same while CO fell, and the Fick equation allows only those two ways to balance.

· A forearm vein samples one bed only and is usually less desaturated than true mixed venous blood, so the calculated difference would be too narrow and the calculated output too high.

Distinctions that matter

· Reduced output with a normal VO2 widens the arteriovenous difference, while a raised VO2 at a normal output widens it as well, so the two must be separated by knowing VO2.

· Low mixed venous saturation reports the adequacy of delivery relative to demand, not the arterial oxygenation, which is normal at 97% here.

Common misconceptions

· Dividing VO2 by a content difference still expressed per decilitre. That gives an answer ten times too large.

· Treating a low ejection fraction and a low cardiac output as the same finding. A dilated ventricle can hold a near-normal stroke volume at a low ejection fraction by filling to a very large EDV.

Where it is taught: Section C01, Fick paragraphs and the worked example on calculating output by two routes of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the Frank-Starling curve and define the three determinants

A competent response contains

· The curve of SV against EDV rises steeply and then flattens, and the normal heart works on the ascending limb.

· Further stretch is resisted by stiff myocardium and by the pericardium.

· Raised contractility shifts the whole curve up and to the left, giving a larger SV at the same EDV; reduced contractility shifts it down and to the right.

· Preload is strictly end-diastolic wall stress, indexed in practice by EDV, and is the stretch applied to the muscle before it contracts.

· Afterload is systolic wall stress, approximated by aortic pressure, and is the load the ventricle must overcome to open the aortic valve and eject.

· Contractility is the strength of contraction at a fixed preload and afterload, indexed by the slope of the end-systolic pressure-volume relationship, and is set by myofilament Ca2+ delivery and responsiveness.

· End-diastolic pressure, estimated for the left ventricle by pulmonary capillary wedge pressure, tracks volume only when compliance is normal, so a stiff ventricle shows a high filling pressure at a normal EDV.

· Optimal sarcomere length is about 2.2 micrometres.

· Stretch increases force by improving filament overlap and by raising the Ca2+ sensitivity of troponin C, and the Ca2+ sensitivity effect matters more.

· The Frank-Starling mechanism also keeps right and left ventricular outputs matched beat to beat.

Distinctions that matter

· Moving along one curve is a preload change; moving to a different curve is a contractility change.

· Preload acts through EDV, while afterload and contractility act through ESV.

Common misconceptions

· Treating filling pressure and filling volume as interchangeable measures of preload. They separate as soon as compliance falls.

· Explaining the Frank-Starling mechanism by filament overlap alone. Length-dependent Ca2+ sensitivity of troponin C is the larger part.

Where it is taught: Section C02, paragraphs on preload and the Frank-Starling mechanism and on contractility of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C02-BD2

Brain dump 2 · integration and application: Follow three single changes through to stroke volume

A competent response contains

· Raised aortic pressure acts on afterload first: more of each contraction is spent building pressure, so ESV rises and SV and EF fall, the Frank-Starling curve shifts down and right, and the loop becomes taller and narrower with the end-systolic point moving right along the same end-systolic pressure-volume relationship.

· Over the next beats the residual blood adds to normal filling, EDV rises, and the Frank-Starling mechanism restores much of the lost SV, at the cost of a higher filling pressure.

· A litre of saline acts on preload first: EDV rises, ESV is about unchanged, SV rises and EF rises slightly, the heart moves up the same Frank-Starling curve, and the loop widens to the right.

· A coronary occlusion that leaves part of the ventricle contracting weakly acts on contractility first: ESV rises, SV and EF fall, the Frank-Starling curve shifts down and right with a flatter end-systolic pressure-volume relationship, and EDV then rises as residual volume accumulates.

· EF is not a clean index of contractility because it also falls when afterload rises, and it can look near normal in a dilated ventricle with a very large EDV.

· At 200 bpm the extra beats come mostly out of diastole, so filling time shortens, EDV falls, and SV and then CO fall.

· Sympathetic lusitropy and inotropy raise the rate at which output starts to fall, while a stiff ventricle that depends on atrial contraction loses output at lower rates.

Distinctions that matter

· Afterload and reduced contractility both raise ESV, and they are separated by whether the end-systolic pressure-volume relationship itself has changed slope.

· The immediate effect of a change and the net effect after several beats of Frank-Starling adjustment are different answers to different questions.

Common misconceptions

· Expecting a faster rate always to raise cardiac output. Beyond a point the loss of filling time outweighs the extra beats.

· Reading a fall in ejection fraction as proof that contractility has fallen. A rise in afterload lowers ejection fraction with contractility unchanged.

Where it is taught: Section C02, afterload and contractility paragraphs, the prediction table, and the stop and predict box of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C03-BD1

Brain dump 1 · foundational retrieval: Draw the cardiac function and venous return curves

A competent response contains

· The cardiac function curve is the Frank-Starling curve with RAP as the measure of filling: CO rises steeply with RAP and then plateaus.

· The venous return curve falls as RAP rises, and the crossing point is the operating point, where CO equals venous return.

· The two flows must be equal in the steady state because the heart can pump only what returns to it.

· The venous return curve meets the horizontal axis where RAP equals mean systemic filling pressure, about 7 mmHg, which is the pressure that would exist throughout the systemic circulation if the heart stopped and pressures equalised.

· Venous return = (MSFP - RAP) / RVR, where RVR is the resistance to venous return, most of which lies in the veins.

· The slope of the curve is 1 divided by RVR.

· The curve plateaus once RAP falls below atmospheric pressure, because the large veins entering the thorax collapse.

· Stressed volume is the blood above what fills the vessels without stretching them, and only stressed volume generates pressure.

· The veins hold about two thirds of the blood.

· Venoconstriction converts unstressed volume into stressed volume, so MSFP rises and the curve shifts right, parallel to itself; transfusion does the same, while hemorrhage and venodilation shift it left.

Distinctions that matter

· A change in MSFP shifts the venous return curve parallel to itself, while a change in resistance rotates it about the same x-intercept.

· The cardiac function curve answers what the heart can do with a given filling pressure, and the venous return curve answers what the vasculature will deliver at that pressure.

Common misconceptions

· Treating total blood volume as what sets MSFP. Only the stressed portion generates pressure, which is why venous tone changes MSFP without any change in volume.

· Reading the venous return curve as falling without limit at low RAP. Venous collapse at the thoracic inlet flattens it.

Where it is taught: Section C03, opening paragraphs and the venous return curve paragraphs of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C03-BD2

Brain dump 2 · integration and application: Name the problem from cardiac output and right atrial pressure

A competent response contains

· When CO and RAP change in the same direction the primary problem is vascular, in volume or venous tone; when they change in opposite directions the problem is in the heart.

· Patient A: low CO with low RAP is inadequate filling, a vascular cause, with the venous return curve shifted left by a fall in MSFP, as in hemorrhage or venodilation.

· Patient B: low CO with high RAP is pump failure, with the cardiac function curve shifted down and right by reduced contractility.

· Patient C: three days of vomiting lowers blood volume and MSFP, shifting the venous return curve left, so CO and RAP both fall together, a vascular cause.

· Patient D: a large anterior infarction lowers contractility and shifts the cardiac function curve down and right, so CO falls while RAP rises, a cardiac cause.

· A fall in TPR rotates the venous return curve up about the same x-intercept and removes afterload so the cardiac function curve shifts up, so CO rises while RAP stays about the same.

· Because both curves move together, resistance changes show up as a change in flow with little change in filling pressure, which is why an arteriovenous fistula or a widely dilated muscle bed raises output.

Distinctions that matter

· The direction of RAP relative to CO is the clue that separates a cardiac cause from a vascular one.

· Volume and venous tone act on MSFP and move CO and RAP the same way; contractility moves them in opposite ways; TPR moves CO alone.

Common misconceptions

· Reading any low cardiac output as heart failure. A low RAP alongside it points to the vascular compartment instead.

· Expecting arteriolar constriction to raise RAP. It moves both curves in the same sense, so RAP changes little and CO falls.

Where it is taught: Section C03, the operating point table and the paragraph on the direction of right atrial pressure of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C03-RT01

Retrieval exercise: Rebuild the hemorrhage sequence on the curves

A competent response contains

· Step 1: blood loss reduces stressed volume.

· Step 2: MSFP falls and the venous return curve shifts left.

· Step 3: the new intersection sits at a lower RAP and a lower CO.

· Step 4: the lower MAP reduces baroreceptor firing.

· Step 5: sympathetic venoconstriction raises MSFP toward normal.

· Step 6: sympathetic inotropy shifts the cardiac function curve up.

· Step 7: CO partly recovers while RAP stays low.

· Blood loss lowers MSFP because it removes stressed volume, the blood above what fills the vessels without stretching them.

· The sensors are the carotid sinus and aortic arch baroreceptors.

· Venoconstriction restores the venous return curve by raising MSFP; inotropy restores the cardiac function curve by shifting it up and to the left.

· RAP stays low because the venous return curve is still shifted left and the upward shift of the cardiac function curve pulls RAP down further.

Distinctions that matter

· The primary event here is a shift of the venous return curve, and the reflex response moves both curves.

· Partial recovery of output is not recovery of volume; the operating point still sits at a low filling pressure.

Common misconceptions

· Expecting RAP to return to normal as output recovers. Reflex inotropy raises output by pumping the atrium emptier, so RAP falls rather than rises.

· Thinking a loss of about 10% of blood volume should drop arterial pressure sharply. The reflex holds pressure nearly constant at that point, so the operating point moves while the cuff reading barely changes.

Where it is taught: Section C03, flow diagram on hemorrhage, with the reflex detail in C05 of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-BD1

Brain dump 1 · foundational retrieval: Write the law of Laplace and the determinants of oxygen demand

A competent response contains

· Wall stress = (P x r) / (2 x h), where P is intraventricular pressure, r is chamber radius and h is wall thickness.

· Wall stress is the force each unit of wall must develop.

· In systole wall stress is the afterload, and it is a major determinant of myocardial O2 demand.

· Higher pressure raises wall stress in direct proportion.

· A larger radius raises wall stress in direct proportion, so a dilated ventricle must develop more force per unit of wall to produce the same pressure.

· A thicker wall lowers wall stress, because the load is shared among more muscle.

· Myocardial O2 demand is set mainly by heart rate, which is the number of tension-generating cycles per minute, by contractility, through the ATP used in Ca2+ cycling and cross-bridge activity, and by systolic wall stress.

· Developing pressure costs far more O2 than ejecting volume, so pressure work is much more expensive than volume work.

· The rate-pressure or double product is heart rate times systolic pressure.

· The heart already extracts about 70 to 75% of the O2 delivered to it at rest, so higher demand must be met almost entirely by higher coronary flow, and most left ventricular flow occurs in diastole.

Distinctions that matter

· Pressure and radius raise wall stress while thickness lowers it, so the same equation explains both the problem of dilation and the benefit of hypertrophy.

· Wall stress is force per unit of wall cross-section, not the pressure inside the chamber.

Common misconceptions

· Treating afterload as simply the aortic pressure. Aortic pressure is the practical approximation; the load itself is systolic wall stress, which also depends on radius and thickness.

· Assuming higher demand can be met by extracting more O2. Extraction is already about 70 to 75% at rest, so flow has to rise.

Where it is taught: Section C04, opening paragraph and the paragraphs on determinants of myocardial O2 demand of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-BD2

Brain dump 2 · integration and application: Remodeling and the ischemic cost of a faster rate

A competent response contains

· The first patient has pressure overload, signalled by raised systolic wall stress, with sarcomeres added in parallel so the wall thickens and the cavity stays normal or small, which is concentric hypertrophy.

· Raising systolic pressure from 120 to 200 mmHg increases wall stress by about two thirds, and thickening the wall from 1.0 to about 1.7 cm returns stress toward normal.

· The costs are a stiff chamber that needs a high filling pressure, which is diastolic dysfunction with an S4, and capillary growth that lags myocyte growth.

· Because capillary supply lags, the subendocardium becomes vulnerable to ischemia even with normal coronary arteries.

· Angiotensin II and catecholamines help drive the remodeling alongside the mechanical load.

· The second patient has volume overload, signalled by raised diastolic wall stress, with sarcomeres added in series so the chamber enlarges, which is eccentric hypertrophy.

· While compensation holds, the wall thickens enough to keep the thickness to radius ratio about normal, so wall stress stays near normal and the chamber delivers a large stroke volume; when it fails, radius grows without matching thickness, the ratio falls, and wall stress and the O2 cost of each beat rise in muscle that is already weakened.

· The third patient's demand rises because heart rate is one of the three main determinants of myocardial O2 demand, and the rate-pressure product doubles.

· On the supply side, most left ventricular coronary flow occurs in diastole, and a faster rate shortens diastole, so the time available for supply falls while demand rises.

Distinctions that matter

· Both remodeling patterns aim to restore normal wall stress; they differ in whether the overload acts in systole or in diastole.

· Concentric hypertrophy usually fails on the filling side with an S4, while decompensated dilation fails on the ejection side with an S3.

Common misconceptions

· Reading hypertrophy as purely harmful. The thicker wall lowers wall stress and is a corrective response; the harm comes from stiffness and from capillary supply lagging.

· Explaining tachycardia-induced ischemia on the demand side only. Shortened diastole cuts coronary supply at the same time.

Where it is taught: Section C04, paragraphs on remodeling under chronic overload and on myocardial O2 demand of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-CP01

Comparison table: Concentric against eccentric remodeling

A competent response contains

· Typical cause: concentric from pressure overload such as hypertension or aortic stenosis; eccentric from volume overload such as aortic or mitral regurgitation, and from a failing dilated ventricle.

· Signal: concentric follows raised systolic wall stress; eccentric follows raised diastolic wall stress.

· Sarcomere addition: in parallel, giving wider cells, in concentric hypertrophy; in series, giving longer cells, in eccentric hypertrophy.

· Thickness to radius ratio: increased in concentric hypertrophy; about normal in compensated eccentric hypertrophy and decreased in decompensated dilation.

· Wall stress: returns toward normal in concentric hypertrophy; near normal while eccentric hypertrophy is compensated, then rises as the chamber dilates.

· Chamber stiffness and filling pressure: concentric gives a stiff chamber needing a high filling pressure; eccentric keeps compliance higher and accommodates a large volume until it decompensates.

· Typical consequence: concentric gives diastolic dysfunction with an S4; eccentric gives a large stroke volume when compensated and systolic dysfunction with an S3 when decompensated.

Distinctions that matter

· The thickness to radius ratio is the quickest discriminator, because it rises in concentric hypertrophy and holds or falls in eccentric hypertrophy.

· Compensated and decompensated eccentric hypertrophy are different answers in the wall stress and consequence rows, so the row needs both states.

Common misconceptions

· Pairing regurgitation with concentric hypertrophy. Regurgitation is a volume load and adds sarcomeres in series.

· Assuming eccentric hypertrophy always means a weak heart. While the thickness to radius ratio holds, wall stress is near normal and stroke volume is large.

Where it is taught: Section C04, the concentric against eccentric remodeling table of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the baroreceptor reflex arc

A competent response contains

· Baroreceptors are stretch-sensitive nerve endings in the adventitia of the carotid sinus, a dilation at the origin of the internal carotid artery, and of the aortic arch.

· Carotid sinus afferents run in the carotid sinus nerve, the nerve of Hering, to the glossopharyngeal nerve (CN IX); aortic arch afferents run in the vagus (CN X).

· Both end in the nucleus tractus solitarius of the dorsal medulla, and the afferents release glutamate there.

· A rise in pressure raises vagal output and lowers sympathetic output; a fall does the opposite.

· The nucleus tractus solitarius excites vagal preganglionic neurons in the nucleus ambiguus and the dorsal motor nucleus of the vagus, and through the caudal ventrolateral medulla it inhibits the sympathetic premotor neurons of the rostral ventrolateral medulla that drive the intermediolateral cell column.

· Effectors: SA node (rate), AV node (conduction), ventricular myocardium (contractility, sympathetic only since the vagus has little direct effect on ventricular force), arterioles through alpha-1 receptors (resistance), veins (venous tone and MSFP), adrenal medulla (catecholamine release), and juxtaglomerular cells through beta-1 receptors (renin release).

· Carotid receptors are nearly silent below about 50 to 60 mmHg and fire maximally near 180 mmHg, with the steepest response around normal MAP, and firing also rises with the rate of pressure change.

· Aortic receptors operate about 30 mmHg higher than carotid receptors.

· The receptors reset within 1 to 2 days, so the reflex corrects short-term changes only; long-term pressure depends on renal control of volume.

Distinctions that matter

· Carotid sinus afferents travel in CN IX and aortic arch afferents in CN X, and the two sites have different operating ranges.

· Arterial baroreceptors sense pressure, while atrial and pulmonary low-pressure receptors sense volume and act mainly through the vagus on heart rate, renal sympathetic activity and ADH release.

Common misconceptions

· Treating the baroreflex as the long-term controller of blood pressure. Resetting within 1 to 2 days means it buffers change rather than setting the level.

· Expecting vagal activity to reduce ventricular contractility. Vagal innervation of ventricular muscle is sparse, so contractility is a sympathetic effect.

Where it is taught: Section C05, paragraphs on sensors and afferent pathways and on central integration and effectors of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C05-BD2

Brain dump 2 · integration and application: Run the reflex forwards and backwards in four situations

A competent response contains

· Blood loss of about 10%: firing falls, vagal output falls, sympathetic output rises, so heart rate, contractility, arteriolar tone and venous tone all rise and MAP is held nearly unchanged.

· Arteriolar constriction in that patient is concentrated in skin, gut and kidney, which gives cool pale skin and low urine output, and the fall in stroke volume narrows the pulse pressure.

· An alpha-1 vasoconstrictor drug: MAP rises, firing rises, vagal output rises and sympathetic output falls, so heart rate and contractility fall, which is reflex bradycardia, while arteriolar tone stays high because the drug is acting directly on the vessels.

· Carotid sinus massage: stretching the receptors mimics a rise in pressure, so firing rises, vagal output rises and sympathetic output falls; the SA node slows, AV nodal conduction lengthens, and arterioles dilate, so MAP falls.

· Slowed AV nodal conduction is why carotid pressure can interrupt a tachycardia whose circuit runs through the AV node.

· The older man with the tight collar has carotid sinus hypersensitivity, so pressure on the neck or head turning produces an exaggerated vagal response with bradycardia and syncope.

· Below a MAP of about 80 mmHg, peripheral chemoreceptors add sympathetic drive; below about 60 mmHg the brainstem ischemic response produces an intense sympathetic discharge.

Distinctions that matter

· A drug that raises pressure directly produces a reflex fall in heart rate, while a drug that lowers pressure directly produces a reflex rise, so the heart rate change reports the reflex and not the drug.

· Carotid massage and a fall in arterial pressure drive the same loop in opposite directions.

Common misconceptions

· Expecting arterial pressure to fall measurably after a loss of about 10% of blood volume. The reflex holds it, and tachycardia with cool skin and low urine output appears first.

· Reading reflex bradycardia after a vasoconstrictor as the drug acting on the heart. It is the baroreflex responding to the higher pressure.

Where it is taught: Section C05, paragraphs on predicting the response and on carotid sinus massage of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C05-RT01

Retrieval exercise: Rebuild the reflex response to a fall in pressure

A competent response contains

· Step 1: arterial pressure falls.

· Step 2: carotid sinus and aortic arch stretch decreases.

· Step 3: afferent firing in CN IX and CN X to the nucleus tractus solitarius falls.

· Step 4: vagal output falls and rostral ventrolateral medulla sympathetic output rises.

· Step 5: heart rate, contractility, arteriolar resistance and venous tone rise.

· Step 6: CO and TPR rise, so MAP returns toward its set point.

· Carotid sinus afferents travel in CN IX and aortic arch afferents in CN X, and both reach the nucleus tractus solitarius.

· The four effector variables that rise are heart rate, contractility, arteriolar resistance and venous tone.

· This is negative feedback because the response opposes the original change: the rising pressure restores baroreceptor firing and switches the response off.

· Reversed for a rise in pressure: stretch increases, afferent firing rises, vagal output rises and sympathetic output falls, so heart rate, contractility, arteriolar resistance and venous tone fall, CO and TPR fall, and MAP returns toward its set point.

Distinctions that matter

· The loop closes at the receptor, which is why restored pressure is what ends the response rather than any separate off switch.

· Falling pressure works through both limbs at once, withdrawing vagal tone and adding sympathetic tone.

Common misconceptions

· Drawing the pathway as sympathetic only. Withdrawal of vagal tone is the faster part of the heart rate response.

· Writing that reduced baroreceptor firing inhibits the rostral ventrolateral medulla. Reduced firing removes the inhibition, so sympathetic output rises.

Where it is taught: Section C05, the baroreceptor reflex cycle diagram of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C06-BD1

Brain dump 1 · foundational retrieval: Build the volume hormone table from memory

A competent response contains

· Angiotensin II: formed from angiotensin I by angiotensin-converting enzyme, stimulated by renin release, acting at AT1 receptors through Gq, causing arteriolar and efferent arteriolar constriction, proximal Na+/H+ exchange, aldosterone and ADH release, thirst and norepinephrine release.

· Aldosterone: from the zona glomerulosa, stimulated by angiotensin II and by a high plasma K+, acting at the intracellular mineralocorticoid receptor through gene transcription, raising ENaC and Na+/K+-ATPase so Na+ reabsorption rises and K+ and H+ secretion rise.

· ADH: from the supraoptic and paraventricular nuclei and released from the posterior pituitary, stimulated by high osmolality and by a large fall in volume or pressure, acting at V2 receptors through Gs and cAMP in the kidney and at V1 receptors through Gq in vessels, inserting aquaporin-2 for water reabsorption and constricting vessels at high levels.

· ANP: from atrial myocytes, stimulated by atrial stretch, acting at NPR-A guanylyl cyclase through cGMP, causing natriuresis, a raised GFR and vasodilation, and inhibiting renin, aldosterone and ADH.

· BNP: mainly from ventricular myocytes, stimulated by ventricular stretch and wall stress, acting through the same NPR-A and cGMP route, with the same actions, and BNP with inactive NT-proBNP marks ventricular strain.

· RAAS sequence: low perfusion pressure, low macula densa NaCl or beta-1 input; juxtaglomerular cells release renin; renin cleaves angiotensinogen to angiotensin I; pulmonary ACE forms angiotensin II; arteriolar constriction, proximal Na+ reabsorption, thirst and ADH follow; aldosterone raises ENaC and Na+/K+-ATPase in principal cells; Na+ and water retention raise volume and pressure.

· The three stimuli for renin are low renal perfusion pressure with less arteriolar stretch, low macula densa NaCl delivery, and beta-1 stimulation; angiotensin II and ANP inhibit release.

· Renin comes from juxtaglomerular cells of the afferent arteriole, and the cleavage of angiotensinogen is the rate-limiting step.

· ACE is most abundant on pulmonary capillary endothelium and also inactivates the vasodilator bradykinin.

· Angiotensin II constricts the efferent arteriole more than the afferent, preserving filtration pressure when renal perfusion is low.

· Timing: the baroreceptor reflex acts in seconds, angiotensin II in minutes, and aldosterone over hours to days because it works through new protein synthesis.

Distinctions that matter

· Aldosterone retains Na+ with water and so changes volume, while ADH retains water alone and so changes osmolality.

· ANP and BNP differ in their source and stimulus, atrial stretch against ventricular stretch and wall stress, not in their actions.

Common misconceptions

· Treating renin as the hormone that acts on vessels and nephron. Renin is an enzyme, and angiotensin II and aldosterone carry the actions.

· Expecting aldosterone to act within minutes. Its effect takes hours because it works through transcription of ENaC and Na+/K+-ATPase.

Where it is taught: Section C06, the hormonal regulators table, the RAAS paragraphs and flow diagram, and the final paragraph on timing of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C06-BD2

Brain dump 2 · integration and application: Three patients, three hormone patterns

A competent response contains

· Patient A: renin rises from low renal perfusion pressure, low macula densa NaCl delivery and beta-1 stimulation, so angiotensin II and aldosterone rise; ADH rises because the fall in volume is large enough, on the order of 10% or more, and angiotensin II also stimulates its release; ANP and BNP fall because atrial and ventricular stretch fall.

· Patient B: osmoreceptors in the anterior hypothalamus detect the rise in osmolality at a change of about 1%, so ADH rises; renin, angiotensin II and aldosterone change little because volume and renal perfusion are normal, and the natriuretic peptides change little.

· Patient C: aldosterone is high from the adenoma, so renin and angiotensin II are suppressed by the resulting volume expansion and higher pressure; ANP rises with atrial stretch; ADH is not the driver.

· Patient A retains Na+ with water through aldosterone and angiotensin II, so extracellular and plasma volume rise; patient B retains water alone through ADH and aquaporin-2 insertion in the collecting duct, which corrects osmolality without adding much volume.

· The difference follows from the target: aldosterone raises ENaC and Na+/K+-ATPase so Na+ is reabsorbed, while ADH inserts aquaporin-2 so water follows the medullary gradient.

· In patient C, Na+ retention raises pressure, and the lumen-negative potential created by Na+ reabsorption drives K+ secretion, so plasma K+ falls.

· There is no edema because within days the higher pressure and ANP increase Na+ and water excretion enough to halt the expansion, which is aldosterone escape.

· Long-term pressure control rests on renal pressure natriuresis: higher pressure increases Na+ and water excretion until volume and pressure return to the level the kidney defends.

· In heart failure natriuretic peptide levels are high but the renal response is blunted, so they cannot overcome RAAS and sympathetic drive.

· Inappropriate ADH secretion retains water rather than Na+, which dilutes plasma Na+ and causes hyponatremia rather than hypertension.

Distinctions that matter

· Osmolality needs about a 1% change to release ADH, while volume or pressure needs a change on the order of 10% or more, which is why osmoregulation dominates day to day.

· High aldosterone gives hypertension with hypokalemia and no edema, while heart failure gives high aldosterone with edema, because the kidney in heart failure is also seeing a low perfusion signal.

Common misconceptions

· Expecting high aldosterone to cause edema. Aldosterone escape halts the expansion, so pressure and K+ change but interstitial volume does not keep rising.

· Assuming high natriuretic peptide levels mean the kidney is excreting salt briskly. In heart failure the renal response to them is blunted.

Where it is taught: Section C06, the ADH and natriuretic peptide paragraphs, the final paragraph on pressure natriuresis, and the chapter integration section of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C06-PR01

Prediction exercise: Predict the response to renal artery narrowing

A competent response contains

· Renin release from the affected kidney rises; the other two stimuli at that cell are reduced stretch of the afferent arteriole from low perfusion pressure and reduced NaCl delivery to the macula densa.

· Circulating angiotensin II rises, because renin cleaving angiotensinogen to angiotensin I is the rate-limiting step and pulmonary ACE then forms angiotensin II.

· Efferent arteriolar tone rises more than afferent tone, which preserves glomerular filtration pressure while renal perfusion is low.

· Aldosterone secretion rises under angiotensin II, and transcription raises the number of apical epithelial Na+ channels and of basolateral Na+/K+-ATPase in principal cells.

· Urinary K+ and H+ secretion rise, because Na+ reabsorption leaves the lumen electrically negative, which drives K+ secretion from principal cells and H+ secretion from intercalated cells.

· Plasma volume rises as Na+ and water are retained, and systemic arterial pressure rises over the next days.

· Renin release falls in the normal kidney once systemic pressure has risen, but the affected kidney still sees a low pressure beyond the narrowing, so it keeps releasing renin and the cycle continues.

Distinctions that matter

· Low perfusion pressure at the juxtaglomerular cell is not the same as low systemic pressure, which is why renin stays high while the arm cuff reads hypertensive.

· Angiotensin II acts within minutes while aldosterone takes hours, because one binds a Gq-coupled receptor and the other works through transcription.

Common misconceptions

· Expecting the rise in systemic pressure to switch off renin from the narrowed side. The narrowing keeps the pressure at that juxtaglomerular cell low.

· Expecting glomerular filtration to collapse in the affected kidney. Preferential efferent constriction by angiotensin II holds filtration pressure up.

Where it is taught: Section C06, the renin-angiotensin-aldosterone paragraphs and flow diagram of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C07-BD1

Brain dump 1 · foundational retrieval: What standing does and how the body answers it

A competent response contains

· Orthostatic hypotension is a sustained fall in systolic pressure of at least 20 mmHg, or diastolic pressure of at least 10 mmHg, within 3 minutes of standing or of head-up tilt to at least 60 degrees.

· Standing raises foot venous pressure to about 90 mmHg in a motionless adult.

· Roughly half a litre of blood pools in the leg and pelvic veins within seconds, and plasma filters into the leg interstitium over minutes.

· Venous return falls, so SV, CO and MAP fall in turn.

· The baroreceptor reflex raises heart rate by about 10 to 20 bpm and constricts splanchnic and muscle arterioles and veins, restoring MAP within about a minute with systolic pressure slightly lower and diastolic pressure slightly higher.

· The skeletal muscle pump drives blood upward, and venous valves make that flow one way.

· Renin and ADH add support over minutes.

· Symptoms appear once MAP at head level falls below the lower limit of cerebral autoregulation, about 60 mmHg: lightheadedness, blurred vision, weakness, cognitive slowing and syncope.

· A coat-hanger ache of the neck and shoulders is the pattern specific to the neurogenic forms.

Distinctions that matter

· Venous pooling happens in seconds while plasma filtration into the leg tissue takes minutes, so the two phases of the fall have different time courses.

· The pressure criteria define the sign, and the symptoms depend on cerebral perfusion, so a patient can meet the criteria without symptoms if autoregulation holds.

Common misconceptions

· Reading a transient dip on standing as orthostatic hypotension. The definition requires a sustained fall within 3 minutes.

· Expecting both systolic and diastolic criteria to be met. Either one is enough.

Where it is taught: Section C07, the definition paragraph and the paragraph on why standing threatens arterial pressure of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C07-BD2

Brain dump 2 · integration and application: Separate the mechanisms from two sets of vital signs

A competent response contains

· Patient A: systolic fall is 112 minus 88, which is 24 mmHg, and the heart rate rise is 128 minus 92, which is 36 bpm; the definition is met, the reflex arc is intact, and the mechanism is volume depletion.

· Patient B: systolic fall is 162 minus 118, which is 44 mmHg, and the heart rate rise is 100 minus 96, which is 4 bpm; the definition is met, the afferent and central limbs are working but the efferent sympathetic limb is not, and the mechanism is neurogenic orthostatic hypotension from autonomic failure.

· The heart rate rise is the single number that separates them: a large rise means an intact reflex working against too little volume, and little or no rise means the efferent sympathetic pathway cannot release enough norepinephrine.

· Patient B is hypertensive lying down partly because denervated vessels are hypersensitive to circulating pressors, and supine hypertension is common in autonomic failure.

· In long-standing diabetes the nerve injury mechanisms are aldose reductase converting glucose to sorbitol with depletion of NADPH, which leaves less glutathione regeneration and more oxidative stress, accumulation of advanced glycation end products, and vasa nervorum disease adding ischemia.

· Long fibres fail first and vagal loss often precedes sympathetic loss, so a fixed resting tachycardia appears before splanchnic vasoconstrictor loss causes orthostatic hypotension.

· Postural tachycardia syndrome is a heart rate rise of at least 30 bpm in adults on standing with symptoms but no significant fall in pressure.

· In vasovagal syncope sympathetic tone is suddenly withdrawn while vagal output surges, so pressure and heart rate fall together.

· Volume depletion also arises from hemorrhage, vomiting, diarrhoea, diuresis and burns, and from adrenal insufficiency, where aldosterone loss causes renal Na+ wasting and cortisol loss reduces vascular responsiveness to catecholamines.

Distinctions that matter

· A fall in pressure with a brisk heart rate rise is a volume problem, while a larger fall with a flat heart rate is an efferent nerve problem.

· Postural tachycardia syndrome has the tachycardia without the pressure fall, and vasovagal syncope has both falling together, so neither matches either of these patients.

Common misconceptions

· Taking the size of the pressure fall as the discriminator. The heart rate response is what separates the mechanisms.

· Assuming autonomic failure means low pressure at all times. Supine hypertension is common in these patients.

Where it is taught: Section C07, the mechanisms paragraphs, the table distinguishing the main mechanisms, and the worked example on two sets of orthostatic vital signs of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C07-CA01

Clinical application: Lightheadedness on standing after two weeks in bed

A competent response contains

· Recognize: systolic pressure fell 128 minus 96, which is 32 mmHg, diastolic fell 76 minus 60, which is 16 mmHg, and the heart rate rose 104 minus 78, which is 26 bpm at 3 minutes standing, so both pressure criteria are met within the time limit and this is orthostatic hypotension.

· Explain: prolonged bed rest shifts blood toward the thorax and stretches the atria, so ANP rises and ADH falls, and the resulting diuresis lowers plasma volume within days; the ventricle becomes smaller and stiffer, baroreflex sensitivity falls, and the leg muscle pump weakens with disuse.

· Differentiate: volume loss through vomiting or bleeding gives the same intact reflex pattern but with an external route of loss and usually a raised urea to creatinine ratio and thirst, whereas this patient lost volume by renal excretion during recumbency; failure of the sympathetic efferent pathway would show little or no heart rate rise with supine hypertension, and the 26 bpm rise here is what separates this patient from that.

· Investigate: a heart rate rise on standing of roughly 20 bpm or more, confirming an intact efferent limb; a supine pressure that is normal rather than high, here 128/76 mm Hg; a urea to creatinine ratio within the reference range, here 14, arguing against brisk external volume loss; a normal plasma Na+ at 139 mEq/L, since water was lost with Na+ rather than water alone; the weight loss of 3 kg and the history of 15 days of immobility; and the coat-hanger ache, which raises neurogenic forms and is checked against the heart rate response.

· Decide: three of the four defences are weakened, since plasma volume and MSFP are low, baroreflex sensitivity is reduced, and the muscle pump is deconditioned, while the reflex arc itself, the afferent, central and efferent sympathetic pathway, remains intact.

· Justify: the lower plasma volume reduces stressed volume, so MSFP falls and the venous return curve shifts left, which moves the operating point to a lower CO at a lower RAP; pooling of roughly half a litre on standing shifts it further left, and because venous return sets what the heart can eject, the intact reflex can raise heart rate, contractility and vascular tone but cannot create the filling volume that the stiffer, smaller ventricle needs, so MAP falls despite a brisk heart rate rise.

Distinctions that matter

· An intact reflex with a brisk heart rate rise separates a filling problem from autonomic failure, and the supine pressure separates them again.

· Deconditioning weakens both the volume and the pump, while the nerve pathway stays intact, which is why the pattern looks like volume depletion without any external fluid loss.

Common misconceptions

· Reading the normal urea to creatinine ratio and normal plasma Na+ as evidence against volume depletion. Recumbency-induced diuresis removes Na+ with water, so osmolality and the ratio can stay normal.

· Attributing the coat-hanger ache alone to autonomic failure here. The heart rate rise of 26 bpm shows the efferent sympathetic limb is working.

Where it is taught: Section C07, mechanisms paragraph on prolonged bed rest and the mechanism table, with the venous return reasoning in C03 of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C01-Q01

Board-style question: Reading the arteriovenous oxygen difference

Correct answer: C. It has fallen from about 5.0 L/min to about 3.1 L/min, because each litre of blood now gives up more oxygen

Why it is right: With VO2 fixed at 250 mL/min, output varies inversely with the arteriovenous O2 difference, so widening it from 50 to 80 mL per litre lowers calculated output from 5.0 to about 3.1 L/min, and the low mixed venous content shows tissues extracting more from each litre because delivery has fallen.

A is wrong: A rise in output would narrow the arteriovenous difference, since less extraction per litre is needed, and extraction here has clearly increased rather than decreased.

B is wrong: Output cannot be unchanged when the denominator of the Fick relationship has changed while the numerator and arterial content have not; the mixed venous value is what carries the information.

D is wrong: Mixed venous content is the second term of the content difference in the Fick relationship, so sampling it from the pulmonary artery is exactly what makes the estimate possible.

A competent response contains

· Fick: CO = VO2 / (CaO2 - CvO2), with content differences converted from mL/dL to mL per litre by multiplying by 10.

· Last year the difference was 20 minus 15, which is 5 mL/dL or 50 mL/L, so CO was 250 divided by 50, which is 5.0 L/min.

· Today the difference is 20 minus 12, which is 8 mL/dL or 80 mL/L, so CO is 250 divided by 80, which is about 3.1 L/min.

· A widening arteriovenous O2 difference at an unchanged VO2 means output has fallen and tissues are extracting more O2 from each litre of blood.

Distinctions that matter

· A falling mixed venous content reports greater extraction, which follows from reduced delivery rather than from any change in arterial oxygenation.

Common misconceptions

· Reading a wider arteriovenous difference as better oxygen delivery. It means each litre is giving up more because fewer litres are arriving.

Where it is taught: Section C01 of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-Q02

Board-style question: Wall stress in a dilating ventricle

Correct answer: A. Systolic wall stress has risen in proportion to the radius, because wall stress varies with pressure times radius divided by twice the thickness

Why it is right: In the law of Laplace wall stress rises in direct proportion to chamber radius at a fixed pressure and thickness, so a 40% larger radius raises systolic wall stress by about 40%, and systolic wall stress is one of the three main determinants of myocardial O2 demand.

B is wrong: Myocardial O2 extraction at rest is already about 70 to 75%, not about 25%, and it is extraction being near maximal that forces demand to be met by coronary flow instead.

C is wrong: A larger radius raises systolic wall stress rather than lowering it, and pressure work costs far more O2 than volume work, so the reasoning is reversed on both counts.

D is wrong: Most left ventricular coronary flow occurs in diastole rather than systole, and coronary flow is the supply side of the balance, not the source of the O2 cost of a beat.

A competent response contains

· Wall stress = (P x r) / (2 x h), so with pressure and thickness unchanged a 40% larger radius raises systolic wall stress by about 40%.

· Systolic wall stress is one of the three main determinants of myocardial O2 demand, alongside heart rate and contractility.

· The dilated ventricle must develop more force per unit of wall to produce the same pressure, so each beat costs more O2.

Distinctions that matter

· Radius and thickness act on wall stress in opposite directions, so dilation without matching thickening raises the O2 cost of every beat.

Common misconceptions

· Assuming a larger chamber shares the load better. A larger radius raises wall stress in direct proportion.

Where it is taught: Section C04 of the CV-6 Cardiac Output and Its Regulation notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-07

CV-7 Conduction System and the Electrocardiogram

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-07-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the conduction pathway

A competent response contains

· Order: SA node, atrial muscle with the anterior interatrial band to the left atrium, AV node, bundle of His, right and left bundle branches with anterior and posterior left fascicles, subendocardial Purkinje fibers, ventricular muscle.

· SA node lies under the epicardium of the right atrium at the junction with the superior vena cava, at the upper end of the sulcus terminalis.

· AV node lies in the lower interatrial septum in the triangle of Koch, bounded by the coronary sinus opening, the septal tricuspid leaflet, and the tendon of Todaro.

· The His bundle pierces the fibrous skeleton and runs along the lower border of the membranous septum.

· Velocities: SA node about 0.05 m/s, atrial muscle about 0.3 m/s, AV node about 0.05 m/s and slowest, Purkinje fibers about 1.5 to 4 m/s and fastest, ventricular muscle about 0.3 to 0.5 m/s.

· Intrinsic rates: SA node about 100 bpm denervated and 60 to 100 bpm at rest with vagal tone, AV junction about 40 to 60 bpm, Purkinje about 20 to 40 bpm, and atrial and ventricular muscle have none normally.

· Upstroke: L-type Ca2+ current in the SA and AV nodes, fast Na+ current in atrial muscle, Purkinje fibers, and ventricular muscle.

· Conduction slows in the AV node because its cells are small, poorly coupled by few gap junctions, rest near -60 mV where fast Na+ channels are inactivated, and depolarize through slow L-type Ca2+ channels.

· The fibrous skeleton is an electrical insulator, so the AV node and His bundle are the only normal route from atria to ventricles.

· Purkinje fibers activate apex and septum before the base and drive the wall from endocardium to epicardium; the right bundle runs through the moderator band to the anterior papillary muscle so it tenses its chordae first.

Distinctions that matter

· Conduction velocity and intrinsic rate are separate properties: Purkinje fibers conduct fastest but pace slowest.

· Gap junctions couple all myocytes, while the conduction system imposes the order and timing on that coupling.

Common misconceptions

· Students place the AV node in the AV septum generally; it sits in the lower interatrial septum within the triangle of Koch.

· Writing that nodal cells use fast Na+ channels misses why they are slow; their upstroke is L-type Ca2+ and their resting voltage already inactivates most Na+ channels.

Where it is taught: Section C01 of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C01-BD2

Brain dump 2 · integration and application: What the AV delay buys and what happens without it

A competent response contains

· AV delay totals about 0.13 s: about 0.09 s in the node and about 0.04 s in the penetrating bundle, reached about 0.03 s after the impulse leaves the SA node.

· The delay lets atrial systole finish before the AV valves close, adding up to about 20% of resting filling, raising end-diastolic volume and stroke volume through the Frank-Starling mechanism.

· That contribution matters more when the ventricle is stiff and when the rate is fast, because filling time is short.

· At an atrial rate of 400 per minute the node blocks most impulses because its refractory period outlasts its action potential, so only impulses arriving at a recovered node conduct.

· The ventricular response is therefore slower than the atrial rate and irregularly irregular, with a narrow QRS because the His-Purkinje system is still doing the distributing.

· An accessory pathway is fast Na+ dependent muscle without decremental conduction, so it removes both the delay and the filter and passes far more impulses to the ventricle.

· Myocyte to myocyte spread, as in bundle branch block or a ventricular ectopic beat, makes contraction less synchronous.

Distinctions that matter

· Decremental conduction is progressive slowing with faster impulse arrival; refractoriness outlasting the action potential is what actually blocks impulses in flutter and fibrillation. Both come from time-dependent Ca2+ channel recovery.

· Atrial kick raises filling volume; it does not change the rate at which the ventricle is activated.

Common misconceptions

· Students say the atria push in most of ventricular filling. Most filling is passive, and the atrial contribution is up to about 20% at rest.

· Describing the ventricular rate in atrial fibrillation as regular misses that conduction depends on when each impulse finds the node recovered.

Where it is taught: Section C01, final paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the SA nodal action potential

A competent response contains

· Maximum diastolic potential is about -60 mV and the upstroke begins at about -40 mV.

· Phase 4: HCN channels open on repolarization and carry the funny current If, mainly Na+ inward; delayed rectifier K+ current is switching off; late in phase 4, T-type Ca2+ current and sarcoplasmic reticulum Ca2+ release extruded by the Na+/Ca2+ exchanger add inward current.

· Phase 0: L-type Ca2+ channels open at about -40 mV, giving a slow upstroke of small amplitude.

· Phase 3: L-type channels inactivate and delayed rectifier K+ efflux returns the cell to about -60 mV.

· Nodal cells have very few inward rectifier K+ channels (IK1), so nothing clamps them near the K+ equilibrium potential and the membrane drifts to threshold after every beat.

· Working myocytes rest near -85 to -90 mV because IK1 holds them there.

· The nodal tracing has no phase 1 notch and no phase 2 plateau; refractoriness still outlasts repolarization because Ca2+ channel recovery is time dependent.

Distinctions that matter

· If is an inward Na+ current through HCN channels, not a K+ current, and it switches on with hyperpolarization rather than depolarization.

· T-type Ca2+ current belongs to late phase 4; L-type Ca2+ current carries phase 0.

Common misconceptions

· Calling -60 mV a resting potential: it is the most negative point of a continuously drifting membrane, so the term is maximum diastolic potential.

· Drawing a fast tall upstroke for the SA node. The Ca2+ upstroke is slow and small, which is why nodal conduction velocity is about 0.05 m/s.

Where it is taught: Section C02, opening paragraph and bullets of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C02-BD2

Brain dump 2 · integration and application: Vagal and sympathetic drive at the node

A competent response contains

· Rate is set by the slope of phase 4, the maximum diastolic potential, and the threshold.

· Acetylcholine at M2 receptors acts through Gi: the alpha subunit lowers cAMP and reduces If, flattening the slope; the beta-gamma subunits open IKACh so the maximum diastolic potential becomes more negative; reduced Ca2+ current raises the effective threshold. The cell starts lower, climbs more slowly, and has further to climb.

· Norepinephrine at beta-1 receptors acts through Gs: cAMP rises and binds HCN channels directly to increase If, and protein kinase A phosphorylates L-type Ca2+ channels, so phase 4 steepens and threshold is reached sooner.

· In the AV node, vagal activity slows conduction and lengthens the PR interval, and sympathetic activity speeds conduction and shortens it.

· Strong vagal discharge, as in the baroreceptor reflex or a vasovagal episode, can produce a sinus pause or transient AV block.

· An L-type Ca2+ channel blocker slows sinus rate and AV conduction and reduces contractility, with little effect on the ventricular Na+ upstroke, because nodal phase 0 and myocyte contraction both depend on Ca2+ entry.

· A selective If blocker slows the rate alone, because HCN channels are confined to the pacemaker phase 4 current and play no part in contraction.

Distinctions that matter

· Vagal slowing works on three variables at once; sympathetic acceleration works mainly on the slope of phase 4.

· Both an If blocker and a Ca2+ channel blocker slow the sinus rate, but only the Ca2+ blocker changes AV conduction and contractility.

Common misconceptions

· Saying acetylcholine hyperpolarizes nodal cells by reducing If. The hyperpolarization comes from IKACh opening; reduced If flattens the slope.

· Treating cAMP as working only through protein kinase A. cAMP binds HCN channels directly and shifts their activation to less negative voltages.

Where it is taught: Section C02, paragraphs on rate and autonomic control, and the stop and predict box of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C02-RT01

Retrieval exercise: Rebuild the sympathetic acceleration sequence

A competent response contains

· Norepinephrine binds beta-1 receptors on nodal cells.

· Gs activates adenylyl cyclase and cAMP rises.

· cAMP binds HCN channels directly and If increases; the binding shifts HCN activation to less negative voltages, so the channels open earlier in repolarization.

· Protein kinase A phosphorylates L-type Ca2+ channels, increasing Ca2+ current and making the upstroke reach threshold more readily.

· Phase 4 steepens and threshold is reached sooner.

· Rate rises and AV conduction speeds, which shortens the PR interval.

· The redrawn slope is steeper and reaches the same threshold in less time, from the same maximum diastolic potential.

Distinctions that matter

· cAMP acts on HCN channels without protein kinase A; protein kinase A is the route to the L-type Ca2+ channel.

· Sympathetic drive changes the slope, not the threshold voltage itself.

Common misconceptions

· Drawing a more negative maximum diastolic potential with sympathetic drive. That is the vagal effect through IKACh.

Where it is taught: Section C02, flow diagram on sympathetic acceleration of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C03-BD1

Brain dump 1 · foundational retrieval: Blood supply of the conduction system

A competent response contains

· SA nodal artery: proximal right coronary artery in about 60% of hearts, circumflex in about 40%.

· AV nodal artery: from the artery crossing the crux near the origin of the posterior descending artery, so the right coronary artery in about 80% (right dominance).

· Bundle of His: dual supply from the AV nodal artery and the first septal perforator of the left anterior descending artery.

· Right bundle branch: septal perforators of the left anterior descending artery, which supply the anterior two thirds of the septum.

· Left anterior fascicle: septal perforators of the left anterior descending artery.

· Left posterior fascicle: left anterior descending septal branches and the posterior descending artery, and it is broad, which is why isolated posterior fascicular block is uncommon.

· The His bundle and the left posterior fascicle each have more than one source, so a single artery occlusion is less likely to silence them.

Distinctions that matter

· The SA nodal artery and the AV nodal artery have different percentages and different anatomical origins, so keep the two figures apart.

· Above the node the supply is nodal and usually right coronary; below the node it is septal and usually left anterior descending.

Common misconceptions

· Assuming the right coronary artery always feeds both nodes. It feeds the SA node in about 60% and the AV node in about 80%, and dominance decides the AV figure.

Where it is taught: Section C03, first paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C03-BD2

Brain dump 2 · integration and application: Why inferior and anterior infarcts block differently

A competent response contains

· Inferior infarct with ST elevation in II, III, and aVF: right coronary artery occlusion, SA and AV nodal tissue at risk, block at the AV node, junctional escape that is narrow at about 40 to 60 bpm, usually transient.

· Anterior infarct with ST elevation in V1 to V4: left anterior descending occlusion, loss of the right bundle and left anterior fascicle, block at the His bundle or bundle branches, ventricular escape that is wide at about 20 to 40 bpm, often persistent with a worse prognosis.

· Non-necrotic mechanisms in the inferior infarct: ischemia depresses the Ca2+ dependent nodal action potential, adenosine released from ischemic tissue acts on A1 receptors to open the same K+ channels as acetylcholine, and vagal afferents in the inferoposterior wall drive reflex bradycardia and hypotension.

· That reflex is the Bezold-Jarisch reflex.

· Nodal tissue usually survives and part of the cause is reflex and metabolic, which is why inferior block recovers.

· In a left dominant heart the circumflex supplies the AV nodal artery, so a circumflex occlusion can cause AV nodal block; because the circumflex feeds the SA node in about 40%, a lateral infarct can cause sinus node dysfunction.

Distinctions that matter

· Level of block decides escape rate and QRS width: nodal block leaves a pacemaker above the bundle branches, so the escape is narrow and faster.

· Transient against persistent follows mechanism: reflex and metabolic depression recovers, necrosis does not.

Common misconceptions

· Reading any complete heart block after infarction as the same problem. The artery sets the level, and the level sets the escape and the prognosis.

· Attributing inferior infarct bradycardia only to nodal ischemia and leaving out adenosine and the vagal reflex.

Where it is taught: Section C03, paragraphs on inferior and anterior infarction and the comparison table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw and label one normal beat

A competent response contains

· P wave is atrial depolarization, under 0.12 s and under 2.5 mm in lead II.

· PR interval runs from the onset of P to the onset of QRS and covers atrial, AV nodal, and His-Purkinje conduction, normally 0.12 to 0.20 s.

· The PR segment is flat because the AV node and His bundle are too small to register, so their delay shows only as elapsed time.

· QRS is ventricular depolarization, 0.06 to 0.10 s, with 0.12 s or more abnormal.

· ST segment corresponds to the plateau, phase 2, and is isoelectric because no voltage gradient exists while the whole ventricle sits at the plateau.

· T wave is repolarization, phase 3, in the same direction as the QRS and inverted in aVR.

· QT interval covers depolarization plus repolarization and approximates ventricular action potential duration; QTc is under about 0.45 s in men and 0.46 s in women.

· U wave is late repolarization, small normally and prominent in hypokalemia.

· Atrial repolarization is hidden within the QRS.

· The T wave is upright where the QRS is upright because epicardial action potentials are shorter, so repolarization runs epicardium to endocardium, the reverse of depolarization.

· At 25 mm/s a small box is 0.04 s and a large box is 0.20 s; 10 mm equals 1 mV.

Distinctions that matter

· PR interval includes the P wave; the PR segment is only the flat part after it.

· The ECG records electrical events, not contraction, and the nodes are inferred from timing rather than seen.

Common misconceptions

· Explaining the upright T wave by saying repolarization travels the same way as depolarization. It travels the opposite way, and the opposite sign of repolarization makes the deflection upright again.

· Calling an isoelectric ST segment a sign of electrical silence. The ventricle is depolarized throughout the plateau, with no gradient to record.

Where it is taught: Section C04, paragraphs on the waveform and the table of normal values of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C04-BD2

Brain dump 2 · integration and application: Work a strip from the numbers

A competent response contains

· Rate is 300 divided by 3 large boxes, which is 100 bpm.

· PR is 6 small boxes times 0.04 s, which is 0.24 s, above the normal 0.12 to 0.20 s, so this is first degree AV block.

· QRS is 3 small boxes times 0.04 s, which is 0.12 s, at the upper limit and abnormal, so ventricular activation is not purely through the His-Purkinje system.

· QT is 8 small boxes times 0.04 s, which is 0.32 s.

· RR is 3 large boxes times 0.20 s, which is 0.60 s; the square root of 0.60 is about 0.77, so QTc is about 0.32 divided by 0.77, or about 0.42 s, which is normal.

· Axis: lead I positive with aVF negative places the axis between 0 and -90 degrees, and a positive lead II keeps it between 0 and -30 degrees, so the axis is normal.

· Rhythm: an upright P in leads I and II before every QRS at a rate of 100 bpm is sinus rhythm, with first degree AV block and a prolonged QRS alongside it.

Distinctions that matter

· A positive lead II with I positive and aVF negative means a normal axis; a negative lead II in the same situation means left axis deviation.

· Rate normality and rhythm are separate judgments from interval normality: the rhythm here is sinus even though the PR and QRS are abnormal.

Common misconceptions

· Reading an abnormal PR as a rhythm diagnosis. Every P conducts here, so this is first degree block within sinus rhythm.

· Comparing the raw QT with the normal limit. QT must be corrected for rate before judging it.

Where it is taught: Section C04, paragraphs on rate, intervals, and axis, and the worked example of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the AV block table from memory

A competent response contains

· First degree: PR constant and over 0.20 s with every P conducted, site the AV node, QRS narrow, low risk of progression.

· Mobitz I: PR lengthens until a P fails to conduct and then resets, site the AV node, QRS narrow, low risk.

· Mobitz II: PR of conducted beats constant with sudden dropped P waves, site the His bundle or bundle branches, QRS often wide, high risk of progression to complete block.

· Third degree: PR varies because the atria and ventricles are dissociated, site either nodal or infranodal, QRS narrow if the escape is junctional and wide if it is ventricular, outlook depends on the escape.

· In Mobitz I the PR increment is largest in the beat after the pause, so RR intervals shorten before the dropped beat and the complexes form groups.

· In third degree block the P waves and QRS complexes march independently, which is AV dissociation.

· Causes: fibrosis, myocardial infarction, Lyme carditis from Borrelia burgdorferi, sarcoidosis, and maternal anti-Ro/SSA antibodies injuring the fetal AV node.

· First degree block also arises from athletic vagal tone, nodal-slowing drugs, and degeneration.

Distinctions that matter

· Gradual PR change locates the block in the AV node; all-or-none failure locates it in His-Purkinje tissue.

· Mobitz II and third degree are not distinguished by QRS width alone: in Mobitz II some P waves still conduct with a fixed PR, while in third degree none conduct.

Common misconceptions

· Expecting RR intervals to lengthen before the dropped beat in Mobitz I. They shorten, because each PR increment is smaller than the one before.

· Treating first degree block as a dropped beat. Every P conducts, just slowly.

Where it is taught: Section C05, paragraphs on the degrees of block and the AV block table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-BD2

Brain dump 2 · integration and application: Gradual failure against all-or-none failure

A competent response contains

· AV nodal cells depolarize through slow L-type Ca2+ channels whose recovery is time dependent; His-Purkinje cells use fast Na+ channels.

· Time-dependent recovery means the node conducts each closely spaced impulse more slowly, which is decremental conduction, so failure is gradual.

· His-Purkinje tissue conducts at nearly full speed or not at all, so infranodal block appears suddenly.

· Mobitz I: each successive impulse arrives at a less recovered node, so the PR lengthens until one impulse is blocked, after which the node recovers and the cycle restarts.

· Mobitz II: diseased His or bundle branch tissue fails without warning, so conducted beats keep a constant PR and P waves drop abruptly.

· Escape after nodal block is junctional, narrow, about 40 to 60 bpm, and uses the normal bundle branches; escape after infranodal block is a distal Purkinje or ventricular pacemaker, wide, about 20 to 40 bpm, spreading myocyte to myocyte, and liable to fail.

· In 2:1 block the PR cannot be compared from beat to beat because only every other P conducts, so no trend is visible; a wide QRS favors an infranodal site.

· When the last fascicle of a bifascicular block fails, the result is complete infranodal block with a wide escape at about 20 to 40 bpm, slow enough to cause syncope.

Distinctions that matter

· Decremental conduction is a normal property of the AV node that becomes visible as Wenckebach when the node is stressed or diseased; it is not a property of His-Purkinje tissue.

· QRS width reports where ventricular activation starts, which is why it helps locate a block the PR cannot.

Common misconceptions

· Assuming 2:1 block with a narrow QRS must be nodal. A narrow QRS makes a nodal site more likely but does not settle it from the surface tracing.

· Describing the junctional escape as unreliable. It is the faster and more dependable of the two because it still uses the bundle branches.

Where it is taught: Section C05, opening paragraph, block descriptions, and the final paragraph on fascicular block of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-CP01

Comparison table: Right against left bundle branch block

A competent response contains

· Septal depolarization: RBBB keeps the normal left to right direction because the left bundle is intact; LBBB reverses it to right to left.

· V1: RBBB shows rSR prime; LBBB shows a deep wide QS or rS.

· I and V6: RBBB shows a wide slurred S wave with the septal q wave kept; LBBB shows a broad notched R wave with the septal q lost.

· Septal q wave in I, aVL, V5, and V6: present in RBBB, absent in LBBB.

· Second heart sound: RBBB gives wide splitting; LBBB gives paradoxical splitting.

· QRS duration: 0.12 s or more in complete block of either type, and about 0.10 to 0.12 s with the same morphology in incomplete block.

· Causes: RBBB from right ventricular strain such as pulmonary embolism or atrial septal defect, and from septal infarction, and it can occur in a normal heart; LBBB almost always means structural disease such as ischemia, hypertension, aortic stenosis, or cardiomyopathy, and it can mask a new infarct.

· In either block the affected ventricle is activated late by myocyte to myocyte spread, the terminal QRS points toward that ventricle, and ST-T changes point away from it.

Distinctions that matter

· The septal q wave is the quickest discriminator: it survives in RBBB and disappears in LBBB, because LBBB reverses the direction of septal activation.

· RBBB can be a normal variant; LBBB implies structural heart disease.

Common misconceptions

· Reading ST-T changes in a bundle branch block as ischemia. They are secondary to the abnormal depolarization and point opposite the terminal QRS.

· Thinking incomplete block has a different pattern. The morphology is the same, only the QRS is 0.10 to 0.12 s.

Where it is taught: Section C05, bundle branch paragraph and the RBBB against LBBB table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-BD1

Brain dump 1 · foundational retrieval: Mechanisms of tachyarrhythmia, and the premature beats

A competent response contains

· Abnormal automaticity: a steep phase 4 in tissue outside the SA node, from catecholamines, stretch, ischemia, or electrolyte disturbance.

· Triggered activity: early afterdepolarizations during a prolonged plateau, or delayed afterdepolarizations in Ca2+ overloaded cells.

· Reentry: the commonest cause of sustained tachycardia, needing two pathways, unidirectional block in one of them, and conduction slow enough that the blocked limb recovers before the impulse returns.

· Wavelength is conduction velocity times refractory period, and the circuit must exceed the wavelength for reentry to persist.

· Slow conduction from ischemia or scar, and short refractoriness from ischemia or atrial remodeling, both shorten the wavelength and make reentry easier to sustain.

· PAC: early abnormal P wave, narrow QRS, and a pause less than compensatory because the premature impulse resets the SA node.

· PVC: early wide QRS with no preceding P wave and a T wave discordant with the QRS, with a fully compensatory pause because retrograde block in the AV node protects the sinus rhythm from being reset.

· A PVC falling on the peak of the T wave, the vulnerable period, can start ventricular fibrillation.

Distinctions that matter

· The pause distinguishes the two premature beats and both pauses follow from what reaches the SA node: a PAC resets it, a PVC does not.

· Early afterdepolarizations arise during repolarization from a long plateau; delayed afterdepolarizations arise after repolarization from Ca2+ overload.

Common misconceptions

· Calling slow conduction harmful only. Reentry needs it, which is why ischemic and scarred tissue is arrhythmogenic.

· Saying a PVC resets the sinus node. The retrograde impulse is blocked in the AV node, so the sinus rhythm continues on schedule.

Where it is taught: Section C06, opening paragraph, reentry cycle, and premature beat paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-BD2

Brain dump 2 · integration and application: Read four strips back to their mechanisms

A competent response contains

· Patient 1: atrial fibrillation from pulmonary vein triggers and a dilated fibrotic atrium sustaining wavelets at about 350 to 600 per minute; the node conducts only impulses that find it recovered, so the ventricular response is far slower than the atrial rate and irregularly irregular with a narrow QRS.

· Patient 1 loses atrial kick from ventricular filling, and stasis in the left atrial appendage promotes embolic stroke.

· Patient 2: atrial flutter, macroreentry around the tricuspid annulus through the cavotricuspid isthmus, with an atrial rate near 300 per minute and 2:1 nodal conduction giving a regular ventricular rate near 150 bpm.

· Patient 3: paroxysmal supraventricular tachycardia, usually AV nodal reentry through dual nodal pathways, regular at 150 to 250 bpm with a narrow QRS and the retrograde P wave buried in or just after the QRS, sometimes as a pseudo R prime in V1; it starts and stops abruptly and vagal slowing of the node can terminate it.

· Patient 4: ventricular tachycardia, three or more consecutive ventricular beats over 100 bpm, usually scar reentry, with a wide QRS and AV dissociation.

· The other two confirming findings in patient 4 are capture beats and fusion beats.

· Narrow QRS means the ventricles were activated through the His-Purkinje system; wide QRS means ventricular origin or aberrant conduction.

Distinctions that matter

· Flutter with 2:1 conduction is regular near 150 bpm, while fibrillation is irregularly irregular; the atrial rate and the filtering node explain both.

· AV nodal reentry and ventricular tachycardia are both regular, and QRS width plus AV dissociation separate them.

Common misconceptions

· Reading a ventricular rate of 130 in atrial fibrillation as the atrial rate. The atria are firing several hundred times per minute and the node blocks most impulses.

· Expecting a visible P wave in AV nodal reentry. The atria and ventricles activate almost together, so it is hidden in or just after the QRS.

Where it is taught: Section C06, atrial arrhythmia paragraph, worked example on nodal reentry, and the tachyarrhythmia table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-RT01

Retrieval exercise: Rebuild the reentrant circuit

A competent response contains

· Step 1: a premature impulse reaches a branch point.

· Step 2: one limb is still refractory, giving unidirectional block.

· Step 3: the impulse travels slowly down the other limb.

· Step 4: the blocked limb recovers during that slow passage.

· Step 5: the impulse returns up the recovered limb.

· Step 6: it re-enters the first limb and the loop repeats.

· Blocking one limb, or lengthening refractoriness, stops the circuit.

· In AV nodal reentry the fast pathway has a long refractory period and the slow pathway a short one. A premature atrial beat finds the fast pathway refractory, descends the slow pathway with a long PR, then travels retrograde up the recovered fast pathway while also descending the His bundle, and re-enters the slow pathway at 150 to 250 bpm.

· Because atria and ventricles activate almost together, the retrograde P wave sits inside or just after the narrow QRS, sometimes appearing as a pseudo R prime in V1.

Distinctions that matter

· Unidirectional block is required at the start, and slow conduction is required to keep the circuit alive; one without the other does not sustain reentry.

· The fast pathway conducts quickly but recovers slowly, which is exactly why a premature beat is blocked there first.

Common misconceptions

· Drawing both limbs as conducting normally at the outset. Without block in one limb the impulse meets itself and extinguishes.

· Expecting a long PR during the tachycardia. The long PR belongs to the initiating premature beat going down the slow pathway.

Where it is taught: Section C06, reentry cycle and the worked example on nodal reentry of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-PR01

Prediction exercise: Predict the effect of a low plasma potassium

A competent response contains

· IKr falls, because low plasma K+ reduces IKr and macrolides block IKr directly.

· Less repolarizing K+ current lengthens the ventricular action potential, so the QT interval lengthens.

· The U wave becomes prominent, which is characteristic of hypokalemia.

· During the prolonged plateau, L-type Ca2+ channels have time to recover and reopen.

· That inward Ca2+ current produces early afterdepolarizations, which fire premature beats.

· Uneven repolarization across the ventricle supports reentry, so the tracing shows a polymorphic QRS that twists about the baseline on a background of long QT, which is torsades de pointes, and it can degenerate into ventricular fibrillation.

Distinctions that matter

· Low K+ reduces IKr rather than increasing K+ efflux, which is why hypokalemia lengthens repolarization instead of shortening it.

· Early afterdepolarizations start the rhythm; dispersed repolarization and reentry keep it going.

Common misconceptions

· Expecting a low plasma K+ to make the cell fire less because K+ currents are repolarizing. Reduced IKr prolongs the plateau and invites Ca2+ channel reopening.

· Treating a prominent U wave as a normal variant in this setting. It is the expected finding in hypokalemia.

Where it is taught: Section C06, torsades paragraph and the long QT flow diagram, with the U wave in the C04 table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C07-BD1

Brain dump 1 · foundational retrieval: Preexcitation and the failing sinus node

A competent response contains

· The accessory pathway exists because the fibrous AV rings incompletely separate atrial from ventricular muscle during development, leaving a bundle of Kent made of fast Na+ dependent muscle that usually lacks decremental conduction.

· Sequence: the atria depolarize normally; the impulse crosses the pathway without nodal delay; part of the ventricle activates early so the PR is under 0.12 s; slow cell to cell spread forms the delta wave; the impulse arriving through the AV node activates the rest of the ventricle rapidly; the result is a wide fusion QRS with secondary ST-T changes.

· On the complex, the pathway produced the short PR and the delta wave, and normal His-Purkinje conduction produced the rest of the QRS.

· Preexcitation alone is the WPW pattern; with episodes of tachycardia it is WPW syndrome. Ebstein anomaly is an association.

· Vagal slowing of the AV node enlarges the delta wave and sympathetic drive can hide it, and a concealed pathway conducts only retrogradely, showing no delta wave while still supporting reentry.

· Sick sinus syndrome is intrinsic failure of the SA node, usually from degenerative fibrosis in older adults, and less often from SA nodal artery ischemia, amyloidosis, sarcoidosis, or inherited channel defects.

· Fewer pacemaker cells give inappropriate sinus bradycardia and chronotropic incompetence, a poor rate rise with exercise; sick sinus syndrome often coexists with AV conduction disease and atrial fibrillation.

Distinctions that matter

· The pathway bypasses the AV node but enters ventricular muscle outside the Purkinje network, which is why early activation is slow and inscribes a delta wave.

· The WPW pattern describes the resting tracing; WPW syndrome adds the arrhythmia.

Common misconceptions

· Saying the whole QRS is preexcited. It is a fusion complex, with the pathway and the AV node each activating part of the ventricle.

· Expecting every accessory pathway to show a delta wave. A concealed pathway conducts retrogradely only.

Where it is taught: Section C07, first paragraph, the WPW flow diagram, and the sick sinus paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C07-BD2

Brain dump 2 · integration and application: The dangerous pathway and the long pause

A competent response contains

· A refractory period near 0.25 s allows about 240 impulses per minute or more, since 60 divided by 0.25 is 240.

· The AV node would not pass that many because its refractory period is long and time dependent, lengthening further as impulses arrive faster.

· The tracing is very fast, irregular, and wide, with a QRS shape that varies beat to beat as the proportion of fusion between pathway and node changes.

· Short diastole and impulses landing in the vulnerable period allow degeneration to ventricular fibrillation.

· Slowing the AV node does not slow this ventricular rate and can shift more conduction onto the pathway.

· Orthodromic reentry descends the AV node and returns up the pathway, so the QRS is narrow and the delta wave disappears; antidromic reentry descends the pathway and gives a wide, fully preexcited QRS.

· A pause that is an exact multiple of the P to P interval means SA exit block: the impulse forms but cannot leave the node.

· A pause that is not a multiple means sinus arrest: the node fails to fire at all.

· The pause follows the tachycardia because the rapid atrial rhythm overdrive suppresses the node; a healthy node recovers within about 1.5 s, while a fibrotic node with few pacemaker cells recovers slowly, so nothing fires until it or an escape pacemaker reaches threshold.

Distinctions that matter

· Orthodromic tachycardia is narrow and antidromic is wide, and the direction of travel through the pathway is what decides this.

· Sinus arrest against SA exit block is settled by arithmetic on the pause, not by its length.

Common misconceptions

· Assuming nodal slowing controls the ventricular rate in every atrial fibrillation. With an accessory pathway the node is not the limiting route.

· Reading a post-tachycardia pause as a separate new problem. It is overdrive suppression unmasking a diseased node, the tachy-brady form of sick sinus syndrome.

Where it is taught: Section C07, accessory pathway paragraph, sick sinus paragraph, tachy-brady paragraph, and the stop and predict box of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C07-CA01

Clinical application: Palpitations in a young man with a wide, irregular tachycardia

A competent response contains

· Recognize: atrial fibrillation conducted to the ventricles through an accessory pathway, identified on the earlier tracing by a PR of 0.10 s with a delta wave, the slurred upstroke, and a QRS of 0.13 s with discordant ST-T changes.

· Explain: the pathway is fast Na+ dependent muscle without decremental conduction and with a short refractory period, so it passes atrial impulses at a rate the AV node would block; QRS width and shape vary because the degree of fusion between pathway conduction and nodal conduction changes from beat to beat.

· Differentiate: atrial fibrillation conducted through the node is irregularly irregular but narrow and slower; ventricular tachycardia is regular and uniform in QRS shape with AV dissociation, capture beats, and fusion beats, while this rhythm is irregular with QRS complexes that change shape.

· Investigate: on the sinus rhythm tracing, measure the PR interval, which is under 0.12 s, look for a delta wave at the QRS onset, measure the QRS, which is 0.12 s or more, and note ST-T changes directed opposite the QRS; the earlier episodes of abrupt regular palpitations fit orthodromic reentry through the same pathway.

· Decide: the dangerous property is the pathway's short, non-decremental refractory period, which permits a very fast ventricular response, and this rhythm can degenerate into ventricular fibrillation.

· Justify: with a pathway refractory period near 0.25 s about 240 impulses per minute or more reach the ventricle, while the AV node's long time-dependent refractoriness would block most of them; slowing the node therefore does not reduce the rate and can shift still more conduction onto the pathway, and short diastole with impulses landing in the vulnerable period is what allows fibrillation.

Distinctions that matter

· Varying QRS shape points to variable fusion through two routes, which separates this from the uniform wide complexes of scar reentry.

· The hemodynamic picture comes from rate and lost filling time, not from loss of the conduction system itself.

Common misconceptions

· Classifying every irregular wide-complex tachycardia as ventricular tachycardia. Preexcited atrial fibrillation is irregular with changing QRS morphology.

· Assuming the AV node still governs the ventricular rate once a pathway is conducting.

Where it is taught: Section C07, accessory pathway paragraphs, with reentry and fibrillation mechanisms in C06 of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C03-Q01

Board-style question: Complete block after an inferior infarction

Correct answer: B. Ischemia of the atrioventricular node with adenosine release and reflex vagal discharge

Why it is right: Right coronary occlusion starves the AV node, and ischemia of its Ca2+ dependent conduction, adenosine acting on A1 receptors to open the same K+ channels as acetylcholine, and the vagal reflex from the inferoposterior wall together produce nodal block with a narrow junctional escape at 40 to 60 bpm.

A is wrong: Necrosis of the His bundle and proximal bundle branches follows anterior infarction from the left anterior descending artery and gives a wide escape at about 20 to 40 bpm, not a 0.08 s QRS at 44/min.

C is wrong: An accessory pathway bypasses the AV node and speeds ventricular activation, producing a short PR with a delta wave rather than bradycardia with dissociated P waves.

D is wrong: Sinoatrial node fibrosis would slow or stop the P waves, but the P waves here are 84/min and normal, and the problem is that they are not conducted.

A competent response contains

· ST elevation in II, III, and aVF marks an inferior infarct, usually from proximal right coronary artery occlusion, which supplies the AV nodal artery in about 80% of hearts.

· Independent P waves at 84/min with QRS complexes at 44/min are complete block with AV dissociation, and the 0.08 s QRS shows the escape pacemaker is junctional, so the block sits in the AV node.

Distinctions that matter

· Escape QRS width locates the block: narrow means the pacemaker is above the bundle branches.

Common misconceptions

· Reading any complete block after infarction as infranodal. The lead group and the narrow escape point to the node.

Where it is taught: Section C03 of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-Q02

Board-style question: Locating the level of block

Correct answer: D. His bundle and bundle branches

Why it is right: His-Purkinje cells depolarize through fast Na+ channels and conduct at nearly full speed or not at all, so a diseased bundle gives a constant PR with P waves that drop without warning, and the wide rSR prime QRS shows existing bundle branch disease.

A is wrong: Sinoatrial node failure would produce missing P waves and pauses, whereas the P waves here arrive on time and some are simply not conducted.

B is wrong: Atrial muscle disease would alter P wave shape and duration and does not interrupt conduction to the ventricles in this pattern.

C is wrong: Atrioventricular nodal block conducts decrementally, so the PR would lengthen progressively before a beat dropped and the conducted QRS would usually be narrow.

A competent response contains

· A constant PR in every conducted beat with sudden non-conducted P waves is Mobitz II, which is infranodal.

· A QRS of 0.14 s with rSR prime in V1 is right bundle branch block, confirming that the disease lies in the bundle branches.

Distinctions that matter

· Constant PR with abrupt dropped beats is all-or-none His-Purkinje failure, while a lengthening PR before the drop is nodal.

Common misconceptions

· Taking a normal PR interval as evidence that conduction below the node is intact. His-Purkinje tissue conducts at nearly full speed until it fails outright.

Where it is taught: Section C05 of the CV-7 Conduction System and the Electrocardiogram notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-08

CV-8 Coronary Circulation

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-08-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the coronary arteries and their territories

A competent response contains

· Both coronary arteries are the first branches of the aorta and arise from the aortic sinuses: the right coronary artery from the right sinus, the left main from the left sinus.

· Each artery runs in a surface groove and supplies the muscle on either side of that groove, so the grooves predict the territories.

· The coronary sulcus circles the heart between atria and ventricles, the anterior and posterior interventricular grooves mark the septum, and the crux is where the coronary sulcus meets the posterior interventricular groove.

· Right coronary artery branches in order: conus branch, sinoatrial nodal artery in about 60% of hearts, right marginal artery at the acute margin, then at the crux the atrioventricular nodal artery in about 80% and the posterior descending artery in right dominant hearts.

· Right coronary artery territory: right atrium, most of the right ventricle, and both nodes in most hearts.

· Left main divides into the left anterior descending and circumflex arteries, and sometimes a ramus intermedius, so its occlusion threatens most of the left ventricle.

· Left anterior descending artery runs in the anterior interventricular groove to the apex, with septal perforators to the anterior two-thirds of the septum carrying the His bundle and both bundle branches, and diagonals to the anterolateral wall; leads V1 to V4.

· Circumflex artery runs in the left coronary sulcus with obtuse marginal branches to the lateral and posterolateral wall and branches to the left atrium, and gives the sinoatrial nodal artery in about 40%; leads I, aVL, V5, V6.

· Posterior descending artery runs in the posterior interventricular groove and supplies the inferior wall, the posterior one-third of the septum, and the posteromedial papillary muscle; leads II, III, aVF.

· Surface relationships: the anterior surface is mostly right ventricle, the inferior surface is mostly left ventricle, and the left surface is the lateral wall.

· The left anterior descending artery is occluded most often in infarction, followed by the right coronary artery and then the circumflex, because it feeds the largest share of left ventricular muscle.

Distinctions that matter

· The sinoatrial nodal percentage (about 60% right coronary) and the atrioventricular nodal percentage (about 80% right coronary) are different figures from different origins, so keep them apart.

· Territory is set by the groove the artery occupies, while dominance is a separate question about which artery reaches the crux and gives the posterior descending artery.

Common misconceptions

· Writing that the left coronary artery supplies the whole left ventricle. The inferior wall and the posterior third of the septum belong to the posterior descending artery, which comes from the right coronary artery in most hearts.

· Assuming the artery most often occluded is the one supplying the most important structures. It is the one supplying the largest mass of left ventricular muscle.

Where it is taught: Section C01, paragraphs on each artery and the table of courses and territories of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C01-BD2

Brain dump 2 · integration and application: Work backward from the tracing to the artery

A competent response contains

· Patient 1, V1 to V4: left anterior descending artery, anterior left ventricle and apex with the anterior two-thirds of the septum, bundle branches and His bundle at risk, so block is infranodal and the escape is wide and slow.

· Patient 2, II, III, aVF: the artery giving the posterior descending artery, usually the right coronary artery, losing the inferior wall and the posterior one-third of the septum, with the atrioventricular node at risk, so block is at the node with a narrow junctional escape; sinus bradycardia is also common because the right coronary artery feeds the sinoatrial node in about 60%.

· Patient 3, I, aVL, V5, V6: circumflex artery, lateral and posterolateral wall and left atrium; conduction is usually spared unless the heart is left dominant, when the circumflex supplies the atrioventricular nodal artery.

· Coronary arteries are functional rather than absolute end arteries, so a stenosis developing over months allows small anastomoses to enlarge into useful collaterals and the muscle beyond can stay alive.

· A sudden thrombus gives no time for collateral growth, so the territory of the blocked artery infarcts.

· The anastomoses are best developed near the apex and within the septum.

· The right ventricle is supplied by the right coronary artery through the conus and right marginal branches, so right ventricular infarction follows a proximal right coronary occlusion and is read in right-sided leads; the left anterior descending artery supplies almost no free right ventricular wall.

Distinctions that matter

· Lead group names the wall, and the wall names the artery, but the conduction consequence depends on which structures sit inside that territory.

· Collaterals protect against a slowly developing stenosis and not against an acute occlusion, which is why the time course matters more than the final degree of narrowing.

Common misconceptions

· Calling the coronary arteries true end arteries. They are functional end arteries, with anastomoses that can enlarge over months.

· Expecting a circumflex occlusion to cause atrioventricular nodal block routinely. It does so only in a left dominant heart.

Where it is taught: Section C01, why the anatomy matters and the final paragraph on collaterals of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-BD1

Brain dump 1 · foundational retrieval: Dominance and the supply of the papillary muscles

A competent response contains

· Coronary dominance describes which artery gives rise to the posterior descending artery, not which artery is larger.

· Right dominant: posterior descending artery from the right coronary artery, in most hearts, about 70% to 85%.

· Left dominant: posterior descending artery from the circumflex, about 5% to 15%.

· Codominant: both the right coronary artery and the circumflex reach the posterior interventricular groove, about 5% to 20%.

· The posterior descending artery supplies the inferior wall of the left ventricle, the posterior one-third of the septum, and the posteromedial papillary muscle.

· The artery that gives the posterior descending artery usually also gives the atrioventricular nodal artery, which is why dominance decides whether an inferior infarct brings nodal block.

· In systole the papillary muscles pull on the chordae tendineae and keep the mitral leaflets from prolapsing into the left atrium.

· The anterolateral papillary muscle lies where left anterior descending diagonal and circumflex obtuse marginal territories overlap, so its supply is dual and it rarely ruptures.

· The posteromedial papillary muscle usually depends on the posterior descending artery alone, so its supply is single and it shares the fate of whichever artery gives the posterior descending artery.

· Papillary tips lie deep in the cavity, at the end of the coronary branches and within the subendocardium, which makes them highly sensitive to ischemia.

Distinctions that matter

· Dominance is about the source of one artery; it is not a statement about how much muscle each coronary artery supplies, since the left coronary artery supplies most of the left ventricle in nearly every heart.

· Single against dual supply, not size or position alone, is what makes one papillary muscle vulnerable and the other safe.

Common misconceptions

· Saying most hearts are left dominant because the left coronary artery supplies more muscle. Most hearts are right dominant because the right coronary artery gives the posterior descending artery.

· Treating both papillary muscles as equally at risk. The posteromedial muscle has one source and the anterolateral has two.

Where it is taught: Section C02, opening paragraph, the dominance table, and blood supply of the papillary muscles of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-BD2

Brain dump 2 · integration and application: Sudden decompensation four days after an inferior infarct

A competent response contains

· The posteromedial papillary muscle has ruptured, supplied by the posterior descending artery, which came from the right coronary artery in this right dominant inferior infarct.

· Dead muscle keeps its tensile strength for about the first day; neutrophils infiltrate and from about day 3 macrophages remove necrotic myocytes before granulation tissue and collagen replace them, so systolic traction tears softened tissue, most often about 2 to 7 days after infarction.

· Sudden mitral incompetence ejects much of each stroke volume into a normal-sized, relatively stiff left atrium, so left atrial pressure rises steeply and pulmonary edema is abrupt.

· Forward stroke volume falls at the same time, which causes hypotension and can progress to cardiogenic shock.

· The murmur can be short and soft because the gradient across the valve collapses before systole ends, so loudness does not track severity here.

· Post-infarct ventricular septal rupture also causes sudden decompensation with a new systolic murmur, and the measurement that separates the two is an oxygen saturation step-up from right atrium to right ventricle, which occurs in septal rupture and not in papillary rupture.

· Ischemic papillary muscle dysfunction without rupture is more common and causes milder regurgitation.

· In a left dominant heart the circumflex gives the posterior descending artery, so the same event follows a circumflex occlusion and the tracing shows an inferolateral infarct, with the atrioventricular node also at risk.

Distinctions that matter

· A chronic stiff left atrium accommodates regurgitation at lower pressure, while an acute lesion in a normal-sized atrium sends pressure straight to the lungs.

· Murmur intensity reflects the pressure gradient and its duration, not the volume of regurgitation.

Common misconceptions

· Expecting rupture during the first hours. Strength is preserved early and falls once enzymatic removal of dead myocytes is under way.

· Using a loud murmur as the sign of severe acute mitral regurgitation. The murmur here is often short and soft.

Where it is taught: Section C02, paragraphs on papillary supply and timing of rupture, the flow diagram, and manifestations explained by mechanism of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-RT01

Retrieval exercise: Rebuild the chain from inferior infarct to mitral regurgitation

A competent response contains

· Step 1: a thrombus occludes the artery giving the posterior descending artery.

· Step 2: the posteromedial papillary muscle loses its only supply, and it is the one at risk because the anterolateral muscle keeps a dual supply from diagonal and obtuse marginal branches.

· Step 3: coagulative necrosis of the papillary muscle.

· Step 4: neutrophils and then macrophages soften it over days 2 to 7, removing necrotic myocytes before collagen can replace them.

· Step 5: systolic traction tears the muscle or one of its heads.

· Step 6: the chordae lose their anchor and the mitral leaflets flail.

· Step 7: acute severe mitral regurgitation, with left atrial pressure rising steeply and forward stroke volume falling.

· In a left dominant heart only the occluded artery changes, from the right coronary artery to the circumflex; the rest of the chain is identical.

Distinctions that matter

· The chain depends on single supply, not on infarct size, which is why the largest infarcts are not the ones that rupture papillary muscle.

· Necrosis and softening are separate stages, and the delay before rupture belongs to the softening stage.

Common misconceptions

· Putting rupture immediately after necrosis. The enzymatic softening step is what sets the day 2 to 7 window.

· Naming the right coronary artery as the cause in every case. The artery that gives the posterior descending artery is the one that matters.

Where it is taught: Section C02, flow diagram from inferior infarct to papillary muscle rupture of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-CP01

Comparison table: Anterolateral against posteromedial papillary muscle

A competent response contains

· Wall: the anterolateral muscle sits on the anterolateral wall, the posteromedial muscle on the inferior wall.

· Supply: the anterolateral muscle is dual, from left anterior descending diagonal and circumflex obtuse marginal branches; the posteromedial muscle is single, usually from the posterior descending artery alone.

· Infarct at risk: an anterolateral infarct threatens the first muscle only if more than one artery is involved, while an inferior infarct threatens the posteromedial muscle directly.

· Rupture frequency: in commonly quoted series the posteromedial muscle ruptures about 6 to 12 times as often as the anterolateral muscle.

· A left anterior descending occlusion leaves the anterolateral muscle perfused by circumflex branches, and the posteromedial muscle lies outside the left anterior descending territory in most hearts, which is why the largest infarcts rarely rupture a papillary muscle.

· In a left dominant heart the posteromedial muscle depends on the circumflex, because the circumflex then gives the posterior descending artery; the anterolateral supply is unchanged.

· Rupture of either muscle removes the anchor for the chordae, the leaflets flail, and acute severe mitral regurgitation follows with abrupt pulmonary edema and falling forward stroke volume.

· Both muscles have tips lying deep in the cavity within the subendocardium, at the end of the coronary branches, so both are sensitive to ischemia even though only one usually ruptures.

Distinctions that matter

· Dual against single supply is the whole explanation for the difference in rupture risk.

· Dominance changes which artery feeds the posteromedial muscle but never makes its supply dual.

Common misconceptions

· Assuming the larger infarct carries the larger rupture risk. Risk follows loss of the only supply.

· Thinking a left dominant heart protects the posteromedial muscle. It simply moves the responsibility to the circumflex.

Where it is taught: Section C02, blood supply of the papillary muscles and the stop and predict box of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-CA01

Clinical application: Breathlessness and a new murmur five days after chest pain

A competent response contains

· Recognize: rupture of the posteromedial papillary muscle, supplied by the posterior descending artery from the right coronary artery, causing acute severe mitral regurgitation.

· Explain: that muscle depends on the posterior descending artery alone, so an inferior infarct removes its only supply, and rupture waits until neutrophils and then macrophages have softened the dead muscle from about day 3, giving the usual 2 to 7 day window rather than day one, when tensile strength is still preserved.

· Differentiate: ventricular septal rupture gives the same sudden decompensation with a new systolic murmur but produces a saturation step-up from right atrium to right ventricle, and ischemic papillary dysfunction without rupture is more common and causes milder regurgitation without this degree of pulmonary edema and hypotension.

· Investigate: right atrial 62% and right ventricular 63% show no step-up, which rules out septal rupture; a septal rupture would have shown a clear rise in saturation from right atrium to right ventricle because oxygenated left ventricular blood enters the right ventricle.

· Decide: the subendocardium of the inferior wall became ischemic first, and the ventricular rate of 44/min with dissociated P waves came from block at the atrioventricular node, supplied by the atrioventricular nodal artery from the right coronary artery in about 80% of hearts.

· Justify: the regurgitant volume enters a normal-sized, relatively stiff left atrium, so left atrial pressure rises steeply and pulmonary edema is abrupt, while forward stroke volume falls and causes hypotension; the murmur is short and soft because the gradient across the valve collapses before systole ends.

Distinctions that matter

· The saturation step-up is the single measurement that separates the two post-infarct ruptures.

· Acute mitral regurgitation into a compliant chronically dilated atrium behaves quite differently from the same lesion in a normal atrium.

Common misconceptions

· Grading severity by the loudness of the murmur in acute mitral regurgitation.

· Reading the narrow QRS escape at 44/min as infranodal disease. A narrow escape places the block at the node, which fits the inferior territory.

Where it is taught: Section C02, papillary supply, rupture timing, and manifestations; nodal supply in Section C01 of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C03-BD1

Brain dump 1 · foundational retrieval: Build the cardiac venous drainage table from memory

A competent response contains

· Great cardiac vein: anterior interventricular groove with the left anterior descending artery, then the left coronary sulcus with the circumflex; drains the anterior and lateral left ventricle, anterior septum, and left atrium; becomes the coronary sinus.

· Middle cardiac vein: posterior interventricular groove with the posterior descending artery; drains the inferior wall and posterior septum; enters the right end of the coronary sinus.

· Small cardiac vein: inferior margin with the right marginal artery, then the right coronary sulcus; drains the inferior right ventricle and right atrium; enters the right end of the coronary sinus.

· Anterior cardiac veins: cross the right coronary sulcus from the anterior right ventricle; drain the anterior right ventricle; open directly into the right atrium, bypassing the sinus.

· Smallest cardiac veins (venae cordis minimae): lie within the myocardium; drain the inner myocardium near each chamber; open directly into the chambers, mostly the atria, bypassing the sinus.

· The coronary sinus is a wide, thin-walled channel in the posterior coronary sulcus between the left atrium and left ventricle.

· It opens into the right atrium between the opening of the inferior vena cava and the tricuspid orifice, partly guarded by the valve of the coronary sinus.

· Its ostium forms one border of the triangle that locates the atrioventricular node.

· About 75% of total coronary flow returns through the sinus, mainly from the left ventricle, while most right ventricular venous blood returns through the anterior cardiac veins.

· Each named vein shadows an artery, which is why the venous map can be rebuilt from the arterial map.

Distinctions that matter

· Three veins empty into the coronary sinus and two bypass it, so sinus flow is not the whole of coronary venous return.

· The coronary sinus drains mainly the left ventricle; the right ventricle sends most of its venous blood straight into the right atrium.

Common misconceptions

· Assuming all cardiac venous blood passes through the coronary sinus. About a quarter of it does not.

· Placing the coronary sinus opening at the tricuspid orifice itself. It opens between that orifice and the inferior vena cava.

Where it is taught: Section C03, paragraphs on the coronary sinus and tributaries, and the table of cardiac veins of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C03-BD2

Brain dump 2 · integration and application: Read the venous side as a measurement

A competent response contains

· Coronary sinus blood is about 25% to 30% saturated because the myocardium extracts about 70% of delivered oxygen at rest, far more than the whole body average, so it is among the most deoxygenated blood in the body.

· Arterial blood carries about 20 mL O2/dL and coronary sinus blood about 5 to 6 mL O2/dL.

· A coronary sinus sample reflects left ventricular oxygen extraction and lactate release, because most of what it carries comes from the left ventricle.

· A dilated coronary sinus is explained by a persistent left superior vena cava, which drains into it.

· The sinus begins where the great cardiac vein meets the oblique vein of the left atrium, a remnant of the embryonic left superior vena cava, and the sinus itself develops from the left horn of the sinus venosus.

· The finding is usually harmless because the blood still reaches the right atrium.

· The normal physiological shunt of about 2% of cardiac output comes from smallest cardiac veins opening into the left heart and from bronchial venous blood draining into the pulmonary veins.

· That shunted blood never passes alveoli, so it enters the systemic circulation still deoxygenated and lowers arterial PO2 slightly below alveolar PO2, which is one reason the alveolar to arterial PO2 difference is not zero in a healthy person.

Distinctions that matter

· The low coronary sinus saturation is a normal consequence of high extraction, not a sign of disease.

· A persistent left superior vena cava dilates the sinus without creating a shunt, while the smallest cardiac veins draining into the left heart do create a small true shunt.

Common misconceptions

· Reading a dilated coronary sinus as obstruction to venous drainage. The usual cause is extra inflow from a persistent left superior vena cava.

· Expecting arterial PO2 to equal alveolar PO2 in a healthy person. A small physiological shunt always keeps a gradient.

Where it is taught: Section C03, the coronary sinus paragraph and physiological consequences of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw coronary flow across the cardiac cycle

A competent response contains

· Left coronary flow falls sharply, and can briefly reverse, as systole begins, because intramyocardial pressure in the inner wall rises to about cavity pressure and squeezes the vessels shut.

· Flow surges in early diastole when that compression is released, and about 70% to 80% of left ventricular flow occurs in diastole.

· Resting coronary flow is about 225 mL/min, about 4% to 5% of cardiac output, for an organ that is about 0.5% of body weight.

· Left ventricular coronary perfusion pressure is about aortic diastolic pressure minus left ventricular end-diastolic pressure, because diastolic cavity pressure is the back pressure on the inner vessels.

· With 120/80 mm Hg and an end-diastolic pressure of 8 mm Hg, perfusion pressure is about 80 minus 8, or about 72 mm Hg.

· Hypotension and aortic regurgitation lower the first term; heart failure and a stiff hypertrophied ventricle raise the second.

· The right ventricle is perfused throughout the cycle because its systolic pressure is only about 25 mm Hg, well below aortic pressure; pulmonary hypertension that raises right ventricular pressure toward systemic levels removes that advantage.

· Arterial blood carries about 20 mL O2/dL and coronary sinus blood about 5 to 6 mL O2/dL, so resting extraction is about 70% to 75% for the heart against about 25% for the whole body.

· Four myocyte features forcing high extraction: continuous contraction, almost all ATP from oxidative phosphorylation with mitochondria filling about one-third of cell volume, small creatine phosphate stores with little anaerobic capacity, and about one capillary per myocyte with cytoplasmic myoglobin easing diffusion.

Distinctions that matter

· Aortic systolic pressure drives flow to most organs, while left ventricular coronary flow is driven by aortic diastolic pressure against end-diastolic cavity pressure.

· High extraction is a feature of the resting heart, not a sign of stress, and it is what removes extraction as a reserve.

Common misconceptions

· Drawing peak coronary flow at peak aortic pressure. Peak left coronary flow is in early diastole, when compression is released.

· Using mean arterial pressure in the perfusion pressure estimate. The diastolic value is the relevant one for the left ventricle.

Where it is taught: Section C04, diastolic perfusion of the left ventricle and high oxygen extraction of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-BD2

Brain dump 2 · integration and application: Four patients with ischemia and normal arteries

A competent response contains

· Wall stress is proportional to cavity pressure times radius divided by wall thickness, by the law of Laplace.

· Aortic stenosis: systolic cavity pressure, muscle mass, and end-diastolic pressure all rise, so demand rises and perfusion pressure falls; the failing term is supply against a raised demand.

· Aortic regurgitation at 150/40 mm Hg with an end-diastolic pressure of 25 mm Hg: perfusion pressure is about 40 minus 25, or about 15 mm Hg, against a normal of about 70 mm Hg, and the dilated ventricle has a large radius and high wall stress.

· Heart rate 160/min: diastole shortens far more than systole, so diastolic perfusion time per minute falls while each extra beat raises demand.

· Hemoglobin 6 g/dL: arterial oxygen content falls, and because extraction is already near its limit the fall cuts delivery directly, so flow is the only compensation available.

· In all four patients the subendocardium becomes ischemic first.

· Four reasons the subendocardium loses out: it lies at the end of the penetrating branches that enter from the epicardium, it receives almost no systolic flow, it faces left ventricular end-diastolic pressure in diastole, and it carries the highest wall stress and oxygen demand with the least remaining vasodilator reserve because lower resting tone already spends part of it.

· Hypertrophy lowers wall stress by increasing wall thickness, but capillaries lag behind muscle growth and the stiff wall raises end-diastolic pressure, so the inner wall remains vulnerable.

· Exertion in these patients produces ST depression, because ischemia is confined to the inner layer rather than being transmural.

Distinctions that matter

· Ischemia can arise with normal coronary arteries whenever perfusion pressure, perfusion time, or oxygen content falls, or demand rises.

· ST depression points to inner wall ischemia while ST elevation points to transmural injury, so the pattern follows the layer involved.

Common misconceptions

· Assuming chest pain on exertion always means a coronary stenosis. Valve disease, tachycardia, hypertrophy, and anemia all produce it.

· Treating hypertrophy as protective because it reduces wall stress. Capillary density and end-diastolic pressure move the wrong way.

Where it is taught: Section C04, subendocardial vulnerability, the Laplace paragraph, and the paragraph on ischemia with normal arteries of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-RT01

Retrieval exercise: Rebuild the tachycardia sequence with the arithmetic

A competent response contains

· Step 1: heart rate rises, from exercise, fever, anemia, or arrhythmia.

· Step 2: diastole shortens far more than systole.

· Step 3: diastolic perfusion time per minute falls.

· Step 4: each extra beat raises oxygen demand, because every contraction costs oxygen and more contractions occur each minute.

· Step 5: the subendocardium exhausts its vasodilator reserve.

· Step 6: inner wall ischemia appears as angina with ST depression.

· At 70/min the cycle lasts about 0.86 s, systole about 0.30 s and diastole about 0.56 s, giving about 39 s of diastole per minute.

· At 150/min the cycle lasts 0.40 s, systole shortens only to about 0.22 s and diastole is about 0.18 s, giving about 27 s of diastole per minute, a fall of about 30%.

· Tachycardia lowers supply and raises demand at the same time, which is why it is poorly tolerated once a stenosis limits the rise in flow.

Distinctions that matter

· Systole shortens only slightly as rate rises, so almost the whole reduction in cycle length comes out of diastole, which is the perfusion window for the left ventricle.

· Perfusion time per minute, not the length of a single diastole, is the quantity that matters for total flow.

Common misconceptions

· Assuming systole and diastole shorten proportionally. Diastole takes almost all the loss.

· Thinking a faster rate helps coronary flow by increasing the number of diastolic intervals. Total diastolic seconds per minute fall.

Where it is taught: Section C04, flow diagram on tachycardia and the worked example on diastolic time per minute of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-PR01

Prediction exercise: Predict the effect of a fast atrial rhythm in aortic stenosis

A competent response contains

· Diastole in each cycle shortens, and total diastolic seconds per minute fall, because systole shortens only slightly as rate rises.

· Left ventricular end-diastolic pressure rises, because a stiff hypertrophied ventricle filling in less time needs a higher pressure for the same volume.

· Coronary perfusion pressure falls, since it is about aortic diastolic pressure minus end-diastolic pressure and the second term has risen.

· Oxygen demand rises, because wall stress is proportional to cavity pressure times radius divided by wall thickness and aortic stenosis keeps systolic cavity pressure high, while the extra beats each cost oxygen.

· Subendocardial vasodilator reserve falls toward zero, because lower resting tone in that layer has already spent part of it.

· The subendocardium becomes ischemic first and the electrocardiogram shows ST depression.

Distinctions that matter

· Rate affects supply and demand in the same direction of harm, which is why a fast rhythm is dangerous in aortic stenosis.

· Hypertrophy reduces wall stress but raises end-diastolic pressure, so it does not protect the inner wall.

Common misconceptions

· Expecting metabolic vasodilation to make up the difference. Reserve in that layer is partly spent at rest and runs out first.

· Looking for ST elevation. Inner wall ischemia without transmural injury gives ST depression.

Where it is taught: Section C04, subendocardial vulnerability, the tachycardia flow diagram, and the paragraph on aortic valve disease of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C05-BD1

Brain dump 1 · foundational retrieval: Rebuild metabolic control of coronary flow

A competent response contains

· By the Fick principle, myocardial oxygen consumption equals coronary flow times the arteriovenous oxygen content difference.

· Skeletal muscle can widen that difference in exercise, but the myocardium already extracts about 70% to 75% at rest and can reach only about 80% to 90%, so flow is the only term left to change.

· Adenosine pathway: oxygen demand exceeds supply; ATP breakdown outpaces regeneration and AMP accumulates; 5 prime nucleotidase converts AMP to adenosine; adenosine binds arteriolar A2A receptors coupled to Gs; cAMP rises and activates protein kinase A; K+ channels open, including ATP-sensitive channels; hyperpolarization closes L-type Ca2+ channels, cytosolic Ca2+ falls, myosin light chain kinase activity drops, the arteriole relaxes, resistance falls, and flow rises until supply matches demand.

· Adenosine is cleared within seconds by cellular uptake and adenosine deaminase, which makes this a fast negative feedback loop.

· Other mediators acting in parallel: low PO2 and ATP opening ATP-sensitive K+ channels directly, CO2, H+, lactate, K+ released by repeated depolarization, and endothelial nitric oxide released by shear stress.

· Blocking adenosine receptors blunts but does not abolish the rise in flow, which is the evidence that other mediators contribute.

· Sympathetic alpha-1 constriction is mild, while beta-1 stimulation raises cardiac work and metabolite release, so sympathetic activation raises coronary flow overall.

· Autoregulation holds flow nearly constant over a perfusion pressure range of about 60 to 140 mm Hg.

· Coronary flow reserve is maximal flow during full vasodilation divided by resting flow, normally about 3 to 5.

Distinctions that matter

· Local metabolic control overrides neural control in this bed, which is why beta-1 effects on work raise flow despite alpha-1 constriction.

· Adenosine is the clearest single mediator but not the only one, so receptor blockade blunts rather than abolishes the response.

Common misconceptions

· Expecting the heart to meet higher demand by extracting more oxygen. Extraction is nearly maximal at rest.

· Treating sympathetic activation as a coronary vasoconstrictor overall. The rise in work and metabolite release dominates the mild alpha-1 effect.

Where it is taught: Section C05, opening paragraphs, the adenosine flow diagram, and the paragraph on parallel mediators of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C05-BD2

Brain dump 2 · integration and application: Work the numbers on a stenosis, then explain steal

A competent response contains

· Arteriovenous difference is 20 minus 6, which is 14 mL O2/dL.

· Resting consumption is 2.25 dL/min times 14 mL/dL, about 32 mL O2/min, and extraction is 14 divided by 20, which is 70%.

· Doubling demand gives about 63 mL O2/min; raising extraction to about 85%, or 17 mL/dL, at unchanged flow yields only about 38 mL/min.

· Meeting the new demand at the resting extraction needs 63 divided by 14, about 4.5 dL/min, which is double the resting flow, so any lesion that caps flow causes ischemia under stress.

· With a 75% diameter stenosis, resting flow in that territory is normal because the downstream arterioles dilate to offset the stenosis, so arteriolar tone is already low and part of the flow reserve is spent at rest, while maximal flow is reduced.

· Maximal flow starts to fall at about 50% diameter narrowing, and resting flow is preserved until about 80% or more.

· The patient is comfortable sitting because resting flow is adequate, and has pain climbing stairs because the arterioles are already dilated, maximal flow falls short of demand, and the subendocardium becomes ischemic first, which is the mechanism of stable angina.

· Given a vasodilator, normal beds dilate while the bed beyond the stenosis is already maximally dilated, so flow shifts toward normal myocardium and can fall in the ischemic bed, which is coronary steal.

· The notes use adenosine and dipyridamole, which blocks cellular adenosine uptake and so raises adenosine levels, and this uneven redistribution is what makes a perfusion defect visible during pharmacologic stress imaging.

Distinctions that matter

· Resting flow and maximal flow fail at different degrees of narrowing, which is why a patient can be symptom free at rest with a severe lesion.

· Steal is a redistribution of flow rather than a fall in total coronary flow.

Common misconceptions

· Reading normal resting flow as evidence of a mild stenosis. Normal resting flow is maintained by spending reserve.

· Expecting a vasodilator to help the ischemic bed. The stenosed bed cannot dilate further, so flow moves away from it.

Where it is taught: Section C05, the Fick worked example and the paragraphs on autoregulation, flow reserve, and steal of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild why ALCAPA fails only after birth

A competent response contains

· In ALCAPA the left main coronary artery arises from the pulmonary trunk while the right coronary artery arises normally from the aorta.

· Before birth, high pulmonary vascular resistance and the ductus arteriosus keep pulmonary artery pressure about equal to aortic pressure, so the anomalous artery fills antegrade at systemic pressure.

· The fetus tolerates it because pulmonary artery blood is only modestly less saturated than ascending aortic blood and the fetal myocardium is adapted to low oxygen tension, so the heart develops normally.

· After birth, pulmonary vascular resistance falls, the ductus closes, and pulmonary artery pressure falls over days to weeks toward about 25/10 mm Hg.

· The left coronary artery then carries low-pressure blood about 70% to 75% saturated, which cannot supply a left ventricle perfused mainly in diastole against end-diastolic pressure and extracting about 70% of its oxygen.

· Collaterals from the right coronary artery, still at aortic pressure, fill the left coronary branches, but because the pulmonary trunk is the lowest-pressure outlet, blood flows backward up the left coronary artery into it, so the artery becomes a drain.

· That is a form of coronary steal, and it leaves the left ventricle ischemic despite growing collateral flow, with anterolateral ischemia and infarction often at 2 to 3 months.

· Self-reinforcing cycle: low-pressure flow and steal leave the ventricle ischemic; the ischemic ventricle contracts poorly and dilates; the larger radius raises wall stress and demand; papillary ischemia and annular dilation cause mitral regurgitation; end-diastolic pressure rises and perfusion pressure falls further; ischemia deepens and drives further dilation.

· Each step in that cycle either raises demand (larger radius and wall stress, regurgitant volume load) or lowers supply (higher end-diastolic pressure, lower perfusion pressure), which is why the infant ventricle can deteriorate quickly.

Distinctions that matter

· The anomalous origin is present from the start, and what changes after birth is pulmonary artery pressure, so the lesion fails because of the normal birth transition.

· Collateral development is not protective here, because the collateral flow drains into the low-pressure pulmonary trunk.

Common misconceptions

· Expecting the infant to be ischemic from birth. Pulmonary artery pressure is still high in the first days.

· Assuming growing right coronary collaterals will rescue the left ventricle. They feed the steal.

Where it is taught: Section C06, paragraphs on fetal and postnatal physiology, the flow diagram, and the self-reinforcing cycle of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C06-BD2

Brain dump 2 · integration and application: The infant who looks well at birth and sick at ten weeks

A competent response contains

· Symptoms begin around 2 to 3 months because that is when pulmonary artery pressure has reached its low postnatal level, and not in the first days, when it is still high.

· Feeding is an infant's main exertion, so ischemia appears with feeds or crying as pallor, sweating, irritability, and breathlessness, described as infant angina.

· Examination and imaging show heart failure: tachypnea, poor weight gain, and cardiomegaly on the chest film, with a dilating poorly contracting ventricle.

· The electrocardiogram shows an anterolateral infarct pattern with abnormal Q waves and ST and T changes in leads I, aVL, V5, and V6, which is almost unheard of in a normal infant.

· Mitral regurgitation arises from anterolateral papillary muscle ischemia and from annular dilation.

· What separates this from dilated cardiomyopathy and myocarditis is a regional infarct pattern in the left coronary territory rather than global dysfunction, and onset after the first weeks rather than from birth.

· A minority develop extensive right coronary collaterals and survive into childhood or adulthood with a dilated right coronary artery, a continuous murmur, mitral regurgitation, exercise-induced ischemia, or sudden arrhythmic death.

· In those patients pulmonary artery saturation can be slightly higher than right ventricular saturation, because the coronary system creates a small left to right shunt into the pulmonary trunk.

· A large ventricular septal defect keeps pulmonary artery pressure high, which can delay or mask the ischemia.

· The variable that decides perfusion is pulmonary artery pressure relative to the pressure needed to perfuse the left ventricle, not the anomalous origin itself.

Distinctions that matter

· Regional against global dysfunction is what points to a coronary cause rather than a myopathic or inflammatory one.

· Most infants deteriorate in the first year, while a minority with rich collaterals present much later, so the same anatomy gives two very different courses.

Common misconceptions

· Reading a normal first few weeks as evidence against the diagnosis. The timing is the expected pattern.

· Assuming pulmonary artery saturation can never exceed right ventricular saturation. The coronary shunt into the trunk makes it possible here.

Where it is taught: Section C06, manifestations explained by mechanism and variability and distinctions of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C01-Q01

Board-style question: Locating the occlusion from the complication

Correct answer: C. Right coronary artery supplying the atrioventricular node

Why it is right: The right coronary artery runs to the crux, where it gives the atrioventricular nodal artery in about 80% of hearts and the posterior descending artery in right dominant hearts, so its occlusion produces an inferior infarct in II, III, and aVF together with nodal block and a narrow junctional escape at 40 to 60 bpm.

A is wrong: A left anterior descending occlusion shows in V1 to V4 and damages the bundle branches, giving a wide escape at about 20 to 40 bpm rather than a 0.08 s QRS.

B is wrong: A circumflex occlusion shows in I, aVL, V5, and V6 and spares the atrioventricular node unless the heart is left dominant, which the inferior lead pattern here does not support.

D is wrong: A left main occlusion threatens most of the left ventricle and gives widespread anterior and lateral changes with profound hypotension, not isolated inferior ST elevation.

A competent response contains

· ST elevation in II, III, and aVF marks the inferior wall, the territory of the posterior descending artery, which comes from the right coronary artery in most hearts.

· P waves at 82/min with dissociated QRS complexes at 46/min of 0.08 s duration are complete block with a narrow junctional escape, which places the block at the atrioventricular node, supplied by the atrioventricular nodal artery from the right coronary artery in about 80% of hearts.

Distinctions that matter

· A narrow escape complex places the pacemaker above the bundle branches and so places the block at the node.

Common misconceptions

· Reading every complete block after infarction as infranodal. The lead group and the narrow escape point to the node.

Where it is taught: Section C01 of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-Q02

Board-style question: Ischemia with unobstructed arteries

Correct answer: A. A coronary perfusion pressure of about 14 mm Hg with high wall stress in a dilated ventricle, which exhausts subendocardial reserve

Why it is right: Her aortic diastolic pressure of 40 mm Hg against an end-diastolic pressure of 26 mm Hg leaves a perfusion pressure of about 14 mm Hg, and the dilated ventricle has high wall stress and oxygen demand, so the subendocardium, which already spends part of its vasodilator reserve at rest and receives almost no systolic flow, becomes ischemic on exertion.

B is wrong: Myocardial extraction is about 70% to 75% at rest rather than 25%, so a failure to rise above 25% does not describe this heart; the whole-body figure is the one near 25%.

C is wrong: The subepicardium is the layer that keeps perfusion best, and it is the subendocardium that receives almost no systolic flow, so this reverses the gradient within the wall.

D is wrong: Diversion of flow into the pulmonary trunk belongs to an anomalous left coronary artery arising from the pulmonary artery, a congenital lesion of infancy, not to aortic regurgitation.

A competent response contains

· Coronary perfusion pressure is about aortic diastolic pressure minus left ventricular end-diastolic pressure, so 40 minus 26 is about 14 mm Hg against a normal of about 70 mm Hg.

· Aortic regurgitation lowers aortic diastolic pressure and dilates the ventricle, and by the law of Laplace the larger radius raises wall stress and oxygen demand, so the subendocardium exhausts its reserve first.

Distinctions that matter

· Ischemia follows the balance of perfusion pressure, perfusion time, oxygen content, and demand, and a normal angiogram does not exclude it.

Common misconceptions

· Requiring a stenosis before accepting ischemia. Valve disease alone can lower supply and raise demand enough to produce it.

Where it is taught: Section C04 of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C06-Q03

Board-style question: Why the infant deteriorates at two months

Correct answer: D. A fall in pulmonary vascular resistance and pulmonary artery pressure, which left the anomalous artery with low-pressure desaturated blood and reversed collateral flow into the pulmonary trunk

Why it is right: The postnatal fall in pulmonary vascular resistance and closure of the ductus drop pulmonary artery pressure over days to weeks, leaving the anomalous artery with low-pressure desaturated blood, and because the pulmonary trunk is then the lowest-pressure outlet, right coronary collateral flow runs backward up the left coronary artery into it, so the left ventricle stays ischemic and infarcts in the anterolateral territory.

A is wrong: Atherosclerosis does not develop over weeks in an infant, and the lesion here is a congenital origin rather than a narrowing.

B is wrong: The foramen ovale carries blood from right atrium to left atrium and plays no part in perfusing the left ventricular muscle, which depends on the coronary arteries.

C is wrong: Myocardial extraction is already near maximal and does not rise to create the problem; the failure is in the pressure and oxygen content delivered to the anomalous artery.

A competent response contains

· Before birth, pulmonary artery pressure about equals aortic pressure, so the anomalous left coronary artery fills antegrade at systemic pressure with only modestly desaturated blood.

· After birth, pulmonary vascular resistance falls and the ductus closes, so pulmonary artery pressure falls toward about 25/10 mm Hg over days to weeks, the left coronary artery receives low-pressure blood about 70% to 75% saturated, and right coronary collateral flow drains backward into the pulmonary trunk as coronary steal, giving an anterolateral infarct pattern in I, aVL, V5, and V6 at about 2 to 3 months.

Distinctions that matter

· The anatomy does not change after birth; pulmonary artery pressure does, and that is what decides whether the left ventricle is perfused.

Common misconceptions

· Treating collateral growth as protective here. Collateral flow drains into the low-pressure pulmonary trunk instead of perfusing muscle.

Where it is taught: Section C06 of the CV-8 Coronary Circulation notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-09

CV-9 Blood Distribution to Body Regions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-09-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the aorta and label every branch with its level

A competent response contains

· The ascending aorta gives only the right and left coronary arteries and ends at the sternal angle, the T4 to T5 disc.

· The arch passes back over the left lung root and returns to T4, giving the brachiocephalic trunk, the left common carotid and the left subclavian arteries from right to left.

· The brachiocephalic trunk divides behind the right sternoclavicular joint into the right subclavian and right common carotid arteries.

· Each common carotid divides at about C4, where the carotid sinus holds baroreceptors.

· The descending thoracic aorta runs from T4 to T12 and gives posterior intercostal arteries to spaces 3 to 11, subcostal arteries, and bronchial, esophageal and superior phrenic branches; the first two intercostal spaces are supplied from the costocervical trunk.

· One large anterior segmental medullary artery, usually on the left, reinforces the anterior spinal artery over the lower cord.

· The aorta passes the diaphragm at T12 and bifurcates at L4, near the umbilicus; the median sacral artery leaves the posterior aorta at the bifurcation.

· Abdominal branches with levels: inferior phrenic (paired) about T12 to the diaphragm, celiac trunk T12 to the foregut as far as the major papilla with the liver and spleen, superior mesenteric L1 to the midgut from the distal duodenum to the proximal two thirds of the transverse colon, renal and middle suprarenal (paired) about L1, gonadal (paired) about L2, inferior mesenteric L3 to the hindgut from the distal third of the transverse colon to the anal canal above the pectinate line, and four pairs of lumbar arteries from L1 to L4 to the posterior abdominal wall.

· Paired branches are the inferior phrenic, middle suprarenal, renal, gonadal and lumbar arteries; the three gut arteries and the median sacral artery are unpaired.

· The gonadal arteries arise high because the gonads develop high and then descend.

· The pancreatic head and second part of the duodenum sit at the foregut and midgut junction and receive both celiac and superior mesenteric blood, and the spleen is supplied by the celiac trunk because it forms in the dorsal mesogastrium.

Distinctions that matter

· The arch gives three branches but five arteries reach the head and arms, because the brachiocephalic trunk divides again.

· Levels are not symmetrical about the branch they name: the celiac trunk and the diaphragmatic hiatus are both at T12, while the renal arteries sit at about L1 with the superior mesenteric artery just above them.

Common misconceptions

· Writing that the right subclavian and right common carotid arteries arise from the arch. They arise from the brachiocephalic trunk behind the right sternoclavicular joint.

· Giving the aorta a lumbar origin for the gonadal arteries at the level of the gonads. They arise at about L2, well above the pelvis, because the gonads descend from a high position.

Where it is taught: Section C01, paragraphs on the ascending aorta and arch, the descending thoracic aorta, and the table of abdominal aortic branches of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C01-BD2

Brain dump 2 · integration and application: Three patients with a lesion on the aorta

A competent response contains

· Patient 1: the left recurrent laryngeal nerve hooks under the arch beside the ligamentum arteriosum, so an aneurysm or mass there stretches it and causes hoarseness.

· Patient 2: the clamp interrupted the large anterior segmental medullary artery, usually on the left, which reinforces the anterior spinal artery over the lower cord.

· Patient 2: the anterior two thirds of the cord infarcts, giving paraplegia with loss of pain and temperature sensation below the lesion.

· Patient 2: vibration and position sense travel in the posterior columns, which are supplied by the posterior spinal arteries and are therefore spared.

· Patient 3: adult coarctation sits just distal to the origin of the left subclavian artery, so both arms are supplied from above the narrowing at high pressure while the lower body is supplied past it.

· Patient 3: the legs are perfused through collateral channels rather than directly, which is why the femoral pulses are weak and delayed compared with the radial pulse.

· Patient 3: notching comes from enlarged, tortuous posterior intercostal arteries eroding the inferior rib borders at about ribs 3 to 8; the arteries of the first two spaces arise from the costocervical trunk above the narrowing, so they are not enlarged and ribs 1 and 2 are spared.

· Levels and landmarks: the arch and the ligamentum arteriosum at T4, the descending thoracic aorta from T4 to T12 for the medullary artery, and the coarctation just distal to the left subclavian origin at the level of the arch.

Distinctions that matter

· Anterior cord infarction and posterior column loss separate on arterial supply, not on level: the anterior spinal artery carries the pain and temperature pathways and the corticospinal tracts, and the posterior spinal arteries carry the posterior columns.

· High arm pressure with weak femoral pulses locates a narrowing between the arch branches and the abdominal aorta; weak femoral pulses with normal arm pressure points instead to iliac or aortoiliac disease.

Common misconceptions

· Blaming hoarseness on the right recurrent laryngeal nerve. The left nerve is the one that loops under the arch, so aortic disease affects that side.

· Expecting rib notching on all ribs in coarctation. The spaces supplied from above the narrowing are not part of the collateral route.

Where it is taught: Section C01, arch paragraph, descending thoracic aorta paragraph, the paragraph on findings predicted by level, and the coarctation flow diagram of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C01-RT01

Retrieval exercise: Rebuild the collateral route around a coarctation

A competent response contains

· Step 1: pressure in the subclavian artery above the narrowing is high.

· Step 2: the internal thoracic artery descends behind the costal cartilages.

· Step 3: anterior intercostal arteries carry blood around each space.

· Step 4: flow reverses in the enlarged, tortuous posterior intercostal arteries.

· Step 5: blood enters the descending aorta below the coarctation.

· Step 6: the enlarged intercostal arteries notch the inferior rib borders at about ribs 3 to 8.

· The reversed vessel is the posterior intercostal artery; normally it carries blood from the aorta outward, and here the pressure gradient runs from the subclavian system toward the low pressure aorta below the narrowing, so flow runs back toward the aorta.

· Ribs 1 and 2 are spared because their intercostal arteries come from the costocervical trunk, which lies above the narrowing and so carries no collateral load.

Distinctions that matter

· The collateral route is driven by the pressure difference across the coarctation, not by any new vessel: every segment of the path exists normally and simply enlarges.

· Notching is produced by the posterior intercostal arteries, while the internal thoracic and anterior intercostal arteries are the upstream part of the same route.

Common misconceptions

· Drawing flow passing through the narrowing itself. The point of the route is that blood reaches the aorta below the coarctation without crossing it.

· Marking reversed flow in the internal thoracic artery. Flow there runs in its normal direction, downward; the reversal is in the posterior intercostals.

Where it is taught: Section C01, flow diagram on the collateral route around an adult coarctation of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the circle of Willis and its feeding vessels

A competent response contains

· The two internal carotid arteries form the anterior circulation; the two vertebral arteries, each the first branch of the subclavian artery, rise through the transverse foramina from C6 to C1 and join at the lower pons as the basilar artery to form the posterior circulation.

· The internal carotid artery has no branches in the neck, crosses the cavernous sinus beside the abducens nerve, gives the ophthalmic artery, then gives the posterior communicating artery and divides into the anterior and middle cerebral arteries.

· The circle is formed by the anterior communicating artery, the proximal anterior cerebral arteries, the terminal internal carotid arteries, the posterior communicating arteries and the proximal posterior cerebral arteries.

· The middle cerebral artery continues the line of the internal carotid artery and therefore receives most emboli.

· The lenticulostriate branches leave the high pressure middle cerebral trunk at right angles as thin end arteries and supply deep structures including the internal capsule.

· The basilar artery gives the anterior inferior cerebellar and superior cerebellar arteries and then the posterior cerebral arteries; the posterior inferior cerebellar artery arises from the vertebral artery.

· The oculomotor nerve emerges between the posterior cerebral artery and the superior cerebellar artery.

· Territories: anterior cerebral artery to the medial hemisphere with the leg represented more than the arm, middle cerebral artery to the lateral convexity with the face and arm more than the leg and the language areas, posterior cerebral artery to the occipital cortex, posterior inferior cerebellar artery to the lateral medulla, anterior inferior cerebellar artery to the lateral pons.

· The ophthalmic artery supplies the eye, so a carotid embolus can cause transient monocular blindness.

· The external carotid artery supplies the face, scalp and dura but no brain tissue; its one branch that matters inside the skull is the middle meningeal artery, from the maxillary artery, which runs under the pterion and is torn in epidural hematoma.

· Berry aneurysms form at branch points of the circle, about 90% anteriorly and most often at the anterior communicating junction; one at the origin of the posterior communicating artery compresses the oculomotor nerve and dilates the pupil early.

Distinctions that matter

· The circle is an inflow ring: the communicating arteries connect the two carotid systems with each other and with the vertebrobasilar system, while the cortical branches beyond it connect only through small leptomeningeal channels.

· The vertebral arteries give the posterior inferior cerebellar arteries and the basilar artery gives the anterior inferior and superior cerebellar arteries, so keep the parent vessel with each cerebellar artery.

Common misconceptions

· Drawing branches of the internal carotid artery in the neck. It gives none until it is inside the skull.

· Treating the external carotid artery as irrelevant once it supplies no brain. Its middle meningeal branch under the pterion causes epidural hematoma.

Where it is taught: Section C02, paragraphs on the internal carotid and vertebrobasilar systems and on the circle of Willis of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C02-BD2

Brain dump 2 · integration and application: Place four deficits on the vascular map

A competent response contains

· Patient 1: anterior cerebral artery, anterior circulation; it supplies the medial hemisphere where the leg is represented, along with frontal areas that govern continence and behavior.

· Patient 2: posterior inferior cerebellar artery, posterior circulation; lateral medullary infarction gives vertigo, dysphagia, hoarseness, ipsilateral Horner syndrome, ataxia and ipsilateral loss of facial pain and temperature sensation with contralateral loss over the body.

· Patient 2: findings on opposite sides of the face and the body are crossed findings, which localize the lesion to the brainstem, because the facial sensory nucleus is ipsilateral while the ascending body pathway has already crossed.

· Patient 3: posterior cerebral artery, posterior circulation; homonymous hemianopia with macular sparing follows occipital infarction.

· Patient 4: a lenticulostriate artery, anterior circulation, giving a lacunar infarct in the internal capsule and a pure motor hemiparesis.

· Patient 4: chronic hypertension causes hyaline arteriolosclerosis in these thin end arteries, which also produces lacunar infarcts and Charcot-Bouchard microaneurysms.

· An embolus in a cortical middle cerebral branch still infarcts because the circle of Willis acts only upstream; beyond it, cortical branches connect only through small leptomeningeal anastomoses that cannot carry enough flow.

· The circle protects against slow occlusion of an inflow vessel, and a complete symmetric circle exists in only a minority of people.

Distinctions that matter

· Leg more than arm points to the anterior cerebral artery, and face and arm more than leg to the middle cerebral artery, because of where each territory sits on the motor strip.

· A deep lacunar infarct gives pure motor loss without language or visual field loss, while a cortical middle cerebral infarct adds aphasia or neglect.

Common misconceptions

· Expecting a complete circle of Willis to prevent all stroke. It refills territory only when the block is upstream of the ring and develops slowly.

· Reading vertigo, hoarseness and dysphagia as a cortical problem. The crossed sensory pattern places the lesion in the lateral medulla.

Where it is taught: Section C02, paragraph on the circle of Willis, the table of cerebral and brainstem territories, and the lenticulostriate description of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C03-BD1

Brain dump 1 · foundational retrieval: Limb arteries, their landmarks and their pulse points

A competent response contains

· Upper limb chain: subclavian artery, which passes between the anterior and middle scalene muscles with the brachial plexus, becomes the axillary artery at the lateral border of the first rib, becomes the brachial artery at the lower border of teres major, and divides in the cubital fossa at about the radial neck into the radial and ulnar arteries, which rejoin in the connected superficial and deep palmar arches.

· Lower limb chain: external iliac artery becomes the femoral artery at the inguinal ligament, becomes the popliteal artery at the adductor hiatus, and divides at the lower border of popliteus into the anterior tibial

artery, which pierces the interosseous membrane and continues as the dorsalis pedis artery at the ankle, and the posterior tibial artery, which gives the fibular artery; the plantar vessels rejoin in the plantar arch.

· Pulse points: brachial in the medial arm or in the cubital fossa medial to the biceps tendon, radial lateral to flexor carpi radialis, ulnar near the pisiform, femoral at the midinguinal point just below the inguinal ligament, popliteal deep in the fossa with the knee flexed, dorsalis pedis lateral to extensor hallucis longus, posterior tibial behind the medial malleolus.

· The axillary pulse is not routinely felt; the ulnar and popliteal pulses are often faint; the dorsalis pedis pulse is absent in a small number of healthy adults.

· The connected palmar arches let either the radial or the ulnar artery perfuse the whole hand, which is what the Allen test examines.

· Nerve and artery pairs: the posterior circumflex humeral artery with the axillary nerve around the surgical neck of the humerus, the deep brachial artery with the radial nerve in the radial groove, and the subclavian artery with the brachial plexus between the scalenes.

· Femoral triangle order from lateral to medial: nerve, artery, vein, canal.

· The lateral compartment of the leg has no artery of its own and is supplied by perforating branches of the anterior tibial and fibular arteries.

Distinctions that matter

· A name change is a landmark, not a new vessel: the same column of blood becomes axillary, brachial, then radial and ulnar as it passes fixed structures.

· A faint pulse is not the same as an absent one: the ulnar and popliteal pulses are often hard to feel in healthy people, and only the dorsalis pedis is genuinely absent in some normal adults.

Common misconceptions

· Placing the femoral pulse at the pubic tubercle or in the mid thigh. It lies at the midinguinal point just below the inguinal ligament.

· Treating an absent dorsalis pedis pulse as proof of disease. Check the posterior tibial pulse, because a few healthy adults never have a palpable dorsalis pedis.

Where it is taught: Section C03, paragraphs on the upper and lower limb arteries and the table of limb landmarks and pulse points of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C03-BD2

Brain dump 2 · integration and application: Localize four limb lesions from the examination

A competent response contains

· Patient 1: the first abnormal pulse is the popliteal, so the obstruction lies upstream in the superficial femoral artery, most often in the adductor canal; the calf aches first because claudication appears one level below the narrowing.

· Patient 2: the ankle-brachial index is 70 divided by 140, which is 0.5.

· Patient 2: ankle pressure normally equals or slightly exceeds arm pressure because the pulse wave amplifies as it travels distally, so an index below about 0.9 means pressure has been lost across a stenosis; 0.5 indicates significant disease.

· Patient 3: the brachial artery is injured by a supracondylar humeral fracture, the forearm flexors become ischemic, and the healed result is Volkmann contracture.

· Patient 4: the medial circumflex femoral artery supplies the femoral head, so a displaced intracapsular neck fracture risks avascular necrosis.

· Emboli lodge most often at the common femoral bifurcation in the lower limb.

· Scapular anastomoses bypass an axillary block above the origin of the subscapular artery but not below it, so a proximal block is better tolerated.

· Peripheral arterial disease favors the superficial femoral artery in the adductor canal and the popliteal artery; aortoiliac disease gives buttock and thigh claudication and femoropopliteal disease gives calf claudication.

Distinctions that matter

· Palpating in sequence places an obstruction between the last normal pulse and the first weak one, so the level of the first abnormal pulse is the finding that localizes it.

· Claudication level and pulse level give the same answer from different directions: muscles ache one level below the narrowing while pulses fail at and below it.

Common misconceptions

· Expecting ankle pressure to be lower than arm pressure in health. The pulse wave amplifies distally, so normal ankle pressure matches or exceeds arm pressure.

· Attributing a cold pulseless hand after a supracondylar fracture to the radial artery. The brachial artery is the vessel at that level.

Where it is taught: Section C03, paragraphs on the limb arteries, the peripheral arterial disease paragraph, and the stop and predict box of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw venous return and the portal system

A competent response contains

· The two brachiocephalic veins unite behind the right first costal cartilage to form the superior vena cava, which receives the azygos vein at about T4.

· The inferior vena cava forms at L5, ascends to the right of the aorta, and passes the diaphragm at T8.

· The right gonadal vein drains into the inferior vena cava; the left gonadal vein drains into the left renal vein at a right angle.

· The left renal vein crosses in front of the aorta below the superior mesenteric artery, so it passes between those two vessels; the third part of the duodenum crosses in the same angle.

· The splenic vein and the superior mesenteric vein unite behind the neck of the pancreas to form the portal vein.

· Blood from the gut, spleen, pancreas and gallbladder passes through the liver sinusoids before reaching the inferior vena cava, which is the basis of first pass metabolism.

· The liver receives about a quarter of cardiac output, about three quarters of it portal, yet the portal and arterial sources each supply about half its oxygen.

· Portal pressure is about 9 mm Hg against near 0 mm Hg in the hepatic veins, and the portal system has no valves, so any rise in resistance is transmitted back to the gut and spleen.

· Portosystemic anastomoses: lower esophagus, left gastric vein to esophageal veins draining to the azygos vein, giving varices that can bleed massively; umbilicus, paraumbilical veins to superficial epigastric veins, giving caput medusae; anorectal junction, superior rectal vein to middle and inferior rectal veins, giving anorectal varices; retroperitoneum, colic veins to lumbar and renal veins, giving silent varices.

Distinctions that matter

· The portal system is defined by having a second capillary bed at its far end: gut blood passes the liver sinusoids before any of it reaches the inferior vena cava.

· Portal blood carries most of the liver's flow but only about half its oxygen, so flow share and oxygen share are separate figures.

Common misconceptions

· Drawing both gonadal veins into the inferior vena cava. Only the right does; the left enters the left renal vein.

· Putting the portal vein in front of the pancreas. It forms behind the pancreatic neck from the splenic and superior mesenteric veins.

Where it is taught: Section C04, paragraphs on the superior and inferior venae cavae and the portal system, and the table of portosystemic anastomoses of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-BD2

Brain dump 2 · integration and application: Four patients with an obstructed vein

A competent response contains

· Patient 1: the superior vena cava is obstructed, giving facial, neck and arm swelling with distended chest wall veins.

· Patient 1: if the block lies above the entry of the azygos vein, blood returns through the azygos vein into the superior vena cava below the block; if it lies below that entry, flow reverses in the azygos vein and passes through the lumbar veins into the inferior vena cava.

· Patient 2: the left renal vein is compressed between the superior mesenteric artery and the aorta, the nutcracker phenomenon, causing hematuria and back pressure in the left gonadal vein.

· Patient 2: the left gonadal vein drains into the left renal vein at a right angle and has a longer course, while the right drains directly into the inferior vena cava, which is why varicocele is usually left sided.

· Patient 3: a new isolated right varicocele suggests obstruction of the inferior vena cava or a retroperitoneal mass, because the right gonadal vein drains straight into the cava and has no equivalent compression point.

· Patient 4: the umbilical site connects paraumbilical veins with superficial epigastric veins, giving caput medusae.

· Patient 4: the portal system has no valves, so intrahepatic resistance raises pressure back through the portal and splenic veins and the spleen becomes congested and enlarges; congestive splenomegaly sequesters platelets, which lowers the platelet count.

· Sites of raised portal pressure: prehepatic, for example portal vein thrombosis; intrahepatic, for example cirrhosis, with schistosomiasis acting presinusoidally; posthepatic, for example Budd-Chiari syndrome, right heart failure or constrictive pericarditis.

· The hepatic venous pressure gradient is normally up to about 5 mm Hg, and varices form at about 10 mm Hg.

Distinctions that matter

· Left sided varicocele is explained by venous drainage route and length, while a new right sided varicocele is explained by something obstructing the cava, so the side of the finding changes what you look for.

· Prehepatic, intrahepatic and posthepatic causes all raise portal pressure, but only the intrahepatic and posthepatic groups raise the gradient across the liver.

Common misconceptions

· Expecting a blocked vein to stop venous return. Veins are richly interconnected, so flow reroutes through channels that are normally tiny and then visibly dilate.

· Reading a low platelet count in cirrhosis as bone marrow failure alone. The congested spleen sequesters platelets.

Where it is taught: Section C04, superior vena cava paragraph, inferior vena cava and left renal vein paragraph, portal hypertension classification, and the anastomosis table of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-RT01

Retrieval exercise: Rebuild the cycle that keeps portal pressure high

A competent response contains

· Step 1: fibrosis, regenerative nodules and stellate cell contraction raise intrahepatic resistance.

· Step 2: nitric oxide and other vasodilators that escape the liver dilate the splanchnic arterioles.

· Step 3: portal inflow rises against the high resistance.

· Step 4: low effective arterial volume activates the renin angiotensin aldosterone system and the sympathetic nerves.

· Step 5: the kidneys retain sodium and water, expanding volume.

· Step 6: the extra volume increases splanchnic inflow, which keeps portal pressure high and closes the loop.

· The three intrahepatic changes raising resistance are fibrosis, nodules and stellate cell contraction.

· The mediator dilating the splanchnic arterioles is nitric oxide.

· At step 4 the body senses a low effective arterial volume, because splanchnic dilatation has enlarged the vascular space even though total volume is normal or high.

· Pressure equals resistance times inflow, and cirrhosis raises both.

· The sodium and water retention at step 5 is also what feeds ascites.

Distinctions that matter

· Resistance starts the problem and inflow sustains it, which is why pressure does not settle once the liver is scarred.

· Effective arterial volume and total blood volume are different quantities, and in cirrhosis they move in opposite directions.

Common misconceptions

· Treating portal hypertension as resistance alone. Splanchnic dilatation raises inflow, and pressure is the product of the two.

· Explaining the sodium retention as dehydration. It is a response to a low effective arterial volume in a dilated circulation.

Where it is taught: Section C04, cycle diagram on why portal pressure stays high in cirrhosis of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C04-CA01

Clinical application: Vomiting blood in a man with long-standing liver disease

A competent response contains

· Recognize: esophageal varices at the lower esophagus, where the left gastric vein meets esophageal veins, and those esophageal veins drain into the azygos vein.

· Explain: the raised resistance lies inside the liver, so portal blood takes the low resistance portosystemic routes upstream of the obstruction instead of crossing the sinusoids to the inferior vena cava, and the valveless portal system transmits the pressure back to the gut and spleen; the distended abdomen with shifting dullness is ascites from sodium and water retention driven by splanchnic dilatation and a low effective arterial volume, and the caput medusae is the umbilical anastomosis between paraumbilical and superficial epigastric veins opening under the same pressure.

· Differentiate: the anorectal anastomosis links the superior rectal vein with the middle and inferior rectal veins and produces anorectal varices, which are not the same as internal hemorrhoids, and hemorrhoids are not more frequent in portal hypertension; bleeding from raised portal pressure arises at a named anastomotic site, comes with splenomegaly, thrombocytopenia and ascites, and follows a raised hepatic venous pressure gradient, whereas a bleed from a gastric wall lesion arises at a single mucosal site with no portosystemic features.

· Investigate: the hepatic venous pressure gradient, which is normally up to about 5 mm Hg and is raised here at 14 mm Hg with varices expected above about 10 mm Hg; portal pressure, normally about 9 mm Hg and raised in this man; hepatic vein pressure, normally near 0 mm Hg; and a platelet count and spleen size, both of which reflect portal congestion.

· Decide: the gradient of 14 mm Hg sits well above the normal ceiling of about 5 mm Hg and above the varix threshold of about 10 mm Hg, and the raised resistance in this man is intrahepatic, from cirrhosis.

· Justify: an intrahepatic block raises portal pressure while hepatic vein pressure stays near 0 mm Hg, so the gradient across the liver is wide; a prehepatic cause such as portal vein thrombosis raises portal pressure with a normal gradient because the liver itself is not the resistance, and a posthepatic cause such as Budd-Chiari syndrome, right heart failure or constrictive pericarditis raises the pressure downstream of the sinusoids as well, so the measured hepatic vein pressure rises with it; the valveless portal system then carries the pressure back to the gut and spleen and opens the anastomoses.

Distinctions that matter

· A raised portal pressure is not the same finding as a raised gradient across the liver, and the difference between them is what separates prehepatic from intrahepatic and posthepatic causes.

· Varices, ascites, splenomegaly and thrombocytopenia are four expressions of the same raised pressure rather than four separate problems.

Common misconceptions

· Calling the bleeding site a hemorrhoidal one. The anorectal anastomosis gives anorectal varices, and hemorrhoids are a separate entity.

· Explaining the ascites as low albumin alone. The notes trace it to sodium and water retention driven by splanchnic dilatation and a low effective arterial volume.

Where it is taught: Section C04, portal system paragraph, portal hypertension classification, the cirrhosis cycle diagram, the anastomosis table, and the stop and predict box of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-BD1

Brain dump 1 · foundational retrieval: Map the watershed zones and explain why they fail first

A competent response contains

· A watershed is tissue at the far edge of two adjacent arterial territories, supplied by the most distal branches of each.

· Pressure falls along every arterial tree because each segment has resistance, so the border receives the lowest perfusion pressure of either tree, and the collaterals there, the leptomeningeal anastomoses in the brain and the marginal artery in the colon, are too small to rescue either side.

· Cerebral blood flow stays nearly constant across a mean arterial pressure of about 60 to 150 mm Hg; below that range flow falls in proportion to pressure and the border zones fail first.

· Chronic hypertension shifts the autoregulatory range upward, so infarction occurs at pressures that would be safe for others.

· Cortical border zones: between the anterior and middle cerebral arteries as a strip over the upper convexity, and between the middle and posterior cerebral arteries in the parieto-occipital region; the internal border zone lies in deep white matter.

· Bilateral anterior to middle cerebral border infarction weakens both proximal arms with the face, hands and legs spared, described as man in a barrel; bilateral middle to posterior cerebral border infarction can produce features of Balint syndrome.

· Colonic sites: the splenic flexure, between the superior and inferior mesenteric arteries, and the rectosigmoid junction; the rectum has additional internal iliac supply and is relatively protected.

· Colonic hypoperfusion causes ischemic colitis with left sided pain and bloody diarrhea, and the mucosa fails first because countercurrent exchange in the villi leaves the tips hypoxic.

· Regions that fail first in low flow: cerebral border zones, at the distal ends of two territories, giving bilateral proximal arm weakness; splenic flexure and rectosigmoid, with weak marginal connections, giving ischemic colitis; subendocardium, farthest from the epicardial arteries and perfused mainly in diastole, giving subendocardial infarction; liver zone 3, last on the sinusoid, giving centrilobular necrosis; renal outer medulla, with a low oxygen tension from vasa recta countercurrent exchange and a high transport demand, giving acute tubular necrosis.

· Only the brain and colon sites are true boundaries between two territories; the other three are the distal ends of a single supply path.

Distinctions that matter

· A border zone between two arterial trees is not the same as the far end of one supply path, although both fail early in low flow.

· Hypoperfusion gives mucosal or mural bowel infarction, while occlusion of a vessel gives transmural infarction.

Common misconceptions

· Expecting autoregulation to protect border zones at any pressure. Once arterioles are maximally dilated, flow follows pressure.

· Assuming a hypertensive patient tolerates low pressure as well as anyone else. The shifted range means injury begins at a higher pressure.

Where it is taught: Section C05, opening paragraph, the cortical and colonic border zone paragraphs, and the table of regions that fail first in low flow of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-BD2

Brain dump 2 · integration and application: Read a pattern of injury back to a mechanism

A competent response contains

· Bloody stools with left sided pain: the colonic border zones at the splenic flexure, between the superior and inferior mesenteric arteries, and the rectosigmoid junction, where the marginal connections are weak; the injury is ischemic colitis, mucosal or mural rather than transmural.

· Weakness of both shoulders and proximal arms with normal grip: the cortical border zones between the anterior and middle cerebral arteries over the upper convexity on both sides, described as man in a barrel.

· Rising aminotransferases: liver zone 3, last on the sinusoid, giving centrilobular necrosis.

· Rising creatinine with granular casts: the renal outer medulla, which has a low oxygen tension from vasa recta countercurrent exchange and a high transport demand, giving acute tubular necrosis.

· What they share: every organ fell below its autoregulatory range at the same time, so the regions that fail first in low flow were injured together, which is global hypoperfusion.

· An embolus or thrombus fills one artery's territory, usually on one side and in one organ, and occlusion gives transmural bowel infarction rather than mucosal injury.

· Strip like infarcts along border zones, often bilateral and in several organs at once, point to global hypoperfusion.

· A severe carotid stenosis already lowers the pressure delivered to that hemisphere, so a modest systemic fall takes the ipsilateral border zones below their threshold while other patients stay within range.

Distinctions that matter

· Number of territories and number of organs is the discriminator: one territory on one side points to an embolus, several border zones in several organs point to a fall in pressure.

· Depth of bowel injury separates the two mechanisms: hypoperfusion injures mucosa or wall, occlusion kills the full thickness.

Common misconceptions

· Reading four organ findings as four unrelated complications. They are one mechanism expressed in the regions that fail first.

· Expecting arm weakness from a border zone infarct to include the hands. The strip over the convexity spares face, hands and legs.

Where it is taught: Section C05, flow diagram from hypotension to watershed infarct, the border zone table, the paragraph on embolus against global hypoperfusion, and the stop and predict box of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-CP01

Comparison table: Global hypoperfusion against occlusion of one artery

A competent response contains

· Mechanism: global hypoperfusion lowers systemic pressure until arterioles are maximally dilated and flow follows pressure; occlusion of a single artery blocks one supply route while systemic pressure is maintained.

· Position within a territory: global hypoperfusion injures the distal border of two territories; occlusion kills the whole territory of the blocked vessel, border and centre alike.

· Sides: global hypoperfusion is often bilateral and roughly symmetrical; occlusion is usually one sided.

· Organs: global hypoperfusion injures several organs at once, in brain, bowel, liver and kidney; occlusion involves one organ.

· Cerebral lesion shape: global hypoperfusion gives narrow strips along the border zones, and bilateral anterior to middle cerebral strips weaken both proximal arms with the face, hands and legs spared; occlusion gives a wedge filling one cortical territory with face and arm or leg weakness according to the artery.

· Bowel wall: hypoperfusion gives mucosal or mural infarction, because the villus tips are hypoxic first; occlusion gives transmural infarction.

· Causes: global hypoperfusion from shock, cardiac arrest, or hypotension during surgery; occlusion from a cardiac embolus, local thrombosis on a plaque, or vasculitis.

Distinctions that matter

· Pressure is adequate in occlusion and inadequate in global hypoperfusion, which is why one produces a single wedge and the other produces thin bilateral strips.

· The mucosa fails first in hypoperfusion because countercurrent exchange in the villi leaves the tips hypoxic, so depth of injury reports mechanism.

Common misconceptions

· Describing a watershed infarct as a small stroke of the same kind as an embolic one. Its position at the far edge of two territories is what names it.

· Expecting one mechanism to explain findings in four organs when the arteries involved are unconnected. Several organs at once means the pressure fell.

Where it is taught: Section C05, flow diagram, border zone table, colonic paragraph, and the final paragraph on embolus against global hypoperfusion of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C06-BD1

Brain dump 1 · foundational retrieval: Build the flow and extraction table from memory

A competent response contains

· Gut and liver: about 25% of resting cardiac output, extraction about 15 to 20%, flow serving absorption and processing.

· Kidneys: about 20% of output, extraction about 7 to 10%, flow serving filtration.

· Resting muscle: about 15% of output, extraction near the body average at rest and 75 to 90% in exercise, flow serving metabolism.

· Brain: about 15% of output, extraction about 30 to 35%, flow serving a constant demand.

· Heart: about 4 to 5% of output, extraction about 70 to 75%, flow serving metabolism.

· Skin: about 5 to 6% of output, very low extraction, flow serving heat loss.

· Ranked by share: gut and liver, kidneys, then resting muscle and brain, then skin, then heart. Ranked by extraction: heart, brain, gut and liver, kidneys, skin.

· The rankings differ because flow does not simply follow oxygen need: the kidneys and gut use blood for filtration and processing rather than for oxygen, while the heart takes a small share and strips most of the oxygen from it.

· Systemic organs are perfused in parallel at the same aortic pressure, so each organ's own arteriolar resistance sets its share of cardiac output.

· Arterial blood carries about 20 mL O2/dL. For the whole body, 5 L/min with an arteriovenous difference of 5 mL/dL gives about 250 mL O2/min, extraction about 25%, and mixed venous saturation about 75% with a PO2 of about 40 mm Hg.

· Fick principle: oxygen consumption equals blood flow times the arteriovenous oxygen difference; extraction is that difference divided by arterial content.

· Low extraction means reserve, because the organ can take more oxygen from the blood it already receives; high extraction means extra oxygen can come only from extra flow.

Distinctions that matter

· Share of cardiac output and oxygen extraction are independent figures, and the heart sits at opposite ends of the two rankings.

· Flow per gram is a third separate figure: the kidneys have the highest flow per gram of any large organ while resting muscle has one of the lowest.

Common misconceptions

· Assuming the organ with the highest flow uses the most oxygen. Renal flow serves filtration, and renal venous blood stays highly saturated.

· Treating organs as though they were in series. Parallel perfusion at a common pressure is what lets each organ set its own share.

Where it is taught: Section C06, opening paragraphs, the table of resting flow and O2 extraction, and the paragraph on individual organs of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C06-BD2

Brain dump 2 · integration and application: Work the numbers, then explain two organs under strain

A competent response contains

· Kidney: arteriovenous difference is 20 minus 18.5, which is 1.5 mL/dL; consumption is 11 times 1.5, which is 16.5 mL O2/min; extraction is 1.5 divided by 20, which is 7.5%.

· Heart: arteriovenous difference is 20 minus 6, which is 14 mL/dL; consumption is 2.25 times 14, which is 31.5 mL O2/min; extraction is 14 divided by 20, which is 70%.

· The kidneys receive about five times the coronary flow and use about half as much oxygen, because renal flow serves filtration of plasma while coronary flow serves the metabolism of working muscle.

· If demand doubles, the kidney can double extraction from 7.5% to 15% without any change in flow.

· The heart would need about 140% extraction, which is impossible because it cannot remove more oxygen than the blood delivers, so coronary flow must roughly double instead.

· Coronary flow rises through metabolic vasodilation, mediated by adenosine and nitric oxide, which is why a fixed stenosis that cannot allow that rise produces ischemia.

· The brain is about 2% of body weight, uses about 20% of resting oxygen, has no oxygen store, and loses consciousness within about 5 to 10 seconds of no flow.

· A rise in arterial PCO2 dilates cerebral arterioles and raises flow; hyperventilation lowers PCO2 and constricts them.

· The kidneys filter about 20% of their plasma, so their venous blood stays highly saturated, and their oxygen use tracks the filtered sodium load.

· The coronary sinus is only 25 to 30% saturated at rest, which is the same statement as a resting extraction of about 70 to 75%.

Distinctions that matter

· Extraction has a ceiling and flow does not, which is why a high extraction organ depends on vasodilation while a low extraction organ has reserve in the blood it already receives.

· Oxygen consumption and blood flow are separate quantities linked only through the arteriovenous difference, which is exactly what the Fick principle states.

Common misconceptions

· Reading high renal flow as high renal oxygen use. Renal venous blood returns nearly fully saturated.

· Expecting the heart to meet extra demand by extracting more. At about 70% extraction there is almost nothing left to take.

Where it is taught: Section C06, worked example using the Fick principle, the extraction table, and the paragraph on individual organs of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C06-PR01

Prediction exercise: Predict the redistribution of cardiac output in hemorrhage

A competent response contains

· Arterial pressure falls and baroreceptor firing falls with it, because the stretch receptors in the carotid sinus and aortic arch are less distended.

· Sympathetic outflow rises, acting on alpha-1 receptors on arteriolar smooth muscle.

· Flow falls through the skin, the splanchnic bed, the kidneys and resting muscle, because alpha-1 mediated constriction raises resistance in those beds.

· Venoconstriction moves blood out of the splanchnic reservoir, the main blood reservoir of the body, toward the heart, which supports venous return.

· Cerebral and coronary arterioles stay dilated, because local metabolic control rather than sympathetic drive sets their diameter.

· The share of cardiac output reaching the brain and heart rises, because those beds keep their resistance low while the others raise theirs.

· Oxygen extraction rises in the constricted beds, since they must take more oxygen from less flow; the kidneys, skin and splanchnic bed have the reserve to do this because their resting extraction is low.

· If low flow continues, those organs are injured: the renal outer medulla gives acute tubular necrosis, liver zone 3 gives centrilobular necrosis, and the colonic border zones give ischemic colitis.

Distinctions that matter

· Low extraction beds are the ones that can be constricted in shock, while the heart depends on flow, so resting extraction predicts which beds are sacrificed.

· Venoconstriction changes venous return by moving stored blood, while arteriolar constriction changes the distribution of whatever output there is.

Common misconceptions

· Expecting sympathetic drive to constrict every bed. Cerebral and coronary arterioles remain under local control, which is how the share shifts toward them.

· Treating the redistribution as harmless because the brain and heart are protected. The constricted beds tolerate only a brief fall, and prolonged low flow injures them.

Where it is taught: Section C06, flow diagram on redistribution of cardiac output in hemorrhage, the extraction table, and the border zone table in C05 of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C01-Q01

Board-style question: Weight loss and vomiting with an aortic landmark

Correct answer: C. Superior mesenteric artery at the level of the first lumbar vertebra

Why it is right: The superior mesenteric artery leaves the aorta at L1 and crosses in front of the third part of the duodenum, so loss of retroperitoneal fat narrows the aortomesenteric angle and compresses the duodenum, with relief when position changes take weight off that angle.

A is wrong: The celiac trunk arises at T12 and supplies the foregut as far as the major papilla; it runs above the duodenum and does not cross its third part.

B is wrong: The inferior mesenteric artery arises at L3 and supplies the hindgut from the distal third of the transverse colon to the anal canal, well below and beyond the duodenum.

D is wrong: The renal arteries arise at about L1 and run laterally to the kidneys rather than across the midline in front of the duodenum.

A competent response contains

· The superior mesenteric artery arises from the aorta at L1 and passes in front of the third part of the duodenum.

· Loss of retroperitoneal fat narrows the angle between the aorta and the superior mesenteric artery, which can obstruct the duodenum, and compression of the left renal vein in the same angle is the nutcracker phenomenon.

Distinctions that matter

· The third part of the duodenum and the left renal vein both cross in the aortomesenteric angle, so one narrowed angle explains two different presentations.

Common misconceptions

· Placing the superior mesenteric artery at T12 with the celiac trunk. The celiac trunk is at T12 and the superior mesenteric artery is at L1.

Where it is taught: Section C01 of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C02-Q02

Board-style question: Dizziness while working with one arm raised

Correct answer: A. Reversal of flow down the left vertebral artery into the dilated arteries of the exercising arm

Why it is right: Arm exercise dilates the arterioles of the arm and lowers pressure further in a subclavian artery already narrowed proximal to the vertebral origin, so flow reverses down the left vertebral artery into the arm and the posterior circulation is drained, which is why vertigo and imbalance appear only during arm use.

B is wrong: Carotid embolization to the middle cerebral artery would produce a sudden fixed deficit of face and arm weakness or language loss on one side, not symptoms that come only while the arm is working.

C is wrong: A cervical rib compressing the subclavian artery between the scalenes affects the arm and the brachial plexus and does not drain the posterior circulation, so it would not cause vertigo.

D is wrong: Occlusion of the posterior inferior cerebellar artery gives a fixed lateral medullary syndrome with dysphagia, hoarseness and crossed sensory loss, and examination between episodes would not be normal.

A competent response contains

· A systolic difference of 30 mm Hg between the arms with a weak delayed pulse on the low side places a stenosis in that subclavian artery.

· The vertebral artery is the first branch of the subclavian artery, so a stenosis proximal to its origin leaves vertebral pressure below basilar pressure and arm exercise reverses flow down the vertebral artery into the arm.

· The posterior circulation loses blood during arm use, giving vertigo and imbalance alongside arm claudication.

Distinctions that matter

· A stenosis proximal to the vertebral origin steals from the posterior circulation, while one distal to it causes arm claudication alone.

Common misconceptions

· Reading dizziness with arm use as a carotid problem. The symptoms are posterior circulation symptoms and they follow the arm, which is what points to the subclavian artery.

Where it is taught: Section C02 of the CV-9 Blood Distribution to Body Regions notes.

NPLEX-P1-CARD-09-01-C05-Q03

Board-style question: Weak arms after a prolonged fall in blood pressure

Correct answer: D. Perfusion pressure at the distal ends of the anterior and middle cerebral territories falling below the autoregulatory range

Why it is right: Pressure falls along every arterial tree, so the distal ends of the anterior and middle cerebral territories sit at the lowest perfusion pressure; once the systemic pressure drops below the autoregulatory range these border zones infarct as narrow strips over the convexity, sparing the face, hands and legs and leaving proximal arm weakness.

A is wrong: Emboli large enough to block both anterior cerebral arteries would infarct the whole medial surface of both hemispheres and weaken the legs more than the arms.

B is wrong: Sagittal sinus thrombosis causes venous congestion with headache, seizures and hemorrhagic infarction rather than narrow symmetrical arterial border strips after an arrest.

C is wrong: Lenticulostriate arteriolosclerosis produces deep lacunar infarcts in the internal capsule with a pure motor hemiparesis on one side, not bilateral cortical strips.

A competent response contains

· After prolonged cardiac arrest, arterioles are maximally dilated and flow falls in proportion to pressure, so perfusion fails first where pressure is lowest.

· The border between the anterior and middle cerebral territories lies over the upper convexity, and bilateral infarction there weakens both proximal arms with the face, hands and legs spared.

Distinctions that matter

· Strips along a border, present on both sides, point to a fall in pressure; a wedge filling one territory points to occlusion of that artery.

Common misconceptions

· Treating bilateral symmetrical infarcts as two separate emboli. Symmetry at the far edge of two territories is the signature of global hypoperfusion.

Where it is taught: Section C05 of the CV-9 Blood Distribution to Body Regions notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-10

CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-10-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the lymphatic capillary and how lymph forms

A competent response contains

· Lymphatic capillaries, also called initial lymphatics, begin as blind-ended tubes near blood capillary beds.

· The endothelium is thin, the basal lamina is incomplete or absent, and there are no pericytes and no smooth muscle.

· Neighbouring endothelial cells overlap like roof shingles, held by loose discontinuous junctions, so the free inner edge of each overlap swings inward as a flap, the primary lymphatic valve.

· Anchoring filaments tie the outer surface of the endothelium to the collagen and elastic fibres of the surrounding tissue.

· Outward Starling forces are capillary hydrostatic pressure and interstitial oncotic pressure; inward forces are plasma oncotic pressure and interstitial hydrostatic pressure.

· The outward forces slightly exceed the inward ones, so a small net volume is filtered, and some albumin escapes with it that cannot be moved back into blood against its concentration gradient.

· Steps: net capillary filtration adds fluid and protein to the interstitium; the tissue swells and anchoring filaments pull the lymphatic capillary open; the overlapping flaps part and interstitial fluid enters; rising pressure inside the vessel closes the flaps; lymph moves on into valved collecting lymphatics.

· The vessel opens because swelling tissue pulls the anchoring filaments taut, so the capillary is pulled open rather than compressed; it closes because internal pressure presses the flaps shut, which makes flow one way.

· The gaps admit protein, lipoprotein, bacteria, and whole cells, which is the opposite of a continuous blood capillary built to retain them.

Distinctions that matter

· The flap valve of the lymphatic capillary is formed by overlapping endothelial cells; the bicuspid valves further along belong to collecting lymphatics.

· Tissue swelling opens a lymphatic capillary and compresses a blood capillary, and the anchoring filaments are the reason for the difference.

Common misconceptions

· Drawing a continuous basal lamina and tight junctions on a lymphatic capillary. Both are what blood capillaries use to keep protein in, and the lymphatic needs protein to get out.

· Saying filtration happens because plasma oncotic pressure is low. Plasma oncotic pressure is the main inward force, and net filtration is small because the outward forces exceed it only slightly.

Where it is taught: Section C01, opening paragraph, structure of the lymphatic capillary, and the formation of lymph flow diagram of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C01-BD2

Brain dump 2 · integration and application: What stops working when protein cannot get back

A competent response contains

· Protein leaves the interstitium only by lymph, because no mechanism moves it back into blood against its concentration gradient, so a blocked drain leaves protein behind.

· Interstitial protein rises, interstitial oncotic pressure rises toward plasma levels, the gradient that holds fluid inside vessels is erased, and net filtration increases, so the limb swells even with normal capillary pressure.

· Lymph protein is about 2 g/dL from most tissues, about 6 g/dL from the liver, and about 3 to 4 g/dL from the intestine.

· Hepatic lymph is the most protein-rich because hepatic sinusoids leak albumin freely.

· A duct draining externally loses dietary long-chain fat and fat-soluble vitamins, albumin, and lymphocytes.

· Enterocytes package long-chain fatty acids as triglyceride in chylomicrons, which are too large for blood capillaries, so they enter the lacteal of each villus and make intestinal lymph milky, which is chyle.

· Most dietary fat therefore reaches blood through the thoracic duct and bypasses the liver on first pass, while medium-chain fatty acids are small enough to enter portal blood.

· No lymphatic capillaries are present in epidermis, cartilage, cornea, bone marrow, or brain and spinal cord parenchyma, which drain along perivascular routes to meningeal lymphatics.

Distinctions that matter

· Lymphedema fails at the drain while the other kinds of edema push too much fluid out, which is why the fluid here is protein-rich.

· Chylomicron size, not the type of fatty acid alone, decides which route absorbed fat takes.

Common misconceptions

· Saying macrophages or the venous end of the capillary clear interstitial protein. The lymphatic is the only route back.

· Expecting all absorbed fat to enter the portal vein. Long-chain fat in chylomicrons reaches the blood at the neck veins.

Where it is taught: Section C01, paragraphs on what the lymphatics return, and the final paragraph on tissues without lymphatics of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C01-RT01

Retrieval exercise: Rebuild the steps that form lymph

A competent response contains

· Step 1: net capillary filtration adds fluid and protein to the interstitium.

· Step 2: the tissue swells and the anchoring filaments pull the lymphatic capillary open.

· Step 3: the overlapping endothelial flaps part and interstitial fluid enters.

· Step 4: rising pressure inside the vessel closes the flaps.

· Step 5: lymph moves on into valved collecting lymphatics.

· The anchoring filaments pull outward on the vessel wall as the surrounding tissue swells, so the vessel is held open instead of being squeezed shut.

· The flaps close because pressure inside the vessel now exceeds interstitial pressure and presses them against the overlap, so flow can only go forward.

· Resting lymph flow is about 120 mL per hour, about 100 mL of it through the thoracic duct, which is 2 to 3 L per day.

· That daily volume matches net filtration because a net filtration pressure of about 0.3 mmHg acting on a large filtration coefficient filters about 2 mL per minute, and whatever is not reabsorbed must leave as lymph.

Distinctions that matter

· Opening the vessel depends on forces outside it; closing it depends on pressure inside it.

· Flow through the thoracic duct is most of total lymph flow but not all of it, since the right lymphatic duct carries the rest.

Common misconceptions

· Drawing the tissue squeezing the lymphatic capillary closed as it swells. The anchoring filaments make swelling open it.

Where it is taught: Section C01, formation of lymph flow diagram and the paragraph on what the lymphatics return of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the thoracic duct from end to end

A competent response contains

· The cisterna chyli is a variable sac anterior to about L1 to L2 beside the aorta, formed where the right and left lumbar trunks and the intestinal trunk join.

· Lumbar trunks carry lymph from the lower limbs, pelvis, abdominal wall, kidneys, adrenals, and gonads after the iliac and lumbar nodes; the intestinal trunk carries lymph including chyle from the stomach, intestines, liver, pancreas, and spleen after the preaortic nodes.

· The duct ascends with the aorta through the aortic hiatus at about T12.

· In the posterior mediastinum it lies on the vertebral bodies behind the esophagus, with the aorta on its left and the azygos vein on its right.

· At about T4 to T6 it crosses to the left behind the esophagus and ascends along the left edge of the esophagus.

· At about C7 it arches laterally behind the left carotid sheath and ends at the left venous angle, the junction of the left internal jugular and left subclavian veins, where a terminal valve keeps blood out.

· Near its end it receives the left jugular, left subclavian, and often the left bronchomediastinal trunks.

· The right lymphatic duct, about 1 cm long, is formed by the right jugular, subclavian, and bronchomediastinal trunks, which may instead open into the veins separately, and ends at the right venous angle.

· It drains only the right head and neck, right upper limb, and right thorax including the right breast and right lung; the thoracic duct drains everything else.

· Much of the left lower lobe of the lung drains against that pattern, to right tracheobronchial nodes.

· The course and termination of both ducts vary between individuals.

Distinctions that matter

· The duct begins on the right at the cisterna chyli and ends on the left at the venous angle, and the crossing at about T4 to T6 is where that changes.

· Only four regions use the right lymphatic duct, so anything outside them returns by the thoracic duct.

Common misconceptions

· Drawing the duct in front of the esophagus. It lies behind the esophagus on the vertebral bodies.

· Expecting a right lymphatic duct in every body. The three right trunks often open into the veins separately.

Where it is taught: Section C02, paragraphs on trunks and the cisterna chyli, the course of the thoracic duct, and the right lymphatic duct of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-BD2

Brain dump 2 · integration and application: Work backward from the node to the primary site

A competent response contains

· Left supraclavicular node: abdominal lymph ascends the thoracic duct to the left venous angle, so gastric carcinoma can appear there first, the Virchow node.

· Lumbar or para-aortic nodes near the renal vessels: testis or ovary, so a testicular or ovarian tumour.

· Superficial inguinal nodes: scrotum, vulva, anal canal below the pectinate line, and superficial lower limb, so a scrotal or vulval skin lesion or a low anal carcinoma.

· Axillary nodes: upper limb and lateral breast, which is most breast lymph, so a breast carcinoma or an arm infection.

· Popliteal nodes: lateral foot and posterolateral calf, draining onward to deep inguinal nodes, so a lesion of the lateral foot.

· The testes descend from the posterior abdominal wall and bring their lymphatics with them, so testicular cancer goes to para-aortic nodes, while scrotal skin is body wall and drains to superficial inguinal nodes.

· The anal canal above the pectinate line is hindgut and drains to internal iliac nodes, with the upper rectum going to inferior mesenteric nodes; below the pectinate line it is ectoderm and drains to superficial inguinal nodes.

· Most of the breast drains to axillary nodes and the medial part drains to parasternal nodes along the internal thoracic vessels.

· The medial hand and forearm drain first to cubital, also called supratrochlear, nodes before the axilla, so an elbow node can enlarge first.

· Each region drains to a predictable first node group, which is where infection causes tender lymphadenitis and where carcinoma usually metastasizes first.

Distinctions that matter

· Skin over an organ and the organ itself often drain to different node groups, which is the point of the scrotum against testis comparison.

· The pectinate line marks a change in embryological origin, and the node groups follow that origin rather than the position in the canal.

Common misconceptions

· Assuming a testicular tumour spreads to inguinal nodes because the testis sits in the scrotum. Lymphatics follow the path of descent, not the final position.

· Sending all breast lymph to the axilla. The medial part goes to parasternal nodes.

Where it is taught: Section C02, first node groups table and the paragraph on development and the classic findings of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-PR01

Prediction exercise: Predict the effect of a block low in the thoracic duct

A competent response contains

· Lymph outflow from both lower limbs and the pelvis falls, because their lumbar trunks drain into the cisterna chyli below the block and have no other route onward, so lower body swelling can follow.

· Lymph outflow from the abdominal viscera, including intestinal chyle, falls for the same reason, and chylous ascites can follow because the obstructed chyle has nowhere to go.

· The left upper limb, left head and neck, and left thorax still drain normally, because the left jugular, left subclavian, and left bronchomediastinal trunks join the duct near its end, above the block.

· The right head and neck, right upper limb, and right thorax still drain normally through the right lymphatic duct, which is a separate route to the right venous angle.

· Pressure inside the duct below the obstruction rises, and the intrinsic pump can generate about 50 to 100 mmHg against an obstruction.

· The obstructed vessels dilate, so the valve leaflets no longer meet, lymph refluxes between contractions, and forward flow falls even though the muscle is still contracting.

Distinctions that matter

· Where a trunk joins the duct relative to the block decides whether its region is affected, so the level of the block matters more than the size of the region.

· A high pump pressure and a failing pump are not a contradiction: pressure rises first, and valve incompetence from dilation is what finally defeats it.

Common misconceptions

· Expecting the whole body to stop draining because the thoracic duct carries most lymph. Everything joining above the block, and everything using the right duct, is unaffected.

· Assuming an obstructed lymphatic simply stops. It pumps harder first, and fails when dilation separates the valve leaflets.

Where it is taught: Section C02, stop and predict box, with the valve and intrinsic pump paragraph in Section C03 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C03-BD1

Brain dump 1 · foundational retrieval: Lay out the whole lymphatic pump

A competent response contains

· Lymph begins in an interstitium slightly below atmospheric pressure, about -3 mmHg in subcutaneous tissue, and ends in veins a few mmHg above atmospheric, in the legs against gravity.

· Valves allow flow in only one direction, so any compression of a valved segment pushes lymph toward the veins rather than back and forth.

· The flap valves of the lymphatic capillaries are the first set; collecting lymphatics carry bicuspid endothelial valves every few millimetres, and the segment between two valves is a lymphangion.

· Cycle: lymph enters through the open upstream valve; filling stretches the smooth muscle of the wall; stretch triggers contraction and pressure rises; the upstream valve closes and the downstream valve opens; lymph is ejected and stretches the next lymphangion; the emptied segment relaxes and refills from upstream.

· Stretch from filling increases the rate and force of contraction, so the pump works harder when more lymph arrives, much as stretch increases cardiac stroke volume through the Frank-Starling mechanism.

· Sympathetic stimulation increases the activity, endothelial nitric oxide relaxes the muscle between contractions, and inflammation depresses it.

· Extrinsic pumps: skeletal muscle contraction, passive movement, pulsation of neighbouring arteries since deep lymphatics share arterial sheaths, and external compression.

· Exercise can raise lymph flow about 10 to 30 times, and immobility lowers it.

· During inspiration the descending diaphragm raises abdominal pressure around the cisterna chyli while intrapleural pressure falls around the thoracic duct, so lymph is drawn upward, and valves block backflow during expiration.

Distinctions that matter

· The intrinsic pump is the lymphangion's own smooth muscle; the extrinsic pumps are everything outside the vessel that compresses it.

· Valves do not move lymph, they decide the direction of whatever force is applied.

Common misconceptions

· Looking for a central pump for lymph. There is none, and the lymphangions plus external compression do the work.

· Treating breathing as a thoracic effect only. It works because abdominal pressure rises while intrathoracic pressure falls at the same moment.

Where it is taught: Section C03, opening paragraph, valves and the intrinsic pump, the lymphangion pump cycle, and extrinsic pumps of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C03-BD2

Brain dump 2 · integration and application: Why the pump runs out of room

A competent response contains

· Subcutaneous interstitial fluid pressure is normally about -3 mmHg, largely because the lymphatics keep pumping fluid out.

· As interstitial pressure rises from about -6 mmHg to slightly above 0 mmHg, lymph flow increases about twentyfold.

· Above about 1 to 2 mmHg flow plateaus, because the tissue pressure that fills the lymphatic capillaries also compresses the larger collecting vessels.

· Once filtration exceeds that maximum flow, edema accumulates.

· Safety factor one: low compliance of the negative-pressure interstitium, so a small fluid gain raises interstitial pressure steeply, worth about 3 mmHg.

· Safety factor two: lymph flow can rise 10 to 50 times, worth about 7 mmHg.

· Safety factor three: that extra flow washes protein out faster than it leaks in, lowering interstitial oncotic pressure, worth about 7 mmHg, and it depends on the second factor existing.

· Together they let capillary pressure rise about 17 mmHg before significant edema appears.

· Removing the regional nodes and vessels loses the second and third factors at once, so leaking protein accumulates and interstitial oncotic pressure rises instead of falling, and the tissue gains protein-rich fluid even with a normal capillary pressure.

· Lymph finally empties against central venous pressure, so right heart failure and superior vena cava obstruction both slow lymph outflow and raise capillary filtration at the same time, which contributes to effusions and ascites.

Distinctions that matter

· The same rise in tissue pressure that fills the initial lymphatics compresses the collecting vessels, which is why the curve has a ceiling rather than rising without limit.

· The third safety factor is a consequence of the second, so losing lymph flow removes both at once.

Common misconceptions

· Expecting lymph flow to keep rising with tissue pressure. It plateaus at about 1 to 2 mmHg, and edema is what happens past that point.

· Treating the safety factors as independent. Protein washout exists only while lymph flow can increase.

Where it is taught: Section C03, paragraph on interstitial pressure and outflow pressure, and the worked example on safety factors of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C03-RT01

Retrieval exercise: Rebuild the lymphangion pump cycle

A competent response contains

· Step 1: lymph enters through the open upstream valve.

· Step 2: filling stretches the smooth muscle of the wall.

· Step 3: stretch triggers contraction and pressure rises.

· Step 4: the upstream valve closes and the downstream valve opens.

· Step 5: lymph is ejected and stretches the next lymphangion.

· Step 6: the emptied segment relaxes and refills from upstream.

· The contraction was triggered by stretch, and the stretch came from lymph filling the segment, so output rises with filling.

· The ejected lymph passes into the next lymphangion downstream and stretches it, which triggers that segment to contract in turn, so contraction runs in sequence toward the veins.

· An obstructed thoracic duct can reach about 50 to 100 mmHg.

· If a chronically obstructed vessel dilates, the valve leaflets no longer meet, lymph refluxes between contractions, and the pump fails.

Distinctions that matter

· Only one valve of a lymphangion is open at any step, which is what makes the segment a pump rather than a tube.

· Pump failure in chronic obstruction comes from valve incompetence after dilation, not from weak muscle.

Common misconceptions

· Drawing both valves open during contraction. The upstream valve must close before lymph can be ejected forward.

· Assuming a dilated lymphatic moves more lymph because it holds more. Dilation separates the leaflets and lets lymph run backward.

Where it is taught: Section C03, the lymphangion pump cycle and the valves and intrinsic pump paragraph of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C04-BD1

Brain dump 1 · foundational retrieval: Build the lymphoid organs from their compartments

A competent response contains

· Primary lymphoid organs are bone marrow, where B cells mature, and thymus, where T cells mature; lymphocytes there generate antigen receptors and are tested against self.

· Secondary lymphoid organs are lymph nodes, which sample lymph, spleen, which samples blood, and mucosa-associated lymphoid tissue, which samples mucosal surfaces; naive lymphocytes meet antigen there and recirculating lymphocytes pass through each.

· Lymph node: afferent lymph enters the subcapsular sinus, percolates through macrophage-lined sinuses, and leaves at the hilum through efferent vessels; B cells sit in cortical follicles drawn by CXCL13 acting on CXCR5 around follicular dendritic cells displaying intact antigen; germinal centres hold proliferating B cells where activation-induced cytidine deaminase drives somatic hypermutation and class switching and higher-affinity cells are selected by antigen and follicular helper T cells; the paracortex holds T cells, dendritic cells, and high endothelial venules through which naive lymphocytes enter using L-selectin and CCR7; medullary cords hold plasma cells secreting antibody.

· Follicular hyperplasia reflects a B-cell response and paracortical hyperplasia a T-cell response, often viral.

· Spleen: white pulp has T cells sheathing the central arteriole in the periarteriolar lymphoid sheath beside B-cell follicles, with a marginal zone that traps blood-borne antigen; red pulp is where blood empties into splenic cords and must squeeze through narrow slits between sinusoidal endothelial cells, so stiff, aged, or antibody-coated red cells are removed and inclusions are pitted; the red pulp holds about one third of platelets; the spleen has no afferent lymphatics.

· Thymus: lies in the anterior superior mediastinum, develops from third pharyngeal pouch endoderm, and involutes into fat after puberty; each lobule has a dark cortex packed with thymocytes behind a blood-thymus barrier and a pale medulla containing Hassall corpuscles, which are whorls of keratinized epithelial cells; it has no follicles and no afferent lymphatics.

· MALT: Peyer patches, largest in the ileum, plus appendix, bronchial tissue, and tonsils, which have a partial capsule; it is unencapsulated; M cells transcytose luminal antigen inward to the underlying lymphoid cells; B cells there switch to IgA under TGF-beta, home back to mucosa through alpha4 beta7 integrin binding MAdCAM-1, and secrete dimeric IgA carried across the epithelium by the polymeric immunoglobulin receptor.

· T-cell maturation: a marrow precursor enters near the corticomedullary junction; a double-negative cell rearranges TCR beta; pre-TCR signalling makes it double positive; TCR alpha rearrangement completes the alpha beta receptor; in the cortex weak binding to self MHC survives while no binding dies by neglect; class II binding keeps CD4 and class I binding keeps CD8; in the medulla strong binding to AIRE-driven self antigen causes deletion or a regulatory fate; naive, MHC-restricted, self-tolerant T cells leave.

· The first six steps are cortical, the strong-binding deletion step is medullary, and cortical positive selection makes T cells useful while medullary negative selection makes them safe.

· In the marrow, pro-B cells rearrange heavy chain genes with RAG1 and RAG2, a functional heavy chain with a surrogate light chain forms the pre-B cell receptor, and after light chain rearrangement an immature B cell whose IgM binds self strongly undergoes receptor editing and then deletion if editing fails; survivors mature in the spleen into naive B cells bearing IgM and IgD.

Distinctions that matter

· Nodes have afferent lymphatics and spleen and thymus do not, which is why nodes sample lymph while the spleen samples blood.

· Positive selection is about usefulness and happens in the cortex; negative selection is about safety and happens in the medulla.

Common misconceptions

· Drawing follicles in the thymus. The thymus has cortex and medulla and no follicles at all.

· Placing plasma cells in the germinal centre. Antibody-secreting plasma cells sit in the medullary cords.

Where it is taught: Section C04, opening paragraphs, the thymus paragraph, the T-cell maturation flow diagram, the node, spleen, and MALT paragraphs, and the compartments table of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C04-BD2

Brain dump 2 · integration and application: Read a biopsy and a blood film back to a compartment

A competent response contains

· Child one has follicles and germinal centres but an empty paracortex, so T cells are missing; the spleen's periarteriolar lymphoid sheath would also be empty while its follicles and marginal zone are intact; this fits DiGeorge syndrome from the 22q11.2 deletion with a hypoplastic thymus.

· Child two has a normal paracortex but no follicles, no germinal centres, and no plasma cells in the medullary cords, so B cells are missing; the spleen's follicles and marginal zone would be empty while the periarteriolar sheath is intact; this fits X-linked, also called Bruton, agammaglobulinemia.

· Serum antibody of every class is low in child two because the plasma cells of the medullary cords are the source of secreted antibody, and without B cells there are no plasma cells to make any isotype.

· Child three has an enlarged paracortex with normal follicles, which is paracortical hyperplasia, a T-cell response and usually viral.

· In the thymic cortex a T-cell receptor must bind self peptide and self MHC weakly to survive, since no binding dies by neglect; in the medulla it must avoid binding AIRE-driven self antigen strongly, since strong binding causes deletion or diversion to regulatory T cells.

· Child one has lost both of those processes, and the organ in which they are lost is the thymus.

· After splenectomy: Howell-Jolly bodies, thrombocytosis, and overwhelming sepsis from encapsulated bacteria.

· The three organisms are Streptococcus pneumoniae, Haemophilus influenzae type b, and Neisseria meningitidis.

· The red cell findings appear because red pulp cords and the narrow sinusoidal slits are where inclusions are pitted and aged or coated cells are culled, and thrombocytosis follows because the red pulp normally holds about one third of platelets.

· The infection risk follows because encapsulated organisms need splenic clearance and marginal zone IgM, which is T-independent antibody against polysaccharide.

· AIRE mutation in the thymic medulla causes autoimmune polyendocrine syndrome type 1, with hypoparathyroidism, adrenal insufficiency, and chronic mucocutaneous candidiasis.

Distinctions that matter

· An immunodeficiency that removes one cell type empties that cell's compartment in every lymphoid organ, so reading a node tells you what the spleen will look like.

· Hyperplasia of a compartment and depletion of a compartment both point to the same cell line, one responding and one absent.

Common misconceptions

· Reading absent germinal centres as a T-cell problem because follicular helper T cells are needed for them. Here the paracortex and T-cell numbers are normal and the B cells themselves are missing.

· Attributing post-splenectomy infection risk to loss of all antibody. The gap is in clearance of encapsulated organisms and in marginal zone IgM against polysaccharide.

Where it is taught: Section C04, the node paragraph, the spleen paragraph, the thymus paragraph, the T-cell maturation flow diagram, and the compartments table of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-BD1

Brain dump 1 · foundational retrieval: Causes of lymphedema and how it hardens

A competent response contains

· Too much filtration produces edema through raised capillary pressure, low plasma oncotic pressure, or increased capillary permeability.

· In lymphedema filtration is normal and the drain fails; because protein leaves the interstitium only by lymph, the retained fluid is rich in protein, and secondary causes far outnumber primary ones.

· Lymph node dissection removes nodes and the vessels running through them: arm after axillary dissection, leg after pelvic or inguinal dissection.

· Radiation causes fibrosis that narrows and obliterates vessels, often years later, in axilla, pelvis, or groin, especially alongside surgery.

· Tumour blocks vessels and nodes with tumour emboli, in metastatic carcinoma and lymphoma, and gives peau d'orange over a breast carcinoma.

· Filariasis: adult worms dilate lymphatics and make valves incompetent, and inflammation and infection scar them, affecting legs, scrotum, and breast in endemic tropical regions.

· Primary developmental causes leave too few or malfunctioning vessels or valves: Milroy disease from VEGFR3 mutation, FOXC2 mutation, and Turner syndrome.

· The organisms are Wuchereria bancrofti and Brugia species; adult worms live in the lymphatics for years while microfilariae circulate in blood, typically at night.

· Vessel damage comes from living worms dilating the lymphatics, from inflammation provoked by dying worms and their Wolbachia endosymbionts, and from repeated bacterial infection adding scarring; the result is leg lymphedema progressing to elephantiasis, hydrocele, and sometimes chyluria.

· Cycle: lymphatic outflow falls; protein-rich fluid accumulates and holds water; stagnant protein and lipid provoke chronic inflammation; fibroblasts deposit collagen and fat expands; fibrosis damages the remaining lymphatics and cellulitis recurs; each infection and increment of fibrosis reduces outflow further.

· At the fibrosis step, chronic inflammation with CD4 T cells and macrophages leads TGF-beta to stimulate fibroblasts to lay down collagen, and lymph stasis promotes fat growth.

· Early lymphedema is soft and pitting and improves with elevation, although its high protein content holds water so swelling returns when the limb hangs down; later, fibrosis and fat do not move under pressure, so the limb is firm and nonpitting and no longer shrinks with elevation, and the skin thickens with hyperkeratosis and papillomatous change.

· Inability to pinch a skin fold at the base of the second toe is the Stemmer sign.

· Lost immune cell traffic and protein-rich fluid favour recurrent streptococcal cellulitis, and lymphangiosarcoma is a rare late complication, classically in the arm years after mastectomy with axillary dissection.

Distinctions that matter

· Every cause of lymphedema removes, obstructs, or disables lymphatic pathways, and the downstream course is the same whichever one it was.

· Pitting against nonpitting here is a matter of time rather than of cause, because fibrosis and fat take months to years to develop.

Common misconceptions

· Explaining nonpitting late lymphedema by the protein content of the fluid. Protein holds water but still pits; what does not move under a finger is collagen and fat.

· Expecting filariasis to damage lymphatics only through the worms themselves. Inflammation from dying worms and repeated bacterial infection do much of the scarring.

Where it is taught: Section C05, opening paragraph, the causes of lymphedema table, the filariasis paragraph, the paragraph on pitting to nonpitting swelling, and the self-perpetuating cycle of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-BD2

Brain dump 2 · integration and application: Reason out a milky effusion

A competent response contains

· The thoracic duct has been injured, and a left-sided effusion places the injury above the crossing at about T4 to T6, because above the crossing the duct runs along the left edge of the esophagus while below it lies behind the esophagus to the right of midline.

· Pleural triglyceride above about 110 mg/dL supports chylothorax, below about 50 mg/dL makes it unlikely, and chylomicrons in the fluid confirm it.

· Fasting lymph is clear and scant, so the fluid was thin while the patient was not eating; eating fat sends chylomicrons through the duct and turns the fluid milky.

· Chyle stays turbid after centrifugation, which separates it from the fluid of an infected pleural space, where turbidity settles out.

· Pseudochylothorax, a chronic cholesterol effusion inside a thickened pleura, has cholesterol above about 200 mg/dL and no chylomicrons, so measuring cholesterol and looking for chylomicrons settles that comparison, and an acute onset argues against it as well.

· A continuing leak depletes albumin, lymphocytes, dietary fat, and fat-soluble vitamins, because the duct normally returns escaped plasma protein, carries recirculating lymphocytes from the nodes, and delivers absorbed long-chain fat as chylomicrons along with the vitamins dissolved in it.

· Surgical trauma and malignancy, especially lymphoma, are the leading causes of chylothorax.

Distinctions that matter

· The side of a chylous effusion locates the injury relative to the T4 to T6 crossing, which is the one piece of anatomy the question turns on.

· Turbidity that persists after centrifugation means suspended chylomicrons, while turbidity from cells and debris settles.

Common misconceptions

· Taking a milky effusion as proof of chyle without a triglyceride level. A chronic cholesterol effusion also looks milky and has no chylomicrons.

· Expecting a duct injury to leak the same fluid whatever the patient eats. Fat intake is what loads the duct with chylomicrons.

Where it is taught: Section C05, the chylothorax paragraph and the worked example on a milky effusion after esophagectomy of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-CP01

Comparison table: Three kinds of swelling that look alike

A competent response contains

· What accumulates: lymphedema holds protein-rich interstitial fluid that is later replaced by collagen and fat; heart failure or low albumin edema holds low-protein interstitial fluid; myxedema holds glycosaminoglycan rather than free fluid.

· Protein content: high in lymphedema because protein leaves only by lymph, low in the filtration edemas, and not applicable in myxedema since it is not free fluid.

· Mechanism: lymphedema is failure of the drain with normal filtration; heart failure raises capillary hydrostatic pressure and low albumin lowers plasma oncotic pressure, both raising filtration; myxedema is deposition in the tissue rather than a Starling problem.

· Pitting: lymphedema pits early and becomes nonpitting once fibrosis and fat develop, because those do not move under pressure; filtration edema stays pitting because it is free fluid; myxedema is nonpitting because it is glycosaminoglycan.

· Elevation: early lymphedema improves with elevation and returns when the limb hangs down because its protein holds water, late lymphedema does not improve at all, filtration edema improves with elevation, and myxedema does not change.

· Skin over time: in lymphedema the skin thickens with hyperkeratosis and papillomatous change and cellulitis recurs; in filtration edema the skin is stretched and shiny without that thickening; in myxedema the tissue is doughy and does not indent.

· Pointer: the Stemmer sign, inability to pinch a skin fold at the base of the second toe, points to lymphedema; a raised jugular venous pressure or a low serum albumin points to the filtration edemas; nonpitting swelling without a drainage history points to myxedema.

Distinctions that matter

· Two kinds of swelling here are nonpitting for different reasons: lymphedema because fibrosis and fat have replaced the fluid, myxedema because the material was never free fluid.

· Protein content is the single feature that separates lymphedema from every filtration edema, and everything that follows comes from it.

Common misconceptions

· Treating nonpitting as a single finding with one cause. It has to be read with the history and the skin.

· Calling lymphedema a high-pressure problem. Capillary pressure is normal and the drain has failed.

Where it is taught: Section C05, opening paragraph and the paragraph on pitting to nonpitting swelling of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C05-CA01

Clinical application: A firm, swollen arm four years after breast surgery

A competent response contains

· Recognize: secondary lymphedema of the right arm. The axillary lymph node dissection removed the nodes and the vessels running through them, and the axillary radiation caused fibrosis that narrowed and obliterated the remaining vessels, which is why the swelling began months after surgery and progressed over years.

· Explain: protein leaves the interstitium only by lymph, so a failed drain leaves protein behind and the retained fluid is protein-rich. Interstitial oncotic pressure then rises instead of falling, the gradient holding fluid inside vessels is eroded, and the tissue gains fluid even though capillary pressure, plasma albumin, and capillary permeability are all normal.

· Differentiate: heart failure edema is a raised capillary pressure problem, is low in protein, stays pitting, and would be bilateral with a raised jugular venous pressure, which is normal here; low albumin edema is also low in protein and pitting and would be generalized, and her albumin is 4.1 g/dL. This arm is unilateral, firm, nonpitting, with thickened skin and a positive Stemmer sign, on the side of the dissection and radiation.

· Investigate: the normal jugular venous pressure and normal heart sounds exclude a raised central venous pressure, the serum albumin of 4.1 g/dL excludes hypoalbuminemia, the normal left arm confirms a regional rather than a systemic cause, and the skin fold at the base of the second finger that cannot be pinched up is the Stemmer sign applied to the hand, which points to lymphatic failure.

· Decide: retained protein and lipid drove chronic inflammation with CD4 T cells and macrophages, TGF-beta stimulated fibroblasts to lay down collagen, and lymph stasis promoted fat growth. Collagen and fat now occupy the tissue, and neither moves under a finger nor drains with elevation.

· Justify: the safety factors against edema are a stiff negative-pressure interstitium worth about 3 mmHg, a 10 to 50 fold rise in lymph flow worth about 7 mmHg, and protein washout by that extra flow worth about 7 mmHg. Removing the nodes and vessels loses the second and third at once, so about 14 mmHg of the roughly 17 mmHg of protection is gone and interstitial protein rises rather than falling. Each episode of cellulitis then enters the cycle: lost immune cell traffic and protein-rich fluid invite streptococcal infection, infection and fibrosis damage the lymphatics that remain, outflow falls further, and the next episode starts from a worse baseline.

Distinctions that matter

· Surgery and radiation are separate insults with different timing, one immediate and one developing over years, and this history contains both.

· The change from pitting to nonpitting marks a change in what is in the tissue, not a change in how severe the drainage failure is.

Common misconceptions

· Reading a unilateral swollen arm as venous obstruction by default. The dissection and radiation history, the protein-rich picture, and the Stemmer sign point to the lymphatics.

· Treating recurrent cellulitis as a complication that sits outside the mechanism. It is part of the cycle that makes the lymphedema progress.

Where it is taught: Section C05, opening paragraph, the causes of lymphedema table, the paragraph on pitting to nonpitting swelling, and the self-perpetuating cycle, with the safety factors in Section C03 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C04-Q01

Board-style question: Which compartment the organ lacks

Correct answer: C. Arrest of B-cell maturation in the bone marrow

Why it is right: B cells populate the cortical follicles and germinal centres and become the plasma cells of the medullary cords, so an arrest of B-cell maturation in the marrow empties all three of those compartments while leaving the paracortex and its high endothelial venules normal, and with no plasma cells every antibody class is low.

A is wrong: Failed positive selection in the thymic cortex would empty the paracortex and the splenic periarteriolar lymphoid sheath and lower circulating T cells, but this boy's paracortex and T-cell count are normal.

B is wrong: Failed entry through high endothelial venules would leave the paracortex poorly populated as well, since that is the route naive lymphocytes of both lines take out of the blood, and the paracortex here is normal.

D is wrong: Absent M cells would impair sampling of luminal antigen at mucosal surfaces, which affects mucosal IgA responses rather than removing follicles, germinal centres, and plasma cells from a lymph node.

A competent response contains

· Absent follicles, absent germinal centres, and absent plasma cells in the medullary cords, with a normal paracortex and normal T-cell numbers, place the defect in the B-cell line.

· Plasma cells in the medullary cords are the source of secreted antibody, so their absence explains IgG, IgA, and IgM all being low, and X-linked agammaglobulinemia is the arrest of B-cell maturation in the marrow.

Distinctions that matter

· An immunodeficiency empties the compartment belonging to the missing cell type in every lymphoid organ, so a node biopsy localizes the defect to a cell line.

Common misconceptions

· Reading absent germinal centres as a T-cell defect. Germinal centres need follicular helper T cells, but here the T-cell compartment is intact and the B cells are missing.

Where it is taught: Section C04 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

NPLEX-P1-CARD-10-01-C02-Q02

Board-style question: Where a testicular tumour spreads first

Correct answer: A. Lumbar (para-aortic) nodes near the renal vessels

Why it is right: The testes develop on the posterior abdominal wall and drag their lymphatic vessels with them as they descend, so the first node group for a testicular tumour is the lumbar or para-aortic group near the renal vessels.

B is wrong: Superficial inguinal nodes drain scrotal skin, vulva, the anal canal below the pectinate line, and the superficial lower limb, and the scrotal skin here is normal.

C is wrong: Internal iliac nodes drain pelvic viscera and the anal canal above the pectinate line, not the testis.

D is wrong: Popliteal nodes drain the lateral foot and posterolateral calf and pass lymph onward to deep inguinal nodes, so they have no role in testicular drainage.

A competent response contains

· The testis descends from the posterior abdominal wall and brings its lymphatics with it, so testicular lymph drains to lumbar, also called para-aortic, nodes near the renal vessels.

· Scrotal skin is body wall and drains to superficial inguinal nodes, which is why the normal overlying skin matters in this case.

Distinctions that matter

· Lymphatic drainage follows the path of embryological descent rather than the adult position of the organ.

Common misconceptions

· Sending testicular lymph to inguinal nodes because the testis sits in the scrotum. The organ and the skin over it drain to different node groups.

Where it is taught: Section C02 of the CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-11

CV-11 Hemodynamics and Blood Flow Regulation

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-11-01-C01-BD1

Brain dump 1 · foundational retrieval: Write the flow equations and Poiseuille's law from memory

A competent response contains

· Q = ΔP / R, the hydraulic form of Ohm's law; rearranged, R = ΔP / Q.

· Resistance is calculated rather than measured because only the pressure difference and the flow can be measured directly.

· For the whole systemic circulation the driving pressure is mean arterial pressure minus right atrial pressure, which is near 0 mm Hg, the flow is cardiac output, and the resistance is total peripheral resistance, so MAP = CO x TPR.

· Resting values: MAP about 93 mm Hg, cardiac output about 5 L/min, total peripheral resistance about 18 to 19 mm Hg per L/min.

· Pressure rises only if cardiac output or total peripheral resistance rises, and a fall in one must be offset by a rise in the other to hold pressure steady.

· Poiseuille's law: $R = 8\eta L / (\pi r^4)$, where η is viscosity, L is vessel length and r is internal radius.

· Resistance is directly proportional to viscosity and to length, and inversely proportional to the fourth power of the radius.

· The fourth power arises because laminar flow has a parabolic velocity profile: a wider vessel has both a larger cross-sectional area, which goes with r squared, and a larger central core of fluid far from the slowing effect of the wall, so velocity in that core rises as well.

· Halving the radius raises resistance 16-fold; widening the radius by about 19% halves resistance and doubles flow, because 1.19 to the fourth power is about 2.

· Length is fixed in the adult, but it explains why long diffuse disease resists more than a short lesion of the same caliber.

· Assumptions: a rigid tube, laminar flow and a Newtonian fluid. Passive vessels distend as pressure rises, so flow rises more than in proportion; autoregulating vessels constrict, so flow stays nearly constant; turbulent flow needs a larger gradient for the same flow. Use the law for direction and relative size of change.

Distinctions that matter

· Resistance and flow move in opposite directions for the same change in radius, so state which one the question asks for before doing the arithmetic.

· Viscosity, length and the pressure gradient change flow only in direct proportion, while radius changes it by the fourth power.

Common misconceptions

· Writing that a doubled pressure gradient halves resistance. In a rigid tube resistance is unchanged and flow doubles; it is the vessel wall, not the gradient, that sets resistance.

· Treating the fourth power as a property of area alone. Area accounts for only two of the four powers, and the velocity profile accounts for the other two.

Where it is taught: Section C01, opening paragraph and the paragraph on Poiseuille's law of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C01-BD2

Brain dump 2 · integration and application: Work out what single changes do to resistance and flow

A competent response contains

· Radius halved: resistance increases 16-fold and flow falls to one sixteenth. Example, arteriolar constriction by sympathetic alpha-1 stimulation or a severe focal stenosis.

· Radius doubled: resistance falls to one sixteenth and flow increases 16-fold. Example, metabolic dilation of muscle arterioles during exercise.

· Radius widened about 19%: resistance halves and flow doubles. Example, a small dilation of a coronary arteriole when myocardial oxygen demand rises.

· Viscosity doubled: resistance doubles and flow halves. Example, severe polycythemia or a hyperviscosity state; anemia does the reverse.

· Length doubled: resistance doubles and flow halves. Example, long diffuse arterial disease compared with a short focal lesion.

· At a hematocrit of 68% viscosity is raised two to three times, so resistance rises in proportion and flow falls.

· Oxygen delivery is flow times oxygen content, so when flow falls by more than content rises, delivery drops; sluggish flow also favors thrombosis.

· Other causes of raised viscosity: large plasma proteins such as an IgM paraprotein in Waldenström macroglobulinemia or a high fibrinogen, rigid sickled cells, and extreme leukocytosis.

· Blood is non-Newtonian: at low flow red cells form rouleaux and viscosity rises, which matters in low-flow states and small veins.

· In vessels below about 0.3 mm a cell-poor plasma layer at the wall lowers apparent viscosity, the Fåhræus-Lindqvist effect, which matters in arterioles and capillaries.

· Whole blood at a hematocrit of 40 to 45% is roughly 3 to 4 times as viscous as water, and hematocrit is the main determinant.

Distinctions that matter

· Hematocrit sets viscosity, which changes resistance in direct proportion, while radius changes it by the fourth power, so a modest vasoconstriction outweighs a large change in hematocrit.

· Rouleaux formation raises apparent viscosity in slow large-vessel flow, while the cell-poor wall layer lowers it in the smallest vessels, so the two effects belong to different parts of the circuit.

Common misconceptions

· Assuming more red cells always means more oxygen reaching the tissue. Once viscosity rises steeply, the loss of flow costs more than the extra content gains.

· Treating blood as a Newtonian fluid at all flow rates. Its apparent viscosity depends on shear rate and on vessel caliber.

Where it is taught: Section C01, the table on predicting a single change and the paragraph on viscosity of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C01-PR01

Prediction exercise: Predict the effect of a focal arterial stenosis

A competent response contains

· Resistance of the narrowed segment rises 16-fold, because resistance varies with the inverse fourth power of the radius.

· Flow through the segment falls toward one sixteenth of its former value if the gradient does not change, from $Q = \Delta P / R$.

· Area falls to one quarter, because area goes with the radius squared, so velocity inside the stenosis rises about 4-fold, from $v = Q / A$ and the continuity principle.

· Reynolds number is $\rho v D / \eta$, so a 4-fold rise in velocity with a halved diameter roughly doubles it, and the jet breaks into eddies as it enters the wider vessel beyond.

· A bruit is heard, because turbulent eddies vibrate the vessel wall; the same disturbance can be felt as a thrill.

· As the stenosis becomes critical, flow, velocity and the Reynolds number all fall, so the bruit softens or disappears, which is why a quiet carotid does not exclude severe stenosis.

· A long diffuse diseased segment resists more than a short lesion of the same caliber, because resistance is directly proportional to length.

· Turbulence also wastes energy, so a larger pressure gradient is needed for the same flow, which adds to the effective resistance of the segment.

Distinctions that matter

· Flow through the whole limb falls while velocity inside the narrowed segment rises, because flow and velocity are linked through area rather than being the same quantity.

· A loud bruit reports a high-velocity jet, not the severity of the narrowing, so loudness and severity part company once the lesion becomes critical.

Common misconceptions

· Reading a louder bruit as a worse stenosis in every case. A critical lesion can be silent because the jet has lost its velocity.

· Expecting velocity to fall inside a narrowing because flow has fallen. The same flow must cross a much smaller area, so velocity rises.

Where it is taught: Section C01, Poiseuille paragraph and table; Section C03, paragraphs on continuity and on laminar and turbulent flow of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C01-Q01

Board-style question: Arteriolar dilation and flow

Correct answer: C. Flow approximately doubles

Why it is right: Flow at a constant pressure gradient varies with the fourth power of the radius, and 1.19 to the fourth power is about 2, which is why a local dilation of roughly 19% can double flow to one muscle without any change in arterial pressure or in flow elsewhere.

A is wrong: A 19% rise in flow would follow if flow varied in direct proportion to radius, which applies to neither resistance nor flow.

B is wrong: A 50% rise follows from scaling flow with cross-sectional area, which accounts for only two of the four powers of the radius.

D is wrong: A 16-fold rise follows from doubling the radius, not from widening it by about 19%.

A competent response contains

· Resistance varies with the inverse fourth power of the radius, so flow at a fixed pressure gradient varies with the fourth power of the radius.

· A radius 1.19 times normal gives 1.19 to the fourth power, which is about 2, so flow approximately doubles.

Distinctions that matter

· A small change in radius produces a large change in flow, which is why arterioles rather than arterial pressure set regional flow.

Common misconceptions

· Scaling flow in direct proportion to radius, or in proportion to area, both of which understate the change.

Where it is taught: Section C01 of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C02-BD1

Brain dump 1 · foundational retrieval: Series and parallel resistance, and the pressure profile

A competent response contains

· In series: R total = R1 + R2 + ..., the same flow passes through every segment, and the segment with the highest resistance takes the largest pressure drop because ΔP = Q x R.

· In parallel: 1 / R total = 1 / R1 + 1 / R2 + ..., every branch sees the same pressure difference, flows add, and conductances add.

· Total parallel resistance is lower than the smallest single branch, and n identical branches of resistance R give R / n.

· Worked in series: 10 + 20 + 20 = 50 units. Worked in parallel: 1/10 + 1/20 + 1/20 = 0.2, so the total is 5 units.

· Aorta and large arteries: about 95 to 100 mm Hg and pulsatile, small share of resistance, aorta about 2.5 cm², and all systemic arteries hold about 13% of blood volume.

· Small arteries and arterioles: pressure falls from about 85 to 90 mm Hg down to about 30 to 35 mm Hg, the largest share of resistance with arterioles dominating, arterioles about 40 cm², about 7% of volume together with the capillaries.

· Capillaries: about 30 to 35 mm Hg at the arterial end and 10 to 15 mm Hg at the venous end, moderate resistance because so many lie in parallel, about 2,500 cm².

· Venules and veins: about 10 to 15 mm Hg falling to near 0 at the right atrium, small share of resistance, venules about 250 cm² and the venae cavae about 8 cm², and systemic veins hold about 64% of blood volume.

· Arterioles hold the largest adjustable share because they are relatively few, they are narrow, and they carry thick smooth muscle relative to their lumen, so their radius can be changed over a wide range.

· Capillaries are individually narrower than arterioles but there are billions in parallel, which gives only a moderate combined resistance, and they have no smooth muscle, so their caliber cannot be adjusted.

· Terminal arterioles and precapillary sphincters decide how many capillaries are open, and the steep arteriolar pressure drop also damps the arterial pulse so capillary flow is nearly steady.

· The heart holds about 7% and the lungs about 9% of blood volume.

Distinctions that matter

· Resistances add in series while conductances add in parallel, so adding a branch raises total resistance in one arrangement and lowers it in the other.

· Having the narrowest lumen and having the largest share of resistance are different claims: capillaries are narrower, arterioles resist more, and the number of vessels in parallel is what reconciles the two.

Common misconceptions

· Adding parallel resistances directly. Total parallel resistance must come out lower than the smallest branch, which is a quick check on the arithmetic.

· Naming the capillaries as the main site of resistance because they are the narrowest vessels. Their enormous parallel number keeps their combined resistance moderate.

Where it is taught: Section C02, opening paragraph, the paragraph on why arterioles set total peripheral resistance, and the table on pressure, area, velocity and volume of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C02-BD2

Brain dump 2 · integration and application: Adding and removing parallel paths, and beds placed in series

A competent response contains

· An arteriovenous fistula is a low-resistance path added in parallel, so total peripheral resistance falls.

· Mean arterial pressure would fall if cardiac output did not change, so baroreceptors raise sympathetic output and cardiac output rises to defend pressure.

· Rapid diastolic runoff through the fistula lowers diastolic pressure and widens pulse pressure.

· A very large fistula keeps cardiac output high over time and can cause high-output heart failure.

· A mid-thigh amputation removes a parallel branch, so total peripheral resistance rises. With branches of 10, 20 and 20 units the total is 5 units, and removing the 10-unit branch raises it to 10 units.

· Sequence when a large bed dilates: one large bed dilates, such as exercising muscle; total peripheral resistance falls; mean arterial pressure would fall if cardiac output were unchanged; baroreceptors raise sympathetic output; cardiac output rises while gut, kidney and inactive muscle constrict; mean arterial pressure is defended and flow shifts to the dilated bed.

· Because systemic organs branch from the aorta in parallel, each sees nearly the same arterial pressure and its flow equals mean arterial pressure divided by its own resistance, so one organ can raise its flow by dilating without starving the others.

· Beds in series: gut and spleen draining to the liver through the portal vein, glomerular capillaries to peritubular capillaries, and the hypothalamus to the anterior pituitary.

· In a series arrangement, raised downstream resistance raises upstream capillary pressure.

· In cirrhosis the fibrotic liver raises resistance downstream of the splanchnic capillaries, so splanchnic capillary pressure rises, which is portal hypertension.

· The pulmonary and systemic circuits are in series with each other, so the two ventricles must have equal average outputs.

Distinctions that matter

· Adding a parallel path lowers total peripheral resistance while removing one raises it, and neither change by itself tells you what happens to pressure until the reflex change in cardiac output is included.

· Organs in parallel are protected from one another, while organs in series are not, which is why only the series beds develop upstream capillary hypertension.

Common misconceptions

· Expecting mean arterial pressure to fall and stay low after a fistula is created. The baroreceptor reflex raises cardiac output, so mean pressure is largely defended while diastolic pressure falls.

· Treating portal hypertension as a problem of blood volume alone. It follows from the series arrangement of splanchnic capillaries and the liver.

Where it is taught: Section C02, opening paragraph, final paragraph, the flow figure on opening a parallel bed and the stop and predict box of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C03-BD1

Brain dump 1 · foundational retrieval: Velocity against total cross-sectional area

A competent response contains

· v = Q / A, so velocity is flow divided by cross-sectional area.

· The same cardiac output passes through every level of the series circuit, so velocity at each level varies inversely with the total cross-sectional area at that level.

· Mean velocity is about 33 cm/s in the aorta and about 0.3 mm/s in the capillaries.

· The combined capillary area is about 1,000 times larger than the aortic area, which is why capillary velocity is lowest.

· On the sketch, total cross-sectional area rises to a peak at the capillaries and falls again toward the venae cavae, while velocity is the mirror image of that curve.

· A red cell spends about 1 to 3 seconds in a systemic capillary, and that transit time allows diffusion of gases, nutrients and wastes.

· Velocity rises again in the veins because total area falls, but it stays below aortic values because the venae cavae together have a larger area than the aorta, about 8 cm^2 against about 2.5 cm^2.

· Locally the continuity principle means a narrowed segment carries the same flow through a smaller area, so velocity rises there.

Distinctions that matter

· Flow is volume per time and velocity is distance per time, and area is what links them, so a low capillary velocity does not mean low capillary flow.

· The whole-circuit relationship between velocity and total area is a different use of the same equation from the local rise in velocity through a stenosis.

Common misconceptions

· Saying capillary velocity is low because each capillary is narrow. A narrow tube carrying the same flow would have a higher velocity; it is the number of capillaries in parallel that lowers it.

· Expecting venous velocity to match aortic velocity because the same cardiac output returns. Total venous area is larger, so velocity stays lower.

Where it is taught: Section C03, opening paragraph and the paragraph on the continuity principle of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C03-BD2

Brain dump 2 · integration and application: Turbulence at the bedside

A competent response contains

· Re = ρvD / η, where ρ is density, v is velocity, D is diameter and η is viscosity. Rising velocity, rising diameter and falling viscosity all raise it.

· Density barely varies in the body, so velocity, diameter and viscosity are the clinical variables.

· Turbulence appears above a Reynolds number of about 2,000 in a straight smooth tube, and at lower values at branches, curves and abrupt changes in caliber.

· Turbulent flow forms eddies that vibrate the vessel wall, which is what makes a murmur, a bruit or a palpable thrill, and the eddies waste energy, so a larger pressure gradient is needed for the same flow.

· Anemia sequence: hemoglobin and hematocrit fall; viscosity falls and resistance falls; low oxygen content dilates arterioles so total peripheral resistance falls further; cardiac output and stroke volume rise; ejection velocity through normal valves rises; higher velocity and lower viscosity raise the Reynolds number; a soft systolic flow murmur appears across normal valves.

· Anemia acts on velocity and viscosity at once, which makes it the classic cause of a flow murmur, while polycythemia raises viscosity and tends to abolish flow murmurs.

· A carotid stenosis is audible because the jet is fast and the caliber changes abruptly; a critical stenosis can be silent because flow, velocity and the Reynolds number have all fallen.

· An arteriovenous fistula connects a high-pressure to a low-pressure vessel, giving a very high velocity and a continuous bruit.

· Just below systolic pressure the cuff leaves a narrowed segment through which blood spurts at each peak, and that turbulence is the first Korotkoff sound; once cuff pressure falls below diastolic pressure flow is laminar throughout and the sound disappears, which is phase V.

· Other causes of raised velocity and flow murmurs: pregnancy, fever, hyperthyroidism and exercise. A dilated aortic root raises the Reynolds number through diameter.

· Shear stress is low and oscillating at branch points and bifurcation walls, which reduces endothelial nitric oxide and favors atherosclerosis, while steady laminar high shear keeps the endothelium quiescent and antithrombotic.

· Jets beyond a stenotic valve can cause post-stenotic dilation.

· The simplified Bernoulli relation estimates the gradient in mm Hg as about 4 times the peak velocity in m/s squared.

Distinctions that matter

· Low viscosity raises the Reynolds number while high viscosity lowers it, so anemia favors turbulence for two reasons at once and polycythemia opposes it.

· Velocity rather than flow determines whether turbulence occurs, which is why a critically narrowed vessel with very low flow can be silent.

Common misconceptions

· Taking a quiet carotid as evidence against severe stenosis. Loss of the bruit can mean the lesion has become critical.

· Treating every systolic murmur as valve disease. Anemia, fever, pregnancy, hyperthyroidism and exercise produce murmurs across structurally normal valves.

Where it is taught: Section C03, paragraph on laminar and turbulent flow, the table on factors favoring turbulence, the flow figure on anemia and the final paragraph on Korotkoff sounds and shear stress of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C03-RT01

Retrieval exercise: Rebuild the sequence from anemia to a flow murmur

A competent response contains

· Step 1: hemoglobin and hematocrit fall.

· Step 2: viscosity falls and resistance falls. Falling viscosity lowers resistance, which raises flow, and it also raises the Reynolds number directly, so the two effects reach the murmur through different variables.

· Step 3: low oxygen content dilates arterioles, so total peripheral resistance falls further. The dilating stimulus is the low oxygen content of the blood reaching the tissue.

· Step 4: cardiac output and stroke volume rise, and stroke volume is the component that rises most.

· Step 5: ejection velocity through normal valves rises.

· Step 6: higher velocity and lower viscosity raise the Reynolds number.

· Step 7: a soft systolic flow murmur is heard across normal valves.

· The same reasoning explains flow murmurs in pregnancy, fever and hyperthyroidism.

Distinctions that matter

· Raised velocity and lowered viscosity are two separate routes into the Reynolds number, and anemia supplies both, which is why its murmur is so reliable.

· The murmur arises across structurally normal valves, so it reports the state of the blood and the output rather than the valve.

Common misconceptions

· Expecting lower viscosity to quiet the circulation because resistance has fallen. Lower viscosity raises the Reynolds number and the higher output raises velocity, so turbulence becomes more likely.

· Putting the arteriolar dilation before the fall in viscosity. The viscosity change follows directly from the lower hematocrit, and the dilation is a response to low oxygen content.

Where it is taught: Section C03, the flow figure on anemia to a flow murmur and the table on factors favoring turbulence of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C04-BD1

Brain dump 1 · foundational retrieval: Compliance, the arterial reservoir and pulse pressure

A competent response contains

· C = ΔV / ΔP, and its reciprocal is stiffness, also called elastance.

· Systemic veins are about 24 times as compliant as the corresponding arteries, roughly 8 times as distensible with about 3 times the volume, which is why they hold most of the blood.

· During ejection the elastin-rich large arteries stretch and store part of the stroke volume; during diastole they recoil and push that stored volume onward.

· That reservoir function turns pulsatile output into near-continuous capillary flow, keeps diastolic pressure from falling to zero, and lowers the peak pressure the ventricle must generate.

· Pulse pressure is systolic minus diastolic pressure, normally about 40 mm Hg, and is roughly proportional to stroke volume divided by arterial compliance.

· Diastolic pressure depends on how fast the stored volume runs off, governed by the time constant R x C.

· Low resistance, low compliance or a long diastole each let diastolic pressure fall lower.

· Mean arterial pressure is about diastolic pressure plus one third of the pulse pressure at resting heart rates.

· That estimate loses accuracy at fast heart rates, because systole then takes a larger share of each cycle.

Distinctions that matter

· Compliance and resistance are separate properties of the arterial system: resistance with cardiac output sets mean pressure, while compliance with stroke volume sets pulse pressure.

· Distensibility is a fractional change in volume while compliance is an absolute change per unit pressure, which is why the venous figures of about 8 and about 24 differ.

Common misconceptions

· Treating the arterial reservoir as a store of blood volume. The veins hold the volume; the arteries store part of each stroke volume for a fraction of a second and return it in diastole.

· Assuming low compliance means low mean pressure. Stiffening changes the shape of the pressure wave while mean pressure stays set by cardiac output and total peripheral resistance.

Where it is taught: Section C04, first two paragraphs of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C04-BD2

Brain dump 2 · integration and application: Predict the pressure pattern in a stiff aorta

A competent response contains

· Pulse pressure is roughly stroke volume divided by compliance, so halving compliance with the same 70 mL stroke volume roughly doubles pulse pressure, from 40 mm Hg to about 80 mm Hg.

· The runoff time constant R x C is also halved, so pressure falls faster and further before the next beat, which lowers diastolic pressure.

· Mean arterial pressure is set by cardiac output and total peripheral resistance, neither of which has changed, so it stays about 93 mm Hg.

· A pressure of about 120/80 mm Hg in the younger person becomes something like 150/65 mm Hg in the older one, so systolic pressure rises, diastolic pressure falls and mean pressure barely moves.

· Wall changes with age: elastin fragments, collagen increases and is cross-linked by advanced glycation end products, and the media may calcify.

· Hypertension, diabetes and chronic kidney disease accelerate those changes.

· Faster pulse wave velocity returns the reflected wave in late systole, so it augments the systolic peak instead of supporting diastolic pressure.

· Isolated systolic hypertension is the result and is the most common form in older adults, with diastolic pressure levelling off or falling after about age 50 to 60. The 2017 ACC/AHA guideline uses a systolic of 130 mm Hg or higher with a diastolic below 80 mm Hg; other guidelines use 140 and 90 mm Hg.

· Consequences: higher afterload drives concentric left ventricular hypertrophy, and lower diastolic pressure reduces coronary perfusion, which for the left ventricle occurs mainly in diastole, so the subendocardium is most exposed.

· The wider pulsatile pressure reaches the low-resistance brain and kidney, contributing to lacunar infarcts, white matter disease and declining renal function, and stiffness itself rises with stretch, so hypertension and stiffening reinforce each other.

· Widens: aortic regurgitation, through a large stroke volume plus diastolic backflow; hyperthyroidism, through a larger stroke volume with lower total peripheral resistance; arteriovenous fistula, through low resistance and rapid diastolic runoff; exercise, through a larger and faster stroke volume, mainly raising systolic pressure.

· Narrows: hemorrhage, aortic stenosis and cardiac tamponade, all through a small stroke volume, with reflex constriction holding diastolic pressure up in hemorrhage.

Distinctions that matter

· A wide pulse pressure from stiffening comes with a normal mean pressure, while a wide pulse pressure from a large stroke volume or a low resistance usually comes with other signs of a high output state.

· Systolic and diastolic pressure can move in opposite directions, which is why mean arterial pressure can look unchanged in a patient whose pressure wave has altered considerably.

Common misconceptions

· Explaining isolated systolic hypertension by a raised cardiac output. Stroke volume and total peripheral resistance are unchanged; it is compliance that has fallen.

· Treating a low diastolic pressure in an older adult as reassuring. It reduces the pressure driving left ventricular coronary flow during diastole.

Where it is taught: Section C04, the table on pulse pressure, the paragraph on aging and isolated systolic hypertension, the flow figure and the worked example of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the capillary and its Starling forces

A competent response contains

· Jv = Kf x [(Pc - Pi) - σ(πc - πi)], where Kf is the filtration coefficient, Pc is capillary hydrostatic pressure, Pi is interstitial hydrostatic pressure, σ is the reflection coefficient, πc is plasma oncotic pressure and πi is interstitial oncotic pressure.

· Teaching values: Pc about 30 mm Hg at the arterial end and 10 mm Hg at the venous end, Pi about -3 mm Hg, πc about 28 mm Hg, πi about 8 mm Hg.

· Pc pushes fluid out, a negative Pi adds to filtration, πc pulls fluid in and πi pulls it out.

· Arterial end: (30 + 3) - (28 - 8) = 33 - 20 = +13 mm Hg, so fluid filters out.

· Venous end: (10 + 3) - 20 = -7 mm Hg, so fluid is reabsorbed.

· Filtration slightly exceeds reabsorption, and the lymphatics return about 2 to 3 L per day along with the protein that has leaked out.

· Albumin supplies about 80% of plasma oncotic pressure, because it is the most abundant plasma protein and is poorly permeant.

· The reflection coefficient is near 1 in continuous capillaries and low in hepatic sinusoids; the filtration coefficient is high in glomeruli and low in the brain.

· In the revised glycocalyx model most capillaries filter along their whole length, and both models predict edema identically.

· Small solutes cross the capillary wall freely, so water movement is set by hydrostatic pressure and by the oncotic pressure of poorly permeant proteins.

Distinctions that matter

· Hydrostatic and oncotic pressures are not two versions of the same force: one is a mechanical push and the other is an osmotic pull created by proteins the wall holds back.

· Kf reports how leaky the wall is to water and σ reports how well it holds back protein, so a wall can change in one without the other.

Common misconceptions

· Taking a negative interstitial pressure as a sign that something has gone wrong. Loose tissue normally sits at about -3 mm Hg, and that slight suction adds to filtration.

· Leaving interstitial oncotic pressure out of the arithmetic. At about 8 mm Hg it cancels nearly a third of the plasma oncotic pull.

Where it is taught: Section C05, first two paragraphs of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-BD2

Brain dump 2 · integration and application: Work out the edema in four patients

A competent response contains

· Patient 1, nephrotic syndrome: πc falls, the fluid is a transudate, the swelling pits, and it is generalized, appearing first where tissue pressure is low, such as periorbital and scrotal tissue, and in dependent areas.

· Patient 1 arithmetic with πc at 16 mm Hg: the arterial end becomes 33 - (16 - 8) = 33 - 8 = +25 mm Hg, up from +13 mm Hg; the venous end becomes 13 - 8 = +5 mm Hg, so a site that formerly reabsorbed now filters.

· Patient 1 is generalized because the fall in plasma oncotic pressure affects every capillary in the body at once.

· Patient 2, right heart failure: Pc rises because venous pressure is raised, the fluid is a transudate, the swelling pits, and it appears in dependent areas with ascites and hepatic congestion. Reduced effective arterial volume activates the renin-angiotensin-aldosterone system and ADH, whose sodium and water retention raises venous pressure further.

· Patient 3, deep burn: permeability rises, so Kf rises, σ falls and πi rises; the fluid is a protein-rich exudate and the swelling is localized to the injured forearm.

· Patient 4, nodes removed and radiated: lymphatic removal fails and πi rises; capillary forces are normal at first; the fluid is protein rich and the swelling is localized to the arm.

· Patient 4 stops pitting because chronic protein-rich fluid drives fibrosis and fat deposition, and fibrosed tissue no longer indents.

· Safety factors: interstitial pressure rises quickly and opposes further filtration, lymph flow increases many-fold, and lymph washes interstitial protein out, which lowers πi.

· Guyton and Hall estimate those factors allow Pc to rise about 17 mm Hg before significant edema appears.

· Above an interstitial pressure of about 0 mm Hg the interstitium becomes highly compliant, so large volumes accumulate for little further rise in pressure.

· Transudates have a specific gravity below about 1.012 and exudates above about 1.020, so the fluid points back to the mechanism.

· Protein washout blunts the early change in nephrotic syndrome, which is why plasma oncotic pressure must fall substantially before edema appears.

Distinctions that matter

· Low plasma oncotic pressure produces generalized edema while raised capillary pressure and lymphatic failure produce edema where the pressure problem is, so distribution points at the term.

· Pitting reports fluid that moves under pressure, so a transudate keeps pitting while protein-rich fibrosed lymphedema stops.

Common misconceptions

· Expecting lymphedema to show abnormal Starling pressures at the capillary. The forces start normal, and the failure is in removal.

· Assuming any fall in albumin causes edema. The safety factors, protein washout in particular, must be overcome first.

Where it is taught: Section C05, the paragraph on mechanisms of edema, the edema table, the safety factor paragraph, the worked example on hypoalbuminemia and the stop and predict box of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-CP01

Comparison table: The four mechanisms of edema side by side

A competent response contains

· Raised capillary hydrostatic pressure: Pc rises; heart failure, deep vein thrombosis, portal hypertension, renal volume overload, prolonged standing and dihydropyridine arteriolar dilation; transudate; low protein content with a specific gravity below about 1.012; pits and keeps pitting; localized to the obstructed or dependent territory.

· Low plasma oncotic pressure: πc falls; nephrotic syndrome, cirrhosis, protein malnutrition such as kwashiorkor, and protein-losing enteropathy; transudate; low protein content with a specific gravity below about 1.012; pits; generalized, appearing first periorbitally and in dependent areas.

· Increased permeability: Kf rises, σ falls and πi rises; inflammation, burns, sepsis, ARDS and angioedema; exudate; protein rich with a specific gravity above about 1.020; pits early; localized in inflammation and widespread in sepsis.

· Lymphatic obstruction: removal fails and πi rises; node dissection or radiation, tumor, and filariasis from Wuchereria bancrofti; protein rich; specific gravity in the exudate range; pits at first and stops pitting once the tissue fibroses; localized to the drainage territory of the blocked lymphatics.

· Edema appears whenever filtration exceeds lymphatic removal, so each mechanism is a different way of crossing that same threshold.

Distinctions that matter

· Protein content separates the mechanisms that leave the wall intact from the two that do not: raised Pc and low πc give a transudate, while increased permeability and lymphatic obstruction give protein-rich fluid.

· Increased permeability lets plasma protein across an abnormal wall, while lymphatic obstruction traps protein that crossed a normal one.

Common misconceptions

· Treating all edema as pitting. Fibrosed chronic lymphedema does not pit.

· Assuming cirrhosis acts through one term. Portal hypertension raises splanchnic Pc while a low albumin lowers πc, which is why ascites forms so readily.

Where it is taught: Section C05, the paragraph on mechanisms of edema and the edema table of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-CA01

Clinical application: Abdominal swelling and ankle edema in a man with liver disease

A competent response contains

· Recognize: capillary hydrostatic pressure is raised in the splanchnic bed by portal hypertension, and plasma oncotic pressure is lowered body-wide by the albumin of 2.1 g/dL, so Pc is up where the liver drains and πc is down everywhere.

· Explain: the gut and splenic capillaries drain through the portal vein into the hepatic sinusoids, so they lie in series with the liver. A fibrotic cirrhotic liver raises resistance downstream of those capillaries, and raised downstream resistance raises upstream capillary pressure, which is portal hypertension; the dilated abdominal wall veins are collateral flow around that obstruction. The failing liver also synthesizes less albumin, and since albumin supplies about 80% of plasma oncotic pressure, πc falls and net filtration rises in every capillary, including those of the ankles.

· Differentiate: right heart failure would raise the jugular venous pressure and congest the liver from the venous side, and the jugular venous pressure here is not raised and the lungs are clear. Lymphatic obstruction would give protein-rich fluid in one drainage territory that eventually stops pitting, whereas this fluid has a specific gravity of 1.009, which is a transudate, and the ankle edema pits and is bilateral.

· Investigate: serum albumin, which is 2.1 g/dL against a reference range of 3.5 to 5.0 g/dL; the specific gravity of the ascitic fluid, which is 1.009 and therefore below the transudate threshold of about 1.012; jugular venous pressure, which is not raised; shifting dullness and dilated abdominal wall veins, which indicate free fluid and portal collateral flow; and the pitting character of the ankle edema with a slow refill.

· Decide: the fluid collects in the abdomen first because the raised capillary hydrostatic pressure is confined to the splanchnic bed drained by the portal vein, while the low oncotic pressure acts everywhere and adds the dependent ankle edema. The specific gravity of 1.009 marks a transudate, so the capillary wall is intact and protein has not leaked across it.

· Justify: the abdominal fluid reflects a high Pc term in the splanchnic capillaries added to a low πc term, while the ankle edema reflects the low πc term with the help of gravity on Pc. The safety factors are overwhelmed because two terms move at once and both move in the direction of filtration, the peritoneal cavity offers little opposing hydrostatic pressure, and once interstitial pressure passes about 0 mm Hg the tissue becomes highly compliant, so large volumes accumulate for little further rise in pressure.

Distinctions that matter

· Ascites with a normal jugular venous pressure points to a pressure problem upstream of the liver, while ascites with a raised jugular venous pressure points to one downstream of it.

· Two Starling terms moving together explain a distribution that neither term alone would give: abdominal fluid from the regional Pc and bilateral ankle edema from the body-wide πc.

Common misconceptions

· Attributing all of this patient's fluid to the low albumin. That would predict generalized edema with no preference for the abdomen, and it misses the series arrangement that raises splanchnic capillary pressure.

· Reading the dilated abdominal wall veins as a sign of heart failure. They are portal collaterals, which tells you where the raised pressure sits.

Where it is taught: Section C05, mechanisms of edema and the edema table, with the series bed and portal hypertension paragraph in C02 of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-BD1

Brain dump 1 · foundational retrieval: The four layers of control over arteriolar tone

A competent response contains

· Contraction: cytosolic Ca2+ rises, binds calmodulin, and the complex activates myosin light chain kinase, which phosphorylates the myosin light chains; relaxation follows a fall in Ca2+ or activation of myosin light chain phosphatase.

· Constrictors act through Gq receptors, where IP3 releases Ca2+, or by depolarization that opens L-type Ca2+ channels; dilators raise cGMP or cAMP, or open K+ channels and hyperpolarize the cell.

· Metabolic control: adenosine, CO2, H+, K+, lactate, phosphate and low PO2 accumulate when activity rises or flow falls short, and they dilate arterioles partly by opening K+ channels. Flow then washes them out, which is negative feedback. Active hyperemia is the rise in flow with raised tissue activity; reactive hyperemia is the surge after an occlusion, proportional to its duration.

· Myogenic control, the Bayliss response: stretch opens mechanosensitive cation channels, the cell depolarizes, L-type Ca2+ channels open, and the arteriole constricts; less stretch relaxes it.

· Endothelial control: shear stress and the agonists acetylcholine, bradykinin, histamine and substance P activate eNOS, and the nitric oxide pathway relaxes the smooth muscle, with the dilation spreading upstream to feeding arteries as flow-mediated dilation. Prostacyclin raises cAMP, hyperpolarizing factors matter in small arterioles, and endothelin-1 constricts through ETA receptors. With a damaged endothelium acetylcholine constricts instead, because it reaches muscarinic receptors on the smooth muscle with no nitric oxide to oppose it.

· Neural control: sympathetic norepinephrine acts on alpha-1 receptors, which are Gq coupled, and sets basal tone. Withdrawal of that tone is the main neural dilator mechanism, because systemic arterioles have no parasympathetic supply.

· Hormonal constrictors: angiotensin II at AT1 and high-dose vasopressin at V1, both Gq coupled.

· Hormonal dilators: epinephrine at beta-2 receptors, which are Gs coupled and raise cAMP, dilating muscle and liver arterioles at physiologic levels, while at high levels it constricts through alpha-1; natriuretic peptides raise cGMP; histamine and bradykinin dilate through nitric oxide and also open venular gaps, which causes inflammatory edema.

· Metabolic and myogenic control together give autoregulation over a mean arterial pressure of roughly 60 to 70 up to 150 to 160 mm Hg in brain and kidney, and the kidney adds tubuloglomerular feedback.

· Chronic hypertension shifts the autoregulatory range upward, so a fall in pressure that a normotensive person tolerates can cause hypoperfusion.

· Tone itself is sustained partial contraction of the arteriolar smooth muscle, which is what makes both dilation and further constriction possible from the resting state.

Distinctions that matter

· Metabolic control raises flow when the tissue needs more, while myogenic control defends flow against a change in pressure, so they answer different questions even though both act on the same smooth muscle.

· Epinephrine's effect depends on dose and on which receptor predominates in the bed, so the same hormone dilates muscle arterioles at physiologic levels and constricts at high levels.

Common misconceptions

· Expecting parasympathetic fibers to dilate systemic arterioles. They do not supply them, and withdrawal of sympathetic tone is the neural dilator.

· Reading a constrictor response to acetylcholine as an error. It is what happens when the endothelium is damaged and the nitric oxide route is lost.

Where it is taught: Section C06, first three paragraphs and the nitric oxide flow figure of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-BD2

Brain dump 2 · integration and application: Which control wins in which bed

A competent response contains

· Coronary: dominant control is metabolites, namely adenosine, nitric oxide and low O2; hypoxia dilates; the alpha-1 effect is overridden by metabolism.

· Cerebral: dominant control is PaCO2 acting through pH, plus autoregulation; hypoxia dilates below a PaO2 of about 50 mm Hg; sympathetic effect is minor.

· Skeletal muscle: sympathetic at rest and metabolites in exercise; hypoxia dilates; sympathetic effect is strong at rest and overridden in active muscle.

· Skin: sympathetic thermoregulation; hypoxia has little role; sympathetic effect is strong, especially at the arteriovenous anastomoses.

· Pulmonary: hypoxic vasoconstriction, with recruitment and distension as output rises; hypoxia constricts; sympathetic effect is minor.

· Kidney: myogenic control and tubuloglomerular feedback; hypoxia has a minor role; strong sympathetic stimulation constricts.

· The heart extracts about 70 to 75% of arterial oxygen at rest, so there is little extraction reserve and more demand must be met by more flow; metabolic dilation therefore overrides alpha-1 constriction in the coronary bed.

· Hyperventilation lowers PaCO2; CO2 crosses the blood-brain barrier and its removal raises perivascular pH, which constricts cerebral arterioles, so cerebral flow falls despite a normal arterial pressure. Plasma H+ crosses slowly, which is why the CO2 level rather than the blood pH drives the response. Cerebral flow roughly doubles as PaCO2 rises from 40 to 80 mm Hg.

· In exercising muscle, local metabolites and nitric oxide override sympathetic constriction, which is functional sympatholysis, and flow rises about 20-fold.

· After hemorrhage, skin flow serves heat loss rather than metabolism, so strong sympathetic constriction makes the skin pale and cool, while the coronary and cerebral beds are governed locally and escape that constriction.

· Low alveolar PO2 inhibits oxygen-sensitive K+ channels in pulmonary arterial smooth muscle, which depolarizes the cell and opens L-type Ca2+ channels, so the vessel constricts and blood is diverted toward better ventilated alveoli. When the hypoxia involves the whole lung the response is global and raises pulmonary vascular resistance, giving pulmonary hypertension.

· A long-standing hypertensive has an autoregulatory range shifted upward, so a mean pressure that sits comfortably inside a normotensive person's range can fall below the hypertensive person's lower limit, and cerebral and renal flow fall.

Distinctions that matter

· Hypoxia dilates every systemic bed and constricts the pulmonary bed, because the systemic response matches flow to tissue demand while the pulmonary response matches perfusion to ventilation.

· Beds governed locally, the heart and brain, escape sympathetic constriction, while skin, gut, kidney and resting muscle do not, which is what lets the body defend pressure by redistributing flow.

Common misconceptions

· Expecting the heart to meet extra demand by extracting more oxygen. Extraction is already about 70 to 75% at rest, so flow has to rise.

· Explaining hyperventilation lightheadedness by low oxygen. The PaO2 is high; it is the low PaCO2 and the resulting alkaline perivascular pH that constrict cerebral arterioles.

Where it is taught: Section C06, the paragraph on dominant control in each bed, the table of beds and the stop and predict box of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-RT01

Retrieval exercise: Rebuild the nitric oxide pathway

A competent response contains

· Step 1, endothelium: shear stress, or acetylcholine and bradykinin, act on the endothelial cell.

· Step 2, endothelium: Ca2+-calmodulin and phosphorylation activate eNOS.

· Step 3, endothelium: eNOS converts L-arginine to nitric oxide and citrulline.

· Step 4, smooth muscle: nitric oxide diffuses across and activates soluble guanylyl cyclase.

· Step 5, smooth muscle: cGMP rises and activates protein kinase G.

· Step 6, smooth muscle: cytosolic Ca2+ falls and myosin light chain phosphatase is activated.

· Step 7, smooth muscle: the myosin light chains are dephosphorylated and the arteriole dilates.

· The enzyme in the endothelium is eNOS, its substrate is L-arginine, and its products are nitric oxide and citrulline.

· In the smooth muscle, nitric oxide activates soluble guanylyl cyclase, which makes cGMP, which activates protein kinase G.

· Nitrate drugs release nitric oxide inside the smooth muscle cell, so they enter the pathway at step 4 and bypass eNOS, which is why they still work when the endothelium is dysfunctional.

· The dilation spreads upstream to feeding arteries, which is flow-mediated dilation.

Distinctions that matter

· The pathway crosses two cells: everything up to the production of nitric oxide happens in the endothelium, and everything after it happens in the smooth muscle.

· Nitric oxide lowers Ca2+ and also activates myosin light chain phosphatase, so it reduces myosin light chain phosphorylation from both directions.

Common misconceptions

· Placing eNOS in the smooth muscle. It is endothelial, which is why endothelial damage removes the physiologic route to dilation while nitrates still work.

· Treating cGMP as acting directly on the contractile proteins. It works through protein kinase G.

Where it is taught: Section C06, the paragraph on endothelial control and the nitric oxide flow figure of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-Q02

Board-style question: Regional flow during hyperventilation

Correct answer: A. A rise in perivascular pH that constricts cerebral arterioles and reduces cerebral blood flow

Why it is right: Removing CO2 raises perivascular pH, and cerebral arterioles constrict as pH rises, so cerebral blood flow falls even though mean arterial pressure sits well inside the autoregulatory range. This is the same relationship that roughly doubles cerebral flow as PaCO2 rises from 40 to 80 mm Hg, running in the opposite direction.

B is wrong: Mean arterial pressure here is about 93 mm Hg, comfortably above the lower autoregulatory limit of roughly 60 to 70 mm Hg, so autoregulatory failure does not apply.

C is wrong: Inhibition of oxygen-sensitive potassium channels is the mechanism of hypoxic vasoconstriction in pulmonary arterial smooth muscle, and in any case the PaO2 is 108 mm Hg.

D is wrong: Loss of myogenic tone with a rise in cerebral blood flow is the opposite of what a low PaCO2 produces.

A competent response contains

· Hyperventilation lowers PaCO2 to 22 mm Hg, CO2 leaves the brain across the blood-brain barrier, and perivascular pH rises.

· An alkaline perivascular pH constricts cerebral arterioles, so cerebral blood flow falls while arterial pressure stays normal.

Distinctions that matter

· Cerebral flow follows PaCO2 rather than arterial pH, because CO2 crosses the blood-brain barrier readily and plasma H+ crosses slowly.

Common misconceptions

· Looking for a fall in arterial pressure to explain the symptom. The pressure is 128/76 mm Hg and the problem is local arteriolar constriction.

Where it is taught: Section C06 of the CV-11 Hemodynamics and Blood Flow Regulation notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-12

CV-12 Exercise and the Cardiovascular System

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-12-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the two inputs that start the exercise response

A competent response contains

· Heart rate rises within the first second of exercise, too fast for circulating metabolites to explain, so the trigger is neural.

· Two inputs converge on the medullary cardiovascular centers: central command (feedforward) and the exercise pressor reflex (feedback from muscle).

· Central command comes from the motor cortex and subcortical locomotor regions, which signal the medullary centers in parallel with recruiting motor units.

· Its output scales with the effort commanded rather than the force produced, and it resets the arterial baroreflex to a higher set point so the reflex still buffers beat to beat changes without opposing the rise in pressure.

· Mechanoreflex: group III thinly myelinated afferents, excited by stretch and tension of contracting muscle, strongest in the first seconds, producing an early heart rate rise largely by reducing vagal outflow.

· Metaboreflex: group IV unmyelinated C fiber afferents, excited by H+, lactate, K+, inorganic phosphate, ATP, bradykinin, and prostaglandins, building over tens of seconds, strongest when flow is too low for metabolism, producing sympathetic activation and a rise in arterial pressure.

· The metaboreflex works as an error signal: inadequate flow lets metabolites accumulate and the reflex raises arterial pressure to push blood into the underperfused muscle.

· Observation 1: when muscles are partly weakened by a neuromuscular blocking drug, the same force takes more effort and heart rate and blood pressure rise more, which shows the response follows commanded effort. Central command also explains the anticipatory rise before a race.

· Observation 2: trapping metabolites in the arm with a cuff after a sustained handgrip keeps blood pressure high while heart rate returns toward baseline, which shows the heart rate response depends mostly on central command and the mechanoreflex.

· Both arms converge on the same brainstem centers and interact.

Distinctions that matter

· Central command is feedforward and scales with commanded effort; the exercise pressor reflex is feedback and scales with what is happening in the muscle.

· The metaboreflex raises pressure more than rate, while central command and the mechanoreflex account for most of the rate response.

Common misconceptions

· Saying accumulating metabolites start the response. They build over tens of seconds, while heart rate rises within the first second.

· Saying central command switches the baroreflex off. It resets the set point upward, so buffering continues around a higher pressure.

Where it is taught: Section C01, opening paragraphs, the exercise pressor reflex table, and the metaboreflex paragraph of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C01-BD2

Brain dump 2 · integration and application: Why the rate rise is vagal first and sympathetic second

A competent response contains

· Vagal acetylcholine acts on M2 receptors; Gi beta-gamma subunits open IKACh channels directly, and acetylcholinesterase clears acetylcholine within milliseconds, so stopping vagal firing raises heart rate within one or two beats.

· Norepinephrine acts on beta-1 receptors through Gs, cAMP, and protein kinase A, and is cleared mainly by reuptake, so its effects build and fade over seconds.

· Vagal withdrawal produces most of the early rise, up to roughly 100 bpm in a young adult.

· Above that, sympathetic drive raises rate further, increases contractility, and constricts veins and arterioles in inactive tissues.

· Venous return rises through sympathetic venoconstriction of the splanchnic and cutaneous veins, which raises mean systemic filling pressure, the skeletal muscle pump driving blood past venous valves toward the heart, and the respiratory pump as deeper breathing lowers intrathoracic pressure.

· Greater filling raises stroke volume through the Frank-Starling mechanism.

· Transplanted heart: denervated, so the resting rate sits near the intrinsic SA node rate and barely changes at onset; cardiac output rises first through the Frank-Starling response, and rate climbs over minutes as circulating catecholamines rise, to a lower peak than normal.

· Beta-1 blockade blunts the sympathetic part of the rate response but spares vagal withdrawal, so the early rise is largely preserved and the later rise is blunted.

· Heart failure: an exaggerated exercise pressor reflex adds excess sympathetic vasoconstriction during exercise.

Distinctions that matter

· Speed of each limb follows its signaling and its clearance: a directly gated channel with enzymatic clearance acts in milliseconds, a second messenger cascade with reuptake acts over seconds.

· Raising cardiac output needs venous return to rise with it, so the venous and pump mechanisms are part of the response rather than a side note.

Common misconceptions

· Treating the early rise in heart rate as sympathetic. It is mostly loss of vagal restraint.

· Expecting a denervated heart to be unable to raise cardiac output. It raises stroke volume by the Frank-Starling mechanism and raises rate later through circulating catecholamines.

Where it is taught: Section C01, paragraphs on vagal withdrawal, venous return, and the final paragraph on denervation, blockade, and heart failure of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C01-RT01

Retrieval exercise: Rebuild the onset sequence

A competent response contains

· Step 1: central command signals the medullary centers as motor units are recruited.

· Step 2: vagal outflow falls and heart rate rises within one or two beats, because IKACh channels close and acetylcholine is cleared within milliseconds.

· Step 3: group III mechanoreceptors reinforce the early heart rate rise.

· Step 4: sympathetic outflow builds over seconds and raises rate, raises contractility, and raises venous tone at the same time.

· Step 5: the muscle pump drives blood past venous valves toward the heart and the respiratory pump lowers intrathoracic pressure, so venous return and preload rise.

· Step 6: stroke volume and cardiac output rise as active arterioles dilate.

· Step 7: if flow lags metabolism, group IV afferents trigger the metaboreflex and raise arterial pressure.

· Central command starts the response and muscle feedback matches it to the working muscle.

Distinctions that matter

· The rate response and the stroke volume response arrive on different timescales: rate within a beat or two, stroke volume once venous return has risen.

· The metaboreflex is a correction that only comes in when perfusion is not keeping up, not a routine step in every bout.

Common misconceptions

· Putting sympathetic activation first. Vagal withdrawal precedes it.

· Leaving out the pumps and assuming a stronger heart alone raises cardiac output. Output cannot rise unless venous return rises with it.

Where it is taught: Section C01, flow diagram on the cardiovascular response at the onset of exercise of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C02-BD1

Brain dump 1 · foundational retrieval: Write the Fick principle and fill in the numbers

A competent response contains

· Fick principle: VO2 equals cardiac output times the arteriovenous O2 difference, which is arterial minus mixed venous O2 content. It holds because O2 is not stored.

· Arterial O2 content is about 1.34 mL O2 per gram of hemoglobin times hemoglobin concentration times saturation, plus 0.003 mL/dL per mmHg of dissolved PO2, giving about 20 mL/dL at a hemoglobin of 15 g/dL and a saturation of 98%.

· Healthy lungs hold arterial content near 20 mL/dL throughout exercise, so the arteriovenous difference widens from the venous side.

· Heart rate: about 70/min at rest and about 50/min in an athlete, about 195/min at maximum untrained, about 185 to 195/min in an elite athlete.

· Stroke volume: about 70 mL at rest and about 100 mL in an athlete, about 110 mL at maximum untrained, about 160 mL or more in an elite athlete.

· Cardiac output: about 5 L/min at rest, about 20 to 22 L/min at maximum untrained, about 30 to 35 L/min in an elite athlete.

· Mixed venous O2 content: about 15 mL/dL at rest (saturation about 75%), about 5 mL/dL at maximum untrained, about 3 to 4 mL/dL in an elite athlete.

· Arteriovenous O2 difference: about 5 mL/dL at rest, about 15 mL/dL at maximum untrained, about 16 to 17 mL/dL in an elite athlete.

· VO2: about 250 mL/min at rest (3.5 mL/kg/min, 1 MET), about 3 L/min (35 to 45 mL/kg/min) at maximum untrained, about 5 to 6 L/min (70 to 85 mL/kg/min) in an elite athlete.

· Extraction rises because metabolites recruit closed capillaries, enlarging the exchange surface and shortening diffusion distance; tissue PO2 falls onto the steep part of the dissociation curve; and higher PCO2, H+, and temperature shift the curve to the right.

· The right shift from CO2 and H+ is the Bohr effect.

Distinctions that matter

· Arterial content stays about the same while venous content falls, so the widening difference is a venous event.

· Capillary recruitment changes the exchange surface; the right shift changes hemoglobin's affinity. They are separate contributions to the same rise in extraction.

Common misconceptions

· Expecting arterial O2 content or saturation to rise with exercise in a healthy person. It is already nearly fully saturated at rest.

· Treating dissolved O2 as negligible to the point of leaving it out of the content equation; it is small but part of the calculation.

Where it is taught: Section C02, opening paragraphs, the table of typical values at rest and maximum, and the extraction paragraph of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C02-BD2

Brain dump 2 · integration and application: Work VO2max from the numbers, then take the hemoglobin away

A competent response contains

· Rest: 70/min times 70 mL is about 4.9 L/min; the difference is 20 minus 15, or 5 mL/dL, which is 50 mL/L, so VO2 is about 245 mL/min, roughly 3.3 mL/kg/min.

· Maximum: 195/min times 110 mL is about 21.5 L/min; the difference is 20 minus 5, or 15 mL/dL, which is 150 mL/L, so VO2max is about 3,200 mL/min, about 43 mL/kg/min.

· Flow rose about 4.4 fold and extraction about threefold.

· With a hemoglobin of 9 g/dL, arterial content is about 12 mL/dL, so even at a venous content of 4 mL/dL the maximal difference is 8 mL/dL, which is 80 mL/L.

· Holding a VO2 of about 2,400 mL/min would then need about 30 L/min of cardiac output, which is beyond the untrained maximum, so VO2max falls.

· Because each liter of blood carries less O2, heart rate is higher at any fixed submaximal workload.

· In a healthy person at sea level, VO2max is limited mainly by maximal cardiac output rather than by the lungs or the muscle: raising arterial O2 content raises VO2max and lowering it reduces VO2max, so maximal stroke volume is the main determinant.

· Stroke volume plateaus at about 40% to 60% of VO2max in untrained people because shortened diastole limits filling, so further rises in cardiac output come from heart rate; trained endurance athletes may keep raising stroke volume.

· Cardiac output rises about 5 to 6 L/min for each 1 L/min rise in VO2.

Distinctions that matter

· Anemia lowers the ceiling on the arteriovenous difference rather than the ability to extract: venous content still falls, but arterial content started lower.

· VO2 at a fixed submaximal workload is set by the work, so anemia shows up as a higher heart rate at that workload rather than a lower VO2.

Common misconceptions

· Mixing units: content in mL/dL must be converted to mL/L before multiplying by cardiac output in L/min.

· Naming the lungs as the limit to VO2max in healthy people at sea level. Arterial saturation is maintained, and maximal cardiac output is the limiting step.

Where it is taught: Section C02, worked example on the Fick equation and the paragraphs on heart rate, stroke volume, and VO2max of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-BD1

Brain dump 1 · foundational retrieval: Rebuild the distribution of cardiac output

A competent response contains

· Skeletal muscle: about 1.2 L/min (20%) at rest, about 22 L/min (about 87%) at maximum, controlled by metabolites and nitric oxide.

· Coronary: about 0.25 L/min (4%) at rest, about 1 L/min (4%) at maximum, controlled metabolically and mainly by adenosine.

· Brain: about 0.75 L/min (13%) at rest, about 0.75 L/min (3%) at maximum, controlled by autoregulation and PCO2.

· Skin: about 0.5 L/min (9%) at rest, about 0.6 L/min at maximum and higher during heavy submaximal work, controlled by hypothalamic thermoregulation.

· Splanchnic: about 1.4 L/min (24%) at rest, about 0.3 L/min at maximum, controlled by sympathetic constriction.

· Kidneys: about 1.1 L/min (19%) at rest, about 0.25 L/min at maximum, controlled by sympathetic constriction and angiotensin II.

· Muscle flow rises from about 3 to 4 mL/100 g/min at rest to about 50 to 80 mL/100 g/min in strenuous exercise, up to about 15 to 25 times resting.

· Interstitial K+ from repeated action potentials hyperpolarizes vascular smooth muscle through inward rectifier K+ channels and the Na+/K+-ATPase.

· Adenosine, inorganic phosphate, H+, CO2, and low PO2 add to the dilation; low PO2 makes red cells release ATP, which stimulates endothelial nitric oxide.

· Shear stress releases nitric oxide upstream, spreading dilation to the feed arteries, and blocking any single mediator only partly reduces the response.

· Extra cardiac output goes to active muscle, the heart, and, as body temperature rises, the skin.

Distinctions that matter

· Coronary flow quadruples while its percentage share stays about 4%, so a constant share is not a constant flow.

· Brain flow is held constant in absolute terms, so its percentage share falls as total output rises.

Common misconceptions

· Reading a falling percentage share as a falling flow. Cerebral flow is unchanged in L/min.

· Naming one mediator as the cause of muscle hyperemia. Several act together, which is why blocking one has only a partial effect.

Where it is taught: Section C03, distribution table and the paragraph on active skeletal muscle of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-BD2

Brain dump 2 · integration and application: Why total resistance falls while some beds constrict

A competent response contains

· Sympathetic nerves keep firing during exercise, but local metabolites blunt alpha-adrenergic constriction in active muscle, especially through alpha-2 receptors, which is functional sympatholysis, so inactive beds constrict while active muscle is protected.

· Some residual constriction is needed, because the dilating capacity of all skeletal muscle exceeds maximal cardiac output.

· Because the beds lie in parallel, their conductances add, so total peripheral resistance falls steeply in dynamic exercise even though other beds constrict.

· The myocardium already extracts about 70% to 75% of delivered O2 at rest, so higher demand has to be met by flow, which rises about three- to fourfold through mainly adenosine-mediated dilation.

· A coronary stenosis that is silent at rest therefore causes angina on exertion, because exercise raises myocardial O2 demand and the stenosis limits the required rise in flow.

· Cerebral flow stays nearly constant through autoregulation and PCO2 sensitivity.

· Splanchnic and renal flow fall to about one quarter of resting values at maximal effort, and splanchnic venoconstriction moves reservoir blood centrally.

· Renal filtration fraction rises because renal blood flow falls more than GFR does.

Distinctions that matter

· Sympathetic outflow rising everywhere and flow rising only in active muscle are reconciled by local control at the arteriole, not by selective nerve traffic.

· A bed meeting demand by extraction has no reserve left, which is why the coronary bed must answer with flow.

Common misconceptions

· Saying sympathetic outflow to active muscle is switched off. It continues, and its vasoconstrictor effect is blunted locally.

· Expecting total peripheral resistance to rise because most beds constrict. Parallel conductances add, and the enormous muscle conductance dominates.

Where it is taught: Section C03, functional sympatholysis paragraph, the flow diagram, and the paragraph on the other organs of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-RT01

Retrieval exercise: Rebuild how flow concentrates in working muscle

A competent response contains

· Step 1: sympathetic outflow rises to all vascular beds.

· Step 2: gut, kidney, and inactive muscle arterioles constrict through alpha receptors.

· Step 3: in active muscle, K+, adenosine, H+, CO2, low PO2, and nitric oxide dilate the arterioles.

· Step 4: the same metabolites blunt alpha constriction, which is functional sympatholysis, acting especially at alpha-2 receptors.

· Step 5: resistance falls in active muscle and rises in inactive beds.

· Step 6: the parallel beds share one arterial pressure, so flow shifts to active muscle.

· Flow through a bed equals the shared perfusion pressure divided by that bed's resistance, so each bed's share depends on its resistance relative to the others.

Distinctions that matter

· Constriction elsewhere and dilation in muscle both push flow the same way, but only the dilation explains the fall in total peripheral resistance.

· A shared perfusion pressure is what makes resistance the deciding variable for each bed's share.

Common misconceptions

· Treating the redistribution as a central decision about where to send blood. The brainstem raises outflow globally, and the arterioles settle the distribution locally.

· Omitting the shared arterial pressure from the explanation, which leaves no reason why one bed's dilation changes another bed's flow.

Where it is taught: Section C03, flow diagram on how flow concentrates in working muscle of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-PR01

Prediction exercise: Predict what heat does to a steady run

A competent response contains

· Skin blood flow rises on the hot day, because the hypothalamus withdraws cutaneous vasoconstrictor tone and activates sympathetic cholinergic fibers that cause active vasodilation and sweating, carrying heat to the skin.

· Blood available to fill the heart falls, because blood pools in compliant skin veins and sweat loss reduces plasma volume.

· Stroke volume falls over the 60 minutes as filling falls.

· Heart rate drifts upward to hold cardiac output, which is cardiovascular drift.

· Flow to the leg muscles falls, because muscle now shares cardiac output with skin, which lowers the sustainable workload.

· At maximal effort skin flow falls again, which favors muscle but limits heat loss, so core temperature rises further; above 40 °C (104 °F) core temperature reaches the range of exertional heat stroke.

· Core temperature rises toward 38 to 39 °C (100.4 to 102.2 °F) in sustained heavy exercise even without heat stress.

Distinctions that matter

· Cardiovascular drift is a fall in stroke volume with a compensating rise in heart rate, so cardiac output is roughly held rather than raised.

· Skin flow rises during heavy submaximal work and falls again at maximal effort, so the direction depends on the intensity.

Common misconceptions

· Reading a rising heart rate at a constant workload as rising cardiac output. It is compensation for a falling stroke volume.

· Assuming skin flow keeps rising as intensity rises. At maximal effort it is sacrificed to muscle perfusion.

Where it is taught: Section C03, skin blood flow paragraph and the stop and predict box of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-BD1

Brain dump 1 · foundational retrieval: Pressure in dynamic exercise against pressure in static exercise

A competent response contains

· Mean arterial pressure equals cardiac output times total peripheral resistance, ignoring the small central venous pressure.

· Dynamic exercise: heart rate rises largely with work rate, stroke volume rises, cardiac output rises severalfold, total peripheral resistance falls, systolic pressure rises progressively to about 160 to 200 mmHg at peak, diastolic pressure stays about the same or falls slightly, and pulse pressure widens.

· Systolic pressure rises in dynamic exercise because a larger stroke volume is ejected faster; diastolic pressure holds or falls because blood runs off through dilated muscle beds.

· The main pressor drive in dynamic exercise is central command with the mechanoreflex.

· Static exercise: heart rate rises moderately, stroke volume changes little or falls slightly, cardiac output rises modestly and mainly through heart rate, total peripheral resistance is unchanged or rises, and systolic and diastolic pressure both rise sharply.

· The main pressor drive in static exercise is the metaboreflex with central command.

· A sustained isometric contraction raises intramuscular pressure and compresses the muscle's own vessels, reducing flow above a moderate fraction of maximal force and nearly stopping it at high force, so metabolites accumulate and drive a strong sympathetic pressor response graded by relative force and duration.

· A Valsalva maneuver during maximal lifting adds intrathoracic pressure to the aorta.

· Dynamic exercise imposes a volume load and the pressure-volume loop widens more than it rises; static exercise imposes a pressure load and the loop becomes taller.

· By the Laplace relationship, wall stress is proportional to pressure times radius divided by wall thickness, so a pressure load raises wall stress and O2 demand.

· Arm exercise at the same VO2 raises heart rate and pressure more than leg exercise, because less muscle dilates while the inactive leg beds stay constricted.

· Most real activities mix the two patterns; rowing adds a large static component to a volume load.

Distinctions that matter

· Both forms raise cardiac output, and the direction of the resistance change is what separates their pressure responses.

· Pulse pressure widens in dynamic exercise because diastolic pressure holds while systolic rises; in static exercise both rise, so pulse pressure changes much less.

Common misconceptions

· Expecting diastolic pressure to rise in dynamic exercise. Runoff through dilated muscle beds holds it about the same or lowers it slightly.

· Treating a handgrip as a mild version of running. It loads the ventricle in the opposite way, with pressure rather than volume.

Where it is taught: Section C04, dynamic and static exercise paragraphs and the table of acute responses of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-BD2

Brain dump 2 · integration and application: Read pressures and work out resistance

A competent response contains

· Mean arterial pressure equals diastolic pressure plus one third of the pulse pressure; total peripheral resistance equals mean arterial pressure divided by cardiac output.

· Rest: 120/80 mmHg gives a pulse pressure of 40 and a mean arterial pressure of about 93 mmHg; 93 divided by 5 gives a total peripheral resistance of about 19 mmHg per L/min.

· Running: 180/76 mmHg gives a pulse pressure of 104 and a mean arterial pressure of about 111 mmHg; 111 divided by 20 gives about 5.5. Cardiac output quadrupled while mean arterial pressure rose only about 17 mmHg, because resistance fell to under one third of resting.

· Handgrip: 165/110 mmHg gives a pulse pressure of 55 and a mean arterial pressure of about 128 mmHg; 128 divided by 6.5 gives about 20. Cardiac output rose about 30%, yet mean arterial pressure rose about 35 mmHg because resistance did not fall.

· The one third formula is least accurate at high heart rates, because diastole shortens, so true running mean arterial pressure lies nearer the midpoint of systolic and diastolic, which still leaves total peripheral resistance far below resting.

· The rate-pressure product, heart rate multiplied by systolic pressure, indexes myocardial O2 demand.

· Pressure work costs more O2 than volume work, so a static effort can raise the rate-pressure product and provoke ischemia at a lower heart rate than running.

· A fall in systolic pressure as dynamic workload rises is abnormal and means cardiac output cannot offset the falling resistance, as in severe left ventricular dysfunction, extensive ischemia, or severe aortic stenosis.

· An exaggerated rise suggests resistance is not falling and may precede resting hypertension.

Distinctions that matter

· Mean arterial pressure and total peripheral resistance move independently: pressure can rise modestly while resistance falls steeply, because cardiac output is the third term.

· The rate-pressure product reflects demand, while coronary flow reflects supply, which is why pressure work is the more provocative load.

Common misconceptions

· Reading a higher pressure as a higher resistance. During running, pressure is up and resistance is sharply down.

· Applying the one third formula without comment at a heart rate of 180/min, where the relative duration of diastole has changed.

Where it is taught: Section C04, worked example on calculating total peripheral resistance and the final paragraph on the rate-pressure product and abnormal responses of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-CP01

Comparison table: Dynamic against static exercise

A competent response contains

· Heart rate: large rise with work rate in dynamic exercise, moderate rise in static exercise.

· Stroke volume: rises in dynamic exercise, little change or a slight fall in static exercise.

· Cardiac output: large rise in dynamic exercise, modest rise in static exercise and mainly from heart rate.

· Total peripheral resistance: falls in dynamic exercise, unchanged or higher in static exercise.

· Systolic pressure: rises progressively in dynamic exercise, rises sharply in static exercise.

· Diastolic pressure: unchanged or slightly lower in dynamic exercise, so pulse pressure widens; rises sharply in static exercise, so pulse pressure widens much less.

· Main pressor drive: central command and the mechanoreflex in dynamic exercise, the metaboreflex and central command in static exercise.

· Load on the left ventricle: volume load in dynamic exercise, with the pressure-volume loop widening more than it rises; pressure load in static exercise, with the loop becoming taller.

· A sustained handgrip raises afterload, so the murmurs of mitral regurgitation, aortic regurgitation, and ventricular septal defect get louder, while the murmur of hypertrophic cardiomyopathy softens.

Distinctions that matter

· Volume load and pressure load are the summary of the whole table, and the direction of the resistance change is what produces them.

· Raising afterload increases regurgitant and shunt murmurs because more blood takes the low pressure route, and softens the obstructive murmur of hypertrophic cardiomyopathy because the ventricle holds a larger cavity.

Common misconceptions

· Writing that cardiac output barely changes in static exercise. It rises modestly, mainly through heart rate.

· Expecting every murmur to behave the same way with handgrip. The obstructive murmur of hypertrophic cardiomyopathy moves in the opposite direction.

Where it is taught: Section C04, table of acute responses and the final paragraph on handgrip and murmurs of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-BD1

Brain dump 1 · foundational retrieval: Eccentric and concentric remodeling, and the trained heart

A competent response contains

· Adult myocytes rarely divide, so the heart adapts to a chronic load by enlarging existing cells.

· Eccentric hypertrophy adds sarcomeres in series, enlarging the cavity with proportional wall thickening, and is the response to a volume load.

· Concentric hypertrophy adds sarcomeres in parallel, thickening the wall relative to cavity size, and is the response to a pressure load.

· Both patterns bring wall stress back toward normal, which follows from the Laplace relationship.

· The endurance-trained left ventricle is remodeled eccentrically, with mild symmetric wall thickening, a normal wall-to-cavity ratio, an enlarged right ventricle and atria, and normal or enhanced diastolic filling.

· Plasma volume expands within days through renin-angiotensin-aldosterone, vasopressin, and albumin, faster than red cell mass, so hemoglobin concentration may fall slightly by dilution rather than from anemia.

· With resting cardiac output still about 5 L/min, a stroke volume of about 100 mL needs only about 50 bpm.

· The slowing comes mainly from vagal tone, with a contribution from a lower intrinsic SA node rate.

· Vagal tone also explains benign sinus arrhythmia, first degree AV block, and Mobitz I block, all of which disappear with exercise.

· At the same submaximal workload, the trained person reaches the same VO2 and cardiac output with a lower heart rate, a larger stroke volume, and less lactate.

· At maximum, cardiac output, extraction, and VO2max rise, but maximal heart rate does not; greater muscle capillary and mitochondrial density raise extraction and fat oxidation, and VO2max typically rises about 15% to 20% in previously sedentary people.

Distinctions that matter

· Training raises stroke volume and VO2max while leaving resting cardiac output and maximal heart rate about the same.

· A slightly lower hemoglobin concentration in a trained athlete is dilution from plasma volume expansion, not iron deficiency.

Common misconceptions

· Explaining athletic bradycardia as a weaker heart. The rate is lower because each beat ejects more.

· Reading first degree AV block or Mobitz I in a trained athlete as disease. Both are vagally mediated and resolve with exercise.

Where it is taught: Section C05, opening paragraph, the endurance-trained heart paragraphs, and the flow diagram from volume load to higher VO2max of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-BD2

Brain dump 2 · integration and application: Separate physiologic growth from sarcomere disease

A competent response contains

· Physiologic hypertrophy is driven by insulin-like growth factor 1 through PI3K and Akt, with matching capillary growth and no fibrosis.

· Pathologic hypertrophy is driven by wall stress and neurohormones, namely angiotensin II, endothelin 1, and norepinephrine at alpha-1 receptors, through Gq-coupled receptors and calcineurin and NFAT signaling.

· Pathologic growth reactivates fetal genes, including beta-myosin heavy chain and the natriuretic peptides, and brings fibrosis, capillary insufficiency, and diastolic and then systolic dysfunction.

· Hypertrophic cardiomyopathy is usually autosomal dominant with variable penetrance, caused by sarcomere gene mutations, most often MYH7 for beta-myosin heavy chain and MYBPC3 for cardiac myosin binding protein C.

· Exertional dyspnea: the stiff ventricle fills poorly, so filling pressures are high.

· Angina with normal epicardial arteries: a large muscle mass is supplied through diseased intramural arterioles.

· Sudden death in a young athlete: myocyte disarray with fibrosis is a substrate for reentrant ventricular arrhythmias.

· Dynamic outflow obstruction comes from the septum together with systolic anterior motion of the mitral valve, and it worsens with lower preload (standing, Valsalva strain, dehydration), lower afterload, greater contractility, or tachycardia, because each makes the ventricle smaller or more forceful.

· The murmur therefore gets louder with standing or Valsalva and softer with squatting or handgrip, which raise preload and afterload and enlarge the cavity.

· After about three months of detraining the athlete's heart regresses, while hypertrophic cardiomyopathy does not, which is why detraining separates them.

· The same muscle mass is physiologic or pathologic according to the stimulus and the signaling pathway behind it.

Distinctions that matter

· Both conditions enlarge myocytes, and the pathway (IGF-1 with PI3K and Akt against Gq with calcineurin and NFAT) decides whether capillaries and interstitium keep pace.

· Anything that shrinks the cavity worsens the gradient, so maneuvers are judged by their effect on ventricular size rather than on pressure alone.

Common misconceptions

· Expecting angina to mean epicardial coronary disease. In hypertrophic cardiomyopathy the large mass and diseased intramural arterioles are enough.

· Predicting that handgrip makes every murmur louder. It raises afterload, enlarges the cavity, and softens this murmur.

Where it is taught: Section C05, paragraphs on physiologic versus pathologic hypertrophy, on hypertrophic cardiomyopathy, on dynamic obstruction, and the stop and predict box of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-CP01

Comparison table: Athlete's heart against hypertrophic cardiomyopathy

A competent response contains

· Wall thickness: symmetric and usually 12 mm or less in the athlete; often asymmetric septal and 15 mm or more in hypertrophic cardiomyopathy.

· Cavity: enlarged in the athlete, often above 55 mm; small or normal in hypertrophic cardiomyopathy, often below 45 mm.

· Diastolic function: normal or enhanced in the athlete; impaired relaxation in hypertrophic cardiomyopathy.

· Histology: orderly myocytes with no fibrosis in the athlete; myocyte disarray with fibrosis in hypertrophic cardiomyopathy.

· Outflow tract: no obstruction in the athlete; dynamic obstruction in many patients with hypertrophic cardiomyopathy.

· Electrocardiogram: bradycardia with isolated high voltage in the athlete; deep T wave inversion and pathologic Q waves in hypertrophic cardiomyopathy.

· Peak VO2: high in the athlete; normal or reduced in hypertrophic cardiomyopathy.

· Detraining for about three months: the athlete's heart regresses, hypertrophic cardiomyopathy does not.

· Family history: usually negative in the athlete; often hypertrophic cardiomyopathy or sudden death in affected families.

· Thresholds are approximate, and in the 13 to 15 mm gray zone the features are weighed together.

Distinctions that matter

· Wall thickness alone does not separate the two in the gray zone; the cavity size, diastolic filling, and family history carry the decision.

· A large cavity with proportional thickening is eccentric remodeling, while a small cavity with a thick asymmetric wall is not a training pattern at all.

Common misconceptions

· Using high QRS voltage as evidence of disease. Isolated high voltage is common in trained athletes.

· Treating the thresholds as fixed cutoffs rather than approximations that are weighed together.

Where it is taught: Section C05, athlete's heart against hypertrophic cardiomyopathy table of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-CA01

Clinical application: Near syncope in a young athlete during a hot race

A competent response contains

· Recognize: hypertrophic cardiomyopathy from sarcomere protein gene mutations, most often MYH7 for beta-myosin heavy chain and MYBPC3 for cardiac myosin binding protein C, usually autosomal dominant with variable penetrance.

· Explain: the stiff ventricle fills poorly, so filling pressures rise and exertional dyspnea follows; a large muscle mass supplied through diseased intramural arterioles causes angina even when the epicardial arteries are normal, and exercise raises myocardial O2 demand.

· Differentiate: 18 mm is well above the 12 mm or less of a trained heart; the septum is 18 mm against a posterior wall of 10 mm, so the hypertrophy is asymmetric rather than symmetric; the cavity at 42 mm is small rather than above 55 mm; relaxation is impaired rather than normal or enhanced; deep lateral T wave inversions and pathologic Q waves are not the isolated high voltage of an athlete; and the sudden death of a maternal uncle at 31 years fits a family history of the disease.

· Investigate: measure septal and posterior wall thickness, expecting 15 mm or more and asymmetric rather than 12 mm or less and symmetric; measure the cavity dimension, expecting below 45 mm rather than above 55 mm; assess diastolic filling, expecting impaired relaxation rather than normal or enhanced filling; read the electrocardiogram for deep T wave inversion and pathologic Q waves rather than isolated high voltage; look for dynamic outflow obstruction; take the family history for hypertrophic cardiomyopathy and sudden death; and note that the athlete's pattern would regress after about three months of detraining while this will not.

· Decide: dehydration from running in the heat lowered preload, and the heat also lowered afterload while catecholamines raised contractility and rate, each of which made the ventricle smaller and more forceful and worsened the outflow gradient. The risk of sudden death comes from reentrant ventricular arrhythmias arising in disarrayed, fibrotic myocardium.

· Justify: lower preload, lower afterload, greater contractility, and tachycardia all reduce ventricular size during systole, and a smaller cavity brings the septum and the mitral leaflet closer together, so obstruction worsens. Standing lowers preload and makes the murmur louder; a sustained handgrip raises afterload, enlarges the cavity, and makes it softer.

Distinctions that matter

· Chest tightness here is a supply and demand problem within an abnormal myocardium rather than epicardial stenosis.

· Maneuvers are read by their effect on ventricular size: smaller cavity means louder murmur in this condition and the opposite in regurgitant lesions.

Common misconceptions

· Attributing the findings to training because the patient is an athlete with a slow pulse. Asymmetry, the small cavity, impaired relaxation, the electrocardiogram, and the family history all point away from adaptation.

· Expecting the murmur to behave like a regurgitant murmur with handgrip. Raising afterload softens this one.

Where it is taught: Section C05, paragraphs on hypertrophic cardiomyopathy and dynamic outflow obstruction, the comparison table, and the stop and predict box of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C02-Q01

Board-style question: Accounting for the rise in oxygen uptake

Correct answer: B. An approximately fourfold rise in cardiac output together with an approximately threefold rise in O2 extraction

Why it is right: Both terms of the Fick equation rise: cardiac output goes from 5 to 21 L/min as heart rate and stroke volume increase, and extraction goes from 5 to 15 mL/dL as capillaries are recruited, tissue PO2 falls onto the steep part of the dissociation curve, and higher PCO2, H+, and temperature shift the curve to the right.

A is wrong: Arterial O2 content is 20 mL/dL at rest and 20 mL/dL at peak in this man, so pulmonary gas exchange contributed nothing to the rise.

C is wrong: The arteriovenous difference clearly changed, from 5 to 15 mL/dL, so attributing the rise to flow alone contradicts the numbers given.

D is wrong: Cardiac output rose from 5 to 21 L/min, so it did not stay at its resting value and extraction cannot be the only contributor.

A competent response contains

· VO2 equals cardiac output times the arteriovenous O2 difference.

· Cardiac output rose from 5 to 21 L/min, about fourfold, and the arteriovenous difference rose from 5 to 15 mL/dL, about threefold, so VO2 rose from about 250 mL/min to about 3.2 L/min.

Distinctions that matter

· Arterial content is held constant by healthy lungs, so the whole widening of the difference comes from the venous side.

Common misconceptions

· Expecting arterial O2 content to rise with exercise. It is already nearly fully saturated at rest.

Where it is taught: Section C02 of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C03-Q02

Board-style question: Regional flow during submaximal cycling

Correct answer: A. Coronary blood flow rises about three- to fourfold, because myocardial O2 extraction is already near maximal at rest

Why it is right: Resting myocardial extraction of about 70% to 75% leaves almost no extraction reserve, so the rise in myocardial O2 demand during exercise has to be met by a three- to fourfold rise in coronary flow, produced mainly by adenosine-mediated dilation.

B is wrong: Renal blood flow falls to about one quarter of its resting value at high workloads through sympathetic constriction and angiotensin II, and filtration fraction rises because flow falls more than GFR.

C is wrong: Cerebral blood flow is held nearly constant by autoregulation and PCO2 sensitivity, so its absolute flow stays about 0.75 L/min while its percentage share falls.

D is wrong: Splanchnic blood flow falls from about 1.4 to about 0.3 L/min, and splanchnic venoconstriction moves reservoir blood centrally, because sympathetic constriction dominates that bed.

A competent response contains

· The myocardium already extracts about 70% to 75% of delivered O2 at rest, so extra demand has to be met by raising flow.

· Coronary flow rises about three- to fourfold through mainly adenosine-mediated dilation, from about 0.25 to about 1 L/min.

Distinctions that matter

· A bed that already extracts nearly all the O2 it receives has no extraction reserve, so flow is its only route to meeting demand.

Common misconceptions

· Reading an unchanged percentage share of cardiac output as unchanged flow. The coronary share stays about 4% while flow quadruples.

Where it is taught: Section C03 of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C04-Q03

Board-style question: Hemodynamics of a sustained handgrip

Correct answer: D. Total peripheral resistance is unchanged or higher, and systolic and diastolic pressure both rise sharply

Why it is right: Rising intramuscular pressure restricts the contracting muscle's own flow, so metabolites accumulate and the metaboreflex with central command produces widespread sympathetic vasoconstriction; total peripheral resistance stays the same or rises while cardiac output rises modestly, so both systolic and diastolic pressure climb sharply and the ventricle faces a pressure load.

A is wrong: Falling resistance with a slightly lower diastolic pressure and a markedly higher stroke volume is the dynamic exercise pattern, not the response to a sustained handgrip.

B is wrong: A fourfold rise in cardiac output with a widened pulse pressure describes vigorous running; in static exercise output rises only modestly and mainly through heart rate.

C is wrong: Heart rate rises moderately during a handgrip rather than falling, because central command resets the baroreflex upward so the reflex does not oppose the rise in pressure.

A competent response contains

· A sustained isometric contraction compresses the muscle's own vessels, metabolites accumulate, and the metaboreflex with central command drives a strong sympathetic pressor response.

· Cardiac output rises modestly through heart rate, stroke volume changes little, and total peripheral resistance is unchanged or rises, so systolic and diastolic pressure both rise sharply.

Distinctions that matter

· The pressure response in static exercise comes from resistance failing to fall, while in dynamic exercise it comes from a large rise in output against falling resistance.

Common misconceptions

· Treating a handgrip as a small version of running. It loads the ventricle with pressure rather than volume.

Where it is taught: Section C04 of the CV-12 Exercise and the Cardiovascular System notes.

NPLEX-P1-CARD-12-01-C05-Q04

Board-style question: Reading an echocardiogram in a trained cyclist

Correct answer: C. The enlarged cavity with symmetric wall thickening and enhanced diastolic filling

Why it is right: An enlarged cavity above 55 mm with proportional symmetric thickening and enhanced early filling is the eccentric pattern of a repeated volume load, and it is the opposite of the small, stiff, asymmetrically thickened ventricle of hypertrophic cardiomyopathy.

A is wrong: A resting rate of 46/min reflects vagal tone and a large stroke volume, but bradycardia alone does not exclude hypertrophic cardiomyopathy.

B is wrong: Isolated high QRS voltage is common in trained athletes and can also occur with pathologic hypertrophy, so on its own it carries little weight.

D is wrong: A wall thickness of 11 mm sits below the 12 mm limit for an athlete, but thickness taken on its own leaves the gray zone unresolved and ignores the cavity and diastolic findings that decide the case.

A competent response contains

· Endurance training remodels the left ventricle eccentrically: an enlarged cavity, mild symmetric wall thickening, a normal wall-to-cavity ratio, an enlarged right ventricle and atria, and normal or enhanced diastolic filling.

· Hypertrophic cardiomyopathy gives a small or normal cavity, often below 45 mm, with asymmetric septal thickening of 15 mm or more and impaired relaxation.

Distinctions that matter

· Cavity size with diastolic function is the discriminator, because hypertrophic cardiomyopathy produces a small stiff cavity while training produces a large compliant one.

Common misconceptions

· Taking bradycardia or high voltage as evidence of adaptation. Both can appear in either condition and neither settles the question.

Where it is taught: Section C05 of the CV-12 Exercise and the Cardiovascular System notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-13

CV-13 Cardiovascular Biochemistry and Nutrition

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-13-01-C01-BD1

Brain dump 1 · foundational retrieval: Build the lipoprotein table from memory

A competent response contains

· Chylomicron: made by enterocytes, carries apo B-48 with C-II and E borrowed from HDL, delivers dietary triglyceride to muscle and adipose tissue, and its remnant returns to the liver.

· VLDL: made by the liver, carries apo B-100 with C-II and E from HDL, and carries liver-made triglyceride to tissues.

· IDL: formed from VLDL in plasma, carries apo B-100 and E, and is either taken up by the liver, about half of it, or converted to LDL.

· LDL: formed from IDL, carries apo B-100 only, and is the main plasma carrier of cholesterol to the liver and peripheral cells.

· HDL: made by liver and intestine as lipid-poor apo A-I, carries A-I, and runs reverse cholesterol transport while lending C-II and E to other particles.

· As lipoprotein lipase removes triglyceride, particles become smaller, denser, and proportionally richer in cholesterol, so density rises from chylomicrons through VLDL, IDL, and LDL to HDL, which has the most protein relative to lipid.

· Apo B-48 is needed for chylomicron assembly and lacks the LDL receptor-binding region; apo B-100 is the LDL receptor ligand; apo C-II activates lipoprotein lipase; apo E mediates hepatic uptake of remnants and IDL; apo A-I activates LCAT and accepts cholesterol through ABCA1.

· LCAT is a plasma enzyme that esterifies free cholesterol on the HDL surface so it moves into the core; ACAT esterifies cholesterol inside cells for storage.

· Apo B never leaves its particle, so each chylomicron carries one B-48 and each VLDL, IDL, and LDL one B-100.

Distinctions that matter

· Apo B-48 and apo B-100 come from the same APOB gene; intestinal RNA editing changes one cytosine to uracil and creates a stop codon at about 48% of full length.

· LCAT works outside the cell on HDL and ACAT works inside the cell, so the two esterification steps serve opposite purposes: keeping the efflux gradient open against storing excess.

Common misconceptions

· Writing that HDL is dense because it is fat-rich. It is the most protein-rich and lipid-poor particle, which is exactly why it is the densest.

· Treating apo C-II and apo E as native to chylomicrons and VLDL. HDL lends them, which is why HDL loss slows triglyceride clearance as well as reverse transport.

Where it is taught: Section C01, opening paragraph and the lipoprotein table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C01-BD2

Brain dump 2 · integration and application: Three broken steps in lipid transport

A competent response contains

· Patient 1 without apo C-II: lipoprotein lipase cannot be activated, so chylomicron and VLDL triglyceride is not hydrolyzed and chylomicrons accumulate; triglyceride rises most, and refrigerated plasma shows a creamy top layer because chylomicrons are the largest and lightest particle.

· Patient 2 without ABCA1: lipid-poor apo A-I cannot collect free cholesterol and phospholipid from cells, so HDL is nearly absent, which is Tangier disease; cholesterol carried on HDL is very low and plasma shows neither a cream layer nor marked turbidity.

· Patient 3 with apo E2 homozygosity: remnants and IDL bind hepatic receptors poorly and accumulate, raising cholesterol and triglyceride to similar levels, and plasma is turbid rather than creamy because the particles are remnants rather than intact chylomicrons.

· Particles upstream cannot move on because each step needs the one before it: no activator means no hydrolysis, no acceptor means no efflux, and no receptor ligand means no uptake.

· High triglyceride with low HDL arises because CETP transfers cholesteryl ester from HDL to abundant apo B particles in exchange for triglyceride, and hepatic lipase then removes that triglyceride, leaving small HDL that is cleared quickly.

· Normal plasma is free of chylomicrons after a 12-hour fast, so finding them in a fasting sample means clearance has failed.

Distinctions that matter

· Chylomicrons float and VLDL stays suspended, so the appearance of refrigerated plasma separates a clearance defect in the exogenous pathway from overproduction in the endogenous one.

· Apo E2 homozygosity usually needs a second metabolic load such as obesity, diabetes, hypothyroidism, or alcohol before lipids rise.

Common misconceptions

· Expecting severe atherosclerosis whenever triglyceride is very high. Chylomicrons are too large to enter the intima efficiently, so type I disease adds little atherosclerotic risk.

· Treating a low HDL as an independent second disorder when triglyceride is high. The CETP and hepatic lipase sequence links the two.

Where it is taught: Section C01, HDL and reverse cholesterol transport paragraph, with the Fredrickson table in C02 of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C01-RT01

Retrieval exercise: Rebuild the exogenous lipoprotein pathway

A competent response contains

· Step 1: enterocytes resynthesize absorbed fat into triglyceride and microsomal triglyceride transfer protein loads it, with cholesteryl ester and fat-soluble vitamins, onto apo B-48 as a chylomicron.

· Step 2: the particle travels through lacteals and the thoracic duct, because chylomicrons are too large to enter blood capillaries, which is why dietary fat reaches peripheral tissues before the liver.

· Step 3: HDL donates apo C-II and apo E.

· Step 4: apo C-II activates lipoprotein lipase on the luminal surface of capillary endothelium, which hydrolyzes core triglyceride and releases fatty acids.

· Step 5: those fatty acids enter adipose tissue, skeletal muscle, and heart.

· Step 6: the liver takes up the remnant through apo E.

· Lipoprotein lipase deficiency stops the pathway at step 4, so chylomicrons accumulate; normal plasma is clear of them after a 12-hour fast.

· Endogenous pathway: the liver secretes VLDL on apo B-100, lipoprotein lipase converts VLDL to IDL, about half of IDL is taken up through apo E, hepatic lipase converts the rest to LDL, and LDL loses apo E and is cleared through apo B-100 binding the LDL receptor, mostly in the liver, with adrenal cortex and gonads taking some for steroid synthesis.

· Insulin raises adipose lipoprotein lipase after a meal, while the cardiac enzyme has a lower Km and keeps supplying the heart during fasting.

Distinctions that matter

· Lipoprotein lipase acts on capillary endothelium and makes IDL from VLDL; hepatic lipase acts at the liver and makes LDL from IDL.

· LDL lasts days in plasma while chylomicrons are cleared within hours, which is why fasting lipid studies measure LDL reliably.

Common misconceptions

· Drawing chylomicrons entering the portal vein. They enter lymph and reach the blood through the thoracic duct, bypassing the first pass through the liver.

· Saying LDL is secreted by the liver. The liver secretes VLDL, and LDL is formed in plasma from IDL.

Where it is taught: Section C01, exogenous pathway flow diagram and the paragraph that follows it of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C01-Q01

Board-style question: Reverse cholesterol transport fails at its first step

Correct answer: A. Transfer of free cholesterol and phospholipid from the cell membrane to lipid-poor apolipoprotein A-I

Why it is right: ABCA1 moves free cholesterol and phospholipid from the cell membrane onto lipid-poor apo A-I, so losing it leaves HDL almost absent and traps cholesteryl ester in macrophages and tonsillar tissue, which gives the orange tonsils and the neuropathy of Tangier disease.

B is wrong: Loss of LCAT would also lower HDL cholesterol, but the esterification step comes after efflux and would not fill macrophages with ester or give orange tonsils.

C is wrong: Lipoprotein lipase failure accumulates chylomicrons and raises triglyceride above 1,000 mg/dL with a creamy plasma layer, which this triglyceride of 210 mg/dL does not match.

D is wrong: Failure of apo E mediated remnant uptake raises both cholesterol and triglyceride to similar levels with palmar xanthomas, not an isolated HDL of 3 mg/dL.

A competent response contains

· Lipid-poor apo A-I collects free cholesterol and phospholipid from cells through ABCA1, and ABCA1 is absent in Tangier disease, in which HDL is nearly absent.

· Peripheral cells cannot degrade the sterol ring, so cholesterol that cannot be exported stays stored as cholesteryl ester, which is why macrophages and tonsillar tissue fill with it.

Distinctions that matter

· An efflux defect lowers HDL cholesterol without raising LDL, which separates it from the disorders that accumulate apo B particles.

Common misconceptions

· Reading a very low HDL as a consequence of the raised triglyceride alone. Here the first step of efflux is missing, and the stored ester in macrophages points to it.

Where it is taught: Section C01, HDL and reverse cholesterol transport paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C02-BD1

Brain dump 1 · foundational retrieval: Build the Fredrickson table from memory

A competent response contains

· Type I familial chylomicronemia, autosomal recessive: lipoprotein lipase or apo C-II deficiency raises chylomicrons with triglyceride often above 1,000 mg/dL, causing pancreatitis, eruptive xanthomas, and lipemia retinalis with little added atherosclerosis.

· Type IIa familial hypercholesterolemia, autosomal dominant: a defective LDL receptor, or abnormal apo B-100, or gain-of-function PCSK9, raises LDL with high cholesterol and normal triglyceride, causing premature atherosclerosis, tendon xanthomas, xanthelasma, and early corneal arcus.

· Type IIb combined: hepatic overproduction of apo B particles, usually polygenic, raises LDL and VLDL with both lipids up and premature atherosclerosis.

· Type III dysbetalipoproteinemia: apo E2 homozygosity plus a second metabolic factor raises remnants and IDL with both lipids raised to similar levels, giving palmar and tuberoeruptive xanthomas and premature coronary and peripheral disease.

· Type IV familial hypertriglyceridemia: hepatic VLDL overproduction raises triglyceride with cholesterol near normal, causing pancreatitis if very high and linked to insulin resistance.

· Type V: VLDL overproduction with impaired clearance raises both VLDL and chylomicrons, with pancreatitis and eruptive xanthomas.

· Desirable adult values: total cholesterol below 200 mg/dL, LDL below 100 mg/dL, HDL at least 40 mg/dL in men and 50 mg/dL in women, fasting triglyceride below 150 mg/dL.

· Secondary causes to exclude first: diabetes, hypothyroidism, nephrotic syndrome, and alcohol.

Distinctions that matter

· High triglyceride points to chylomicrons or VLDL, and high cholesterol with a normal triglyceride points to LDL, so the lipid pattern names the particle before the defect.

· Type I and type IV both raise triglyceride, but one is a clearance failure of chylomicrons and the other an overproduction of VLDL.

Common misconceptions

· Treating the Fredrickson types as single diseases. They describe patterns of raised lipoproteins, and several molecular causes can give the same type.

· Calling a pattern familial before excluding diabetes, hypothyroidism, nephrotic syndrome, and alcohol.

Where it is taught: Section C02, opening paragraph and the Fredrickson table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C02-BD2

Brain dump 2 · integration and application: Name the type from the numbers

A competent response contains

· Patient 1: type I, chylomicrons accumulating, lipoprotein lipase or apo C-II deficiency, settled by the creamy floating layer over a clear infranatant in a fasting sample; the modest cholesterol rise is the cholesterol each chylomicron carries.

· Patient 2: type IV, VLDL accumulating, hepatic VLDL overproduction, settled by uniform turbidity with triglyceride raised and cholesterol near normal; the high triglyceride, low HDL, small dense LDL pattern is the dyslipidemia of the metabolic syndrome.

· Patient 3: type III, remnants and IDL accumulating, apo E2 homozygosity with a second metabolic load, settled by palmar xanthomas with cholesterol and triglyceride raised to similar levels.

· Patient 4: type IIa, LDL accumulating, a defective LDL receptor, settled by tendon xanthomas with a very high cholesterol, a normal triglyceride, and a first-degree relative with early coronary disease.

· Pancreatitis in patient 1: pancreatic lipase hydrolyzes chylomicron triglyceride in pancreatic capillaries, and the released free fatty acids injure acinar cells.

· Atherosclerotic risk is not greatly raised in patient 1 because chylomicrons are too large to enter the intima efficiently.

· In patient 4 the hepatocytes have too few receptors, so they behave as if cholesterol-poor and keep synthesizing cholesterol and secreting VLDL, which adds still more LDL.

Distinctions that matter

· Which lipid is higher, and the appearance of refrigerated plasma, separate the triglyceride disorders from each other before any genetic test.

· Palmar and tuberoeruptive xanthomas belong to remnant accumulation, while tendon xanthomas and xanthelasma belong to LDL accumulation.

Common misconceptions

· Reading a high plasma cholesterol as a signal that will switch off hepatic synthesis. The hepatocyte senses its own sterol, not the plasma level, so receptor loss leaves synthesis running.

· Assuming any very high triglyceride causes atherosclerosis. Particle size decides whether the lipid can be retained in the intima.

Where it is taught: Section C02, paragraph on mechanisms and the worked example on classifying a dyslipidemia of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C02-CP01

Comparison table: Type I against type IIa against type III

A competent response contains

· Defect and inheritance: type I is lipoprotein lipase or apo C-II deficiency, autosomal recessive; type IIa is a defective LDL receptor, or abnormal apo B-100, or gain-of-function PCSK9, autosomal dominant; type III is apo E2 homozygosity plus a second metabolic factor.

· Particle: type I accumulates chylomicrons, type IIa accumulates LDL, type III accumulates remnants and IDL.

· Lipid pattern: type I has triglyceride often above 1,000 mg/dL with a modest cholesterol rise; type IIa has high cholesterol with a normal triglyceride; type III has cholesterol and triglyceride raised to similar levels.

· Refrigerated plasma: type I shows a creamy floating layer over clear plasma; type IIa leaves plasma clear; type III leaves plasma turbid from remnant particles.

· Skin and eye findings: type I gives eruptive xanthomas and lipemia retinalis; type IIa gives tendon xanthomas on the Achilles and hand extensor tendons, xanthelasma, and early corneal arcus; type III gives palmar and tuberoeruptive xanthomas.

· Atherosclerotic risk: little added risk in type I because chylomicrons are too large to enter the intima efficiently; premature atherosclerosis in type IIa because LDL is retained in the intima; premature coronary and peripheral disease in type III because remnants are also retained.

· What else must be present: type I and type IIa express with the defect alone, while most apo E2 homozygotes have near-normal lipids until obesity, diabetes, hypothyroidism, or alcohol adds a second load.

Distinctions that matter

· Particle size decides atherosclerotic risk, which is why the highest triglyceride of the three carries the lowest arterial risk.

· Type III needs a second metabolic factor to appear, so a normal lipid panel does not exclude the genotype.

Common misconceptions

· Ranking risk by triglyceride level. The retained particle, not the lipid number, sets arterial risk.

· Expecting xanthomas to look the same across the types. Their site is the clue: tendons in type IIa, palmar creases in type III, eruptive skin lesions in type I.

Where it is taught: Section C02, Fredrickson table, mechanism paragraph, and worked example of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-BD1

Brain dump 1 · foundational retrieval: Cholesterol synthesis and the four controls on the reductase

A competent response contains

· Three cytosolic acetyl-CoA form HMG-CoA through cytosolic HMG-CoA synthase; the mitochondrial isoform of that synthase instead makes HMG-CoA for ketogenesis.

· HMG-CoA reductase, in the endoplasmic reticulum membrane, reduces HMG-CoA to mevalonate using 2 NADPH, and this is the rate-limiting and main regulated step.

· Order after that: mevalonate, isoprene units, farnesyl pyrophosphate, squalene, lanosterol, cholesterol.

· Farnesyl pyrophosphate is the branch point, also supplying ubiquinone, dolichol, and protein lipid anchors, so blocking the reductase lowers these as well.

· Control 1, transcription: SREBP-2 raises reductase transcription when cell sterol is low.

· Control 2, degradation: accumulating sterols make the enzyme bind Insig, which leads to ubiquitination and proteasomal degradation.

· Control 3, phosphorylation: AMPK phosphorylates and inactivates the enzyme when energy is short.

· Control 4, hormones: insulin and thyroid hormone increase activity or expression, while glucagon and glucocorticoids decrease them.

Distinctions that matter

· Sterols act at two different levels, raising degradation of existing enzyme and lowering transcription of new enzyme, which is why the response is both fast and sustained.

· AMPK reports energy status while sterols report cholesterol status, so the enzyme can be switched off for either reason.

Common misconceptions

· Naming HMG-CoA synthase as the regulated step. The reductase that makes mevalonate is rate-limiting.

· Treating the pathway as making only cholesterol. Ubiquinone, dolichol, and protein lipid anchors branch off at farnesyl pyrophosphate.

Where it is taught: Section C03, synthesis and control of HMG-CoA reductase of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-BD2

Brain dump 2 · integration and application: Follow one LDL particle into the cell and out again

A competent response contains

· Apo B-100 binds the LDL receptor, which also binds apo E.

· The receptor's NPXY tail motif gathers it into clathrin-coated pits, and dynamin pinches the vesicle off.

· In the acidified endosome LDL is released from the receptor; the receptor recycles to the surface and LDL goes to the lysosome.

· In the lysosome apo B-100 is degraded and acid lipase frees cholesterol, which leaves through NPC1 and NPC2.

· Blocked steps named in the notes: a defective receptor or defective apo B-100 in familial hypercholesterolemia, gain-of-function PCSK9 which diverts receptors to the lysosome and shortens their life, and NPC1 or NPC2 failure which traps cholesterol in Niemann-Pick type C.

· The released cholesterol suppresses SREBP processing, speeds reductase degradation, and activates ACAT to store the excess as cholesteryl ester.

· Hypothyroidism lowers LDL receptor expression, so clearance falls and plasma LDL rises, which makes it a common secondary cause of high LDL.

· Macrophage scavenger receptors such as SR-A and CD36 take up oxidized LDL and are not downregulated by cell cholesterol, so the cell keeps loading until it is full of cholesteryl ester and becomes a foam cell.

Distinctions that matter

· The receptor recycles and the ligand is degraded, so one receptor can clear many particles unless PCSK9 sends it to the lysosome.

· The LDL receptor is regulated by cell sterol and the scavenger receptor is not, which is the whole reason foam cells form.

Common misconceptions

· Saying the receptor is degraded with its ligand. It releases LDL in the acidic endosome and returns to the surface.

· Treating free cholesterol as the storage form. ACAT esterifies it, because excess free cholesterol is toxic to membranes.

Where it is taught: Section C03, LDL receptor-mediated endocytosis and the paragraph on feedback of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-RT01

Retrieval exercise: Rebuild SREBP activation

A competent response contains

· Step 1: endoplasmic reticulum membrane cholesterol falls.

· Step 2: SCAP releases Insig and escorts SREBP-2 to the Golgi.

· Step 3: site-1 protease then site-2 protease cleave SREBP-2.

· Step 4: the freed N-terminal transcription factor enters the nucleus.

· Step 5: it binds sterol regulatory elements.

· Step 6: transcription of the LDL receptor and HMG-CoA reductase rises.

· Step 7: more LDL uptake and more synthesis restore cell sterol.

· When sterol is high, cholesterol-bound SCAP binds Insig and the whole complex stays in the endoplasmic reticulum, so both genes are turned down.

· SREBP-1c mainly controls fatty acid synthesis genes and responds to insulin.

· This is why inhibiting hepatic HMG-CoA reductase lowers plasma LDL: less cell sterol releases SREBP-2, LDL receptor number rises, and clearance increases, which requires receptor genes able to respond.

Distinctions that matter

· SCAP is the cholesterol sensor and escort, Insig is the anchor that holds the complex in the endoplasmic reticulum, and the two proteases do the cutting in a fixed order.

· SREBP-2 serves cholesterol genes and SREBP-1c serves fatty acid genes, so insulin and sterol signals act through different family members.

Common misconceptions

· Describing SREBP as a free cytosolic transcription factor. It starts as a membrane-bound protein and only its cleaved N-terminal fragment reaches the nucleus.

· Expecting the LDL receptor gene and the reductase gene to move in opposite directions. The same signal raises both.

Where it is taught: Section C03, SREBP activation flow diagram and its caption of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C03-Q02

Board-style question: Why one child responds poorly

Correct answer: C. Most of the plasma effect of the drug comes from SREBP-2 driven production of new LDL receptors, which her cells cannot make

Why it is right: Inhibiting the reductase lowers hepatocyte sterol, which releases SREBP-2 and raises LDL receptor transcription, and that extra receptor number is what clears LDL from plasma; a child with two null alleles cannot make the receptor, so only the reduced VLDL output is left.

A is wrong: Hepatic synthesis is not suppressed in this child: the liver takes up little LDL cholesterol, so SREBP-2 stays active and synthesis runs high.

B is wrong: Her apolipoprotein B-100 is normal; the defect is in the receptor gene, and an abnormal apo B-100 is a separate cause of the same lipid pattern.

D is wrong: The drug acts before the farnesyl pyrophosphate branch point, so it lowers ubiquinone as well as sterols rather than acting only on the non-sterol branch.

A competent response contains

· Most of the plasma LDL lowering produced by an HMG-CoA reductase inhibitor comes from lower hepatic sterol releasing SREBP-2, which raises LDL receptor number and increases clearance.

· With two null receptor alleles there is no functional receptor gene to turn on, so only the fall in VLDL output remains, and hepatic synthesis stays high because the liver receives little LDL cholesterol.

Distinctions that matter

· Blocking synthesis and clearing plasma LDL are two different effects, and the second depends entirely on receptor genes that can respond.

Common misconceptions

· Assuming a high plasma cholesterol suppresses hepatic synthesis. The hepatocyte senses its own endoplasmic reticulum sterol, which stays low without receptor uptake.

Where it is taught: Section C03, paragraph on how lowering liver cholesterol lowers plasma LDL, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-BD1

Brain dump 1 · foundational retrieval: Cardiac fuels and the carnitine shuttle

A competent response contains

· Fatty acids enter through the carnitine shuttle and then beta-oxidation, and their share rises in fasting, the resting adult state, and diabetes.

· Glucose enters through glycolysis and then pyruvate dehydrogenase, and its share rises with high insulin after a carbohydrate meal, in ischemia where glycolysis is anaerobic, and in the fetal heart.

· Lactate is converted by lactate dehydrogenase to pyruvate, and its share rises in exercise.

· Ketone bodies enter through succinyl-CoA:3-ketoacid CoA transferase to acetyl-CoA, and their share rises in prolonged fasting and diabetic ketoacidosis.

· At rest about 60 to 90% of cardiac ATP comes from fatty acids, which arrive bound to albumin or are released by cardiac lipoprotein lipase, enter mainly through CD36, and are activated to fatty acyl-CoA in the cytosol.

· Carnitine shuttle: CPT-I on the outer membrane forms acylcarnitine, a translocase exchanges it into the matrix for carnitine, and CPT-II re-forms acyl-CoA for beta-oxidation; short- and medium-chain fatty acids need no carnitine.

· Malonyl-CoA inhibits CPT-I; acetyl-CoA carboxylase makes it, insulin and high glucose raise it, and AMPK inhibits the carboxylase and so lowers it.

· The liver makes ketone bodies but cannot use them because it lacks the transferase that heart, muscle, and brain have.

· Creatine kinase carries high-energy phosphate from mitochondria to myofibrils as phosphocreatine, which also buffers ATP for the first seconds of a shortfall.

· The heart stores only seconds of ATP, mitochondria fill about a third of a ventricular myocyte, O2 extraction is already high at rest, so greater demand is met mainly by raising coronary flow.

Distinctions that matter

· Carnitine or CPT-II defects can cause cardiomyopathy, while short- and medium-chain fatty acids bypass the shuttle entirely.

· Malonyl-CoA is a signal at the heart rather than a building block: it reports carbohydrate abundance by closing the door to fatty acid entry.

Common misconceptions

· Saying the heart raises O2 extraction to meet demand. Extraction is already high at rest, so flow has to rise.

· Assuming all fatty acids need carnitine. Only long-chain acyl groups do.

Where it is taught: Section C04, opening paragraph, cardiac fuels table, and the carnitine shuttle paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-BD2

Brain dump 2 · integration and application: Work out the oxygen cost of each fuel

A competent response contains

· Glucose consumes 6 O2, which is 12 oxygen atoms, and yields about 31 ATP, so about 2.6 ATP per oxygen atom.

· Palmitate gives 8 acetyl-CoA worth 80 ATP, 7 NADH worth 17.5, and 7 FADH2 worth 10.5, totaling 108, minus 2 for activation, so 106 ATP.

· Palmitate consumes 23 O2, which is 46 oxygen atoms, so about 2.3 ATP per oxygen atom.

· Palmitate yields over three times as much ATP per molecule, while glucose yields roughly 10% more ATP for the same O2.

· When O2 is limited but not absent, shifting toward glucose oxidation gets more work from the same coronary flow, which is why fuel selection matters in ischemia.

· Glucose and insulin raise malonyl-CoA through acetyl-CoA carboxylase, and malonyl-CoA inhibits CPT-I, so long-chain fatty acyl groups cannot enter the matrix.

· In the other direction, active fatty acid oxidation raises acetyl-CoA and NADH, which activate pyruvate dehydrogenase kinase and so inactivate pyruvate dehydrogenase, and raises citrate, which inhibits phosphofructokinase-1; this is the glucose-fatty acid cycle.

Distinctions that matter

· ATP per molecule and ATP per oxygen atom give opposite answers, and only the second one matters when flow is the limiting factor.

· Fatty acids suppress glucose use at two points, one in glycolysis through citrate and one at the gateway to the TCA cycle through pyruvate dehydrogenase.

Common misconceptions

· Concluding that fatty acids are always the better cardiac fuel because they yield more ATP per molecule. Per oxygen atom they yield less.

· Reporting absolute ATP totals as fixed. They depend on the P/O ratios used, and older texts give higher numbers without changing the comparison.

Where it is taught: Section C04, worked example on oxygen cost and the carnitine shuttle paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-RT01

Retrieval exercise: Rebuild energy failure in ischemia

A competent response contains

· Step 1: flow stops and myocardial O2 is exhausted within seconds.

· Step 2: oxidative phosphorylation, the TCA cycle, and beta-oxidation halt.

· Step 3: phosphocreatine falls first, then ATP, while AMP and phosphate rise.

· Step 4: glycogenolysis and anaerobic glycolysis make lactate, yielding only 2 ATP per glucose and 3 per glucose unit from glycogen, with lactate dehydrogenase regenerating the NAD+ glycolysis needs.

· Step 5: lactate and H+ accumulate, and acidosis inhibits phosphofructokinase-1, which limits the one remaining ATP source.

· Step 6: phosphate and H+ weaken cross-bridges and contraction fails within about a minute, which is reversible and spares the remaining ATP.

· Step 7: pump failure causes K+ loss and Ca2+ overload.

· Step 8: injury becomes irreversible after about 20 to 40 minutes.

· Ion sequence: the Na+/K+-ATPase slows, K+ leaks out through ATP-sensitive channels, intracellular Na+ rises, the Na+/Ca2+ exchanger extrudes less Ca2+ while Na+/H+ exchange adds more Na+, cytosolic Ca2+ rises, and SERCA failure impairs relaxation.

· Troponin is released once the membrane is irreversibly damaged, and lactate appearing in coronary sinus blood marks ischemia because the normal heart takes lactate up.

Distinctions that matter

· Losing contraction early is protective rather than part of the damage, because it conserves what ATP is left.

· Raised local extracellular K+ depolarizes ischemic cells, which links energy failure to conduction slowing and arrhythmia.

Common misconceptions

· Expecting anaerobic glycolysis to sustain the heart. Two ATP per glucose is far short of demand, and its own acid product shuts it down.

· Treating contractile failure as the moment of cell death. Injury is still reversible for roughly 20 to 40 minutes.

Where it is taught: Section C04, flow diagram from coronary occlusion to energy failure and the paragraph that follows of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C04-PR01

Prediction exercise: Predict what happens when the carnitine shuttle fails

A competent response contains

· Long-chain fatty acyl-CoA accumulates in the cytosol, because CPT-II cannot re-form acyl-CoA in the matrix and the shuttle backs up.

· Beta-oxidation falls, because its substrate never reaches the matrix.

· Mitochondrial acetyl-CoA and NADH fall, so pyruvate dehydrogenase kinase is less active and pyruvate dehydrogenase activity rises, leaving the myocyte more dependent on glucose oxidation.

· ATP production falls during the fast, when fatty acids should be supplying 60 to 90% of resting cardiac ATP, and the chronically energy-starved ventricle develops cardiomyopathy.

· Medium-chain fatty acids are oxidized normally, because short- and medium-chain fatty acids enter the matrix without carnitine.

· Glucose raises malonyl-CoA through acetyl-CoA carboxylase, which inhibits CPT-I and shifts the heart further toward glucose oxidation, so glucose both supplies fuel and reduces the futile attempt at fatty acid entry.

Distinctions that matter

· A block at CPT-II leaves the cytosolic activation steps intact, so the problem is transport into the matrix rather than failure to activate the fatty acid.

· Carnitine and CPT-II defects both cause cardiomyopathy, and both spare medium-chain fatty acid oxidation.

Common misconceptions

· Expecting ketone bodies to rescue the heart here. Hepatic ketogenesis also depends on fatty acid oxidation, so the alternative fuel is limited at the same time.

· Assuming all fatty acids are blocked. Chain length decides whether the shuttle is needed.

Where it is taught: Section C04, carnitine shuttle paragraph and the cardiac fuels table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-BD1

Brain dump 1 · foundational retrieval: Thiamine pyrophosphate and the enzymes that need it

A competent response contains

· Thiamine is absorbed mainly in the jejunum and converted by thiamine pyrophosphokinase to thiamine pyrophosphate; stores are small and turnover is fast, so deficiency can appear within weeks.

· The thiazolium carbanion of the coenzyme attacks the carbonyl carbon of an alpha-keto acid, releasing CO2 and carrying the acyl fragment, which is the chemistry both oxidative decarboxylation and transketolase need.

· Pyruvate dehydrogenase E1 converts pyruvate to acetyl-CoA, and deficiency lets pyruvate accumulate and become lactate.

· Alpha-ketoglutarate dehydrogenase E1 converts alpha-ketoglutarate to succinyl-CoA, and deficiency slows the TCA cycle, cutting NADH and ATP from every fuel.

· Branched-chain alpha-ketoacid dehydrogenase catabolizes leucine, isoleucine, and valine, and deficiency lets those keto acids accumulate, which is minor in acquired deficiency.

· Transketolase transfers two carbons in the pentose phosphate pathway, and deficiency lowers red cell activity, which is used as a test.

· The dehydrogenase complexes also need lipoic acid, CoA from B5, FAD from B2, and NAD+ from B3.

· The test is red cell transketolase activity: a large rise when thiamine pyrophosphate is added to the sample indicates deficiency.

· Tissues with high aerobic glucose use are hit first, which is why the nervous system and the heart show the earliest effects.

Distinctions that matter

· Two of the four enzymes sit on the main fuel route and two do not, which is why lactic acidosis and low red cell transketolase appear together.

· Thiamine pyrophosphate is one of five cofactors the dehydrogenase complexes need, so the complexes can fail for reasons other than thiamine.

Common misconceptions

· Naming thiamine itself as the coenzyme. Thiamine pyrophosphokinase has to add the pyrophosphate first.

· Expecting slow onset because the vitamin is water-soluble. Stores are small and turnover fast, so weeks are enough.

Where it is taught: Section C05, opening paragraph and the table of thiamine pyrophosphate enzymes of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-BD2

Brain dump 2 · integration and application: Explain every finding in wet beriberi

A competent response contains

· Mechanism in order: low thiamine pyrophosphate blocks pyruvate and alpha-ketoglutarate dehydrogenase; aerobic ATP falls while pyruvate and lactate accumulate; metabolic vasodilation lowers systemic vascular resistance; venous return and cardiac output rise; sympathetic and renin-angiotensin-aldosterone activation retain Na+ and water; the ATP-starved myocardium dilates and fails.

· Warm extremities and a bounding pulse with a wide pulse pressure come from low resistance with a high stroke volume; tachycardia comes from sympathetic drive; raised jugular venous pressure, edema, and pulmonary congestion come from salt and water retention; the enlarged heart on radiograph comes from dilation of the failing myocardium; the raised lactate comes from the enzyme block itself.

· Lactate is high with normal oxygenation because pyruvate dehydrogenase cannot move pyruvate into the TCA cycle, so pyruvate accumulates and is reduced to lactate; the problem is fuel entry, not O2 delivery.

· Shoshin beriberi is the fulminant form, with severe lactic acidosis and circulatory collapse.

· Settings: alcohol use disorder, which impairs intake, absorption, storage, and conversion to the coenzyme; polished rice diets; persistent vomiting or bariatric surgery; and long diuretic use.

· A glucose load increases flux through thiamine pyrophosphate enzymes, so it can unmask marginal stores.

· Separating the other high-output states: severe anemia shows a low hemoglobin, thyrotoxicosis shows a low TSH with a high free T4 or T3, and an arteriovenous fistula has a continuous bruit with a heart rate that falls on compression.

· Ordinary low-output failure gives cool extremities and a narrow pulse pressure.

Distinctions that matter

· Low resistance and a failing myocardium come from the same metabolic block, so this is one disease rather than two coexisting problems.

· Dry beriberi is the peripheral neuropathy, and Wernicke encephalopathy with Korsakoff syndrome are the central forms of the same deficiency.

Common misconceptions

· Reading a raised lactate as proof of poor oxygen delivery. Here the cell has O2 but cannot feed fuel into the TCA cycle to use it.

· Expecting cool extremities in heart failure. High-output failure runs warm with a wide pulse pressure.

Where it is taught: Section C05, wet beriberi flow diagram, the paragraph on findings, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-CA01

Clinical application: Breathlessness and warm hands in a malnourished man

A competent response contains

· Recognize: thiamine deficiency causing wet beriberi, a high-output heart failure; the warm extremities and the wide pulse pressure of 134/48 mm Hg with a bounding pulse show output is high rather than low.

· Explain: pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase lose thiamine pyrophosphate, so aerobic ATP falls and pyruvate and lactate accumulate; the accumulating metabolites drive peripheral vasodilation that lowers systemic vascular resistance and raises venous return and output, while the same ATP shortage leaves the myocardium unable to sustain that work, so it dilates and fails, and sympathetic and renin-angiotensin-aldosterone activation retains salt and water.

· Differentiate: severe anemia is excluded by a hemoglobin of 13.1 g/dL, thyrotoxicosis by a normal TSH, an arteriovenous fistula by the absence of a continuous bruit and a heart rate that would fall on compression, and ordinary low-output failure by his warm extremities and wide pulse pressure rather than cool extremities and a narrow pulse pressure.

· Investigate: red cell transketolase activity measured before and after adding thiamine pyrophosphate, where a large rise in activity on adding the coenzyme indicates deficiency; the lactate itself is the test that reflects the blocked entry of pyruvate into the TCA cycle.

· Decide: lactate is raised despite 97% saturation because the block is at fuel entry rather than O2 delivery, and with a lactate of 6.2 mmol/L and a low diastolic pressure the fulminant form to watch for is Shoshin beriberi with severe lactic acidosis and circulatory collapse.

· Justify: the thiazolium carbanion of thiamine pyrophosphate attacks the carbonyl carbon of an alpha-keto acid and releases CO2, which is what makes oxidative decarboxylation possible; without it pyruvate

accumulates behind the block and is reduced to lactate, and the nervous system and heart fail first because they depend most on high aerobic glucose use and have the least metabolic slack.

Distinctions that matter

· High-output failure and low-output failure are separated at the bedside by skin temperature and pulse pressure, before any laboratory test.

· His alcohol history explains the deficiency, but the diagnosis rests on the pattern of findings and the enzyme block, not on the history alone.

Common misconceptions

· Looking for a cause of poor oxygen delivery to explain the lactate. Oxygenation is normal and the defect is the coenzyme.

· Attributing the edema and raised jugular venous pressure to a failing pump alone. Sympathetic and renin-angiotensin-aldosterone driven salt and water retention is part of the mechanism.

Where it is taught: Section C05, wet beriberi flow diagram, the paragraph on findings, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C05-Q03

Board-style question: A high lactate with normal oxygenation

Correct answer: D. Pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase

Why it is right: Transketolase and both of these dehydrogenase complexes use thiamine pyrophosphate, so a large rise in red cell transketolase activity on adding the coenzyme shows thiamine deficiency; the block at pyruvate dehydrogenase raises lactate and the block at alpha-ketoglutarate dehydrogenase cuts aerobic ATP, which together give the warm, bounding, high-output picture of wet beriberi.

A is wrong: The carnitine palmitoyltransferases need no thiamine, and their failure causes cardiomyopathy during fasting rather than lactic acidosis with a transketolase response.

B is wrong: Methionine synthase and cystathionine beta-synthase use B12 and B6 and would raise homocysteine, not lactate.

C is wrong: Lysyl oxidase and prolyl hydroxylase need copper and ascorbate and would weaken collagen and elastin, giving bleeding or fragile vessels rather than this circulatory picture.

A competent response contains

· Persistent vomiting after bariatric surgery is one of the listed settings for thiamine deficiency, and a sharp rise in red cell transketolase activity when thiamine pyrophosphate is added confirms it.

· The two dehydrogenase complexes that fail first are pyruvate dehydrogenase, which lets pyruvate accumulate and become lactate, and alpha-ketoglutarate dehydrogenase, which slows the TCA cycle.

Distinctions that matter

· The transketolase test identifies the missing coenzyme, while the clinical picture of warm extremities and a wide pulse pressure identifies which organ system is failing.

Common misconceptions

· Taking the normal oxygen saturation as evidence against a metabolic cause of the lactate. Here the cell cannot use the O2 it has.

Where it is taught: Section C05 of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C06-BD1

Brain dump 1 · foundational retrieval: Draw the methionine cycle and its two exits

A competent response contains

· Cycle in order: methionine is activated by ATP to S-adenosylmethionine, which donates a methyl group and becomes S-adenosylhomocysteine, and hydrolysis of that yields homocysteine.

· Remethylation exit: methionine synthase uses B12 as methylcobalamin and N5-methyltetrahydrofolate as its methyl donor, regenerating tetrahydrofolate; MTHFR makes that methyl donor; the liver also remethylates through betaine-homocysteine methyltransferase.

· Transsulfuration exit: cystathionine beta-synthase commits homocysteine to cystathionine and cystathionine gamma-lyase completes the step to cysteine, both using pyridoxal phosphate from B6.

· Transsulfuration is the only exit that removes homocysteine permanently; remethylation returns it to methionine.

· Homocysteine leaves only through these enzymes, so loss of B12, folate, B6, or the enzymes themselves makes it accumulate.

· The methylfolate trap: in B12 deficiency, folate is stuck as N5-methyltetrahydrofolate because methionine synthase cannot use it, so both deficiencies cause megaloblastic anemia.

· Methylmalonic acid separates them: it is raised only in B12 deficiency, because B12 is also the cofactor for methylmalonyl-CoA mutase.

Distinctions that matter

· Folate and B12 act on the same step from different sides, one as the methyl donor and one as the cofactor, which is why their blood pictures look alike.

· B6 serves the exit that disposes of homocysteine, so B6 deficiency raises homocysteine without a megaloblastic anemia.

Common misconceptions

· Treating remethylation as disposal. It recycles homocysteine back to methionine, so only transsulfuration clears it.

· Using megaloblastic anemia to tell folate from B12 deficiency. Methylmalonic acid is the value that separates them.

Where it is taught: Section C06, methionine cycle diagram and the paragraph on remethylation and transsulfuration of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C06-BD2

Brain dump 2 · integration and application: Two ways micronutrients let a vessel fail

A competent response contains

· Homocysteine oxidizes readily and generates reactive oxygen species that injure endothelium; it reduces nitric oxide; it promotes smooth muscle proliferation; and it favors thrombosis, for example by raising tissue factor and impairing the thrombomodulin and protein C pathway.

· Normal fasting total homocysteine is about 5 to 15 micromol per litre, and a mild elevation is best read as a risk marker, because whether it causes atherosclerosis or only accompanies it is unsettled.

· The infant has Menkes disease: copper-dependent lysyl oxidase has failed, so elastin and collagen cannot be cross-linked, the aldehyde-derived covalent cross-links including desmosine in elastin are missing, and the aorta and large arteries lose strength and recoil and become tortuous and aneurysmal.

· The older man has scurvy: prolyl and lysyl hydroxylases have failed because ascorbate is needed to keep their iron as Fe2+, so hydroxyproline cannot stabilize the collagen triple helix at 37 degrees C (98.6 degrees F) and hydroxylysine cannot provide glycosylation and cross-link sites; weak collagen in capillary basement membranes and perivascular tissue gives the bleeding.

· Location: the hydroxylases act inside the cell in the rough endoplasmic reticulum, while lysyl oxidase acts extracellularly after secretion.

· Menkes disease is X-linked recessive from ATP7A, so copper cannot leave the enterocyte and serum copper and ceruloplasmin are low; acquired copper deficiency follows bariatric surgery or excess zinc; Wilson disease from ATP7B is the opposite, with copper retained in liver and brain.

· Other scurvy findings are perifollicular hemorrhages, bleeding gums, hemarthrosis, subperiosteal bleeding, poor wound healing, and corkscrew hairs, and stores last about 1 to 3 months.

Distinctions that matter

· Vitamin C is needed to build the helix and lay down cross-link sites, while copper is needed to form the cross-links themselves, which is why one causes bleeding and the other causes fragile tortuous arteries.

· Menkes and Wilson disease are both copper transport defects in the same family of proteins with opposite consequences.

Common misconceptions

· Explaining the bleeding of scurvy as a platelet or clotting problem. The collagen supporting small vessels is weak.

· Treating a mildly raised homocysteine as established cause of atherosclerosis. Homocystinuria shows the extreme, but a mild rise is a marker.

Where it is taught: Section C06, paragraph on vascular injury and the final two paragraphs on vitamin C and copper of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C06-CP01

Comparison table: Sorting out a raised homocysteine

A competent response contains

· Folate deficiency: remethylation is affected, methylmalonic acid is normal, and megaloblastic anemia is the other finding.

· Vitamin B12 deficiency: remethylation is affected, methylmalonic acid is raised, and there is megaloblastic anemia with neurologic signs.

· Vitamin B6 deficiency: transsulfuration is affected, methylmalonic acid is normal, and there is sideroblastic anemia with neuropathy, classically with isoniazid.

· Cystathionine beta-synthase deficiency, homocystinuria: transsulfuration is affected, methylmalonic acid is normal, inheritance is autosomal recessive, and homocysteine is often above 100 micromol per litre with raised methionine, downward lens dislocation, a marfanoid habitus, and early thrombosis.

· MTHFR C677T variant: methyl donor formation is affected, methylmalonic acid is normal, and the rise is mild and appears mainly with low folate.

· Chronic kidney disease: clearance is reduced, methylmalonic acid may be raised, and the clue is a low glomerular filtration rate.

· Normal fasting total homocysteine is about 5 to 15 micromol per litre.

Distinctions that matter

· Methylmalonic acid is the single value that separates B12 deficiency from folate deficiency, because B12 also serves methylmalonyl-CoA mutase.

· A very high homocysteine with a raised methionine points to the transsulfuration enzyme itself, since the remethylation causes leave methionine low or normal.

Common misconceptions

· Expecting a megaloblastic anemia with every cause of raised homocysteine. B6 deficiency and cystathionine beta-synthase deficiency affect the other exit.

· Reading the MTHFR variant as a strong independent cause. It gives a mild rise, mainly when folate is low.

Where it is taught: Section C06, table of causes of raised homocysteine of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-BD1

Brain dump 1 · foundational retrieval: Map each electrolyte onto the action potential and the tracing

A competent response contains

· Action potential to ECG: phase 4 rest near -90 mV is set by K+, phase 0 is the Na+ upstroke and gives the QRS, phase 2 is the Ca2+ plateau and corresponds to the ST segment, and phase 3 is K+ repolarization and gives the T wave.

· Hyperkalemia: phase 3 is faster and the resting membrane depolarizes with Na+ channel inactivation, so the tracing shows a peaked T wave, then a flat P, a long PR, a wide QRS, and finally a sine wave, with ventricular fibrillation and asystole as the rhythm risks.

· Hypokalemia: phase 3 is slower, so T waves flatten or invert, the ST segment sags, U waves appear with a long QU interval, and the risks are torsades and other tachyarrhythmias with increased glycoside toxicity.

· Hypocalcemia: phase 2 is longer, so the ST segment and QT are long with a normal T wave, and torsades is possible though uncommon, alongside tetany.

· Hypercalcemia: phase 2 is shorter, so the ST segment and QT are short, with bradyarrhythmia and AV block at high levels.

· Hypomagnesemia: repolarization is long, mostly through low K+ and Ca2+, giving a long QT often with hypokalemic features and a risk of torsades de pointes.

· Ranges: K+ 3.5 to 5.0 mEq/L, total calcium 8.5 to 10.5 mg/dL with ionized calcium about 4.5 to 5.3 mg/dL, and Mg2+ about 1.7 to 2.2 mg/dL.

· Add 0.8 mg/dL to total calcium for each 1 g/dL that albumin is below 4.0 g/dL; hemolysis falsely raises a potassium result; and plasma magnesium can be normal despite real cellular depletion.

· Kidney failure often combines hyperkalemia and hypocalcemia, giving a long ST segment followed by a peaked T wave.

Distinctions that matter

· Potassium acts on phase 3 and the T wave first, while calcium acts on phase 2 and the length of the ST segment, so the two disorders change different parts of the tracing.

· Hypokalemia and hyperkalemia both act on K+ conductance but in opposite directions, which is why one slows phase 3 and the other speeds it.

Common misconceptions

· Expecting a low extracellular K+ to speed repolarization because K+ efflux is repolarizing. Low K+ reduces IK1 and IKr conductance, so phase 3 slows.

· Treating a normal tracing as reassurance in hyperkalemia. The level at which each change appears varies widely and a normal ECG does not exclude danger.

Where it is taught: Section C07, opening paragraph, the action potential diagram, and the electrolyte table of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-BD2

Brain dump 2 · integration and application: Why one potassium goes high and another will not come up

A competent response contains

· The first change is a tall, narrow, peaked T wave, because higher extracellular K+ increases IKr and inward rectifier conductance and speeds phase 3; repolarizing currents respond at moderate levels before anything else.

· QRS widening needs enough resting depolarization to inactivate a large fraction of fast Na+ channels, which slows phase 0 and conduction; the order is a flattening P wave, then a lengthening PR, then a widening QRS, then a sine wave pattern that can lead to ventricular fibrillation or asystole.

· In the woman with refractory hypokalemia, measure magnesium: Mg2+ normally blocks K+ exit through renal ROMK channels, so hypomagnesemia causes continued urinary K+ loss and the hypokalemia persists until magnesium is restored.

· The two other consequences of magnesium deficiency given in the notes are impaired parathyroid hormone release and action, which causes hypocalcemia, and a predisposition to torsades de pointes.

· Torsades arises when a long QT allows early afterdepolarizations, driven by reactivation of L-type Ca2+ channels, to reach threshold, often after a short-long-short beat sequence; the QRS axis appears to twist around the baseline.

· Low K+ increases cardiac glycoside toxicity because K+ competes with these drugs at the Na+/K+-ATPase.

· Hypomagnesemia also promotes torsades by lowering K+ and Ca2+ and by removing its own limit on inward Ca2+ current, and its causes include alcohol use disorder, diarrhea, loop and thiazide diuretics, and long-term proton pump inhibitors.

Distinctions that matter

· Hyperkalemia changes repolarization first and depolarization later, which is why the T wave is the early sign and the wide QRS is the late one.

· A potassium that will not correct is a magnesium problem at the renal tubule, not a dosing problem.

Common misconceptions

· Reading a wide QRS as the first sign of hyperkalemia. Peaked T waves come first, at lower levels.

· Assuming a normal plasma magnesium rules out depletion. Plasma can be normal while cells are depleted.

Where it is taught: Section C07, potassium paragraphs, the calcium and magnesium paragraph, and the stop and predict box of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-PR01

Prediction exercise: Predict the tracing in kidney failure

A competent response contains

· Phase 3 is faster and the T wave becomes tall, narrow, and peaked, because the raised extracellular K+ increases IKr and inward rectifier conductance.

· Phase 2 is longer and the ST segment lengthens, because slower Ca2+-dependent inactivation of L-type channels at a low ionized calcium prolongs the plateau.

· The QT interval is long, and the hypocalcemia is driving it through the lengthened phase 2, while the T wave itself stays normal in shape from the calcium effect and peaked from the potassium effect; with an albumin of 4.0 g/dL no correction to total calcium is needed, so the value of 6.8 mg/dL is a true hypocalcemia.

· The resting membrane potential becomes less negative, and a growing fraction of fast Na+ channels is inactivated, because the resting potential lies near the K+ equilibrium potential and a raised extracellular K+ shifts it upward.

· As K+ rises further the P wave flattens, the PR interval lengthens, and the QRS widens, because slowed phase 0 slows conduction through atrium, node, and ventricle.

· The tracing becomes a sine wave pattern, and the rhythms to expect are ventricular fibrillation and asystole.

· This combination is the classic kidney failure tracing: a long ST segment followed by a peaked T wave.

Distinctions that matter

· The two abnormalities act on different parts of the tracing, so the ST segment reports the calcium and the T wave reports the potassium.

· Albumin correction only applies when albumin is below 4.0 g/dL, so here the measured total calcium stands as reported.

Common misconceptions

· Expecting a long QT to come from the hyperkalemia. Raised K+ speeds phase 3 and shortens repolarization; the long segment here is the calcium plateau.

· Assuming the level of potassium predicts the finding reliably. Thresholds vary widely between patients.

Where it is taught: Section C07, potassium paragraphs, calcium and magnesium paragraph, and the electrolyte table with its caption of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

NPLEX-P1-CARD-13-01-C07-Q04

Board-style question: A potassium that will not correct

Correct answer: B. Magnesium depletion has removed its normal block of renal ROMK channels, so potassium continues to be lost in the urine

Why it is right: Magnesium normally blocks K+ exit through renal ROMK channels, so while magnesium is depleted the kidney keeps losing potassium and replacement cannot raise the plasma level; her thiazide and long-term proton pump inhibitor are both listed causes of magnesium loss, and low magnesium also explains the long corrected QT and the polymorphic ventricular tachycardia.

A is wrong: Impaired parathyroid hormone release does follow severe magnesium deficiency and lengthens phase 2 through hypocalcemia, but it explains a long ST segment rather than why the potassium itself will not rise.

C is wrong: Her extracellular potassium is low, not raised, and a raised potassium would speed phase 3 and peak the T waves instead of flattening them and adding U waves.

D is wrong: Alkalosis increasing albumin binding of calcium causes tetany with a normal total calcium; it does not cause renal potassium loss.

A competent response contains

· Mg2+ normally blocks K+ exit through renal ROMK channels, so hypomagnesemia causes ongoing urinary K+ loss and a hypokalemia that persists until magnesium is restored.

· Thiazide diuretics and long-term proton pump inhibitors are both listed causes of magnesium loss, and low magnesium also promotes torsades through low K+ and Ca2+ and by removing its limit on inward Ca2+ current.

Distinctions that matter

· Flat T waves with U waves and a long corrected QT is the hypokalemic pattern, and the failure to correct points past the potassium to the magnesium.

Common misconceptions

· Treating a persistently low potassium as inadequate replacement. The renal leak continues while magnesium is low.

Where it is taught: Section C07, calcium and magnesium paragraph of the CV-13 Cardiovascular Biochemistry and Nutrition notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-14

CV-14 Cardiovascular Genetics

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-14-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the LDL receptor pathway and its feedback

A competent response contains

· Each LDL particle carries one apolipoprotein B-100, which is the ligand for the LDL receptor.

· The liver clears about 70% of plasma LDL, so hepatic receptor number sets the plasma LDL concentration.

· Sequence: apoB-100 binds the receptor, the complex is internalized in clathrin-coated pits, the ligand is released in the acidified endosome while the receptor recycles to the surface, and LDL is hydrolyzed in the lysosome to free cholesterol.

· The receptor also binds apolipoprotein E on intermediate-density lipoprotein, so scarce receptors leave more IDL to become LDL.

· When endoplasmic reticulum cholesterol is low, SCAP escorts SREBP-2 to the Golgi, where proteases release the active fragment.

· The active SREBP-2 fragment switches on HMG-CoA reductase, the LDL receptor, and PCSK9.

· When endoplasmic reticulum cholesterol is high, INSIG anchors SREBP-2 so it never reaches the Golgi.

· PCSK9 is secreted by hepatocytes, binds the receptor, and directs it to the lysosome instead of letting it recycle, so PCSK9 activity sets how many trips each receptor makes.

· HMG-CoA reductase is the rate-limiting enzyme of cholesterol synthesis.

Distinctions that matter

· Receptor recycling and receptor degradation are different fates for the same protein, and PCSK9 decides between them.

· The receptor has two ligands: apoB-100 on LDL and apoE on IDL, which is why defects in the two proteins behave differently.

Common misconceptions

· Treating SREBP-2 as responding to plasma cholesterol. It responds to cholesterol in the endoplasmic reticulum membrane, which is why plasma can be loaded while the pathway stays switched on.

· Saying the receptor is consumed with each LDL particle. It is released in the endosome and recycles unless PCSK9 sends it to the lysosome.

Where it is taught: Section C01, first two paragraphs of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C01-BD2

Brain dump 2 · integration and application: Why plasma LDL stays high while the liver cell starves

A competent response contains

· Familial hypercholesterolemia is autosomal dominant, with heterozygotes about 1 in 250.

· Locus heterogeneity: loss-of-function LDLR variants on chromosome 19p in most cases, APOB variants in the receptor-binding region, and gain-of-function PCSK9 variants, which are the rarest.

· Familial defective apoB-100 is usually milder because apoE-mediated clearance of IDL is preserved.

· Heterozygotes have LDL cholesterol about twice normal from birth, usually above about 190 mg/dL (4.9 mmol/L) in untreated adults against a desirable value below about 100 mg/dL (2.6 mmol/L), with normal triglycerides, which is Fredrickson type IIa.

· Little LDL reaches the hepatocyte endoplasmic reticulum, so SREBP-2 stays active, synthesis continues, and VLDL secretion feeds future LDL.

· Reduced receptor number lowers clearance of both LDL and IDL, and the surviving IDL becomes LDL, so production rises as clearance falls.

· Cholesterol delivered by a route that bypasses the receptor suppresses synthesis normally in familial hypercholesterolemia fibroblasts, which proves that uptake is defective and feedback is intact.

· Lowering hepatic synthesis upregulates the normal allele in a heterozygote, which is why it works there and does little in a receptor-negative homozygote.

· Macrophage scavenger receptors take up oxidized LDL and are not downregulated, so foam cells form, producing xanthomas, corneal arcus, and premature atherosclerosis.

· Tendon xanthomas at the Achilles and hand extensors and corneal arcus before about age 45 fit the heterozygous phenotype, with coronary disease often by the fourth or fifth decade in men.

Distinctions that matter

· Defective uptake against defective feedback: the fibroblast experiment separates them, and only uptake is broken.

· Gain of function in PCSK9 raises LDL while loss of function in PCSK9 gives lifelong low LDL and markedly lower coronary risk, so the direction of the functional change matters more than the gene name.

Common misconceptions

· Saying the high plasma LDL should switch off hepatic synthesis. The hepatocyte cannot sense cholesterol it never takes up.

· Expecting the receptor pathway to limit foam cell formation. Scavenger receptors on macrophages are not downregulated by cell cholesterol.

Where it is taught: Section C01, paragraphs on familial hypercholesterolemia and on lost feedback of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C01-RT01

Retrieval exercise: Rebuild the familial hypercholesterolemia sequence

A competent response contains

· Step 1: an LDLR, APOB, or gain-of-function PCSK9 variant.

· Step 2: fewer working receptor trips at the hepatocyte surface.

· Step 3: less LDL and IDL cleared, and more IDL becomes LDL.

· Step 4: low endoplasmic reticulum cholesterol keeps SREBP-2 active and synthesis high.

· Step 5: plasma LDL is high from birth.

· Step 6: foam cells give xanthomas, corneal arcus, and premature atherosclerosis.

· Fewer receptors raise production as well as lowering clearance, because the receptor also clears IDL and unretrieved IDL is converted to LDL.

· SREBP-2 stays active and keeps switching on HMG-CoA reductase, the rate-limiting enzyme of cholesterol synthesis.

· The five classes of LDL receptor variant fail at synthesis, transport from endoplasmic reticulum to Golgi, ligand binding, clustering in coated pits, and recycling.

· Transport from endoplasmic reticulum to Golgi is the most common class.

Distinctions that matter

· Reduced clearance and continued production are two separate contributions to the same high LDL, and both follow from the one receptor defect.

· The five variant classes are defined by the step in the receptor's life cycle that fails, not by severity.

Common misconceptions

· Writing the sequence as a single clearance problem. Production stays high because feedback never sees the cholesterol.

· Placing the lipid deposits before the lifelong elevation. Deposits in skin, tendon, cornea, and artery follow decades of high LDL.

Where it is taught: Section C01, flow diagram on pathogenesis of familial hypercholesterolemia of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C02-BD1

Brain dump 1 · foundational retrieval: Sarcomere genes to the hypertrophic cardiomyopathy phenotype

A competent response contains

· Hypertrophic cardiomyopathy is left ventricular thickening not explained by abnormal loading such as hypertension or aortic stenosis.

· It affects about 1 in 500 adults and in most familial cases is an autosomal dominant disease of the sarcomere.

· MYH7 encodes beta-myosin heavy chain and MYBPC3 encodes cardiac myosin-binding protein C, and together they account for most genotype-positive cases.

· Less common genes are troponin T and troponin I, alpha-tropomyosin, the myosin light chains, and cardiac actin.

· MYH7 missense variants make a mutant myosin head built into the thick filament beside normal myosin, which is a dominant negative effect.

· MYBPC3 variants are usually truncating, and nonsense-mediated decay lowers the amount of protein, which is haploinsufficiency, with later and milder disease on average.

· Testing finds a variant in only about a third to a half of patients, so a negative test does not exclude a genetic cause.

· Sequence: variant, then more myosin heads cycling or higher Ca2+ sensitivity, then hypercontractility with high ATP use and impaired relaxation, then an energy deficit that activates hypertrophic and fibrotic signaling, then hypertrophy with myocyte disarray, fibrosis, and small vessel disease, then a stiff nondilated ventricle with an arrhythmic substrate.

· Many variants release myosin heads from their energy-saving folded resting state, and thin filament variants raise Ca2+ sensitivity.

· Adult wall thickness of about 15 mm or more is diagnostic, against about 11 mm or less normally, often with asymmetric septal hypertrophy and a small cavity, and the ventricle is not dilated.

· The hallmark is myocyte disarray, with short branched myocytes in whorls, accompanied by interstitial fibrosis and thick-walled intramural arteries.

Distinctions that matter

· Dominant negative against haploinsufficiency: MYH7 puts a faulty protein into the filament, while MYBPC3 simply supplies less protein, which is why MYBPC3 disease tends to be later and milder.

· Hypertrophy here follows from the motor itself, not from load, which is what separates it from hypertensive or valvular thickening.

Common misconceptions

· Describing the ventricle as weak. It is hypercontractile and relaxes poorly, and the problem is diastolic.

· Taking a negative gene panel as evidence against the diagnosis. Only about a third to a half of patients have a detectable variant.

Where it is taught: Section C02, first three paragraphs and the flow diagram of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C02-BD2

Brain dump 2 · integration and application: Two relatives, one variant, and a murmur that moves

A competent response contains

· Each child of a carrier has a 50% risk because the disease is autosomal dominant.

· Relatives with the same variant range from sudden death to a normal echocardiogram, which is variable expressivity with incomplete penetrance.

· Penetrance is age dependent, with hypertrophy usually appearing in adolescence or early adulthood, so a genotype-positive relative with a normal study needs repeated evaluation over time.

· With MYBPC3 hypertrophy sometimes does not appear until middle age.

· A second sarcomere variant raises severity, and male sex, hypertension, and modifier genes raise penetrance.

· The stiff ventricle fills at high pressure, which causes exertional dyspnea and a fourth heart sound.

· Filling depends on atrial contraction, so atrial fibrillation is poorly tolerated.

· About a third of patients have dynamic outflow obstruction at rest: the septum narrows the outflow tract and flow drags the anterior mitral leaflet toward it, which is systolic anterior motion, adding mitral regurgitation.

· Standing and the strain of a Valsalva maneuver reduce ventricular volume, so obstruction and the murmur increase; squatting, leg raising, and handgrip raise volume or afterload, so the murmur softens.

· The murmur of aortic stenosis responds in the opposite direction to the same volume changes.

· Angina occurs with normal coronary arteries because demand is high and the intramural vessels are narrow.

· Phenocopies: athlete's heart with an enlarged cavity, normal relaxation, and regression with detraining; hypertensive hypertrophy; Fabry disease from X-linked GLA; Friedreich ataxia; and Noonan syndrome.

Distinctions that matter

· Penetrance is whether any disease appears and expressivity is how severe it is; an age-dependent penetrance is why one evaluation never settles a relative's status.

· A dynamic murmur that grows when the ventricle empties separates obstructive hypertrophic cardiomyopathy from fixed valvular stenosis.

Common misconceptions

· Telling a genotype-positive adolescent with a normal echocardiogram that he is unaffected. Hypertrophy may still appear, and with MYBPC3 it can be delayed to middle age.

· Expecting a louder murmur with a fuller ventricle. A larger cavity moves the septum and leaflet apart and softens the murmur.

Where it is taught: Section C02, paragraphs on hemodynamics and on penetrance, and the stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C02-CA01

Clinical application: Collapse during a basketball game in a 17-year-old

A competent response contains

· Recognize: hypertrophic cardiomyopathy, autosomal dominant, most often from MYH7 encoding beta-myosin heavy chain or MYBPC3 encoding cardiac myosin-binding protein C.

· Explain: a mutant myosin head built into the thick filament beside normal myosin acts as a dominant negative, or a truncating MYBPC3 variant lowers protein through nonsense-mediated decay; heads are released from their folded resting state and thin filament variants raise Ca2+ sensitivity, so the heart is hypercontractile, wastes ATP, and relaxes poorly, and the energy deficit activates hypertrophic and fibrotic signaling that produces hypertrophy, myocyte disarray, fibrosis, and small vessel disease without any abnormal load.

· Differentiate: athlete's heart has an enlarged cavity with normal relaxation and regresses with detraining, while this boy has a small cavity and a 20 mm septum; hypertensive hypertrophy requires raised blood pressure, and his is 118/72 mm Hg; Fabry disease is X-linked from GLA; Friedreich ataxia brings ataxia and sensory loss; and Noonan syndrome brings the Turner-like features and pulmonary valve dysplasia.

· Investigate: the murmur becomes louder when ventricular volume falls on standing and softer with squatting or handgrip; the electrocardiogram shows high voltage with deep narrow Q waves from the thick septum; wall thickness of about 15 mm or more in an adult is diagnostic against about 11 mm or less normally; and tissue shows myocyte disarray with interstitial fibrosis and thick-walled intramural arteries.

· Decide: myocyte disarray and patchy interstitial fibrosis slow and scatter conduction and support reentry, so ventricular arrhythmia can be the first event, and relative risk depends on inheriting the variant, on age, and on modifiers, since each first-degree relative has a 50% chance of carrying it.

· Justify: standing lowers venous return and ventricular volume, bringing the septum and the anterior mitral leaflet closer during systole, so obstruction and turbulence increase, while squatting and handgrip raise volume or afterload and move them apart; relatives can carry the variant with a normal echocardiogram

because penetrance is incomplete and age dependent, which is why evaluation is repeated rather than done once.

Distinctions that matter

· Angina with normal coronary arteries here comes from high demand and narrow intramural vessels, not from epicardial obstruction.

· Sudden death in this disease comes from the arrhythmic substrate in the tissue, not from the outflow gradient itself.

Common misconceptions

· Reading normal left ventricular emptying as evidence against the diagnosis. The ventricle empties well and fails to fill.

· Calling a murmur that changes with posture a functional murmur. The change in loudness is the mechanism of the obstruction showing itself.

Where it is taught: Section C02, all paragraphs, the flow diagram, and the stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C03-BD1

Brain dump 1 · foundational retrieval: Marfan syndrome from gene to dissection

A competent response contains

· The aortic media is built from elastic lamellae with smooth muscle between them, held by collagen and microfibrils.

· Marfan syndrome affects about 1 in 5,000, about a quarter of cases are de novo, and it is autosomal dominant.

· The gene is FBN1 on chromosome 15 at 15q21, encoding fibrillin-1.

· Fibrillin-1 forms microfibrils that scaffold elastin and anchor the lens zonule, and microfibrils also sequester latent TGF-beta.

· Missense variants act as dominant negatives and truncating variants act by haploinsufficiency.

· Fibrillin deficiency increases TGF-beta signaling, matrix metalloproteinase activity, and matrix loss.

· Sequence: FBN1 variant with abnormal or reduced fibrillin-1, then weak microfibrils with excess TGF-beta signaling, then medial degeneration, then dilation of the root at the sinuses of Valsalva, then a larger radius raising wall tension, then aortic regurgitation followed by dissection or rupture.

· By Laplace's law wall tension rises with radius, so dilation feeds further dilation.

· The lesion is medial degeneration: elastic fiber fragmentation, smooth muscle loss, and proteoglycan pools; the older term cystic medial necrosis is misleading because there is neither a true cyst nor necrosis.

· The stretched annulus causes aortic regurgitation, and myxomatous mitral valves prolapse.

· Pleiotropy gives upward ectopia lentis, tall stature, arachnodactyly, pectus deformity, scoliosis, and joint laxity.

· Homocystinuria mimics the habitus but displaces the lens downward and causes thrombosis and intellectual disability.

· Ascending aortic dissection is the classic cause of death in untreated Marfan syndrome.

Distinctions that matter

· Upward against downward lens displacement separates Marfan syndrome from homocystinuria in a patient with the same body habitus.

· Fibrillin-1 has a structural job and a signaling job, and the aortic disease needs both the weakened scaffold and the excess TGF-beta.

Common misconceptions

· Taking cystic medial necrosis literally. There is no cyst and no necrosis, only fragmented elastin, lost smooth muscle, and pools of proteoglycan.

· Treating the root dilation as a fixed size. Laplace's law makes a wider root generate more wall tension, so the process accelerates.

Where it is taught: Section C03, first two paragraphs and the Marfan flow diagram of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C03-BD2

Brain dump 2 · integration and application: Three families, three failing wall components

A competent response contains

· Patient 1 is Marfan syndrome: FBN1, fibrillin-1, autosomal dominant, with weak microfibrils and excess TGF-beta producing medial degeneration worst at the aortic root, so the root dilates and the ascending aorta dissects.

· Patient 2 is Loeys-Dietz syndrome: usually TGFBR1 or TGFBR2, also SMAD3, TGFB2, and TGFB3, autosomal dominant, with aneurysms and dissections occurring younger, at smaller diameters, and throughout tortuous arteries.

· The Loeys-Dietz paradox is that the variants reduce receptor function in cells yet aortic TGF-beta signaling is increased.

· Patient 3 is vascular Ehlers-Danlos syndrome: COL3A1, type III collagen, autosomal dominant; type III collagen is a homotrimer abundant in arteries, bowel, uterus, and skin, so medium-sized arteries rupture or dissect without a preceding aneurysm and the sigmoid colon and gravid uterus can rupture.

· Most COL3A1 variants replace a glycine in the Gly-X-Y repeat, a position only glycine is small enough to fill, so the triple helix misfolds.

· In vascular Ehlers-Danlos syndrome the skin is thin and translucent but not markedly hyperextensible, and median survival is about 50 years.

· Arithmetic for a three-chain collagen: (1/2) to the third power is 1/8, so about 12.5% of molecules are fully normal with a glycine substitution, against about 50% of the usual amount all normal with a null allele.

· Arithmetic for a four-subunit channel such as KCNQ1: (1/2) to the fourth power is 1/16, about 6% working channels with a dominant negative subunit, against about 50% with a null allele.

· This is why dominant negative missense variants are often more severe than null alleles, and why null COL3A1 alleles cause later, milder disease.

· Clues outside the vessels: upward ectopia lentis with tall stature and arachnodactyly in Marfan; hypertelorism, bifid uvula or cleft palate, and craniosynostosis with no ectopia lentis in Loeys-Dietz; thin skin with bruising and colon and uterine rupture in vascular Ehlers-Danlos.

Distinctions that matter

· Rupture without a preceding aneurysm points to collagen failure, while progressive dilation before dissection points to loss of the elastic scaffold.

· Absence of ectopia lentis with hypertelorism and a bifid uvula separates Loeys-Dietz from Marfan syndrome, which share aortic disease and skeletal features.

Common misconceptions

· Assuming a null allele is always worse than a missense variant. For a multimer the mutant subunit spoils normal partners, so half the protein all normal beats one eighth of it fully normal.

· Grouping all three as the same disease because all three are autosomal dominant. The failing wall component predicts which vessels fail and how.

Where it is taught: Section C03, paragraphs on Loeys-Dietz and vascular Ehlers-Danlos, the comparison table, and the worked example of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C03-CP01

Comparison table: The three heritable disorders of the arterial wall

A competent response contains

· Gene and protein: FBN1 and fibrillin-1; TGFBR1 or TGFBR2 receptors, with SMAD3, TGFB2, and TGFB3 also reported; COL3A1 and type III collagen.

· All three are autosomal dominant.

· Arterial pattern: aortic root dilation with ascending dissection in Marfan; widespread aneurysms with tortuosity, younger and at smaller diameters, in Loeys-Dietz; rupture or dissection of medium-sized arteries in vascular Ehlers-Danlos.

· An aneurysm precedes the event in Marfan and Loeys-Dietz, and rupture occurs without a preceding aneurysm in vascular Ehlers-Danlos.

· Tissue level: medial degeneration with fragmented elastic fibers, lost smooth muscle, and proteoglycan pools in Marfan; reduced receptor function in cells with paradoxically increased aortic TGF-beta signaling in Loeys-Dietz; misfolded type III collagen helix from a glycine substitution in the Gly-X-Y repeat in vascular Ehlers-Danlos.

· Eye: upward ectopia lentis in Marfan; no ectopia lentis in Loeys-Dietz; no characteristic lens finding in vascular Ehlers-Danlos.

· Face, mouth, and skull: no characteristic findings listed for Marfan; hypertelorism, bifid uvula or cleft palate, and craniosynostosis in Loeys-Dietz; no characteristic findings listed for vascular Ehlers-Danlos.

· Skin, bowel, and uterus: no characteristic findings listed for Marfan or Loeys-Dietz; thin translucent skin that is not markedly hyperextensible, easy bruising, and sigmoid colon and gravid uterine rupture in vascular Ehlers-Danlos.

· Skeleton and joints: tall stature, arachnodactyly, pectus deformity, scoliosis, and joint laxity in Marfan; no separate skeletal list in the notes for the other two beyond the craniofacial findings in Loeys-Dietz.

· Median survival in vascular Ehlers-Danlos is about 50 years.

Distinctions that matter

· Marfan and Loeys-Dietz both dilate before they dissect, and the discriminator is age, diameter, how widespread the aneurysms are, and the craniofacial findings.

· The lens is the fastest separator between Marfan and Loeys-Dietz, and the skin, bowel, and uterus are the fastest separators for vascular Ehlers-Danlos.

Common misconceptions

· Waiting for a Marfan-sized aortic diameter in Loeys-Dietz. Dissection occurs at smaller diameters and younger ages.

· Expecting markedly hyperextensible skin in vascular Ehlers-Danlos. The skin is thin and translucent rather than stretchy.

Where it is taught: Section C03, comparison table on heritable disorders of the arterial wall of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-BD1

Brain dump 1 · foundational retrieval: Repolarizing currents and the long QT genotypes

A competent response contains

· Phase 3 repolarization depends on the rapid delayed rectifier current IKr from KCNH2, also called hERG, and the slow current IKs from KCNQ1 with its accessory subunit KCNE1; IK1 finishes repolarization.

· Repolarization reserve means the two currents share the work, so losing part of one may barely lengthen the resting QT interval.

· The four situations that unmask it are hypokalemia, bradycardia, a sympathetic surge, and an IKr-blocking drug, which is much of the explanation for incomplete penetrance.

· Congenital long QT syndrome affects about 1 in 2,000.

· LQT1 is the most common: KCNQ1 with loss of IKs, triggered by exercise and especially swimming, with a broad-based T wave.

· LQT2: KCNH2 with loss of IKr, triggered by startle and emotion with hypokalemia dangerous, and a low, notched or bifid T wave.

· LQT3: SCN5A gain of function producing a late Na+ current, with events at rest or in sleep, and a late T wave after a long isoelectric ST segment.

· The pattern across the three is that the K+ channel genes lose function while the Na+ channel gene gains function.

· Channelopathies cause sudden death in a structurally normal heart, so diagnosis rests on the electrocardiogram, family history, and genetic testing.

· Less outward K+ current or an added late inward Na+ current prolongs phases 2 and 3 and lengthens the QT interval.

Distinctions that matter

· IKr and IKs are different currents from different genes, and which one is lost predicts the trigger.

· Loss of function in a repolarizing current and gain of function in a depolarizing current reach the same long QT interval from opposite directions.

Common misconceptions

· Taking a normal resting QT interval as proof that a relative is unaffected. Repolarization reserve can hide a partial current loss until something else stresses it.

· Assuming every long QT gene loses function. SCN5A in LQT3 gains function by leaving a late Na+ current switched on.

Where it is taught: Section C04, first paragraph, the ventricular action potential diagram, and the long QT genotype table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-BD2

Brain dump 2 · integration and application: Why the trigger, the ear, and the fever all follow from the current

A competent response contains

· Faints while swimming: LQT1, because IKs is increased by sympathetic stimulation and normally shortens the action potential during exercise, so losing it leaves repolarization unsupported exactly when it is needed.

· Collapses at a fire alarm: LQT2, because IKr dominates resting repolarization and is the current lost, and startle and emotion are the triggers; low extracellular K+ reduces IKr further, so hypokalemia is dangerous.

· Found dead in bed: LQT3, because a late Na+ current matters most at slow rates, so events occur at rest or in sleep.

· All three lead to early afterdepolarizations and torsades de pointes.

· Romano-Ward syndrome is the common autosomal dominant form with normal hearing.

· Jervell and Lange-Nielsen syndrome is autosomal recessive, from two abnormal alleles of KCNQ1 or KCNE1, and combines congenital sensorineural deafness with marked QT prolongation, often above 500 ms, and childhood sudden death.

· The stria vascularis secretes K+ into endolymph through KCNQ1 and KCNE1 channels to generate the endocochlear potential, so with no working channel hearing fails.

· One working allele suffices for the ear, so heterozygous parents hear normally but may have a mildly long QT interval.

· SCN5A shows allelic heterogeneity: gain of function causes LQT3, while loss of function causes Brugada syndrome and conduction disease, and some variants produce both.

Distinctions that matter

· The trigger reports which current is missing, because each current is recruited in a different physiological state.

· The ear needs only one working allele while the ventricle is affected by one bad allele, which is why the deafness is recessive and the long QT is dominant.

Common misconceptions

· Treating the triggers as coincidence. Exercise, startle, and rest each match the current that normally carries repolarization in that state.

· Assuming one gene gives one disease. SCN5A gives long QT when it gains function and Brugada syndrome when it loses function.

Where it is taught: Section C04, paragraphs on long QT triggers, Romano-Ward and Jervell and Lange-Nielsen, and the final paragraph on SCN5A of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-RT01

Retrieval exercise: Rebuild the Brugada sequence

A competent response contains

· Step 1: reduced peak Na+ current.

· Step 2: the transient outward current Ito deepens the epicardial phase 1 notch.

· Step 3: the epicardial dome is lost while the endocardium keeps its plateau.

· Step 4: the transmural gradient produces coved ST elevation in V1 and V2.

· Step 5: phase 2 reentry gives a closely coupled beat.

· Step 6: polymorphic ventricular tachycardia degenerates to ventricular fibrillation.

· The gene is SCN5A, loss of function, autosomal dominant with incomplete penetrance, found in only about 20% to 25% of patients.

· Right ventricular outflow epicardium has a large Ito and the endocardium does not, so a weaker upstroke lets Ito pull the epicardial membrane below L-type Ca2+ activation only there, which creates the transmural gradient.

· The type 1 electrocardiogram shows coved ST elevation of at least 2 mm in V1 or V2 running into a negative T wave.

· Fever, vagal tone, and Na+ channel blockers unmask the pattern; fever does it because some variant channels inactivate faster when warm, cutting Na+ current further.

· Events cluster at rest, in sleep, and with fever, and slowed outflow conduction probably contributes.

· It is most common in men and more often seen in Southeast Asia, and the larger male epicardial Ito helps explain the male predominance.

· Brugada syndrome causes ventricular fibrillation in a structurally normal heart.

Distinctions that matter

· The ST elevation here comes from a transmural repolarization gradient, not from the current of injury of ischemia.

· Loss of Na+ current is the first event and Ito is the current that exploits it, so the two are not interchangeable in the sequence.

Common misconceptions

· Expecting a positive gene test in most patients. SCN5A variants are found in only about a fifth to a quarter.

· Assuming fever is unrelated to the channel. Some variant channels inactivate faster when warm, so warming worsens the defect.

Where it is taught: Section C04, Brugada paragraph, flow diagram, and stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-PR01

Prediction exercise: Predict the effect of a low plasma potassium in LQT2

A competent response contains

· IKr falls, because the rapid delayed rectifier current drops when extracellular K+ is low.

· The margin left by repolarization reserve is used up, because her IKs cannot cover for an already reduced IKr.

· The ventricular action potential lengthens and the QT interval lengthens, because less outward K+ current prolongs phases 2 and 3.

· L-type Ca2+ channels have time to recover and reopen during the prolonged plateau.

· An early afterdepolarization appears, arising during repolarization in phases 2 and 3 from that inward Ca2+ current.

· The rhythm is torsades de pointes, and it can degenerate into ventricular fibrillation.

· Her risk rises further with an IKr-blocking drug, because it removes the same current that her variant and her low K+ have already reduced.

Distinctions that matter

· Low plasma K+ reduces IKr rather than increasing K+ efflux, which is why hypokalemia lengthens repolarization instead of shortening it.

· The early afterdepolarization starts the rhythm, and uneven repolarization across the wall is what sustains it.

Common misconceptions

· Expecting a normal baseline QT interval to protect her. Repolarization reserve only hides the defect until a second hit arrives.

· Treating the variant, the low K+, and the drug as three separate problems. All three subtract from the same current.

Where it is taught: Section C04, first paragraph on repolarization reserve and the LQT2 row of the genotype table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-BD1

Brain dump 1 · foundational retrieval: Embryology to the characteristic heart defect

A competent response contains

· The endocardial cushions fuse to divide the atrioventricular canal, close the ostium primum, and form parts of the atrioventricular valves.

· Neural crest cells build the aorticopulmonary septum that divides the truncus arteriosus.

· Down syndrome is trisomy 21, about 95% from nondisjunction, mostly maternal meiosis I with advanced maternal age, about 4% from Robertsonian translocation, and about 1% from mosaicism.

· About 40% to 50% of patients with Down syndrome have a heart defect, classically an atrioventricular septal defect with an ostium primum defect, an inlet ventricular septal defect, and a common valve, from failed cushion fusion; other findings are hypotonia, a flat face, and duodenal atresia.

· Large shunts plus a tendency to early pulmonary vascular disease bring pulmonary hypertension sooner in Down syndrome.

· Turner syndrome is 45,X in about half of patients and otherwise mosaic or structural X changes, and it favors left-sided lesions: bicuspid aortic valve in about a fifth to a third, coarctation in about a tenth, and later aortic dilation and dissection; other findings are short stature, webbed neck, and ovarian insufficiency.

· One hypothesis in Turner syndrome is that fetal jugular lymphedema, which leaves the webbed neck, compresses the ascending aorta and reduces left heart flow.

· Coarctation causes arm hypertension, weak delayed femoral pulses, and rib notching from collaterals.

· The 22q11.2 deletion, called DiGeorge or velocardiofacial syndrome, occurs in about 1 in 4,000, removes about 3 million base pairs, is usually de novo, and is transmitted as autosomal dominant.

· Loss of one TBX1 copy disturbs the pharyngeal arches and the neural crest, giving conotruncal defects: interrupted aortic arch type B, truncus arteriosus, tetralogy of Fallot, and a right-sided arch.

· Third and fourth pouch failure causes thymic hypoplasia with T cell deficiency and parathyroid hypoplasia with hypocalcemia, summarized as CATCH-22, along with cleft palate.

· Williams syndrome is a 7q11.23 microdeletion including ELN; elastin also tells smooth muscle to stop proliferating, so haploinsufficiency thickens the media, causing supravalvular aortic stenosis at the sinotubular junction and peripheral pulmonary and renal artery stenosis, with elfin facies, hypercalcemia, and a friendly personality.

· Noonan syndrome is autosomal dominant through the RAS pathway, often PTPN11, with dysplastic pulmonary valve stenosis and hypertrophic cardiomyopathy, Turner-like features, a normal karyotype, and involvement of both sexes.

Distinctions that matter

· Cushion disease gives central atrioventricular defects, while neural crest disease gives outflow and arch defects, which is why Down syndrome and the 22q11.2 deletion have different lesions.

· Turner syndrome and Noonan syndrome share the physical habitus but differ in karyotype, in which side of the heart is affected, and in which sexes are involved.

Common misconceptions

· Explaining the Williams arterial lesion as a missing structural protein alone. Elastin also signals smooth muscle to stop proliferating, so losing one copy thickens the media.

· Expecting a ventricular septal defect as the classic Down syndrome lesion. The classic lesion is a complete atrioventricular septal defect from failed cushion fusion.

Where it is taught: Section C05, opening paragraph and the paragraphs on each syndrome of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-BD2

Brain dump 2 · integration and application: Read the findings back to the mechanism

A competent response contains

· Child 1 is Down syndrome, trisomy 21, with an atrioventricular septal defect from failed endocardial cushion fusion leaving a common atrioventricular valve.

· Child 1 reaches pulmonary hypertension sooner because a large shunt is combined with a tendency to early pulmonary vascular disease.

· Child 2 is the 22q11.2 deletion, with loss of one TBX1 copy disturbing the pharyngeal arches and neural crest to give interrupted aortic arch type B.

· In child 2 the immune and parathyroid findings travel with the cardiac findings because the third and fourth pharyngeal pouches fail alongside the arch and crest derivatives, giving thymic hypoplasia with T cell deficiency and parathyroid hypoplasia with hypocalcemia.

· Child 3 is Turner syndrome, 45,X or mosaic, with coarctation; the lesion gives arm hypertension, weak delayed femoral pulses, and rib notching from collaterals, and her aorta is at risk of later dilation and dissection.

· Child 4 is Williams syndrome, a 7q11.23 microdeletion including ELN, with supravalvular aortic stenosis at the sinotubular junction from a thickened media.

· In child 4 an ELN variant alone causes only familial supravalvular aortic stenosis, because the neighboring genes in the deletion account for the facies, the calcium abnormality, and the personality.

· Child 5 is Noonan syndrome, autosomal dominant through the RAS pathway and often PTPN11, with dysplastic pulmonary valve stenosis and hypertrophic cardiomyopathy and a normal karyotype.

· A syndrome has one cause, an association is a nonrandom clustering without a single cause, and a sequence is a cascade from one defect.

· Examples from the notes: Down syndrome as a syndrome, VACTERL as an association, and Potter sequence as a sequence.

· VACTERL needs at least three features, is usually sporadic with low recurrence risk, and reflects an early mesodermal disturbance, most often with a ventricular septal defect.

· Holt-Oram syndrome from TBX5 gives septal defects with upper limb anomalies, and Fanconi anemia gives radial defects with marrow failure.

Distinctions that matter

· An association names a pattern and not a cause, which is why its recurrence risk is low and no single test confirms it.

· Radial or upper limb defects with a septal defect should raise a single-gene look-alike rather than stopping at the association.

Common misconceptions

· Treating VACTERL as a diagnosis with a genetic test. It is a clustering of features with no single cause.

· Reading hypocalcemia and infection in a child with an arch anomaly as unrelated problems. They come from the same pouch failure.

Where it is taught: Section C05, paragraphs on each syndrome, the paragraph on VACTERL, and the syndrome table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-CP01

Comparison table: Genetic syndromes and their cardiovascular lesions

A competent response contains

· Down syndrome: trisomy 21; atrioventricular septal defect; failed endocardial cushion fusion; hypotonia, flat face, duodenal atresia.

· Turner syndrome: 45,X or mosaic; bicuspid aortic valve and coarctation; reduced left heart flow, with fetal jugular lymphedema compressing the ascending aorta as one hypothesis; short stature, webbed neck, ovarian insufficiency.

· 22q11.2 deletion syndrome: microdeletion of about 3 million base pairs including TBX1; interrupted arch type B, truncus arteriosus, tetralogy of Fallot; disturbed pharyngeal arches and neural crest; thymic hypoplasia, hypocalcemia, cleft palate.

· Williams syndrome: 7q11.23 deletion including ELN; supravalvular aortic stenosis with peripheral pulmonary and renal artery stenosis; elastin haploinsufficiency allowing smooth muscle proliferation and a thickened media; elfin facies, hypercalcemia, friendly personality.

· VACTERL: a sporadic association with no single cause; ventricular septal defect most often; early mesodermal disturbance; vertebral, anal, tracheoesophageal, renal, and radial limb defects.

· Noonan syndrome: autosomal dominant RAS pathway, often PTPN11; dysplastic pulmonary valve stenosis and hypertrophic cardiomyopathy; RAS pathway signaling in valve and myocardial development; Turner-like features with a normal karyotype in both sexes.

Distinctions that matter

· Down syndrome and the 22q11.2 deletion sit at opposite ends of the heart: cushions in the middle against outflow and arch at the exit.

· Turner syndrome and Williams syndrome both narrow the left-sided outflow, but one is a valve and arch problem and the other is a thickened media above the valve.

Common misconceptions

· Filling the Noonan row with Turner karyotype findings. The karyotype is normal and both sexes are affected.

· Giving VACTERL a genetic basis in the second column. It is an association, so the entry is that it is sporadic with no single cause.

Where it is taught: Section C05, table of genetic syndromes and their characteristic cardiovascular lesions of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-Q01

Board-style question: A newborn with hypocalcemia and an arch anomaly

Correct answer: C. Loss of one copy of TBX1 disturbing the pharyngeal arches and the neural crest

Why it is right: The deletion removes one copy of TBX1, which disturbs the pharyngeal arches and the neural crest that build the aorticopulmonary septum and the arch, giving interrupted aortic arch type B, while failure of the third and fourth pharyngeal pouches produces thymic hypoplasia with T cell deficiency and parathyroid hypoplasia with hypocalcemia.

A is wrong: Trisomy 21 with failed cushion fusion gives an atrioventricular septal defect with a common valve, together with hypotonia, a flat face, and duodenal atresia, and it does not explain the absent thymus or the low calcium.

B is wrong: Elastin haploinsufficiency in Williams syndrome thickens the arterial media and gives supravalvular aortic stenosis with hypercalcemia rather than an interrupted arch with hypocalcemia.

D is wrong: A GAA repeat expansion in FXN lowers frataxin and causes Friedreich ataxia with concentric hypertrophy later in childhood, not a neonatal arch interruption with thymic and parathyroid failure.

A competent response contains

· Hypocalcemia, absent thymus, cleft palate, and interrupted aortic arch type B are the 22q11.2 deletion.

· Loss of one TBX1 copy disturbs the pharyngeal arches and the neural crest, and failure of the third and fourth pouches explains the thymic and parathyroid findings.

Distinctions that matter

· Conotruncal and arch defects follow neural crest disease, while central atrioventricular defects follow cushion disease.

Common misconceptions

· Treating the cardiac, immune, and endocrine findings as separate conditions in one infant. They share one developmental origin.

Where it is taught: Section C05 of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C06-BD1

Brain dump 1 · foundational retrieval: Dystrophin, frataxin, and mitochondrial DNA

A competent response contains

· Dystrophin links actin through beta-dystroglycan and alpha-dystroglycan to laminin in the basement membrane, protecting the sarcolemma during contraction.

· DMD sits at Xp21 and is the largest human gene, so about a third of cases are new variants; Duchenne and Becker dystrophies are X-linked recessive.

· Duchenne dystrophy affects about 1 in 3,500 to 5,000 males, and deletions make up about two thirds of variants.

· Reading frame rule: in-frame deletions make a shortened, partly functional dystrophin, which is Becker, and frameshifts create a premature stop with no dystrophin, which is Duchenne; the rule is correct in about 90% of cases.

· Without dystrophin, contraction tears the membrane, Ca2+ enters, and proteases and mitochondrial Ca2+ overload cause necrosis.

· Creatine kinase is very high, with Gowers sign and calf pseudohypertrophy.

· Cardiac fibrosis starts subepicardially in the inferolateral left ventricle, producing dilated cardiomyopathy by the late teens, with tall R waves in V1, lateral Q waves, and sinus tachycardia; Becker cardiomyopathy can outpace limb weakness.

· Female carriers can develop cardiomyopathy when random or skewed X inactivation leaves many myocytes expressing only the mutant allele.

· Friedreich ataxia is autosomal recessive from a GAA repeat expansion in the first intron of FXN, normal up to about 33 repeats and disease alleles about 66 to more than 1,000; the expansion silences transcription.

· Frataxin is needed for mitochondrial iron-sulfur cluster assembly, which aconitase and complexes I to III depend on, so energy failure, mitochondrial iron accumulation, and oxidative damage hit dorsal root ganglia and spinocerebellar tracts, the heart, and beta cells.

· Friedreich cardiomyopathy is concentric hypertrophy without extensive disarray, with later fibrosis and dilation, and is the usual cause of death.

· Mitochondrial DNA is about 16,600 base pairs and encodes 13 respiratory chain subunits, 22 transfer RNAs, and 2 ribosomal RNAs, and is maternally inherited.

· Heteroplasmy, random segregation, and the oocyte bottleneck make mutant fractions vary between tissues and siblings, and disease appears where a tissue's threshold is exceeded, lowest in brain, retina, muscle, and heart.

· Kearns-Sayre syndrome is a large, usually sporadic mitochondrial DNA deletion with onset before age 20, giving progressive external ophthalmoplegia, pigmentary retinopathy, and conduction block, high CSF protein, or ataxia; His-Purkinje block can progress to complete heart block and muscle shows ragged red fibers.

Distinctions that matter

· Deletion size does not predict severity; the reading frame does.

· Membrane fragility causes dilation, energy failure causes hypertrophy, and a mitochondrial DNA deletion favors the conduction system.

Common misconceptions

· Assuming a larger deletion means Duchenne. A large in-frame deletion still makes a partly working protein and gives Becker dystrophy.

· Treating Friedreich ataxia as a coding variant. The GAA expansion sits in the first intron and silences transcription rather than altering the protein sequence.

Where it is taught: Section C06, dystrophin paragraphs, worked example, Friedreich paragraph, mitochondrial paragraph, and the disorder table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C06-BD2

Brain dump 2 · integration and application: Work the frame, the carrier, and the threshold

A competent response contains

· Boy A: 522 divided by 3 is 174 with no remainder, so 174 codons are removed, the downstream frame is intact, a shortened dystrophin with both binding ends is made, and the disease is Becker.

· Boy B: 197 divided by 3 leaves a remainder of 2, so the frame shifts, a premature stop appears, and no dystrophin is made, which is Duchenne.

· Boy A has the larger deletion and the milder disease, because the reading frame rather than the amount of sequence lost decides whether a protein is made.

· A premature stop codon causes the transcript to undergo nonsense-mediated decay.

· The carrier develops cardiomyopathy because random X inactivation leaves some myocytes expressing only the mutant DMD allele, and if enough do, through chance or skewed inactivation, those cells are damaged with each contraction.

· Her heart cannot replace them because cardiomyocytes are permanent cells, so the damaged myocytes are replaced by fibrosis.

· The two sisters differ because heteroplasmy, random segregation, and the oocyte bottleneck give them different mutant fractions, and disease appears only where a tissue's threshold is exceeded.

· Thresholds are lowest in brain, retina, muscle, and heart, because those tissues depend most heavily on oxidative energy supply.

· Membrane fragility gives a dilated ventricle because repeated myocyte loss and fibrosis leave a weak wall that stretches.

· Failure of energy supply gives a thick ventricle, as concentric hypertrophy in Friedreich ataxia.

· A large mitochondrial DNA deletion shows itself in the conduction system as progressive His-Purkinje block that can reach complete heart block.

Distinctions that matter

· The reading frame rule is arithmetic on nucleotides, not a judgment about deletion size, and it holds in about 90% of cases.

· X inactivation and heteroplasmy are different mechanisms that produce the same result, which is cell to cell variation within one person.

Common misconceptions

· Calling a female carrier of an X-linked recessive disease immune to the phenotype. Enough myocytes expressing only the mutant allele produce cardiomyopathy.

· Expecting siblings who inherit the same mitochondrial variant to be equally affected. The mutant fraction they receive and keep differs.

Where it is taught: Section C06, worked example on the reading frame rule, the carrier sentence, the mitochondrial paragraph, and the stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C06-Q02

Board-style question: Progressive block in a young adult

Correct answer: A. A large deletion of mitochondrial DNA removing respiratory chain subunits and transfer RNAs

Why it is right: A large mitochondrial DNA deletion removes respiratory chain subunits and transfer RNAs, so oxidative energy supply fails in the tissues with the lowest threshold, giving progressive external ophthalmoplegia, pigmentary retinopathy, and His-Purkinje conduction block before age 20, with ragged red fibers on muscle biopsy, and the deletion is usually sporadic so relatives are unaffected.

B is wrong: A frameshift DMD deletion abolishing dystrophin causes Duchenne dystrophy in males, with very high creatine kinase, calf pseudohypertrophy, and dilated cardiomyopathy from inferolateral fibrosis, not ophthalmoplegia with retinopathy.

C is wrong: A GAA repeat expansion in FXN causes Friedreich ataxia with ataxia, sensory loss, and concentric hypertrophy rather than progressive external ophthalmoplegia and ragged red fibers.

D is wrong: A dominant negative MYH7 variant causes hypertrophic cardiomyopathy with myocyte disarray and a thick nondilated ventricle, and it does not affect the eyes, the retina, or skeletal muscle histology in this way.

A competent response contains

· Ophthalmoplegia with ptosis, pigmentary retinopathy, and progressive conduction block before age 20 is Kearns-Sayre syndrome.

· Ragged red fibers point to a mitochondrial DNA disorder, and the lesion is a large, usually sporadic deletion, which is why the mother and brothers are unaffected.

Distinctions that matter

· A large mitochondrial DNA deletion favors the His-Purkinje system, which separates it from the dilated and hypertrophic phenotypes of the other inherited myocardial diseases.

Common misconceptions

· Expecting every mitochondrial disorder to show maternal transmission. A large deletion of this kind is usually sporadic.

Where it is taught: Section C06 of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C07-BD1

Brain dump 1 · foundational retrieval: Why the adult heart grows by hypertrophy

A competent response contains

· Fetal cardiomyocytes divide, but around birth they largely leave the cell cycle, and many become polyploid or binucleate.

· Cell cycle exit is maintained by low cyclin and cyclin-dependent kinase activity, the inhibitors p21 and p27, and retinoblastoma protein repression of E2F.

· Carbon-14 dating shows renewal of only about 1% per year in young adults, and no confirmed adult cardiac stem cell rebuilds muscle, so the heart is a permanent tissue.

· Skeletal muscle regenerates from satellite cells and smooth muscle can divide, but cardiac muscle has no such reserve.

· By Laplace's law, ventricular wall stress is proportional to pressure times radius divided by twice the wall thickness.

· Pressure overload from hypertension or aortic stenosis adds sarcomeres in parallel and thickens the wall, which is concentric hypertrophy; volume overload from regurgitation adds sarcomeres in series and enlarges the cavity, which is eccentric hypertrophy; both tend to return wall stress to normal.

· Normal heart weight is about 250 to 300 g in women and about 300 to 350 g in men.

· Stretch is sensed by integrins, Z disc proteins, and stretch-activated channels.

· Angiotensin II, endothelin-1, and norepinephrine act through Gq, phospholipase C, Ca2+, and protein kinase C, and sustained Ca2+ activates calcineurin, which dephosphorylates NFAT so it enters the nucleus and activates hypertrophic genes with GATA4 and MEF2.

· Exercise hypertrophy uses IGF-1 through PI3K and Akt instead.

· The fetal gene program raises ANP in the ventricle, BNP in proportion to stretch, beta-myosin, and skeletal alpha-actin, and lowers alpha-myosin and SERCA2a, with a shift toward glucose use.

· Consequences: slower myosin saves energy, less SERCA2a slows relaxation, and the natriuretic peptides raise cGMP to oppose the renin-angiotensin-aldosterone system.

· BNP below about 100 pg/mL makes heart failure an unlikely cause of acute dyspnea.

Distinctions that matter

· Concentric and eccentric hypertrophy differ in where sarcomeres are added, which is why one thickens the wall and the other enlarges the cavity.

· The IGF-1 route through PI3K and Akt and the Gq route through calcineurin and NFAT both enlarge myocytes, and the route decides whether the fetal gene program switches on.

Common misconceptions

· Describing adult cardiac hypertrophy as an increase in cell number. The myocytes enlarge, since they have left the cell cycle.

· Reading a raised BNP as a hormone that worsens the picture. It raises cGMP and opposes the renin-angiotensin-aldosterone system, and it is useful as a measure of stretch.

Where it is taught: Section C07, paragraphs on cell cycle exit, Laplace's law, signaling, and the comparison table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C07-BD2

Brain dump 2 · integration and application: Two thick hearts and one scar

A competent response contains

· The rower: IGF-1 through PI3K and Akt, the fetal gene program not activated, capillaries keeping pace with no fibrosis, and regression when training stops.

· The hypertensive man: Gq with calcineurin and NFAT, the fetal gene program activated with ANP, BNP, and beta-myosin, capillaries lagging with interstitial fibrosis, and hypertrophy that may progress to failure.

· Pathologic hypertrophy progresses because capillaries lag, SERCA2a falls, and collagen accumulates, so ischemia and mishandled Ca2+ kill myocytes.

· The ventricle dilates despite starting thick because lost myocytes are not replaced, the survivors are overloaded, and a larger radius raises wall stress again.

· Repair after infarction: macrophages clear dead tissue and release TGF-beta, fibroblasts convert into contractile myofibroblasts, type III collagen is laid down, and it is later replaced by cross-linked type I collagen.

· The scar is strong but cannot contract and cannot conduct, so surviving myocytes hypertrophy to carry the load and border-zone strands support reentrant ventricular tachycardia.

· Stunned and hibernating myocardium can recover; scar cannot.

· Every disease here that kills myocytes ends in fibrosis, hypertrophy of the survivors, and an arrhythmic substrate, because cardiomyocytes are permanent cells.

· The same repair follows myocyte loss in the muscular dystrophies and in Friedreich ataxia.

Distinctions that matter

· Physiologic and pathologic hypertrophy both enlarge existing myocytes, and the signaling route rather than the amount of thickening decides the outcome.

· Recoverable myocardium and scar look alike on a wall motion study and differ entirely in whether the cells are alive.

Common misconceptions

· Reading dilation as a different disease from the earlier thickening. It is the same cycle after enough myocytes have been lost.

· Treating scar as inert. It cannot contract or conduct, and its border zone is where reentrant ventricular tachycardia arises.

Where it is taught: Section C07, comparison table, maladaptive cycle, and the final paragraph on repair of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C07-RT01

Retrieval exercise: Rebuild the maladaptive hypertrophy cycle

A competent response contains

· Step 1: sustained load raises wall stress.

· Step 2: myocytes hypertrophy and fetal genes switch on.

· Step 3: capillaries lag, SERCA2a falls, and collagen accumulates.

· Step 4: ischemia and Ca2+ mishandling kill myocytes.

· Step 5: the survivors are overloaded and the ventricle dilates.

· Step 6: a larger radius raises wall stress again, closing the loop.

· The genes switched on are ANP in the ventricle, BNP in proportion to stretch, beta-myosin heavy chain, and skeletal alpha-actin, with alpha-myosin and SERCA2a switched down.

· Less SERCA2a slows relaxation, and lagging capillaries with accumulating collagen reduce oxygen supply to a thicker wall.

· The survivors have more work because the lost myocytes are gone and the same load is shared among fewer cells.

· By Laplace's law wall stress is proportional to pressure times radius divided by twice wall thickness, so a dilated cavity raises wall stress and sends the loop round again.

· Renewal is only about 1% per year and no confirmed adult cardiac stem cell rebuilds muscle, so lost myocytes are replaced by scar and the loop cannot be broken by replacing cells.

Distinctions that matter

· Hypertrophy at first lowers wall stress, and it is the later loss of myocytes and the dilation that raise it again.

· The fetal gene program is an adaptation that saves energy and slows relaxation at the same time, so it is neither purely protective nor purely harmful.

Common misconceptions

· Drawing the cycle as a straight line ending in failure. Each turn regenerates the stimulus that started it.

· Expecting new myocytes to replace the lost ones over time. Renewal of about 1% per year cannot keep pace.

Where it is taught: Section C07, cycle diagram on maladaptive pathologic hypertrophy of the CV-14 Cardiovascular Genetics notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-15

CV-15 Heart and Lung Interactions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-15-01-C01-BD1

Brain dump 1 · foundational retrieval: Build the two circuits side by side

A competent response contains

· The two circuits are in series, so each ventricle ejects about 5 L/min at rest and any pressure difference between them reflects a difference in resistance.

· Pulmonary artery pressure is about 25/8 mmHg with a mean near 15 mmHg; systemic arterial pressure is about 120/80 mmHg with a mean near 93 mmHg.

· Left atrial pressure is about 5 mmHg and right atrial pressure about 2 mmHg.

· Pulmonary capillary hydrostatic pressure is about 7 mmHg, with some sources giving about 10 mmHg; systemic capillary pressure is about 17 mmHg.

· Resistance at 5 L/min is about 2 Wood units in the pulmonary circuit and about 18 Wood units in the systemic circuit, a Wood unit being 1 mmHg per L/min.

· The pulmonary pressure drop is about 10 mmHg against about 90 mmHg systemically, so with equal flow the resistance is about one ninth.

· Pulmonary resistance is spread across arteries, capillaries, and veins; systemic resistance sits mainly in the arterioles.

· Pulmonary artery walls are about one third as thick as the aorta's and pulmonary arterioles have little smooth muscle, so caliber is set mainly by the pressures inside and around the vessels.

· The pulmonary circuit holds about 450 mL of blood and is highly compliant.

· Bronchial arteries from the aorta supply the airways at systemic pressure and carry about 1 to 2% of cardiac output.

· Low pressure has two consequences: the thin right ventricle tolerates a sudden rise in afterload poorly, and a capillary pressure far below the plasma oncotic pressure of about 28 mmHg keeps the alveoli dry.

· Pulmonary hypertension is a sustained resting mean pulmonary artery pressure above about 20 mmHg.

Distinctions that matter

· Flow is the same in both circuits; only the pressure drop and therefore the resistance differ.

· Bronchial arteries supply lung tissue at systemic pressure, while pulmonary arteries carry the whole cardiac output at low pressure for gas exchange.

Common misconceptions

· Writing that the pulmonary circuit carries less blood flow than the systemic circuit because its pressures are lower. The circuits are in series and carry the same output.

· Placing pulmonary resistance in the arterioles by analogy with the systemic bed. Pulmonary arterioles have little muscle and resistance is distributed across the whole bed.

Where it is taught: Section C01, first paragraph, the pressure table, and the paragraph on consequences of low pressure of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C01-BD2

Brain dump 2 · integration and application: Work the pulmonary resistance numbers through exercise and disease

A competent response contains

· At rest, pulmonary vascular resistance is (15 minus 5) divided by 5 L/min, which is 2 Wood units.

· In exercise it is (22 minus 10) divided by 15 L/min, which is 0.8 Wood units, so flow tripled while resistance fell by more than half.

· Had resistance stayed at 2 Wood units, mean pulmonary artery pressure would have approached 40 mmHg.

· The two passive mechanisms are recruitment, the opening of capillaries that were closed or unperfused at rest and mostly apical, and distension, the widening of capillaries already open.

· Recruitment dominates at low pressure and distension at higher pressure.

· Resistance varies inversely with the fourth power of the radius, so modest widening lowers resistance greatly.

· Capillary transit time falls from about 0.8 s to about 0.3 s, which is still enough for oxygen to equilibrate.

· Together these mechanisms spare the right ventricle, limit filtration, and enlarge the gas exchange surface.

· At high lung volume the alveolar vessels, the capillaries in the alveolar walls, are stretched and compressed, so resistance rises, as in hyperinflation or high positive end-expiratory pressure.

· At low lung volume the extra-alveolar vessels lose radial traction and narrow, as in atelectasis; they are pulled open as the lung expands.

· Total pulmonary vascular resistance is therefore lowest near functional residual capacity and rises at both extremes.

· Recruitment and distension are a reserve. After removal of one healthy lung the remaining bed recruits and distends, so resting pressure rises little.

· When bed is lost to emphysema, emboli, or arterial remodeling, the remaining vessels are already recruited at rest, so further flow raises pressure steeply and the first signs are pulmonary hypertension and dyspnea on exercise.

Distinctions that matter

· Recruitment adds parallel channels; distension widens existing ones. Both are passive responses to pressure, not active dilation.

· Alveolar and extra-alveolar vessels respond to lung volume in opposite directions, which is why the resistance curve against lung volume is U shaped.

Common misconceptions

· Attributing the fall in resistance during exercise to vasodilator release. The pulmonary bed has little smooth muscle and the change is passive.

· Expecting resting pulmonary artery pressure to reveal early pulmonary vascular disease. The reserve hides it until exercise uses the reserve up.

Where it is taught: Section C01, paragraph on why resistance falls as flow rises, the worked example, and the paragraphs on lung volume and lost reserve of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw hypoxic pulmonary vasoconstriction from sensor to constriction

A competent response contains

· Hypoxia inhibits the voltage-gated K+ channels Kv1.5 and Kv2.1 that hold the resting potential of pulmonary artery smooth muscle negative.

· The membrane depolarizes and L-type Ca2+ channels open.

· Calcium enters through those channels and is also released from the sarcoplasmic reticulum, activating calmodulin and myosin light chain kinase.

· Rho-kinase inhibits myosin light chain phosphatase and so raises calcium sensitivity.

· Endothelial nitric oxide falls and endothelin-1 rises, adding to the constriction.

· The stimulus is mainly alveolar PO2, which diffuses directly to the small muscular arteries, and the response becomes strong below about 70 mmHg.

· The response persists in an isolated denervated lung, which shows it is local rather than neural.

· Its purpose is to divert blood away from poorly ventilated alveoli toward better ventilated lung, preserving ventilation-perfusion matching, since blood sent to an unventilated alveolus is wasted.

· The upstream oxygen sensor is debated; the leading proposal is a change in mitochondrial reactive oxygen species and redox state.

Distinctions that matter

· Alveolar PO2 rather than arterial or mixed venous PO2 is the main stimulus, because it reaches the small arteries by direct diffusion.

· Calcium entry raises calcium concentration while Rho-kinase raises sensitivity to the calcium already present; both contribute to the same contraction.

Common misconceptions

· Saying hypoxia opens K+ channels in these cells. It inhibits them, and inhibition is what depolarizes the membrane.

· Treating the response as reflex or sympathetic. An isolated denervated lung still constricts.

Where it is taught: Section C02, opening paragraph, the paragraph on ion channel events, and the hypoxic pulmonary vasoconstriction flow diagram of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-BD2

Brain dump 2 · integration and application: Regional against global hypoxia, and what chronic hypoxia leaves behind

A competent response contains

· Pneumonia is regional hypoxia: constriction diverts blood to healthy lung, limits the shunt, and protects arterial oxygen, while mean pulmonary artery pressure barely changes because the rest of the bed recruits.

· Rapid ascent to 4,000 m is global hypoxia: every region constricts at once, pulmonary vascular resistance rises, and nothing is gained in matching.

· Long-standing COPD with chronic hypoventilation is also global, with hypoxia plus loss of capillary bed from emphysema, giving mild to moderate pulmonary hypertension and, over years, chronic cor pulmonale.

· Hypercapnia and acidosis strengthen the constriction.

· On rapid ascent, constriction is uneven, so less constricted regions are overperfused at high pressure, which is the accepted explanation for high-altitude pulmonary edema.

· Sustained hypoxia stabilizes hypoxia-inducible factors HIF-1 alpha and HIF-2 alpha.

· These drive medial hypertrophy, muscularization of arterioles that normally lack muscle, and intimal and adventitial fibrosis, which add a fixed component to the reversible constriction.

· HIF also raises erythropoietin, and the resulting polycythemia raises viscosity and therefore resistance.

· In the fetus, constriction in the fluid-filled lung keeps pulmonary vascular resistance high and sends right ventricular output through the ductus arteriosus.

· At birth, lung expansion, rising alveolar PO2, and nitric oxide lower resistance sharply; failure of that fall is persistent pulmonary hypertension of the newborn.

Distinctions that matter

· Regional hypoxia uses the response as intended and barely raises pressure; global hypoxia raises resistance with no matching benefit.

· Constriction is reversible with oxygen, while HIF-driven remodeling and polycythemia are not, which is why chronic pulmonary hypertension becomes fixed.

Common misconceptions

· Expecting mean pulmonary artery pressure to rise substantially in lobar pneumonia. The unaffected bed recruits and absorbs the diverted flow.

· Treating high-altitude pulmonary edema as a left heart problem. It follows uneven constriction and overperfusion of the remaining open regions.

Where it is taught: Section C02, paragraphs on regional and global hypoxia and on remodeling, and the stop and predict box of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-RT01

Retrieval exercise: Rebuild the hypoxic pulmonary vasoconstriction sequence

A competent response contains

· Step 1: regional alveolar PO2 falls below about 70 mmHg.

· Step 2: Kv channels in pulmonary artery smooth muscle are inhibited; while open they carry K+ outward and hold the resting potential negative.

· Step 3: the membrane depolarizes and L-type Ca2+ channels open.

· Step 4: calcium from those channels and from sarcoplasmic reticulum release drives contraction, with Rho-kinase raising calcium sensitivity by inhibiting myosin light chain phosphatase.

· Step 5: the small arteries and arterioles of that region constrict.

· Step 6: blood is diverted to ventilated lung, preserving ventilation-perfusion matching.

· In a hypoxic systemic bed the opposite happens: adenosine acts on A2 receptors, falling ATP opens ATP-sensitive K+ channels, the membrane hyperpolarizes, and the vessel dilates, with lactate, H+, CO2, nitric oxide, and prostacyclin adding to the dilation.

Distinctions that matter

· Both beds use K+ channels, but closing Kv channels depolarizes pulmonary muscle while opening KATP channels hyperpolarizes systemic muscle.

· The pulmonary response matches blood to ventilated alveoli; the systemic response matches blood to tissue oxygen demand.

Common misconceptions

· Drawing the systemic vessel as constricting as well. Systemic arterioles dilate in hypoxia, which is the whole point of the contrast.

Where it is taught: Section C02, flow diagram on hypoxic pulmonary vasoconstriction and the comparison table of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C02-CP01

Comparison table: Pulmonary against systemic arterioles

A competent response contains

· Local hypoxia: pulmonary arterioles constrict, systemic arterioles dilate.

· Ion channel event: pulmonary Kv channels close and the membrane depolarizes; systemic KATP channels open and the membrane hyperpolarizes.

· Other mediators: pulmonary constriction gains from less endothelial nitric oxide, more endothelin-1, and Rho-kinase; systemic dilation gains from adenosine at A2 receptors, lactate, H+, CO2, nitric oxide, and prostacyclin.

· Purpose: the pulmonary response sends blood to ventilated alveoli, the systemic response sends blood to tissue that needs oxygen.

· Acidosis and a high CO2 tension constrict pulmonary arterioles, strengthening the hypoxic response, and dilate systemic arterioles.

· Global hypoxia gives pulmonary hypertension in the lung, and systemic vasodilation offset by reflex sympathetic tone in the body.

· Pulmonary arterioles have little smooth muscle and share resistance with arteries, capillaries, and veins; systemic arterioles are heavily muscled and carry most of the systemic resistance.

Distinctions that matter

· The direction of the membrane potential change is the quickest way to keep the two beds apart: depolarization in the lung, hyperpolarization in the tissues.

· Carbon dioxide and acidosis move the two beds in opposite directions as well, following the same logic as oxygen.

Common misconceptions

· Assuming any vessel dilates when its environment is hypoxic. In the lung, low alveolar oxygen is a signal that ventilation has failed, not that the tissue needs more flow.

· Treating the pulmonary bed as a resistance bed governed by arteriolar tone like the systemic bed.

Where it is taught: Section C02, table on pulmonary versus systemic arteriolar responses, with wall structure in section C01 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-BD1

Brain dump 1 · foundational retrieval: One breath, four chambers, and the second heart sound

A competent response contains

· Transmural pressure is the pressure inside a chamber minus the pressure outside it, and the heart and great veins sit inside the pleural pressure, so one change in pleural pressure changes the filling pressure of every chamber at once.

· Pleural pressure is about -5 cmH2O at functional residual capacity and about -7.5 cmH2O at end inspiration.

· Venous return is driven by mean systemic filling pressure, about 7 mmHg, minus right atrial pressure.

· In spontaneous inspiration, right atrial pressure falls relative to the extrathoracic veins and the descending diaphragm raises abdominal pressure and pushes blood toward the thorax, so venous return and right ventricular stroke volume rise in the same breath.

· Left ventricular stroke volume falls slightly for three reasons: the expanded pulmonary vessels hold more blood, the fuller right ventricle bows the septum to the left, and the more negative pleural pressure slightly raises left ventricular afterload.

· The extra right ventricular output reaches the left heart a few beats later, in early expiration.

· P2 is delayed by the longer right ventricular ejection and by the low impedance of the expanded pulmonary vessels, which lets blood keep moving forward after right ventricular pressure starts to fall.

· A2 comes slightly earlier because left ventricular stroke volume is slightly smaller.

· The result is physiologic splitting of S2 in inspiration only, best heard at the left upper sternal border because P2 is soft.

· The normal inspiratory fall in systolic pressure is less than 10 mmHg.

Distinctions that matter

· The two ventricles respond to the same breath in opposite directions, and the lag between them is what the pulse and the heart sounds record.

· Ventricular interdependence is a mechanical effect through the septum and pericardium, separate from the volume the pulmonary vessels hold.

Common misconceptions

· Saying inspiration raises the output of both ventricles. Right output rises within the breath and left output falls slightly until the extra volume arrives.

· Explaining the split by saying A2 is delayed. A2 is slightly early and P2 is delayed.

Where it is taught: Section C03, first two paragraphs, the inspiration flow diagram, and the paragraph on P2 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-BD2

Brain dump 2 · integration and application: Exaggerated and reversed breathing effects

A competent response contains

· Pulsus paradoxus is an inspiratory fall in systolic pressure greater than 10 mmHg, an exaggeration of the normal dip rather than a reversal of it.

· In tamponade, pericardial fluid fixes the total volume of the two ventricles, so inspiratory right ventricular filling pushes the septum into the left ventricle and left stroke volume falls far more than normal; the same compression equalizes diastolic pressures and explains hypotension and distended neck veins.

· In severe asthma or COPD the pericardium is normal, and very large pleural pressure swings raise right heart filling and left ventricular afterload instead. Massive pulmonary embolism and severe hypovolemia can also produce it.

· It is less common in constrictive pericarditis, where the rigid pericardium shields the heart from the pleural pressure change, and jugular venous pressure rises on inspiration rather than falling.

· A ventilator raises pleural pressure during inspiration and positive end-expiratory pressure keeps it raised, so right atrial pressure rises, the gradient for venous return narrows, and cardiac output falls.

· The fall is largest in hypovolemia, because mean systemic filling pressure is already low.

· High lung volumes also compress alveolar vessels and can create zone 1 lung.

· The raised pressure around the left ventricle lowers its afterload, so in a failing left ventricle with high filling pressures stroke volume may be maintained, while with low filling pressures the loss of preload dominates.

· The strain phase of a Valsalva maneuver acts the same way, lowering venous return and pulse pressure and triggering a baroreflex rise in heart rate.

· Physiologic splitting: split in inspiration only, because inspiration delays P2 and advances A2.

· Wide splitting: split in both phases and wider in inspiration, from delayed right ventricular emptying, as in right bundle branch block or pulmonic stenosis.

· Fixed splitting: a wide split unchanged by breathing, as in atrial septal defect, because shunt flow falls as venous return rises and right ventricular volume stays constant.

· Paradoxical splitting: split in expiration and single in inspiration, because A2 is delayed until after P2, as in left bundle branch block or severe aortic stenosis.

Distinctions that matter

· Tamponade works through a fixed pericardial volume; severe airflow obstruction works through the size of the pleural pressure swing.

· Spontaneous and positive pressure breathing move pleural pressure in opposite directions, so every downstream effect reverses.

Common misconceptions

· Reading the name pulsus paradoxus as a reversal of the normal pattern. It is the normal direction, exaggerated beyond 10 mmHg.

· Expecting positive pressure ventilation to lower cardiac output in every patient. With a failing left ventricle and high filling pressures the fall in afterload can hold stroke volume up.

Where it is taught: Section C03, pulsus paradoxus paragraph, positive pressure ventilation paragraph, and the S2 splitting table of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-RT01

Retrieval exercise: Rebuild the inspiration sequence through to S2

A competent response contains

· Step 1: pleural pressure falls.

· Step 2: venous return and right ventricular stroke volume rise, helped by the fall in right atrial pressure relative to the extrathoracic veins and by the descending diaphragm raising abdominal pressure.

· Step 3: right ventricular ejection lasts longer, delaying P2, with the low impedance of the expanded pulmonary vessels adding to the delay.

· Step 4: the pulmonary vessels hold more blood and the septum bows left.

· Step 5: left stroke volume falls slightly, so A2 comes slightly earlier; the three reasons are the pooled pulmonary volume, the septal shift, and the small rise in left ventricular afterload.

· Step 6: S2 splits and systolic pressure dips by less than 10 mmHg.

· In expiration each step reverses: pleural pressure rises, venous return and right stroke volume fall, P2 moves earlier, the pulmonary vessels hold less blood, left stroke volume rises, and S2 becomes single.

· The extra right ventricular output reaches the aorta a few beats later, in early expiration.

Distinctions that matter

· The split closes in expiration because P2 moves earlier, not because A2 moves later.

· Right and left stroke volume change within the same breath but not at the same time, and the lag is what produces the pressure dip.

Common misconceptions

· Putting the rise in left ventricular output in the same breath as the rise in right ventricular output. It arrives a few beats later.

Where it is taught: Section C03, flow diagram on inspiration, ventricular output, and S2 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-BD1

Brain dump 1 · foundational retrieval: Starling forces at the pulmonary capillary and the safety margin

A competent response contains

· Outward forces: capillary hydrostatic pressure about 7 mmHg, interstitial hydrostatic pressure about -8 mmHg acting as suction, and interstitial oncotic pressure about 14 mmHg, totaling about 29 mmHg.

· Inward force: plasma oncotic pressure about 28 mmHg.

· Net outward filtration is about 1 mmHg, and the lymphatics carry that filtrate away.

· Three defences hold off edema as capillary pressure rises: increased lymph flow, loss of interstitial suction, and dilution of interstitial protein.

· Alveolar flooding begins once capillary pressure approaches plasma oncotic pressure of about 28 mmHg, so the safety margin is about 21 mmHg.

· Interstitial edema forms first and the alveoli flood second.

· A low plasma albumin lowers plasma oncotic pressure and narrows the margin.

· Chronic elevation widens the margin, with lymphatics able to enlarge about tenfold, so some patients with mitral stenosis tolerate a capillary pressure of 40 mmHg.

· Permeability injury, as in ARDS, lets protein cross the barrier, so protein-rich edema forms at a normal capillary pressure.

Distinctions that matter

· Hydrostatic edema comes from a raised capillary pressure with an intact barrier; permeability edema comes from an injured barrier at a normal pressure, and the protein content of the fluid separates them.

· Interstitial oncotic pressure in the lung is high enough to matter, which is why the pulmonary Starling balance is not simply capillary pressure against plasma oncotic pressure.

Common misconceptions

· Treating interstitial hydrostatic pressure as positive. In the lung it is negative and pulls fluid out of the capillary.

· Expecting the same capillary pressure to flood every patient's alveoli. The margin is narrowed by low albumin and widened by years of raised pressure.

Where it is taught: Section C04, paragraph on Starling forces and the safety margin of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-BD2

Brain dump 2 · integration and application: Which failing chamber, and which side of the circuit caused it

A competent response contains

· COPD is group 3 pulmonary hypertension from lung disease and hypoxia, through hypoxic constriction, remodeling, and loss of capillary bed, and the term cor pulmonale applies.

· Mitral stenosis is group 2, from backward transmission of left atrial pressure, and cor pulmonale does not apply because the cause is left heart disease.

· The five groups: 1 pulmonary arterial hypertension, including idiopathic, BMPR2, and connective tissue disease, with endothelial dysfunction and plexiform lesions; 2 left heart disease; 3 lung disease or hypoxia, including COPD, interstitial disease, and sleep apnea; 4 chronic thromboembolism with organized clots obstructing arteries; 5 multifactorial, for example sarcoidosis.

· Groups 1, 3, and 4 load the right ventricle from the lung side, so cor pulmonale applies to them.

· Cor pulmonale is right ventricular hypertrophy, dilation, or failure from pulmonary hypertension of lung, pulmonary vascular, or ventilatory origin, and right heart failure from left heart disease is excluded even though it is the most common cause of right heart failure overall.

· The right ventricle meets slowly rising afterload with concentric hypertrophy, since by the law of Laplace a thicker wall lowers wall stress.

· It later dilates, the tricuspid valve leaks, and systemic venous pressure rises, giving distended jugular veins, a nutmeg liver, and dependent edema.

· Pulmonary hypertension gives a loud P2 and a parasternal heave, and the ECG shows right axis deviation, a tall R wave in V1, and P pulmonale in lead II.

· In mitral stenosis, pulmonary capillary wedge pressure rises, normally about 6 to 12 mmHg, and it exceeds left ventricular diastolic pressure because the obstruction sits at the valve.

· Anything that shortens diastole makes her worse: exercise, fever, pregnancy, and rapid atrial fibrillation.

· Dyspnea comes from stiff lungs and stimulation of juxtacapillary J receptors, and lying flat shifts blood into the thorax, giving orthopnea.

· The chest film shows Kerley B lines and upper lobe vessel prominence, and chronic congestion leaves hemosiderin-laden macrophages; flooded alveoli give basal crackles and act as shunt.

· In massive pulmonary embolism, clot obstructs the bed and platelet thromboxane A2 and serotonin constrict the rest.

· A thin unhypertrophied right ventricle cannot generate a mean pressure much above about 40 mmHg, and obstruction of about 60% or more of the circulation typically causes acute failure or sudden death.

Distinctions that matter

· Cor pulmonale is defined by where the pulmonary hypertension came from, not by the appearance of the failing right ventricle.

· Wedge pressure separates group 2 from the rest: it is raised when the left heart is the source and normal when the lung is.

Common misconceptions

· Calling any right ventricular failure with raised venous pressure cor pulmonale. Left heart disease is excluded by definition.

· Expecting a previously normal right ventricle to generate high pressures acutely. Hypertrophy takes time, which is why sudden obstruction is so poorly tolerated.

Where it is taught: Section C04, cor pulmonale paragraphs, the pulmonary hypertension group table, the left heart failure paragraph, and the stop and predict box of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-PR01

Prediction exercise: Predict the course of a massive pulmonary embolus

A competent response contains

· Right ventricular afterload rises, from mechanical obstruction by clot plus constriction of the remaining bed by platelet thromboxane A2 and serotonin.

· The right ventricle dilates and its output falls, because a thin unhypertrophied wall cannot generate a mean pressure much above about 40 mmHg.

· The septum shifts into the left ventricle, since the pericardium limits the space the two ventricles share.

· Left ventricular filling falls, so left ventricular output and systemic arterial pressure fall.

· Right coronary perfusion falls with arterial pressure while right ventricular wall stress stays high, so supply drops as demand rises.

· Right ventricular ischemia weakens contraction further, closing the spiral, which is why hypotension in pulmonary embolism signals high mortality.

· Arterial PO_2 falls while arterial PCO_2 is normal or low: the blocked region becomes dead space, diverted blood overperfuses the rest and creates low ventilation-perfusion regions, atelectasis and sometimes a right-to-left shunt through a patent foramen ovale add to the hypoxemia, and hyperventilation clears the extra carbon dioxide.

Distinctions that matter

· Obstruction and vasoconstriction both raise afterload, and the constriction is why the hemodynamic effect can exceed what the clot volume alone suggests.

· The hypoxemia here is mainly low ventilation-perfusion rather than true shunt, which is why it usually responds to oxygen.

Common misconceptions

· Expecting a raised arterial PCO2 because a large part of the lung has become dead space. Chemoreceptor-driven hyperventilation keeps it normal or low.

· Treating the fall in left ventricular output as left ventricular failure. The left ventricle is underfilled, not weak.

Where it is taught: Section C04, massive embolism paragraph and the right ventricular failure spiral, with the hypoxemia of embolism in section C05 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the upright lung and its three zones

A competent response contains

· Pressure rises about 1 cmH2O per centimetre of depth, and 1 mmHg is about 1.36 cmH2O.

· The apex sits about 20 cm above the pulmonic valve and the base about 10 cm below it.

· With a pulmonary artery pressure of 25/8 mmHg, apical pressure is about 10 mmHg in systole and below zero in diastole, while basal pressure is about 33/16 mmHg.

· Blood flow per unit volume therefore rises steeply from apex to base.

· Capillaries in the alveolar walls are also exposed to alveolar pressure, which stays near atmospheric in quiet breathing, and comparing arterial, alveolar, and venous pressure defines the three zones.

· Zone 1: alveolar greater than arterial greater than venous, no flow, no driving pressure, alveolar dead space, normally absent and seen at the apex only with low arterial or high alveolar pressure.

· Zone 2: arterial greater than alveolar greater than venous, intermittent flow mainly in systole, driving pressure is arterial minus alveolar, in the upper and middle lung.

· Zone 3: arterial greater than venous greater than alveolar, continuous flow, driving pressure is arterial minus venous, in the lower lung.

· In zone 2 the outflow end of the capillary collapses to alveolar pressure, so venous pressure does not affect flow, like a waterfall whose flow does not depend on the drop below it.

· Zone 1 is normally absent because apical systolic pressure, about 10 mmHg, still exceeds alveolar pressure.

· Moving down the lung, arterial and venous pressure rise with depth while alveolar pressure stays constant.

Distinctions that matter

· The zones are defined by the order of three pressures, not by fixed anatomical levels, so they move when any of the three changes.

· Zone 1 wastes ventilation while a shunt wastes perfusion; they are opposite failures of matching.

Common misconceptions

· Treating the zones as fixed thirds of the lung. The boundaries shift with posture, arterial pressure, and alveolar pressure.

· Using arterial minus venous pressure as the driving pressure in zone 2. The relevant downstream pressure there is alveolar.

Where it is taught: Section C05, first paragraphs and the table of zones of the upright lung of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C05-BD2

Brain dump 2 · integration and application: Make zone 1 appear, then work the ventilation and perfusion ratios

A competent response contains

· Zone 1 appears when pulmonary artery pressure falls, as in hemorrhage or shock, or when alveolar pressure rises, as in positive pressure ventilation with positive end-expiratory pressure.

· A positive end-expiratory pressure of 15 cmH2O is about 11 mmHg, which exceeds an apical systolic pressure of about 10 mmHg, so apical capillaries carry no flow.

· Zone 1 alveoli are dead space, so arterial PCO2 rises unless ventilation increases.

· Exercise does the reverse, recruiting apical capillaries and converting zone 2 to zone 3.

· Pleural pressure is about -10 cmH2O at the apex and about -2.5 cmH2O at the base at functional residual capacity, because of the weight of the lung.

· Apical alveoli are therefore more expanded and sit on the flat, less compliant part of the pressure-volume curve, so they take in less of each breath, while basal alveoli expand more.

· Ventilation rises toward the base, but less steeply than perfusion does.

· Average ventilation-perfusion ratio is about 0.8, from alveolar ventilation of about 4 L/min over a cardiac output of about 5 L/min.

· At the apex the ratio is about 3, with alveolar PO2 about 130 mmHg and PCO2 about 28 mmHg; at the base it is about 0.6, with PO2 about 90 mmHg and PCO2 about 42 mmHg.

· The high apical PO2 is the usual explanation for the upper lobe location of reactivation tuberculosis, since Mycobacterium tuberculosis is an obligate aerobe.

· High ratio units cannot make up for low ratio units in oxygen because their blood is already nearly saturated on the flat top of the dissociation curve, and the apex also receives little flow.

· They can unload extra carbon dioxide, because the carbon dioxide curve is nearly linear, so mismatch gives hypoxemia with a normal or low arterial PCO2.

· Normal regional mismatch plus the small anatomical shunt produce the normal alveolar to arterial PO2 difference of about 5 to 15 mmHg in young adults.

· Supine, the apex to base differences largely disappear and dependent posterior regions receive more flow.

Distinctions that matter

· Oxygen and carbon dioxide behave differently in mismatch because of the shape of their dissociation curves, which is why hypoxemia appears while PCO2 stays normal or low.

· Apical alveoli are the most expanded but the least well ventilated per breath, since compliance and not resting volume sets the share of each breath they take.

Common misconceptions

· Expecting hyperventilation of healthy lung to correct the hypoxemia of mismatch. Blood leaving those units is already nearly fully saturated.

· Reading a high apical ventilation-perfusion ratio as good gas exchange. The apex contributes little to arterial blood because it receives little flow.

Where it is taught: Section C05, paragraphs on zone 1, on ventilation and ratios from apex to base, and on the supine lung of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-BD1

Brain dump 1 · foundational retrieval: Shunt, and the work the pulmonary endothelium does

A competent response contains

· A shunt is blood that reaches the systemic arteries without passing ventilated alveoli, a ventilation-perfusion ratio of zero.

· Anatomical shunts: right-to-left cardiac connections, pulmonary arteriovenous malformations, and the normal 2% from bronchial and Thebesian veins.

· Intrapulmonary shunts are perfused alveoli that are collapsed or filled, as in atelectasis, lobar pneumonia, pulmonary edema, and ARDS.

· Angiotensin-converting enzyme, a zinc peptidase abundant in the pulmonary capillaries and also called kininase II, removes a terminal dipeptide from angiotensin I to form angiotensin II.

· The same enzyme inactivates bradykinin, so blocking it lets bradykinin accumulate, which explains the dry cough and the angioedema of ACE inhibition.

· In one passage the lung removes much of the serotonin, prostaglandins E2 and F2 alpha, and leukotrienes, and about 30% of norepinephrine.

· Epinephrine, angiotensin II, and vasopressin pass through largely unchanged.

· The capillaries filter small venous clots, fat, and air.

· Small emboli rarely infarct because bronchial arteries keep the tissue alive and endothelial fibrinolysis dissolves clots; infarction is more likely when bronchial flow is also poor, as in left heart failure.

· A clot crossing a patent foramen ovale, present in about a quarter of adults, is a paradoxical embolism.

Distinctions that matter

· Anatomical shunt bypasses the alveoli entirely; intrapulmonary shunt perfuses alveoli that are present but not ventilated.

· The lung clears some mediators in a single pass and leaves others untouched, so the pulmonary endothelium is selective rather than a general filter.

Common misconceptions

· Treating all shunt as congenital or cardiac. Lobar pneumonia and alveolar edema are common causes.

· Attributing the cough of ACE inhibition to a lack of angiotensin II. It follows accumulation of bradykinin, which the same enzyme normally destroys.

Where it is taught: Section C06, first paragraph, the nonrespiratory functions paragraphs, and the final paragraph on filtration of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-BD2

Brain dump 2 · integration and application: Separate the causes of hypoxaemia with oxygen and arithmetic

A competent response contains

· High altitude: low inspired PO2, normal alveolar to arterial difference, PCO2 low from hyperventilation, corrected by 100% oxygen.

· Opioid overdose: hypoventilation, normal alveolar to arterial difference with a high PCO2, corrected by 100% oxygen.

· Interstitial fibrosis: diffusion limitation, raised difference that worsens with exercise, PCO2 normal or low, corrected by 100% oxygen.

· COPD exacerbation: low ventilation-perfusion regions, raised difference, PCO2 variable and often raised in severe disease, corrected by 100% oxygen.

· Dense lobar pneumonia: shunt with a ratio of zero, raised difference, PCO2 normal or low, little improvement on 100% oxygen.

· Shunted blood never meets alveolar gas, and blood from ventilated alveoli is already about 97 to 98% saturated, so raising its PO2 adds only dissolved oxygen at 0.003 mL/dL per mmHg.

· Mixing combines oxygen content rather than partial pressure, and the steep part of the dissociation curve turns the diluted content into a low PO2.

· On room air, ventilated blood carries about 19.9 mL/dL, so arterial content is 0.7 times 19.9 plus 0.3 times (content minus 5), which gives about 17.8 mL/dL, a saturation near 88% and a PaO2 near 55 mmHg.

· On 100% oxygen with an alveolar PO2 of about 660 mmHg, ventilated blood carries 20.1 bound plus about 2.0 dissolved, or 22.1 mL/dL, so arterial content rises only to about 20.0 mL/dL and PaO2 to about 100 mmHg, against more than 500 mmHg in a normal lung.

· Arterial PCO2 stays normal or low because chemoreceptor-driven hyperventilation unloads the extra carbon dioxide.

· A left-to-right shunt causes no cyanosis at first because the flow direction keeps oxygenated blood in the pulmonary circuit.

· The extra pulmonary flow, and in a ventricular defect or a patent ductus the systemic-level pressure, drives endothelial injury, medial hypertrophy, and intimal fibrosis.

· The six steps: the shunt raises pulmonary flow and often pressure; endothelial injury, medial hypertrophy, and intimal fibrosis follow; pulmonary vascular resistance and right-sided pressures rise progressively; right-sided pressures exceed left-sided pressures; the shunt reverses to right-to-left; late cyanosis, clubbing, and polycythemia appear.

· That end state is Eisenmenger syndrome, and by then the remodeled vessels no longer relax.

Distinctions that matter

· Only shunt resists 100% oxygen; low ventilation-perfusion regions respond, which is why oxygen helps in COPD and embolism but not in dense consolidation.

· Alveolar dead space wastes ventilation and raises PCO2, while shunt wastes perfusion and lowers PO2.

Common misconceptions

· Averaging partial pressures when two streams of blood mix. Content mixes, and the dissociation curve then sets the resulting partial pressure.

· Expecting a shunt to raise PCO2 because blood bypasses the alveoli. Hyperventilation clears carbon dioxide easily because its curve is nearly linear.

Where it is taught: Section C06, paragraph on 100% oxygen, the worked shunt example, the hypoxemia table, and the Eisenmenger flow diagram of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-CP02

Comparison table: Mechanisms of arterial hypoxaemia

A competent response contains

· Low inspired PO2: difference normal, PCO2 low from hyperventilation, corrected by oxygen, example high altitude.

· Hypoventilation: difference normal, PCO2 high, corrected by oxygen, example opioid overdose.

· Diffusion limitation: difference increased and worse with exercise, PCO2 normal or low, corrected by oxygen, example interstitial fibrosis.

· Low ventilation-perfusion regions: difference increased, PCO2 normal or low and sometimes raised in severe airflow obstruction, corrected by oxygen, examples COPD, asthma, pulmonary embolism.

· Shunt with a ratio of zero: difference increased, PCO2 normal or low, little improvement with oxygen, examples right-to-left cardiac shunt, lobar pneumonia, alveolar edema.

· Alveolar dead space with a very high ratio: ventilation is wasted rather than perfusion, so PCO2 rises unless ventilation increases, example zone 1 lung created by positive end-expiratory pressure.

Distinctions that matter

· The two mechanisms with a normal alveolar to arterial difference are low inspired oxygen and hypoventilation, and the PCO2 separates those two.

· The response to 100% oxygen separates shunt from everything else.

Common misconceptions

· Expecting the alveolar to arterial difference to be raised in hypoventilation. The alveolar PO2 is low, and the blood follows it.

· Grouping dead space with shunt. Dead space wastes ventilation and raises PCO2, the mirror image of shunt.

Where it is taught: Section C06, table on mechanisms of arterial hypoxemia, with dead space in section C05 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C06-CA01

Clinical application: Hypoxaemia that does not lift on oxygen

A competent response contains

· Recognize: shunt hypoxemia from dense right lower lobe consolidation, with a ventilation-perfusion ratio of zero in the affected units; the two pointers are the PaO2 rising only from 54 to 71 mmHg on close to 100% oxygen and the raised alveolar to arterial difference alongside a low PaCO2.

· Explain: shunted blood never meets alveolar gas, and blood from the ventilated left lung is already about 97 to 98% saturated, so extra alveolar oxygen adds only dissolved oxygen at 0.003 mL/dL per mmHg; the two streams mix by content, and because the mixed content sits on the steep part of the dissociation curve the resulting PO2 stays low.

· Differentiate: in opioid overdose the alveolar to arterial difference is normal and the PaCO2 is high, whereas his difference is raised and his PaCO2 is 31 mmHg; in a COPD exacerbation the hypoxemia comes from low ventilation-perfusion regions rather than a ratio of zero, so oxygen corrects it, while his PaO2 barely moved.

· Investigate: a wedge pressure of 9 mmHg in the normal range of about 6 to 12 mmHg and a mean pulmonary artery pressure of 18 mmHg exclude a raised left atrial pressure; a normal heart size with unilateral lobar consolidation and a clear left lung fits a lung source; the failure of PaO2 to rise on 100% oxygen measures the shunt itself.

· Decide: hypoxic pulmonary vasoconstriction is constricting the arterioles of the consolidated lobe and diverting blood to the ventilated left lung, and without that response more blood would pass through unventilated alveoli and his PaO2 would be lower still.

· Justify: the stimulus is the low alveolar PO2 of the consolidated units, which becomes strong below about 70 mmHg and acts locally by diffusing to the small arteries; hypoxia inhibits Kv channels, the membrane depolarizes, L-type Ca2+ channels open, and calcium with Rho-kinase drives contraction; mean pulmonary artery pressure stays near normal because the hypoxia is regional and the rest of the bed recruits closed capillaries and distends open ones to absorb the diverted flow.

Distinctions that matter

· A normal wedge pressure with severe hypoxemia places the problem in the lung rather than the left heart.

· Regional hypoxia protects arterial oxygen with little pressure cost, while global hypoxia raises pulmonary artery pressure and gains nothing.

Common misconceptions

· Reading the small rise in PaO2 as inadequate oxygen delivery. It is the expected behaviour of a shunt and it is how a shunt is identified.

· Expecting a whole constricted lobe to produce pulmonary hypertension. Recruitment and distension in the remaining bed keep pressure near normal.

Where it is taught: Section C06, shunt paragraphs, worked example, and hypoxemia table, with regional hypoxia in C02 and recruitment in C01 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C01-Q01

Board-style question: Why pulmonary pressure barely rises in exercise

Correct answer: C. Opening of previously unperfused capillaries together with widening of those already open

Why it is right: A small rise in pulmonary artery pressure opens capillaries that were closed or unperfused at rest, adding parallel channels, and widens those already open, and since resistance varies inversely with the fourth power of the radius this lowers resistance enough to let flow triple with a rise in mean pressure of only about 7 mmHg.

A is wrong: Pulmonary arterioles have little smooth muscle and little resting sympathetic tone to withdraw, and exercise raises rather than lowers sympathetic activity.

B is wrong: The bronchial circulation carries only about 1 to 2% of cardiac output at systemic pressure and supplies the airways, so it cannot absorb a 10 L/min rise in output.

D is wrong: A rise in alveolar pressure compresses alveolar wall capillaries and would raise pulmonary vascular resistance rather than lower it.

A competent response contains

· Pulmonary vascular resistance fell from 2 Wood units at rest to 0.8 Wood units in exercise, because flow tripled while the pressure drop rose only from 10 to 12 mmHg.

· The fall is passive: recruitment of closed or unperfused capillaries, mostly apical, and distension of those already open.

Distinctions that matter

· Recruitment dominates at low pressure and distension at higher pressure, and both are responses to pressure rather than active dilation.

Common misconceptions

· Looking for a vasodilator mechanism in a bed whose arterioles have very little smooth muscle.

Where it is taught: Section C01 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C04-Q02

Board-style question: Alveolar flooding at a normal wedge pressure

Correct answer: A. Injury to the alveolar capillary barrier that allows protein to cross at a normal capillary pressure

Why it is right: Sepsis injures the alveolar capillary barrier so protein crosses into the interstitium and alveoli, which removes the inward oncotic force that normally holds fluid in, and edema forms even though the capillary hydrostatic pressure and the wedge pressure are normal.

B is wrong: Backward transmission of a raised left atrial pressure would raise the wedge pressure above 12 mmHg and usually enlarge the heart, and the fluid would be protein poor.

C is wrong: A low plasma oncotic pressure narrows the safety margin but still needs a rise in capillary pressure to flood the alveoli, and it does not let plasma protein cross into the airway.

D is wrong: Lymphatic drainage is a defence that is overwhelmed in this situation rather than the primary fault, and obstructed lymphatics alone would not produce fluid with the protein content of plasma.

A competent response contains

· A wedge pressure of 8 mmHg is within the normal range of about 6 to 12 mmHg, so left atrial pressure is not raised.

· Edema fluid with a protein concentration close to that of plasma shows the alveolar capillary barrier is leaking protein, which is permeability edema at a normal capillary pressure.

Distinctions that matter

· Hydrostatic edema has a raised wedge pressure and protein-poor fluid; permeability edema has a normal wedge pressure and protein-rich fluid.

Common misconceptions

· Assuming any diffuse alveolar edema means a failing left ventricle. The wedge pressure and the protein content settle it.

Where it is taught: Section C04 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C05-Q03

Board-style question: Dead space created at the apex

Correct answer: D. Alveolar pressure at the apex now exceeding arterial pressure, so those ventilated alveoli carry no blood flow

Why it is right: Blood loss lowers pulmonary artery pressure and the positive end-expiratory pressure of about 11 mmHg raises alveolar pressure above the apical arterial pressure, creating zone 1 lung, and those ventilated but unperfused alveoli are dead space, so at a fixed minute ventilation the arterial PCO2 climbs.

A is wrong: Hypoxic constriction would redirect blood flow rather than create dead space, and it lowers perfusion to poorly ventilated units, which does not raise PCO2.

B is wrong: Loss of radial traction narrows extra-alveolar vessels at low lung volume, and this patient's lung volume is raised rather than lowered by the applied pressure.

C is wrong: Collapsed alveoli with continued perfusion are shunt, which lowers PO2 and leaves PCO2 normal or low because hyperventilation clears carbon dioxide.

A competent response contains

· A positive end-expiratory pressure of 15 cmH2O is about 11 mmHg, which exceeds the apical systolic pressure produced by a pulmonary artery pressure of 18/6 mmHg over a 20 cm column.

· Alveolar pressure above arterial pressure is zone 1: those alveoli are ventilated but carry no blood flow, so they are dead space and arterial PCO2 rises unless ventilation increases.

Distinctions that matter

· Zone 1 wastes ventilation and raises PCO2, whereas shunt wastes perfusion and lowers PO2.

Common misconceptions

· Looking for a cause of hypoxemia when the measurement that changed is the PCO2. Dead space is a ventilation problem.

Where it is taught: Section C05 of the CV-15 Heart and Lung Interactions notes.

NPLEX-P1-CARD-15-01-C03-Q04

Board-style question: A large inspiratory fall in systolic pressure

Correct answer: B. Pericardial fluid fixes the combined ventricular volume, so inspiratory right ventricular filling displaces the septum into the left ventricle

Why it is right: With the pericardium full of fluid the two ventricles share a fixed total volume, so the normal inspiratory rise in right ventricular filling can only be accommodated by displacing the septum into the left ventricle, and left ventricular stroke volume therefore falls far more than the normal few millimetres of mercury.

A is wrong: Large pleural pressure swings explain pulsus paradoxus in severe asthma or COPD, but her lungs are clear and the effusion with equalized diastolic pressures identifies the pericardium as the cause.

C is wrong: Pulmonary arterioles constrict in response to low alveolar oxygen rather than to inspiration, and inspiration expands the pulmonary vessels so they hold more blood rather than constricting.

D is wrong: A rigid pericardium shielding the heart describes constrictive pericarditis, in which pulsus paradoxus is less common and jugular venous pressure rises on inspiration.

A competent response contains

· A fall in systolic pressure of 22 mmHg on inspiration is pulsus paradoxus, an exaggeration of the normal dip of less than 10 mmHg.

· A large circumferential effusion with equal ventricular diastolic pressures is tamponade, in which the pericardial fluid fixes the combined volume of the two ventricles.

Distinctions that matter

· In tamponade the pericardium does the work; in severe airflow obstruction the size of the pleural pressure swing does.

Common misconceptions

· Reading the finding as a reversal of the normal inspiratory change. It is the normal direction, exaggerated.

Where it is taught: Section C03 of the CV-15 Heart and Lung Interactions notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-16

CV-16 Immune Responses in Cardiovascular Disease

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-16-01-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild atheroma formation as an inflammatory response

A competent response contains

· Atherosclerosis is a chronic inflammatory and healing response of the intima of large and medium arteries to endothelial injury, sustained by LDL retained in the wall.

· Healthy endothelium under laminar shear stress releases nitric oxide and prostacyclin and displays few adhesion molecules.

· High LDL, hypertension, smoking, hyperglycemia, and disturbed flow cause endothelial dysfunction: the intact endothelium becomes more permeable to LDL, makes less nitric oxide, and activates NF-kappa B.

· NF-kappa B induces selectins for leukocyte rolling, VCAM-1 which binds VLA-4, and MCP-1 which draws cells into the intima; M-CSF converts monocytes into macrophages.

· VCAM-1 is selective for mononuclear cells, so early lesions contain monocytes and T cells rather than neutrophils; the T cells are mainly Th1 and their IFN-gamma activates macrophages.

· LDL trapped on intimal proteoglycans is oxidized; oxidized LDL no longer fits the LDL receptor, which SREBP shuts down as cell cholesterol rises.

· Oxidized LDL enters macrophages through scavenger receptors SR-A and CD36, which are not down-regulated by cell cholesterol, so uptake has no brake.

· ACAT esterifies the excess cholesterol into droplets to make foam cells, and efflux to HDL runs through ABCA1 and ABCG1.

· Foam cells form the fatty streak, already present in many children.

· PDGF from macrophages, endothelium, and platelets draws medial smooth muscle cells into the intima, and TGF-beta makes them lay down collagen over a necrotic core of lipid, cholesterol clefts, debris, and tissue factor.

· Plaques form first at the coronary and renal ostia and the carotid and aortic bifurcations, where flow is disturbed rather than laminar.

Distinctions that matter

· Endothelial dysfunction is not endothelial denudation: the layer is still intact but leaky, less protective, and activated.

· The LDL receptor is cholesterol regulated and the scavenger receptor is not, which is why macrophages keep loading until they become foam cells.

Common misconceptions

· Writing that neutrophils start the lesion. VCAM-1 selects monocytes and lymphocytes, which is why early plaques are mononuclear.

· Treating plaque as passive lipid accumulation. Smooth muscle recruitment and collagen synthesis make it a healing response, and it is inflammation that undoes that repair.

Where it is taught: Section C01, opening paragraphs, the recruitment and foam cell paragraphs, and the atheroma flow figure of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C01-BD2

Brain dump 2 · integration and application: Why the smaller plaque is the dangerous one

A competent response contains

· Lesion A is the vulnerable plaque and is more likely to cause infarction; lesion B with its thick cap and severe stenosis more typically causes stable angina.

· Macrophage matrix metalloproteinases MMP-1, MMP-8, MMP-13, and MMP-9, along with cathepsins, degrade cap collagen.

· Th1 IFN-gamma blocks collagen synthesis by smooth muscle cells, and those cells undergo apoptosis, so the cap is not repaired.

· Rupture usually begins at the macrophage-rich shoulders of the plaque, exposes tissue factor from the necrotic core, and thrombus forms within minutes.

· A minority of coronary thrombi form over plaque erosion, where endothelium is lost over an intact plaque with no cap tear.

· A thin cap is classically under about 65 micrometers, with a large lipid core, many macrophages and T cells, sparse apoptotic smooth muscle cells, and often only mild to moderate stenosis.

· Rheumatoid arthritis and lupus accelerate atherosclerosis because the same cytokines, IL-1 beta and IL-6, are already circulating.

· An hs-CRP of 4 mg/L is in the higher risk band (below 1 mg/L lower risk, 1 to 3 mg/L average, above 3 mg/L higher) and reflects plaque inflammation.

· The route to CRP is oxidized LDL on Toll-like receptors priming pro-IL-1 beta and NLRP3, cholesterol crystals damaging lysosomes, NLRP3 caspase-1 releasing active IL-1 beta, IL-1 beta inducing IL-6, and IL-6 driving hepatic CRP.

Distinctions that matter

· Degree of stenosis predicts angina; composition and inflammation predict rupture. They are separate questions about the same plaque.

· Rupture exposes the core and its tissue factor, while erosion simply loses the endothelial surface; both end in thrombus.

Common misconceptions

· Assuming the tightest lesion causes the infarct. Many infarcts arise from lesions of only mild to moderate stenosis.

· Reading hs-CRP as a measure of plaque size. It reports inflammatory activity through the IL-1 beta and IL-6 route.

Where it is taught: Section C01, paragraph on the fibrous cap, the stable against vulnerable plaque table, and the stop and predict box of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C01-RT01

Retrieval exercise: Rebuild the atheroma sequence and the inflammasome branch

A competent response contains

· Step 1: endothelial dysfunction from LDL, hypertension, smoking, hyperglycemia, and disturbed flow.

· Step 2: NF-kappa B induces selectins, VCAM-1, and MCP-1.

· Step 3: monocytes and Th1 cells enter the intima; selectins and VCAM-1 made them adhere and MCP-1 drew them in.

· Step 4: retained LDL is oxidized.

· Step 5: scavenger receptor uptake creates foam cells and the fatty streak; SR-A and CD36 are not down-regulated by cell cholesterol, so there is no brake on uptake.

· Step 6: PDGF and TGF-beta build a smooth muscle and collagen cap.

· Step 7: metalloproteinases and IFN-gamma thin the cap.

· Step 8: rupture exposes tissue factor and thrombus forms.

· Inflammasome branch: oxidized LDL on Toll-like receptors, macrophage makes pro-IL-1 beta and NLRP3, cholesterol crystals damage lysosomes, NLRP3 inflammasome assembles, caspase-1 cleaves pro-IL-1 beta, active IL-1 beta is released, IL-1 beta induces IL-6, and the liver produces CRP.

· hs-CRP bands: below 1 mg/L lower risk, 1 to 3 mg/L average, above 3 mg/L higher risk.

Distinctions that matter

· Priming and activation are two separate signals: Toll-like receptors supply the pro-form and the sensor, and crystals supply the trigger.

· IL-1 beta acts locally on the plaque while IL-6 carries the signal to the liver, which is why a blood test reflects wall inflammation.

Common misconceptions

· Putting cap formation before foam cells. Smooth muscle recruitment is the repair response to an established lipid lesion.

· Describing CRP as made by the plaque. It is a hepatic acute phase protein induced by IL-6.

Where it is taught: Section C01, atheroma flow figure and the IL-1 beta and CRP paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C02-BD1

Brain dump 1 · foundational retrieval: Rheumatic fever from the throat to the valve

A competent response contains

· Sequence: group A streptococcal pharyngitis in a susceptible child; antibodies and T cells form against M protein and group A carbohydrate; about 2 to 3 weeks later they cross-react with myosin and valve proteins; activated valve endothelium admits CD4 T cells; pancarditis with Aschoff bodies and verrucae on the lines of closure; repeated attacks heal by fibrosis into chronic mitral stenosis.

· The organism is Streptococcus pyogenes, the infection is pharyngitis rather than skin, and the usual age is about 5 to 15 years.

· The M protein is an alpha-helical coiled coil resembling cardiac myosin and tropomyosin, and antibody to the N-acetylglucosamine of the group A carbohydrate cross-reacts with valve laminin.

· Cross-reactive antibody activates valve endothelium and raises VCAM-1, letting CD4 T cells enter and sustain inflammation with IFN-gamma and IL-17.

· The disease is classed as type II hypersensitivity, but chronic valve damage is driven largely by T cells.

· Risk is higher with certain HLA class II alleles and rheumatogenic strains, and the organism is absent from the heart lesions.

· The Aschoff body is fibrinoid collagen surrounded by T cells, plasma cells, and Anitschkow cells, activated macrophages with a wavy ribbon of chromatin.

· Aschoff bodies occur in all three layers, so the carditis is a pancarditis: fibrinous pericarditis resolves, myocarditis is the main cause of acute death, and valvulitis leaves lasting damage.

· Acute verrucae are about 1 to 2 mm along the lines of closure, mitral far more often than aortic.

· Repeated attacks give leaflet thickening, commissural fusion, and short fused chordae, producing fish-mouth mitral stenosis.

Distinctions that matter

· The latent period separates rheumatic carditis from infective endocarditis: here the organism has gone before the heart is injured.

· Mimicry follows throat infection only, while poststreptococcal glomerulonephritis follows throat or skin infection because it is an immune complex disease.

Common misconceptions

· Looking for bacteria in the valve lesion. The lesions are sterile and the injury is immune.

· Calling acute rheumatic valvulitis a stenosis. The acute lesion usually causes mitral regurgitation, and stenosis is the scar of repeated attacks.

Where it is taught: Section C02, opening paragraphs, the mimicry flow figure, and the pathology paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C02-BD2

Brain dump 2 · integration and application: Work a suspected case from timing and serology

A competent response contains

· Major criteria present: carditis (new murmur) and polyarthritis, which are migratory and in large joints.

· Minor criteria present: fever of 38.7 degrees C (101.7 degrees F), which meets the low-risk threshold of 38.5 degrees C (101.3 degrees F) or higher, and an ESR of 72 mm/h, which meets the threshold of 60 mm/h or higher.

· Two major criteria with evidence of recent streptococcal infection meet the requirement for an initial attack.

· The prolonged PR interval cannot be counted here, because carditis is already serving as a major criterion.

· Serology: antistreptolysin O rises after about one week and peaks at about 3 to 6 weeks; anti-DNase B peaks later and persists longer, which helps when chorea appears late. A rising titer is stronger evidence than a single value.

· The joints recover because the arthritis is a transient exudative synovitis that heals without deformity; the valve scars because valve inflammation heals by fibrosis.

· The child with involuntary writhing movements months later has Sydenham chorea, from antibodies cross-reacting with basal ganglia neurons, and it can support the diagnosis alone after titers have fallen.

· The child with skin sores, dark urine, and red cell casts has poststreptococcal glomerulonephritis, an immune complex disease, which is why it can follow either pharyngitis or impetigo.

· The site of infection decides the answer because mimicry of cardiac proteins follows throat infection, whereas immune complexes can form after infection at either site.

· Recurrences after new streptococcal infections are common and each adds valve scarring.

Distinctions that matter

· Higher risk populations use lower thresholds: fever 38.0 degrees C (100.4 degrees F), ESR 30 mm/h, and monoarthritis or polyarthralgia counted as major.

· A negative throat culture does not argue against the diagnosis, because the disease appears weeks after the organism has cleared.

Common misconceptions

· Counting the prolonged PR interval alongside carditis. It is excluded when carditis is a major criterion so that one process is not counted twice.

· Calling late chorea unrelated because titers have fallen. Anti-DNase B persists longer and chorea can carry the diagnosis by itself.

Where it is taught: Section C02, Jones criteria table, serology paragraph, and the stop and predict box of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-BD1

Brain dump 1 · foundational retrieval: Organize the vasculitides by immune effector

A competent response contains

· Immune complex group: polyarteritis nodosa, IgA vasculitis, and cryoglobulinemic vasculitis.

· ANCA group: granulomatosis with polyangiitis, microscopic polyangiitis, and eosinophilic granulomatosis with polyangiitis.

· Granulomatous T-cell group: giant cell arteritis and Takayasu arteritis.

· Immune complexes form in slight antigen excess, escape clearance because they are small, and deposit where flow is turbulent or filtered, as in glomeruli and dermal venules.

· They activate the classical pathway, C5a recruits neutrophils, and neutrophil enzymes and oxidants cause fibrinoid necrosis; complement may fall and immunofluorescence shows granular immunoglobulin and complement.

· Nuclear dust from dying neutrophils gives leukocytoclastic vasculitis, and in dermal venules the lesion is palpable purpura: raised, nonblanching lesions on dependent skin.

· ANCA are IgG against PR3, giving cytoplasmic staining, or MPO, giving perinuclear staining because ethanol fixation lets cationic MPO move to the nucleus.

· Infection, TNF, or C5a primes neutrophils and moves these enzymes to the surface; ANCA bind by Fab and engage Fc-gamma receptors on the same cell.

· The neutrophil then releases oxidants, proteases, and extracellular traps onto endothelium and activates the alternative pathway for more C5a; because the neutrophil delivers the injury, lesions are pauci-immune and complement is normal.

· In giant cell and Takayasu arteritis, adventitial dendritic cells activate CD4 T cells, IFN-gamma recruits macrophages and giant cells into the media around a fragmented internal elastic lamina, intimal hyperplasia narrows the lumen, there are no deposits, and IL-6 drives a very high ESR and CRP.

Distinctions that matter

· Granular deposits mean immune complexes, few or no deposits mean ANCA or T-cell disease, and linear deposits mean antibody against fixed type IV collagen in anti-GBM disease.

· Immune complex disease is often complement consuming while ANCA disease uses the alternative pathway locally and leaves serum complement normal.

Common misconceptions

· Grouping vasculitis by vessel size when the question is about serology. Size predicts the organs involved; mechanism predicts the complement level and the immunofluorescence.

· Treating pauci-immune as meaning no antibody involved. ANCA is the antibody; it simply acts on the neutrophil instead of depositing in the wall.

Where it is taught: Section C03, immune complex paragraphs, the ANCA and granulomatous paragraphs, and the vasculitis table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-BD2

Brain dump 2 · integration and application: Name the vasculitis from complement and serology

A competent response contains

· Patient 1 has mixed cryoglobulinemic vasculitis: hepatitis C drives clonal B cells to make IgM rheumatoid factor that binds IgG, and these complexes consume the classical pathway, so C4 is very low; the complexes precipitate below 37 degrees C (98.6 degrees F).

· Patient 2 has microscopic polyangiitis: MPO-ANCA with pulmonary hemorrhage and crescentic glomerulonephritis and no granulomas; a pauci-immune biopsy settles it.

· Patient 3 has IgA vasculitis: complexes of galactose-deficient IgA1 deposit in skin, gut, joints, and mesangium after a respiratory infection, and mesangial IgA settles it.

· Patient 4 has polyarteritis nodosa: medium and small muscular arteries, in some cases through hepatitis B antigen-antibody complexes, ANCA negative, with lungs spared.

· C4 is very low in patient 1 because IgM-IgG complexes drive the classical pathway hard.

· Complement stays normal in patient 3 because IgA uses the alternative and lectin pathways rather than the classical pathway.

· Patient 2 has no deposits because the primed neutrophil, armed by ANCA, delivers the injury directly rather than complexes settling in the wall.

· Patient 4 has renal artery lesions with infarcts and renin-dependent hypertension but no glomerulonephritis, because the disease is in muscular arteries rather than in glomerular capillaries; mesenteric lesions cause postprandial pain and vasa nervorum lesions cause mononeuritis multiplex.

· The 74-year-old has giant cell arteritis and the 29-year-old has Takayasu arteritis; both are type IV granulomatous T-cell disease of large arteries with IL-6 driving a high ESR.

· Giant cell arteritis involves cranial carotid branches and the vertebral arteries, giving temporal headache, jaw claudication, and vision loss from posterior ciliary artery occlusion; Takayasu arteritis involves the aorta and its main branches in patients under 40, often women, giving weak arm pulses and an arm blood pressure difference.

Distinctions that matter

· Low C4 with near normal C3 points to classical pathway consumption by IgM-IgG cryoglobulins, whereas IgA vasculitis spares C4 even though complexes are deposited.

· Segmental lesions of different ages with microaneurysms and no lung involvement separate polyarteritis nodosa from the ANCA diseases.

Common misconceptions

· Expecting glomerulonephritis in polyarteritis nodosa. The lesions are in muscular arteries, so the kidney shows infarcts and renin-dependent hypertension instead.

· Reading normal complement as evidence against immune complex disease. IgA vasculitis has deposits with normal complement.

Where it is taught: Section C03, immune complex and ANCA paragraphs, the vasculitis table, and the worked example on using complement of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-CP01

Comparison table: Immune complex, ANCA, and granulomatous vasculitis side by side

A competent response contains

· Type and effector: immune complex disease is type III with neutrophils recruited by C5a; ANCA disease is antibody directed at primed neutrophils, which deliver the injury; large vessel arteritis is type IV with CD4 T cells, macrophages, and giant cells.

· Immunofluorescence: granular immunoglobulin and complement in immune complex disease; few or no deposits, described as pauci-immune, in ANCA disease; no deposits in granulomatous arteritis.

· Complement: may fall in immune complex disease because the classical pathway is consumed, with very low C4 in cryoglobulinemia and normal complement in IgA vasculitis which uses the alternative and lectin pathways; normal in ANCA disease, where the alternative pathway is activated locally; normal in granulomatous arteritis.

· Markers: hepatitis B or C serology, rheumatoid factor and cryoglobulins, and serum complement for the complex diseases; PR3-ANCA and MPO-ANCA for the ANCA diseases; a very high ESR and CRP driven by IL-6 for the granulomatous diseases.

· Vessels and tissues: small vessels and dermal venules and glomeruli, with medium muscular arteries in polyarteritis nodosa; small vessels of lung and kidney with upper airway granulomas in granulomatosis with polyangiitis; the aorta and its large branches, cranial carotid branches, and vertebral arteries in the granulomatous group.

· Examples: polyarteritis nodosa, IgA vasculitis, and cryoglobulinemic vasculitis; granulomatosis with polyangiitis, microscopic polyangiitis, and eosinophilic granulomatosis with polyangiitis; giant cell arteritis and Takayasu arteritis.

· Findings: palpable purpura with leukocytoclastic change, arthralgia, abdominal pain, and nephritis in complex disease; crescentic glomerulonephritis with pulmonary hemorrhage, and sinus and nasal granulomas when PR3-ANCA is present; headache, jaw claudication, vision loss, polymyalgia rheumatica, weak arm pulses, and an arm blood pressure difference in the granulomatous group, with the skin and kidney spared.

Distinctions that matter

· The immunofluorescence pattern is the fastest discriminator: granular means complexes, pauci-immune means ANCA, and no deposits with giant cells means T-cell disease.

· Within the ANCA group, PR3 with granulomas of the sinuses and lungs means granulomatosis with polyangiitis, MPO without granulomas means microscopic polyangiitis, and asthma with eosinophilia driven by IL-5 means eosinophilic granulomatosis with polyangiitis.

Common misconceptions

· Expecting a low complement in every vasculitis with deposits. IgA vasculitis has normal complement because of the pathway it uses.

· Expecting skin purpura and casts in giant cell arteritis. It is a large artery disease, so the findings follow the branches involved.

Where it is taught: Section C03, all paragraphs and the vasculitis by immune mechanism table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C04-BD1

Brain dump 1 · foundational retrieval: Three immune routes into the heart

A competent response contains

· Dressler syndrome: necrosis of a transmural infarct releases sequestered myocardial proteins, antibodies form against them, and immune complexes are thought to deposit on pericardium and pleura; it appears weeks to months later, and the delay is the time to mount that response.

· Early post-infarct pericarditis within about 1 to 3 days is a local fibrinous reaction to necrosis rather than an autoimmune process.

· Dressler findings: fever, pleuritic pain relieved by sitting forward, a friction rub, pleural and pericardial effusions, and high ESR and CRP, with diffuse ST elevation and PR depression rather than the regional changes of the infarct.

· Post-pericardiotomy syndrome has the same mechanism.

· Libman-Sacks endocarditis: immune complex and complement injury of valve endothelium in lupus, aided by antiphospholipid antibodies that promote thrombosis, gives sterile verrucous vegetations of fibrin, platelets, and mononuclear cells, sometimes with hematoxylin bodies.

· Those vegetations can sit on either leaflet surface, the mitral valve most often, heal by fibrosis that causes regurgitation, and can embolize fragments to the brain.

· Congenital heart block: maternal IgG against Ro (SSA), often with La (SSB), crosses the placenta on the neonatal Fc receptor from the second trimester; the mother may have lupus, Sjogren syndrome, or no symptoms.

· Antibody binds Ro and La exposed on apoptotic fetal myocytes, macrophages release TNF and TGF-beta, and fibrosis and calcification replace the AV node at about 18 to 24 weeks.

· Anti-Ro also inhibits L-type and T-type Ca2+ channels, and since AV nodal cells depolarize by Ca2+ current, conduction slows even before scarring.

· The three routes differ: antigens exposed by injury in Dressler syndrome, systemic autoimmunity in Libman-Sacks endocarditis, and maternal antibody crossing the placenta in congenital heart block.

Distinctions that matter

· Timing separates the two post-infarct pericarditides: days means a local reaction to necrosis, weeks to months means an autoimmune response.

· Diffuse ST elevation with PR depression is pericardial and involves the whole surface, while the infarct gives changes confined to a lead group.

Common misconceptions

· Treating Libman-Sacks vegetations as infected. They are sterile, and their position on either leaflet surface is part of what identifies them.

· Expecting the mother of an affected newborn to be unwell. She may have no symptoms and still carry anti-Ro.

Where it is taught: Section C04, Dressler and Libman-Sacks paragraphs, the vegetation table, and the congenital heart block paragraphs of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C04-BD2

Brain dump 2 · integration and application: Explain the newborn and sort four sets of vegetations

A competent response contains

· The atria beat normally at 142 per minute while the ventricles follow a slow escape rhythm at 50 per minute with no relation between them: complete heart block with AV dissociation, below the normal fetal rate of about 110 to 160 bpm.

· The immediate effect is inhibition of L-type and T-type Ca2+ channels by anti-Ro, which slows conduction in a node that depolarizes by Ca2+ current; the permanent effect is fibrosis and calcification of the node.

· Macrophages responding to antibody bound on apoptotic fetal myocytes release TNF and TGF-beta, which drive that fibrosis.

· The node is injured at about 18 to 24 weeks of gestation.

· By 8 months the rash, cytopenias, and hepatitis of neonatal lupus have resolved as maternal IgG is cleared over about 6 to 8 months, but the fibrosed AV node cannot regenerate, so the block is permanent.

· Risk is about 1 to 2% per anti-Ro positive pregnancy and about 15 to 20% after an affected child.

· Rheumatic vegetations: molecular mimicry, small at 1 to 2 mm, on the lines of closure, no organisms, and fibrosis with stenosis after repeated attacks.

· Libman-Sacks vegetations: immune complexes with antiphospholipid antibodies, small to medium at 1 to 4 mm, on either or both leaflet surfaces, no organisms, healing to fibrosis and regurgitation.

· Infective vegetations: bacteria or fungi, large and friable, on the upstream surface of the closure line, organisms present, with destruction, perforation, and abscess.

· Nonbacterial thrombotic vegetations: a hypercoagulable state, often cancer, small to medium, on the lines of closure, no organisms, little valve damage but bland emboli.

Distinctions that matter

· Antibody effects that depend on continued antibody presence resolve when IgG is catabolized; effects that have already become scar do not.

· Only infective vegetations contain organisms, and position on the leaflet separates the three sterile types.

Common misconceptions

· Expecting the bradycardia to lift with the rash. The channel block is reversible but the nodal scar is not.

· Calling every sterile vegetation Libman-Sacks. Rheumatic and nonbacterial thrombotic vegetations sit on the lines of closure, while Libman-Sacks lesions can sit on either surface.

Where it is taught: Section C04, congenital heart block flow figure and final paragraph, and the vegetation comparison table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C04-RT01

Retrieval exercise: Rebuild the pathway to congenital heart block

A competent response contains

· Step 1: the mother has IgG anti-Ro, often with anti-La.

· Step 2: FcRn, the neonatal Fc receptor, carries the IgG across the placenta from the second trimester.

· Step 3: antibody binds Ro and La exposed on apoptotic fetal myocytes and also inhibits L-type and T-type Ca2+ channels.

· Step 4: macrophages release TNF and TGF-beta.

· Step 5: fibrosis and calcification replace the AV node at about 18 to 24 weeks.

· Step 6: permanent complete heart block with a slow escape rhythm, well below the normal fetal rate of about 110 to 160 bpm.

· The fibrosis step makes the block permanent, because nodal conduction tissue cannot regenerate once it has been replaced by scar and calcium.

Distinctions that matter

· Antibody binding and channel inhibition act while the antibody is present; fibrosis outlives it.

· AV nodal cells depend on Ca2+ current for their upstroke, which is why an antibody that blocks Ca2+ channels affects this site in particular.

Common misconceptions

· Treating the antibody as directly cytotoxic to nodal tissue. The lasting damage comes from the macrophage response and the fibrosis it drives.

Where it is taught: Section C04, pathogenesis of congenital heart block flow figure of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the rejection table from memory

A competent response contains

· Hyperacute: minutes to hours, preformed anti-ABO IgM or anti-HLA IgG, type II, with small vessel thrombosis and necrosis.

· Acute cellular: weeks to months and mostly in the first year, CD8 and Th1 T cells, type IV, with perivascular and interstitial lymphocytes and myocyte damage.

· Antibody-mediated: days to years, donor-specific anti-HLA antibody, type II, with endothelial swelling, intravascular macrophages, and capillary C4d.

· Allograft vasculopathy: months to years, T cells, antibody, and nonimmune injury, mixed mechanism, with diffuse concentric intimal thickening.

· Direct allorecognition: recipient T cells see intact donor HLA on donor dendritic cells, the passenger leukocytes carried in the graft; alloreactive T cells are numerous, so this drives the strong early response.

· Indirect allorecognition: recipient antigen-presenting cells display processed donor HLA peptides on recipient class II molecules, so it persists after passenger leukocytes are gone and gives B cells the CD4 help to make donor-specific antibody, which is why it matters in antibody-mediated and chronic rejection.

· Preformed antibody binds endothelium, activates the classical pathway, and small vessels thrombose at once.

· In cellular rejection CD8 T cells kill myocytes with perforin, granzymes, and Fas ligand, and Th1 IFN-gamma activates macrophages and raises graft HLA expression; it is often clinically silent.

· C4d marks the capillaries because it binds covalently to tissue where C4 was cleaved, so it remains as a footprint of classical pathway activation.

· Allograft vasculopathy gives diffuse concentric intimal thickening of epicardial and intramyocardial arteries with an intact internal elastic lamina and little lipid or calcium.

· Hyperacute rejection is now rare because of ABO matching and crossmatch screening for anti-HLA antibody.

Distinctions that matter

· Timing follows the effector: preformed antibody needs no time, naive T cells need costimulation, IL-2, and clonal expansion, and vessel remodeling takes years.

· Cellular rejection is interstitial and lymphocytic while antibody-mediated rejection is microvascular and complement marked, which is why C4d is the discriminator.

Common misconceptions

· Reading antibody-mediated rejection as a late process only. It occurs from days to years, whenever donor-specific antibody appears.

· Assuming higher biopsy grade always means more illness. Widespread microvascular injury compromises hemodynamics more than cellular rejection of similar grade.

Where it is taught: Section C05, allorecognition paragraph, the four types, and the rejection table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C05-BD2

Brain dump 2 · integration and application: Three recipients and a denervated heart

A competent response contains

· Recipient 1 has acute cellular rejection: CD8 and Th1 T cells, and 3 weeks fits the time naive T cells need for costimulation, IL-2, and clonal expansion.

· Recipient 2 has antibody-mediated rejection: donor-specific anti-HLA antibody on capillary endothelium with classical pathway activation, which C4d records; indirect allorecognition supplied the CD4 help for the antibody to appear after transplantation.

· Recipient 3 has cardiac allograft vasculopathy, the chronic form, which takes months to years of remodeling.

· Recipient 2 is more unwell because microvascular injury is widespread and compromises perfusion of the whole graft, more than cellular rejection of similar grade.

· Allograft vasculopathy differs from atherosclerosis by being diffuse and concentric rather than focal and eccentric, by leaving the internal elastic lamina intact, and by containing little lipid or calcium.

· Immune drivers are indirect allorecognition, IFN-gamma, and donor-specific antibody; nonimmune drivers are ischemia-reperfusion injury, cytomegalovirus, hyperlipidemia, and hypertension; endothelium releases PDGF and TGF-beta that drive smooth muscle into the intima.

· Angiography understates it because diffuse narrowing leaves no normal reference segment to compare against.

· All three have a resting rate near the intrinsic SA nodal rate of about 90 to 110 bpm, because the graft is denervated and has no vagal tone.

· Exercise raises the rate slowly, through circulating catecholamines rather than sympathetic nerves.

· Recipient 3 has no angina because the graft has no sensory afferents, so vasculopathy presents as heart failure, arrhythmia, silent infarction, or sudden death.

· Thrombosis within minutes of reperfusion is hyperacute rejection, from preformed anti-ABO IgM or anti-HLA IgG.

Distinctions that matter

· C4d with new donor-specific antibody separates antibody-mediated from cellular rejection even when both show infiltrating cells.

· Denervation explains both the high resting rate and the absence of angina, so the same fact answers two different questions.

Common misconceptions

· Expecting chest pain to warn of graft coronary disease. The graft is denervated, so the presentation is painless.

· Reading a normal coronary angiogram as excluding allograft vasculopathy. The disease is diffuse, so there is no narrower segment to see.

Where it is taught: Section C05, the four types, the denervation paragraph, the rejection table, and the worked example on matching biopsies of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-BD1

Brain dump 1 · foundational retrieval: Viral myocarditis in three phases

A competent response contains

· Myocarditis is myocardial inflammation with myocyte injury not explained by ischemia.

· Phase 1, days: coxsackievirus B enters myocytes through the coxsackievirus and adenovirus receptor with CD55, viral protease 2A cleaves dystrophin and kills myocytes, and type I interferons, NK cells, and

macrophages limit spread. Adenovirus, parvovirus B19, HHV-6, influenza, and SARS-CoV-2 are other causes.

· Phase 2, from about 1 to 2 weeks: CD8 T cells kill infected myocytes and CD4 T cells release IFN-gamma and TNF; this clears virus but causes most of the muscle loss, and released myosin with molecular mimicry can trigger autoimmunity.

· Phase 3: most patients recover, and a minority develop dilated cardiomyopathy from persistent virus or from autoimmunity, with fibrosis and remodeling over months to years.

· Diagnosis: the Dallas criteria require an inflammatory infiltrate with nonischemic myocyte damage on biopsy, but biopsy can miss patchy disease, so MRI is used.

· On MRI, late gadolinium enhancement is subepicardial or midwall, unlike the subendocardial start of infarction.

· Necrosis raises troponin.

· Myocardial and pericardial inflammation cause chest pain and ST changes outside a single coronary territory.

· Muscle loss causes heart failure, and fulminant disease can recover substantially because many myocytes are dysfunctional rather than dead.

· Patchy scar supports reentry, so ventricular arrhythmias and sudden death occur, sometimes in young athletes.

· A viral prodrome often precedes the cardiac symptoms.

Distinctions that matter

· Direct viral killing belongs to the first days; the immune phase from 1 to 2 weeks accounts for most myocyte loss.

· Enhancement that starts subepicardially is inflammatory, while infarction starts subendocardially where perfusion is last and least.

Common misconceptions

· Assuming the virus does most of the damage. The CD8 response that clears it causes most of the muscle loss.

· Reading ST changes with a raised troponin as an infarct. The changes here cross coronary territories and the arteries are normal.

Where it is taught: Section C06, opening paragraph, the three phases, the flow figure, and the clinical paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-BD2

Brain dump 2 · integration and application: Separate four patients with myocarditis

A competent response contains

· Patient 1 has viral myocarditis in the immune phase: CD8 T cells with perforin, granzymes, and Fas ligand, with CD4 cells releasing IFN-gamma and TNF; biopsy shows lymphocytes with myocyte necrosis.

· Patient 2 has hypersensitivity myocarditis, a dose-independent type IV drug reaction in which Th2 cytokines such as IL-5 recruit eosinophils; biopsy shows T cells and eosinophils with limited necrosis, and sulfonamides, beta-lactams, methyldopa, and clozapine are examples.

· Patient 3 has hypereosinophilic myocarditis, from hypereosinophilic syndrome or eosinophilic granulomatosis with polyangiitis; biopsy shows eosinophil granule injury with necrosis and mural thrombus.

· Patient 4 has myocarditis after mRNA COVID-19 vaccination, which clusters in young males after the second dose, begins within days, and is usually mild; the mechanism is not established and MRI shows subepicardial enhancement.

· Patient 3 is the one who ends with restrictive physiology: eosinophil major basic protein and cationic protein kill myocytes and injure endocardium, mural thrombus organizes, and endomyocardial fibrosis, also called Loffler endocarditis, stiffens the ventricle.

· Patient 2 should subside once the drug is stopped, because the reaction depends on continued exposure to the drug, and it is dose independent.

· The fifth patient has giant cell myocarditis: T cells with giant cells and wide necrosis and no granulomas, associated with autoimmunity and thymoma, with rapid heart failure, arrhythmias, and a poor prognosis.

· Eosinophilia with asthma and a neuropathy also raises eosinophilic granulomatosis with polyangiitis, where IL-5 drives the eosinophils and MPO-ANCA is found in about 40%.

· The effector cell on histology is what predicts the course in each form.

Distinctions that matter

· Drug hypersensitivity and hypereosinophilic disease both bring eosinophils, but one subsides when exposure ends and the other scars the endocardium.

· Giant cell myocarditis has giant cells and wide necrosis without granulomas, which separates it from granulomatous myocarditis.

Common misconceptions

· Expecting a drug reaction in the heart to be dose related. Hypersensitivity myocarditis is a type IV reaction and is dose independent.

· Grouping all eosinophilic myocardial disease together. The granule proteins of a sustained hypereosinophilia are what produce endomyocardial fibrosis.

Where it is taught: Section C06, other forms paragraph and the table comparing forms of myocarditis of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-PR01

Prediction exercise: Predict what the T-cell phase does to the myocardium

A competent response contains

· Virus in the myocardium falls, because CD8 T cells kill infected myocytes and clear the infection.

· Surviving myocytes fall further than in the first days, because immune killing accounts for most of the muscle loss.

· Serum troponin rises, because ongoing myocyte necrosis releases it.

· Dystrophin and the myocyte membrane are already compromised, because viral protease 2A cleaved dystrophin in the first phase and the cell cannot maintain its membrane.

· Exposure to cardiac myosin rises and so does the chance of autoimmunity, because myosin released from dying myocytes can cross-react by molecular mimicry.

· In the minority who do not recover, the chamber dilates and the wall thins as lost muscle is replaced by fibrosis and the ventricle remodels into a dilated cardiomyopathy.

· The risk of reentrant ventricular arrhythmia rises, because patchy fibrosis between surviving muscle creates slow conduction and the circuit conditions reentry needs.

Distinctions that matter

· Clearing the virus and preserving the muscle are not the same goal, which is why the phase that cures the infection is the phase that costs the most myocytes.

· Dilatation reflects remodeling after muscle loss, not continuing infection.

Common misconceptions

· Expecting troponin to fall once the virus is being cleared. Necrosis is greatest while the T-cell response is at its peak.

· Assuming every patient progresses to dilated cardiomyopathy. Most recover, and progression needs persistent virus or autoimmunity.

Where it is taught: Section C06, the three phases, the flow figure from viral infection to dilated cardiomyopathy, and the clinical paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C07-BD1

Brain dump 1 · foundational retrieval: HHV-8 latency and the cytokines of myxoma

A competent response contains

· Kaposi sarcoma is a low-grade vascular tumor caused by human herpesvirus 8, a gamma-herpesvirus spread mainly by saliva and also sexually.

· HHV-8 latently infects spindle cells of lymphatic endothelial type.

· LANA keeps the viral episome in the cell and inhibits p53 and Rb.

· v-cyclin pairs with CDK6 to drive the cell from G1 into S phase.

· v-FLIP activates NF-kappa B and blocks apoptosis.

· A viral G protein-coupled receptor induces VEGF, and viral IL-6 signals through gp130 without needing the IL-6 receptor alpha chain.

· Histology: spindle cells with slit-like vascular spaces, extravasated red cells, and hemosiderin, progressing from patches to plaques to nodules in skin, oral mucosa, gut, lymph nodes, and lungs; LANA staining confirms HHV-8.

· The same virus causes primary effusion lymphoma and multicentric Castleman disease.

· Myxoma is the most common primary cardiac tumor in adults, rhabdomyoma is the most common in children, and metastases outnumber both.

· About 90% of myxomas arise in the atria and roughly 80% in the left atrium, as a pedunculated gelatinous mass on the septum near the fossa ovalis, with stellate cells in a glycosaminoglycan-rich matrix.

· Most are sporadic in middle-aged women; familial cases occur in Carney complex from PRKAR1A mutation, and are younger and multiple with spotty skin pigmentation and endocrine tumors.

Distinctions that matter

· Latent genes keep the infected cell alive and dividing while lytic genes supply paracrine growth and angiogenic signals to neighbouring cells.

· Viral IL-6 bypasses the IL-6 receptor alpha chain, so it can act on cells that would not respond to the host cytokine.

Common misconceptions

· Calling Kaposi sarcoma a tumor of blood vessel endothelium. The spindle cell resembles lymphatic endothelium.

· Treating a cardiac mass as most likely primary. Metastases to the heart are more common than either primary tumor.

Where it is taught: Section C07, Kaposi sarcoma paragraph and the atrial myxoma paragraphs of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C07-BD2

Brain dump 2 · integration and application: Failed surveillance, and one cytokine imitating three diseases

A competent response contains

· CD8 T cells and NK cells normally hold HHV-8 in check, and the CD8 response depends on CD4 help.

· The patient with untreated HIV and a CD4 count of 90 cells per microlitre has AIDS-associated Kaposi sarcoma, the most aggressive form and AIDS-defining: CD4 loss removes the help the CD8 response needs, and HIV Tat further stimulates the infected cells.

· The transplant recipient on a calcineurin inhibitor has lost T-cell activation to the drug, so latent virus is no longer controlled.

· The 68-year-old man of Mediterranean descent has classic Kaposi sarcoma, which is indolent on the lower legs; endemic African disease can involve lymph nodes in children.

· Early lesions depend on viral and cytokine signals, so restoring CD4 counts or reducing immunosuppression often makes them regress, after a possible transient flare from immune reconstitution, while advanced monoclonal lesions may not respond.

· The HIV protein that adds to the proliferation is Tat.

· The woman has an atrial myxoma, and IL-6 secreted by the myxoma cells explains the whole systemic picture.

· IL-6 to fever: COX-2 and PGE2 in the hypothalamus raise the thermal set point. IL-6 to the liver: CRP and fibrinogen raise the ESR, and hepcidin produces the anemia of inflammation. IL-6 on B cells: polyclonal hypergammaglobulinemia. IL-6 driven catabolism and anorexia: weight loss and fatigue.

· The positional symptoms come from the pedunculated mass prolapsing into the mitral orifice in diastole, which obstructs filling and causes dyspnea or syncope with a tumor plop in early diastole.

· The mass is friable, so fragments embolize systemically.

· All the systemic findings resolve after the mass is removed, because the source of IL-6 is gone.

Distinctions that matter

· Immune surveillance failure explains why the same virus produces indolent disease in one host and aggressive disease in another.

· A tumor can produce fever, anemia, and a high ESR through a cytokine alone, with no infection and no autoimmunity.

Common misconceptions

· Expecting all Kaposi lesions to regress when immunity returns. Advanced monoclonal lesions may not, and a transient flare can come first.

· Investigating the myxoma picture as infection or vasculitis without looking at the heart. Sterile cultures with negative autoantibodies and normal complement point away from both.

Where it is taught: Section C07, immune surveillance paragraph and the myxoma paragraphs with the IL-6 flow figure of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C07-CA01

Clinical application: Fever, weight loss, and a high ESR in a woman with positional breathlessness

A competent response contains

· Recognize: a left atrial myxoma, pedunculated and attached to the interatrial septum near the fossa ovalis, which is the characteristic site; about 90% of myxomas are atrial and roughly 80% left atrial.

· Explain: myxoma cells secrete IL-6. In the hypothalamus COX-2 and PGE2 raise the set point and cause fever; in the liver CRP and fibrinogen raise the ESR and hepcidin produces the anemia of inflammation, which fits a normal to raised ferritin with a normocytic anemia; on B cells IL-6 drives polyclonal hypergammaglobulinemia; IL-6 driven catabolism and anorexia cause the weight loss and fatigue.

· Differentiate: infective endocarditis would have organisms and positive blood cultures, and its vegetations sit on valve leaflets rather than on the septum; a systemic vasculitis would be expected to show ANCA or complement consumption with purpura, casts, or a neuropathy; lupus would show antinuclear antibody and low complement and would give sterile valve vegetations on the leaflets rather than a pedunculated septal mass.

· Investigate: look at the mass itself for a pedicle arising near the fossa ovalis, mobility with diastolic prolapse toward the mitral orifice, and a gelatinous friable surface; histology shows stellate cells in a glycosaminoglycan-rich matrix; ask about younger age, multiple tumors, spotty skin pigmentation, and endocrine tumors for Carney complex with PRKAR1A mutation.

· Decide: the positional breathlessness comes from the mobile mass prolapsing into the mitral orifice in diastole and obstructing ventricular filling, with the early diastolic sound being a tumor plop; the arm weakness episodes are systemic emboli, permitted by the friable surface of the mass.

· Justify: a benign tumor mimics chronic infection or malignancy because one cytokine reproduces the acute phase response, and all the systemic findings, the fever, anemia, high ESR, and raised immunoglobulins, resolve after the mass is removed because the source of IL-6 is gone.

Distinctions that matter

· Position settles the lesion: a pedunculated mass on the interatrial septum is a myxoma, while vegetations sit on valve leaflets.

· Anemia of inflammation through hepcidin is not iron deficiency, which is why the ferritin is not low.

Common misconceptions

· Reading fever with a high ESR and raised immunoglobulins as proof of infection or autoimmunity. A cytokine-secreting tumor produces the same profile.

· Expecting obstruction from a cardiac mass to be constant. A mobile pedunculated tumor obstructs only when it falls into the orifice, which is why the symptoms are positional.

Where it is taught: Section C07, atrial myxoma paragraphs and the flow figure on how a myxoma causes constitutional symptoms of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C02-Q01

Board-style question: Carditis three weeks after a sore throat

Correct answer: B. Antibodies and T cells raised against streptococcal M protein and group A carbohydrate that also recognize cardiac myosin and valve proteins

Why it is right: Antibodies and T cells raised against M protein and the group A carbohydrate cross-react with cardiac myosin and valve laminin; cross-reactive antibody activates valve endothelium and raises VCAM-1, letting CD4 T cells enter and sustain the carditis with IFN-gamma and IL-17.

A is wrong: Bacterial invasion with endocardial destruction describes infective endocarditis, which produces large friable vegetations with organisms present, whereas rheumatic lesions are sterile and the organism has already cleared.

C is wrong: Immune complex deposition with classical pathway activation is the mechanism of poststreptococcal glomerulonephritis, which is why that disease can follow skin infection as well as pharyngitis.

D is wrong: Antineutrophil cytoplasmic antibodies act on primed neutrophils in the ANCA vasculitides and produce pauci-immune small vessel injury, not post-streptococcal carditis.

A competent response contains

· Migratory large joint arthritis, carditis, and a rising antistreptolysin O titer about 3 weeks after untreated pharyngitis is acute rheumatic fever.

· The mechanism is molecular mimicry: the M protein is an alpha-helical coiled coil resembling cardiac myosin and tropomyosin, and antibody to the group A carbohydrate cross-reacts with valve laminin.

Distinctions that matter

· The organism is absent from the heart lesions, and the 2 to 3 week latent period is the time taken to build the adaptive response.

Common misconceptions

· Looking for infection in the valve. The valve lesion of rheumatic fever is sterile.

Where it is taught: Section C02 of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-Q02

Board-style question: Purpura, casts, and a very low C4

Correct answer: A. IgM rheumatoid factor bound to IgG forming complexes that deposit in vessel walls and consume the classical pathway

Why it is right: Hepatitis C drives clonal B cells to produce IgM rheumatoid factor that binds IgG; these complexes precipitate below 37 degrees C (98.6 degrees F), deposit in vessel walls, and consume the classical pathway, which is why C4 is very low and the biopsy shows granular deposits.

B is wrong: Antibodies against myeloperoxidase cause pauci-immune injury with normal complement and almost no immunoglobulin deposition, which does not fit granular deposits with a very low C4.

C is wrong: Antibody against type IV collagen in the basement membrane gives linear rather than granular immunofluorescence, as in anti-GBM disease.

D is wrong: Dendritic cell activation of CD4 T cells with medial granulomas describes giant cell and Takayasu arteritis, which leave no deposits and affect large arteries rather than dermal venules and glomeruli.

A competent response contains

· Palpable purpura with red cell casts, a very low C4 with near normal C3, rheumatoid factor, and hepatitis C is mixed cryoglobulinemic vasculitis.

· Granular immunoglobulin and complement in dermal venules confirms an immune complex mechanism.

Distinctions that matter

· A very low C4 points to classical pathway consumption by IgM-IgG complexes, which separates this from IgA vasculitis where complement is normal.

Common misconceptions

· Treating purpura with casts as one disease. The complement level and the immunofluorescence pattern are what identify the mechanism.

Where it is taught: Section C03 of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C05-Q03

Board-style question: Hypotension five months after transplantation

Correct answer: C. Donor-specific antibody binding capillary endothelium with classical pathway activation

Why it is right: Donor-specific anti-HLA antibody binds capillary endothelium and activates the classical pathway, leaving covalently bound C4d, and the widespread microvascular injury causes more hemodynamic compromise than cellular rejection of similar grade; indirect allorecognition provided the CD4 help for the antibody to develop after transplantation.

A is wrong: CD8 T cells recognizing intact donor HLA on passenger dendritic cells cause acute cellular rejection, which shows perivascular and interstitial lymphocytes with myocyte damage and no C4d.

B is wrong: PDGF and TGF-beta driven smooth muscle migration into the intima is cardiac allograft vasculopathy, which develops over months to years as diffuse concentric intimal thickening rather than acute capillary injury.

D is wrong: Preformed anti-ABO IgM causes hyperacute rejection within minutes to hours of reperfusion with small vessel thrombosis and necrosis, not an illness beginning 5 months later.

A competent response contains

· Capillary C4d with endothelial swelling, intravascular macrophages, few interstitial lymphocytes, and a newly detected donor-specific anti-HLA antibody is antibody-mediated rejection.

· C4d binds covalently where C4 was cleaved, so it marks tissue in which the classical pathway was activated.

Distinctions that matter

· Antibody-mediated rejection is microvascular and complement marked, while cellular rejection is interstitial and lymphocytic.

Common misconceptions

· Assuming antibody-mediated rejection must occur immediately. It occurs from days to years, once donor-specific antibody appears.

Where it is taught: Section C05 of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-Q04

Board-style question: A young man two weeks after a viral illness

Correct answer: D. CD8 T cells killing infected myocytes with perforin and granzymes

Why it is right: From about 1 to 2 weeks CD8 T cells kill infected myocytes with perforin, granzymes, and Fas ligand while CD4 cells release IFN-gamma and TNF; this clears the virus but accounts for most of the muscle loss, and it is the phase this patient has reached.

A is wrong: Cleavage of dystrophin by viral protease 2A is the phase 1 mechanism acting over the first days, before the T-cell response has expanded.

B is wrong: Eosinophil major basic protein injures the myocardium in hypereosinophilic syndrome and eosinophilic granulomatosis with polyangiitis, which present with eosinophilia and endocardial disease rather than after a brief viral illness.

C is wrong: Occlusion of an epicardial coronary artery produces necrosis beginning subendocardially within one artery territory, whereas the angiogram is normal and the enhancement here is subepicardial and crosses territories.

A competent response contains

· A viral prodrome, a raised troponin, ST changes crossing coronary territories, normal arteries, and subepicardial late gadolinium enhancement is viral myocarditis.

· At about 12 days the patient is in the immune phase, which begins from about 1 to 2 weeks.

Distinctions that matter

· Direct viral killing dominates the first few days; the T-cell phase accounts for most myocyte loss from about 1 to 2 weeks.

Common misconceptions

· Attributing all myocyte loss in myocarditis to the virus itself. The response that clears it causes most of the damage.

Where it is taught: Section C06 of the CV-16 Immune Responses in Cardiovascular Disease notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-17

CV-17 Cardiovascular Pathogens

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-17-01-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild the four endocarditis cocci

A competent response contains

· Staphylococcus aureus: grapelike clusters because it divides in several planes, catalase positive, coagulase positive, beta-hemolytic.

· S. aureus virulence factors: bound coagulase (clumping factor) binds fibrinogen, free coagulase activates prothrombin so fibrin coats the organism, fibronectin-binding proteins allow attachment to and entry into endothelial cells, protein A binds the Fc region of IgG, and pore-forming alpha toxin with tissue-degrading enzymes destroys leaflets within days.

· Viridans streptococci: chains because they divide in one plane, catalase negative, alpha-hemolytic from hydrogen peroxide oxidizing hemoglobin, optochin resistant and bile insoluble, which separates them from Streptococcus pneumoniae.

· Viridans virulence factor: dextran, a glucan polymer made from dietary sucrose, binds the platelet-fibrin deposit on an already damaged valve, and the organism becomes buried in it and shielded from neutrophils.

· Enterococcus faecalis: pairs and short chains, catalase negative, Lancefield group D antigen, grows in 6.5% NaCl and in bile, hydrolyzes esculin; surface adhesins and biofilm plus intrinsic resistance to many antibiotics.

· Staphylococcus epidermidis: clusters, catalase positive, coagulase negative, novobiocin sensitive, low virulence, and its one important factor is a polysaccharide adhesin that builds biofilm excluding neutrophils and antibodies.

· S. aureus acts on normal valves and destroys them acutely; viridans streptococci and enterococci act on damaged valves over weeks to months; S. epidermidis acts on prosthetic valves, pacemaker leads, and catheters indolently.

· Arrangement (clusters against chains) and catalase separate staphylococci from streptococci and enterococci, and coagulase separates S. aureus from S. epidermidis.

Distinctions that matter

· Catalase is a staphylococcal property; streptococci and enterococci are catalase negative, so catalase comes before coagulase in the reasoning.

· Adhesion mechanism decides which valve an organism can take: fibrinogen and fibronectin binding works on near-normal endothelium, dextran binding needs a platelet-fibrin deposit, and biofilm needs a foreign surface.

Common misconceptions

· Treating alpha hemolysis as partial lysis by a toxin. It is a greenish discoloration produced when the organism's hydrogen peroxide oxidizes hemoglobin.

· Calling S. epidermidis harmless because it is a skin commensal. On a prosthetic valve its biofilm makes it the classic cause of early prosthetic valve endocarditis, although a single positive culture is often contamination.

Where it is taught: Section C01, organism subsections and the table of gram-positive cocci of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C01-BD2

Brain dump 2 · integration and application: Two patients, two valves, two time courses

A competent response contains

· Patient one: Staphylococcus aureus from skin flora entering the venous blood, so the tricuspid valve is reached first; clumping factor and fibronectin-binding proteins attach it to minimally injured or normal endothelium, and alpha toxin with tissue-degrading enzymes perforates the leaflet within days.

· Patient two: viridans streptococci from mouth flora entering during the brief bacteremia of dental work; turbulent flow across the bicuspid valve had already injured the endothelium and laid down a sterile platelet-fibrin deposit, and dextran binds the organism into it.

· Viridans streptococci invade poorly and are weakly toxic, so they need an abnormal valve and give subacute disease over weeks to months.

· Protein A binds the Fc region of IgG and holds the antibody in an orientation that cannot opsonize or fix complement; bound coagulase and free coagulase coat the organism in fibrin, which hinders phagocytosis.

· A sterile platelet-fibrin deposit gives a poorly invasive organism a surface it can stick to and then be buried in, out of reach of neutrophils.

· Patient one is the one expected to have an abscess and perforation, because S. aureus destroys tissue directly with toxins and enzymes.

Distinctions that matter

· Acute against subacute follows from the organism's toxins, not from how long the patient waited to be seen.

· Right-sided disease in injection drug use follows the route of entry: skin organisms reach the right heart first through venous blood.

Common misconceptions

· Expecting a damaged valve in every case of endocarditis. S. aureus can colonize a normal valve.

· Assuming fibrin around the organism helps the host. Coagulase-generated fibrin shields the bacterium from phagocytes.

Where it is taught: Section C01, S. aureus and viridans streptococci subsections and the stop and predict box of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C01-CP01

Comparison table: The four gram-positive cocci of infective endocarditis

A competent response contains

· Arrangement: S. aureus and S. epidermidis in clusters; viridans streptococci in chains; E. faecalis in pairs and short chains.

· Catalase: positive for both staphylococci, negative for viridans streptococci and E. faecalis.

· Other bench tests: S. aureus coagulase positive and beta-hemolytic; viridans streptococci alpha-hemolytic, optochin resistant, bile insoluble; E. faecalis group D antigen, grows in 6.5% NaCl and bile, hydrolyzes esculin; S. epidermidis coagulase negative and novobiocin sensitive.

· Virulence factors: clumping factor, fibronectin-binding proteins, protein A, alpha toxin and tissue-degrading enzymes for S. aureus; dextran for viridans streptococci; adhesins, biofilm, and intrinsic drug resistance for E. faecalis; polysaccharide adhesin and biofilm for S. epidermidis.

· Attachment: S. aureus binds fibrinogen and fibronectin on minimally injured endothelium; viridans streptococci bind an existing platelet-fibrin deposit; E. faecalis uses surface adhesins and biofilm; S. epidermidis builds biofilm on a foreign surface.

· Valve required: normal or minimally injured for S. aureus; scarred or congenitally abnormal for viridans streptococci; usually abnormal for E. faecalis; prosthetic valve or device for S. epidermidis.

· Portal of entry: skin, including injection drug use, for S. aureus with the tricuspid valve reached first; mouth with dental bacteremia for viridans streptococci; urinary or gastrointestinal tract, often after instrumentation in older adults, for E. faecalis; skin at the time of surgery or device insertion for S. epidermidis.

· Course: acute with abscess and perforation for S. aureus; subacute over weeks to months for viridans streptococci and E. faecalis; indolent, usually within the first year after valve surgery, for S. epidermidis.

Distinctions that matter

· Both staphylococci are catalase positive, so coagulase is the test that tells them apart.

· E. faecalis and viridans streptococci are both catalase negative, and growth in 6.5% NaCl and bile with esculin hydrolysis marks the enterococcus.

Common misconceptions

· Filling the valve row with abnormal for every organism. The S. aureus row is the exception that explains acute endocarditis on a normal valve.

· Listing biofilm for every organism. Biofilm is the defining factor for S. epidermidis on devices and contributes for E. faecalis, but it is not what viridans streptococci or S. aureus rely on.

Where it is taught: Section C01, table of gram-positive cocci of infective endocarditis of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C02-BD1

Brain dump 1 · foundational retrieval: Why blood cultures stay negative

A competent response contains

· Culture-negative endocarditis is endocarditis by clinical and echocardiographic findings with no growth in routine blood cultures.

· The commonest cause is antibiotics given before the cultures were drawn, which suppresses growth of any organism.

· HACEK: Haemophilus species, mainly H. parainfluenzae; Aggregatibacter; Cardiobacterium hominis; Eikenella corrodens; Kingella kingae. They are fastidious gram-negative rods and coccobacilli of the normal oropharyngeal flora, need enriched media and added CO2, grow slowly, and tend to form large friable vegetations that embolize. Modern automated systems usually recover them within about 5 days.

· Coxiella burnetii is a small gram-negative obligate intracellular bacterium that multiplies inside the acidic phagolysosome of macrophages; a spore-like small-cell form survives drying, so infection follows inhaling dust or aerosols contaminated with the placenta and amniotic fluid of cattle, sheep, and goats, with no vector involved.

· Coxiella does not grow in routine culture, so diagnosis rests on serology, with high titers against phase I antigen marking chronic infection, and chronic endocarditis occurs with damaged or prosthetic valves, vascular grafts, or weak immunity, often with small or absent vegetations.

· Bartonella species are fastidious, facultatively intracellular gram-negative rods. B. quintana is carried by the human body louse, and B. henselae comes from cats through scratches or cat fleas. Diagnosis uses serology, PCR, or silver-stained valve tissue.

· Streptococcus gallolyticus is a group D streptococcus of the colon that is bile-esculin positive but does not grow in 6.5% NaCl, which is what separates it from Enterococcus.

Distinctions that matter

· Fastidious means slow growth on enriched media; obligate intracellular means no growth on cell-free routine media at all, which is why HACEK eventually grows and Coxiella does not.

· Coxiella needs no vector, while Bartonella quintana needs the body louse, so the exposure history separates them.

Common misconceptions

· Reading culture-negative as unusual organism first. Prior antibiotics are the commonest explanation.

· Expecting large vegetations in Q fever endocarditis. Vegetations are often small or absent, which is one reason the diagnosis is missed.

Where it is taught: Section C02, opening paragraph, HACEK and Coxiella subsections, and the table of causes of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C02-BD2

Brain dump 2 · integration and application: Match the exposure to the organism and read the bench results

A competent response contains

· Goat farm with kidding: Coxiella burnetii, inhaled from aerosols of birth products; it is an obligate intracellular organism that multiplies in the macrophage phagolysosome and does not grow on routine media, so serology is used, with phase I titers marking chronic infection.

· Living outdoors with body lice: Bartonella quintana, a fastidious gram-negative rod that grows very slowly, identified by serology, PCR, or silver stain of valve tissue.

· Poor dental health with large friable embolizing vegetations: a HACEK organism from oropharyngeal flora, fastidious and slow growing and needing enriched media with added CO2.

· Recent antibiotic: any organism with its growth suppressed, which is the commonest cause of culture-negative endocarditis.

· The second patient's isolate is Streptococcus gallolyticus: chains with a negative catalase place it among streptococci and enterococci, bile esculin hydrolysis marks a group D organism, and failure to grow in 6.5% NaCl excludes Enterococcus.

· Its source is the colon, and a colorectal adenoma or carcinoma disrupting the mucosal barrier is what lets it enter the blood, so its isolation calls for evaluation of the colon even without bowel symptoms.

· Had the isolate grown in 6.5% NaCl it would be Enterococcus, with a urinary or gastrointestinal source, often after instrumentation in an older adult.

Distinctions that matter

· Bile esculin positivity is shared by Enterococcus and S. gallolyticus; the salt tolerance test is the one that separates them.

· Exposure history does most of the work in culture-negative endocarditis, because each organism has a characteristic source: livestock aerosols, body lice, cats, or the mouth.

Common misconceptions

· Treating S. gallolyticus bacteremia as a finding about the heart alone. It is also a finding about the colon.

· Expecting Coxiella to come from an animal bite or a tick. Transmission is by inhalation, with no vector.

Where it is taught: Section C02, table of causes, the S. gallolyticus subsection, and the worked example of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C02-Q01

Board-style question: A group D organism from the bench tests

Correct answer: C. Streptococcus gallolyticus

Why it is right: Streptococcus gallolyticus is a group D streptococcus of the colon that hydrolyzes esculin in bile but, unlike Enterococcus, does not grow in 6.5% NaCl, and it reaches the blood through a colorectal adenoma or carcinoma that has broken the mucosal barrier.

A is wrong: Enterococcus faecalis is also bile-esculin positive but grows in 6.5% NaCl and in bile, which this isolate did not.

B is wrong: Streptococcus mitis is a viridans streptococcus, so it is alpha-hemolytic rather than nonhemolytic and is not bile-esculin positive; it follows dental bacteremia on a damaged valve.

D is wrong: Staphylococcus epidermidis grows in clusters and is catalase positive, and it infects prosthetic valves and devices by biofilm rather than native valves in this pattern.

A competent response contains

· Chains with a negative catalase place the isolate among the streptococci and enterococci.

· Esculin hydrolysis in bile marks a group D organism, which leaves Enterococcus and Streptococcus gallolyticus.

· Failure to grow in 6.5% NaCl excludes Enterococcus, leaving S. gallolyticus, whose source is the colon.

Distinctions that matter

· Both group D organisms are bile-esculin positive, and only Enterococcus grows in 6.5% NaCl.

Common misconceptions

· Stopping at group D and naming Enterococcus, which is the commoner group D isolate but does not fit the salt result.

Where it is taught: Section C02 of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C03-BD1

Brain dump 1 · foundational retrieval: The two human forms and every route of transmission

A competent response contains

· Trypanosoma cruzi is a flagellated protozoan and the cause of Chagas disease, also called American trypanosomiasis.

· Trypomastigote: extracellular, flagellated, circulates in blood, invades cells, and does not divide in humans.

· Amastigote: intracellular, rounded, no external flagellum, divides by binary fission in the cytoplasm after the parasite escapes its vacuole.

· Multiplication in the human happens only inside host cells, and the favored cells are macrophages, cardiac myocytes, smooth muscle, and neurons of the enteric plexuses.

· The vector is the triatomine bug, also called the reduviid or kissing bug, which lives in the cracks of poorly built houses in rural Mexico, Central America, and South America and feeds at night, often on the face.

· The infective metacyclic trypomastigotes are in the bug's feces, not its saliva, and the bug defecates during or soon after the blood meal, so the feces must be rubbed into the bite wound, conjunctiva, or mucosa; the bite alone does not transmit.

· Other routes: blood transfusion, organ transplant, across the placenta, food or juice contaminated with bugs or their feces, and laboratory accidents. Oral outbreaks deliver a large inoculum and tend to cause severe acute myocarditis.

· The acute phase lasts several weeks to about 2 months, is usually mild, and shows many trypomastigotes on a stained blood smear.

· Entry through the conjunctiva gives painless swelling of one eyelid, the Romana sign; entry through skin gives a local inflammatory nodule, a chagoma.

Distinctions that matter

· T. cruzi trypomastigotes circulate but do not divide, unlike the African trypanosomes, which multiply in the blood.

· The parasite is in the feces rather than the saliva, which is why scratching the bite matters more than the bite.

Common misconceptions

· Saying the bug injects the parasite as it bites. The feces deposited beside the bite are rubbed in afterwards.

· Describing the amastigote as flagellated. It is rounded with no external flagellum, and it is the only dividing form in the human.

Where it is taught: Section C03, opening paragraph, the life cycle, and the paragraph on the two human forms of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C03-BD2

Brain dump 2 · integration and application: Predict chronic Chagas disease from the cells the parasite lives in

A competent response contains

· She has been in the indeterminate phase, which is lifelong in most people, with few circulating parasites, so diagnosis depends on serology rather than a blood smear.

· About 20 to 30% of infected people develop chronic disease over decades.

· Persisting amastigotes and the inflammation they provoke destroy myocytes, which are replaced by fibrosis.

· Cardiac result: dilated cardiomyopathy with reduced contractility and enlarged chambers, a thin apical aneurysm that can hold thrombus, and conduction disease, typically right bundle branch block with left anterior fascicular block.

· Loss of myenteric neurons leaves the esophagus and colon unable to relax and propel their contents, so they dilate into megaesophagus and megacolon.

· The organs dilate because the neurons that coordinate relaxation and propulsion are gone, so contents accumulate behind a segment that will not relax and the wall stretches over time.

· Trypomastigotes circulate and amastigotes persist in tissue for life, so transfusion, transplant, and placental transmission occur far from any vector, which is why blood banks and transplant programs screen for infection.

Distinctions that matter

· Cell tropism predicts the lesion: cardiac myocytes give cardiomyopathy and conduction disease, enteric neurons give megaesophagus and megacolon.

· The acute phase is diagnosed by seeing the parasite; the indeterminate and chronic phases are diagnosed by serology.

Common misconceptions

· Expecting chronic disease in everyone infected. Most people stay in the indeterminate phase.

· Attributing the chronic lesions to the parasite alone. Myocyte destruction and the T-cell response together drive the fibrosis.

Where it is taught: Section C03, paragraphs on transmission without the bug and on the phases of disease of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C03-RT01

Retrieval exercise: Rebuild the Trypanosoma cruzi life cycle

A competent response contains

· Step 1: the bug ingests blood trypomastigotes during a blood meal.

· Step 2: in the bug midgut, epimastigotes multiply.

· Step 3: in the hindgut, metacyclic trypomastigotes form and pass in the feces.

· Step 4: the feces are rubbed into the bite wound, the conjunctiva, or a mucosal surface.

· Step 5: trypomastigotes invade host cells and become amastigotes.

· Step 6: amastigotes divide by binary fission in the cytoplasm.

· Step 7: amastigotes become trypomastigotes and the cell ruptures.

· Step 8: blood trypomastigotes infect new cells or a feeding bug.

· Steps 1 to 3 happen in the insect and steps 4 to 8 in the human.

· The two multiplying steps are the epimastigote in the bug midgut and the amastigote inside the human cell.

· Parasite numbers in blood rise because each invaded cell releases many trypomastigotes when it ruptures, even though the circulating form itself never divides.

Distinctions that matter

· Three forms, three places: epimastigote in the insect gut, amastigote inside human cells, trypomastigote in blood and in insect feces.

· Only the metacyclic trypomastigote in the hindgut feces is infective to humans.

Common misconceptions

· Putting the dividing form in the bloodstream. Division in the human happens only intracellularly.

· Leaving out the rubbing-in step, which is where transmission actually occurs.

Where it is taught: Section C03, life cycle diagram of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C04-BD1

Brain dump 1 · foundational retrieval: Two tick-borne bacteria from the ground up

A competent response contains

· Borrelia burgdorferi is a spirochete: a long, thin, helical bacterium whose flagella lie in the periplasm between the inner and outer membranes, and their rotation turns the whole cell like a corkscrew so it can move through connective tissue.

· Its outer membrane lacks classic lipopolysaccharide and is rich in lipoproteins, and its genome is a linear chromosome plus many linear and circular plasmids that carry most of the surface protein genes.

· Rickettsia rickettsii is a small gram-negative coccobacillus and is obligate intracellular because it imports ATP and other metabolites from host cytoplasm and cannot make enough of its own coenzyme A and NAD+.

· Stains: Borrelia with Giemsa, Wright, or silver; Rickettsia with Giemsa or Gimenez. Borrelia is too thin to see well on Gram stain and Rickettsia stains poorly with it.

· Borrelia vectors are Ixodes ticks, Ixodes scapularis in the eastern United States and Ixodes pacificus in the West; Rickettsia rickettsii is carried by Dermacentor ticks, the American dog tick and the Rocky Mountain wood tick.

· Reservoirs: the white-footed mouse for Borrelia, with larval ticks acquiring the organism from it and deer feeding adult ticks without being a significant reservoir; the tick itself for Rickettsia rickettsii, which passes it to its offspring through the eggs.

· Nymphal Ixodes ticks transmit most human Lyme disease.

· Despite the name, most United States cases of Rocky Mountain spotted fever occur in southeastern and south-central states.

Distinctions that matter

· Borrelia is extracellular and invades tissue mechanically; Rickettsia rickettsii is obligate intracellular in endothelium.

· Borrelia needs a mammalian reservoir, while the Dermacentor tick is both vector and reservoir for Rickettsia rickettsii.

Common misconceptions

· Placing spirochete flagella on the outside of the cell. They are periplasmic, which is what makes the whole cell rotate.

· Naming deer as the Lyme reservoir. Deer feed adult ticks; the white-footed mouse is the reservoir.

Where it is taught: Section C04, Borrelia and Rickettsia subsections and the comparison table of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C04-BD2

Brain dump 2 · integration and application: Explain the block and explain the rash

A competent response contains

· Patient one: Borrelia burgdorferi from an Ixodes tick, with the expanding lesion of erythema migrans; spread through blood carries the spirochete to the heart, where spirochetes with a lymphocyte and plasma cell infiltrate involve the myocardium and conduction tissue.

· Lyme carditis most often produces atrioventricular block at the level of the AV node, it can change degree within hours as the inflammatory infiltrate shifts, and it usually resolves as inflammation subsides.

· Patient two: Rickettsia rickettsii from a Dermacentor tick, infecting the endothelium of small vessels of the skin, brain, lungs, and kidneys.

· OmpA and OmpB bind endothelial cells and trigger uptake, the bacterium lyses the phagosome and multiplies in the cytoplasm, then polymerizes host actin into a tail that pushes it into the next endothelial cell, so infection travels along small vessels without re-entering the blood.

· Injured endothelium becomes leaky and procoagulant, which gives edema, loss of intravascular volume with hypotension, and platelet consumption; the rash begins as macules on the wrists and ankles a few days after fever, spreads to the trunk and onto the palms and soles, and becomes petechial as vessel damage progresses.

· Early tick removal works because Borrelia in an unfed tick gut expresses OspA, which anchors it to the gut, and it must switch to OspC, multiply, and migrate to the salivary glands before it can be injected, which usually takes 36 to 48 hours of attachment.

Distinctions that matter

· Borrelia damages conduction tissue by direct invasion with an inflammatory infiltrate; Rickettsia damages the vessel wall from inside endothelial cells.

· Lyme block is at the node and usually reversible, which is why its degree can vary within hours.

Common misconceptions

· Thinking the spotted fever rash begins centrally. It starts peripherally on the wrists and ankles and moves inward, reaching palms and soles.

· Assuming any tick bite transmits Borrelia immediately. The surface switch and migration take time, so a tick found and removed on the first day rarely transmits.

Where it is taught: Section C04, Borrelia flow and host paragraph, Rickettsia paragraphs, and the stop and predict box of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C04-RT01

Retrieval exercise: Rebuild the Borrelia surface switch from tick to host

A competent response contains

· Step 1: in the unfed tick gut, Borrelia expresses OspA, which anchors it to the gut.

· Step 2: the blood meal warms the tick and supplies nutrients.

· Step 3: OspA falls, OspC rises, and the spirochetes multiply.

· Step 4: spirochetes migrate to the tick salivary glands.

· Step 5: OspC binds the tick salivary protein Salp15, which shields the spirochete.

· Step 6: injection into the skin, usually after 36 to 48 hours of attachment.

· Step 7: in the host, VlsE variation keeps specific antibody behind.

· The trigger for the switch is the blood meal itself, through warming and nutrients.

· In the host, recombination moves segments of silent vls cassettes into the expressed vlsE gene, so the surface lipoprotein VlsE keeps changing and antibody raised against the earlier version no longer binds.

Distinctions that matter

· OspA is for life in the tick gut and OspC is for transmission and early infection, so the two proteins mark two different environments.

· Antigenic variation through VlsE is a host-phase strategy and has nothing to do with the tick.

Common misconceptions

· Treating the delay in transmission as travel time through the skin. The delay is the surface switch, replication, and migration to the salivary glands inside the tick.

· Describing VlsE change as mutation. It is recombination from silent cassettes into the expressed gene.

Where it is taught: Section C04, Borrelia surface switching flow and the paragraph after it of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C05-BD1

Brain dump 1 · foundational retrieval: Flaviviruses and filoviruses side by side

A competent response contains

· Flaviviruses: yellow fever virus and dengue virus, small spherical enveloped virions with a positive-sense single-stranded RNA genome translated as one polyprotein that proteases cut.

· Filoviruses: Ebola virus and Marburg virus, long filamentous enveloped virions with a negative-sense single-stranded RNA genome, so the virion must carry an RNA-dependent RNA polymerase.

· A positive-sense genome is itself mRNA, so naked genome RNA is infectious; a negative-sense genome is complementary to mRNA and naked genome RNA is not infectious.

· Reservoirs: humans for dengue and monkeys in the forest cycle for yellow fever; fruit bats for filoviruses, proven for Marburg virus with Rousettus bats and strongly suspected for Ebola virus.

· Transmission: Aedes mosquito bite for the flaviviruses, with Aedes aegypti a day-biting urban mosquito; contact with blood and body fluids for filoviruses, with no vector.

· First cells: monocytes and dendritic cells for the flaviviruses, with hepatocytes for yellow fever; macrophages, monocytes, and dendritic cells for filoviruses, with endothelium and hepatocytes later.

· Yellow fever: hepatocytes in the midzone of the lobule, zone 2, die by apoptosis and appear as eosinophilic Councilman bodies; lost hepatocyte function gives jaundice and reduced clotting factor synthesis, which with endothelial damage produces bleeding, including black vomit, and renal failure.

· Dengue: four serotypes DENV-1 to DENV-4 infecting dendritic cells, monocytes, and macrophages; severe dengue is a plasma leak syndrome that peaks as fever falls, with hemoconcentration, low platelets, and shock, and the secreted protein NS1 adds to leak by damaging the endothelial glycocalyx.

Distinctions that matter

· Both families end in endothelial leak and coagulopathy, but by different routes: plasma leak and antibody-assisted entry in dengue, hepatocyte apoptosis in yellow fever, and interferon blockade with cytokine storm in the filoviruses.

· Strand sense predicts what the virion must bring with it, which is why only the negative-sense virion carries a polymerase.

Common misconceptions

· Calling filoviruses arthropod-borne. They spread by direct contact with blood and body fluids, including those of the dead, and can persist for months in semen.

· Placing yellow fever necrosis at the center or edge of the lobule. It is the midzone, zone 2.

Where it is taught: Section C05, opening paragraph, the comparison table, and the yellow fever and dengue subsection of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C05-BD2

Brain dump 2 · integration and application: Why the second dengue infection is the dangerous one

A competent response contains

· Infection with one serotype gives lasting immunity to that serotype but only brief cross-protection against the other three.

· Years later the old antibody still binds the new serotype but cannot neutralize it, and the coated virus attaches to Fc gamma receptors on monocytes and macrophages.

· That route takes the virus in more efficiently than the virus could enter alone, so viral load rises, and entry by that route also dampens interferon responses.

· Cross-reactive memory T cells add a burst of cytokines that worsens the leak.

· An infant with waning maternal antibody has the same situation at a first infection: nonneutralizing antibody is present without protective immunity.

· A rising hematocrit with no external fluid loss means plasma has leaked out of the vessels, leaving the red cells concentrated; falling platelets add bleeding risk to shock.

· The patient gets worse as the fever breaks because that is when the critical phase of plasma leak begins.

· Ebola differs: the glycoprotein GP mediates entry into macrophages, monocytes, and dendritic cells, VP35 blocks type I interferon induction and VP24 blocks interferon signaling, infected dendritic cells fail to activate T cells and bystander lymphocytes die by apoptosis, and infected macrophages release TNF, IL-6, and tissue factor, with tissue factor triggering disseminated intravascular coagulation.

Distinctions that matter

· In dengue the antibody is the problem because it helps the virus in; in Ebola the problem is that interferon is blocked before antibody is ever made.

· Hemoconcentration reports where the plasma went, while thrombocytopenia reports the bleeding risk; they are separate findings.

Common misconceptions

· Expecting the sickest point of dengue to be the fever peak. Severe disease begins as the fever falls.

· Treating any antibody as protective. Nonneutralizing antibody plus Fc gamma receptors is a route of entry.

Where it is taught: Section C05, antibody-dependent enhancement subsection, the worked example on dengue labs, and the Ebola and Marburg subsection of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C05-PR01

Prediction exercise: Predict the course of a second dengue infection

A competent response contains

· The DENV-1 antibody binds DENV-2 but does not neutralize it, because cross-protection between serotypes is brief.

· Virus entry into monocytes and macrophages rises, because the coated virus attaches to Fc gamma receptors, which take it in more efficiently than the virus could enter alone.

· The type I interferon response in those cells falls, because entry by the Fc receptor route dampens interferon signaling, so replication is less restrained.

· Fluid leaves the vessels and the hematocrit rises, because NS1 damages the endothelial glycocalyx and cytokines from cross-reactive memory T cells increase permeability.

· The platelet count falls, which is part of the severe dengue picture of hemoconcentration, low platelets, and shock.

· Blood pressure falls and she becomes lightheaded, because the critical phase of plasma leak begins as the fever falls and intravascular volume drops.

Distinctions that matter

· Rising hematocrit with falling platelets is the signature pair: one marks plasma leak, the other marks bleeding risk.

· Entry by Fc receptor does two things at once, raising viral load and lowering interferon, so both belong in the prediction.

Common misconceptions

· Predicting improvement when the fever falls. The critical phase begins at that point.

· Explaining the raised hematocrit as dehydration from fever. Plasma has leaked into the tissues rather than been lost from the body.

Where it is taught: Section C05, antibody-dependent enhancement subsection and the worked example on dengue labs of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C06-BD1

Brain dump 1 · foundational retrieval: Coxsackie B from gut to dilated heart

A competent response contains

· Myocarditis is inflammation of the myocardium with myocyte injury, and viruses are its most commonly identified cause in high-income countries.

· Coxsackie B virus, serotypes B1 to B6, is an enterovirus in the picornavirus family: small, nonenveloped, icosahedral, with a positive-sense single-stranded RNA genome.

· Having no envelope lets it resist stomach acid, bile, and drying, so it spreads mainly by the fecal-oral route, and also through respiratory secretions, mostly in summer and fall.

· Step 1: fecal-oral or respiratory entry with replication in pharyngeal and intestinal lymphoid tissue. Step 2: viremia carries virus to the heart. Step 3: the virus binds DAF (CD55), then CAR at the intercalated discs. Step 4: the genome is translated into one polyprotein. Step 5: protease 2A cleaves eIF4G and dystrophin. Step 6: the sarcolemma tears, myocytes die and release troponin. Step 7: T-cell inflammation and fibrosis, with possible dilated cardiomyopathy.

· Entry depends on the coxsackie-adenovirus receptor (CAR), a transmembrane protein of the immunoglobulin superfamily in tight junctions and, in the heart, at the intercalated discs; many strains first bind decay-accelerating factor (DAF, CD55), which helps deliver them to CAR.

· CAR is abundant in the developing heart and is expressed again in injured myocardium, which is one reason neonatal coxsackie myocarditis is severe.

· Protease 2A cleaves eIF4G, which shuts off cap-dependent host translation, and cleaves dystrophin, which links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma, so the membrane tears under contractile stress.

· The virus also causes pleurodynia and aseptic meningitis, and it often inflames the pericardium as well, giving myopericarditis with pleuritic chest pain and a friction rub.

Distinctions that matter

· Translation of the genome is cap-independent through an internal ribosome entry site, which is why cleaving eIF4G stops host protein synthesis without stopping viral synthesis.

· DAF is a helper that delivers the virus; CAR is the receptor that gets it in.

Common misconceptions

· Expecting an enveloped virus because it reaches the heart through the blood. It is naked, and that is exactly why it survives the gut.

· Attributing all the damage to the immune response. Direct viral injury dominates the first days, and immune injury follows over weeks.

Where it is taught: Section C06, opening paragraph, the gut to dilated heart flow, and the paragraph on entry and protease 2A of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C06-BD2

Brain dump 2 · integration and application: Explain the findings and judge a biopsy result

A competent response contains

· Troponin appears in the blood because myocytes die when the sarcolemma tears and release their contents.

· The pain is pleuritic with a rub because coxsackie B often inflames the pericardium alongside the myocardium, which is myopericarditis.

· The rhythm is unstable because inflamed myocardium is electrically unstable, giving arrhythmias with ST and T wave changes.

· Contractility falls because myocytes have been lost.

· The protein is dystrophin, which links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma; the inherited disease with the same structural failure is Duchenne muscular dystrophy.

· The two differ in how the link is lost: viral protease 2A cuts the existing dystrophin, while in Duchenne muscular dystrophy the protein is not made properly in the first place, and in both cases the membrane tears under contractile stress.

· Over the following weeks the T-cell response clears virus but also kills infected myocytes, and persisting viral RNA with disrupted dystrophin can leave a dilated ventricle even after the virus is gone.

· A positive parvovirus B19 result does not establish cause, because parvovirus B19 and HHV-6 genomes are among the most frequent viral findings in heart biopsies and are also found in hearts without myocarditis.

· Parvovirus B19 binds P antigen (globoside) on erythroid precursors and endothelium, and it injures small vessels.

Distinctions that matter

· Direct viral injury and immune-mediated injury are two phases of the same illness, days against weeks.

· Finding a viral genome in tissue is not the same as showing that it caused the disease.

Common misconceptions

· Reading troponin release as proof of coronary occlusion. Any process that kills myocytes releases troponin.

· Treating the T-cell response as purely protective. It clears virus and also kills infected myocytes.

Where it is taught: Section C06, paragraph on myocyte loss and the T-cell phase, the table of other myocarditis viruses, and the closing paragraph of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C06-Q02

Board-style question: The torn sarcolemma in coxsackie myocarditis

Correct answer: A. Viral protease cleavage of dystrophin, which links sarcomeric actin to the sarcolemma

Why it is right: Coxsackie B protease 2A cleaves dystrophin, the protein that links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma, and without that link the membrane tears under contractile stress, which is the same structural failure seen in Duchenne muscular dystrophy.

B is wrong: ADP-ribosylation of elongation factor 2 is the mechanism of diphtheria toxin, which arrests protein synthesis in an organism that stays in the pharynx rather than being recovered from stool.

C is wrong: Actin tails that push an organism from cell to cell are how Rickettsia rickettsii spreads through endothelium, and it is a tick-borne bacterium rather than a naked RNA virus.

D is wrong: Nonneutralizing antibody carrying virus in through Fc gamma receptors is antibody-dependent enhancement in dengue, which produces plasma leak rather than a torn sarcolemma.

A competent response contains

· A naked positive-sense single-stranded RNA virus from stool after a summer diarrheal illness, with myopericarditis, is coxsackie B virus.

· Viral protease 2A cleaves dystrophin, which links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma, so the membrane tears under contractile stress.

· The same protease cleaves eIF4G and shuts off cap-dependent host translation, which is a separate effect from the membrane failure.

Distinctions that matter

· Protease 2A has two targets, and only the dystrophin cut explains a torn membrane.

Common misconceptions

· Choosing the host translation shutdown as the cause of the tear. Stopping protein synthesis kills the cell over time but does not explain mechanical failure of the sarcolemma.

Where it is taught: Section C06 of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-BD1

Brain dump 1 · foundational retrieval: Diphtheria toxin and the tuberculous pericardium

A competent response contains

· Corynebacterium diphtheriae is a club-shaped gram-positive rod that lies in V and L patterns, contains metachromatic polyphosphate granules, and forms black colonies on tellurite agar; it colonizes the pharynx without invading deeply.

· Only strains carrying a lysogenic beta-prophage with the tox gene make toxin, and the bacterial repressor DtxR blocks tox transcription when iron is plentiful, so toxin is made when iron is scarce.

· Toxin mechanism: the B portion binds the heparin-binding EGF-like growth factor precursor, the toxin enters an endosome, acidification moves the A fragment into the cytoplasm, and the A fragment ADP-ribosylates elongation factor 2, so protein synthesis stops and the cell dies.

· Locally, dead epithelium, fibrin, and leukocytes form a gray adherent pseudomembrane.

· Absorbed toxin reaches the myocardium and peripheral nerves; myocarditis usually appears 1 to 2 weeks into the illness, often as the throat improves, with myocyte degeneration and necrosis and relatively little inflammation, causing atrioventricular and bundle branch block, arrhythmias, and heart failure, and it is a major cause of death in diphtheria. Demyelinating neuropathy follows later.

· Mycobacterium tuberculosis is an acid-fast rod whose mycolic acid-rich wall resists decolorization and lets it survive inside macrophages.

· It usually reaches the pericardium by retrograde lymphatic spread from mediastinal peritracheal and peribronchial lymph nodes, or by blood.

· The cell-mediated response forms caseating granulomas with a fibrinous, often bloody effusion, and as the inflammation heals the visceral and parietal layers fuse into a thick fibrous, sometimes calcified shell.

Distinctions that matter

· The toxin is an A-B exotoxin: the B portion decides which cells are reached and the A fragment does the damage.

· Diphtheria damages the heart without the bacterium leaving the throat, which separates it from every other organism in this chapter.

Common misconceptions

· Expecting plentiful iron to increase toxin production. DtxR blocks tox transcription when iron is abundant, so toxin is made when iron is scarce.

· Expecting heavy inflammation in diphtheritic myocarditis. There is myocyte degeneration and necrosis with relatively little inflammation, because the toxin kills the cell directly.

Where it is taught: Section C07, diphtheria and tuberculous pericarditis subsections of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-BD2

Brain dump 2 · integration and application: Predict the physiology of a fixed pericardium and a starved aortic media

A competent response contains

· Early diastolic filling is rapid until the ventricles meet the rigid shell, then stops abruptly, which produces a pericardial knock.

· The jugular venous pulse shows a steep y descent, matching that rapid early filling.

· The inspiratory fall in intrathoracic pressure is not transmitted through the shell, so the extra venous return of inspiration cannot enter the right heart and jugular venous pressure rises on inspiration, which is the Kussmaul sign.

· Diastolic pressures equalize in all four chambers because the shell fixes the total volume of the heart.

· Systemic venous congestion gives hepatomegaly, ascites, and edema.

· In syphilis the treponemes persist in the adventitia around the vasa vasorum, the small adventitial arteries that nourish the outer media of the thoracic aorta, and obliterative endarteritis narrows them, so the outer media becomes ischemic and loses elastic fibers and smooth muscle with patchy scarring.

· The ascending aorta and arch fail because they depend most on vasa vasorum; the intima over the scarred media wrinkles into a tree-bark pattern and may calcify.

· A dilated root and arch stretch the valve ring and cause aortic regurgitation, narrow the coronary ostia and cause angina, and a large aneurysm can compress the trachea, esophagus, or left recurrent laryngeal nerve.

· Atherosclerotic aneurysms favor the infrarenal abdominal aorta because atherosclerotic intimal thickening impairs diffusion from the lumen, which matters most where there are few vasa vasorum.

Distinctions that matter

· Constrictive physiology comes from a fixed total volume, which is why all four chambers equalize and why inspiration raises rather than lowers jugular pressure.

· The two aneurysm sites follow two different nutrient routes to the media: vasa vasorum in the thoracic aorta and diffusion from the lumen in the infrarenal aorta.

Common misconceptions

· Expecting jugular venous pressure to fall on inspiration as it normally does. The shell prevents the thoracic pressure change from reaching the right heart.

· Saying the treponeme destroys the aortic wall directly. The wall fails from ischemia of its media.

Where it is taught: Section C07, paragraph on constrictive physiology, the syphilitic aortitis subsection and flow, and the stop and predict box of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-RT01

Retrieval exercise: Rebuild the path from vasa vasorum to aneurysm

A competent response contains

· Step 1: treponemes persist in the adventitia around the vasa vasorum.

· Step 2: lymphocyte and plasma cell cuffs surround those vessels.

· Step 3: intimal thickening, which is obliterative endarteritis, narrows the vasa vasorum.

· Step 4: the outer media becomes ischemic.

· Step 5: elastic fibers and smooth muscle are lost, with patchy scarring.

· Step 6: the weakened ascending aorta and arch dilate.

· Step 7: aneurysm, aortic regurgitation, and coronary ostial narrowing follow.

· Obliterative endarteritis is concentric thickening of the intima of small arteries that narrows or closes the lumen and makes the tissue they supply ischemic; the cuffs around the vessels are lymphocytes and plasma cells.

· The media loses elastic fibers and smooth muscle, so the wall can no longer recoil against systolic pressure and stretches instead.

· On the cross section, the adventitia carries the vasa vasorum, the outer media is the layer that becomes ischemic, and the intima lies against the lumen.

· The loss of blood supply to the wall does the damage, not destruction of tissue by the organism itself.

Distinctions that matter

· The lesion is in the small adventitial vessels, while the failure is in the media they feed.

· The intimal tree-bark change is a consequence of the scarred media underneath it, not a primary intimal disease.

Common misconceptions

· Putting the endarteritis in the aortic intima. It is in the intima of the vasa vasorum.

· Expecting the aneurysm where the aorta is widest rather than where it depends most on vasa vasorum.

Where it is taught: Section C07, syphilitic aortitis flow and the paragraph after it of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-CA01

Clinical application: Chest discomfort and a widened ascending aorta in a man with a remote untreated infection

A competent response contains

· Recognize: Treponema pallidum in tertiary syphilis, usually 10 to 30 years after untreated infection, and the tree-bark intima overlies a media scarred by obliterative endarteritis of the vasa vasorum.

· Explain: the treponemes persist in the adventitia around the vasa vasorum of the thoracic aorta, lymphocyte and plasma cell cuffs form, intimal thickening narrows those vessels, the outer media becomes ischemic, and elastic fibers and smooth muscle are lost with patchy scarring; the leaflets are normal because the leak comes from dilation of the root stretching the valve ring rather than from disease of the leaflets.

· Differentiate: this aneurysm is in the ascending aorta and arch and arises from ischemia of the media through narrowed vasa vasorum, while atherosclerotic aneurysms favor the infrarenal abdominal aorta and arise from intimal thickening that impairs diffusion from the lumen; the feature that explains the difference is the density of vasa vasorum, which is richest in the thoracic aorta and sparse in the infrarenal aorta.

· Investigate: the organism is a thin spirochete seen by dark-field microscopy or silver stain and is invisible on Gram stain because it is too thin to resolve; in the wall, look for loss of elastic fibers and smooth muscle with patchy medial scarring, and in the adventitial vessels look for intimal thickening with lymphocyte and plasma cell cuffs.

· Decide: the exertional chest discomfort comes from narrowing of the coronary ostia by the dilated aortic root, and a large aneurysm of the arch can also compress the trachea, the esophagus, or the left recurrent laryngeal nerve.

· Justify: the outer media of the thoracic aorta is nourished by the vasa vasorum rather than by diffusion from the lumen, so narrowing those vessels starves the media of oxygen; the organism is scarce in the wall and does not destroy it directly, and the structural failure follows the ischemic loss of elastic fibers and smooth muscle, which is why the wall dilates under normal systolic pressure.

Distinctions that matter

· Aortic regurgitation here is a problem of the root rather than the leaflets, which is why the leaflets look normal at surgery.

· The wide pulse pressure and collapsing pulse follow from regurgitant flow back into the ventricle in diastole.

Common misconceptions

· Looking for the organism in the media. It persists in the adventitia around the vasa vasorum.

· Expecting Gram stain to show a spirochete. Treponema pallidum is too thin for it, so dark-field or silver stain is used.

Where it is taught: Section C07, syphilitic aortitis subsection, the flow, and the paragraph on consequences of the CV-17 Cardiovascular Pathogens notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-01-C01-BD1

Brain dump 1 · foundational retrieval: Blood pressure categories and why the kidney sets pressure

A competent response contains

· Normal is below 120/80 mmHg, elevated is 120 to 129 systolic with diastolic below 80, stage 1 is 130 to 139 systolic or 80 to 89 diastolic, stage 2 is 140 or higher systolic or 90 or higher diastolic.

· When the systolic and diastolic values fall in different categories the higher category applies, so 128/86 mmHg is stage 1.

· Diagnosis rests on the average of repeated readings on two or more occasions, ideally confirmed outside the office.

· Mean arterial pressure equals cardiac output multiplied by systemic vascular resistance; in established primary hypertension output is usually normal and resistance is raised.

· Cardiovascular risk rises continuously from about 115 mmHg systolic, so any threshold is a chosen cut point.

· Baroreceptors defend pressure over minutes but reset within a day or two, so they cannot set long-term pressure.

· The kidney sets long-term pressure: a rise in arterial pressure increases sodium and water excretion, which is pressure natriuresis, and volume falls until sodium output again equals sodium intake.

· Because the gain of this feedback is very high, pressure can stay high only if the renal pressure to sodium excretion relationship shifts to the right, so a higher pressure is needed to excrete the daily salt load.

· Every chronic hypertension, primary or secondary, involves such a shift in pressure natriuresis.

Distinctions that matter

· Short-term pressure control is neural and resets; long-term control is renal and does not.

· A raised resistance is the usual finding in established disease, which is not the same as saying resistance started the process.

Common misconceptions

· Treating 140/90 mmHg as the threshold for hypertension under the scheme used here. Stage 1 begins at 130/80 mmHg under the 2017 ACC/AHA guideline.

· Saying baroreceptors keep pressure high in chronic hypertension. They reset to the prevailing pressure within a day or two.

Where it is taught: Section C01, how blood pressure is classified and why pressure stays high of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C01-BD2

Brain dump 2 · integration and application: Office readings, night-time pressure, and two inherited salt channels

A competent response contains

· Patient one has white coat hypertension, high only in the office from an alerting sympathetic response, so the clinic category of stage 1 overstates the true pressure.

· Patient two has masked hypertension, high only outside the office, so a normal clinic reading understates the true pressure.

· Patient three has loss of nocturnal dipping; pressure normally falls about 10 to 20% during sleep, and loss of the fall suggests sleep apnea, kidney disease, or autonomic dysfunction.

· Liddle syndrome: a gain-of-function mutation keeps the epithelial sodium channel of the collecting duct in the membrane, so sodium is reabsorbed regardless of aldosterone.

· Apparent mineralocorticoid excess: deficiency of 11-beta-hydroxysteroid dehydrogenase type 2 lets cortisol occupy the mineralocorticoid receptor.

· Glycyrrhetinic acid in licorice inhibits the same enzyme and reproduces the picture.

· In each one, sodium reabsorption through the epithelial sodium channel makes the lumen more negative, which drives potassium secretion by principal cells and hydrogen ion secretion by alpha-intercalated cells, giving hypokalemia and metabolic alkalosis.

· Volume expansion suppresses renin, and because the drive is not aldosterone itself, aldosterone is also low, so both are low with early-onset hypertension.

· Risk factors acting partly through sympathetic outflow: obesity through leptin, heavy alcohol intake, physical inactivity, and sleep apnea.

· Risk factors acting mainly through volume and renal sodium handling: high sodium with low potassium intake, hyperinsulinemia of obesity, diabetes, and chronic kidney disease.

· Older age acts through aortic stiffening, which raises systolic and pulse pressure and lowers diastolic pressure, giving isolated systolic hypertension.

Distinctions that matter

· White coat hypertension is high in the office only; masked hypertension is high outside it only. The out-of-office reading settles which one it is.

· Low renin with high aldosterone points to the adrenal cortex; low renin with low aldosterone points to a mineralocorticoid effect from something other than aldosterone.

Common misconceptions

· Expecting aldosterone to be high whenever a patient retains sodium and wastes potassium. In Liddle syndrome and apparent mineralocorticoid excess the receptor or channel is active without aldosterone.

· Assigning a stage from a single office reading. The category comes from averaged repeated readings, ideally confirmed outside the office.

Where it is taught: Section C01, paragraphs on measurement and on polygenic and monogenic hypertension, and the risk factor table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C02-BD1

Brain dump 1 · foundational retrieval: Build the secondary hypertension table from memory

A competent response contains

· Renal artery stenosis: abdominal bruit, hypokalemia, small kidney on the affected side, and a fall in glomerular filtration rate when angiotensin II is blocked; renin high, aldosterone high.

· Primary aldosteronism: hypokalemia, metabolic alkalosis, no edema; renin low, aldosterone high.

· Cushing syndrome, licorice, and Liddle syndrome: cushingoid habitus, licorice intake, or early onset with hypokalemia; renin low, aldosterone low or normal.

· Pheochromocytoma: episodic headache, sweating, and palpitations with high plasma or urine metanephrines; renin and aldosterone are not the primary abnormality.

· Coarctation of the aorta: arm hypertension with weak and delayed femoral pulses and rib notching; renin may be raised.

· Obstructive sleep apnea: snoring, daytime sleepiness, loss of nocturnal dipping; renin variable.

· Atherosclerosis causes most renal artery stenosis, in older adults with vascular disease, and narrows the ostium or proximal artery.

· Fibromuscular dysplasia is a noninflammatory, usually medial thickening in women about 15 to 50 years old, affecting the mid to distal artery with a string of beads appearance, and can involve the carotid and vertebral arteries.

· In pheochromocytoma, alpha-1 receptors raise systemic vascular resistance and beta-1 receptors raise cardiac output.

· Secondary hypertension is about 5 to 10% of all cases.

· Look for a cause when hypertension begins before about age 30, begins abruptly, is severe or resistant, or comes with an unexpected finding such as hypokalemia.

· Chronic kidney disease of any cause is a common secondary cause, because fewer nephrons must excrete the same salt load.

Distinctions that matter

· Both renin and aldosterone high points to renal hypoperfusion; low renin with high aldosterone points to the adrenal cortex; both low points to a mineralocorticoid effect from something other than aldosterone.

· Atherosclerotic and fibromuscular renal artery disease differ in age, sex, and the part of the artery involved, so keep the two profiles apart.

Common misconceptions

· Expecting every secondary cause to change renin and aldosterone. In pheochromocytoma the catecholamines are the abnormality and those two values are not the test.

· Assuming hypokalemia means primary aldosteronism. Renal artery stenosis, Cushing syndrome, licorice, and Liddle syndrome all produce it.

Where it is taught: Section C02, opening paragraph, renal artery stenosis, adrenal causes, and the secondary causes table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C02-BD2

Brain dump 2 · integration and application: Reason from the laboratory pattern back to the organ

A competent response contains

· Patient A has renal artery stenosis. The narrowed artery lowers afferent arteriolar pressure, so juxtaglomerular cells release renin as though the whole body were underperfused, and angiotensin II then drives aldosterone, so both are high.

· Patient B has primary aldosteronism, from bilateral adrenal hyperplasia or an aldosterone-producing adenoma. Autonomous aldosterone expands volume, and volume expansion suppresses renin, so renin is low while aldosterone is high.

· Patient B's tubular steps: aldosterone opens more epithelial sodium channels in principal cells, the more negative lumen drives potassium secretion, and alpha-intercalated cells secrete more hydrogen ion, producing hypokalemia with metabolic alkalosis.

· Patient B has no edema because rising pressure and atrial natriuretic peptide restore sodium balance within days, which is aldosterone escape, while potassium wasting continues; many such patients are in fact normokalemic.

· Patient C has a mineralocorticoid effect from cortisol. Glycyrrhetinic acid in licorice inhibits 11-beta-hydroxysteroid dehydrogenase type 2, so cortisol occupies the mineralocorticoid receptor; volume expansion suppresses renin and aldosterone is not the driver, so both are low.

· Patient D has a pheochromocytoma, a catecholamine-secreting chromaffin cell tumor of the adrenal medulla, called a paraganglioma when extra-adrenal.

· Metanephrines stay high between attacks because catechol-O-methyltransferase within the tumor forms them continuously, while catecholamine release is episodic.

· An epinephrine-secreting tumor is almost always adrenal because cortisol from the adrenal cortex induces phenylethanolamine N-methyltransferase, the enzyme that converts norepinephrine to epinephrine.

· Hereditary pheochromocytoma occurs in MEN 2 with RET mutations, von Hippel-Lindau disease, and neurofibromatosis type 1.

· Filtration in a stenotic kidney depends on angiotensin II constricting the efferent arteriole, so blocking angiotensin II dilates that arteriole, drops glomerular capillary pressure, and can sharply lower the glomerular filtration rate.

· Bilateral renal artery disease can cause recurrent flash pulmonary edema.

Distinctions that matter

· Renin separates renal artery stenosis from primary aldosteronism, because one starts with renal hypoperfusion and the other starts with volume expansion.

· Aldosterone separates adrenal cortical excess from a cortisol or channel effect at the mineralocorticoid receptor, since both suppress renin but only one raises aldosterone.

Common misconceptions

· Expecting edema in primary aldosteronism. Aldosterone escape restores sodium balance, so pressure rises without visible fluid retention.

· Reading a normal potassium as evidence against primary aldosteronism. Many patients are normokalemic.

Where it is taught: Section C02, renal artery stenosis, adrenal causes, and the secondary causes table caption of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C02-CP01

Comparison table: Renal artery stenosis against primary aldosteronism

A competent response contains

· Site: renal artery stenosis is a narrowed main renal artery, from atherosclerosis in older adults or fibromuscular dysplasia in younger women; primary aldosteronism is in the adrenal cortex, from bilateral hyperplasia or an aldosterone-producing adenoma.

· Renin: high in renal artery stenosis because reduced afferent arteriolar pressure makes juxtaglomerular cells release renin; low in primary aldosteronism because volume expansion suppresses it.

· Aldosterone: high in stenosis because angiotensin II stimulates the adrenal cortex; high in aldosteronism because secretion is autonomous.

· Potassium and acid-base: both can give hypokalemia with metabolic alkalosis, because in each case aldosterone acts on the collecting duct.

· Clues: stenosis gives an abdominal bruit, a small kidney on the affected side, and a string of beads appearance in fibromuscular disease; aldosteronism gives hypokalemia with no bruit and normal kidneys, often found during work-up of resistant hypertension.

· Edema: neither develops edema. In aldosteronism, rising pressure and atrial natriuretic peptide restore sodium balance within days, which is aldosterone escape.

· Blocking angiotensin II: in stenosis the efferent arteriole dilates and the glomerular filtration rate of that kidney falls sharply, and bilateral disease can cause a large fall; in aldosteronism angiotensin II is not driving the aldosterone, so blocking it does not correct the abnormality.

Distinctions that matter

· The renin level is the single discriminating cell in this table; aldosterone is high in both.

· A high renin state is driven from the kidney, and a low renin state with high aldosterone is driven from the adrenal gland.

Common misconceptions

· Writing that primary aldosteronism causes edema because sodium is retained. Escape resets sodium balance and the extra sodium shows as pressure, not swelling.

· Treating a falling glomerular filtration rate after angiotensin II blockade as a drug toxicity unrelated to anatomy. It reveals that filtration depended on efferent constriction in a stenotic kidney.

Where it is taught: Section C02, renal artery stenosis and adrenal causes paragraphs with the secondary causes table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C03-BD1

Brain dump 1 · foundational retrieval: Hyaline arterioles and the concentrically hypertrophied ventricle

A competent response contains

· Wall stress is proportional to pressure times radius divided by twice the wall thickness, which is the law of Laplace.

· A higher afterload raises systolic wall stress, and myocytes respond by adding sarcomeres in parallel, so the wall thickens and the denominator of that relationship rises, returning stress toward normal.

· Pressure overload gives concentric hypertrophy, a thick wall with little chamber enlargement; volume overload adds sarcomeres in series and dilates the chamber, which is eccentric hypertrophy.

· The normal left ventricular wall is about 1.0 to 1.5 cm and can approach or exceed 2 cm in hypertensive heart disease.

· Hypertrophied myocytes are thicker with enlarged rectangular boxcar nuclei.

· Hyaline arteriolosclerosis: chronic stress injures endothelium, plasma proteins leak into the wall, and smooth muscle cells add basement membrane and matrix.

· The result is a homogeneous, pink, glassy wall with a narrowed lumen, causing slow patchy ischemia of the tissue supplied.

· Arterioles are the resistance vessels, which is why they take the strain.

· The same hyaline lesion is typical of diabetes and appears mildly with normal aging.

Distinctions that matter

· Concentric against eccentric hypertrophy follows from the load: pressure adds sarcomeres in parallel, volume adds them in series.

· Hypertrophy is an adaptation that normalizes systolic wall stress, and the cost appears in diastole.

Common misconceptions

· Saying hypertrophy raises wall stress. Thickening is the term in the denominator, so it lowers stress back toward normal.

· Treating hyaline arteriolosclerosis as specific to hypertension. It also occurs in diabetes and mildly with aging.

Where it is taught: Section C03, hyaline arteriolosclerosis and concentric left ventricular hypertrophy of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C03-BD2

Brain dump 2 · integration and application: Trace chronic pressure into four organs

A competent response contains

· A thick, fibrotic ventricle relaxes slowly and is stiff, so a normal end-diastolic volume requires a higher filling pressure; left atrial and pulmonary venous pressures rise and the lungs congest although the ejection fraction is normal.

· Atrial fibrillation removes atrial contraction, which this stiff ventricle depends on for filling, and the fast rate shortens diastole, so filling falls further.

· Atrial contraction into a stiff ventricle produces a fourth heart sound.

· ECG: tall R waves in V5 and V6 with deep S waves in V1 reflect the greater mass, and lateral ST depression with T wave inversion forms the strain pattern.

· Ischemia in a hypertrophied wall: greater muscle mass, capillary density that lags behind the muscle, and high diastolic pressure compressing subendocardial vessels, all worsened by accelerated coronary atherosclerosis.

· Kidney: benign nephrosclerosis, with hyaline afferent arterioles and thickened interlobular arteries causing glomerulosclerosis, tubular atrophy, and interstitial fibrosis; the kidneys are symmetrically small with a finely granular surface, proteinuria is mild, and glomerular filtration rate falls slowly.

· Hypertension is the second leading cause of end-stage kidney disease after diabetes, with higher risk in carriers of high-risk APOL1 variants, and nephron loss worsens the hypertension by impairing sodium excretion.

· Retina: arteriolar narrowing, copper or silver wiring, arteriovenous nicking, flame hemorrhages, hard exudates, and cotton wool spots, which are nerve fiber layer microinfarcts.

· Arteriovenous nicking occurs because a thickened arteriole compresses a vein where the two share a connective tissue sheath.

· Brain: lipohyalinosis occludes penetrating arteries and causes lacunar infarcts under about 15 mm in the basal ganglia, internal capsule, thalamus, and pons; Charcot-Bouchard microaneurysms weaken the same vessels and rupture causes deep intracerebral hemorrhage, most often in the putamen, then thalamus, pons, and cerebellum.

· The bleeds sit deep because penetrating arteries such as the lenticulostriate branches of the middle cerebral artery receive nearly full parent arterial pressure; amyloid angiopathy affects superficial cortical and leptomeningeal vessels, so its hemorrhages are lobar.

Distinctions that matter

· Heart failure with a preserved ejection fraction is a filling problem, so the ejection fraction cannot exclude it.

· Lacunar infarcts and deep hemorrhages share one vascular bed and two different consequences of the same arteriolar disease.

Common misconceptions

· Expecting a normal ejection fraction to rule out pulmonary congestion. The limiting problem here is diastolic filling pressure.

· Explaining subendocardial ischemia only by coronary atherosclerosis. Mass, capillary supply, and diastolic compression contribute independently.

Where it is taught: Section C03, the paragraph after the pressure overload flow figure and the kidney, retina, and brain paragraphs, with the stop and predict box of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C03-RT01

Retrieval exercise: Rebuild the pressure overload sequence

A competent response contains

· Step 1: sustained hypertension raises left ventricular afterload.

· Step 2: systolic wall stress rises, by the law of Laplace, in which stress is proportional to pressure times radius divided by twice the wall thickness; pressure is the term that rises first.

· Step 3: sarcomeres are added in parallel, giving concentric hypertrophy, so the wall thickens while chamber volume changes little.

· Step 4: the thick, fibrotic wall relaxes slowly and is stiff.

· Step 5: a higher left atrial pressure is needed for filling, because a stiff ventricle needs a greater pressure to reach the same end-diastolic volume.

· Step 6: the left atrium dilates and pulmonary venous pressure rises.

· Step 7: atrial fibrillation and heart failure with preserved ejection fraction follow.

· First drawing: wall about 1.0 to 1.5 cm with a normally sized atrium. Final drawing: wall approaching or above 2 cm, a cavity of similar or slightly smaller size, and a dilated left atrium.

· Very late in the course the ventricle can dilate and lose systolic function, so the ejection fraction falls.

Distinctions that matter

· Hypertrophy succeeds in systole and fails in diastole, which is why the ejection fraction can stay normal while symptoms appear.

· Atrial dilation is a consequence of the filling pressure, which is why atrial fibrillation belongs at the end of this sequence rather than as a separate problem.

Common misconceptions

· Drawing a large dilated chamber for pressure overload. The chamber changes little; dilation belongs to volume overload and to the late failing stage.

· Placing atrial fibrillation before atrial dilation. The raised filling pressure stretches the atrium first.

Where it is taught: Section C03, flow figure on pressure overload to diastolic failure and atrial fibrillation of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-BD1

Brain dump 1 · foundational retrieval: Autoregulation, breakthrough, and the acute arteriolar lesion

A competent response contains

· A hypertensive emergency is severe elevation, often 180/120 mmHg or higher, with new or worsening acute damage to a target organ.

· Equally high pressure without acute organ damage is sometimes called hypertensive urgency.

· Malignant hypertension was the older term for severe hypertension with retinal hemorrhages, exudates, or papilledema, often with acute kidney injury, and is now treated as a form of emergency.

· The rate of rise and the failure of protective mechanisms matter more than the level reached.

· The brain and kidney hold flow nearly constant across a range of pressures by arteriolar constriction and dilation; in the brain this range is roughly 60 to 150 mmHg mean arterial pressure.

· Chronic hypertension shifts the autoregulation curve to the right, so both the lower and the upper limits move up.

· Above the upper limit, arteriolar constriction is forced open, which is breakthrough; capillaries then meet full arterial pressure and the blood-brain barrier leaks.

· The result is vasogenic edema and hypertensive encephalopathy with headache, confusion, visual disturbance, and seizures.

· The edema often favors the posterior circulation, which has less sympathetic innervation, giving posterior reversible encephalopathy syndrome.

· Fibrinoid necrosis is necrosis of the vessel wall into which fibrin and other plasma proteins seep, appearing bright pink and smudgy on routine stains, often with luminal thrombus.

· Surviving arterioles build concentric layers of smooth muscle with reduplicated basement membrane, the onion-skin lesion of hyperplastic arteriolosclerosis, most conspicuous in renal interlobular arteries and afferent arterioles.

· The same lesion occurs in scleroderma renal crisis and other thrombotic microangiopathies.

Distinctions that matter

· Emergency against urgency is decided by acute target organ damage, not by the number on the cuff.

· Hyaline thickening is the chronic lesion and onion-skin hyperplasia with fibrinoid necrosis is the acute severe lesion.

Common misconceptions

· Using a pressure threshold alone to call an emergency. Without acute organ injury the same pressure is urgency.

· Describing autoregulation as protective at any pressure. Beyond the upper limit constriction is overcome and the protection is lost.

Where it is taught: Section C04, opening paragraph, autoregulation and breakthrough, and the acute arteriolar lesion of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-BD2

Brain dump 2 · integration and application: Why the lower pressure did more damage

A competent response contains

· The pregnant woman has a hypertensive emergency; the older woman has severe hypertension without acute organ damage, which is urgency.

· The older woman's autoregulation curve has shifted right over 30 years, so 186/112 mmHg still lies within her range. The previously normotensive woman has an unshifted curve, so 168/108 mmHg exceeds her upper limit, causing breakthrough, vasogenic edema, and encephalopathy.

· Rising creatinine and hematuria come from fibrinoid necrosis and onion-skin narrowing of renal arterioles with glomerular ischemia, which is acute kidney injury.

· Fibrin strands in damaged small vessels shear red cells into schistocytes, giving microangiopathic hemolytic anemia with thrombocytopenia from platelet consumption and a raised lactate dehydrogenase from hemolysis.

· Renal vicious cycle: severe rise in arterial pressure; fibrinoid necrosis and onion-skin narrowing of renal arterioles; glomerular ischemia downstream; juxtaglomerular cells release large amounts of renin; angiotensin II constricts arterioles while pressure natriuresis shrinks plasma volume and adds further renin; pressure rises further and injures more arterioles.

· Renin is very high here because arteriolar injury makes the glomeruli ischemic, which is a direct stimulus to the juxtaglomerular cells, whereas in chronic hypertension renin is variable and often normal or low.

· Pressure natriuresis at a very high pressure contracts plasma volume, and a smaller volume is itself a stimulus to renin, so the two stimuli add.

· Triggers: untreated primary hypertension, renovascular disease, glomerulonephritis, scleroderma renal crisis, preeclampsia, pheochromocytoma, cocaine or amphetamines, and abrupt clonidine withdrawal.

· Abrupt clonidine withdrawal belongs on the list because it causes a rebound sympathetic surge.

· Other acute injuries follow the same logic: sudden afterload can cause left ventricular failure or ischemia, and shear stress can tear the aortic intima, causing dissection.

Distinctions that matter

· The position of the autoregulation curve, not the absolute pressure, decides who breaks through.

· Acute severe hypertension raises renin through renal ischemia and volume contraction, which is the opposite direction from the volume expansion of most chronic hypertension.

Common misconceptions

· Ranking severity by the higher cuff reading. The chronically hypertensive woman tolerates a higher pressure because her curve has shifted.

· Expecting plasma volume to be expanded in a hypertensive emergency. Pressure natriuresis contracts it and that drives renin higher.

Where it is taught: Section C04, autoregulation and breakthrough, the renal vicious cycle figure, the comparison table, and the stop and predict box of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-PR01

Prediction exercise: Predict what a rapid severe rise in pressure does

A competent response contains

· Cerebral arteriolar constriction is overcome and the arterioles are forced open, which is breakthrough of autoregulation; a previously normotensive person has an unshifted curve and breaks through at a lower pressure.

· Barrier permeability rises and brain water content rises, giving vasogenic edema, because capillaries now meet full arterial pressure.

· Renal arterioles show fibrinoid necrosis, bright pink and smudgy with fibrin in the wall, often with luminal thrombus, and surviving arterioles show onion-skin hyperplastic layering.

· Renin release rises sharply, because arteriolar narrowing makes the glomeruli ischemic and pressure natriuresis contracts plasma volume, and both stimulate the juxtaglomerular cells.

· The platelet count falls, the blood film shows schistocytes, and lactate dehydrogenase rises, because fibrin strands in injured small vessels shear red cells and consume platelets.

· The kidney surface shows petechial hemorrhages, the flea-bitten appearance, rather than the symmetrically small, finely granular kidneys of long-standing moderate hypertension.

· The left ventricle has had no time to hypertrophy, so the sudden afterload can cause acute failure with pulmonary edema, or ischemia.

Distinctions that matter

· Vasogenic edema follows barrier leak under pressure, which is not the same as the cytotoxic swelling of ischemic cells.

· Flea-bitten against finely granular kidney surface separates the acute severe lesion from the chronic one.

Common misconceptions

· Expecting the kidneys to be small and granular in an acute emergency. That appearance takes years of benign nephrosclerosis.

· Treating the low platelet count as a separate bleeding disorder. It is consumption within injured small vessels.

Where it is taught: Section C04, autoregulation and breakthrough, the acute arteriolar lesion, the comparison table, and the renal vicious cycle figure of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C05-BD1

Brain dump 1 · foundational retrieval: Pulmonary pressures, the resistance equation, and the five groups

A competent response contains

· Normal pulmonary artery pressure is about 25/8 mmHg with a mean of about 14 mmHg, and pulmonary vascular resistance is about one tenth of systemic resistance or less.

· During exercise, recruitment and distension of capillaries lower resistance as flow rises.

· The thin-walled right ventricle handles a volume load well but tolerates a rise in pressure poorly.

· Pulmonary hypertension is a mean pulmonary artery pressure above 20 mmHg at rest on right heart catheterization, the 2018 World Symposium threshold; many textbooks still quote 25 mmHg.

· Echocardiography raises the suspicion first, estimating systolic pressure from the tricuspid regurgitant jet.

· Mean pulmonary artery pressure equals cardiac output times pulmonary vascular resistance plus left atrial pressure, which is estimated by the wedge pressure.

· Pulmonary vascular resistance is mean pulmonary artery pressure minus wedge pressure, divided by cardiac output, in Wood units; one Wood unit is 1 mmHg per L/min.

· Precapillary: wedge pressure 15 mmHg or less with resistance above 2 Wood units. Isolated postcapillary: wedge pressure above 15 mmHg with resistance of 2 or less. Combined: high wedge pressure and high resistance.

· Group 1, pulmonary arterial hypertension: disease of small pulmonary arteries with endothelial dysfunction, constriction, and proliferative remodeling; idiopathic, heritable through BMPR2, anorexigens and methamphetamine, systemic sclerosis, HIV, portal hypertension, congenital shunts, schistosomiasis.

· Group 2, left heart disease: high left atrial pressure transmitted backward; heart failure with reduced or preserved ejection fraction, mitral and aortic valve disease. This is the most common cause.

· Group 3, lung disease or hypoxia: hypoxic vasoconstriction, capillary loss, remodeling; chronic obstructive pulmonary disease, interstitial lung disease, sleep apnea, hypoventilation, high altitude. This is the next most common.

· Group 4, pulmonary artery obstruction: organized thrombi obstruct arteries and open vessels remodel; chronic thromboembolic pulmonary hypertension.

· Group 5, unclear or multifactorial: sarcoidosis, myeloproliferative disorders.

Distinctions that matter

· Mean pressure alone does not localize the disease; the wedge pressure and the calculated resistance do.

· Group 1 is uncommon but carries the molecular pathogenesis, while groups 2 and 3 carry the numbers of patients.

Common misconceptions

· Treating an echocardiographic estimate as the diagnosis. The definition rests on right heart catheterization.

· Assuming a high mean pressure means disease of the pulmonary arteries. A high wedge pressure with normal resistance places the problem behind the left atrium.

Where it is taught: Section C05, opening paragraph, definition and hemodynamic types, and the classification table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C05-BD2

Brain dump 2 · integration and application: Separate two catheter studies and explain two mechanisms

A competent response contains

· Patient A: (36 minus 10) divided by 5 equals 5.2 Wood units with a normal wedge pressure, so this is precapillary disease; examples are pulmonary arterial hypertension, chronic lung disease, and chronic thromboembolic disease.

· Patient B: (36 minus 28) divided by 5 equals 1.6 Wood units, which is normal, with a high wedge pressure, so this is isolated postcapillary disease; examples are left ventricular failure and mitral stenosis.

· The same mean pressure can be produced by resistance within the lung or by pressure transmitted backward from the left atrium, so the wedge pressure and resistance decide which organ is at fault.

· Hypoxic vasoconstriction in order: low alveolar oxygen closes oxygen-sensitive potassium channels in pulmonary arterial smooth muscle; the cells depolarize; L-type calcium channels open; calcium enters; the muscle contracts.

· Locally this diverts blood away from poorly ventilated alveoli and improves the matching of ventilation and perfusion.

· When the whole lung is hypoxic the response constricts the entire pulmonary bed, raising resistance and driving medial hypertrophy.

· Chronic hypoxemia drives polycythemia, and the higher viscosity raises resistance further.

· In group 2, raised left atrial pressure first raises pulmonary pressure passively, but persistent venous hypertension, classically from long-standing mitral stenosis, remodels the arteries and adds a precapillary component, so the resistance rises as well.

· In group 4 a small minority of emboli organize into fibrous webs instead of lysing, and ventilation-perfusion scanning shows mismatched perfusion defects, whereas group 1 shows no such mismatched defects.

Distinctions that matter

· Systemic arterioles dilate with hypoxia and pulmonary arterioles constrict, which is why lung disease raises only pulmonary resistance.

· A combined picture is a postcapillary disease that has acquired a precapillary component, so both the wedge pressure and the resistance are high.

Common misconceptions

· Calculating resistance from mean pressure divided by cardiac output alone. The wedge pressure must be subtracted first.

· Assuming hypoxic vasoconstriction is always useful. It helps regionally and harms globally.

Where it is taught: Section C05, worked example on precapillary and postcapillary disease and the paragraph after the classification table of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild the pathogenesis of pulmonary arterial hypertension

A competent response contains

· BMPR2 is the gene most often mutated in heritable pulmonary arterial hypertension; it is a receptor of the transforming growth factor beta superfamily, signals through SMAD proteins, and normally restrains smooth muscle proliferation and promotes apoptosis.

· Inheritance is autosomal dominant with incomplete penetrance, so a second hit such as inflammation or hypoxia is thought to be needed.

· Nitric oxide acts through soluble guanylyl cyclase and cGMP, dilating and acting against proliferation, and is decreased.

· Prostacyclin acts through the IP receptor, Gs, and cAMP, dilating and inhibiting platelets, and is decreased while thromboxane A2 rises.

· Endothelin-1 acts through the ETA receptor, Gq, and calcium, constricting and acting as a mitogen, and is increased.

· Sequence: BMPR2 loss or another trigger injures the endothelium; nitric oxide and prostacyclin fall while endothelin-1 and thromboxane rise; small arteries constrict and in situ thrombi form; apoptosis-resistant smooth muscle proliferates; medial hypertrophy, intimal fibrosis, and plexiform lesions develop; pulmonary vascular resistance becomes fixed; the right ventricle hypertrophies and then fails, which is cor pulmonale.

· Structural order: medial hypertrophy with muscularization of normally nonmuscular arterioles first, then concentric intimal fibrosis.

· Plexiform lesions are tufts of slit-like endothelial channels at small artery branch points that resemble a glomerulus, seen in severe idiopathic, heritable, and shunt-related disease.

· Large pulmonary arteries develop atherosclerotic plaques, which do not normally form at pulmonary pressures.

· Acquired triggers and associations: serotonergic anorexigens such as fenfluramine, since serotonin is a smooth muscle mitogen, methamphetamine, systemic sclerosis, HIV, portal hypertension, congenital shunts, and hepatosplenic Schistosoma mansoni infection, in which eggs reach the lungs through portosystemic collaterals and provoke granulomatous arteriolar injury.

· Idiopathic disease is more common in women.

· A large left-to-right shunt remodels the arteries until pulmonary resistance exceeds systemic resistance and the shunt reverses, which is Eisenmenger syndrome.

Distinctions that matter

· Two dilator and antiproliferative signals fall while one constrictor and mitogen rises, so the imbalance acts on both tone and growth.

· Constriction and thrombosis are reversible features, while proliferative remodeling is what makes the resistance fixed.

Common misconceptions

· Treating BMPR2 loss as sufficient on its own. Penetrance is incomplete and a second hit is thought to be required.

· Calling pulmonary arterial hypertension a vasoconstrictive disease only. The remodeling, including plexiform lesions, is the reason resistance stops responding.

Where it is taught: Section C06, genetic and acquired triggers, the endothelial mediator table, and the pathogenesis flow figure of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C06-BD2

Brain dump 2 · integration and application: Explain the signs of the pressure-loaded right heart

A competent response contains

· Loud P2: high pulmonary artery diastolic pressure closes the pulmonic valve forcefully.

· Left parasternal heave and right-sided fourth heart sound: a hypertrophied, stiff right ventricle.

· Holosystolic murmur louder on inspiration with large v waves: tricuspid regurgitation from annular dilation.

· Raised jugular venous pressure, hepatomegaly, and edema: raised systemic venous pressure.

· Right axis deviation, tall R wave in V1, and P pulmonale: right ventricular mass and right atrial enlargement.

· The right ventricle first hypertrophies concentrically, then dilates.

· Dilation stretches the tricuspid annulus, so the valve leaks, and pushes the interventricular septum to the left into a D shape, which limits left ventricular filling.

· When right ventricular pressure nears systemic pressure, its normal systolic coronary perfusion is lost while demand rises, so ischemia accelerates the failure.

· Cor pulmonale is right ventricular hypertrophy, dilation, or failure caused by disease of the lungs or pulmonary vessels, excluding left heart disease and congenital heart disease.

· It is chronic and hypertrophic in chronic obstructive pulmonary disease or pulmonary arterial hypertension, and acute and dilated after a massive pulmonary embolism.

· Systemic pressure can be low because the right ventricle cannot push enough blood across the high resistance, so left atrial return falls, and the shifted septum restricts left ventricular filling; low preload lowers stroke volume.

· Exertional syncope means cardiac output cannot rise across a fixed resistance, so it marks advanced disease.

· A dilated pulmonary artery can compress the left recurrent laryngeal nerve, causing hoarseness.

Distinctions that matter

· Hypertrophy gives the heave and the fourth heart sound, while dilation gives the regurgitant murmur and the venous congestion, so the findings date the stage.

· The left ventricle underfills here rather than failing, which is why systemic pressure falls with normal left ventricular structure.

Common misconceptions

· Reading a low systemic pressure as left ventricular disease. The limitation is right ventricular output and restricted left ventricular filling.

· Treating cor pulmonale as any right ventricular failure. It excludes failure caused by left heart disease and by congenital heart disease.

Where it is taught: Section C06, the right ventricle and cor pulmonale, the table of findings and mechanisms, and the stop and predict box of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C06-CA01

Clinical application: Progressive breathlessness in a young woman with a loud P2

A competent response contains

· Recognize: resistance is (52 minus 9) divided by 3.4, which is about 12.6 Wood units, with a wedge pressure of 9 mmHg, so this is precapillary pulmonary hypertension; with no mismatched perfusion defects and no lung or connective tissue disease, she belongs to group 1, pulmonary arterial hypertension, and the family history points to the heritable form.

· Explain: a loss-of-function BMPR2 mutation removes a transforming growth factor beta superfamily brake that normally restrains smooth muscle proliferation and promotes apoptosis through SMAD signaling; endothelial injury then lowers nitric oxide and prostacyclin and raises endothelin-1 and thromboxane, so small arteries constrict and thrombose in situ, apoptosis-resistant smooth muscle proliferates, and medial hypertrophy, intimal fibrosis, and plexiform lesions fix the resistance.

· Differentiate: left heart disease is excluded by the wedge pressure of 9 mmHg, which shows left atrial pressure is normal; chronic lung disease is excluded by clear lungs, no smoking history, and no hypoxic lung disease; chronic thromboembolic disease is excluded by the absence of mismatched perfusion defects on ventilation-perfusion scanning.

· Investigate: a family history and genetic testing for BMPR2 for the heritable form; skin tightening, Raynaud phenomenon, and connective tissue serology for systemic sclerosis; HIV testing; signs of portal hypertension and liver disease; travel or residence history for schistosomiasis; an anorexigen or methamphetamine exposure history; echocardiography for a congenital shunt, right ventricular hypertrophy and dilation, septal flattening, and tricuspid regurgitation.

· Decide: the right ventricle is failing; the raised jugular venous pressure with large v waves, the hepatomegaly, the ankle edema, and the inspiratory holosystolic murmur of tricuspid regurgitation show failure and dilation rather than hypertrophy alone; right ventricular failure from pulmonary vascular disease is cor pulmonale.

· Justify: output is low because the right ventricle cannot push blood across a fixed resistance of about 12.6 Wood units and because its dilation shifts the septum left and restricts left ventricular filling, so left ventricular preload and stroke volume fall despite normal left ventricular structure; she fainted on exertion because output cannot rise across that fixed resistance, so cerebral perfusion falls with exercise; she is hoarse because a dilated pulmonary artery compresses the left recurrent laryngeal nerve.

Distinctions that matter

· A normal wedge pressure with a high resistance localizes the disease to the pulmonary arteries, which is the single most useful line in the catheter report.

· Exertional syncope and a falling cardiac output reflect a fixed resistance, not a failing left ventricle.

Common misconceptions

· Reading a low systemic pressure in severe pulmonary arterial hypertension as left ventricular failure.

· Assuming any pulmonary hypertension with right heart failure is chronic thromboembolic disease. The perfusion scan separates them.

Where it is taught: Section C05, definition and hemodynamic types with the worked example, and Section C06, pathogenesis and the right ventricle and cor pulmonale of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C04-Q01

Board-style question: Arteriolar lesion in an abrupt severe rise in pressure

Correct answer: C. Concentric laminated layers of smooth muscle with fibrin deposition and wall necrosis

Why it is right: A rapid, severe rise injures arteriolar endothelium, so fibrin and plasma proteins enter the wall and smooth muscle cells die, giving fibrinoid necrosis, while surviving arterioles lay down concentric layers of smooth muscle with reduplicated basement membrane, the onion-skin lesion of hyperplastic arteriolosclerosis.

A is wrong: Homogeneous glassy pink thickening with a narrowed lumen is hyaline arteriolosclerosis, the lesion of years of moderate hypertension, diabetes, and aging, and it does not explain acute kidney injury 4 months after a normal reading.

B is wrong: Subendothelial lipid-laden macrophages under a fibrous cap describe an atherosclerotic plaque, which forms in large and medium arteries rather than in arterioles.

D is wrong: Nodular mesangial matrix expansion with thickened capillary basement membranes is diabetic glomerular disease, which involves the glomerulus rather than the arteriolar wall and develops over years.

A competent response contains

· A previously normotensive patient whose pressure rises abruptly to 212/128 mmHg with encephalopathy, papilledema, acute kidney injury, thrombocytopenia, and schistocytes has a hypertensive emergency.

· The arteriolar lesion of a rapid severe rise is fibrinoid necrosis with concentric onion-skin hyperplastic arteriolosclerosis, most conspicuous in renal interlobular arteries and afferent arterioles.

Distinctions that matter

· Onion-skin layering with fibrinoid necrosis marks acute severe hypertension, while glassy hyaline thickening marks chronic moderate hypertension.

Common misconceptions

· Choosing the chronic hyaline lesion because the patient is hypertensive. The time course and the acute organ injury point to the hyperplastic lesion.

Where it is taught: Section C04 of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

NPLEX-P1-CARD-18-01-C05-Q02

Board-style question: Reading a right heart catheter study

Correct answer: A. Backward transmission of a raised left atrial pressure

Why it is right: A wedge pressure of 26 mmHg with a calculated resistance of 2 Wood units shows that raised left atrial pressure is being transmitted backward, which is group 2 pulmonary hypertension from left heart disease, and her concentric hypertrophy with diastolic dysfunction and left atrial enlargement is the source of that pressure.

B is wrong: Proliferative remodeling with plexiform lesions is group 1 disease, which gives a wedge pressure of 15 mmHg or less with a resistance above 2 Wood units, the opposite of this study.

C is wrong: Hypoxic vasoconstriction with loss of alveolar capillaries is group 3 disease, which is precapillary and requires chronic lung disease or hypoxia, neither of which is present.

D is wrong: Organized thrombus obstructing the pulmonary arteries is group 4 disease, which is also precapillary and shows mismatched perfusion defects rather than a high wedge pressure.

A competent response contains

· Resistance here is (34 minus 26) divided by 4, which is 2 Wood units, a normal value, with a wedge pressure of 26 mmHg.

· A high wedge pressure with normal resistance is isolated postcapillary pulmonary hypertension, and the thick left ventricle with a preserved ejection fraction and a dilated left atrium identifies hypertensive heart disease as the cause.

Distinctions that matter

· The wedge pressure separates disease behind the left atrium from disease within the pulmonary arteries, whatever the mean pulmonary artery pressure is.

Common misconceptions

· Treating any mean pulmonary artery pressure above 20 mmHg as pulmonary arterial disease without calculating resistance.

Where it is taught: Section C05 of the CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-B Congestive Heart Failure

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-02-C01-BD1

Brain dump 1 · foundational retrieval: From a failing left ventricle to alveolar flooding

A competent response contains

· Route: left ventricle with a raised end-diastolic pressure, open mitral valve, left atrium, pulmonary veins, pulmonary capillaries, interstitium, alveoli.

· The pulmonary veins have no valves, so left atrial pressure passes directly back to the pulmonary capillaries; the open mitral valve in diastole is what passes ventricular pressure to the atrium.

· Normal pulmonary capillary pressure is about 7 mmHg against a plasma oncotic pressure of about 28 mmHg, so lymphatics clear the small net filtration easily.

· Pulmonary capillary wedge pressure approximates left atrial pressure and is normally about 6 to 12 mmHg.

· Fluid collects first in the interstitium around vessels and small airways; alveolar flooding follows once capillary pressure exceeds roughly the plasma oncotic pressure and filtration outruns lymph flow.

· A low plasma albumin lowers the oncotic pressure and so lowers the flooding threshold.

· Loss of muscle: ischemic heart disease, the most common cause, through infarction and chronic ischemia.

· Pressure load: systemic hypertension and aortic stenosis, producing concentric hypertrophy and impaired filling.

· Volume load: aortic and mitral regurgitation, which dilate the ventricle.

· Direct muscle damage or stiffening: cardiomyopathies, myocarditis, sustained tachyarrhythmias, alcohol, anthracyclines.

· Mitral stenosis and left atrial myxoma produce every pulmonary sign while the left ventricle stays normal or underfilled, because the obstruction sits at or above the mitral valve.

Distinctions that matter

· A raised wedge pressure reports left atrial pressure, which is not the same as a weak left ventricle: mitral stenosis raises one without the other.

· Interstitial edema and alveolar edema are different stages with different findings, separated by whether filtration has outrun lymphatic clearance.

Common misconceptions

· Saying pulmonary edema begins in the alveoli. Fluid collects in the interstitium first, which is why dyspnea precedes crackles and frothy sputum.

· Treating the flooding threshold as a fixed wedge pressure. It falls with a low albumin and rises when chronic congestion has enlarged the lymphatics.

Where it is taught: Section C01, opening paragraph, causes, and the paragraph on raised left atrial pressure of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C01-BD2

Brain dump 2 · integration and application: Explain three breathing patterns and one blood test

A competent response contains

· Exertional dyspnea: interstitial fluid stiffens the lungs and stimulates juxtacapillary (J) receptors, which drive rapid shallow breathing, and alveolar fluid adds ventilation-perfusion mismatch and hypoxemia.

· Orthopnea: lying flat shifts several hundred millilitres of blood from the legs and splanchnic veins into the thorax, and a ventricle on the flat part of its Frank-Starling curve converts that preload into higher capillary pressure rather than more stroke volume.

· Paroxysmal nocturnal dyspnea: edema fluid from the legs is reabsorbed over one to a few hours while sympathetic support and respiratory centre sensitivity fall during sleep, so the central volume load builds slowly.

· Orthopnea appears within a minute or two because the blood shift is immediate; PND takes hours because it depends on gradual reabsorption of interstitial fluid from the legs.

· Wheeze comes from bronchial wall edema narrowing the airways, which is cardiac asthma.

· Crackles are fine and late inspiratory, from fluid-filled small airways snapping open, and begin at the bases where capillary hydrostatic pressure is highest.

· Forward failure gives prerenal azotemia with a BUN to creatinine ratio above about 20, because enhanced proximal reabsorption and ADH-driven urea reabsorption in the collecting duct accompany the fall in filtration, so urea is retained out of proportion to creatinine.

· Other forward findings: fatigue, cool pale extremities, and confusion in older patients.

· Pleural effusions in this setting are transudates and are often bilateral.

Distinctions that matter

· Orthopnea and PND are both posture related but differ in timing and mechanism: a volume shift against a reabsorption process.

· Creatinine tracks filtration alone, while urea is reabsorbed, which is why the ratio rather than either value identifies the prerenal pattern.

Common misconceptions

· Explaining PND as fluid moving into the chest the moment the patient lies down. That is orthopnea; PND waits for leg edema to be reabsorbed.

· Reading wheeze as evidence of asthma rather than heart failure. Bronchial wall edema wheezes too.

Where it is taught: Section C01, paragraph on dyspnea, orthopnea, and PND, and the final paragraph on crackles and forward findings of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C01-RT01

Retrieval exercise: Rebuild the seven-step congestion sequence

A competent response contains

· Step 1: left ventricular end-diastolic pressure rises.

· Step 2: left atrial and pulmonary venous pressures rise.

· Step 3: capillary filtration exceeds lymphatic clearance, giving interstitial edema.

· Step 4: stiff lungs and J receptor stimulation cause dyspnea on exertion.

· Step 5: lying flat shifts blood into the chest, causing orthopnea.

· Step 6: leg edema is reabsorbed during sleep, causing paroxysmal nocturnal dyspnea.

· Step 7: alveolar flooding gives crackles, hypoxemia, and pulmonary edema.

· At step 3 the capillary hydrostatic pressure now outweighs the plasma oncotic pressure of about 28 mmHg, and the lymphatic route has been exceeded.

· At step 5 several hundred millilitres shift centrally, and a ventricle on the flat part of the Frank-Starling curve gains almost no stroke volume from it, so the pressure appears in the capillaries.

· Long-standing mitral stenosis enlarges the lymphatics so they carry many times normal flow, which moves the flooding step to a much higher pressure, and the chronically raised pressure has also thickened the pulmonary arterioles.

· One day after a large infarct the lymphatics have had no time to adapt, so a wedge pressure that a chronic patient tolerates floods the alveoli.

Distinctions that matter

· The rate of pressure rise matters as much as the level reached, which is why the same wedge pressure means different things acutely and chronically.

· Lymphatic capacity is the variable that decides when interstitial edema becomes alveolar edema.

Common misconceptions

· Writing the sequence as a list of symptoms rather than a sequence of pressures. Each step adds central volume or pressure the ventricle cannot absorb.

· Assuming a chronic patient with a high wedge pressure must be in alveolar edema. Enlarged lymphatics let them tolerate it.

Where it is taught: Section C01, flow diagram on a failing left ventricle to orthopnea and PND of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C02-BD1

Brain dump 1 · foundational retrieval: Right-sided failure from causes to organs

A competent response contains

· Most common cause is left-sided failure: raised pulmonary venous pressure raises pulmonary artery pressure, first passively and then through arteriolar remodeling, and the right ventricle hypertrophies, dilates, and fails, giving biventricular failure.

· Cor pulmonale is right ventricular hypertrophy, dilation, or failure from pulmonary hypertension arising in the lungs or pulmonary vessels, not from left heart or congenital disease.

· Acute cor pulmonale follows massive pulmonary embolism; chronic cor pulmonale follows chronic obstructive pulmonary disease, interstitial lung disease, hypoventilation syndromes, pulmonary arterial hypertension, and chronic thromboembolic disease, through hypoxic pulmonary vasoconstriction and later medial hypertrophy.

· Right ventricular infarction accompanies an inferior infarct from proximal right coronary occlusion and gives hypotension with a raised JVP and clear lungs.

· The right internal jugular vein acts as a manometer for right atrial pressure, normally about 0 to 6 mmHg; with the patient at about 45 degrees, a pulsation more than about 3 to 4 cm above the sternal angle is raised.

· Hepatojugular reflux: abdominal pressure for about 10 to 15 seconds pushes splanchnic blood toward the right heart, and a failing right ventricle cannot accept it, so the JVP stays up while pressure is held.

· Dependent edema needs both raised venous pressure, which increases capillary filtration, and renal sodium and water retention, which supplies the volume.

· Several litres accumulate before swelling is visible, so weight gain comes first; the fluid settles at the ankles in ambulatory patients and the sacrum in bedridden ones.

· It pits because it is a low-protein transudate, unlike lymphedema or the myxedema of hypothyroidism.

· Hepatic acinus: zone 3 is farthest from portal inflow, becomes hypoxic, and atrophies; zone 1 may show fatty change, giving the dark red and tan nutmeg pattern.

· Low output added to congestion gives centrilobular hemorrhagic necrosis; years of congestion give centrilobular fibrosis, called cardiac sclerosis.

· Congestion also produces a tender enlarged liver, ascites, congestive splenomegaly, and bowel wall edema with anorexia.

Distinctions that matter

· Cor pulmonale excludes right ventricular failure caused by left heart disease, which is why the most common cause of right-sided failure is not cor pulmonale.

· Raised venous pressure alone does not make edema; the kidney has to supply the volume, which is why the two requirements are separate.

Common misconceptions

· Calling every raised JVP with ankle edema cor pulmonale. Left-sided failure is the usual cause, and the lungs tell you which.

· Expecting zone 1 to be the hypoxic zone. Zone 1 is closest to portal inflow; zone 3 sits at the end of the gradient and is the one that dies.

Where it is taught: Section C02, causes, the paragraph on neck veins and pitting edema, and the nutmeg liver paragraph of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C02-BD2

Brain dump 2 · integration and application: Three patients with a raised JVP

A competent response contains

· Patient 1: biventricular failure driven by left-sided failure. The left ventricle is failing, pulmonary venous pressure is raised, and the right ventricle has failed against that raised pulmonary artery pressure, so the lungs are congested and the apex is displaced.

· Patient 2: chronic cor pulmonale. The lesion sits in the lungs and pulmonary vessels, so the right ventricle faces a high afterload while left atrial pressure stays normal and the lungs are clear of congestion.

· Patient 3: right ventricular infarction with an inferior infarct from proximal right coronary occlusion. Muscle loss sits in the right ventricle, and the lungs are clear.

· The lung finding follows the position of the lesion in two pumps in series: congestion appears only in the venous bed behind the failing chamber, and only the left heart has the pulmonary veins behind it.

· Patient 2 mechanism: alveolar hypoxia causes hypoxic pulmonary vasoconstriction, and over time the arterioles undergo medial hypertrophy, so the pressure becomes fixed rather than reversible.

· Patient 3 left ventricular filling is reduced for two reasons: right ventricular output falls so less blood crosses the lungs to the left heart, and the dilated right ventricle shifts the septum leftward.

· Volume helps patient 3 because the underfilled left ventricle needs preload and the failing right ventricle must be pushed through the lungs, whereas patient 1 already has a high left atrial pressure and extra volume floods the alveoli.

· Isolated right-sided failure raises systemic venous pressure while pulmonary venous pressure stays normal, which is the full neck veins with clear lungs pattern.

Distinctions that matter

· A raised JVP locates the problem at the right heart but says nothing about the lungs; the lung fields are what separate biventricular failure from isolated right-sided failure.

· Right ventricular infarction and cor pulmonale both give clear lungs, and the difference is lost muscle against a raised afterload.

Common misconceptions

· Assuming any hypotensive patient with an inferior infarct is volume overloaded because the neck veins are full. The left ventricle is underfilled.

· Expecting crackles whenever the JVP is raised. Crackles require raised pulmonary venous pressure, which isolated right-sided failure does not produce.

Where it is taught: Section C02, causes paragraph, the left versus right table, and the stop and predict box of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C03-BD1

Brain dump 1 · foundational retrieval: Two failing ventricles, two sounds, one peptide

A competent response contains

· Ejection fraction is stroke volume divided by end-diastolic volume, normally about 55% to 70%; HFrEF is 40% or lower, mildly reduced is 41% to 49%, and HFpEF is 50% or higher.

· In HFrEF the end-systolic pressure-volume relationship flattens, so the ventricle ejects less and keeps a larger end-systolic volume.

· It adds sarcomeres in series, which is eccentric hypertrophy, and uses the Frank-Starling mechanism at a larger end-diastolic volume.

· Law of Laplace: wall stress is proportional to pressure times radius divided by wall thickness, so a dilated ventricle has high wall stress and high oxygen demand, and the stretched mitral annulus may leak as functional mitral regurgitation.

· Active relaxation needs ATP to return Ca2+ to the sarcoplasmic reticulum through SERCA2a and to release cross-bridges, so ischemia, hypertrophy, and aging slow it.

· Passive stiffness rises with concentric hypertrophy, which is sarcomeres added in parallel, with fibrosis, titin changes, and infiltration such as amyloid, so the end-diastolic pressure-volume relationship steepens and a normal volume needs a high diastolic pressure.

· HFpEF is most common in older adults, women, and patients with hypertension, obesity, diabetes, and atrial fibrillation.

· S3: low-pitched, early diastolic, at the apex, produced when rapid filling from a high-pressure atrium is suddenly halted by a dilated ventricle near its limit of distension; it marks volume overload and high filling pressure, classically in HFrEF, and is normal in children, young adults, athletes, and pregnancy.

· S4: late diastolic, from atrial contraction into a stiff ventricle, so it accompanies HFpEF and disappears in atrial fibrillation.

· BNP is released from stretched ventricular myocytes as proBNP, cleaved into active BNP and inactive NT-proBNP; BNP raises cGMP through the NPR-A receptor and is cleared by neprilysin and NPR-C, while NT-proBNP is cleared mainly by the kidney.

· A BNP below about 100 pg/mL or an NT-proBNP below about 300 pg/mL makes acute heart failure unlikely.

· Levels rise with age, renal failure, atrial fibrillation, and right ventricular strain, and fall with obesity; they are lower in HFpEF because a thick wall lowers wall stress for a given pressure.

Distinctions that matter

· Eccentric and concentric hypertrophy differ in the direction sarcomeres are added, which is why one dilates and the other thickens.

· The S3 reports filling into a distended ventricle and the S4 reports atrial contraction into a stiff one, so the timing in diastole follows the mechanism.

Common misconceptions

· Treating a preserved ejection fraction as a normally functioning ventricle. It empties the fraction it fills, but it fills only at a high pressure.

· Reading any S3 as pathological. It is a normal finding in children, young adults, athletes, and pregnancy.

Where it is taught: Section C03, opening paragraph, HFrEF and HFpEF paragraphs, and the paragraphs on S3 and BNP of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C03-BD2

Brain dump 2 · integration and application: Why the numbers can look reassuring

A competent response contains

· This is heart failure with preserved ejection fraction: her age, long-standing hypertension, obesity, and atrial fibrillation are the classic setting, and hypertension produces concentric hypertrophy with a normal cavity size and a thick wall.

· A normal ejection fraction does not exclude failure because the defect is in filling, not emptying: the steep end-diastolic pressure-volume relationship means a normal volume sits at a high diastolic pressure, which passes back to the pulmonary capillaries.

· BNP is lower than expected for two reasons: a thick wall lowers wall stress for any given pressure, and BNP levels fall with obesity.

· The radiograph shows no cardiomegaly because the cavity is not dilated; a cardiothoracic ratio above 0.5 supports a dilated heart and is often absent in HFpEF, and 0.46 is within normal limits.

· Fast atrial fibrillation takes away the atrial contraction that a stiff ventricle depends on, and it shortens diastole so there is less time to fill, so filling pressure climbs and the lungs flood.

· Radiographic order with wedge pressure: normal at about 6 to 12 mmHg with lower-lobe vessels larger than upper; about 13 to 18 mmHg gives upper-lobe diversion, called cephalization; about 18 to 25 mmHg gives Kerley B lines, peribronchial cuffing, and hazy hila; above about 25 mmHg gives perihilar bat-wing alveolar opacities.

· In failure building over years the lymphatics have enlarged, so the radiographic signs appear at higher pressures, and the signs also lag behind rapid pressure changes.

Distinctions that matter

· Ejection fraction reports emptying and says nothing about the pressure needed to fill, which is the whole of the defect in HFpEF.

· A low BNP argues against acute failure in a thin patient with a dilated ventricle, and argues much less in an obese patient with a thick one.

Common misconceptions

· Using a normal ejection fraction or a normal heart size to rule out heart failure. Neither excludes a stiff ventricle at a high filling pressure.

· Reading a modest BNP as a modest filling pressure. Wall thickness and body habitus both pull the number down.

Where it is taught: Section C03, HFpEF paragraph, BNP paragraph, and the flow diagram on radiographic signs of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C03-CP01

Comparison table: Reduced against preserved ejection fraction

A competent response contains

· Cutoff: HFrEF is an ejection fraction of 40% or lower; HFpEF is 50% or higher, with 41% to 49% called mildly reduced.

· Core defect: HFrEF is reduced contractility; HFpEF is impaired relaxation with increased stiffness.

· Loop: HFrEF has a flatter end-systolic relationship and the loop shifts to the right; HFpEF has a steeper end-diastolic relationship and the loop is narrow at a high diastolic pressure.

· Geometry: HFrEF is dilated with eccentric hypertrophy, sarcomeres added in series, and a normal or thin wall; HFpEF is normal in size with concentric hypertrophy, sarcomeres added in parallel, and a thick wall.

· Causes: HFrEF from myocardial infarction, dilated cardiomyopathy, myocarditis, and chronic volume overload; HFpEF from hypertension, aging, obesity, diabetes, hypertrophic and restrictive cardiomyopathy, and amyloidosis.

· Extra sound: HFrEF gives an S3, from rapid filling halted by a distended ventricle; HFpEF gives an S4, from atrial contraction into a stiff ventricle, when the atria are contracting.

· Dependence on atrial contraction: moderate in HFrEF; high in HFpEF, so atrial fibrillation is poorly tolerated.

· Why filling pressure rises: in HFrEF a large residual volume and continued filling raise end-diastolic pressure; in HFpEF a stiff chamber needs a high pressure even for a normal volume. Both therefore cause the same congestion.

Distinctions that matter

· The two forms differ in whether the ventricle fails to empty or fails to fill, and they do not differ in the congestion they cause.

· An S4 requires organised atrial contraction, which is why it vanishes in atrial fibrillation while an S3 does not.

Common misconceptions

· Putting thick walls with HFrEF and thin walls with HFpEF. Pressure overload thickens and volume overload dilates.

· Assuming HFpEF is the milder condition because the ejection fraction is normal. The filling pressures and the congestion are the same.

Where it is taught: Section C03, HFrEF against HFpEF table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-BD1

Brain dump 1 · foundational retrieval: Lay out the three compensatory systems

A competent response contains

· The baroreceptors and kidney sense arterial filling, not total volume, and cannot tell a failing pump from hemorrhage, so a fall in cardiac output triggers the hemorrhage response.

· Effective arterial blood volume is the portion of blood volume that the arterial baroreceptors and kidney sense as filling the arterial tree; it is low in heart failure even when total volume is expanded, which is why the systems stay switched on.

· Sympathetic: reduced baroreceptor firing raises sympathetic outflow. Beta-1 stimulation raises heart rate and contractility, alpha-1 arteriolar constriction holds arterial pressure, and venoconstriction raises preload.

· Sympathetic harm over months: catecholamines raise oxygen demand and afterload, cause Ca2+ overload, arrhythmias, and myocyte death, and downregulate beta-1 receptors through receptor kinases and beta-arrestin.

· Renin stimuli: low afferent arteriolar pressure, low macula densa NaCl delivery, and renal beta-1 stimulation.

· Angiotensin II at AT1 receptors constricts arterioles, constricts the efferent arteriole more than the afferent, increases proximal Na+ reabsorption, stimulates thirst, stimulates ADH release, and releases aldosterone.

· Aldosterone drives Na+ reabsorption in collecting duct principal cells at the cost of K+ and H+.

· ADH here is driven by baroreceptors rather than osmolality, which is nonosmotic release; it acts at V2 receptors to insert aquaporin-2 and retain free water, so hyponatremia marks advanced disease.

· Retained volume adds little stroke volume because the ventricle sits on the flat part of the Frank-Starling curve, so the extra filling mostly raises venous pressure and causes congestion.

Distinctions that matter

· Total extracellular volume is expanded while effective arterial blood volume is low, and the kidney responds to the second one.

· Osmotic and nonosmotic ADH release have different stimuli; only the nonosmotic route explains retaining water against a low serum sodium.

Common misconceptions

· Describing these systems as a failed or inappropriate response. They are the correct response to the signal they receive, which is arterial underfilling.

· Expecting sodium retention to raise cardiac output. On a flat Frank-Starling curve it mostly raises venous pressure.

Where it is taught: Section C04, opening paragraph and the paragraphs on sympathetic activation, RAAS, and ADH of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-BD2

Brain dump 2 · integration and application: Why blocking hormones works and squeezing harder does not

A competent response contains

· Remodeling is the progressive change in ventricular size, shape, and composition after injury: myocytes enlarge and re-express fetal genes, some die by apoptosis, and angiotensin II and aldosterone drive fibrosis.

· In HFrEF the ventricle becomes larger and more spherical, which raises wall stress and pulls the papillary muscles apart, so functional mitral regurgitation appears and the process becomes self-sustaining.

· Blocking beta-1 receptors, angiotensin II, or aldosterone slows remodeling, which shows that the hormones themselves drive progression.

· Raising contractility alone does not slow remodeling, so the ventricle keeps enlarging; the drive to hypertrophy, apoptosis, and fibrosis is hormonal rather than a matter of how hard the muscle squeezes.

· Natriuretic peptides: ANP from the atria and BNP from the ventricles, acting through NPR-A and cGMP.

· Their three opposing actions are natriuresis, vasodilation, and suppression of renin and aldosterone.

· Three reasons they cannot win: receptor downregulation, clearance by neprilysin, and low renal perfusion, which blunts the natriuretic effect.

· Serum sodium is low because nonosmotic ADH release retains water in excess of sodium.

· Urine sodium is low because angiotensin II and aldosterone avidly reabsorb filtered sodium, so the kidney behaves as if the patient were volume depleted even though both sodium and water are in excess.

Distinctions that matter

· Contractility and remodeling are separate targets: one changes the beat, the other changes the chamber.

· A low serum sodium with a low urine sodium is the signature of a kidney reading low effective arterial volume, not of true depletion.

Common misconceptions

· Assuming a drug that makes the heart contract harder must slow the disease. Remodeling is hormone driven, so it continues.

· Expecting a high urine sodium in a patient who is grossly overloaded with salt and water. Aldosterone and angiotensin II hold it back.

Where it is taught: Section C04, paragraph on remodeling and natriuretic peptides, the compensations table, and the stop and predict box of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-RT01

Retrieval exercise: Rebuild the neurohormonal cycle

A competent response contains

· Step 1: myocardial injury lowers cardiac output and arterial filling.

· Step 2: baroreceptors and the kidney sense low effective arterial volume.

· Step 3: sympathetic activation, renin release, and nonosmotic ADH release.

· Step 4: short-term support through higher heart rate, contractility, vascular tone, and blood volume.

· Step 5: vasoconstriction raises afterload, and salt and water retention raise preload beyond benefit, causing congestion.

· Step 6: catecholamines, angiotensin II, and aldosterone cause myocyte loss, hypertrophy, and fibrosis.

· Step 7: remodeling lowers pump function further, which lowers output again and returns the loop to step 1.

· The genuinely useful step is step 4; harm begins at step 5 with raised afterload and congestion and continues at step 6 with remodeling.

· Vasoconstriction raises afterload, and salt and water retention raise preload.

· Remodeling hormones: catecholamines cause myocyte death and Ca2+ overload, angiotensin II causes hypertrophy and fibrosis, and aldosterone causes myocardial and vascular fibrosis.

· A response designed for hemorrhage, sustained for months, worsens the pump it is trying to support, because the pump cannot use the extra volume and cannot tolerate the extra resistance.

· An AT1 receptor blocker interrupts the loop at step 3 into 5 and at step 6; a mineralocorticoid receptor blocker interrupts the retention at step 5 and the fibrosis at step 6.

Distinctions that matter

· Preload and afterload are raised by different arms of the same response, so each needs naming separately.

· Short-term support and long-term harm are the same mechanisms at different durations, not different mechanisms.

Common misconceptions

· Drawing the loop as a straight line ending in congestion. It closes, which is why the condition progresses on its own.

· Placing the harm at the sensing step. The sensing is accurate for arterial filling; the harm comes from sustaining the response.

Where it is taught: Section C04, cycle diagram on the neurohormonal cycle of heart failure of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C04-PR01

Prediction exercise: Predict what six months of neurohormonal drive does

A competent response contains

· Systemic vascular resistance rises, so afterload and the work of ejection rise: alpha-1 adrenergic constriction plus angiotensin II at AT1 receptors.

· Preload rises and venous pressure behind both ventricles rises: venoconstriction plus angiotensin II and aldosterone driven sodium retention and ADH driven water retention.

· Beta-1 receptors on the myocyte surface fall in number: sustained catecholamine exposure downregulates them through receptor kinases and beta-arrestin.

· Serum potassium falls because aldosterone reabsorbs Na+ in collecting duct principal cells at the cost of K+ and H+; serum sodium falls because nonosmotic ADH acts at V2 receptors to insert aquaporin-2 and retain water in excess of sodium.

· Interstitial collagen rises: angiotensin II and aldosterone drive myocardial fibrosis.

· The ventricle becomes larger and more spherical, and by the law of Laplace the larger radius raises wall stress, which raises oxygen demand and pulls the papillary muscles apart.

· The ability of natriuretic peptides to offset all of this falls: NPR-A downregulation, neprilysin clearance, and low renal perfusion blunt natriuresis and vasodilation.

Distinctions that matter

· Hypokalemia comes from aldosterone and hyponatremia from ADH, so the two electrolyte changes have separate hormones behind them.

· Receptor downregulation reduces the response to catecholamines without reducing the catecholamine level, which is why the heart is both overstimulated and underresponsive.

Common misconceptions

· Expecting sodium to rise because the body is retaining sodium. Water is retained in greater proportion, so the serum sodium falls.

· Treating natriuretic peptides as a brake that eventually stops the cycle. Their effect is too weak to offset the vasoconstrictor systems.

Where it is taught: Section C04, RAAS and ADH paragraphs, remodeling paragraph, and the compensations table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C05-BD1

Brain dump 1 · foundational retrieval: Organise the complications by their cause

A competent response contains

· The three headings are high filling pressures, low forward output, and enlarged, scarred, neurohormonally stimulated chambers.

· Acute pulmonary edema sits under high filling pressure: left atrial pressure rises faster than lymphatics can adapt, triggered by ischemia, a hypertensive surge, a fast arrhythmia, acute valve regurgitation, or a sudden salt load, giving severe dyspnea, diffuse crackles, frothy pink sputum, and hypoxemia; the sympathetic surge it provokes raises afterload further.

· The same picture at a normal wedge pressure points to noncardiogenic edema, as in acute respiratory distress syndrome.

· Cardiogenic shock sits under low forward output: systolic pressure below about 90 mmHg for at least 30 minutes, cardiac index below about 2.2 L/min/m2, and wedge pressure above about 15 to 18 mmHg,

classically after infarction of 40% or more of the left ventricle, with a cold wet patient, low mixed venous oxygen saturation, and lactic acidosis.

· Atrial fibrillation sits under chamber change: atrial stretch and angiotensin II and aldosterone driven fibrosis promote it, it removes the atrial kick and shortens diastole, and persistently fast rates can cause tachycardia-induced cardiomyopathy.

· Ventricular tachycardia and fibrillation cause many sudden deaths: scar supports reentry, fewer repolarizing K+ channels prolong the action potential and favor early afterdepolarizations, catecholamines and Ca2+ overload favor delayed afterdepolarizations, and hypokalemia and hypomagnesemia destabilize repolarization.

· Mural thrombi follow Virchow's triad: stasis in a dilated ventricle, akinetic segment, or fibrillating left atrial appendage; endocardial injury over an infarct; and a procoagulant state. They show lines of Zahn and embolize to the brain, gut, kidneys, spleen, and limbs.

· Cardiorenal syndrome has an arterial mechanism, low output and renal vasoconstriction reducing renal flow, and a venous mechanism, raised central venous pressure raising renal venous and interstitial pressure inside the capsule and lowering the gradient for filtration; creatinine and BUN rise and the added sodium retention worsens congestion.

· Cardiac cachexia comes from TNF alpha and interleukin-6 driven proteolysis, gut congestion with anorexia and malabsorption, and higher energy expenditure, and edema can hide the weight loss.

· Profiles: cardiogenic shock is low output, high wedge pressure, high systemic vascular resistance; hypovolemic shock is low output, low wedge pressure, high resistance; early septic shock is high output, low or normal wedge pressure, low resistance.

Distinctions that matter

· Wedge pressure is what separates cardiogenic from hypovolemic shock, since both have a low output and a high resistance.

· Cardiorenal injury is not purely a flow problem; the venous side acts through pressure inside a non-expandable capsule.

Common misconceptions

· Treating every pulmonary edema as cardiac. A normal wedge pressure with the same picture means the capillary leak is the problem.

· Reading a stable weight as evidence against cachexia. Retained fluid masks the loss of muscle.

Where it is taught: Section C05, paragraphs on acute pulmonary edema and cardiogenic shock, arrhythmias, and mural thrombi, cardiorenal syndrome, and cachexia of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C05-BD2

Brain dump 2 · integration and application: Four deteriorations in one patient

A competent response contains

· First occasion: acute cardiogenic pulmonary edema. Left atrial pressure rose faster than the lymphatics could adapt, so alveolar flooding gives frothy pink sputum and diffuse crackles, and the sympathetic surge raises afterload and worsens it.

· Second occasion: noncardiogenic pulmonary edema, as in acute respiratory distress syndrome. A wedge pressure of 10 mmHg is normal, so the fluid cannot be coming from a raised hydrostatic pressure and must come from increased capillary permeability.

· Third occasion: cardiorenal syndrome.

· Third occasion mechanisms: low output with renal vasoconstriction reduces renal blood flow, and the raised central venous pressure raises renal venous and interstitial pressure.

· Because the kidney sits inside a non-expandable capsule, a rise in interstitial pressure opposes glomerular filtration directly, lowering the net filtration gradient even when arterial pressure is adequate; sodium retention then worsens the congestion that caused it.

· Fourth occasion: embolic stroke from a mural thrombus in the dilated, akinetic left ventricle.

· The three conditions are Virchow's triad: stasis in a dilated ventricle or akinetic segment, endocardial injury over the infarcted area, and a procoagulant state.

· On section the thrombus shows lines of Zahn, alternating pale platelet and fibrin layers with darker red cell layers.

· His weight has not risen despite muscle loss because retained fluid replaces the lost tissue mass; the cachexia itself comes from TNF alpha and interleukin-6 driven proteolysis, gut congestion with anorexia and malabsorption, and higher energy expenditure.

Distinctions that matter

· Cardiogenic and noncardiogenic edema look identical at the bedside and are separated by the filling pressure.

· Congestive kidney injury and low-flow kidney injury can occur in the same patient, and the venous mechanism can dominate while the blood pressure is near normal.

Common misconceptions

· Assuming a rising creatinine in heart failure always means the patient needs more volume. A raised venous pressure can be the cause.

· Expecting a mural thrombus to need a hypercoagulable blood state alone. Stasis and endocardial injury are supplied by the failing, infarcted ventricle.

Where it is taught: Section C05, acute pulmonary edema paragraph, and the paragraph on mural thrombi, cardiorenal syndrome, and cachexia of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C05-CA01

Clinical application: Cold, wet, and oliguric three days after an infarction

A competent response contains

· Recognize: cardiogenic shock. He meets all three criteria, with a systolic pressure of 84 mm Hg (below about 90 mmHg), a cardiac index of 1.8 L/min/m2 (below about 2.2), and a wedge pressure of 26 mmHg (above about 15 to 18), after a large anterior infarct.

· Explain, cold mottled skin: low forward output with reflex alpha-1 and angiotensin II driven vasoconstriction shunts blood away from the skin, giving the cold and clammy picture.

· Explain, crackles with a wedge pressure of 26 mmHg: the failing left ventricle has raised left atrial pressure above the plasma oncotic pressure, so filtration outruns lymphatic clearance and the alveoli flood.

· Explain, mixed venous saturation of 44%: with a low output the tissues extract more oxygen from each unit of blood delivered, so the arteriovenous oxygen difference widens and venous blood returns desaturated.

· Explain, raised lactate: oxygen delivery has fallen below tissue demand, so anaerobic metabolism produces lactic acidosis.

· Differentiate: hypovolemic shock gives a low output with a low wedge pressure and a high systemic vascular resistance; early septic shock gives a high output with a low or normal wedge pressure and a low resistance; his profile is low output, high wedge pressure, high resistance.

· Investigate: look for the cardiorenal pattern, with the raised venous pressure lowering the filtration gradient inside the renal capsule alongside low renal flow, a prerenal BUN to creatinine ratio above about 20, and a low urine sodium from angiotensin II and aldosterone; the sodium of 129 mmol/L points to nonosmotic ADH release at V2 receptors retaining water in excess of sodium.

· Decide: the self-reinforcing mechanism is that hypotension lowers coronary perfusion, which worsens ischemia and lowers output further; the organ is the heart itself, whose perfusion depends on the diastolic arterial pressure it generates.

· Justify: coronary flow occurs mainly in diastole and depends on aortic pressure, so a systolic pressure of 84 mm Hg starves the very muscle that must raise it, and cardiogenic shock classically requires loss of 40% or more of the left ventricle, so there is little reserve muscle left. The sympathetic surge raises heart rate and contractility, which raise oxygen demand, and raises afterload, which opposes ejection, so it worsens the ischemic deficit it is responding to.

Distinctions that matter

· A high wedge pressure with a low output is what makes this cardiac rather than hypovolemic, and a high resistance is what makes it cardiac rather than septic.

· Mixed venous saturation reports the balance between delivery and extraction, which is why it falls here and rises in high-output states.

Common misconceptions

· Reading cold clammy skin as a sign that the vessels have failed. The vasoconstriction is intact and intense; the pump is what has failed.

· Treating the hyponatremia as a separate electrolyte problem. It is the expected consequence of nonosmotic ADH release in severe forward failure.

Where it is taught: Section C05, cardiogenic shock paragraph, with ADH and RAAS in C04 and forward failure findings in C01 of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C06-BD1

Brain dump 1 · foundational retrieval: How a high output still leaves a congested patient

A competent response contains

· High-output heart failure is congestive failure in which cardiac output is above normal but still inadequate; the defining feature is low systemic vascular resistance from arteriolar vasodilation, arteriovenous shunting, or high metabolic demand.

· Mean arterial pressure is approximately cardiac output times systemic vascular resistance.

· Sequence: resistance falls, so arterial pressure falls unless output rises; baroreceptors and the kidney sense arterial underfilling exactly as in low-output failure; sympathetic activation, RAAS, and ADH retain sodium and water; blood volume, venous return, and output rise; the persistent volume and rate load eventually raises filling pressures and dilates the ventricles.

· Baseline: 93 mmHg divided by 5 L/min is about 18.6 mmHg per L/min, which is about 18.6 Wood units.

· A large dialysis fistula adds a low-resistance channel in parallel and lowers total resistance to about 10 Wood units.

· To hold the mean arterial pressure near 90 mmHg at 10 Wood units, output must rise to about 9 L/min, since 90 divided by 10 is 9.

· Tissues gain little because much of the extra flow passes through the fistula without reaching capillaries, so venous blood returns well oxygenated and the heart carries the extra work.

· Compressing the fistula raises resistance and arterial pressure, and the baroreflex then slows the heart, which is the Nicoladoni-Branham sign.

· Causes: severe anemia, through hypoxic vasodilation and low blood viscosity; thyrotoxicosis, through high metabolic rate and heat dissipation with T3 relaxing vascular smooth muscle while increasing SERCA2a, alpha myosin heavy chain, and beta-1 receptors and decreasing phospholamban; arteriovenous fistula, a low-resistance shunt bypassing capillaries; wet beriberi, where thiamine deficiency impairs oxidative metabolism so arterioles dilate and the energy-starved myocardium dilates, with Shoshin beriberi the fulminant form with lactic acidosis; and extensive Paget disease, where hypervascular bone behaves like many shunts.

· The three thiamine pyrophosphate dependent enzymes are pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, and transketolase.

· Pregnancy, sepsis, cirrhosis, severe obesity, and some myeloproliferative disorders also produce high-output states.

Distinctions that matter

· High output and adequate perfusion are not the same thing: output can be above normal while the tissue or the pressure is still short.

· Lowered resistance and raised demand are two separate routes to the same high-output state.

Common misconceptions

· Assuming a high cardiac output rules out heart failure. The congestion comes from the retained volume, not from the output figure.

· Expecting a normal heart to be immune. It sustains the load for a time, and the volume and rate load eventually raises filling pressures.

Where it is taught: Section C06, opening paragraphs, the worked example on the fistula, and the causes paragraph and table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C06-BD2

Brain dump 2 · integration and application: Sort two congested patients by their resistance

A competent response contains

· Patient 1 has low-output failure and patient 2 has high-output failure.

· Skin: high resistance from alpha-1 and angiotensin II driven vasoconstriction makes patient 1 cool and pale, while low resistance from dilated arterioles or shunting makes patient 2 warm and flushed.

· Pulse: a low stroke volume into a constricted arterial tree gives a weak pulse with a narrow pulse pressure in patient 1, while a large stroke volume into a low-resistance tree gives a bounding pulse with a wide pulse pressure in patient 2.

· Oxygen: patient 1 has a wide arteriovenous oxygen difference because tissues extract more from each unit of blood delivered, so mixed venous saturation is low; patient 2 has a narrow difference because of shunting or flow in excess of demand, so mixed venous saturation is normal or high.

· Both patients have the same congestive findings, congestion, edema, neurohormonal activation, and an S3, because in both the kidney senses arterial underfilling and retains sodium and water, so venous pressures rise regardless of the output.

· Severe anemia: pallor, a low hemoglobin, and a flow murmur.

· Thyrotoxicosis: weight loss, heat intolerance, tremor, and atrial fibrillation.

· Arteriovenous fistula: a continuous bruit or thrill over the site, with bradycardia on compression.

· Wet beriberi: alcohol use or a poor diet, with edema, neuropathy, and lactic acidosis.

· Paget disease: bone pain, an enlarging skull, warm skin over affected bone, and a raised alkaline phosphatase.

· A heart that was already diseased has less reserve for the extra volume and rate load, so filling pressures rise and the ventricles dilate sooner than in a previously normal heart.

Distinctions that matter

· The resistance, not the output, is what produces the skin, pulse, and oxygen findings, which is why those findings separate the two forms at the bedside.

· Mixed venous saturation and the arteriovenous oxygen difference report the same thing from opposite ends.

Common misconceptions

· Assuming warm flushed hands rule out heart failure. They point to low resistance, which is the high-output form.

· Expecting a high mixed venous saturation to mean the tissues are well supplied. In a shunt the blood never reached the capillaries.

Where it is taught: Section C06, the low-output against high-output table, the causes table, and the stop and predict box of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C07-BD1

Brain dump 1 · foundational retrieval: Rupture to shock, and why the murmur can be quiet

A competent response contains

· The mitral apparatus has anterior and posterior leaflets held by chordae tendineae anchored to the anterolateral and posteromedial papillary muscles, which keep the leaflets closed in systole.

· The anterolateral muscle has a dual supply from the left anterior descending and circumflex arteries; the posteromedial muscle is usually supplied only by the posterior descending artery, which is why it is the vulnerable one.

· Chordal rupture: myxomatous degeneration or mitral valve prolapse, infective endocarditis, blunt chest trauma, and less often rheumatic disease.

· Papillary muscle rupture complicates myocardial infarction about 3 to 7 days afterwards, when necrotic muscle has been removed by inflammatory cells before collagen has formed, so the tissue is at its weakest.

· The posteromedial muscle ruptures with inferior infarcts; rupture of a whole trunk is usually rapidly fatal, while rupture of one head can be survivable.

· Step 1: a chord or papillary muscle ruptures and a leaflet flails into the left atrium.

· Step 2: a large regurgitant volume enters a small, noncompliant left atrium each systole.

· Step 3: left atrial pressure rises steeply, giving a giant v wave on the pressure tracing, because a normal-sized atrium that has never dilated has low compliance and cannot absorb the volume.

· Step 4: pulmonary capillary pressure rises abruptly, giving acute pulmonary edema.

· Step 5: forward stroke volume falls because the undilated ventricle cannot raise total output enough.

· Step 6: hypotension and cardiogenic shock follow, and sympathetic vasoconstriction diverts still more flow backward.

· The murmur may be early systolic, decrescendo, soft, or absent because atrial pressure approaches ventricular pressure in late systole, so the gradient driving the jet disappears before systole ends.

· Heart size on the radiograph is normal because neither chamber has had time to dilate.

· Ventricular septal rupture gives a loud holosystolic murmur at the left lower sternal border and an oxygen saturation step-up from right atrium to right ventricle.

Distinctions that matter

· A loud murmur reports a maintained pressure gradient, so in acute mitral regurgitation a quiet murmur is the dangerous finding rather than a reassuring one.

· Timing of papillary muscle rupture follows the stage of infarct repair, not the size of the infarct.

Common misconceptions

· Expecting a loud holosystolic murmur in acute mitral regurgitation. That belongs to the chronic lesion and to septal rupture.

· Reading a normal heart size on the radiograph as evidence against a severe valve lesion. Sudden lesions give no time for dilation.

Where it is taught: Section C07, opening paragraph, causes, hemodynamics and findings, and the flow diagram on rupture to shock of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C07-BD2

Brain dump 2 · integration and application: Work the volumes behind a normal ejection fraction

A competent response contains

· Total stroke volume is 120 minus 30, which is 90 mL.

· Ejection fraction is 90 divided by 120, which is 75%.

· Forward stroke volume is 90 minus 50, which is 40 mL.

· Regurgitant fraction is 50 divided by 90, which is about 56%; it is the backward share of total stroke volume.

· Forward cardiac output is 40 mL times 100 per minute, which is 4.0 L/min, below the 5.0 L/min he had before rupture at a heart rate of 72 per minute.

· The ejection fraction rose because the ventricle now empties into the low-pressure left atrium as well as the aorta, so end-systolic volume fell from 50 mL to 30 mL.

· Ejection fraction measures the fraction of the ventricle's contents ejected anywhere, so it overstates forward performance whenever part of the stroke volume goes backward.

· Those 50 mL per beat enter a small, noncompliant left atrium, so atrial pressure rises steeply with giant v waves and the wedge pressure climbs sharply, flooding the alveoli.

· A patient with fifteen years of mitral regurgitation has a dilated, highly compliant left atrium and a dilated left ventricle with eccentric hypertrophy, which this patient lacks.

· A compliant atrium absorbs the same regurgitant volume with only a modest pressure rise, so there is gradual exertional dyspnea rather than sudden pulmonary edema, and the dilated ventricle maintains forward output for years by generating a larger total stroke volume.

Distinctions that matter

· Total and forward stroke volume are different quantities, and ejection fraction is calculated from the first one.

· Compliance, not regurgitant volume alone, decides how high the atrial pressure rises, which is why the acute and chronic lesions present so differently.

Common misconceptions

· Taking a high ejection fraction as proof of good pump performance. In mitral regurgitation it can be high while forward output is failing.

· Assuming chronic mitral regurgitation is simply a milder leak. The adaptations of the atrium and ventricle are what make it tolerable.

Where it is taught: Section C07, hemodynamics paragraph, the worked example on ejection fraction, and the acute against chronic table of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C02-Q01

Board-style question: Hypotension with a raised JVP and clear lungs

Correct answer: B. Reduced left ventricular filling because right ventricular output has fallen and the septum has shifted leftward

Why it is right: Right ventricular infarction lowers right ventricular output so less blood crosses the lungs to the left heart, and the dilated right ventricle shifts the septum leftward, reducing left ventricular filling and therefore stroke volume; systemic venous pressure rises while pulmonary venous pressure does not, which gives the raised JVP with clear lungs.

A is wrong: Alveolar flooding from a rapid rise in pulmonary capillary pressure would give crackles and radiographic opacities, and the lungs are clear by both.

C is wrong: A fall in systemic vascular resistance would give warm flushed extremities with a wide pulse pressure, which belongs to high-output or septic states rather than an occluded right coronary artery.

D is wrong: Acute regurgitation into a small noncompliant left atrium raises pulmonary capillary pressure and causes sudden pulmonary edema, so the lungs would not be clear.

A competent response contains

· ST elevation in II, III, and aVF marks an inferior infarct, usually from proximal right coronary occlusion, which also supplies the right ventricle.

· A raised JVP with a sustained response to abdominal pressure and clear lungs on auscultation and radiograph places the problem in the right heart with normal pulmonary venous pressure.

Distinctions that matter

· Hypotension with a raised JVP and clear lungs separates right ventricular infarction from left-sided or biventricular failure.

Common misconceptions

· Reading full neck veins as volume overload. The left ventricle here is underfilled.

Where it is taught: Section C02 of the CV-18-B Congestive Heart Failure notes.

NPLEX-P1-CARD-18-02-C06-Q02

Board-style question: Congestion with warm extremities on dialysis

Correct answer: A. A large share of the cardiac output returns to the heart without passing through capillaries

Why it is right: A large fraction of the cardiac output passes through the fistula from artery to vein without entering capillaries, so it unloads no oxygen and returns to the heart still saturated, which narrows the arteriovenous oxygen difference and raises mixed venous saturation.

B is wrong: Greater tissue extraction widens the arteriovenous oxygen difference and lowers mixed venous saturation, which is the low-output pattern and the opposite of her figure.

C is wrong: Pulmonary edema limiting alveolar oxygen uptake would lower arterial saturation and, with unchanged extraction, lower rather than raise the mixed venous value.

D is wrong: A reduced hemoglobin concentration lowers oxygen carrying capacity and drives greater fractional extraction, so mixed venous saturation would fall; her finding also has a direct anatomical explanation at the fistula.

A competent response contains

· A large arteriovenous fistula is a low-resistance channel in parallel with the systemic circulation, so systemic vascular resistance falls and cardiac output rises to hold the arterial pressure.

· Warm hands, a bounding pulse with a wide pulse pressure, an output of 9 L/min, and congestion with an S3 are the high-output pattern.

Distinctions that matter

· Mixed venous saturation is high when flow bypasses capillaries and low when tissues extract more from a small delivery.

Common misconceptions

· Assuming a high mixed venous saturation means the tissues are well supplied. Shunted blood never unloaded oxygen.

Where it is taught: Section C06 of the CV-18-B Congestive Heart Failure notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-C Ischemic Heart Disease

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-03-C01-BD1

Brain dump 1 · foundational retrieval: Myocardial oxygen supply and demand from memory

A competent response contains

· Resting coronary flow is about 225 mL/min, about 4 to 5% of cardiac output, and can rise about three to four fold in heavy exercise.

· The myocardium extracts about 70% or more of the oxygen in coronary blood at rest, so extra demand can be met only by raising flow.

· Flow reserve comes from dilation of small resistance arterioles in response to local metabolic signals: chiefly adenosine from ATP breakdown, with low oxygen tension, carbon dioxide, hydrogen ions, and potassium.

· Demand is set by heart rate, contractility, and ventricular wall stress.

· By the law of Laplace, wall stress is proportional to pressure times radius divided by wall thickness, so high afterload or a large chamber raises oxygen use.

· The left ventricle is perfused mainly in diastole, and the driving pressure is approximately aortic diastolic pressure minus left ventricular end-diastolic pressure.

· The subendocardium is ischemic first because systolic compression of intramural vessels is greatest there and penetrating branches reach it last.

· Subendocardial ischemia therefore appears whenever perfusion pressure falls, diastole shortens, or ventricular diastolic pressure rises.

Distinctions that matter

· Ischemia is inadequate flow, which removes oxygen and leaves metabolic waste behind; hypoxemia lowers oxygen content while flow continues.

· Coronary flow reserve is a property of the arterioles, while a fixed stenosis is a property of a large epicardial artery; the arterioles can hide the stenosis at rest.

Common misconceptions

· Expecting the heart to meet higher demand by extracting more oxygen. Extraction is already near maximal at rest, so flow has to rise.

· Treating ischemia as a disease of one artery. It is a mismatch between supply and demand, which is why aortic stenosis and severe anemia cause angina with normal coronary arteries.

Where it is taught: Section C01, first three paragraphs and the oxygen balance table of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C01-BD2

Brain dump 2 · integration and application: Four patients with chest pain and clean coronary arteries

A competent response contains

· Patient 1, aortic stenosis with hypertrophy: both sides. High systolic pressure and hypertrophy raise wall stress and oxygen use per beat, while a long ejection time and a high left ventricular end-diastolic pressure cut the subendocardial driving pressure, and hypertrophy adds muscle with fewer capillaries per myocyte.

· Patient 2, hemoglobin 6.5 g/dL: supply side through oxygen content. Each mL of coronary blood carries less oxygen, so the same flow delivers less oxygen; compensatory tachycardia adds a demand problem and shortens diastole.

· Patient 3, hyperthyroidism at 136/min: demand side mainly. Heart rate and contractility raise oxygen use per minute, and tachycardia also shortens diastole, so perfusion time falls.

· Patient 4, cocaine: supply side through coronary resistance. Vasospasm of an epicardial artery, with added sympathetic rise in rate and blood pressure on the demand side.

· Patients 1 to 3 develop subendocardial ischemia, because that layer is compressed in systole and perfused last, so it suffers first whenever supply falls short.

· Patient 4 can develop ischemia through the whole wall thickness, because spasm of an epicardial artery shuts off flow to the full thickness of its territory rather than trimming the reserve.

Distinctions that matter

· A demand problem spends the reserve from above; a content problem lowers the oxygen carried in a normal flow; a resistance problem blocks flow itself.

· Layer follows mechanism: reduced perfusion pressure or shortened diastole starves the subendocardium, while epicardial occlusion or spasm takes the whole wall.

Common misconceptions

· Concluding that normal coronary arteries rule out myocardial ischemia. Aortic stenosis, severe anemia, hypoxemia, carbon monoxide poisoning, tachycardia, and spasm all cause it with patent arteries.

· Assigning each patient to one row of the oxygen balance table. Most triggers act on several rows at once, and tachycardia almost always acts on both supply and demand.

Where it is taught: Section C01, oxygen balance table and its caption of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C01-PR01

Prediction exercise: Predict what happens when demand rises on a fixed stenosis

A competent response contains

· Oxygen content per mL of blood falls, because hemoglobin is 7.2 g/dL and hemoglobin carries almost all the oxygen.

· Perfusion time falls, because a rate of 134/min shortens diastole, and the left ventricle is perfused mainly in diastole.

· The gradient for subendocardial flow falls, because the driving pressure is aortic diastolic pressure minus left ventricular end-diastolic pressure and the blood pressure is 86/50 mm Hg.

· Oxygen used per minute rises, because fever, tachycardia, and sympathetic drive raise heart rate and contractility.

· Arteriolar reserve distal to the stenosis is already spent, because the arterioles had dilated to keep resting flow normal across a 75% narrowing, so no further dilation is available.

· The subendocardium becomes ischemic first, and the leads facing that territory show ST depression, because the injury current points toward the cavity and away from the surface electrodes.

Distinctions that matter

· Fever and tachycardia act on both sides of the balance at once: more oxygen used per minute and less time to deliver it.

· Anemia lowers oxygen content without changing resistance, so flow can look adequate while delivery is not.

Common misconceptions

· Expecting arteriolar dilation to rescue this situation. The reserve was already used to keep resting flow normal past the stenosis.

· Looking for ST elevation here. Subendocardial ischemia depresses the ST segment, and that depression does not localize the artery.

Where it is taught: Section C01, oxygen balance table; Section C02, paragraphs on stable angina and ST segment movement of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-BD1

Brain dump 1 · foundational retrieval: Stable angina and the moving ST segment

A competent response contains

· Angina pectoris is episodic chest pressure or heaviness from transient ischemia that is not severe or long enough to kill myocytes.

· Ischemic myocytes release adenosine, bradykinin, hydrogen ions, and potassium, which stimulate visceral afferents travelling with the cardiac sympathetic nerves into roughly T1 to T5.

· The same dorsal horn neurons receive somatic input from the chest wall and inner arm, so the pain is referred to the chest, left arm, neck, and jaw.

· Stable angina comes from a fixed stenosis of about 70% or more, and about 90% or more can limit flow at rest.

· By the Poiseuille relationship resistance varies inversely with the fourth power of the radius, so halving the radius raises segment resistance 16 fold.

· At rest the downstream arterioles dilate enough to keep flow normal, which spends the reserve, so flow cannot rise with exertion and the subendocardium becomes ischemic; pain is predictable, brief, and eased by rest.

· Nitroglycerin works because nitric oxide raises cGMP and dilates veins more than arteries, lowering ventricular volume, wall stress, and demand.

· The ST segment is normally isoelectric because all healthy ventricular cells sit at the same plateau voltage, so no gradient exists to record.

· Ischemic cells lose potassium, depolarize partially, and have shorter lower plateaus, so an injury current flows between ischemic and normal muscle.

· Subendocardial ischemia points the ST vector toward the cavity and away from surface electrodes, giving ST depression that does not localize the artery; ischemia through the whole wall thickness points it toward the epicardium, giving ST elevation in the facing leads, which does localize the artery.

Distinctions that matter

· Flow reserve is spent at rest in stable angina, so resting flow is normal while exertional flow is not; this is why the history is exertional rather than constant.

· ST depression reports the layer, not the artery, while ST elevation reports both.

Common misconceptions

· Explaining nitroglycerin by coronary dilation alone. Its main action here is venodilation that lowers ventricular volume, wall stress, and demand.

· Reading an isoelectric ST segment as electrical silence. The whole ventricle is depolarized at the plateau, with no voltage gradient to record.

Where it is taught: Section C02, opening paragraph, stable angina paragraph, and the paragraph on why the ST segment moves of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-BD2

Brain dump 2 · integration and application: Sort three episodes of chest pain by mechanism

A competent response contains

· Patient 1: stable angina from a fixed stenosis of about 70% or more; subendocardial ischemia; ST depression during pain; troponin normal. The reserve is spent at rest, so flow cannot rise with exertion and the inner layer starves first.

· Patient 2: unstable angina, the mildest acute coronary syndrome, from a fissured plaque with a nonocclusive platelet thrombus, sometimes with spasm from thromboxane A2 and serotonin; usually subendocardial ischemia; ST depression or T inversion or a normal tracing; troponin stays below the upper reference limit.

· Patient 3: Prinzmetal (variant) angina from intense spasm of an epicardial artery, with or without plaque, linked to smooth muscle hyperreactivity and reduced endothelial nitric oxide; ischemia through the full wall thickness, so the ST segment rises transiently; troponin normal unless infarction occurs.

· A troponin rise above the assay-specific 99th percentile upper reference limit would make patient 2 a non-ST-elevation myocardial infarction.

· Prinzmetal attacks come at rest, often at night, in younger patients whose main risk factor is smoking, and can be provoked by cocaine, amphetamines, ergot alkaloids, triptans, and hyperventilation.

· Microvascular angina causes exertional ischemia with normal epicardial arteries and belongs in the same differential.

Distinctions that matter

· Unstable angina and non-ST-elevation myocardial infarction share the lesion and the tracing; serial troponin is what separates them.

· Rest pain with ST elevation that comes and goes is spasm, while rest pain with ST depression is an active nonocclusive plaque.

Common misconceptions

· Taking transient ST elevation as proof of infarction. In Prinzmetal angina it resolves with the spasm and troponin stays normal.

· Calling unstable angina a mild problem because troponin is normal. The thrombus can grow and occlude the artery.

Where it is taught: Section C02, unstable, Prinzmetal, and microvascular paragraph, the comparison table, and the stop and predict box of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-RT01

Retrieval exercise: Rebuild the ischemic cascade

A competent response contains

· Step 1: demand exceeds supply, or supply suddenly falls.

· Step 2: anaerobic glycolysis begins and lactate, hydrogen ions, and adenosine accumulate.

· Step 3: relaxation slows, because calcium reuptake needs ATP.

· Step 4: regional contraction weakens.

· Step 5: the ST segment changes from injury currents.

· Step 6: chest pain follows from stimulation of visceral afferents.

· The pump that needs ATP at step 3 is the sarcoplasmic reticulum calcium ATPase, which pulls calcium back out of the cytosol so the muscle can relax.

· At step 5 ischemic cells have lost potassium and are partly depolarized with shorter lower plateaus, so a current flows between ischemic and normal myocardium and displaces the ST segment.

· Because pain is last in the order, ischemia can be silent, especially when diabetic autonomic neuropathy has damaged the afferent fibers.

· Rest restores the balance between supply and demand and the whole cascade reverses without necrosis.

Distinctions that matter

· Relaxation fails before contraction does, because taking calcium back up costs ATP while letting the muscle relax is not passive.

· Electrical change precedes pain, so a normal tracing taken between episodes does not rule the condition out.

Common misconceptions

· Putting pain early in the cascade. It is the last event, which is what makes silent ischemia possible.

· Assuming any episode of ischemia kills muscle. In angina the episode ends before myocytes die.

Where it is taught: Section C02, the ischemic cascade flow diagram and its caption of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C02-CP01

Comparison table: Stable, unstable, and Prinzmetal angina

A competent response contains

· Lesion: stable angina has a fixed stenosis of about 70% or more; unstable angina has a disrupted plaque with a nonocclusive thrombus; Prinzmetal angina is epicardial vasospasm.

· Pattern: stable is exertional, predictable, and relieved by rest; unstable is rest pain, new pain, or crescendo pain; Prinzmetal is spontaneous rest pain, often at night.

· Layer ischemic: subendocardial in stable, usually subendocardial in unstable, and the full wall thickness in Prinzmetal.

· ECG during pain: ST depression in stable; ST depression, T inversion, or a normal tracing in unstable; transient ST elevation in Prinzmetal.

· Troponin: normal in stable; normal in unstable, which is what separates it from non-ST-elevation myocardial infarction; normal in Prinzmetal unless infarction occurs.

· Patient and provoking factors: stable angina in a patient with established coronary disease provoked by exertion, cold, or emotion; unstable angina from plaque disruption with thromboxane A2 and serotonin driving added spasm; Prinzmetal in younger patients whose main risk factor is smoking, provoked by cocaine, amphetamines, ergot alkaloids, triptans, and hyperventilation.

· Main risk: gradual progression in stable angina, infarction in unstable angina, and arrhythmia or infarction in Prinzmetal angina.

Distinctions that matter

· The ischemic layer decides whether the ST segment falls or rises, which is why Prinzmetal angina elevates it and the other two depress it.

· Stable and unstable differ by the state of the plaque rather than by the degree of narrowing; many culprit plaques are only mildly to moderately stenotic.

Common misconceptions

· Grading these three by severity of stenosis. Plaque stability, not size, decides which syndrome a patient has.

· Expecting Prinzmetal angina only in patients with atherosclerosis. Spasm occurs with or without plaque.

Where it is taught: Section C02, comparison table of stable, unstable, and Prinzmetal angina of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C03-BD1

Brain dump 1 · foundational retrieval: From a ruptured cap to a dead myocyte

A competent response contains

· Sequence: the cap ruptures and exposes collagen and a lipid core rich in tissue factor; von Willebrand factor binds platelet glycoprotein Ib and platelets adhere; ADP and thromboxane A2 recruit more platelets and glycoprotein IIb/IIIa binds fibrinogen; tissue factor drives thrombin and fibrin and the thrombus occludes the artery.

· Contraction stops within about 1 to 2 minutes, irreversible injury begins after about 20 to 40 minutes starting in the subendocardium, and necrosis spreads toward the epicardium over about 3 to 6 hours.

· Inside the myocyte: oxidative phosphorylation stops and anaerobic glycolysis yields little ATP while generating lactate and hydrogen ions.

· Failure of the sodium-potassium ATPase lets sodium and water in so the cell swells, potassium leaks out, and calcium accumulates as ATP-dependent pumps fail.

· Calcium floods in and activates phospholipases and proteases.

· The two changes marking irreversible injury are sarcolemmal disruption and amorphous densities in mitochondria.

· Troponin, the MB isoenzyme of creatine kinase, and myoglobin leak into blood.

· If flow returns within roughly 20 minutes the injury is reversible, although the muscle may stay stunned for hours to days.

· Rupture clusters in the early morning and after intense exertion or emotional stress, when an adrenergic surge raises blood pressure, heart rate, and shear stress.

· Most infarctions follow rupture of a thin inflamed cap; fewer follow superficial erosion over an intact plaque, and many culprit plaques were only mildly to moderately stenotic, so infarction is often the first sign of coronary disease.

Distinctions that matter

· Loss of contraction at 1 to 2 minutes is a functional failure in living cells, while irreversible injury at 20 to 40 minutes is cell death; the gap between them is the window for salvage.

· Plaque rupture exposes tissue factor and collagen, so both platelet adhesion and the coagulation cascade start at the same lesion.

Common misconceptions

· Assuming the most stenotic plaque is the one that infarcts. Cap thickness, lipid core size, and inflammation decide rupture, not the degree of narrowing.

· Treating the immediate wall motion abnormality as proof of necrosis. Contraction fails long before the cell dies.

Where it is taught: Section C03, opening paragraph, the plaque rupture flow diagram, and the paragraphs on the ischemic myocyte of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C03-BD2

Brain dump 2 · integration and application: Two infarcts and two reperfused hearts

A competent response contains

· Eight hours of complete occlusion gives a transmural infarct that spans the full or nearly full wall in one artery's territory, with ST elevation, later pathologic Q waves, a troponin rise, and the electrical label of ST-elevation myocardial infarction.

· Thirty minutes of partial occlusion that then opens gives a subendocardial infarct limited to the inner third to half of the wall, with ST depression or T inversion, usually no Q waves, a troponin rise, and the label of non-ST-elevation myocardial infarction.

· The transmural infarct can cause pericarditis and free wall rupture because necrosis reaches the epicardium; the subendocardial infarct rarely ruptures because the outer wall stays intact.

· The electrical and pathologic labels are strongly associated but not identical.

· Reperfused infarcts are often hemorrhagic on gross examination.

· Irreversibly injured myocytes take up a surge of calcium and hypercontract, producing contraction band necrosis with dense eosinophilic cross bands.

· Reactive oxygen species and calcium opening the mitochondrial permeability transition pore kill some cells that might have survived, which is reperfusion injury, so early reperfusion limits infarct size while changing the appearance of the infarct.

· In shock or profound hypoxemia a circumferential subendocardial infarct ignores arterial territories, because the whole inner layer is underperfused rather than one artery being blocked.

Distinctions that matter

· Transmural against subendocardial is a statement about the tissue; ST elevation against no ST elevation is a statement about the tracing, and the two pair up without being the same thing.

· Infarct size is limited by reperfusion, while the appearance of the infarct is altered by it; hemorrhage and contraction bands are features of the reperfused infarct.

Common misconceptions

· Equating every ST-elevation infarct with a transmural one at autopsy. The pairing is an association.

· Treating reperfusion as purely protective. It salvages muscle overall and also kills some borderline cells through oxygen radicals and calcium overload.

Where it is taught: Section C03, paragraphs on transmural and subendocardial infarction and on reperfusion of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C03-RT01

Retrieval exercise: Rebuild the path from plaque rupture to transmural infarction

A competent response contains

· Step 1: the cap ruptures, exposing collagen and a lipid core rich in tissue factor.

· Step 2: von Willebrand factor binds platelet glycoprotein Ib and platelets adhere to the exposed surface.

· Step 3: ADP and thromboxane A2 recruit further platelets, and glycoprotein IIb/IIIa binds fibrinogen to cross-link them.

· Step 4: tissue factor from the lipid core drives thrombin and fibrin generation and the thrombus occludes the artery.

· Step 5: contraction stops within about 1 to 2 minutes.

· Step 6: irreversible injury begins after about 20 to 40 minutes, in the subendocardium first.

· Step 7: necrosis spreads toward the epicardium over about 3 to 6 hours.

· Shading: at 30 minutes only the inner third to half is dying; at 2 hours the necrosis has moved outward into the middle of the wall; by about 6 hours it reaches the epicardium in the territory supplied.

· Collateral flow and reperfusion can stop the wavefront before it becomes transmural.

Distinctions that matter

· Platelet adhesion needs von Willebrand factor and glycoprotein Ib, while platelet aggregation needs glycoprotein IIb/IIIa and fibrinogen; the drawing should separate the two sites.

· The wavefront moves from endocardium outward, so the outer layers have the longest window for salvage.

Common misconceptions

· Drawing the necrosis beginning in the middle of the wall or spreading inward. It starts in the subendocardium and moves out.

· Leaving the lipid core out of the clotting story. Tissue factor in the core is what drives thrombin generation.

Where it is taught: Section C03, flow diagram from plaque rupture to transmural infarction and its caption of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C04-BD1

Brain dump 1 · foundational retrieval: Build the coronary territory table from memory

A competent response contains

· Anterior: left anterior descending artery, proximal if extensive; ST elevation in V1 to V4 and also V5, V6, I, and aVL if extensive; largest infarcts, with shock, bundle branch block, apical thrombus, and aneurysm.

· Lateral: left circumflex artery or an LAD diagonal branch; ST elevation in I, aVL, V5, and V6; reciprocal inferior ST depression.

· Inferior: right coronary artery in most hearts, circumflex if left dominant; ST elevation in II, III, and aVF; bradycardia, atrioventricular block, and papillary muscle rupture.

· Right ventricular: proximal right coronary artery; ST elevation in V4R with inferior changes; hypotension, clear lungs, and raised neck veins.

· Posterior: right coronary or circumflex; no elevation in the standard leads, with a mirror image of ST depression and tall R waves in V1 to V3; occurs with inferior or lateral infarction.

· The LAD supplies the anterior left ventricular wall, the apex, and the anterior two thirds of the septum with much of the bundle branches; the circumflex supplies the lateral wall; the right coronary artery supplies the right ventricle, the inferior left ventricular wall, and usually the sinoatrial and atrioventricular nodes.

· The posterior descending artery arises from the right coronary artery in about 80% of hearts, which is right dominance, and otherwise from the circumflex.

· The LAD is occluded most often, about 40 to 50% of cases, then the right coronary and the circumflex.

· Reciprocal ST depression supports infarction over pericarditis, which elevates the ST segment diffusely without reciprocal change.

· A pathologic Q wave appears because dead myocardium is electrically silent, so an electrode over it records the opposite wall depolarizing away; it is commonly at least 40 ms wide or more than about one quarter of the R wave height.

Distinctions that matter

· Lead groups localize the artery only when the ST segment is elevated; depression from subendocardial ischemia does not localize.

· Posterior infarction is read as a mirror image in the anterior leads rather than as elevation anywhere on the standard tracing.

Common misconceptions

· Assigning the inferior wall to the right coronary artery in every patient. In a left dominant heart the circumflex supplies it.

· Reading ST elevation in many lead groups at once as a large infarct without checking for reciprocal change. Diffuse elevation without reciprocal depression points to pericarditis.

Where it is taught: Section C04, first paragraph, the territory table, and the paragraph on reciprocal change and Q waves of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C04-BD2

Brain dump 2 · integration and application: Read the biomarkers across four days

A competent response contains

· Troponin I or T first rises at about 3 to 4 hours, earlier with high-sensitivity assays, peaks at about 24 to 48 hours, and returns to normal at about 7 to 10 days, with troponin T up to about 14 days.

· The MB isoenzyme of creatine kinase first rises at about 3 to 6 hours, peaks at about 24 hours, and returns to normal at about 48 to 72 hours.

· Myoglobin first rises at about 1 to 3 hours, peaks at about 6 to 12 hours, and returns to normal at about 24 hours, so it is earliest but nonspecific.

· Troponin persists because a small cytosolic pool leaks early and a large myofilament-bound pool is released slowly; the MB isoenzyme is cytosolic and clears quickly.

· On day 4 troponin is still high from the first infarct, so a single value cannot show new necrosis, while the MB isoenzyme had normalized by about 48 to 72 hours, so a new rise in it points to reinfarction.

· If the MB isoenzyme is normal and the pain is pleuritic and positional, fibrinous pericarditis is the more likely explanation.

· Troponin also rises in myocarditis, pulmonary embolism, sepsis, heart failure, tachyarrhythmia, cardiac contusion, and chronic kidney disease.

· Infarction requires a rise or fall of troponin with at least one value above the assay-specific 99th percentile, plus evidence of ischemia.

Distinctions that matter

· Specificity and usefulness for reinfarction run in opposite directions: troponin is the most specific marker but too persistent, while the MB isoenzyme is less specific but clears fast enough to show a second event.

· A high troponin reports myocyte necrosis of any cause; infarction is necrosis from ischemia, which is why the definition asks for a rise and fall with evidence of ischemia.

Common misconceptions

· Reading a single high troponin on day 4 as reinfarction. It can simply be the tail of the first event.

· Taking any troponin rise as an infarction. Pulmonary embolism, sepsis, myocarditis, and chronic kidney disease all raise it.

Where it is taught: Section C04, biomarker paragraph, the biomarker table, and the worked example on recurrent chest pain on day 4 of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C04-Q01

Board-style question: Hypotension with clear lungs after an inferior infarction

Correct answer: C. Right ventricular infarction limiting delivery of blood to the left heart

Why it is right: A proximal right coronary occlusion infarcts the right ventricle as well as the inferior left ventricular wall. The right ventricle cannot pump its volume through the lungs to the left heart, so left ventricular filling and cardiac output fall while the lungs stay clear and systemic venous pressure rises, and ST elevation in V4R is the finding that identifies it.

A is wrong: Papillary muscle rupture with acute mitral regurgitation floods the lungs, so the examination would show widespread crackles rather than clear lung fields, and it appears around days 3 to 7 rather than within 2 hours of the onset of pain.

B is wrong: Blood in the pericardial sac follows free wall rupture of a transmural infarct, usually around days 3 to 7, and would present with pulseless electrical activity rather than a measurable blood pressure 2 hours into the first episode of pain.

D is wrong: Loss of more than 40% of left ventricular muscle causes cardiogenic shock with pulmonary congestion and crackles, and that degree of loss follows a large anterior infarct rather than an inferior one.

A competent response contains

· ST elevation in II, III, and aVF with reciprocal depression in I and aVL is an inferior infarct, usually from right coronary artery occlusion.

· Elevation in V4R adds right ventricular infarction, which occurs with proximal right coronary occlusion.

· A failing right ventricle cannot deliver blood to the left heart, so filling pressures rise behind it while the pulmonary circulation stays dry: hypotension, raised jugular venous pressure, and clear lungs.

· The bradycardia fits right coronary supply of the sinoatrial and atrioventricular nodes together with vagal reflexes from the inferior and posterior wall.

Distinctions that matter

· Hypotension with clear lungs points to the right ventricle, while hypotension with crackles points to loss of left ventricular muscle.

Common misconceptions

· Reading raised neck veins after an infarction as tamponade in every case. Right ventricular infarction raises them without any pericardial blood.

Where it is taught: Section C04, territory table; Section C06, paragraph on pump failure and right ventricular infarction of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C05-BD1

Brain dump 1 · foundational retrieval: Build the healing timeline from memory

A competent response contains

· 0 to 30 minutes: no gross change; no light microscopic change, with reversible changes visible by electron microscopy; arrhythmia.

· 30 minutes to 4 hours: no gross change, pale on tetrazolium stain after about 2 to 3 hours; usually no light microscopic change, with wavy fibers at the border; ventricular fibrillation and sudden death.

· 4 to 24 hours: occasional dark mottling; early coagulative necrosis, edema, hemorrhage, contraction bands, early neutrophils; arrhythmia, left ventricular failure, cardiogenic shock.

· 1 to 3 days: mottling with a yellow-tan center; complete coagulative necrosis, loss of nuclei, brisk neutrophils; fibrinous pericarditis.

· 3 to 7 days: hyperemic border with central softening; disintegrating myocytes with macrophages at the border; free wall, septal, or papillary muscle rupture.

· 7 to 10 days: softest and most yellow with red-tan margins; phagocytosis and early granulation tissue at the margins; rupture risk persists.

· 10 to 14 days: red-gray depressed borders; established granulation tissue with new vessels and collagen; rupture less likely.

· 2 to 8 weeks: gray-white scar forming from the border inward; increasing collagen and decreasing cellularity; aneurysm, mural thrombus, Dressler syndrome.

· Over 2 months: firm white scar; dense collagen; heart failure and scar-related ventricular tachycardia.

· Dead myocytes keep their outlines for several days because ischemia and acidosis denature structural proteins and lytic enzymes alike, so autolysis is delayed; this is coagulative necrosis.

· Adult myocytes have very little capacity to divide, so dead muscle is replaced by scar rather than new muscle.

· Healing advances from the viable margins inward, so a large infarct heals more slowly and its center lags behind its edges.

· Wavy fibers appear at the border within hours because contracting living fibers stretch the noncontracting dead ones.

· Triphenyltetrazolium chloride stains viable muscle brick red through dehydrogenase activity, so an infarct appears as a pale unstained zone after about 2 to 3 hours, before light microscopy shows necrosis.

· Neutrophils arrive near the end of day 1, drawn by chemokines and complement fragments; contraction bands appear at the margins in the first day and throughout reperfused infarcts.

Distinctions that matter

· Neutrophils peak in the first 3 days and macrophages take over from about day 3 to day 7; the cell type present dates the infarct more reliably than the gross appearance.

· Loss of dehydrogenase activity precedes visible necrosis, which is why tetrazolium staining detects an infarct that light microscopy would call normal.

Common misconceptions

· Expecting dead myocytes to disappear quickly. In coagulative necrosis their outlines persist for days because the lytic enzymes were denatured too.

· Looking for granulation tissue in the first week. It appears at the margins around days 7 to 10 and is established by about days 10 to 14.

Where it is taught: Section C05, opening paragraphs and the morphologic timeline table of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C05-BD2

Brain dump 2 · integration and application: Date four infarcts from what the tissue shows

A competent response contains

· Heart 1 is about 3 to 7 days old: macrophage clearance with central softening and a hyperemic border, and the risk is rupture of the free wall, the septum, or a papillary muscle because dead muscle has been removed and collagen has not formed.

· Heart 2 is about 1 to 3 days old: complete coagulative necrosis with brisk neutrophils, and the linked complication is fibrinous pericarditis over a transmural infarct.

· Heart 3 is about 10 to 14 days old: established granulation tissue with new vessels and collagen, and rupture is now less likely while aneurysm and mural thrombus become the concerns as the scar forms.

· Heart 4 is over 2 months old: dense collagen in a firm white scar, and the risks are heart failure and scar-related ventricular tachycardia.

· Heart 1 has the weakest wall. The wall is weakest after macrophages have removed the dead muscle and before collagen has been laid down, about days 3 to 10, which is the window for rupture.

· A pale unstained zone on tetrazolium staining with normal light microscopy means the infarct is younger than the point at which necrosis becomes visible, roughly 2 to 3 hours onward, because loss of dehydrogenase activity comes first.

Distinctions that matter

· Strength of the wall tracks the balance between removal of necrotic muscle and deposition of collagen, which is why the softest stage is also the one that ruptures.

· Each stage depends on the one before, so the appearance of an infarct dates it and the date predicts the complication.

Common misconceptions

· Assuming a bigger or older infarct is more likely to rupture. Rupture belongs to the narrow window when the muscle is gone and the scar is not yet built.

· Dating an infarct by gross appearance alone. The inflammatory and repair cells on microscopy are the more reliable clock.

Where it is taught: Section C05, morphologic timeline table and the closing paragraph on wall strength of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-BD1

Brain dump 1 · foundational retrieval: Complications laid out on a time line

A competent response contains

· First minutes to hours: ventricular fibrillation, the main cause of death in the first hours; inferior infarcts also give sinus bradycardia and atrioventricular block.

· First hours to days: pump failure with pulmonary congestion, cardiogenic shock, and right ventricular infarction from proximal right coronary occlusion.

· About 1 to 3 days: fibrinous or fibrinohemorrhagic pericarditis over a transmural infarct, with pleuritic positional pain and a friction rub.

· About 3 to 7 days, from day 1 to about 2 weeks or later: rupture of the free wall, the septum, or a papillary muscle.

· Weeks: mural thrombus over a large anterior apical infarct, Dressler syndrome, and true aneurysm; months onward, heart failure and scar-related ventricular tachycardia.

· The three electrical changes in ischemic myocytes are a less negative resting potential, slow conduction, and altered refractoriness, so a wavefront can block in one direction and re-enter tissue it has already passed, which is the substrate for reentry.

· Added instability comes from abnormal automaticity, triggered activity, local hyperkalemia, and catecholamines.

· Loss of about 40% or more of the left ventricle causes cardiogenic shock.

· An inferior infarct slows the heart because the right coronary artery supplies the nodes and because vagal reflexes arise from the inferior and posterior wall; bundle branch block from an anterior septal infarct is more ominous because it reflects extensive necrosis below the atrioventricular node.

· Cardiogenic shock cycle: loss of about 40% or more of left ventricular muscle; stroke volume and cardiac output fall; aortic pressure falls while left ventricular diastolic pressure rises; coronary perfusion pressure falls; ischemia extends into surviving muscle; contractility falls further.

· Right ventricular infarction gives hypotension, distended neck veins, and clear lungs, because the right ventricle cannot deliver blood to the left heart.

Distinctions that matter

· Early complications are electrical and hemodynamic, middle ones are mechanical and follow tissue weakness, and late ones follow the scar.

· Inferior bradycardia is partly reflex and metabolic and usually recovers, while block from an anterior septal infarct reflects dead conducting tissue.

Common misconceptions

· Treating every post-infarction bradycardia as the same problem. Site decides level, mechanism, and outlook.

· Expecting rupture in the first day or two. The wall is still held together by necrotic muscle until macrophages clear it.

Where it is taught: Section C06, paragraphs on arrhythmia and pump failure, the cardiogenic shock cycle, and the mechanical complications table; the timeline in Section C05 of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-BD2

Brain dump 2 · integration and application: Four patients deteriorate after an infarction

A competent response contains

· Patient 1: papillary muscle rupture with acute severe mitral regurgitation, giving sudden pulmonary edema and hypotension with an apical murmur that may be soft because pressures equalize across the valve.

· Patient 2: septal rupture with a left-to-right shunt, giving a new holosystolic murmur at the left lower sternal border and an oxygen step-up in the right ventricle; anterior infarcts tear the apical septum and inferior infarcts the basal septum.

· Patient 3: free wall rupture with blood in the pericardial sac, giving tamponade with hypotension, distended neck veins, and muffled heart sounds, and pulseless electrical activity; it is classically a first infarct without hypertrophy or collaterals.

· Patient 4: fibrinous pericarditis over a transmural infarct at about 1 to 3 days, with pleuritic positional pain and a friction rub.

· All three ruptures occur mostly at about 3 to 7 days because macrophages have removed the dead muscle and collagen has not yet formed, so the wall is at its weakest.

· The posteromedial papillary muscle ruptures more often because it usually depends on the posterior descending artery alone, while the anterolateral muscle receives both left anterior descending and circumflex branches; rupture of a whole trunk is often rapidly fatal, and ischemia without rupture causes milder regurgitation.

· Pseudoaneurysm is a contained free wall rupture whose wall is pericardium and thrombus, with a narrow neck and a high risk of later rupture; a true aneurysm is a thin transmural scar over a large anterior apical

infarct, appearing weeks to months later, with a wide neck, dyskinesis in systole, and rare rupture, and it carries heart failure, thrombus, ventricular tachycardia, and persistent ST elevation.

· Dressler syndrome is pericarditis several weeks after infarction with fever, pleuritis, and effusion, caused by exposure of sequestered myocardial antigens with antibodies and immune complexes inflaming the pericardium; the delay fits an adaptive immune response, which separates it from the direct fibrinous pericarditis of days 1 to 3.

Distinctions that matter

· The site that tears decides the syndrome: free wall gives tamponade, septum gives a shunt, and papillary muscle gives acute mitral regurgitation.

· A loud murmur is not a measure of severity in acute mitral regurgitation; a soft murmur can accompany torrential regurgitation once left atrial and ventricular pressures equalize.

Common misconceptions

· Expecting a loud murmur in every papillary muscle rupture. Rapid pressure equalization can make it soft.

· Grouping pericarditis at day 2 with Dressler syndrome. One is direct inflammation over necrosis and the other is a delayed immune reaction weeks later.

Where it is taught: Section C06, paragraph on pericarditis, rupture, and thrombus, the mechanical complications table, and the Dressler paragraph of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-CA01

Clinical application: Sudden breathlessness on the fifth day after an inferior infarction

A competent response contains

· Recognize: rupture of a papillary muscle with acute severe mitral regurgitation; the structure that failed is the posteromedial papillary muscle.

· Explain: by day 5 macrophages have removed the necrotic myocytes and collagen has not yet been laid down, so the tissue is at its weakest between about days 3 and 10, which is the window in which rupture occurs.

· Differentiate: free wall rupture would give tamponade with distended neck veins, muffled heart sounds, and pulseless electrical activity, and the jugular venous pressure here is not raised; septal rupture would give a holosystolic murmur at the left lower sternal border with an oxygen step-up in the right ventricle rather than a soft apical murmur with pulmonary edema; reinfarction would show new ST elevation and a fresh rise of the MB isoenzyme of creatine kinase, and this tracing shows no new ST change.

· Investigate: look for severe mitral regurgitation with a flail or disrupted papillary muscle on imaging, pulmonary edema on the chest radiograph, and a raised left atrial or pulmonary venous pressure; the murmur can be soft because left atrial pressure rises steeply and the pressure difference driving flow across the valve falls, so little turbulence is generated.

· Decide: the posteromedial papillary muscle has failed, because it usually depends on the posterior descending artery alone while the anterolateral muscle receives branches from both the left anterior descending and the circumflex arteries.

· Justify: this was an inferior infarct with ST elevation in II, III, and aVF, and the posterior descending artery arises from the right coronary artery in about 80% of hearts, so the inferior wall and the posteromedial

papillary muscle infarct together; troponin rises again because fresh myocyte necrosis accompanies the rupture, while the absence of new ST elevation shows there is no new coronary occlusion.

Distinctions that matter

· Clear neck veins with flooded lungs point to a lesion that drives blood backward into the left atrium, while raised neck veins with clear lungs point to the right side or to the pericardium.

· Rising troponin without new ST change fits new tissue destruction from a mechanical complication rather than a new occlusion.

Common misconceptions

· Reading sudden pulmonary edema after an infarction as simple pump failure. A new murmur with abrupt onset on day 5 points to a mechanical complication.

· Using murmur loudness to grade the regurgitation. The murmur can be soft when pressures equalize quickly.

Where it is taught: Section C06, paragraph on pericarditis, rupture, and thrombus, the mechanical complications table, and the stop and predict box; Section C04 for the territory and the posterior descending artery of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C06-Q02

Board-style question: Timing of free wall rupture

Correct answer: A. Macrophages have removed the necrotic myocytes and collagen deposition has not yet strengthened the wall

Why it is right: Between about days 3 and 10 the necrotic myocytes have been phagocytosed and granulation tissue has not yet deposited enough collagen to take their place, so the infarcted wall is at its mechanically weakest and can tear under systolic pressure, which is why rupture clusters in that window.

B is wrong: Digestion of a coronary plaque cap by inflammatory enzymes is the mechanism that starts an infarction by causing thrombosis, and it says nothing about the strength of the ventricular wall 5 days later.

C is wrong: A thin fibrous scar that bulges during systole describes a true aneurysm, which forms over weeks to months and rarely ruptures; at 5 days no scar has yet matured.

D is wrong: Slow conduction through dense collagen in a healed infarct sets up reentrant ventricular tachycardia months later and does not weaken the wall or cause a tear.

A competent response contains

· Rupture occurs mostly at about 3 to 7 days, when macrophages have cleared the necrotic myocytes and collagen has not yet formed, so the wall has lost its old strength and gained none.

· A first infarct without hypertrophy or collaterals is the classic setting for free wall rupture, which fills the pericardial sac and causes tamponade with pulseless electrical activity.

Distinctions that matter

· The weakest moment of the wall is defined by the balance between clearance of dead muscle and deposition of collagen, not by the size of the infarct.

Common misconceptions

· Expecting a transmural infarct to be weakest on the first day. Necrotic muscle still holds the wall together until macrophages remove it.

Where it is taught: Section C05, morphologic timeline table and closing paragraph; Section C06, mechanical complications table of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C07-BD1

Brain dump 1 · foundational retrieval: The chronically ischemic heart and its remodeling cycle

A competent response contains

· Chronic ischemic heart disease, also called ischemic cardiomyopathy, is progressive heart failure from ischemic myocardial damage, usually the late result of one or more healed infarcts, sometimes silent, plus ongoing ischemia from diffuse atherosclerosis; less often it develops without a recognized infarct.

· It is one of the most common causes of heart failure with reduced ejection fraction.

· Gross findings: the heart is enlarged and heavy, with a dilated hypertrophied left ventricle, gray-white scars in coronary territories, moderate to severe coronary atherosclerosis, and sometimes patchy endocardial fibrosis and mural thrombi.

· Microscopic findings: hypertrophied surviving myocytes, subendocardial myocyte vacuolization as a sign of chronic sublethal ischemia, and interstitial and replacement fibrosis.

· Regional scars matching coronary territories separate it from other dilated cardiomyopathies, in which fibrosis is diffuse and the coronary arteries are usually normal.

· Remodeling cycle: infarction and ongoing ischemia reduce contracting muscle; stroke volume falls with sympathetic and renin-angiotensin-aldosterone activation; volume retention and eccentric hypertrophy dilate the left ventricle; a larger radius raises wall stress and oxygen demand; subendocardial ischemia, fibrosis, and myocyte loss worsen; contractile function declines further.

· Angiotensin II and aldosterone promote fibrosis.

· In eccentric hypertrophy myocytes add sarcomeres in series, so the chamber lengthens and enlarges and becomes more spherical.

· Scar cannot contract, so surviving muscle carries the whole load, and myocytes continue to die by apoptosis and ischemia.

Distinctions that matter

· Sarcomeres added in series lengthen the cell and dilate the chamber in eccentric hypertrophy, while sarcomeres added in parallel thicken the wall in concentric hypertrophy.

· Regional scar in arterial territories marks an ischemic cause, while diffuse fibrosis with normal arteries marks other dilated cardiomyopathies.

Common misconceptions

· Treating dilation as purely compensatory. It begins as compensation and then raises wall stress and oxygen demand, which perpetuates the injury.

· Expecting a history of a recognized infarct in every patient. Silent infarcts and diffuse ischemia can produce the same heart.

Where it is taught: Section C07, first two paragraphs, the remodeling paragraph, and the remodeling cycle of the CV-18-C Ischemic Heart Disease notes.

NPLEX-P1-CARD-18-03-C07-BD2

Brain dump 2 · integration and application: Work the cost of dilation and sort recoverable muscle from scar

A competent response contains

· Wall stress in a roughly spherical ventricle is proportional to pressure times radius divided by twice the wall thickness.

· Normal ventricle: 120 times 2.5 divided by 1.0 gives 300 in these units.

· Remodeled ventricle: 120 times 3.5 divided by 0.8 gives 525 in these units.

· The ratio is 525 divided by 300, about 1.75, so the dilated wall bears about 75% more stress at the same pressure and uses more oxygen for each beat.

· Hypertrophy partly offsets the rise in stress by adding thickness, but it also adds muscle that the narrowed arteries cannot supply.

· Hibernating myocardium is alive, chronically underperfused, and reduces contraction to match the reduced flow, and it can recover if flow is restored.

· Stunned myocardium is alive and already reperfused, and it stays dysfunctional for hours to days before recovering on its own.

· Scar is dead collagenous tissue receiving flow it cannot use, and restoring flow cannot help it; scar is permanent.

· Dilation pulls the papillary muscles apart and enlarges the mitral annulus, so the leaflets no longer meet, giving functional or secondary mitral regurgitation with a normal valve, and the regurgitant volume adds a volume load to an already failing ventricle.

· At scar borders, surviving myocyte strands interwoven with collagen conduct slowly, creating fixed reentry circuits for monomorphic ventricular tachycardia that can degenerate into fibrillation.

· Examination findings include a displaced apical impulse, a third heart sound from rapid filling of a dilated ventricle, and an apical murmur of functional mitral regurgitation; the electrocardiogram shows Q waves from old infarcts.

Distinctions that matter

· Hibernating and stunned myocardium differ by flow: hibernating muscle is still underperfused, while stunned muscle has already been reperfused and simply needs time.

· Functional mitral regurgitation is a geometry problem of the ventricle, while primary mitral regurgitation is a problem of the leaflets or chordae.

Common misconceptions

· Reading poor contraction as dead muscle. Hibernating and stunned myocardium are alive and can recover function.

· Expecting a thicker wall to solve the stress problem. Hypertrophy lowers stress per unit of muscle while adding oxygen demand the narrowed arteries cannot meet.

Where it is taught: Section C07, worked example on the law of Laplace and the paragraphs on recoverable myocardium, valves, and rhythm of the CV-18-C Ischemic Heart Disease notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-D Valvular Heart Diseases

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-04-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild calcific aortic stenosis and the wall it builds

A competent response contains

· The normal aortic valve has three cusps and an area of about 3 to 4 cm2; symptoms usually begin when the area falls to about 1 cm2 or less, so years pass silently.

· Calcific aortic stenosis is an active process: mechanical stress injures the cusp endothelium, low-density lipoprotein and lipoprotein(a) enter the cusp, inflammation follows, and valve interstitial cells become osteoblast-like and deposit calcium.

· Heaped calcified masses bulge into the sinuses of Valsalva and the commissures stay unfused.

· The calcification is dystrophic, so serum calcium is normal, and the risk factors match those of atherosclerosis.

· Age-related calcific disease on a once-normal valve usually presents after about age 70; a congenitally bicuspid valve, present in about 1 to 2% of people, concentrates stress and presents one to two decades earlier.

· Rheumatic aortic stenosis shows fused commissures and almost always accompanies rheumatic mitral disease.

· Law of Laplace: wall stress is proportional to pressure times radius divided by twice the wall thickness, so a rise in pressure is offset by a thicker wall.

· Pushing a normal stroke volume through a narrow orifice demands a much higher ventricular pressure, so myocytes add sarcomeres in parallel and the wall thickens around a normal or small cavity, which is concentric hypertrophy.

· Ejection fraction is preserved for years, but the stiff wall depends on atrial contraction for filling, which produces an S4, and demand eventually outgrows the capillary supply, giving fibrosis and failure.

Distinctions that matter

· Pressure load gives concentric hypertrophy with a normal cavity, while volume load gives eccentric hypertrophy with a dilated cavity.

· Unfused commissures with heaped nodular calcium point to calcific or bicuspid disease; fused commissures point to rheumatic disease.

Common misconceptions

· Calling calcific aortic stenosis simple mechanical wear. It is an inflammatory, lipid-driven, bone-forming process in the cusp.

· Expecting a raised serum calcium. Dystrophic calcification occurs in injured tissue with normal serum calcium.

Where it is taught: Section C01, first three paragraphs of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C01-BD2

Brain dump 2 · integration and application: Work a Doppler study back to the symptoms

A competent response contains

· Peak gradient is about 4 times velocity squared: 4 times 4.6 squared is about 85 mm Hg.

· With an arterial systolic pressure of 128 mm Hg the ventricle must generate about 213 mm Hg.

· By Laplace with an unchanged radius, thickness must rise in proportion to pressure: 213 divided by 128 is about 1.7, so a 1.0 cm wall must reach about 1.7 cm.

· This study is severe on all three thresholds: area about 1 cm2 or less (0.8 cm2), peak velocity about 4 m/s or more (4.6 m/s), and mean gradient about 40 mm Hg or more.

· Angina with clean coronary arteries: demand is high in a thick wall, a raised ventricular diastolic pressure compresses subendocardial vessels, and prolonged systole shortens diastolic perfusion time.

· Exertional syncope: exercise lowers peripheral resistance while output across a fixed orifice cannot rise, with a ventricular mechanoreceptor reflex and arrhythmias adding to it.

· Dyspnea: high ventricular diastolic pressure is transmitted back to the pulmonary veins.

· The carotid upstroke is weak, slow rising, and late peaking, which is pulsus parvus et tardus, because the orifice limits the rate of rise of aortic pressure; the apical impulse is sustained because a thick wall contracts long and hard.

· A failing ventricle generates less flow, so the murmur becomes softer even as the lesion worsens, which is why loudness does not track severity.

Distinctions that matter

· Jet velocity and gradient report the pressure difference, while valve area reports the orifice itself; a low-flow state can lower the gradient without the orifice improving.

· Angina here reflects supply and demand in a hypertrophied wall rather than fixed coronary obstruction.

Common misconceptions

· Reading a quiet murmur as mild disease. Reduced flow from a failing ventricle softens it.

· Expecting symptoms early. The valve area must fall to roughly a third of normal before the disease declares itself.

Where it is taught: Section C01, Doppler paragraph, findings table, and the worked example of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C01-Q01

Board-style question: Chest pain with clean coronary arteries

Correct answer: C. High oxygen demand in a thick wall whose raised diastolic pressure compresses subendocardial vessels while prolonged systole shortens diastolic perfusion time

Why it is right: Concentric hypertrophy raises oxygen demand, the raised left ventricular diastolic pressure compresses subendocardial vessels, and prolonged ejection shortens diastolic perfusion time, so the subendocardium becomes ischemic even though the epicardial arteries are normal.

A is wrong: Plaque rupture with thrombotic occlusion would produce an acute coronary syndrome, and angiography in this patient shows no epicardial stenosis.

B is wrong: Calcific embolism into a coronary artery is uncommon and would cause an abrupt infarct rather than months of predictable exertional tightness.

D is wrong: Loss of large von Willebrand factor multimers from high shear favours bleeding from intestinal angiodysplasia and does not reduce oxygen carrying capacity or cause angina.

A competent response contains

· The murmur, the weak late carotid upstroke, the soft A2, and the valve area of 0.8 cm2 with a mean gradient of 48 mm Hg establish severe aortic stenosis.

· Angina occurs with normal coronary arteries because demand is high in a 1.6 cm wall, raised diastolic pressure compresses subendocardial vessels, and prolonged systole shortens the time available for diastolic perfusion.

Distinctions that matter

· Ischemia in aortic stenosis is a supply and demand problem within the wall, not a problem of epicardial coronary obstruction.

Common misconceptions

· Assuming exertional chest pain always means coronary atherosclerosis. Severe aortic stenosis produces angina with clean arteries.

Where it is taught: Section C01 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C02-BD1

Brain dump 1 · foundational retrieval: Causes of aortic regurgitation and the wide pulse pressure

A competent response contains

· Root causes: hypertension and aging, Marfan syndrome, tertiary syphilis, ankylosing spondylitis, and ascending aortic dissection; root disease also carries a risk of aneurysm and dissection.

· Leaflet causes: bicuspid valve, endocarditis perforating a cusp, and rheumatic cusp retraction, usually with mitral disease.

· Marfan syndrome: mutant fibrillin-1 (FBN1) releases excess TGF-beta activity, producing cystic medial degeneration.

· Tertiary syphilis: endarteritis of the vasa vasorum scars the aortic media and gives the intima a tree-bark appearance.

· In chronic aortic regurgitation the ventricle receives atrial inflow plus the regurgitant volume, so end-diastolic volume rises and the Frank-Starling mechanism raises total stroke volume; ejecting that volume into a high-pressure aorta adds a pressure load to the volume load.

· Sarcomeres are added in series, the chamber dilates with proportional wall growth (eccentric hypertrophy), and chronic aortic regurgitation produces some of the largest ventricles seen.

· The dilated chamber is compliant, so diastolic pressure stays near normal and patients can be asymptomatic for years until dilation outruns hypertrophy and contractility falls.

· A representative pressure is 170/45 mm Hg: systolic rises with the large rapidly ejected stroke volume, and diastolic falls because the aorta empties both backward into the ventricle and forward into vasodilated tissues.

· Named signs, all from the same wide pulse pressure: water-hammer or Corrigan pulses, head bobbing (de Musset), nail bed capillary pulsation (Quincke), pistol-shot femoral sound (Traube), and a to-and-fro femoral bruit on compression (Duroziez); the apical impulse is hyperdynamic and displaced down and laterally.

· The murmur is high-pitched, blowing, early diastolic, and decrescendo, beginning at A2 where the aorta to ventricle gradient is greatest, heard best at the left sternal border with the patient sitting forward in held expiration, and louder with handgrip because handgrip raises aortic diastolic pressure.

· Low aortic diastolic pressure lowers coronary perfusion pressure and can cause angina.

Distinctions that matter

· Root disease pulls normal cusps apart, while leaflet disease damages the cusps themselves; the distinction decides whether the aorta itself is also at risk.

· Handgrip raises afterload, so it makes aortic regurgitation louder while making the aortic stenosis gradient and murmur smaller.

Common misconceptions

· Treating the named peripheral signs as separate phenomena. Each one is the same wide pulse pressure seen in a different tissue.

· Placing the murmur in mid diastole. It begins at A2, when the pressure difference is largest, and decreases as the two pressures converge.

Where it is taught: Section C02, cause paragraphs and the sections on chronic loading, wide pulse pressure, and the murmur of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C02-BD2

Brain dump 2 · integration and application: Two patients with a leaking aortic valve

A competent response contains

· Patient 1 has a large, dilated, compliant ventricle: diastolic pressure stays near normal, the murmur is long because the gradient persists through diastole, and the pulse pressure is wide with bounding pulses.

· Patient 2 has a normal-sized, stiff ventricle: diastolic pressure rises steeply, aortic and ventricular pressures equalise quickly so the murmur is short and soft, and the pulse pressure is not wide.

· Compliance of the receiving chamber, not regurgitant volume alone, decides whether the patient compensates.

· A severe acute leak can be almost silent because the pressures equalise early in diastole, leaving little gradient and little turbulence to hear.

· The steeply rising ventricular diastolic pressure can close the mitral valve early, and pulmonary edema and shock follow.

· Regurgitation occurs only in diastole, so bradycardia lengthens diastole and increases the regurgitant volume per beat, while tachycardia reduces it; a slower rate therefore worsens the load in both patients and is particularly poorly tolerated in the acute lesion.

· An Austin Flint murmur is an apical mid-to-late diastolic rumble caused by the regurgitant jet partly closing the anterior mitral leaflet; mitral stenosis adds an opening snap and a loud S1, which the Austin Flint murmur lacks.

Distinctions that matter

· Murmur length in regurgitation reports how long a pressure gradient lasts, not how much blood leaks.

· A wide pulse pressure is a feature of the chronic lesion only, because it requires a large total stroke volume that a stiff ventricle cannot deliver.

Common misconceptions

· Expecting a wide pulse pressure in acute aortic regurgitation. Stroke volume is not large and the pressures equalise, so the pulse pressure is normal or narrow.

· Judging severity from how loud or long the murmur is. The quietest murmurs in this chapter belong to the most unstable patients.

Where it is taught: Section C02, murmur paragraph and acute aortic regurgitation paragraph, with the chronic against acute table in section C04 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C03-BD1

Brain dump 1 · foundational retrieval: Mitral stenosis: the chambers that pay and the sounds you hear

A competent response contains

· Normal mitral valve area is about 4 to 6 cm2, so left atrial and ventricular pressures are nearly equal in diastole; a gradient appears below about 2.5 cm2 and severe disease is about 1.5 cm2 or less.

· Mitral stenosis is almost always rheumatic; heavy mitral annular calcification in older adults and congenital malformation are uncommon causes.

· The left atrium hypertrophies and dilates while the left ventricle is underfilled, normal or small, with normal or low diastolic pressure.

· The hallmark is a pulmonary capillary wedge pressure higher than left ventricular diastolic pressure.

· Hemoptysis comes from rupture of bronchial veins that connect with congested pulmonary veins.

· Loud S1: high left atrial pressure holds the leaflets wide open until systole begins, and S1 softens once they calcify.

· Opening snap after S2: fused leaflets dome open abruptly when ventricular pressure falls below atrial pressure.

· The interval from S2 to the opening snap shortens as the disease worsens, because higher left atrial pressure opens the valve sooner after A2.

· Low-pitched apical mid-diastolic rumble from turbulent flow through the orifice, heard with the bell in the left lateral decubitus position; its length rather than its loudness tracks severity.

· Presystolic accentuation comes from atrial contraction boosting flow and is lost in atrial fibrillation.

· Electrocardiogram: a broad notched P wave in lead II with a deep terminal negative P in V1 (P mitrale) from left atrial enlargement, and right axis deviation from pulmonary hypertension with right ventricular hypertrophy.

· The enlarged left atrium, the most posterior chamber, can indent the esophagus causing dysphagia, compress the left recurrent laryngeal nerve causing hoarseness (Ortner syndrome), and elevate the left main bronchus, widening the carinal angle.

Distinctions that matter

· The opening snap is diastolic and belongs to mitral stenosis; the mid-systolic click belongs to mitral valve prolapse.

· Murmur length tracks severity in mitral stenosis, while loudness does not.

Common misconceptions

· Expecting a dilated left ventricle. The obstruction sits upstream, so the ventricle is underfilled and small.

· Reading a shorter S2 to opening snap interval as milder disease. A shorter interval means a higher left atrial pressure and worse stenosis.

Where it is taught: Section C03, hemodynamics paragraph and the findings table of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C03-BD2

Brain dump 2 · integration and application: Why a faster heart rate floods the lungs

A competent response contains

· Flow across an orifice varies with its area times the square root of the pressure difference, so the gradient rises with the square of the flow rate.

· The ventricle fills only in diastole and tachycardia shortens diastole more than systole, so the same volume must cross the valve faster and left atrial pressure rises sharply; the notes give a rise from 70 to 120 per minute as roughly doubling the gradient.

· Her precipitants are fever, tachycardia, and pregnancy; exercise, anemia, and atrial fibrillation do the same in other patients.

· Pregnancy adds blood volume, cardiac output, and heart rate, each of which raises flow across the fixed orifice.

· Her left ventricular end-diastolic pressure and ventricular size are normal or low, because the obstruction lies upstream of the ventricle and the chamber is underfilled rather than overloaded.

· Atrial fibrillation removes the atrial contribution to filling and shortens diastole, so filling worsens twice over.

· Stasis in the left atrial appendage forms mural thrombi that embolise to the brain, kidneys, spleen, and limbs.

· In mitral stenosis the wedge pressure exceeds the left ventricular diastolic pressure, whereas in pulmonary edema from left ventricular failure the raised wedge pressure reflects an equally raised ventricular diastolic pressure.

Distinctions that matter

· Pulmonary edema upstream of a stenotic mitral valve coexists with a normal ventricular filling pressure; pulmonary edema from ventricular failure does not.

· Heart rate matters here through diastolic filling time, which is why rate control changes the gradient without changing the valve.

Common misconceptions

· Assuming any patient with pulmonary edema has a failing left ventricle. In mitral stenosis the ventricle is protected.

· Treating atrial fibrillation as a rhythm problem alone. It also removes atrial kick and shortens diastole, which is why it so often precipitates the first episode of edema.

Where it is taught: Section C03, paragraphs on heart rate and on atrial fibrillation, and the stop and predict box of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C03-RT01

Retrieval exercise: Rebuild the backward transmission of pressure

A competent response contains

· Step 1: fused, narrowed mitral orifice.

· Step 2: left atrial pressure rises to drive flow into the ventricle.

· Step 3: the left atrium hypertrophies and dilates while the left ventricle stays small.

· Step 4: pulmonary venous pressure rises, giving dyspnea, edema, and hemoptysis.

· Step 5: reactive arteriolar constriction and then remodelling of the arteriolar walls produce pulmonary hypertension.

· Step 6: right ventricular hypertrophy and dilation follow, and the dilated right ventricle stretches the tricuspid annulus so tricuspid regurgitation appears.

· Step 7: right heart failure, with distended neck veins, hepatomegaly, and peripheral edema.

· Pressures are raised in the pulmonary veins, pulmonary capillaries, pulmonary arteries, right ventricle, right atrium, and left atrium, and are normal or low in the left ventricle.

· The left ventricle is spared because it sits downstream of the obstruction and simply receives less blood.

Distinctions that matter

· Pulmonary hypertension here begins as reversible arteriolar constriction and becomes fixed once the walls remodel.

· Tricuspid regurgitation in this sequence is functional, caused by right ventricular dilation rather than by disease of the tricuspid leaflets.

Common misconceptions

· Drawing the left ventricle as loaded. Every chamber upstream of the narrowed valve carries the load and the ventricle does not.

Where it is taught: Section C03, backward transmission flow diagram of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-BD1

Brain dump 1 · foundational retrieval: The mitral apparatus and how each part fails

A competent response contains

· The apparatus is the annulus, leaflets, chordae, papillary muscles, and the ventricular wall that holds them.

· Primary mitral regurgitation: mitral valve prolapse, which is the most common primary cause in developed countries, rheumatic disease, endocarditis, and papillary muscle or chordal rupture.

· Secondary or functional mitral regurgitation: a dilated ventricle stretches the annulus and displaces the papillary muscles, tethering normal leaflets; ischemic papillary muscle dysfunction causes transient regurgitation.

· The posteromedial papillary muscle ruptures most often after infarction because it usually has a single arterial supply.

· Functional regurgitation feeds itself: the added regurgitant volume dilates the ventricle further, which stretches the annulus and tethers the leaflets more.

· Left atrial pressure is low, so backflow begins at the onset of systole before the aortic valve opens, and there is no true isovolumetric contraction.

· The regurgitant volume returns to the ventricle in early diastole, so both the atrium and the ventricle are volume loaded.

· The left atrium dilates and becomes compliant, holding a large volume at modest pressure, and often fibrillates; the ventricle develops eccentric hypertrophy.

· The murmur is holosystolic, high-pitched, and blowing, running from S1 through A2, loudest at the apex and radiating to the left axilla.

· Handgrip and squatting make it louder, Valsalva and standing soften it, and a ventricular septal defect murmur is heard at the left lower sternal border instead.

· S1 is soft, early A2 widens the split of S2, and rapid early diastolic inflow from the loaded atrium produces an S3 that reflects volume rather than proving failure.

Distinctions that matter

· Primary disease is of the leaflets or chordae; secondary disease is of the ventricle, with normal leaflets tethered out of position.

· An S3 in chronic mitral regurgitation reports a large early diastolic inflow, so it is not by itself evidence of heart failure.

Common misconceptions

· Expecting an isovolumetric phase in mitral regurgitation. Blood escapes into the atrium as soon as ventricular pressure exceeds atrial pressure.

· Attributing all mitral regurgitation to leaflet disease. A dilated ventricle alone can make normal leaflets incompetent.

Where it is taught: Section C04, cause paragraph, loading paragraph, and murmur paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-BD2

Brain dump 2 · integration and application: Why a normal ejection fraction misleads

A competent response contains

· Total stroke volume is 210 minus 75, which is 135 mL.

· Ejection fraction is 135 divided by 210, which is about 64%, a value that looks normal.

· Regurgitant volume is 60% of 135, which is 81 mL, so forward stroke volume is 135 minus 81, which is 54 mL.

· With end-systolic volume at 105 mL, total stroke volume is 105 mL, ejection fraction is 105 divided by 210, which is 50%, regurgitant volume is 63 mL, and forward stroke volume is 42 mL.

· The ventricle unloads into a low-pressure atrium, so ejection fraction overstates contractility and a low-normal value already signals impairment; the chapter notes about 60% or less as the guideline threshold suggesting dysfunction in chronic mitral regurgitation.

· In aortic stenosis the ventricle ejects against a high pressure, so a preserved ejection fraction is a more honest report of contractility and a fall in it indicates late disease.

· Forward stroke volume is the measurement that reports how much blood actually reaches the tissues.

Distinctions that matter

· Total stroke volume and forward stroke volume are different quantities, and only the second one perfuses the body.

· Ejection fraction depends on the load the ventricle ejects against, which is why the same number means different things in mitral regurgitation and in aortic stenosis.

Common misconceptions

· Reading an ejection fraction of 50% in chronic mitral regurgitation as mildly reduced. In this lesion it reports substantial loss of contractility.

· Assuming a large total stroke volume means good cardiac output. Most of it may be moving backward.

Where it is taught: Section C04, worked example on ejection fraction, with the threshold noted in the verify list of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-CP01

Comparison table: Chronic against acute regurgitation

A competent response contains

· Receiving chamber: chronic disease dilates it and makes it compliant, the ventricle in aortic regurgitation and both the atrium and the ventricle in mitral regurgitation; in acute disease it is normal in size and stiff.

· Filling pressure: chronic disease keeps it near normal or moderately raised for years; acute disease raises it steeply, as left ventricular diastolic pressure in aortic regurgitation and as left atrial pressure with tall v waves in mitral regurgitation.

· Murmur: chronic disease gives a long murmur, early diastolic and decrescendo in aortic regurgitation and holosystolic in mitral regurgitation; acute disease gives a short, soft murmur that is sometimes nearly absent.

· Pulse pressure in aortic regurgitation: wide with bounding pulses when chronic, not wide when acute.

· Presentation: chronic disease is silent for years and then brings exertional dyspnea and failure; acute disease brings sudden pulmonary edema and shock.

· Typical causes: chronic aortic regurgitation from root dilation, a bicuspid valve, or rheumatic disease, and chronic mitral regurgitation from prolapse, rheumatic disease, or ventricular dilation; acute disease from endocarditis, aortic dissection, or trauma, and in the mitral valve from papillary muscle or chordal rupture.

· Compliance of the receiving chamber, not regurgitant volume alone, decides whether the patient compensates.

Distinctions that matter

· Acute and chronic lesions can leak the same volume and look completely different, because a compliant chamber absorbs the volume at low pressure.

· A short soft murmur in regurgitation suggests early equalisation of pressures, which marks the acute and more dangerous lesion.

Common misconceptions

· Grading severity by murmur intensity in regurgitation. Loudness tracks the gradient and the duration of flow, not the volume that leaks.

· Expecting bounding pulses in every patient with aortic regurgitation. They require the large stroke volume of the chronic lesion.

Where it is taught: Section C04, chronic against acute regurgitation table, with the acute paragraphs in C02 and C04 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C04-CA01

Clinical application: Sudden breathlessness four days after an infarction

A competent response contains

· Recognize: acute severe mitral regurgitation from rupture of the posteromedial papillary muscle, part of the mitral apparatus, leaving a flail posterior leaflet.

· Explain: the left atrium is normal in size and stiff, so the regurgitant volume raises atrial pressure steeply, producing tall v waves and a wedge pressure of 34 mm Hg that is transmitted straight to the pulmonary capillaries, giving edema within minutes; a chronically dilated compliant atrium would have absorbed the same volume at modest pressure.

· Differentiate: chronic mitral regurgitation has a dilated atrium and ventricle, a long loud holosystolic murmur radiating to the axilla, an S3, and years of exertional symptoms; acute aortic regurgitation gives a short early diastolic murmur with a steeply raised ventricular diastolic pressure and no tall atrial v waves, and the cardiac silhouette here is normal because there has been no time to dilate.

· Investigate: on echocardiography look for a flail posterior leaflet with a ruptured chordal or papillary attachment, a normal-sized left atrium and left ventricle, inferior wall hypokinesis, and a severe regurgitant jet; on the catheter tracing look for tall systolic v waves on the wedge trace with a high mean wedge pressure.

· Decide: the murmur is soft and short because left atrial pressure rises so high during systole that it approaches ventricular pressure late in systole, leaving little gradient and little turbulence, so nothing about severity can be inferred from its loudness.

· Justify: the posteromedial papillary muscle usually has a single arterial supply, which makes it vulnerable in an inferior infarct supplied by the artery that was occluded, and the late treatment allowed full necrosis; a small stiff atrium cannot accommodate the regurgitant volume, so pressures equalise late in systole and pulmonary pressures rise instead.

Distinctions that matter

· The normal cardiac silhouette and normal chamber sizes with severe regurgitation date the lesion as acute.

· Tall v waves on the wedge tracing localise the problem to a stiff left atrium receiving systolic backflow.

Common misconceptions

· Taking a soft murmur as evidence of a mild leak in a shocked patient. In acute mitral regurgitation the most severe lesions can be nearly silent.

· Expecting the anterolateral papillary muscle to rupture. The posteromedial muscle usually has one arterial supply and is the one that fails.

Where it is taught: Section C04, cause paragraph, acute mitral regurgitation paragraph, and the chronic against acute table of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C05-BD1

Brain dump 1 · foundational retrieval: How a vegetation forms and what the organism decides

A competent response contains

· A vegetation is a mass of platelets and fibrin with organisms and inflammatory cells attached to a valve or the endocardium; organisms grow inside an avascular fibrin mesh that neutrophils reach poorly, so the vegetation sheds them continuously.

· Pathogenesis in order: turbulence or prior damage injures valve endothelium; platelets and fibrin form a sterile nidus; transient bacteremia delivers organisms; adhesins attach them to the deposit; organisms multiply under fibrin, shielded from phagocytes; the vegetation grows, destroys tissue, and sheds emboli and organisms.

· Virulent Staphylococcus aureus can invade intact endothelium and skip the first two steps, which is why it infects normal valves.

· Collagen and tissue factor are exposed by turbulence, a scarred or congenitally abnormal valve, a prosthesis, or a catheter; transient bacteremia follows chewing, dental work, procedures, injection drug use, or skin infection.

· Viridans streptococci attach through adhesins and dextran; S. aureus binds fibrinogen and fibronectin through surface proteins such as clumping factor.

· Acute disease: S. aureus, usually a normal valve, days, high fever with rigors, large destructive vegetations with ring abscesses and metastatic infection, and embolic signs prominent; portals are skin, injection drug use, and catheters.

· Subacute disease: viridans streptococci of low virulence, a previously damaged valve (rheumatic, congenital, bicuspid, or prolapse with regurgitation), weeks to months, low-grade fever with fatigue, weight loss, anemia, splenomegaly, and clubbing when long-standing, with immunologic signs dominating because prolonged bacteremia allows immune complexes to form.

· Enterococci after genitourinary or gastrointestinal procedures and Streptococcus gallolyticus with colon cancer usually follow a subacute course.

· Left-sided valves face higher pressures and turbulent jets, so the mitral valve is most often involved, closely followed by the aortic.

· In people who inject drugs the tricuspid valve is most often involved because skin organisms and injected particles reach it first, with S. aureus dominating and Pseudomonas and Candida less often.

· Right-sided vegetations embolise to the lungs as multiple, often cavitating septic emboli; left-sided fragments cause infarcts and abscesses in the brain, spleen, kidneys, and limbs and seed arterial walls as mycotic aneurysms.

Distinctions that matter

· Virulence sets the tempo: a destructive organism needs no abnormal valve, while a low-virulence organism needs one.

· Side of the heart predicts where the emboli land, which is why pulmonary lesions point to a right-sided vegetation.

Common misconceptions

· Assuming endocarditis always requires a damaged valve. S. aureus invades intact endothelium.

· Expecting immunologic signs in acute disease. Immune complexes need weeks of bacteremia to form, so they belong to subacute disease.

Where it is taught: Section C05, pathogenesis flow diagram and the paragraphs on virulence and on valve involvement of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C05-BD2

Brain dump 2 · integration and application: Read the peripheral signs and the laboratory panel

A competent response contains

· Painful red-purple nodules on the finger pads are Osler nodes, caused by immune complex small vessel vasculitis and more typical of subacute disease.

· Painless flat red or hemorrhagic macules on the palms are Janeway lesions, caused by septic microemboli with small dermal abscesses and more typical of acute disease.

· Retinal hemorrhages with a pale centre are Roth spots, caused by a fibrin-platelet plug from immune vasculitis or microemboli, and painless.

· The sorting rule is that embolic lesions are painless while immune complex lesions hurt, because the vessel wall itself is inflamed.

· Red cell casts with low complement are immune complex glomerulonephritis from deposition in the glomeruli, and her 7-week illness is exactly the prolonged bacteremia that allows immune complexes to form.

· Her anemia is anemia of chronic inflammation, driven by hepcidin, alongside a raised erythrocyte sedimentation rate and C-reactive protein; rheumatoid factor may be positive in subacute disease.

· Continuous shedding of organisms makes the bacteremia persistent, so cultures drawn at different times grow the same organism, which tells you the vegetation itself is seeding the blood.

· A subacute course on a previously abnormal aortic valve points to viridans streptococci.

· Duke criteria: positive blood cultures and echocardiographic evidence of a vegetation, abscess, or new regurgitation are major criteria; a predisposing valve condition, fever of 38.0 °C (100.4 °F) or higher, and embolic or immunologic phenomena are minor criteria, so her cultures are major and her valve, fever, Osler nodes, Janeway lesions, Roth spots, glomerulonephritis, and rheumatoid factor fall among the minor criteria.

Distinctions that matter

· Osler nodes and Janeway lesions sit on the same hands and separate by pain, which follows from vasculitis against embolism.

· Persistent bacteremia with the same organism in separated cultures is what distinguishes endocarditis from a transient bacteremia.

Common misconceptions

· Treating the peripheral signs as specific. Splinter hemorrhages and conjunctival petechiae in particular are nonspecific.

· Explaining the anemia as blood loss or hemolysis. It is the hepcidin-driven anemia of chronic inflammation.

Where it is taught: Section C05, peripheral signs table and the final paragraph on cultures and criteria of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C05-Q02

Board-style question: A painful nodule on the finger pad

Correct answer: A. Immune complex deposition with small vessel vasculitis in the digital pad

Why it is right: Weeks of low-grade bacteremia with a low-virulence organism allow immune complexes to form and deposit in small vessels, and vasculitis of the digital vessels inflames the vessel wall, which is why these nodules are raised and tender.

B is wrong: Septic microemboli with small dermal abscesses produce Janeway lesions, which are flat and painless and sit on the palms and soles.

C is wrong: Fibrin and platelet plugs in retinal arterioles produce Roth spots, which are retinal hemorrhages with pale centres and cause no pain in the fingers.

D is wrong: A mycotic aneurysm is a weakened arterial wall seeded by organisms, typically in a larger artery, and it does not present as multiple tender nodules on finger pads.

A competent response contains

· A 6-week illness on a rheumatic valve with viridans streptococci is subacute endocarditis, in which prolonged bacteremia allows immune complexes to form.

· Tender nodules on the finger pads are Osler nodes from immune complex small vessel vasculitis, and the same mechanism explains her glomerulonephritis, low complement, and positive rheumatoid factor.

Distinctions that matter

· Painful lesions in endocarditis are immune complex lesions; painless lesions are embolic.

Common misconceptions

· Using Osler nodes and Janeway lesions interchangeably. One is painful and immune, the other painless and embolic.

Where it is taught: Section C05 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C06-BD1

Brain dump 1 · foundational retrieval: The two sterile vegetations

A competent response contains

· Nonbacterial thrombotic endocarditis, also called marantic endocarditis, consists of small fibrin-platelet thrombi about 1 to 5 mm across, bland and loosely attached, along the lines of closure of otherwise normal mitral and aortic leaflets, with no organisms and little inflammation.

· It arises when hypercoagulability meets the normal wear at the line of closure.

· The key association is advanced malignancy, especially mucin-producing adenocarcinoma of the pancreas or lung; tumour tissue factor and mucins, which engage selectins on platelets and leukocytes, also cause migratory thrombophlebitis (Trousseau syndrome).

· Other settings are disseminated intravascular coagulation, sepsis, antiphospholipid syndrome, and indwelling catheters.

· There is usually no murmur, but fragments cause infarcts in the brain, kidneys, and spleen, and multiple strokes in different arterial territories in a patient with cancer are the classic presentation.

· A sterile lesion can also become the nidus for infective endocarditis.

· Libman-Sacks endocarditis occurs in systemic lupus erythematosus, often with antiphospholipid antibodies.

· Its warty vegetations are about 1 to 4 mm and can sit on either or both leaflet surfaces, including the ventricular undersurface of the mitral valve, and on the chordae and mural endocardium.

· They show fibrinoid necrosis with inflammation from immune complex injury, and healing fibrosis deforms the leaflets, most often causing mitral regurgitation.

· Emboli and the thrombotic tendency of antiphospholipid syndrome add further risk.

· Vegetations on both leaflet surfaces, rather than on the closure line alone, point to Libman-Sacks.

Distinctions that matter

· Both lesions are sterile, but only Libman-Sacks is inflammatory and only Libman-Sacks scars the leaflet enough to leave regurgitation.

· Position separates them: closure lines for nonbacterial thrombotic endocarditis, either or both surfaces for Libman-Sacks.

Common misconceptions

· Expecting a murmur. Nonbacterial thrombotic endocarditis usually produces none, because the leaflet is not deformed and nothing obstructs or leaks.

· Assuming a sterile vegetation stays sterile. It makes an excellent nidus for organisms.

Where it is taught: Section C06, paragraphs on nonbacterial thrombotic and Libman-Sacks endocarditis and the vegetation comparison table of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C06-BD2

Brain dump 2 · integration and application: Strokes in three territories with negative cultures

A competent response contains

· This is nonbacterial thrombotic endocarditis, pointed to by a mucin-producing pancreatic adenocarcinoma and by small bland nodules on the closure lines of normal leaflets with no fever and no murmur.

· Mechanism: tumour tissue factor and mucins engaging selectins on platelets and leukocytes create a hypercoagulable state, and fibrin-platelet thrombi form where the leaflets meet and wear.

· Infarcts appear in unrelated territories because fragments are loosely attached and shower repeatedly into the systemic circulation from a left-sided valve.

· There is no murmur because the vegetations are small and do not obstruct the orifice or prevent leaflet closure.

· Negative cultures do not prove sterility: fastidious organisms and prior antibiotics cause culture-negative infective endocarditis.

· Findings favouring infection: persistent fever, valve destruction, new regurgitation, and immunologic signs such as glomerulonephritis or Osler nodes.

· Findings keeping you with a sterile lesion: known cancer, lupus, or a coagulopathy, with bland infarcts and no destruction.

· If organisms seeded the nodules they would enlarge and become bulky and friable, destroy the leaflet with perforation or abscess, produce new regurgitation with a murmur, and shed septic rather than bland emboli.

Distinctions that matter

· Embolic burden can be identical in sterile and infected vegetations; destruction of the valve is what separates them.

· Multiple strokes in different arterial territories point to a cardiac or proximal aortic source showering repeatedly rather than to disease of one artery.

Common misconceptions

· Taking negative blood cultures as proof that a vegetation is sterile.

· Expecting a hypercoagulable patient with valve nodules to be febrile. Absence of fever is typical of the sterile lesion.

Where it is taught: Section C06, nonbacterial thrombotic endocarditis paragraph and the closing paragraph on culture-negative disease of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C06-CP02

Comparison table: Four kinds of valve vegetation

A competent response contains

· Size and consistency: infective vegetations are large, bulky, and friable; nonbacterial thrombotic about 1 to 5 mm; Libman-Sacks about 1 to 4 mm and warty; acute rheumatic fever about 1 to 2 mm warty nodules.

· Position: infective lesions sit on closure lines and can spread to chordae and the wall; nonbacterial thrombotic on the lines of closure; Libman-Sacks on either or both surfaces, chordae, and mural endocardium; rheumatic on the lines of closure.

· Organisms: present in infective disease, absent in the other three.

· Inflammation and damage: marked in infective disease with perforation and abscess; minimal with no damage in nonbacterial thrombotic disease; fibrinoid necrosis with scarring and mitral regurgitation in Libman-Sacks; part of a pancarditis in acute rheumatic fever, with scarring after recurrences.

· Emboli: septic in infective disease, bland and often causing multiple infarcts in nonbacterial thrombotic disease, bland in Libman-Sacks, and uncommon in acute rheumatic fever.

· Setting: bacteremia; cancer, disseminated intravascular coagulation, and other hypercoagulable states; lupus with antiphospholipid antibodies; and 2 to 4 weeks after group A streptococcal pharyngitis.

· Size and destruction separate infected from sterile vegetations, involvement of both leaflet surfaces points to Libman-Sacks, and the clinical setting separates nonbacterial thrombotic from rheumatic lesions.

Distinctions that matter

· Three of the four are sterile, so the discriminating features are size, destruction, position, and the clinical setting rather than the presence of a vegetation.

· Rheumatic and nonbacterial thrombotic vegetations both sit on closure lines and are small, which is why the history decides between them.

Common misconceptions

· Treating any vegetation as evidence of infection.

· Assuming a sterile vegetation leaves the valve normal. Libman-Sacks and rheumatic lesions both scar.

Where it is taught: Section C06, comparison of valve vegetations table, with the rheumatic lesion in C08 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C07-BD1

Brain dump 1 · foundational retrieval: Myxomatous degeneration and the mid-systolic click

A competent response contains

· Mitral valve prolapse is systolic billowing of one or both mitral leaflets above the annulus into the left atrium, affecting about 2 to 3% of adults and usually asymptomatic.

· The lesion is myxomatous degeneration: the spongiosa expands with proteoglycans and glycosaminoglycans while the collagen of the fibrosa becomes thin and disorganised, driven by valve interstitial cells overproducing matrix-degrading enzymes.

· The leaflets become thick, redundant, and hooded between chordal attachments, the chordae elongate, thin, and may rupture, the posterior leaflet is most often involved, and the annulus may dilate.

· Marfan syndrome: deficient fibrillin-1 microfibrils fail to hold TGF-beta inactive, and excess signalling disrupts the valve matrix. Ehlers-Danlos syndromes: defective collagen weakens the fibrosa. Autosomal dominant polycystic kidney disease: abnormal polycystins, with berry aneurysms as an associated finding.

· Click mechanism: as the ventricle empties, the papillary muscles and annulus move closer together, the redundant leaflets lose support and billow into the atrium, and the slack chordae suddenly tighten, producing a sharp mid-systolic click at the apex.

· If coaptation then fails, a late systolic murmur of mitral regurgitation follows the click to S2.

· Complications follow from weak tissue: progressive mitral regurgitation as leaflets stretch, acute severe regurgitation when a thinned chorda ruptures and leaves a flail segment, infective endocarditis on prolapsing leaflets especially with regurgitation, platelet-fibrin thrombi on the atrial surface that embolise and cause transient ischemic attacks or stroke, and atrial and ventricular arrhythmias linked to papillary muscle traction and adjacent fibrosis; sudden cardiac death is rare.

Distinctions that matter

· Myxomatous degeneration adds proteoglycan and weakens collagen, which is the opposite of the fibrosis and fusion of rheumatic disease.

· The click comes from chordae snapping taut, while the murmur comes from leaflets failing to meet; a patient can have the click without the murmur.

Common misconceptions

· Expecting a thin, shrunken valve. The leaflets are thick, redundant, and hooded.

· Placing the click in early systole. It occurs in mid systole, once the ventricle has emptied enough for the leaflets to lose support.

Where it is taught: Section C07, first two paragraphs, the click paragraph, and the complications paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C07-BD2

Brain dump 2 · integration and application: Tell four systolic findings apart at the bedside

A competent response contains

· A smaller ventricle reaches the critical size sooner, so the click moves toward S1 and the murmur lengthens; a larger ventricle delays the click toward S2 and shortens the murmur.

· Smaller ventricle: standing and the strain phase of Valsalva. Larger ventricle: squatting, passive leg raise, and sustained handgrip.

· Mitral valve prolapse: standing and Valsalva move the click earlier and lengthen the murmur; squatting and handgrip delay the click and shorten the murmur, because the effect depends on chamber size.

· Hypertrophic cardiomyopathy: standing and Valsalva make the murmur louder and squatting and handgrip make it softer, because a smaller cavity worsens outflow obstruction.

· Aortic stenosis: standing and Valsalva make the murmur softer because less preload means less flow, and handgrip also softens it by raising afterload and lowering the gradient.

· Chronic mitral regurgitation: handgrip and squatting make it louder by raising afterload, and standing and Valsalva soften it by reducing preload.

· Prolapse and hypertrophic cardiomyopathy both worsen with a smaller ventricle; the mid-systolic click with a late systolic murmur at the apex separates prolapse from the crescendo-decrescendo murmur of outflow obstruction.

· The click is systolic and moves with ventricular size, while the opening snap of mitral stenosis is diastolic and moves with left atrial pressure.

· The patient with prolapse is the one at risk of sudden severe regurgitation, which requires a thinned chorda to rupture and leave a flail leaflet segment.

Distinctions that matter

· Manoeuvres separate lesions by whether the finding depends on flow across the valve or on the size of the cavity.

· Handgrip raises afterload, which increases regurgitant murmurs and the aortic regurgitation murmur while reducing the aortic stenosis gradient.

Common misconceptions

· Expecting everything to get louder with standing. Standing reduces preload, so flow murmurs soften while cavity-dependent findings intensify.

· Confusing the click with the opening snap. One is systolic and depends on ventricular size, the other diastolic and depends on atrial pressure.

Where it is taught: Section C07, manoeuvre paragraph, with the aortic stenosis findings table in C01 and the mitral regurgitation murmur paragraph in C04 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C07-PR01

Prediction exercise: Predict what happens when a patient stands up

A competent response contains

· Standing reduces venous return, so left ventricular end-diastolic volume falls.

· The ventricle is smaller throughout systole, so it reaches the critical size at which the redundant leaflets lose support earlier in systole.

· The click moves earlier, closer to S1, because the chordae snap taut sooner.

· The late systolic murmur lengthens, because coaptation fails for a greater part of systole.

· The murmur of hypertrophic cardiomyopathy becomes louder, because a smaller cavity worsens dynamic outflow obstruction.

· The murmur of aortic stenosis becomes softer, because reduced preload means less flow across a fixed orifice.

Distinctions that matter

· Prolapse and hypertrophic obstruction both intensify with a smaller ventricle, while a fixed obstruction such as aortic stenosis depends on flow and therefore softens.

· The click changes its timing while the murmur changes its length, and both follow from the same change in chamber size.

Common misconceptions

· Expecting the click and the murmur to move in the same direction on the timeline. An earlier click leaves more of systole for the murmur, so the murmur gets longer.

· Assuming any murmur that softens on standing is aortic stenosis. Mitral regurgitation softens too, and the timing separates them.

Where it is taught: Section C07, manoeuvre paragraph and the stop and predict box of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C08-BD1

Brain dump 1 · foundational retrieval: The chronic rheumatic valve and which valves it takes

A competent response contains

· Acute rheumatic fever follows group A streptococcal pharyngitis by about 2 to 4 weeks.

· Antibodies and T cells against streptococcal antigens, particularly M protein, cross-react with cardiac myosin and valve tissue, which is molecular mimicry, causing a pancarditis with Aschoff bodies and tiny sterile vegetations along the lines of closure.

· A single attack may heal with little damage, but each new streptococcal throat infection can reinflame already scarred valves.

· Healing replaces inflamed tissue with fibrous tissue and brings new vessels into normally avascular leaflets, so damage accumulates and becomes evident years to decades after the first attack.

· The four changes of the chronic lesion: thickened, fibrotic, later calcified leaflets; fused commissures; thickened, shortened, fused chordae; and leaflet neovascularisation.

· Seen from the atrium the stenotic mitral valve has a rigid fish-mouth or buttonhole orifice, and Aschoff bodies are rarely found because scar has replaced them.

· Commissural fusion causes the stenosis, while retracted leaflets and short fused chordae prevent closure, so combined stenosis and regurgitation is common; degenerative disease by contrast usually causes isolated aortic stenosis or isolated mitral regurgitation.

· The mitral valve is involved in nearly all cases, alone in about two-thirds and with the aortic valve in about one-quarter; tricuspid involvement almost always accompanies left-sided disease and the pulmonic valve is rarely affected.

· The usual explanation for the left-sided preference is that left-sided valves close against systemic pressure and scar more after each episode.

· Rheumatic aortic disease almost never occurs without mitral disease, so isolated aortic disease in an older adult suggests a calcific or bicuspid valve.

· Rheumatic disease remains the leading cause of mitral stenosis worldwide, especially where streptococcal pharyngitis goes untreated.

Distinctions that matter

· Commissural fusion is the signature of rheumatic disease, while heaped nodular calcium with unfused commissures is the signature of degenerative disease.

· Acute rheumatic carditis is active inflammation; chronic rheumatic heart disease is the scar left behind.

Common misconceptions

· Describing the chronic valve as infected. The organism is long gone and the damage is immune scarring.

· Expecting Aschoff bodies in the chronic lesion. They are rarely found once fibrosis has replaced them.

Where it is taught: Section C08, all four paragraphs of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C08-BD2

Brain dump 2 · integration and application: Two narrowed valves with different histories

A competent response contains

· Patient 1 has chronic rheumatic mitral stenosis with aortic involvement, confirmed on the valve by fused commissures and thickened, shortened, fused chordae, with leaflet fibrosis and neovascularisation.

· Patient 2 has age-related calcific aortic stenosis, confirmed by heaped calcified masses bulging into the sinuses of Valsalva with unfused commissures.

· Rheumatic aortic disease almost never occurs without mitral disease, so a normal mitral valve effectively excludes it in patient 2.

· Commissures are fused in patient 1 and unfused in patient 2.

· Patient 1 can have both narrowing and leakage of the same valve because commissural fusion narrows the orifice while retracted leaflets and short fused chordae prevent the leaflets from meeting.

· She is at particular risk of subacute infective endocarditis caused by viridans streptococci, because her scarred valve exposes collagen and tissue factor and supports a sterile platelet-fibrin deposit that a low-virulence organism can colonise.

· Pregnancy brings more blood volume, higher cardiac output, and a faster heart rate; each raises flow across a fixed orifice, and because the gradient rises with the square of the flow rate and tachycardia shortens diastole, left atrial pressure climbs steeply and pulmonary edema follows.

· Classic decompensation in pregnancy is a recognised consequence of rheumatic mitral stenosis.

Distinctions that matter

· The pattern of valves involved is as informative as the appearance of any single valve, because rheumatic disease takes the mitral valve first.

· Rheumatic disease commonly gives stenosis and regurgitation together, while degenerative disease usually gives one lesion alone.

Common misconceptions

· Reading isolated aortic stenosis in an older adult as old rheumatic disease. Without mitral involvement it is calcific or bicuspid.

· Expecting a tight mitral valve to tolerate pregnancy because the patient has been well. Volume, output, and rate all rise and the orifice cannot.

Where it is taught: Section C08, final two paragraphs, with calcific disease in C01 and the heart rate paragraph in C03 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C08-RT02

Retrieval exercise: Rebuild the path from a sore throat to a fused valve

A competent response contains

· Step 1: group A streptococcal pharyngitis.

· Step 2: molecular mimicry causes acute rheumatic carditis, with antibodies and T cells against M protein cross-reacting with cardiac myosin and valve tissue.

· Step 3: valvulitis with small vegetations about 1 to 2 mm on the lines of closure.

· Step 4: healing by fibrosis and neovascularisation; new vessels in a normally avascular leaflet give inflammatory cells continuing access, so each later infection can reach and reinflame the valve.

· Step 5: recurrent infections reinflame scarred valves, and each recurrence adds further fibrosis and deformity.

· Step 6: commissural fusion with thick, short, fused chordae.

· Step 7: fish-mouth mitral stenosis, often with mitral regurgitation and aortic disease.

· Chronic rheumatic heart disease is the scar of repeated immune injury, not an ongoing infection of the valve.

Distinctions that matter

· Neovascularisation is a healing change, not an acute one, and it is what makes an avascular leaflet accessible to later immune injury.

· Fusion of the commissures causes the stenosis, while shortening of the chordae prevents closure.

Common misconceptions

· Expecting damage after one attack. Accumulation across recurrences is what produces the chronic lesion.

· Calling the chronic valve infected and looking for organisms on it.

Where it is taught: Section C08, flow diagram from pharyngitis to chronic valve disease of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C09-BD1

Brain dump 1 · foundational retrieval: Serotonin, two filters, and the right-sided plaques

A competent response contains

· Carcinoid tumours are well-differentiated neuroendocrine tumours of enterochromaffin cells, most often in the ileum.

· Serotonin is made from tryptophan by tryptophan hydroxylase and aromatic amino acid decarboxylase; monoamine oxidase converts serotonin to 5-hydroxyindoleacetic acid, which is measured in the urine.

· Other mediators are histamine, kallikrein, which generates bradykinin, tachykinins, and prostaglandins.

· Sequence: a serotonin-secreting tumour, often ileal; metastasis to the liver; serotonin enters the hepatic veins, bypassing hepatic monoamine oxidase; high serotonin reaches the right heart; 5-HT2B signalling activates valve interstitial cells; fibrous plaques form on the tricuspid and pulmonic valves; lung monoamine oxidase lowers serotonin before the left heart.

· A tumour confined to the bowel drains through the portal vein, so hepatic monoamine oxidase inactivates its products and no valve disease follows; liver metastases release products straight into the hepatic veins and the right atrium.

· The lung protects the left heart because pulmonary endothelium takes serotonin up through its transporter and degrades it with monoamine oxidase.

· Left-sided lesions appear with a patent foramen ovale or other right-to-left shunt, with very high levels, or with a bronchial carcinoid draining into the pulmonary veins; ovarian carcinoids drain outside the portal system and can cause the syndrome without liver metastases.

· The lesions are pearly white plaques of myofibroblasts in glycosaminoglycan and collagen laid on right-sided endocardium, without destroying the valve beneath.

· Serotonin acts on 5-HT2B receptors of valve interstitial cells, partly through TGF-beta, to drive proliferation and matrix production; 5-HT2B agonists such as fenfluramine and certain ergot-derived dopamine agonists produce similar plaques.

· Tricuspid leaflets are fixed open, so tricuspid regurgitation is the most common lesion and tricuspid stenosis can also occur; pulmonic cusps thicken and retract, so pulmonic stenosis often coexists with regurgitation, and the right ventricle fails under combined volume and pressure loads.

Distinctions that matter

· The plaque sits on the surface and fixes the leaflet, which is why the valve beneath is intact and the lesion is not destructive.

· Two filters stand between the tumour and the left heart: the liver and the lung, and the lesion appears only when one of them is bypassed.

Common misconceptions

· Expecting valve disease from any carcinoid tumour. A bowel tumour without liver metastases usually causes none.

· Assuming a right-sided lesion means the organism or the lesion started there. The side affected is decided by which filter the products escape.

Where it is taught: Section C09, first three paragraphs and the carcinoid flow diagram of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C09-BD2

Brain dump 2 · integration and application: A flushing patient with a murmur that changes with breathing

A competent response contains

· Flushing and bronchospasm with wheezing come from kinins, histamine, and tachykinins; watery diarrhea and cramps come from serotonin.

· The holosystolic murmur at the left lower sternal border that increases on inspiration is tricuspid regurgitation; the systolic ejection murmur at the left upper sternal border is pulmonic stenosis.

· It increases on inspiration because inspiration raises venous return to the right heart and therefore the regurgitant flow.

· Large jugular v waves, a pulsatile liver, edema, and ascites follow from tricuspid regurgitation and right heart failure.

· The rash and memory change are pellagra: tryptophan is diverted to serotonin synthesis, leaving too little for niacin, and the deficiency gives dermatitis, diarrhea, and dementia.

· Urinary 5-hydroxyindoleacetic acid over 24 hours is high; it is the monoamine oxidase metabolite of serotonin.

· The tumour must have metastasised to the liver, because liver metastases release products into the hepatic veins and bypass hepatic monoamine oxidase.

· His left-sided valves should be spared, because pulmonary endothelium takes up and degrades serotonin before the blood reaches the left heart.

· The patient with a bronchial carcinoid and no liver metastases develops left-sided lesions, because a bronchial tumour drains into the pulmonary veins and delivers its products to the left atrium without passing the liver or the pulmonary capillary bed, while the right heart receives systemic venous blood with little serotonin.

Distinctions that matter

· Different mediators produce different symptoms: serotonin drives the diarrhea and the valve plaques, while kinins, histamine, and tachykinins drive the flushing and wheeze.

· A right-sided murmur that increases on inspiration is distinguished from a left-sided murmur by that respiratory change alone.

Common misconceptions

· Explaining the flushing by serotonin alone. Kinins, histamine, and tachykinins are responsible.

· Expecting the valve lesion to be on the side where the tumour sits. The route of venous drainage decides it.

Where it is taught: Section C09, symptom and sign paragraph and the stop and predict box of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C09-Q03

Board-style question: Valve disease in a patient with liver metastases

Correct answer: D. Tricuspid regurgitation with pulmonic stenosis from fibrous plaques that fix the leaflets

Why it is right: Serotonin that bypasses hepatic monoamine oxidase through liver metastases reaches the tricuspid and pulmonic valves first and drives plaque formation through 5-HT2B receptors, fixing the tricuspid leaflets open to give regurgitation while the retracted pulmonic cusps give stenosis, often with regurgitation.

A is wrong: Warty vegetations on both surfaces of the leaflets with mitral regurgitation describe Libman-Sacks endocarditis in lupus, not a serotonin-driven plaque.

B is wrong: Heaped calcified cusps with unfused commissures describe age-related calcific aortic stenosis, and the left-sided valves here are normal.

C is wrong: Commissural fusion with a fish-mouth orifice describes chronic rheumatic mitral stenosis, which follows streptococcal infection rather than a neuroendocrine tumour.

A competent response contains

· Flushing, wheeze, and diarrhea with a raised urinary 5-hydroxyindoleacetic acid and an ileal tumour with liver metastases establish carcinoid syndrome with cardiac involvement.

· Serotonin reaching the right heart acts on 5-HT2B receptors of valve interstitial cells, producing pearly fibrous plaques that fix the tricuspid leaflets open and thicken and retract the pulmonic cusps.

Distinctions that matter

· Carcinoid plaques lie on the surface and fix the leaflets, which is why tricuspid regurgitation predominates with pulmonic stenosis alongside.

Common misconceptions

· Expecting left-sided lesions. Pulmonary endothelium takes up and degrades serotonin before the blood reaches the left heart.

Where it is taught: Section C09 of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C10-BD1

Brain dump 1 · foundational retrieval: Two kinds of prosthesis and the ways each one fails

A competent response contains

· Mechanical valves are usually bileaflet pyrolytic carbon, last decades, and click crisply, but their foreign surface and abnormal flow make them highly thrombogenic.

· Bioprosthetic valves are porcine valves or bovine pericardium fixed in glutaraldehyde, which crosslinks collagen, kills the cells, and lowers antigenicity; the tissue is far less thrombogenic but nonviable and unable to repair itself, so structural deterioration commonly occurs within about 10 to 20 years and faster in younger patients.

· Transcatheter valves are bioprosthetic tissue mounted on a metal frame.

· Virchow's triad on a mechanical valve: the foreign surface adsorbs plasma proteins and activates platelets and the contact pathway (endothelial or surface injury), flow around the hinges creates zones of stasis and high shear (abnormal flow), and hypercoagulability or interrupted anticoagulation completes it.

· Mitral prostheses carry greater thrombotic risk than aortic ones because mitral flow is slower and the left atrium is often large and fibrillating.

· Thrombus can immobilise a leaflet, giving a new murmur, muffled clicks, heart failure, and shock, or it can embolise.

· Pannus is a slower fibrous ingrowth from the sewing ring that narrows the orifice over months to years.

· High-velocity turbulent flow fragments red cells, worst through a paravalvular leak, which is a gap between the sewing ring and the tissue.

· Laboratory findings: schistocytes including helmet cells from shear fragmentation, high lactate dehydrogenase and unconjugated bilirubin from released cell contents and heme breakdown, low or absent haptoglobin because it is consumed binding free hemoglobin, reticulocytosis from marrow compensation, and hemoglobinuria from filtered free hemoglobin.

· Tubular cells store filtered hemoglobin iron as hemosiderin and shed it, so chronic urinary iron loss can cause iron deficiency.

· Bioprosthetic tissue undergoes dystrophic calcification as calcium binds the phospholipids of devitalised cells and altered collagen, faster in children and young adults, with collagen fatigue from repeated flexing adding to it, so cusps stiffen giving stenosis or tear giving regurgitation, sometimes suddenly.

Distinctions that matter

· Mechanical valves fail by thrombosis and shear; bioprosthetic valves fail by calcification and tearing of nonviable tissue.

· Thrombosis presents over hours to days while pannus narrows the orifice over months to years.

Common misconceptions

· Assuming a bioprosthetic valve needs no monitoring because it is less thrombogenic. Its tissue cannot repair itself and deteriorates structurally.

· Expecting the hemolysis to be immune. It is mechanical destruction by shear.

Where it is taught: Section C10, first two paragraphs, the hemolysis paragraph, and the calcification paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C10-BD2

Brain dump 2 · integration and application: Read the panel and name the complication

A competent response contains

· This is mechanical intravascular hemolysis, most likely from a paravalvular leak, a gap between the sewing ring and the tissue through which a high-velocity jet shears red cells apart.

· Schistocytes and helmet cells mark the shearing itself; high lactate dehydrogenase marks release of cell contents; raised unconjugated bilirubin marks heme breakdown; absent haptoglobin marks its consumption binding free hemoglobin; reticulocytosis marks marrow compensation; hemoglobinuria and hemosiderinuria mark filtration of free hemoglobin and storage of its iron in tubular cells that are then shed.

· A negative direct antiglobulin test shows the destruction is mechanical rather than antibody-mediated, and a normal platelet count separates this from disseminated intravascular coagulation, thrombotic thrombocytopenic purpura, and hemolytic uremic syndrome, which also produce schistocytes; autoimmune hemolysis is the immune process the antiglobulin test excludes.

· She can become iron deficient because iron is lost continuously in the urine as hemoglobin and shed hemosiderin, so it is not recycled as it would be in extravascular hemolysis.

· Prosthetic valve endocarditis begins at the sewing ring and often spreads into the annulus, causing ring abscess, dehiscence, paravalvular leak, and heart block.

· Early disease, within about the first year and most often the first 2 months, reflects perioperative contamination, and the expected organism is a coagulase-negative staphylococcus, especially Staphylococcus epidermidis, alongside S. aureus, gram-negative bacilli, and Candida.

· It depends on a polysaccharide biofilm that binds synthetic material and shields the organism from neutrophils, complement, and antibiotics.

· Laboratory features of that organism: gram-positive cocci in clusters, catalase positive, coagulase negative, and novobiocin sensitive.

· Late disease after valve replacement resembles native valve endocarditis in its organisms.

Distinctions that matter

· Mechanical and microangiopathic hemolysis both give schistocytes, and the platelet count plus the valve finding separate them.

· Early and late prosthetic valve endocarditis differ by organism, because early disease reflects contamination at surgery.

Common misconceptions

· Reading schistocytes as proof of a microangiopathy. A prosthetic jet shears cells just as effectively, with a normal platelet count.

· Assuming hemolysis always raises iron stores. Intravascular hemolysis loses iron in the urine and can cause deficiency.

Where it is taught: Section C10, hemolysis paragraph, the hemolysis flow diagram, and the prosthetic valve endocarditis paragraph of the CV-18-D Valvular Heart Diseases notes.

NPLEX-P1-CARD-18-04-C10-Q04

Board-style question: Anemia three years after valve replacement

Correct answer: B. Shear fragmentation of red cells by a high-velocity jet through a paravalvular leak

Why it is right: The echocardiogram shows a gap between the sewing ring and the annulus with a high-velocity jet, and shear stress through that leak fragments red cells inside the circulation, which accounts for the helmet cells, the high lactate dehydrogenase and bilirubin, the absent haptoglobin, and the hemoglobin and hemosiderin in the urine.

A is wrong: Warm autoantibody hemolysis would give a positive direct antiglobulin test, and hers is negative; antibody-coated cells are also cleared in the spleen rather than fragmented.

C is wrong: Platelet consumption with fibrin strands in the microvasculature would lower the platelet count, and hers is 240,000/mm3.

D is wrong: Hepcidin-driven iron restriction causes anemia without hemolysis, so it would not produce fragmented cells, absent haptoglobin, or hemoglobinuria.

A competent response contains

· A high-velocity jet through a paravalvular leak shears red cells, giving schistocytes and helmet cells with high lactate dehydrogenase, raised unconjugated bilirubin, absent haptoglobin, and hemoglobinuria with hemosiderinuria.

· A negative direct antiglobulin test shows the destruction is mechanical, and a normal platelet count excludes the consumptive microangiopathies.

Distinctions that matter

· Intravascular hemolysis loses iron in the urine, which is why chronic iron deficiency can follow despite ongoing red cell destruction.

Common misconceptions

· Taking schistocytes as evidence of a microangiopathy without checking the platelet count and the prosthesis.

Where it is taught: Section C10 of the CV-18-D Valvular Heart Diseases notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-E Primary Myocardial Diseases

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-05-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild dilated cardiomyopathy from its central defect

A competent response contains

· The central defect is loss of contractile force: myocytes generate less tension, so stroke volume falls and end-systolic volume rises.

· Compensation runs through the Frank-Starling mechanism and neurohormonal activation, with salt and water retention raising end-diastolic volume to restore some stroke volume.

· Sarcomeres are added in series, which is eccentric hypertrophy, so myocytes lengthen and the cavity enlarges while the wall stays normal or thin even though total heart weight rises.

· Law of Laplace: wall stress is proportional to pressure times radius divided by wall thickness.

· A wider, thinner ventricle must develop more wall tension for the same pressure, which raises oxygen demand, injures more myocytes, and drives further dilation, so the ejection fraction falls, usually below about 40%.

· Familial disease accounts for about 30% to 50% of cases, usually autosomal dominant, most often truncating variants of TTN (titin), also lamin A/C and X-linked dystrophin.

· Alcohol: ethanol and acetaldehyde damage mitochondria and contractile proteins, and the picture may improve with abstinence.

· Prior myocarditis: myocyte loss and fibrosis after viral or immune injury.

· Doxorubicin: dose-dependent, through iron-dependent free radical injury and damage to topoisomerase II beta in myocytes.

· Peripartum: late pregnancy to about 5 months postpartum, with oxidative stress activating cathepsin D, which cleaves prolactin to an antiangiogenic 16 kDa fragment.

· Chagas disease: chronic Trypanosoma cruzi infection with inflammation and fibrosis, often an apical aneurysm.

· Thiamine deficiency (wet beriberi): loss of thiamine pyrophosphate, the cofactor for pyruvate and alpha-ketoglutarate dehydrogenase, lowers ATP, and vasodilation adds high-output failure.

· Hemochromatosis: iron catalyzes free radical (Fenton) injury, giving a dilated pattern and sometimes a restrictive one.

Distinctions that matter

· Eccentric hypertrophy adds sarcomeres in series and enlarges the cavity; concentric hypertrophy adds them in parallel and thickens the wall at the expense of the cavity.

· Heart weight and wall thickness move in different directions here: the heart is heavier while the wall is normal or thin, because the muscle is spread around a much larger cavity.

Common misconceptions

· Writing that the wall thickens in dilated disease. Growth is in series, so the chamber widens rather than the wall thickening.

· Treating dilation as purely compensatory. Larger radius with a thin wall raises wall stress, which is what makes the process self-perpetuating.

Where it is taught: Section C01, first two paragraphs and the causes table of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C01-BD2

Brain dump 2 · integration and application: Explain every finding in a large, weak ventricle

A competent response contains

· Both murmurs are functional regurgitation: dilation stretches the mitral and tricuspid annuli, pulling the leaflets apart so they fail to coapt, giving holosystolic murmurs with structurally normal valves. The apical murmur is mitral and the one at the left lower sternal border is tricuspid.

· The early diastolic extra sound is an S3, produced when blood rushing into an overfilled ventricle in early diastole is abruptly decelerated.

· Low output explains fatigue, and high filling pressures explain dyspnea and orthopnea.

· Raised jugular venous pressure and peripheral edema come from high right-sided filling pressures and volume retention.

· Stasis in dilated, poorly contracting chambers favors mural thrombus and systemic or pulmonary emboli.

· Stretched, fibrotic myocardium supports atrial fibrillation and ventricular tachycardia.

· Chest radiography shows a globular, enlarged heart.

· Ischemic cardiomyopathy produces the same dilated, low ejection fraction ventricle from prior infarcts and hibernating myocardium, but the wall motion loss is regional rather than diffuse, and the history carries infarction or coronary disease.

· Histology in idiopathic or familial disease is nonspecific: hypertrophied myocytes with large nuclei, attenuated fibers, and interstitial fibrosis.

Distinctions that matter

· Functional regurgitation comes from annular geometry, not from leaflet or chordal disease, so the valve itself looks normal.

· Diffuse against regional wall motion loss is the feature that separates primary dilated disease from ischemic cardiomyopathy.

Common misconceptions

· Reading a holosystolic apical murmur in a dilated heart as primary mitral valve disease. The leaflets are normal and the annulus has been stretched.

· Attributing the S3 to valve closure. It is the sound of rapid early filling being abruptly decelerated by an overfilled ventricle.

Where it is taught: Section C01, paragraphs on findings and on distinguishing ischemic cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C01-RT01

Retrieval exercise: Rebuild the remodeling spiral

A competent response contains

· Step 1: myocyte injury or a defective sarcomere protein.

· Step 2: contractility and stroke volume fall.

· Step 3: volume retention and series sarcomeres enlarge the cavity.

· Step 4: a larger radius and thin wall raise wall stress, by the law of Laplace.

· Step 5: higher oxygen demand injures more myocytes, which returns the loop to step 1.

· The two processes that enlarge the cavity are neurohormonally driven salt and water retention, which raises end-diastolic volume, and eccentric hypertrophy adding sarcomeres in series.

· Wall stress is proportional to pressure times radius divided by wall thickness: radius is rising and wall thickness is falling, so both moving variables push wall stress up.

· Higher wall tension costs oxygen because developing and maintaining tension is what consumes myocardial ATP, so demand rises in muscle whose supply has not changed.

· Each turn of the loop enlarges the ventricle further and lowers the ejection fraction further.

Distinctions that matter

· Stroke volume can be partly restored by a larger end-diastolic volume while the ejection fraction still falls, because the denominator grows faster than the numerator.

· The loop is driven by geometry as well as by the original injury, which is why it continues after the initial insult has passed.

Common misconceptions

· Drawing the spiral as a straight line that ends in dilation. It is a closed loop, and that is why the disease progresses.

· Leaving wall thickness out of the Laplace step. A thinner wall raises wall stress as much as a wider radius does.

Where it is taught: Section C01, the remodeling spiral cycle of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-BD1

Brain dump 1 · foundational retrieval: Rebuild hypertrophic cardiomyopathy from gene to tissue to filling

A competent response contains

· Autosomal dominant with variable penetrance, affecting about 1 in 500 people, and the most common inherited heart disease.

· The genes most often involved are MYH7, encoding beta-myosin heavy chain, and MYBPC3, encoding myosin-binding protein C.

· Mutant sarcomeres are hypercontractile and use energy inefficiently, and the myocardium hypertrophies with no pressure overload on it.

· Hypertrophy is usually asymmetric and thickest in the interventricular septum, and the cavity is small.

· Histology: myocyte disarray, interstitial and replacement fibrosis, and thickened intramural coronary arteries.

· The main defect is diastolic: a thick, fibrotic wall is stiff and relaxes slowly, so filling needs high pressure and depends on atrial contraction.

· Atrial contraction into a stiff ventricle produces an S4, and atrial fibrillation is poorly tolerated because the atrial contribution to filling is lost.

· Systolic function is preserved, so the ejection fraction is normal or high.

· Exertional angina occurs with normal epicardial coronaries because thick muscle with narrowed intramural arteries outgrows its blood supply.

Distinctions that matter

· Hypertrophy here is independent of load, unlike the concentric hypertrophy of pressure overload, which is why no hypertension or valve lesion is needed to explain it.

· A normal or high ejection fraction coexists with heart failure, because the failure is one of filling rather than emptying.

Common misconceptions

· Reading a high ejection fraction as a healthy ventricle. The ejection fraction is a fraction of a small end-diastolic volume, and filling is the problem.

· Explaining the angina by coronary atherosclerosis. The epicardial arteries are normal and the mismatch comes from muscle mass and thickened intramural vessels.

Where it is taught: Section C02, first two paragraphs of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-BD2

Brain dump 2 · integration and application: Reason through obstruction, syncope, and sudden death

A competent response contains

· Sequence: septal hypertrophy narrows the left ventricular outflow tract; systolic flow accelerates through the narrow tract; drag pulls the anterior mitral leaflet toward the septum; leaflet contact obstructs outflow in mid systole; the malcoapting leaflet leaks, giving posteriorly directed mitral regurgitation.

· The rule: anything that reduces left ventricular volume (less preload or less afterload) or raises contractility brings leaflet and septum closer and worsens obstruction; anything that enlarges the ventricle separates them and relieves it.

· Standing from squatting drops venous return, so the cavity is smaller and the murmur is louder; squatting raises venous return and afterload, so the cavity is larger and the murmur is softer.

· The murmur is a harsh crescendo-decrescendo systolic murmur at the left lower sternal border that does not radiate well to the carotids.

· The carotid pulse rises briskly, then falls as obstruction develops, and may be bifid.

· Sudden death: myocyte disarray and patchy fibrosis create regions of slow, nonuniform conduction that support reentrant ventricular tachycardia and fibrillation, and this is the most commonly identified cause of sudden cardiac death in young athletes in the United States.

· Exertion adds catecholamines, a small cavity, and ischemia.

· Exertional syncope can come from ventricular arrhythmia or from obstruction that worsens as heart rate and contractility rise.

· The athlete's heart has symmetric physiologic hypertrophy, a normal or large cavity, normal diastolic filling, and regression with rest from training.

· The electrocardiogram often shows left ventricular hypertrophy voltage and deep, narrow septal Q waves.

Distinctions that matter

· The obstruction is dynamic rather than fixed, so it tracks cavity size and contractility beat to beat, which is why maneuvers change the murmur.

· Aortic stenosis responds the opposite way to Valsalva and standing, because a fixed orifice depends on the volume crossing it rather than on leaflet position.

Common misconceptions

· Saying anything that increases flow makes the murmur louder. Raising preload enlarges the cavity, moves the leaflet away from the septum, and softens it.

· Attributing the mitral regurgitation to primary valve disease. The leaflet is displaced toward the septum and therefore fails to coapt.

Where it is taught: Section C02, the obstruction flow, the maneuver table, the worked example, and the paragraph on sudden death of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-CP01

Comparison table: Hypertrophic cardiomyopathy against the athlete's heart

A competent response contains

· What drives the hypertrophy: sarcomere protein mutations that make sarcomeres hypercontractile and energy inefficient, independent of load, against physiologic adaptation to training.

· Symmetry and site: asymmetric, thickest in the interventricular septum, against symmetric thickening.

· Cavity size: small left ventricular cavity, against a normal or large cavity.

· Diastolic filling: impaired, with a stiff slowly relaxing wall and dependence on atrial contraction, against normal filling.

· After rest from training: hypertrophy persists in the disease, while the athlete's hypertrophy regresses.

· Tissue architecture: myocyte disarray with interstitial and replacement fibrosis and thickened intramural coronary arteries, against normal architecture.

· Risk of exertional arrhythmia and sudden death: high in the disease, because disarray and fibrosis create a reentry substrate, and not raised in the athlete's heart.

Distinctions that matter

· Cavity size and diastolic filling do most of the work in separating the two, because both conditions can show a thick wall.

· Regression with rest is a feature of adaptation, not of disease.

Common misconceptions

· Using wall thickness alone to decide. Thickness overlaps between the two, so symmetry, cavity size, and filling are what separate them.

· Assuming the athlete's thick wall also carries an arrhythmia risk. The architecture is normal, so the reentry substrate is absent.

Where it is taught: Section C02, paragraph on sudden death and the athlete's heart, with the opening paragraph on genetics and histology of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-PR01

Prediction exercise: Predict what the strain phase of Valsalva does in hypertrophic cardiomyopathy

A competent response contains

· Venous return to the right heart falls, because raised intrathoracic pressure during strain impedes systemic venous inflow.

· Left ventricular end-diastolic volume falls, because less blood reaches the left ventricle to fill it.

· The distance between the anterior mitral leaflet and the septum falls, because a smaller cavity brings the two closer together.

· Systolic anterior motion begins earlier and is greater, because the leaflet starts nearer the septum and drag from accelerated flow pulls it across sooner.

· The outflow murmur becomes louder, because the dynamic obstruction is greater.

· The murmur of aortic stenosis becomes softer, because a fixed orifice murmur depends on the volume crossing it and less blood is being ejected.

· Squatting and passive leg raising both raise venous return and enlarge the ventricle, separating leaflet from septum, so the murmur becomes softer; squatting also raises afterload, which enlarges the cavity further.

Distinctions that matter

· Valsalva and standing separate the two murmurs precisely because one obstruction is dynamic and the other is fixed.

· Afterload and preload both act through cavity size here, which is why handgrip and leg raising change the murmur in the same direction.

Common misconceptions

· Predicting that both murmurs change in the same direction with Valsalva. They move opposite ways, and that is the point of the maneuver.

· Thinking the leaflet moves because of the murmur. Flow acceleration and drag move the leaflet, and the murmur reports the resulting obstruction.

Where it is taught: Section C02, the obstruction flow, the maneuver table and its caption of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C03-BD1

Brain dump 1 · foundational retrieval: Rebuild restrictive cardiomyopathy and its causes

A competent response contains

· The wall stiffens either because something is deposited in it or because myocardium and endocardium are replaced by fibrous tissue.

· Compliance is the change in ventricular volume per change in filling pressure, so a stiff ventricle has low compliance and its filling curve is steep.

· A small increase in volume therefore produces a steep rise in diastolic pressure.

· Early diastolic filling is rapid and then stops abruptly when the rigid wall reaches its limit, which gives the dip-and-plateau or square root pattern on a ventricular pressure tracing.

· Contraction is relatively preserved, so cavity size and ejection fraction are near normal until late disease.

· Amyloidosis is the most common cause: extracellular AL light chain or transthyretin amyloid, with low electrocardiogram voltage despite thick walls, Congo red staining with apple-green birefringence, myeloma in AL disease, and older men in wild-type transthyretin disease.

· Sarcoidosis: noncaseating granulomas and scar, with heart block and ventricular arrhythmia, and lung and node disease.

· Hemochromatosis: intracellular iron and fibrosis, with diabetes, skin pigmentation, and liver disease, and it may be dilated instead.

· Radiation: interstitial and endocardial fibrosis after chest irradiation, and it may coexist with constriction.

· Endomyocardial fibrosis: fibrosis of the apex and inflow tracts in children and young adults in tropical Africa and Asia.

· Loeffler endocarditis: eosinophil-driven endocardial injury, then thrombus, then fibrosis, in hypereosinophilic syndrome with marked blood eosinophilia.

· The three stages in Loeffler endocarditis are eosinophil granule proteins such as major basic protein damaging the endocardium, mural thrombus forming on the injured surface, and organization into fibrosis that obliterates the apex.

· Restrictive cardiomyopathy is the least common of the three cardiomyopathies.

Distinctions that matter

· Infiltrative causes thicken the wall, while endomyocardial causes obliterate the cavity from the inside.

· Compliance and contractility are separate properties: compliance is lost here while contractility is largely intact.

Common misconceptions

· Expecting a low ejection fraction. Emptying is preserved and filling is what fails, so the ejection fraction stays near normal until late.

· Describing filling as uniformly slow. Early filling is rapid and then halts abruptly, which is what makes the dip-and-plateau shape.

Where it is taught: Section C03, first paragraph, the causes table, and the Loeffler paragraph of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C03-BD2

Brain dump 2 · integration and application: Trace high filling pressures to the bedside and read the ECG mismatch

A competent response contains

· High filling pressures are transmitted backward into the atria, so both atria dilate markedly.

· Atrial dilation predisposes to atrial fibrillation and to thrombus.

· Right-sided congestion often dominates: raised jugular venous pressure that may rise on inspiration, which is the Kussmaul sign, with hepatomegaly, ascites, and peripheral edema, along with dyspnea from pulmonary congestion.

· Cardiac output is fixed because stroke volume cannot be raised by filling the ventricle more, so exercise tolerance is poor.

· Low voltage sequence: amyloid fibrils deposit between myocytes; the echocardiogram shows a thick, often speckled wall; deposits add mass but no electrically active cells; myocytes atrophy and are separated by protein; less depolarizing muscle gives low QRS voltage.

· In true hypertrophy, as in hypertrophic cardiomyopathy or hypertension, added muscle raises voltage, so the same thickness would give tall complexes.

· The mismatch of low QRS voltage in the limb leads with increased wall thickness is the classic clue to cardiac amyloidosis, along with Q waves without infarction (pseudoinfarct pattern) and conduction disease.

· Endomyocardial biopsy shows pink extracellular material that stains with Congo red and shows apple-green birefringence under polarized light.

· Constrictive pericarditis produces a similar picture, but the restriction comes from the pericardium rather than the myocardium.

Distinctions that matter

· Measured wall thickness and electrical mass are different things, which is the whole basis of the amyloid clue.

· Restrictive cardiomyopathy and constrictive pericarditis share the physiology of failed filling and raised venous pressure, and they are separated by where the restriction sits.

Common misconceptions

· Reading a thick wall as evidence of hypertrophy. In amyloidosis the thickness is extracellular protein, not myocytes.

· Expecting stroke volume to rise with exercise through greater filling. The ventricle cannot accept more volume, so output is fixed.

Where it is taught: Section C03, second paragraph, the amyloid flow, and the paragraph on the voltage mismatch of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C04-BD1

Brain dump 1 · foundational retrieval: Sort the three cardiomyopathies by which half of the cycle fails

A competent response contains

· The first question is which half of the cardiac cycle fails, emptying or filling.

· Dilated disease: the ventricle cannot empty, contractility is lost, end-systolic volume rises, and the ejection fraction falls, which is systolic dysfunction and heart failure with reduced ejection fraction.

· Hypertrophic disease: the ventricle cannot fill, contraction is preserved, and the stiff wall requires high filling pressure, which is diastolic dysfunction and heart failure with preserved ejection fraction.

· Restrictive disease: the ventricle cannot fill for the same reason, with preserved contraction and a preserved ejection fraction.

· Hypertrophic disease is stiff because of thick, disorganized muscle; restrictive disease is stiff because of infiltration or fibrosis.

· Dilated loop: lower contractility flattens the end-systolic pressure-volume relationship, so the loop shifts right to larger volumes and narrows.

· Hypertrophic loop: the loop shifts left to smaller volumes and the end-diastolic pressure-volume relationship is steep.

· Restrictive loop: a steep end-diastolic pressure-volume relationship with a small loop.

· The end-systolic pressure-volume relationship represents contractility and the end-diastolic pressure-volume relationship represents compliance, so a flatter end-systolic line means weaker contraction and a

steeper end-diastolic curve means a stiffer ventricle in which every filling volume costs more pressure, and that pressure is what the lungs and systemic veins feel.

· All three share one consequence: scarred, stretched, or disorganized myocardium supports reentry, so ventricular arrhythmia and sudden death occur in each.

Distinctions that matter

· Contractility and compliance are read from different lines on the pressure-volume diagram, which is why systolic and diastolic failure look different there.

· Hypertrophic and restrictive disease share the physiology of failed filling and differ in what makes the wall stiff.

Common misconceptions

· Equating a preserved ejection fraction with a normal heart. Filling pressure is high in both diastolic diseases, which is what produces the symptoms.

· Drawing the dilated loop as larger in every dimension. It shifts to larger volumes and narrows, because stroke volume is reduced.

Where it is taught: Section C04, first paragraph, the comparison table, and the paragraph on pressure-volume reasoning of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C04-BD2

Brain dump 2 · integration and application: Classify four sets of measurements

A competent response contains

· Patient A: emptying has failed. An enlarged cavity, a 7 mm wall, and an ejection fraction of 20% give dilated cardiomyopathy, settled by the low ejection fraction with a large cavity. The loop shifts right and narrows with a flatter end-systolic line, and the early diastolic extra sound is an S3.

· Patient B: filling has failed. Asymmetric septal thickening of 21 mm against a 10 mm posterior wall, a small cavity, an ejection fraction of 72%, and the anterior mitral leaflet touching the septum in systole give hypertrophic cardiomyopathy with obstruction, settled by the asymmetric thickening with systolic anterior motion. The loop shifts left with a steep end-diastolic curve, and the expected extra sound is an S4.

· Patient C: filling has failed. A normal cavity, a 16 mm wall, an ejection fraction of 55%, huge atria, and low limb lead voltage give restrictive disease from amyloid, settled by the low voltage with a thick wall and enlarged atria. The loop is small with a steep end-diastolic curve, and an S3 from the abrupt halt to filling is expected, with an S4 if the atria still contract.

· Patient D: filling has failed. A 19 mm wall with tall QRS complexes, a small cavity, an ejection fraction of 70%, and deep narrow septal Q waves give hypertrophic cardiomyopathy, settled by high voltage with a thick wall. The loop shifts left with a steep end-diastolic curve, and the expected extra sound is an S4.

· Patients C and D have opposite voltages because patient C's thickness is extracellular amyloid, which adds mass without electrically active cells while myocytes atrophy, whereas patient D's thickness is muscle, which adds electrical mass and raises voltage.

· The ordered approach is to decide first whether emptying or filling fails, then to use wall thickness and the electrocardiogram to name the cause.

Distinctions that matter

· Ejection fraction separates systolic from diastolic failure; wall thickness with voltage then separates hypertrophic from restrictive disease.

· Asymmetry of thickening and systolic anterior motion are what mark hypertrophic disease, rather than thickness alone.

Common misconceptions

· Naming the disease from wall thickness first. Thickness is shared by hypertrophic and infiltrative disease, so the cycle half and the voltage are what decide.

· Reading an ejection fraction of 55% as excluding significant disease. In restrictive disease emptying is preserved and filling is the problem.

Where it is taught: Section C04, the comparison table and the worked example on classifying echocardiograms of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C04-CP02

Comparison table: The three cardiomyopathies side by side

A competent response contains

· Cavity size: enlarged in all four chambers in dilated disease; small left ventricular cavity in hypertrophic disease; normal or small with greatly enlarged atria in restrictive disease.

· Wall thickness: normal or thin in dilated disease; thick and often asymmetric at the septum in hypertrophic disease; normal or thick in restrictive disease, and thick in amyloid.

· Pattern of hypertrophy: eccentric with series sarcomeres in dilated disease; disarrayed and independent of load in hypertrophic disease; none in restrictive disease, where infiltration or fibrosis does the stiffening.

· Ejection fraction: reduced, usually under about 40%, in dilated disease; normal or high in hypertrophic disease; normal or near normal in restrictive disease.

· Primary dysfunction: systolic in dilated disease; diastolic in hypertrophic disease, with outflow obstruction in many; diastolic in restrictive disease.

· Pressure-volume loop: shifted right and narrow with a flatter end-systolic relationship in dilated disease; shifted left with a steep end-diastolic relationship in hypertrophic disease; a small loop with a steep end-diastolic relationship in restrictive disease.

· Extra heart sound: S3 in dilated disease; S4 in hypertrophic disease; S3 from the abrupt halt to filling in restrictive disease, with an S4 if the atria still contract.

· Electrocardiogram: nonspecific in dilated disease with arrhythmias and bundle branch block; high left ventricular voltage with septal Q waves in hypertrophic disease; low voltage in amyloid.

Distinctions that matter

· The end-diastolic pressure-volume relationship reports compliance and the end-systolic relationship reports contractility, so the two diastolic diseases steepen the first while the dilated ventricle flattens the second.

· Restrictive disease is the one in which a thick wall can coexist with low voltage, because the thickness is not muscle.

Common misconceptions

· Pairing a thick wall with a reduced ejection fraction. Thick-walled disease here preserves emptying and fails at filling.

· Assuming heart sounds alone separate the three. The S3 of dilated disease and the S3 of restrictive disease arise from different mechanisms, so they need the rest of the row to interpret.

Where it is taught: Section C04, the table of the three cardiomyopathies side by side and its caption of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-BD1

Brain dump 1 · foundational retrieval: Rebuild the causes and histology of myocarditis

A competent response contains

· Myocarditis is inflammation of the myocardium with injury or necrosis of myocytes that is not ischemic.

· Most cases in developed countries are viral: coxsackie B virus is the classic cause, with adenovirus, parvovirus B19, human herpesvirus 6, influenza, and SARS-CoV-2 among the others.

· Coxsackie B enters myocytes through the coxsackie-adenovirus receptor, which adenovirus also uses, and enteroviral protease 2A cleaves dystrophin, weakening the myocyte cytoskeleton.

· Myocytes are injured first by the virus and then by the cytotoxic T-cell response that clears it.

· Coxsackie B and other viruses: direct lysis plus T-cell injury, a lymphocytic infiltrate, often after a viral illness.

· Chagas disease from Trypanosoma cruzi: triatomine bug vector in Latin America, amastigotes in myocytes acutely, and chronic dilated cardiomyopathy with conduction block and an apical aneurysm years later, plus megaesophagus and megacolon.

· Lyme disease from Borrelia burgdorferi: weeks to months after the tick bite, with atrioventricular block of varying degree that usually resolves.

· Diphtheria from Corynebacterium diphtheriae: exotoxin ADP-ribosylates elongation factor 2, halting protein synthesis, with myocyte injury and conduction block.

· Drug hypersensitivity: interstitial infiltrate rich in eosinophils, often with rash and fever.

· Systemic lupus erythematosus: immune complex and autoantibody injury, which may coexist with pericarditis and Libman-Sacks endocarditis.

· Giant cell myocarditis: multinucleated giant cells with widespread necrosis in young to middle-aged adults, linked to autoimmunity and thymoma, and often fulminant.

· The defining histology is an inflammatory infiltrate with myocyte necrosis, which are the Dallas criteria.

· Viral infiltrate is mainly lymphocytes, patchy and interstitial, surrounding damaged myocytes; hypersensitivity disease is perivascular and interstitial with many eosinophils; giant cell disease shows multinucleated giant cells through extensive necrosis without the well-formed granulomas of sarcoidosis; diphtheria injures myocytes with relatively little inflammation.

· Because inflammation is often focal, a biopsy can miss it.

Distinctions that matter

· Eosinophils point to hypersensitivity, giant cells without well-formed granulomas point to giant cell disease, and parasites inside myocytes point to Chagas.

· Giant cell myocarditis and cardiac sarcoidosis both show giant cells, and the well-formed granulomas of sarcoidosis are what separate them.

Common misconceptions

· Treating myocarditis as purely a direct viral killing process. The cytotoxic T-cell response that clears the virus also destroys myocytes.

· Reading a negative biopsy as excluding the diagnosis. Involvement is patchy, so sampling can miss the inflamed tissue.

Where it is taught: Section C05, first paragraph, the causes table, and the histology paragraph of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-BD2

Brain dump 2 · integration and application: Explain the findings of myocarditis and its late outcome

A competent response contains

· Troponin rises because myocyte necrosis releases it, and here the necrosis is inflammatory rather than from coronary occlusion.

· Edema and patchy inflammation slow and block conduction and create foci of abnormal automaticity and reentry, which give palpitations, atrioventricular block, ventricular arrhythmias, and occasionally sudden death in a young person.

· The ejection fraction falls because widespread loss of contractile myocytes removes contractile tissue, which can reach fulminant cardiogenic shock.

· Diffuse ST elevation fits widespread myocardial involvement rather than a single coronary territory.

· Chest pain may be ischemic-like or pleuritic; a pleuritic character implies pericardial involvement as well, which is myopericarditis.

· Sequence to dilated cardiomyopathy: the virus enters myocytes through the coxsackie-adenovirus receptor; viral replication injures myocytes directly; cytotoxic T cells kill infected myocytes; exposed cardiac proteins can trigger persistent autoimmunity; lost myocytes are replaced by fibrosis; remodeling produces dilated cardiomyopathy.

· Replacement of lost myocytes by fibrosis is the step that makes the damage permanent, because scar cannot contract.

· Many eosinophils with rash and fever: drug hypersensitivity myocarditis, named by the eosinophil-rich infiltrate with the systemic allergic features.

· Multinucleated giant cells in widespread necrosis: giant cell myocarditis, named by giant cells without well-formed granulomas, often fulminant and linked to autoimmunity and thymoma.

· Parasites inside myocytes: Chagas disease, named by amastigotes of Trypanosoma cruzi in myocytes with exposure in Latin America.

Distinctions that matter

· A raised troponin reports myocyte death, not its mechanism, so inflammation and infarction can look the same on that one result.

· Myocarditis and myopericarditis differ in which layer is inflamed, and the character of the pain is the clue at the bedside.

Common misconceptions

· Reading ST elevation with a raised troponin as necessarily infarction. Diffuse changes with normal coronary arteries point elsewhere.

· Assuming recovery is impossible once the ejection fraction falls. Many patients recover fully, and it is the fibrotic replacement of myocytes that makes the loss permanent.

Where it is taught: Section C05, paragraph on manifestations, the flow from viral infection to dilated cardiomyopathy, and the causes table of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-RT02

Retrieval exercise: Rebuild the path from viral infection to dilated cardiomyopathy

A competent response contains

· Step 1: the virus enters myocytes through the coxsackie-adenovirus receptor, which coxsackie B virus and adenovirus both use.

· Step 2: viral replication injures myocytes directly, and enteroviral protease 2A cleaves dystrophin, weakening the cytoskeleton.

· Step 3: cytotoxic T cells kill infected myocytes while clearing the virus.

· Step 4: cardiac proteins exposed by that injury can trigger persistent autoimmunity, so damage continues after the virus has gone.

· Step 5: lost myocytes are replaced by fibrosis.

· Step 6: remodeling produces dilated cardiomyopathy.

· Fibrous tissue cannot contract, which is the mechanical property it lacks.

· The ejection fraction falls because contractile myocytes have been replaced by noncontractile scar, so the ventricle generates less force, empties less completely, and dilates.

Distinctions that matter

· Viral injury and the immune response are two separate routes to myocyte loss, and both operate in the same illness.

· Autoimmunity explains progression after viral clearance, which is why a resolved infection can still be followed by a failing ventricle.

Common misconceptions

· Stopping the sequence at viral clearance. Autoimmunity and fibrosis are what carry the process on to dilated disease.

· Treating the fibrosis as a repair that restores function. Scar fills the space without contributing contraction.

Where it is taught: Section C05, the flow from viral infection to dilated cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-CA01

Clinical application: A young man with chest pain, a raised troponin, and clean coronary arteries

A competent response contains

· Recognize: acute myocarditis, most likely viral given the preceding febrile illness with sore throat and diarrhea, and the most classic organism is coxsackie B virus, an enterovirus.

· Explain: inflammatory myocyte necrosis releases troponin without coronary occlusion; widespread loss and dysfunction of contractile myocytes give diffusely reduced wall motion and an ejection fraction of 34%;

edema and patchy inflammation slow and block conduction, which produces the intermittent atrioventricular delay; and the same inflamed tissue creates foci of abnormal automaticity and reentry, which produce the wide early beats.

· Differentiate: infarction would give regional wall motion loss in one coronary territory with obstructive disease on angiography, whereas the changes here are diffuse with normal coronary arteries; stress cardiomyopathy would show apical ballooning with a hypercontracting base after a catecholamine surge in an older, usually postmenopausal woman, with a troponin rise small relative to the dysfunction and no lymphocytic infiltrate; giant cell myocarditis would show multinucleated giant cells in widespread necrosis and hypersensitivity myocarditis would show an eosinophil-rich infiltrate, and the biopsy here shows neither, only lymphocytes.

· Investigate: the two features to name are an inflammatory infiltrate and myocyte injury or necrosis, which together are the Dallas criteria; a biopsy without them would not exclude myocarditis because involvement is often focal and sampling can miss the inflamed tissue.

· Decide: it can progress over months to years to dilated cardiomyopathy, with a large four-chamber dilated ventricle, thin or normal walls, a low ejection fraction, an S3, and functional mitral and tricuspid regurgitation.

· Justify: the virus enters myocytes through the coxsackie-adenovirus receptor, viral replication injures them directly, cytotoxic T cells kill infected myocytes, exposed cardiac proteins can trigger persistent autoimmunity, and lost myocytes are replaced by fibrosis that cannot contract; once contractility falls and the cavity enlarges, the law of Laplace means a larger radius with a thinner wall raises wall stress and oxygen demand, which injures more myocytes and drives further dilation.

Distinctions that matter

· Diffuse against regional dysfunction is what separates myocarditis from infarction, and the angiogram confirms it.

· The type of infiltrate names the kind of myocarditis, so lymphocytes, eosinophils, and giant cells point in three different directions.

Common misconceptions

· Taking a markedly raised troponin with ST elevation as proof of coronary occlusion in a young patient.

· Assuming progression to dilated disease is inevitable. Many patients recover fully, and progression depends on how many myocytes are lost and replaced by scar.

Where it is taught: Section C05, all paragraphs and the flow diagram, with the Laplace reasoning in Section C01 of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild stress and tachycardia-induced cardiomyopathy

A competent response contains

· Stress cardiomyopathy is also called takotsubo cardiomyopathy and broken heart syndrome.

· About 90% of patients are women, most of them postmenopausal, and triggers are sudden emotional or physical stress such as bereavement, an argument, surgery, or severe illness.

· The typical pattern is apical ballooning: the apex and mid ventricle are akinetic while the base contracts vigorously, so in systole the ventricle resembles a narrow-necked Japanese octopus pot, the takotsubo.

· Mechanism: acute stress releases epinephrine and norepinephrine; very high epinephrine acts on dense apical beta receptors; beta-2 signaling switches from Gs to Gi and apical contraction falls; cAMP-driven calcium overload and microvascular spasm stun myocytes; the apex balloons while the base hypercontracts; function recovers over days to weeks.

· Tachycardia-induced cardiomyopathy is ventricular dysfunction caused by a persistently fast heart rate, most often atrial fibrillation or flutter with a rapid ventricular response, incessant supraventricular tachycardia, or very frequent premature ventricular contractions.

· A sustained high rate shortens diastole, which is the period of coronary perfusion and of calcium reuptake, while energy demand rises.

· The four myocardial changes are myocardial energy depletion, abnormal calcium handling, beta receptor downregulation, and subendocardial ischemia, which lead to dilation and a reduced ejection fraction resembling dilated cardiomyopathy.

· The shared feature that sets both conditions apart from the three classic cardiomyopathies is reversibility: takotsubo wall motion usually normalizes within weeks, and tachycardia-induced dysfunction usually recovers over weeks to months once the rate normalizes.

· The takotsubo mechanism is not fully settled, and catecholamine excess is the shared feature of all proposed explanations.

Distinctions that matter

· In takotsubo the myocytes are stunned rather than dead, which is why function returns; in dilated cardiomyopathy myocytes have been lost and replaced by scar.

· Rate itself can be the cause of a low ejection fraction, rather than only a consequence of a weak ventricle.

Common misconceptions

· Explaining apical ballooning by coronary occlusion. The akinetic region extends beyond any single coronary territory and the arteries are not obstructed.

· Treating a low ejection fraction in rapid atrial fibrillation as fixed muscle disease. Function often recovers once the rate is controlled.

Where it is taught: Section C06, first paragraph, the takotsubo mechanism flow, and the paragraph on tachycardia-induced cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C06-BD2

Brain dump 2 · integration and application: Separate a stunned ventricle from an infarcted one

A competent response contains

· Against infarction: the coronary arteries are normal, the akinetic region extends beyond the territory of any single coronary artery, the troponin rise is small relative to the large area of dysfunction, and the whole picture follows an acute emotional trigger with a long QT interval and deep T wave inversion.

· Biopsy shows contraction band necrosis, which is the pattern of catecholamine excess rather than of coronary occlusion.

· In six weeks her wall motion should be near normal, because catecholamine-stunned myocytes recover; they were not killed.

· An infarct of the same size would have left a permanent scar with persistent regional akinesis.

· Complications: heart failure and cardiogenic shock from the acutely weak apex and mid ventricle; dynamic outflow obstruction from the hypercontracting base narrowing the outflow tract; apical thrombus forming where blood stagnates against an akinetic apex; and ventricular arrhythmias, favored by the acute injury and the prolonged QT interval.

· The second patient has tachycardia-induced cardiomyopathy: an ejection fraction that recovers from 30% to 55% after rate control shows that the fast rate was the cause rather than primary muscle disease.

· Rate alone lowers the ejection fraction because a sustained high rate shortens diastole, which is the period of coronary perfusion and calcium reuptake, while raising energy demand, producing energy depletion, abnormal calcium handling, beta receptor downregulation, and subendocardial ischemia, with dilation and a fall in ejection fraction.

· Takotsubo mimics acute myocardial infarction at presentation with chest pain, dyspnea, ST elevation or deep T wave inversion, and a raised troponin.

Distinctions that matter

· Stunned myocardium recovers and infarcted myocardium scars, so the same initial tracing can have two very different outcomes.

· A troponin rise disproportionately small for the area of dysfunction is a useful pointer away from infarction.

Common misconceptions

· Expecting permanent dysfunction after takotsubo. Wall motion usually normalizes within weeks.

· Assuming a reduced ejection fraction during a sustained tachycardia means the arrhythmia was caused by a weak ventricle. The sequence can run the other way, and recovery after rate control shows which it was.

Where it is taught: Section C06, paragraph on takotsubo mimicking infarction, the stop and predict box, and the paragraph on tachycardia-induced cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-Q01

Board-style question: A systolic murmur that changes with position

Correct answer: C. Reduced left ventricular volume brings the anterior mitral leaflet closer to the hypertrophied septum

Why it is right: Standing lowers venous return and preload, so the left ventricle is smaller at the start of systole; a smaller cavity places the anterior mitral leaflet closer to the thickened septum, systolic anterior motion begins earlier, and the dynamic obstruction and its murmur both increase.

A is wrong: There is no fixed valvular narrowing in this patient, and in aortic stenosis standing reduces the volume crossing the orifice and makes the murmur softer rather than louder.

B is wrong: Standing reduces venous return rather than increasing it, and a larger end-diastolic volume would separate the leaflet from the septum and soften the murmur.

D is wrong: Contractility is not reduced on standing; sympathetic drive rises, and in this disease greater contractility worsens obstruction by emptying the ventricle further rather than by lengthening ejection.

A competent response contains

· Standing from squatting drops venous return, so left ventricular volume falls and the anterior mitral leaflet sits closer to the hypertrophied septum, worsening dynamic obstruction.

· Asymmetric septal thickening of 22 mm with a small cavity, a high ejection fraction, an S4, a murmur that does not radiate to the carotids, and a family history of sudden death all fit hypertrophic cardiomyopathy.

Distinctions that matter

· The obstruction is dynamic and depends on leaflet position, not on a fixed valve orifice.

Common misconceptions

· Treating every systolic murmur that changes with maneuvers as valvular. Here the obstruction is in the outflow tract and moves with cavity size.

Where it is taught: Section C02 of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C03-Q02

Board-style question: A thick wall with small complexes

Correct answer: A. Extracellular protein deposits add wall thickness without adding electrically active myocytes

Why it is right: Extracellular amyloid increases measured wall thickness while adding no depolarizing cells, and the myocytes that remain atrophy and are separated by protein, so less electrically active muscle is available and QRS voltage falls despite the thick wall.

B is wrong: Myocyte disarray belongs to hypertrophic cardiomyopathy, which shows high left ventricular voltage with septal Q waves rather than low voltage, and the thickening there is muscle.

C is wrong: Series addition of sarcomeres is eccentric hypertrophy, which enlarges the cavity and leaves a normal or thin wall with a reduced ejection fraction, not a thick wall with a preserved one.

D is wrong: The pericardium is not the problem in this patient; pericardial disease limits filling without producing a thick, speckled myocardial wall, and the thick wall here is the finding that needs explaining.

A competent response contains

· A thick, speckled wall with low limb lead voltage, huge atria, a preserved ejection fraction, and right-sided congestion with the Kussmaul sign fit cardiac amyloidosis.

· Amyloid fibrils deposit between myocytes, adding mass without electrically active cells, while myocytes atrophy, so depolarizing muscle falls and QRS voltage drops.

Distinctions that matter

· Wall thickness on imaging reports anatomy; QRS voltage reports how much electrically active muscle is present.

Common misconceptions

· Assuming any thick-walled ventricle must show high voltage. That holds for true hypertrophy and is reversed by infiltration.

Where it is taught: Section C03 of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C06-Q03

Board-style question: Akinesis beyond one coronary territory

Correct answer: D. Contraction band necrosis of myocytes without occlusive vascular disease

Why it is right: Stress cardiomyopathy is driven by a catecholamine surge, and biopsy in this condition shows contraction band necrosis, the histological signature of catecholamine excess, with no obstructive coronary disease to account for an akinetic region that extends beyond a single coronary territory.

A is wrong: An eosinophil-rich interstitial infiltrate indicates hypersensitivity myocarditis, which follows a drug exposure and usually comes with rash and fever rather than an acute emotional trigger and apical ballooning.

B is wrong: Congo red positive material with apple-green birefringence indicates cardiac amyloidosis, which presents with a thick wall, low QRS voltage, huge atria, and chronic right-sided congestion rather than acute akinesis with recovery.

C is wrong: Coagulative necrosis with neutrophils in a wedge-shaped region is the histology of myocardial infarction, which requires occlusive coronary disease and a region confined to one coronary territory, neither of which this patient has.

A competent response contains

· An acute emotional trigger, apical and mid ventricular akinesis with a hypercontracting base, a mildly raised troponin, a long corrected QT interval, and normal coronary arteries in a postmenopausal woman fit stress cardiomyopathy.

· The tissue pattern of catecholamine excess is contraction band necrosis, and there is no occlusive vascular disease to explain the dysfunction.

Distinctions that matter

· Contraction band necrosis reflects calcium overload from catecholamine excess, which is distinct from the coagulative necrosis of ischemic infarction.

Common misconceptions

· Expecting infarct histology because the tracing shows ST elevation. The arteries are open and the akinetic area crosses coronary territories.

Where it is taught: Section C06 of the CV-18-E Primary Myocardial Diseases notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-F Pericardial Disease

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-06-C01-BD1

Brain dump 1 · foundational retrieval: Pericardial layers and the findings of acute pericarditis

A competent response contains

· The pericardium has an outer fibrous sac of dense collagen and an inner serous sac; the parietal serous layer lines the fibrous sac and the visceral layer, the epicardium, covers the heart.

· Between the layers lies about 15 to 50 mL of serous fluid, a plasma ultrafiltrate that lets the heart move with little friction.

· The fibrous and parietal layers receive sensory fibers from the phrenic nerves (C3 to C5); the visceral layer is insensitive to pain.

· Pain is sharp and pleuritic because the inflamed parietal layer is stretched and rubbed with each heartbeat and breath; it worsens lying flat and with deep inspiration and eases on sitting up and leaning forward, which is thought to reduce pressure on the inflamed parietal surface.

· Phrenic innervation refers the pain to the trapezius ridge, a feature ischemic pain lacks.

· The friction rub is fibrin-coated layers grating together: scratchy, up to three components (atrial systole, ventricular systole, early diastolic filling), best heard at the lower left sternal border with the patient leaning forward, and it comes and goes.

· Low-grade fever, leukocytosis, and raised C-reactive protein reflect systemic inflammation.

· Most cases in developed countries are labeled idiopathic and presumed viral; coxsackie B virus is the classic cause, with echovirus, adenovirus, influenza, and HIV as others, in young and middle-aged adults, more often men.

· Raised vascular permeability lets fibrinogen escape and polymerize on both serous surfaces, giving a fibrinous or serofibrinous exudate that usually resolves without scarring.

Distinctions that matter

· Parietal and fibrous layers carry pain; the visceral layer does not, so the pain comes from the outer layers being rubbed.

· A rub can persist even once some fluid has separated the layers, so a rub and an effusion are not mutually exclusive.

Common misconceptions

· Saying the serous fluid is secreted. It is a plasma ultrafiltrate, which is why Starling forces alone can enlarge it.

· Attributing the trapezius pain to the heart itself. It follows the phrenic nerve supply to the fibrous and parietal layers.

Where it is taught: Section C01, first three paragraphs of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C01-BD2

Brain dump 2 · integration and application: Read the tracing back to the inflamed epicardium

A competent response contains

· The pericardium generates no electrical signal; the changes come from inflammation of the superficial myocardium beneath the visceral layer.

· Injured epicardial cells create a current of injury that elevates the ST segment in leads facing the epicardium.

· Because the whole epicardial surface is involved, elevation appears in most limb and precordial leads rather than in one coronary territory.

· Diffuse epicardial injury has no opposite wall to show reciprocal change, so there is no reciprocal depression outside aVR.

· Lead aVR looks into the cavities from above, so it shows the mirror image: ST depression and PR elevation.

· PR depression comes from atrial epicardial injury displacing the PR segment, so the atrial epicardium is involved as well.

· Evolution in order: diffuse ST elevation, then ST returns to baseline with T flattening, then T inversion, then normalization, over days to weeks.

· A troponin rise means the adjacent myocardium is inflamed, which is myopericarditis.

Distinctions that matter

· ST elevation here reports epicardial inflammation, not coronary occlusion, which is why it crosses territories.

· In pericarditis T inversion appears after the ST returns to baseline, while in infarction it often appears while the ST is still elevated.

Common misconceptions

· Expecting reciprocal depression in the opposite leads. With the whole surface injured there is no uninvolved wall to show it.

· Reading PR depression as a conduction problem. It is atrial epicardial injury displacing the segment.

Where it is taught: Section C01, electrocardiogram paragraph, the comparison table, and the stop and predict box of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C01-CP01

Comparison table: Acute pericarditis against ST elevation myocardial infarction

A competent response contains

· Distribution: pericarditis elevates ST diffusely in most leads; infarction elevates ST in one coronary territory.

· Shape: concave upward in pericarditis, often convex or straight in infarction.

· Reciprocal depression: only aVR and sometimes V1 in pericarditis; in the opposite leads in infarction.

· PR segment: depressed in pericarditis and elevated in aVR; usually unchanged in infarction.

· Q waves: absent in pericarditis; they develop with necrosis in infarction.

· T wave inversion: after the ST returns to baseline in pericarditis; often while the ST is still elevated in infarction.

· The shift arises from injured epicardium in pericarditis, and because the injury covers the whole heart there is no opposite wall to show reciprocal change apart from leads looking into the cavity.

Distinctions that matter

· Lead grouping is the first discriminator: one territory points to a vessel, most leads point to a surface.

· A PR shift belongs to pericardial inflammation and is not part of the infarct pattern.

Common misconceptions

· Treating ST shape as decisive on its own. Shape supports the pattern but distribution, PR shift, and Q waves carry more weight.

· Expecting Q waves to develop in pericarditis. There is no transmural necrosis, so they do not appear.

Where it is taught: Section C01, the table of acute pericarditis versus ST elevation myocardial infarction of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C01-Q01

Board-style question: Diffuse ST elevation with PR depression

Correct answer: C. Current of injury from inflammation of the epicardial myocardium across the whole cardiac surface

Why it is right: Inflammation of the superficial myocardium beneath the visceral layer creates a current of injury across the entire epicardial surface, so the ST segment is elevated in most limb and precordial leads, PR depression appears from atrial epicardial injury, and only aVR, which looks into the cavities, shows the mirror image.

A is wrong: A dominant right coronary artery occlusion would elevate the ST segment in the inferior and lateral leads with reciprocal depression in the opposite leads, and it would not produce anterior elevation with PR depression in every affected lead.

B is wrong: The fibrous and parietal pericardial layers generate no electrical signal, which is why the tracing reflects the underlying myocardium rather than the inflamed sac itself.

D is wrong: Beat to beat change in cardiac axis within pericardial fluid is electrical alternans, which alters QRS amplitude or axis on alternate beats; fluid between the heart and the electrodes lowers QRS voltage rather than elevating the ST segment.

A competent response contains

· Diffuse concave ST elevation with PR depression and the mirror image in aVR is acute pericarditis, most often idiopathic and presumed viral after a respiratory or gastrointestinal illness.

· The pericardium is electrically silent, so the ST shift comes from a current of injury in the epicardial myocardium under the visceral layer, which is why it spans coronary territories.

Distinctions that matter

· A coronary occlusion injures the wall supplied by one vessel; epicardial inflammation injures the whole surface.

Common misconceptions

· Looking for a single vessel to explain elevation in inferior, lateral, and anterior leads at once.

Where it is taught: Section C01 of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C02-BD1

Brain dump 1 · foundational retrieval: Build the secondary pericarditis table from memory

A competent response contains

· The exudate reflects how strongly and in what way the outside disease injures the serosa: mild permeability gives serous fluid, greater permeability lets fibrinogen escape and coat the layers, neutrophil-recruiting organisms give pus, damaged or fragile vessels and bleeding tendency add blood, and granulomatous infection gives caseous necrosis.

· The shaggy surface of severe fibrinous pericarditis is described as bread and butter.

· Early post-infarction pericarditis, about 1 to 3 days: necrosis of a transmural infarct inflames the overlying pericardium, giving fibrinous or fibrinohemorrhagic exudate over the infarct.

· Dressler syndrome, weeks after infarction: autoimmune reaction to released cardiac antigens, fibrinous, often with effusion.

· Cardiac surgery: direct trauma, and later a Dressler-like post-pericardiotomy syndrome, fibrinous or hemorrhagic.

· Uremia: retained metabolic toxins injure the serosa, fibrinous, and hemorrhagic with uremic platelet dysfunction.

· Systemic lupus erythematosus: immune complex deposition, serous or fibrinous. Rheumatoid arthritis: immune complexes and rheumatoid inflammation, fibrinous, may later constrict.

· Mediastinal radiation: endothelial and serosal injury, fibrinous acutely with fibrosis later.

· Tuberculosis: granulomatous response to Mycobacterium tuberculosis, caseous and often hemorrhagic. Pyogenic bacteria: spread from pneumonia or empyema, blood, or surgery, purulent. Cancer: tumor implants with fragile vessels and blocked lymphatics, hemorrhagic.

· Fibrinous and serofibrinous exudates are the most common; purulent and caseous exudates are the most likely to organize into constriction.

Distinctions that matter

· Immune complex disease gives serous or fibrinous fluid, while organisms that recruit neutrophils give pus, so the cell response rather than the severity decides between them.

· Hemorrhagic fluid has several routes: vessel damage, fragile tumor vessels, and a bleeding tendency such as uremic platelet dysfunction.

Common misconceptions

· Treating every post-infarction pericarditis as one entity. Timing separates direct inflammation at 1 to 3 days from the immune reaction weeks later.

· Expecting uremic pericarditis to be infectious. It comes from retained toxins.

Where it is taught: Section C02, opening paragraph and the table of causes and their usual exudate of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C02-BD2

Brain dump 2 · integration and application: Four patients, four exudates

A competent response contains

· Patient 1: early post-infarction pericarditis, fibrinous or fibrinohemorrhagic over the infarct, identified by the day 2 timing after a transmural infarct, localized and self-limited.

· Patient 1 would not be expected to have pericarditis with a subendocardial infarct, because the necrosis has to reach the epicardium to inflame the overlying pericardium.

· Patient 2: Dressler syndrome, fibrinous with effusion, identified by the delay of weeks plus fever and pleuritis; the delay fits priming of antibodies and T cells against released cardiac antigens rather than direct injury, and the same picture after cardiac surgery is post-pericardiotomy syndrome.

· Patient 3: uremic pericarditis from retained metabolic toxins, fibrinous and hemorrhagic; diffuse ST elevation is often absent because the inflammation tends not to involve the epicardial myocardium, and uremic platelet dysfunction makes the fluid bloody and tamponade more likely.

· Patient 4: bacterial pericarditis, usually Staphylococcus aureus or Streptococcus pneumoniae, purulent, reaching the sac by spread from pneumonia or empyema, acute and toxic.

· Patient 4 is the most likely of these to heal by dense fibrosis, since purulent and caseous exudates organize into constriction, as does radiation injury.

Distinctions that matter

· Early post-infarction pericarditis is a local response to necrosis; Dressler syndrome is systemic and immune, with fever and pleuritis as well.

· Absence of diffuse ST elevation does not exclude pericarditis, because the tracing depends on epicardial myocardial involvement rather than on the sac.

Common misconceptions

· Relying on the electrocardiogram to rule out pericarditis in renal failure.

· Expecting a purulent exudate to resolve cleanly like a viral fibrinous one. Pus organizes and can constrict.

Where it is taught: Section C02, post-infarction paragraph, the Dressler flow, and the final paragraph on uremia, bacteria, and tuberculosis of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C02-RT01

Retrieval exercise: Rebuild the Dressler syndrome sequence

A competent response contains

· Step 1: infarction or surgery injures myocardium and pericardium.

· Step 2: hidden myocardial antigens are released; they normally sit inside myocytes, so the immune system has not been exposed to them.

· Step 3: antibodies and T cells are primed over weeks, which is an adaptive response and explains the delay.

· Step 4: immune attack on pericardium and pleura.

· Step 5: fever, pleuritic pain, rub, and effusion appear weeks later, with a fibrinous exudate in the sac.

· The delay of weeks is the feature that separates Dressler syndrome from early post-infarction pericarditis at 1 to 3 days.

· The same process after cardiac surgery is called post-pericardiotomy syndrome.

Distinctions that matter

· Direct inflammation from necrosis is immediate and local; an adaptive immune response needs weeks and is systemic.

· Pleural involvement belongs to the immune process, not to early post-infarction pericarditis.

Common misconceptions

· Calling Dressler syndrome a second infarct or reinfarction. The injury has already happened and the attack is immune.

· Expecting it within days. Priming takes weeks, usually given as about 2 to 10 weeks.

Where it is taught: Section C02, the flow on Dressler syndrome as an immune reaction to injury of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C03-BD1

Brain dump 1 · foundational retrieval: The rigid shell and what it does to filling

A competent response contains

· Sequence: fibrinous, purulent, or caseous exudate; granulation tissue organizes the exudate; visceral and parietal layers fuse; dense fibrosis, sometimes with calcification; a rigid shell fixes total cardiac volume.

· Tuberculosis is a major cause worldwide; in developed countries most cases follow idiopathic or viral pericarditis, cardiac surgery, or mediastinal radiation, for example for Hodgkin lymphoma.

· The shell fixes total cardiac volume but the myocardium inside relaxes normally.

· In early diastole the ventricles fill very rapidly, driven by high atrial pressure, until they reach the limit of the shell, where filling stops abruptly.

· The abrupt halt makes the pericardial knock, an early diastolic sound that comes earlier and is higher pitched than an S3.

· On ventricular pressure tracings the same event gives a dip-and-plateau or square root pattern, and the jugular venous pulse shows a steep y descent.

· Diastolic pressures in all four chambers rise and become nearly equal.

· Systolic function is preserved but limited filling lowers stroke volume; high venous pressure gives hepatomegaly, ascites, and edema out of proportion to lung congestion.

· Plain films or CT may show pericardial calcification.

Distinctions that matter

· Filling is not reduced throughout diastole: it is very rapid early and then stops, which is what the dip-and-plateau tracing records.

· The knock is earlier and higher pitched than an S3, which is how the two are told apart at the bedside.

Common misconceptions

· Assuming the myocardium must be stiff for filling to be limited. In constriction the muscle relaxes normally and the sac sets the limit.

· Expecting pulmonary congestion to dominate. Systemic venous signs are out of proportion to lung findings.

Where it is taught: Section C03, first two blocks and the paragraph on the fixed shell of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C03-BD2

Brain dump 2 · integration and application: Kussmaul sign and telling the sac from the muscle

A competent response contains

· Normally inspiration lowers intrathoracic pressure, the fall is transmitted to the right atrium, venous return is drawn in, and the neck veins fall.

· In constriction the jugular venous pressure rises or fails to fall with inspiration, which is Kussmaul sign.

· Mechanism: the rigid shell blocks transmission of intrathoracic pressure to the heart, while descent of the diaphragm still pushes abdominal venous blood toward a right heart that cannot expand, so venous pressure rises.

· Reciprocal filling: the pulmonary veins lie outside the shell, so inspiration lowers their pressure but not left heart pressure, left filling falls, and the extra right ventricular volume pushes the septum left.

· Kussmaul sign is not specific; it also occurs in restrictive cardiomyopathy and right ventricular infarction.

· Constriction against restriction: stiffness in the pericardium against the myocardium; pericardium thickened and possibly calcified against normal; myocardial relaxation normal against impaired; exaggerated reciprocal respiratory variation against little; left and right diastolic pressures nearly equal against left usually higher; pulmonary hypertension usually mild against often marked; natriuretic peptide normal or mildly raised, because the shell limits myocardial stretch, against markedly raised; pericardial knock against a possible S3; low voltage with nonspecific T changes against low voltage with conduction disease suggesting amyloid.

· Restrictive cardiomyopathy comes from a myocardium infiltrated or scarred by amyloid, sarcoidosis, hemochromatosis, or endomyocardial fibrosis, with a normal pericardium.

· Both limit filling with a preserved ejection fraction and both give dip-and-plateau tracings, so the location of the stiffness explains every difference.

Distinctions that matter

· Natriuretic peptide separates the two because the shell prevents the myocardial stretch that drives its release, while an infiltrated myocardium is stretched and releases a great deal.

· Respiratory variation needs a fixed shared space; with a compliant pericardium each stiff ventricle limits only its own filling.

Common misconceptions

· Taking Kussmaul sign as proof of constriction. It appears in restrictive cardiomyopathy and right ventricular infarction too.

· Expecting marked pulmonary hypertension in constriction. It is usually mild, and marked pulmonary hypertension points to restriction.

Where it is taught: Section C03, the Kussmaul sign paragraph, the restrictive cardiomyopathy paragraph, the comparison table, and the stop and predict box of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C03-RT01

Retrieval exercise: Rebuild the path from pericarditis to constriction

A competent response contains

· Step 1: a fibrinous, purulent, or caseous exudate, and purulent and caseous exudates are the most likely to take this path, as is radiation injury.

· Step 2: granulation tissue organizes the exudate.

· Step 3: the visceral and parietal layers fuse and obliterate the pericardial space, so the normal 15 to 50 mL of lubricating fluid is lost.

· Step 4: dense fibrosis forms, sometimes with calcification that can be seen on a plain film or CT.

· Step 5: a rigid shell fixes total cardiac volume, while the myocardium inside still relaxes normally.

· Organization of exudate replaces a compliant sac with a fixed shell, so the pressure-volume relationship loses its reserve volume and pressure rises steeply almost at once.

Distinctions that matter

· Granulation tissue organizing an exudate is ordinary repair by fibrosis; what makes it dangerous here is the space it fuses shut.

· The shell changes the sac, not the muscle, which is the whole basis for separating constriction from restriction.

Common misconceptions

· Expecting every pericarditis to constrict. Fibrinous and serofibrinous exudates from viral disease usually resolve without scarring.

· Thinking calcification is required for constriction. It is common and helpful on imaging but not necessary.

Where it is taught: Section C03, the flow from pericarditis to constriction, with the fluid volume from C01 of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C04-BD1

Brain dump 1 · foundational retrieval: Effusion: the curve, the causes, and the signs

A competent response contains

· A pericardial effusion is fluid in excess of the normal 15 to 50 mL.

· The fibrous pericardium has a J-shaped pressure-volume curve: over the small reserve volume added fluid barely raises pressure, and beyond it collagen fibers straighten and pressure climbs steeply.

· Fluid added over minutes reaches the steep limb at once, so as little as about 150 to 200 mL can compress the heart.

· Fluid collecting over weeks to months lets the sac stretch and remodel, shifting the curve to the right, so a liter or more can accumulate with little rise in pressure.

· Serous effusions are transudates from shifted Starling forces: raised venous pressure in heart failure, or low plasma oncotic pressure in nephrotic syndrome and cirrhosis.

· Hypothyroidism produces a large, slowly forming, protein-rich effusion through increased capillary leak and reduced lymphatic drainage.

· Exudates come from the pericarditis causes in C02; hemorrhagic fluid points to trauma, free wall rupture, aortic dissection, malignancy, tuberculosis, uremia, or a bleeding tendency.

· Findings and mechanisms: muffled distant heart sounds because fluid separates the heart from the chest wall and dampens sound; a rub that may fade because fluid separates the inflamed layers, though it can persist; low-voltage QRS, under 5 mm in all limb leads, because fluid attenuates the signal; electrical alternans because the heart swings within a large effusion and changes its axis on alternate beats; an

enlarged globular water bottle silhouette, usually needing more than about 250 mL to show; an echo-free space on echocardiography where fluid separates the layers.

· Low voltage is not specific: obesity, emphysema, hypothyroidism, and amyloid also cause it. Electrical alternans in this setting suggests a large effusion and possible tamponade.

Distinctions that matter

· Transudate against exudate is decided by mechanism: pressure or oncotic imbalance against increased permeability from inflammation.

· The signs of effusion report how much fluid is present; whether filling is impaired depends on pericardial pressure.

Common misconceptions

· Reading a large silhouette or low voltage as evidence of compression. Both describe fluid, not pressure.

· Expecting a small effusion to be harmless. Over minutes about 150 to 200 mL is enough to reach the steep limb.

Where it is taught: Section C04, first two paragraphs and the table of findings and their mechanism of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C04-BD2

Brain dump 2 · integration and application: Two patients, and why volume is the wrong number

A competent response contains

· Patient A sits far to the right on a curve that has shifted right: four months of slow stretch remodeled and enlarged the sac, so pericardial pressure stayed below right atrial pressure, about 2 to 6 mmHg, and filling continued despite 1,200 mL.

· Patient B sits on the steep limb of an unstretched curve: 200 mL in 20 minutes gave the sac no time to stretch, so pericardial pressure rose past right atrial and right ventricular diastolic pressure.

· The chamber with the lowest pressure stops filling first, which is the thin-walled low-pressure right heart, and output falls.

· Patient A is likely to show an enlarged globular water bottle silhouette on the chest film and low-voltage QRS complexes, possibly with electrical alternans, with distant heart sounds.

· Patient B has too little fluid to enlarge the silhouette, since more than about 250 mL is usually needed, and shows hypotension with distended neck veins.

· The lung fields are clear in patient A because left heart pressures are not raised, so a large chronic effusion can be well tolerated while still producing the effusion signs.

· The rule: the speed of accumulation predicts compression better than the volume, and judging by volume alone would have predicted the opposite outcome in this pair.

Distinctions that matter

· Position on the curve, not volume, decides whether pericardial pressure exceeds right atrial pressure.

· Signs of a large effusion and signs of compression are different sets: patient A has the first and patient B the second.

Common misconceptions

· Expecting the larger effusion to be the dangerous one.

· Expecting pulmonary edema with a large effusion. Left heart filling pressures are not raised, so the lung fields stay clear.

Where it is taught: Section C04, the example on why rate matters more than volume, with the findings table of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C04-PR01

Prediction exercise: Predict the effect of a slowly accumulating effusion

A competent response contains

· The curve shifts to the right, because slow stretch over months lets the sac remodel and enlarge its reserve volume.

· Pericardial pressure stays below right atrial pressure, about 2 to 6 mmHg, so the sac is still on the flat part of its curve.

· Diastolic filling continues and stroke volume is maintained, so blood pressure is normal.

· QRS amplitude falls, giving low voltage in the limb leads, because fluid between the heart and the electrodes attenuates the signal, and hypothyroidism itself also lowers voltage.

· The silhouette is enlarged and globular, the water bottle shape, since the volume is far above the roughly 250 mL needed to show.

· The lung fields are clear, because left heart filling pressures are not raised.

· The same volume in one hour would reach the steep limb of an unstretched curve, pericardial pressure would exceed right heart diastolic pressure, and tamponade with hypotension and distended neck veins would follow.

Distinctions that matter

· A large effusion and a high pericardial pressure are separate things, and only the second impairs filling.

· Low voltage here has two contributors, the fluid and the hypothyroidism, which is part of why the sign is not specific.

Common misconceptions

· Predicting tamponade from the volume alone.

· Expecting hypothyroid effusions to be thin transudates. They are protein-rich and form slowly through capillary leak and reduced lymphatic drainage.

Where it is taught: Section C04, the pressure-volume paragraph, the example, and the findings table of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-BD1

Brain dump 1 · foundational retrieval: Tamponade: definition, causes, and the fall in output

A competent response contains

· Tamponade is a hemodynamic state, not a fluid volume: it exists once pericardial pressure rises enough to impair filling.

· Any effusion can cause it; the classic acute cause is hemopericardium, and nontraumatic causes include malignancy, uremia, and pericarditis.

· Penetrating trauma often enters the right ventricle, the most anterior chamber.

· Free wall rupture after transmural infarction occurs most often about 3 to 7 days after the event, when macrophages have softened the necrotic wall, and causes sudden tamponade with pulseless electrical activity.

· Type A aortic dissection can rupture backward into the sac, because the proximal ascending aorta lies inside the pericardial reflection.

· Sequence: fluid or blood collects faster than the sac can stretch; pericardial pressure exceeds right heart diastolic pressure; the right atrium and ventricle collapse in diastole; diastolic pressures in all chambers equalize with pericardial pressure; preload and stroke volume fall with tachycardia compensating; hypotension, distended neck veins, and distant heart sounds follow.

· The thin-walled, low-pressure right heart is compressed first.

· Right atrial, right ventricular diastolic, pulmonary artery diastolic, and pulmonary capillary wedge pressures become nearly equal to pericardial pressure.

· Beck triad: hypotension from low stroke volume, distended neck veins from high venous pressure, and distant heart sounds from the fluid.

· Sinus tachycardia is the compensating response, and electrical alternans may appear.

Distinctions that matter

· Tamponade is defined by pressure, so a small rapid bleed can cause it and a large chronic effusion may not.

· Equalization of diastolic pressures is the hallmark of a shared tense space, and it happens in constriction as well.

Common misconceptions

· Naming a threshold volume for tamponade. There is none, because the rate of accumulation and the state of the sac decide it.

· Expecting the left ventricle to fail first. The low-pressure right heart is compressed first.

Where it is taught: Section C05, causes paragraph, the flow on rising pericardial pressure, and the equal pressures paragraph of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-BD2

Brain dump 2 · integration and application: Separate fluid under pressure from a fixed shell

A competent response contains

· In tamponade filling occurs mainly during ventricular systole, when ejection briefly lowers intrapericardial volume and pressure.

· The jugular x descent is preserved, because it coincides with that systolic fall in pericardial pressure; the y descent is blunted or absent, because pericardial pressure stays above right atrial pressure after the tricuspid valve opens, so early diastolic filling cannot occur.

· Comparison: onset acute or subacute against chronic; pulsus paradoxus usual against less common; Kussmaul sign usually absent against typical; y descent blunted or absent against steep; filling impaired throughout diastole against mid and late diastole only; diastolic pressures equalize in both; no extra heart sound against a pericardial knock.

· Fluid under pressure compresses the heart throughout diastole, while a shell allows rapid early filling and stops it only once the limit is reached.

· Kussmaul sign is usually absent in tamponade because the fluid still transmits the inspiratory fall in intrathoracic pressure to the right atrium.

· Pulsus paradoxus is measured by inflating the cuff above systolic pressure and deflating slowly during quiet breathing: the first pressure is where Korotkoff sounds appear in expiration only, the second is where they are heard with every beat in both phases, and the difference is the pulsus; more than 10 mmHg is abnormal.

· Mechanism: inspiration lowers intrathoracic pressure, venous return to the right ventricle increases, the right ventricle cannot expand outward against the tense sac, the septum bulges into the left ventricle, the pulmonary veins also hold more blood, left ventricular filling and stroke volume fall, and systolic pressure drops more than 10 mmHg.

· Other causes: severe asthma and COPD through large swings in intrathoracic pressure, and massive pulmonary embolism through right ventricular strain.

Distinctions that matter

· The y descent reports early diastolic filling, which is why it is lost with fluid and steep with a shell.

· Pulsus paradoxus and Kussmaul sign point in opposite directions between the two diseases, because fluid transmits intrathoracic pressure and a rigid shell does not.

Common misconceptions

· Treating pulsus paradoxus as specific for tamponade.

· Expecting Kussmaul sign in tamponade. The fluid still transmits the inspiratory pressure fall to the right atrium.

Where it is taught: Section C05, the paragraph on equal pressures and filling, the pulsus paradoxus flow and example, and the tamponade against constriction table of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-RT01

Retrieval exercise: Rebuild pulsus paradoxus

A competent response contains

· Step 1: inspiration lowers intrathoracic pressure.

· Step 2: venous return to the right ventricle increases.

· Step 3: the right ventricle cannot expand outward, because the tense sac fixes total cardiac volume; the principle linking the ventricles is ventricular interdependence.

· Step 4: the septum bulges into the left ventricle.

· Step 5: the pulmonary veins, which lie in the chest outside the sac, also hold more blood on inspiration.

· Step 6: left ventricular filling and stroke volume fall.

· Step 7: systolic pressure drops, and a fall of more than 10 mmHg is abnormal.

· Total cardiac volume is fixed through the respiratory cycle, so any gain in right heart filling becomes a loss in left heart filling.

Distinctions that matter

· Two things reduce left ventricular filling on inspiration: the septum shifting and blood held in the pulmonary veins.

· The normal small inspiratory fall in systolic pressure is present in everyone; interdependence inside a tense sac is what makes it large.

Common misconceptions

· Thinking the pulse disappears on expiration. The beats lost are the inspiratory ones, which carry the smallest left ventricular stroke volume.

· Expecting the right ventricle to expand outward into the sac. The sac is tense, so the only room available is across the septum.

Where it is taught: Section C05, the flow on pulsus paradoxus in tamponade and the example on measuring it of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-CA01

Clinical application: Sudden hypotension six days after an anterior infarction

A competent response contains

· Recognize: free wall rupture of the infarcted left ventricle has filled the sac with blood, producing acute cardiac tamponade with pulseless electrical activity; day 6 fits rupture, which occurs most often about 3 to 7 days after a transmural infarct, when macrophages have softened the necrotic wall.

· Explain: blood collects faster than the sac can stretch, pericardial pressure exceeds right heart diastolic pressure, the right atrium and right ventricle collapse in diastole, and right atrial, right ventricular diastolic, pulmonary artery diastolic, and pulmonary capillary wedge pressures equalize with pericardial pressure; the ventricles cannot fill, so stroke volume approaches zero while the conduction system keeps producing organized complexes, which is pulseless electrical activity.

· Differentiate: acute left ventricular failure would raise left-sided pressures and give pulmonary congestion with crackles rather than clear lung fields, and it would not show diastolic collapse of the right chambers; constrictive pericarditis is chronic, gives a pericardial knock and a steep y descent with Kussmaul sign, and the acute onset, pulsus paradoxus, and echo-free space here separate this patient from it.

· Investigate: on echocardiography look for an echo-free space around the heart and diastolic collapse of the right atrium and right ventricle; at the bedside look for Beck triad with hypotension, distended neck veins, and distant heart sounds, pulsus paradoxus above 10 mmHg, sinus tachycardia, and possible electrical alternans, with equal diastolic pressures across the chambers.

· Decide: the fibrous pericardium has a J-shaped pressure-volume curve with only a small reserve volume, and bleeding over minutes gave the sac no time to stretch, so the sac is on the steep limb of the curve where pressure climbs sharply, which is why as little as about 150 to 200 mL is enough.

· Justify: inspiration increases right ventricular filling, the right ventricle cannot expand against the tense sac, the septum bulges into the left ventricle, and the pulmonary veins hold more blood, so left ventricular filling and stroke volume fall and systolic pressure drops by more than 10 mmHg; the jugular venous pressure does not rise with inspiration here because fluid still transmits the inspiratory fall in intrathoracic pressure to the right atrium, while the rigid shell of constriction blocks that transmission and produces Kussmaul sign.

Distinctions that matter

· Clear lung fields with distended neck veins and hypotension separate compression of the whole heart from left ventricular pump failure.

· An organized rhythm with no pulse is a filling problem here, not a conduction problem.

Common misconceptions

· Assuming a large volume of blood is needed. Speed decides, and about 150 to 200 mL added acutely is enough.

· Expecting Kussmaul sign in any patient with raised venous pressure and impaired filling. It belongs to the rigid shell, not to fluid.

Where it is taught: Section C05, causes paragraph, both flows, the measurement example, the comparison table, and the stop and predict box of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C05-Q02

Board-style question: Blunted y descent in a tense sac

Correct answer: A. Pericardial pressure remains above right atrial pressure after the tricuspid valve opens, so early diastolic filling cannot occur

Why it is right: Pericardial pressure remains above right atrial pressure once the tricuspid valve opens, so there is no early diastolic emptying of the atrium and the y descent is lost, while the x descent persists because ejection briefly lowers intrapericardial volume and pressure during systole.

B is wrong: A rigid fibrocalcific pericardium permits very rapid early diastolic filling until the shell is reached, which produces a steep y descent and a pericardial knock, neither of which is present here.

C is wrong: Loss of atrial contraction would remove the a wave and the late diastolic contribution to filling rather than the y descent, and this patient is in sinus rhythm.

D is wrong: Impaired myocardial relaxation from amyloid infiltration is restrictive cardiomyopathy, which leaves the pericardium normal, raises left-sided pressure above right, and does not blunt the y descent.

A competent response contains

· Hypotension, distended neck veins, distant heart sounds, pulsus paradoxus above 10 mmHg, and equal diastolic pressures in a patient with malignancy indicate cardiac tamponade.

· Filling occurs mainly during ventricular systole, when ejection lowers intrapericardial volume and pressure, so the x descent survives and the y descent, which reports early diastolic filling, is lost.

Distinctions that matter

· A blunted or absent y descent belongs to tamponade; a steep y descent belongs to constriction.

Common misconceptions

· Reading the absent y descent as a myocardial relaxation problem. The ventricle would relax normally if pericardial pressure allowed it to fill.

Where it is taught: Section C05 of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C06-BD1

Brain dump 1 · foundational retrieval: How tumor reaches the heart and what it does there

A competent response contains

· Autopsy series find cardiac or pericardial metastases in roughly 5 to 10% of patients who die of cancer, and metastases are far more common than primary cardiac tumors such as myxoma.

· Sites in order of frequency: pericardium, then epicardium, then myocardium.

· Direct extension: lung carcinoma, breast carcinoma, esophageal carcinoma, mediastinal tumors.

· Retrograde lymphatic spread through mediastinal nodes: lung and breast carcinoma, lymphoma.

· Hematogenous seeding: melanoma, leukemia, lymphoma, sarcomas.

· Growth along the inferior vena cava into the right atrium: renal cell carcinoma, hepatocellular carcinoma.

· Proximity explains lung and breast, which lie next to the heart and are common, so they account for the largest numbers of cases; wide bloodstream spread explains melanoma and the hematologic cancers.

· Melanoma is less common but involves the heart in a high proportion of patients with widespread disease.

· Leukemia and lymphoma often infiltrate the heart.

Distinctions that matter

· Largest number of cases and highest proportion of patients involved are different measures: lung and breast lead the first, melanoma the second.

· Growth along the inferior vena cava is a route into a chamber rather than onto the serosa, which is why it causes obstruction rather than effusion.

Common misconceptions

· Thinking of myxoma first when a cardiac mass or pericardial effusion appears in a patient with cancer. Metastases greatly outnumber primary tumors.

· Assuming the myocardium is the usual site. The pericardium is the most frequent.

Where it is taught: Section C06, first paragraph and the table of routes of spread of the CV-18-F Pericardial Disease notes.

NPLEX-P1-CARD-18-06-C06-BD2

Brain dump 2 · integration and application: The effusion that keeps coming back

A competent response contains

· Sequence: tumor reaches the pericardium by extension, lymphatics, or blood; serosal implants with fragile new vessels bleed; lymphatic obstruction blocks fluid drainage; a bloody exudate with malignant cells accumulates; pericardial pressure rises until filling fails.

· The two mechanisms are bleeding from fragile leaky tumor vessels on the serosa and blockage of the mediastinal lymphatics that normally drain pericardial fluid.

· The result is a hemorrhagic exudate, and finding malignant cells on cytology of the pericardial fluid confirms tumor as the cause.

· Effusion forms faster than it drains, so it tends to recur and progress, which is why a rapidly growing effusion or one that reaccumulates after drainage raises suspicion of tumor.

· Malignancy is one of the most common causes of nontraumatic tamponade.

· A pericardial problem in a patient with cancer is not always tumor: mediastinal radiation given for the same cancer can later cause radiation pericarditis and constriction, and tumor encasing the heart can behave like constrictive pericarditis.

· Patients usually present with progressive dyspnea, chest discomfort, and the effusion signs in C04.

· Myocardial deposits are less often symptomatic but can disturb conduction and cause arrhythmias, and intracavitary tumor can obstruct flow.

Distinctions that matter

· Tumor can limit filling two ways: fluid under pressure in the sac, or a tumor shell around the heart that imitates constriction.

· Reaccumulation after drainage is the feature that separates a malignant effusion from a self-limited inflammatory one.

Common misconceptions

· Assuming every effusion in a cancer patient is malignant. Radiation pericarditis from earlier treatment is also a cause.

· Expecting a transudate. Tumor vessels bleed and lymphatics are blocked, so the fluid is a hemorrhagic exudate.

Where it is taught: Section C06, the paragraphs on pericardial deposits and clinical picture, the flow to tamponade, and the final paragraph of the CV-18-F Pericardial Disease notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-G Congenital Heart Conditions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-07-C01-BD1

Brain dump 1 · foundational retrieval: Draw the three left-to-right shunts and label the pressures

A competent response contains

· After birth the left ventricle runs at about 120/10 mm Hg and the right ventricle at about 25/5 mm Hg, so a VSD, an ASD or a PDA sends oxygenated blood back toward the lungs.

· Blood crosses a defect toward the side with lower pressure, or, between filling chambers, toward the side that fills more easily.

· VSD and PDA are driven by the systolic and diastolic pressure difference; an ASD is driven mainly by the greater compliance of the thin-walled right ventricle.

· Qp:Qs is the ratio of pulmonary blood flow to systemic blood flow; above 1 means a net left-to-right shunt and below 1 a net right-to-left shunt.

· Systemic arterial blood stays fully saturated in all three because no desaturated blood reaches the aorta, so the child is not cyanotic.

· The cost of a left-to-right shunt is volume: the lungs and the chambers carrying the recirculated blood receive more flow than the body uses.

· A defect before the tricuspid valve (ASD) adds flow at low pressure; a defect after it (VSD, PDA) transmits systemic pressure as well as extra flow, so pulmonary vascular disease develops earlier and more often with VSD and PDA.

Distinctions that matter

· Pressure drives the ventricular and ductal shunts; compliance drives the atrial shunt, which is why atrial pressures only a few mm Hg apart still produce a large shunt.

· Extra flow and transmitted pressure are separate insults to the pulmonary arteries, and where the defect sits relative to the tricuspid valve decides which the lungs receive.

Common misconceptions

· Calling any shunt a cause of cyanosis. A left-to-right shunt leaves systemic saturation normal because the extra blood goes to the lungs, not to the body.

· Describing the ASD shunt as pressure driven. Left and right atrial pressures differ by only a few mm Hg; ventricular compliance is what moves the blood.

Where it is taught: Section C01, first two paragraphs and the shunt categories table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C01-BD2

Brain dump 2 · integration and application: Explain every Eisenmenger finding from the reversal

A competent response contains

· Chronic high flow and pressure injure the pulmonary endothelium: nitric oxide and prostacyclin fall and endothelin-1 rises.

· The arterioles show medial hypertrophy, then intimal proliferation and fibrosis, and finally plexiform lesions, the same changes as in pulmonary arterial hypertension.

· As PVR rises the right ventricle hypertrophies, the shunt shrinks, becomes bidirectional, and finally reverses when right-sided pressure exceeds left-sided pressure.

· Cyanosis appears late because it begins only once desaturated blood reaches the systemic arteries.

· Oxygen barely helps because the shunted blood never passes the alveoli.

· Renal hypoxia raises erythropoietin, giving secondary polycythemia, which raises viscosity and makes the cyanosis more visible.

· Clubbing accompanies chronic hypoxemia; a leading explanation is that platelet clumps bypass the pulmonary capillaries and release growth factors in the fingertips.

· A loud P2 reflects raised pulmonary arterial pressure, and the original murmur softens because the pressure difference across the defect has disappeared.

· An ASD of the same size adds flow at low pressure, so pulmonary vascular disease develops later and in a minority of patients.

· Advanced remodeling, once plexiform lesions have formed, is irreversible.

Distinctions that matter

· A softening murmur in this setting signals worsening disease rather than improvement, because murmur loudness tracks the gradient across the defect.

· Polycythemia here is secondary and appropriate to hypoxia, not a primary marrow disorder.

Common misconceptions

· Expecting supplemental oxygen to correct the cyanosis. The shunted blood bypasses the alveoli entirely.

· Treating the reversal as the start of the disease. The reversal is the end stage of years of pulmonary vascular remodeling.

Where it is taught: Section C01, third and fifth paragraphs and the flow diagram of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C01-RT01

Retrieval exercise: Rebuild the path from shunt to reversal

A competent response contains

· Step 1: a left-to-right shunt overloads the pulmonary circulation.

· Step 2: high flow and pressure injure the pulmonary endothelium.

· Step 3: medial hypertrophy, intimal fibrosis and plexiform lesions raise PVR.

· Step 4: the right ventricle hypertrophies as its pressure rises.

· Step 5: right-sided pressure exceeds left-sided pressure and the shunt reverses.

· Step 6: late cyanosis, clubbing and polycythemia appear.

· Plexiform lesions mark advanced, irreversible remodeling.

· Reversal requires right-sided pressure to exceed left-sided pressure, so the arrow moves left to right early, both ways at the midpoint, and right to left at the end.

· Systemic arterial saturation is normal early, mildly reduced when the shunt is bidirectional, and clearly low after reversal.

Distinctions that matter

· Resistance rises before the shunt direction changes, so a shrinking shunt is an early sign of the same process that later reverses it.

Common misconceptions

· Drawing a sudden switch from left-to-right to right-to-left. The shunt passes through a bidirectional stage as pressures equalize.

Where it is taught: Section C01, flow diagram on Eisenmenger syndrome of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C01-PR01

Prediction exercise: Predict what happens as pulmonary vascular resistance falls after birth

A competent response contains

· Shunt volume increases, because falling PVR lowers the resistance to pulmonary flow while systemic resistance is unchanged.

· Qp:Qs rises further above 1 as pulmonary flow climbs relative to systemic flow.

· Pulmonary blood flow increases, and the recirculated volume returns through the left atrium and left ventricle, so both dilate.

· In a large nonrestrictive defect the murmur is soft or short and does not become loud, because ventricular pressures are nearly equal and there is little gradient to generate turbulence; an apical mid-diastolic rumble from high mitral flow and a loud P2 appear instead.

· By the second month the infant shows heart failure: tachypnea, sweating and fatigue with feeds, poor weight gain, and hepatomegaly.

· Systemic arterial saturation stays normal, because the shunt is left to right and no desaturated blood reaches the aorta.

Distinctions that matter

· Falling PVR explains why a large shunt causes few symptoms at birth and heart failure within one to two months.

· Volume overload and hypoxemia are different burdens; this infant has the first and not the second.

Common misconceptions

· Expecting a large defect to produce a loud murmur. Loudness follows the gradient, so the loudest defects are often the smallest.

· Expecting cyanosis as the shunt grows. More flow to the lungs does not desaturate systemic blood.

Where it is taught: Section C01, second paragraph; Section C02, final paragraph and the small versus large VSD table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C02-BD1

Brain dump 1 · foundational retrieval: Build the ventricular septal defect from the septum outward

A competent response contains

· The interventricular septum has a thick muscular part and a small membranous part sitting just below the aortic valve.

· The membranous septum forms last from fusion of the endocardial cushions, the conotruncal ridges, and the muscular septum, so it is most often left incomplete.

· Most VSDs are membranous or perimembranous; muscular defects lie within the trabeculated septum and may be multiple.

· Left ventricular pressure exceeds right from the start of isovolumetric contraction to the end of ejection, so blood crosses throughout systole.

· The murmur is harsh and holosystolic, starts with S1 and often obscures it, is loudest at the left lower sternal border in the third and fourth intercostal spaces, and frequently comes with a thrill.

· Sustained handgrip raises systemic resistance, increases the left-to-right gradient, and makes the murmur louder.

· Shunted blood leaves the left ventricle in systole and passes almost directly into the pulmonary artery, so the volume load falls on the pulmonary circulation, the left atrium and the left ventricle.

· The right ventricle enlarges mainly when pulmonary pressure rises, not from the shunt volume itself.

· Associations: trisomy 21, fetal alcohol exposure, other chromosomal syndromes, and VSD is one of the four components of tetralogy of Fallot.

· VSD is the most common congenital heart defect at birth when bicuspid aortic valve is not counted.

Distinctions that matter

· A VSD loads the left atrium and left ventricle even though the defect lies between the ventricles, because the shunted blood is routed through the lungs and back to the left heart.

· Membranous and muscular defects differ in location and in how often they close, not in the direction of the shunt.

Common misconceptions

· Expecting right ventricular enlargement from the shunt volume. The blood crosses during systole and leaves through the pulmonary artery rather than filling the right ventricle.

· Placing the membranous septum away from the aortic valve. It sits just below it, which is why a cusp can prolapse into a defect there.

Where it is taught: Section C02, first three paragraphs of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C02-BD2

Brain dump 2 · integration and application: Two infants, two murmurs, two courses

A competent response contains

· Infant A has the smaller restrictive defect: a large pressure gradient persists across it, right ventricular pressure stays low, and the high-velocity jet makes a loud harsh holosystolic murmur with a thrill.

· Infant B has the larger nonrestrictive defect: the gradient is small, right ventricular pressure approaches left ventricular pressure, so the murmur is softer and shorter, with an apical mid-diastolic rumble from high mitral flow and a loud P2.

· In a small defect the shunt size is limited by the defect itself; in a large defect ventricular pressures equalize and the shunt is set by the ratio of pulmonary to systemic resistance.

· Infant A's defect, especially a muscular one, often closes spontaneously in early childhood as surrounding muscle grows or as tricuspid septal leaflet tissue covers a perimembranous opening; endocarditis remains a risk if it persists.

· Infant B has the findings of heart failure as PVR falls, and without correction pulmonary vascular disease can become irreversible within the first few years, followed by Eisenmenger reversal.

· A defect just below the aortic valve can let an aortic cusp prolapse into it, adding aortic regurgitation.

· The high-velocity jet damages the right ventricular endocardium where it strikes, creating a site for endocarditis.

Distinctions that matter

· Murmur loudness reports the pressure gradient, not the size of the hole, so the loudest VSDs are often the smallest.

· Spontaneous closure belongs to small defects; irreversible pulmonary vascular disease belongs to large ones.

Common misconceptions

· Reading a soft murmur as a small defect. A soft murmur with a loud P2 and failure to thrive points to a large defect with nearly equal ventricular pressures.

· Assuming the mid-diastolic rumble means mitral stenosis. It comes from the large volume crossing a normal mitral valve.

Where it is taught: Section C02, small versus large VSD table and final paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-BD1

Brain dump 1 · foundational retrieval: Build the atrial septum, then explain the fixed split S2

A competent response contains

· The septum primum grows toward the endocardial cushions and closes the ostium primum, while perforations in its upper part create the ostium secundum.

· The septum secundum then grows on the right side of the septum primum, leaving the foramen ovale, whose flap valve is the septum primum.

· In the fetus high right atrial pressure pushes the flap open; after birth left atrial pressure rises above right and presses it shut, and the two septa usually fuse over the following months.

· An ASD shunts left to right mainly because the thin-walled right ventricle is more compliant and fills more easily than the left, not because of a large pressure difference.

· The extra volume passes through the right atrium, right ventricle and pulmonary arteries and enlarges them, while the left atrium decompresses through the defect.

· Flow across the defect itself is low velocity and silent.

· The systolic ejection murmur comes from increased flow across a normal pulmonic valve and is heard at the left upper sternal border; a large shunt adds a mid-diastolic rumble from increased tricuspid flow.

· Normally inspiration increases right heart venous return and delays pulmonic closure, so S2 splits in inspiration and narrows in expiration.

· In an ASD the right ventricle already carries a large stroke volume in both phases: in inspiration the extra systemic venous return is offset by less left-to-right shunting, and in expiration the shunt grows again.

· Right ventricular volume, and therefore the delay of P2, stays nearly constant, giving a wide fixed split S2.

Distinctions that matter

· The murmur of an ASD is a pulmonic flow murmur, not the sound of blood crossing the defect.

· Wide splitting and fixed splitting are different statements: the width comes from the large right ventricular stroke volume, the fixity from the offsetting of venous return and shunt flow.

Common misconceptions

· Explaining the left-to-right atrial shunt by atrial pressure alone. Ventricular compliance is the main driver.

· Expecting the defect itself to be audible. Velocity across it is too low to generate turbulence.

Where it is taught: Section C03, first paragraph and the paragraphs on shunt direction and the split S2 of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-BD2

Brain dump 2 · integration and application: A stroke at 34 with no murmur

A competent response contains

· A patent foramen ovale explains the stroke: the septum primum and septum secundum never fused, so the flap is unsealed although no tissue is missing.

· While left atrial pressure exceeds right, the flap is held against the septum secundum, no blood crosses, and there is neither a murmur nor volume overload.

· Transient reversal occurs when right atrial pressure exceeds left, as on release of a Valsalva maneuver, with coughing or straining, or with acute pulmonary embolism.

· A deep vein thrombus can then cross to the left atrium and reach the systemic arteries, causing stroke or limb ischemia; this is a paradoxical embolism.

· A patent foramen ovale is present in about one in four adults, which is why an unexplained stroke in a young adult raises suspicion of a PFO or ASD.

· Adult presentations of an unrecognized ASD: atrial fibrillation or flutter, exercise intolerance, and right heart failure, with Eisenmenger syndrome in a minority.

· The rhythm problem follows right atrial stretch from years of extra volume.

· An ASD is missing septal tissue; a patent foramen ovale is complete tissue that never sealed.

· An ostium primum ASD is the lesion with a cleft anterior mitral leaflet and mitral regurgitation, and it is associated with trisomy 21.

Distinctions that matter

· A patent foramen ovale allows embolism without any resting shunt, while an ASD carries a continuous shunt and enlarges the right heart.

· Secundum and primum defects are both tissue deficits; only the primum defect involves the endocardial cushions and the mitral valve.

Common misconceptions

· Assuming any intracardiac communication must cause a murmur. A PFO is silent because nothing crosses it at rest.

· Expecting an ASD to declare itself in childhood. Many are found only in adulthood, often through an arrhythmia.

Where it is taught: Section C03, atrial septal lesions table, final paragraph, and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-CP01

Comparison table: The three lesions of the atrial septum

A competent response contains

· Secundum ASD: excess resorption of the septum primum or a septum secundum too small to cover the ostium secundum; a true tissue deficit at the fossa ovalis; the most common ASD; fetal alcohol exposure and Holt-Oram syndrome with upper limb defects; no specific valve lesion; ECG right axis deviation with incomplete right bundle branch block; a resting shunt large enough to overload the right heart.

· Primum ASD: the septum primum fails to meet the endocardial cushions, making a partial atrioventricular septal defect; a tissue deficit; associated with trisomy 21; usually a cleft anterior mitral leaflet with mitral regurgitation; ECG left or superior axis deviation; a resting shunt is present.

· Patent foramen ovale: the septum primum and septum secundum fail to fuse, so no tissue is missing and the flap is simply unsealed; present in about one in four adults; no specific syndrome association; no valve lesion; no characteristic ECG change; no resting shunt while left atrial pressure exceeds right.

· Axis direction is the quick separator between the two ASDs: right axis deviation for secundum, left or superior axis deviation for primum.

Distinctions that matter

· Missing tissue against unsealed tissue separates an ASD from a patent foramen ovale and explains why only one of them overloads the right heart.

· The primum defect is an endocardial cushion lesion, which is why it carries a mitral valve abnormality and the secundum defect does not.

Common misconceptions

· Treating a patent foramen ovale as a small ASD. It is unfused complete tissue with no resting shunt.

· Assigning right axis deviation to both ASDs. The primum defect gives a left or superior axis.

Where it is taught: Section C03, table of atrial septal lesions of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-BD1

Brain dump 1 · foundational retrieval: The ductus arteriosus from arch to ligament

A competent response contains

· The ductus arteriosus develops from the distal part of the left sixth aortic arch.

· It joins the pulmonary trunk near its bifurcation, at the root of the left pulmonary artery, to the aortic arch just distal to the origin of the left subclavian artery.

· In the fetus pulmonary vascular resistance is high, so most right ventricular output bypasses the lungs through the ductus into the descending aorta.

· Its smooth muscle is held relaxed by low fetal oxygen tension and by prostaglandin E2, produced by the placenta and the ductus and poorly cleared by the unventilated fetal lungs.

· Closure sequence: first breaths expand the lungs and PVR falls; arterial PO2 rises and constricts ductal smooth muscle; the placenta is removed and the perfused lungs clear PGE2; the ductus closes functionally within the first few days; intimal proliferation and fibrosis form the ligamentum arteriosum.

· Prematurity is the major risk factor, because immature ductal muscle constricts weakly to oxygen and stays sensitive to PGE2.

· Congenital rubella, especially first trimester infection, causes PDA with cataracts and sensorineural deafness.

· Birth at high altitude, where arterial PO2 is low, also predisposes.

· The left recurrent laryngeal nerve hooks under the ligamentum arteriosum, which is why surgery in this region can cause hoarseness.

Distinctions that matter

· Oxygen and prostaglandin are two separate closure signals, and a ductus that misses either one can stay open.

· Functional closure in the first days is muscular; anatomic closure forming the ligamentum arteriosum takes weeks to months.

Common misconceptions

· Placing the ductus proximal to the left subclavian artery. It joins the arch just distal to it, which is what produces differential cyanosis later.

· Naming the sixth arch generally. It is the distal part of the left sixth arch.

Where it is taught: Section C04, first paragraph, closure flow diagram, and the risk factor paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-BD2

Brain dump 2 · integration and application: Pulses, pressures and blue toes

A competent response contains

· After birth aortic pressure exceeds pulmonary arterial pressure in both systole and diastole, so flow runs from aorta to pulmonary artery throughout the cycle, giving a continuous machine-like murmur loudest below the left clavicle and peaking around S2.

· Diastolic pressure falls because the aorta drains into the low-resistance pulmonary circuit during diastole.

· Systolic pressure rises because the extra left ventricular stroke volume is larger, so the pulse pressure widens and the pulses are bounding.

· The recirculated blood returns to the left atrium and left ventricle, so both dilate, and a large PDA causes heart failure.

· In a premature infant whose PVR is still high, the diastolic gradient may be small and the murmur only systolic.

· With Eisenmenger physiology, desaturated pulmonary arterial blood enters the aorta beyond the left subclavian artery, so the lower body receives desaturated blood while the head and arms are supplied from vessels proximal to the ductus.

· The result is differential cyanosis: cyanotic clubbed toes with pink fingers.

· Other complications are infective endarteritis at the ductal jet and, rarely, ductal aneurysm.

Distinctions that matter

· The continuous murmur reflects a gradient in both systole and diastole, which is why a premature infant with high PVR may have only a systolic murmur.

· The wide pulse pressure has two separate causes, diastolic runoff and increased stroke volume, and both should be stated.

Common misconceptions

· Expecting cyanosis in the arms with a reversed PDA. The arch vessels arise proximal to the ductus and carry undiluted left ventricular blood.

· Treating a continuous murmur as evidence of severity. Its presence reports the gradient, and losing it can mean rising pulmonary pressure.

Where it is taught: Section C04, fourth and fifth paragraphs and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-RT01

Retrieval exercise: Rebuild ductal closure after birth

A competent response contains

· Step 1: the first breaths expand the lungs and PVR falls.

· Step 2: arterial PO2 rises and constricts ductal smooth muscle.

· Step 3: the placenta is removed and the perfused lungs clear PGE2.

· Step 4: the ductus closes functionally within the first few days.

· Step 5: intimal proliferation and fibrosis form the ligamentum arteriosum, with anatomic closure taking weeks to months.

· PGE2 is the prostaglandin that falls, and it falls because the placenta is gone and the now perfused lungs clear it.

· Cyclooxygenase inhibition lowers PGE2 and promotes closure; exogenous prostaglandin E1 holds the ductus open.

· The premature ductus is at highest risk because its immature muscle constricts weakly to oxygen and it remains sensitive to PGE2.

Distinctions that matter

· Pharmacologic opening and closing confirm which signal is being manipulated: prostaglandin keeps it open, and blocking prostaglandin synthesis closes it.

Common misconceptions

· Attributing closure to oxygen alone. Rising oxygen and falling PGE2 act together.

Where it is taught: Section C04, flow diagram on ductal closure and the following paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C05-BD1

Brain dump 1 · foundational retrieval: Build tetralogy of Fallot from one developmental error

A competent response contains

· One error explains all four features: anterosuperior and leftward displacement of the infundibular (outlet) septum.

· The displaced septum crowds the right ventricular outflow tract, producing infundibular pulmonary stenosis, often with a small pulmonary valve and arteries.

· Because it no longer lines up with the muscular septum, it leaves a large malalignment VSD.

· The aortic root, shifted rightward, sits above that defect, giving the overriding aorta.

· Right ventricular hypertrophy is the consequence rather than a primary defect, because the right ventricle pumps against the obstruction at systemic pressure through the unrestrictive VSD.

· Neural crest cells migrate into the conotruncal ridges and help build the outflow septum.

· That explains the link with 22q11.2 deletion (DiGeorge syndrome), which also causes thymic and parathyroid hypoplasia and is associated with truncus arteriosus and interrupted aortic arch.

· Other associations are trisomy 21 and fetal alcohol exposure.

· Tetralogy of Fallot is the most common cyanotic congenital heart defect.

Distinctions that matter

· Three features are structural results of the displaced septum; the hypertrophy is a response to the pressure load, so it is the one that is secondary.

· Conotruncal lesions share a neural crest origin, which is why tetralogy, truncus arteriosus and interrupted arch travel with the same deletion.

Common misconceptions

· Listing the four features as four separate developmental mistakes. They follow from a single displacement of the outlet septum.

· Calling the right ventricular hypertrophy a congenital malformation. It develops because the right ventricle works at systemic pressure against the obstruction.

Where it is taught: Section C05, first and second paragraphs of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C05-BD2

Brain dump 2 · integration and application: Why one child with tetralogy is blue and another is not

A competent response contains

· The variable is the severity of right ventricular outflow obstruction, which sets how much blood takes the aortic exit.

· With a large VSD the ventricles share one pressure, so flow divides between the stenotic pulmonary outflow and the VSD into the aorta in inverse proportion to their resistances, the aortic route carrying the resistance of the systemic circulation.

· Severe stenosis diverts deoxygenated blood across the VSD into the aorta and causes early cyanosis; mild stenosis gives a balanced or even left-to-right shunt and little cyanosis.

· The VSD is silent because there is no pressure gradient across it.

· The murmur is a systolic ejection murmur at the left upper sternal border from flow through the narrowed outflow tract, and S2 is single because P2 is soft.

· Pulmonary blood flow is reduced, so pulmonary vascular remodeling and Eisenmenger physiology do not develop.

· Spell loop: crying, feeding or dehydration narrows the outflow or lowers SVR; more right-to-left flow crosses the VSD; arterial PO2 falls and metabolic acidosis develops; hyperpnea increases systemic venous return to the right ventricle; acidosis lowers SVR further and the shunt increases again.

· The murmur softens or disappears during a spell because less blood crosses the obstructed outflow tract.

· Squatting kinks the femoral arteries and raises SVR, so more right ventricular output is pushed through the pulmonary outflow, and it also briefly reduces return of desaturated blood from the legs; the knee-chest position works the same way.

· Chronic hypoxemia causes clubbing and secondary polycythemia, and the right-to-left shunt allows paradoxical emboli and brain abscess.

· Fever or a hot bath lowers SVR and worsens cyanosis, and exercise does the same through muscle arteriolar dilation.

Distinctions that matter

· A softer murmur during a spell means less flow through the outflow tract, so the quieter child is the sicker one.

· The VSD carries the shunt but makes no sound; the murmur belongs entirely to the outflow obstruction.

Common misconceptions

· Expecting Eisenmenger syndrome in tetralogy. Pulmonary blood flow is reduced, so the pulmonary arterioles are not exposed to high flow and pressure.

· Explaining squatting by increased venous return alone. The main effect is a rise in systemic vascular resistance that redirects right ventricular output to the lungs.

Where it is taught: Section C05, third paragraph, the tet spell cycle, and the worked example on squatting of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C06-BD1

Brain dump 1 · foundational retrieval: Bicuspid aortic valve and its four complications

A competent response contains

· Bicuspid aortic valve is the most common congenital cardiac anomaly, present in about 1 to 2% of people and more often in males.

· The valve forms from cushion tissue in the outflow tract; two of the three cusps fail to separate, usually the right and left coronary cusps, leaving two unequal leaflets often with a raphe marking the fused commissure.

· Most cases are sporadic but it clusters in families, and some familial cases involve NOTCH1 mutations; it is common in Turner syndrome (45,X) and in coarctation of the aorta.

· Two cusps must fold and crease more than three to open fully, and they open to an eccentric, often domed orifice with turbulent flow.

· Calcific aortic stenosis: repeated mechanical stress injures the leaflet endothelium, then lipid infiltration, chronic inflammation, transformation of valve interstitial cells toward an osteoblast-like phenotype, and dystrophic calcification.

· Significant stenosis typically appears one to two decades earlier than with a three-cusp valve, often between the 40s and 60s instead of the 70s or 80s.

· Aortic regurgitation: a larger fused cusp may prolapse, or a dilated root may pull the commissures apart, so regurgitation is more typical in younger patients and stenosis in older ones.

· Infective endocarditis: damaged leaflet surfaces and turbulent jets favor platelet and fibrin deposits that circulating bacteria can colonize.

· Ascending aortic aneurysm or dissection: intrinsic medial weakness plus eccentric jet stress.

· The auscultatory clue is an early systolic ejection click from the domed valve snapping open, heard while the leaflets are still mobile and lost as they calcify.

Distinctions that matter

· One structural defect explains four complications, so a single patient may develop several over a lifetime.

· The process of calcification is the same as age-related calcific stenosis; the bicuspid valve only runs it faster.

Common misconceptions

· Treating the bicuspid valve as a stenosis problem only. It also leaks, seeds infection, and accompanies an abnormal aorta.

· Expecting the ejection click to persist. It is lost once the leaflets calcify and stop moving.

Where it is taught: Section C06, first three paragraphs and the complications table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C06-BD2

Brain dump 2 · integration and application: Why the aorta is part of the same disease

A competent response contains

· The media shows smooth muscle loss and elastic fiber fragmentation, an intrinsic wall weakness that accompanies the valve defect.

· The eccentric jet from the two-cusp orifice adds focal wall stress, which is the flow contribution.

· Because the medial weakness is intrinsic, aortic dilation can be present even when the valve works normally.

· The dilated ascending aorta can progress to an aneurysm and carries a higher risk of aortic dissection than in the general population.

· Coexisting Turner syndrome or coarctation of the aorta raises that risk further.

· Endocarditis risk is higher because endothelial injury from turbulent flow allows platelet and fibrin deposits to form, which circulating bacteria can then colonize.

· A younger patient is more likely to present with regurgitation from cusp prolapse or root dilation preventing coaptation, while an older patient presents with stenosis after years of dystrophic calcification.

Distinctions that matter

· Intrinsic medial degeneration and jet-induced wall stress are separate contributions, and only the first explains dilation in a normally functioning valve.

· Age sorts the two valve complications: prolapse and root dilation act early, calcification accumulates late.

Common misconceptions

· Treating the aortic dilation as a consequence of stenosis alone. The wall abnormality travels with the valve defect and can precede any valve dysfunction.

· Assuming a normal valve examination rules out aortic disease in these patients.

Where it is taught: Section C06, third and fourth paragraphs and the complications table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-BD1

Brain dump 1 · foundational retrieval: Coarctation: where it sits and what it does to the pulses

A competent response contains

· The narrowing is localized and almost always juxtaductal, at the junction with the ductus arteriosus or ligamentum arteriosum, just distal to the left subclavian artery.

· A leading explanation is that ductal tissue extends into the aortic wall and constricts along with the ductus after birth.

· It is more common in males, occurs in a substantial minority of girls with Turner syndrome, and is accompanied by a bicuspid aortic valve in most patients.

· The narrowing is a high resistance in series with everything downstream, so pressure upstream rises and pressure and flow beyond it are reduced.

· The arms, supplied by vessels arising proximal to the coarctation, are hypertensive.

· Femoral pulses are weak and delayed relative to the radial pulse, which is radiofemoral delay.

· Leg blood pressure is lower than arm pressure, although normally it is slightly higher.

· The legs may be cool or claudicate with exercise.

· Reduced renal perfusion pressure activates the renin-angiotensin-aldosterone system, which adds to the hypertension.

· The left ventricle hypertrophies against the high afterload, and long-standing proximal hypertension raises the risk of heart failure, aortic dissection, and rupture of cerebral berry aneurysms, which are associated with coarctation.

Distinctions that matter

· Hypertension here is both mechanical, from the fixed resistance, and hormonal, from renal hypoperfusion; naming only one leaves the picture incomplete.

· Normally leg pressure is slightly higher than arm pressure, so an equal reading is already abnormal.

Common misconceptions

· Placing the coarctation proximal to the left subclavian artery. It lies just distal to it, which is why the arms are hypertensive.

· Expecting the femoral pulse to be only weak. It is also delayed, and the delay is a specific clue.

Where it is taught: Section C07, first and third paragraphs of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-BD2

Brain dump 2 · integration and application: The newborn who collapses and the teenager who does not

A competent response contains

· The newborn has a severe narrowing, often with arch hypoplasia, historically called infantile or preductal, and the lower body was being perfused by right ventricular blood flowing through the PDA, which is why lower body cyanosis can be present.

· When the ductus closes in the first weeks, perfusion of the lower body and kidneys collapses, producing shock, acidosis and absent femoral pulses; the acidosis is lactic, from tissue hypoperfusion.

· The teenager has a milder narrowing, historically called adult or postductal, and years have allowed collateral vessels to develop, so the presentation is hypertension found on examination rather than collapse.

· Collateral route: subclavian artery above the coarctation, internal thoracic artery, anterior intercostal arteries, posterior intercostal arteries with flow reversed, descending aorta below the coarctation.

· Flow runs backwards in the posterior intercostal arteries.

· Chest film findings: rib notching along the inferior rib margins, usually from about the third rib downward, the figure 3 sign from the indented contour with dilation above and below, and the enlarged tortuous collateral vessels responsible for the notching.

· The first two ribs are spared because their intercostal arteries arise from the costocervical trunk rather than the aorta.

· Notching takes years to appear and is rarely seen in young children.

· The murmur of flow through the narrowing and the collaterals is heard over the back between the scapulae.

Distinctions that matter

· A ductal-dependent lesion deteriorates when the ductus closes, which separates the neonatal presentation from the chronic one entirely.

· Collaterals are protective in the chronic form and absent in the newborn, which is why one presents with shock and the other with a blood pressure reading.

Common misconceptions

· Expecting rib notching in an infant. Collateral enlargement and bone erosion take years.

· Assuming cyanosis rules out coarctation. In the severe neonatal form the lower body is perfused through the ductus and can be cyanotic.

Where it is taught: Section C07, second paragraph, the collateral flow diagram, and the final paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-RT01

Retrieval exercise: Rebuild the collateral route around a coarctation

A competent response contains

· Step 1: subclavian artery above the coarctation.

· Step 2: internal thoracic artery.

· Step 3: anterior intercostal arteries.

· Step 4: posterior intercostal arteries, with flow reversed.

· Step 5: descending aorta below the coarctation.

· The posterior intercostal arteries enlarge and become tortuous under chronically high collateral flow.

· They run in the costal grooves and erode the inferior margins of the ribs, giving rib notching on the chest film, usually from about the third rib downward.

· The first two ribs are spared because their intercostal arteries arise from the costocervical trunk rather than the aorta.

· Notching takes years to develop, so it is rarely seen in a young child.

Distinctions that matter

· The reversed segment is the posterior intercostal arteries, which is the detail that makes the circuit work: blood must travel from the chest wall back into the aorta.

Common misconceptions

· Drawing the collateral route straight from the subclavian artery to the aorta. It travels through the chest wall by way of the internal thoracic and intercostal vessels.

Where it is taught: Section C07, collateral flow diagram and the following paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-CA01

Clinical application: Hypertension in a 16-year-old with weak femoral pulses

A competent response contains

· Recognize: coarctation of the aorta, juxtaductal and just distal to the left subclavian artery, with a bicuspid aortic valve suggested by the early systolic ejection click and right upper sternal border murmur, and Turner syndrome (45,X) suggested by short stature, webbed neck and broad chest.

· Explain: the narrowing is a fixed high resistance in series, so pressure upstream rises and arm pressure is high, while pressure and flow beyond it fall and leg pressure is low; the femoral pulse is weak because flow and pressure downstream are reduced and delayed because the pulse wave reaches the legs partly through longer collateral paths and through the narrowed segment.

· Differentiate: in a PDA with Eisenmenger physiology the legs are desaturated but pulses are bounding with a wide pulse pressure and the fingers stay pink, with no arm to leg pressure gradient of this kind; in severe neonatal coarctation the infant is in shock with absent femoral pulses when the ductus closes, with no collaterals and no rib notching; here the pulses are present but weak and delayed and the pressure gradient is measurable.

· Investigate: rib notching shows enlarged tortuous posterior intercostal arteries eroding the inferior rib margins, which means long-standing collateral flow, and the indented contour with dilation above and below is the figure 3 sign at the coarctation; record blood pressure in both arms and a leg and palpate radial and femoral pulses together to document the gradient and the delay.

· Decide: the renin-angiotensin-aldosterone system is adding to her hypertension, and she is at risk of heart failure, aortic dissection, and rupture of cerebral berry aneurysms.

· Justify: renal perfusion pressure is low beyond the narrowing, so the kidney responds as it would to hypovolemia and activates renin, raising pressure further proximal to the obstruction; dissection risk comes from the high proximal pressure load acting on an aorta whose media already shows smooth muscle loss and elastic fiber fragmentation as part of the bicuspid valve disease, a risk raised further by Turner syndrome.

Distinctions that matter

· The arm to leg gradient with a delayed femoral pulse is what separates coarctation from the other causes of poor lower body perfusion in this chapter.

· The valve lesion and the aortic wall weakness are part of the same disease as the coarctation, not incidental findings.

Common misconceptions

· Reading the hypertension as primary. A measurable arm to leg gradient with a delayed femoral pulse points to a mechanical cause.

· Treating the renin response as inappropriate. The kidney senses low perfusion pressure and responds normally; the response is what makes the hypertension worse.

Where it is taught: Section C07 throughout, with the bicuspid valve and aortic wall changes in C06 and the PDA reversal in C04 of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C08-BD1

Brain dump 1 · foundational retrieval: Draw the parallel circulations of transposition

A competent response contains

· Normally neural crest cells form a pair of conotruncal ridges that grow in a spiral, so the aorticopulmonary septum twists about 180 degrees, connecting the aorta to the left ventricle and the pulmonary trunk to the right ventricle.

· In transposition the septum grows straight instead of spiraling, so the aorta arises from the right ventricle, usually anterior and to the right, and the pulmonary artery arises from the left ventricle.

· Systemic loop: body to right atrium to right ventricle to aorta and back to the body, carrying desaturated blood.

· Pulmonary loop: lungs to left atrium to left ventricle to pulmonary artery and back to the lungs, carrying highly saturated blood.

· Survival requires mixing through a patent foramen ovale or ASD, a PDA, or a VSD.

· Maternal diabetes before conception is a recognized risk factor, and the lesion is more common in males.

· The circulations are parallel rather than in series because each loop returns blood to itself, so the highly saturated pulmonary loop is useless to the tissues while aortic blood is deeply desaturated.

Distinctions that matter

· Tetralogy and transposition are both conotruncal septation defects, but one is a displacement of the outlet septum and the other a failure of the same septum to spiral.

· A high saturation in the pulmonary loop is not a reassuring finding here, because that blood never reaches the tissues.

Common misconceptions

· Describing transposition as a shunt lesion. The problem is the arrangement of the circuits, and a shunt is what keeps the infant alive.

· Expecting the lungs to fix the saturation. The pulmonary loop recirculates its own blood.

Where it is taught: Section C08, first two paragraphs and the parallel circulations flow diagram of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C08-BD2

Brain dump 2 · integration and application: Why oxygen barely helps and why day three is worse

A competent response contains

· In the fetus this causes little difficulty, because oxygen comes from the placenta and the foramen ovale and ductus are both open.

· After birth, an infant with an intact ventricular septum depends on a foramen ovale or ASD and a PDA; the ductus is the one closing on day three.

· Severe cyanosis appears within hours to days and worsens as the ductus closes, so anything that keeps the ductus open or enlarges the atrial opening improves mixing.

· Supplemental oxygen raises saturation only slightly because it oxygenates blood in the pulmonary loop, which recirculates to the lungs and reaches the aorta only through whatever mixing remains.

· Mixing must go both ways so that volume entering one circuit is returned, otherwise one loop would fill and the other empty.

· An infant with a large VSD has more mixing, so cyanosis is milder, but pulmonary overcirculation brings early heart failure.

· Untreated, most infants die within the first months of life.

· Transposition is the most common cause of cyanosis appearing on the first day of life, while tetralogy of Fallot is the more common cyanotic lesion overall.

· Because the great arteries lie front to back, the upper mediastinal shadow is narrow, and with a rounded heart this gives the egg-on-a-string appearance.

Distinctions that matter

· Most common on the first day against most common overall is a frequent examination distinction between transposition and tetralogy.

· More mixing trades hypoxemia for volume overload, which is why the infant with a VSD looks pinker but fails earlier.

Common misconceptions

· Expecting oxygen to correct the cyanosis. The limitation is anatomic routing, not alveolar oxygenation.

· Treating a closing ductus as unimportant here. It is one of the few mixing sites, so closure makes the infant markedly worse.

Where it is taught: Section C08, third and fourth paragraphs and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C08-CP01

Comparison table: Tetralogy of Fallot against transposition of the great vessels

A competent response contains

· Developmental error: tetralogy from anterosuperior displacement of the infundibular septum; transposition from failure of the aorticopulmonary septum to spiral.

· Associations: tetralogy with 22q11.2 deletion and trisomy 21; transposition with maternal diabetes and male sex.

· Circulation: tetralogy in series with right-to-left flow across the VSD; transposition parallel, requiring mixing.

· Onset of cyanosis: tetralogy variable and set by the degree of outflow obstruction; transposition within hours to days of birth.

· Pulmonary blood flow: reduced in tetralogy; normal or increased in transposition.

· Response to oxygen: little rise in arterial PO2 in either, because in tetralogy the shunted blood bypasses the alveoli and in transposition the oxygenated blood recirculates to the lungs.

· Chest film: boot-shaped heart with dark lung fields in tetralogy; egg on a string with a narrow mediastinum in transposition.

· Survival dependence: transposition with an intact ventricular septum depends on a patent foramen ovale or ASD and a PDA for mixing, while tetralogy does not depend on the ductus unless pulmonary stenosis is severe enough to limit pulmonary flow critically.

Distinctions that matter

· Both are conotruncal defects of outflow tract septation, but they cause cyanosis by different mechanisms: a right-to-left shunt in one, two separated circuits in the other.

· Pulmonary blood flow sorts them quickly: reduced in tetralogy, normal or increased in transposition.

Common misconceptions

· Expecting transposition to have reduced pulmonary flow because the infant is blue. Pulmonary flow is normal or increased; the blood simply recirculates.

· Treating cyanosis in tetralogy as fixed. It varies with outflow obstruction and with systemic vascular resistance.

Where it is taught: Section C08, comparison table, with the tetralogy mechanism in C05 of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C02-Q01

Board-style question: The loud murmur and the quiet one

Correct answer: A. A restrictive defect maintains a large systolic pressure gradient between the ventricles

Why it is right: A restrictive defect limits flow and preserves a large systolic pressure difference between the ventricles, and the high-velocity jet through a small orifice generates the loud harsh holosystolic murmur and the palpable thrill.

B is wrong: Greater volume crossing the septum belongs to the larger defect, which is the second infant, and that infant has the softer murmur, so volume does not explain loudness.

C is wrong: Right ventricular pressure approaching left ventricular pressure describes a large nonrestrictive defect, and it reduces the gradient, which makes a murmur softer rather than louder.

D is wrong: Pulmonary vascular resistance normally falls below systemic resistance in every term infant by this age, so it does not distinguish these two infants or explain the difference in loudness.

A competent response contains

· Murmur loudness reflects the pressure gradient across the defect, not the size of the defect.

· A small restrictive defect keeps left ventricular pressure far above right ventricular pressure throughout systole, so the jet is high velocity and the murmur is loud and harsh with a thrill.

· A large nonrestrictive defect lets ventricular pressures equalize, so the murmur softens while the shunt volume and the clinical burden are greater.

Distinctions that matter

· The loudest VSDs are often the smallest, and the quiet murmur with a loud P2 and failure to thrive is the dangerous one.

Common misconceptions

· Equating murmur loudness with the amount of blood crossing the defect.

Where it is taught: Section C02, small versus large VSD table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C05-Q02

Board-style question: Squatting during a spell

Correct answer: C. Systemic vascular resistance rises, so a greater share of right ventricular output enters the pulmonary outflow tract

Why it is right: The knee-chest position kinks the femoral arteries and raises systemic vascular resistance, so the aortic exit through the VSD becomes the higher resistance route and a greater share of right ventricular output is pushed through the pulmonary outflow tract, raising arterial saturation.

A is wrong: Pulmonary vascular resistance is not the limiting factor in tetralogy; the obstruction is a fixed infundibular narrowing, and the maneuver does not act on the pulmonary arterioles.

B is wrong: Increased venous return from the legs would raise right ventricular preload without changing how output divides between the two exits, and the maneuver actually reduces return of desaturated blood from the legs briefly.

D is wrong: The infundibular obstruction is structural and does not relax with position; nothing in the maneuver widens the outflow tract.

A competent response contains

· The right ventricle has two exits, the fixed high resistance of the stenotic pulmonary outflow and the VSD into the aorta, whose resistance is systemic vascular resistance.

· Flow divides in inverse proportion to resistance, so raising systemic vascular resistance sends a greater share of right ventricular output to the lungs.

· Squatting and the knee-chest position kink the femoral arteries and raise systemic vascular resistance, and they also briefly reduce return of desaturated blood from the legs.

· The murmur softens during a spell because less blood crosses the obstructed outflow tract.

Distinctions that matter

· The maneuver works on the systemic side of the circuit, not on the obstruction itself, which is fixed.

Common misconceptions

· Explaining the maneuver by increased venous return alone, which would not change how right ventricular output divides between its two exits.

Where it is taught: Section C05, tet spell cycle and the worked example on squatting of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-Q03

Board-style question: Blue toes and pink fingers

Correct answer: D. Desaturated pulmonary arterial blood enters the aorta distal to the left subclavian artery

Why it is right: Reversed flow through the ductus delivers desaturated pulmonary arterial blood into the aorta distal to the left subclavian artery, so the descending aorta and the legs receive it while the arch vessels, arising proximal to the ductus, supply the arms with fully saturated blood.

A is wrong: Oxygen extraction occurs in capillaries and lowers venous saturation, not arterial saturation, so it cannot create a difference between arterial readings at the hand and the foot.

B is wrong: The left ventricle does not eject selectively into particular branches; the arch vessels and the descending aorta receive the same ejected blood, and the difference arises only where the ductus adds desaturated blood.

C is wrong: Pulmonary venous blood reaches all systemic arteries through the left ventricle at the same time; sequence of arrival does not produce a sustained saturation gradient.

A competent response contains

· The ductus joins the aorta just distal to the origin of the left subclavian artery.

· With Eisenmenger reversal, desaturated pulmonary arterial blood enters the descending aorta beyond the arch vessels.

· The head and arms are supplied by the brachiocephalic trunk and left subclavian artery, which arise proximal to the ductus and carry undiluted left ventricular blood.

· The result is differential cyanosis with cyanotic clubbed toes and pink fingers.

Distinctions that matter

· Differential cyanosis is an anatomic finding that localizes the communication, not a measure of how severe the hypoxemia is.

Common misconceptions

· Attributing the saturation difference to tissue oxygen extraction. Pulse oximetry reports arterial saturation, which extraction does not change.

Where it is taught: Section C04, final paragraph and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-Q04

Board-style question: A second sound that does not move

Correct answer: B. Right ventricular stroke volume stays nearly constant because inspiratory venous return is offset by reduced shunting

Why it is right: The left-to-right shunt and systemic venous return vary with respiration in opposite directions, so right ventricular stroke volume and the delay in pulmonic valve closure stay nearly constant and the two components of S2 remain equally separated in inspiration and expiration.

A is wrong: Elevated pulmonary arterial pressure would be expected to narrow or alter the split and would come with a low room air saturation if the shunt had reversed, which is not the picture here.

C is wrong: Left ventricular stroke volume is reduced in a large ASD, but an early aortic closure would widen the split variably with respiration rather than fix it, and it does not explain the loss of respiratory variation.

D is wrong: An incomplete right bundle branch block pattern is common in a secundum ASD and contributes to the width of the split, but conduction delay is fixed in both respiratory phases only in appearance and does not explain why the normal inspiratory widening is absent; the respiratory offset of right ventricular volume is what does.

A competent response contains

· Normally inspiration increases systemic venous return to the right heart, delays pulmonic valve closure, and widens the split, which narrows again in expiration.

· In an ASD the right ventricle carries a large stroke volume in both phases of respiration.

· In inspiration the extra systemic venous return is offset by reduced left-to-right shunting, and in expiration the shunt increases again.

· Right ventricular volume, and therefore the timing of P2, stays nearly constant, so the split is wide and fixed.

· The systolic murmur comes from increased flow across a normal pulmonic valve, and normal room air saturation confirms the shunt is still left to right.

Distinctions that matter

· Wide splitting comes from the large right ventricular stroke volume, while fixity comes from the respiratory offset between venous return and shunt flow.

Common misconceptions

· Reading a fixed split as evidence of pulmonary hypertension. Saturation is normal here, so the shunt has not reversed.

Where it is taught: Section C03, paragraph on the fixed split S2 of the CV-18-G Congenital Heart Conditions notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-08-C01-BD1

Brain dump 1 · foundational retrieval: Build the Virchow triad and the two thrombus types

A competent response contains

· Virchow triad: endothelial injury or dysfunction, abnormal flow (stasis or turbulence), and hypercoagulability; each limb can act alone but most thrombi need two or more.

· Endothelial injury exposes collagen and von Willebrand factor for platelet adhesion and tissue factor to start coagulation, and activated endothelium loses its anticoagulant surface; settings include ruptured atherosclerotic plaque, endocardium over a myocardial infarct, vasculitis, valve injury, hypertension, and smoking.

· Abnormal flow disrupts laminar flow so platelets reach the wall, stops activated factors being washed away and diluted, delays inflow of inhibitors, and turbulence itself injures endothelium; settings include immobility, atrial fibrillation, ventricular aneurysm, dilated chambers, aortic aneurysm, hyperviscosity, and sickle cell disease.

· Hypercoagulability is more procoagulant activity or less anticoagulant or fibrinolytic activity in the blood itself.

· Endothelial injury dominates in the heart and arteries; stasis dominates in veins.

· Arterial thrombus: pale gray, platelet rich with fibrin binding the platelets, firmly attached, usually starting on a ruptured plaque in coronary, cerebral, or femoral arteries, and growing back toward the heart from its point of attachment.

· Venous thrombus: soft red (red or stasis thrombus) because coagulation in nearly stagnant blood traps many red cells, grows in the direction of flow toward the heart, and can have a long loosely attached tail that breaks off.

· Most venous thrombi form in the deep veins of the leg, and those above the knee (popliteal, femoral, iliac) are the main source of pulmonary emboli.

· Mural thrombi form on the wall of a heart chamber or the aorta; vegetations are thrombi on valves.

· Normal endothelium keeps blood fluid by expressing thrombomodulin and heparan sulfate, releasing nitric oxide and prostacyclin, and making tissue plasminogen activator.

Distinctions that matter

· Composition follows flow rate, not vessel name: fast flow builds a platelet rich pale thrombus, slow flow builds a red cell rich soft one.

· Arterial thrombi grow back toward the heart against flow from their attachment, while venous thrombi grow toward the heart with flow; both end up pointing centrally for different reasons.

Common misconceptions

· Treating the three limbs as alternatives one of which must be chosen. Most thrombi arise where two or three act together, which is why a first thrombosis often needs an inherited trait plus a trigger.

· Saying stasis matters only because blood sits still. It also prevents dilution and washout of activated factors and delays the arrival of inhibitors.

Where it is taught: Section C01, opening paragraph, the Virchow triad table, and the composition paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C01-BD2

Brain dump 2 · integration and application: Tell a thrombus formed in life from one formed after death

A competent response contains

· The femoral vein mass is a postmortem clot: not attached to the wall, gelatinous, dark red where red cells settled with a yellow chicken fat layer above, and with no lines of Zahn.

· The left atrial mass is a thrombus formed during life: attached to the wall and showing lines of Zahn.

· Lines of Zahn are alternating pale bands of platelets and fibrin and dark bands of red cells, laid down in layers only while blood is moving, so they identify a thrombus formed before death.

· They are most obvious in the heart and aorta and fainter in veins.

· In still blood after death, cells sediment by gravity instead of layering, which gives the dark red and yellow division rather than banding.

· The thrombus that embolized formed in life; a postmortem clot cannot have embolized, so the attached layered mass is the plausible source.

· Fates of a thrombus: propagation (more platelets and fibrin, occluding the vessel or making a longer less stable thrombus), embolization (part or all breaks off and blocks a distant vessel), dissolution (plasmin degrades fibrin in a recent thrombus, while older cross-linked thrombi resist lysis), organization (endothelial cells, smooth muscle cells, and fibroblasts grow in, making granulation tissue then scar), and recanalization (new capillary channels form through the organized thrombus and partly restore flow).

· A fibrous thickening of the vein wall years later is an organized thrombus incorporated into the wall.

· Thrombi can also become infected, giving a septic thrombus or a mycotic aneurysm.

Distinctions that matter

· Attachment and layering are the two features that date a mass; color alone does not, since a venous thrombus formed in life is also red.

· Dissolution depends on the age of the thrombus: recent fibrin is lysed by plasmin, cross-linked fibrin resists it.

Common misconceptions

· Calling a red mass in a vein a postmortem clot because it is red and soft. A venous thrombus formed in life is red and soft too, and attachment plus lines of Zahn separate them.

· Treating recanalization as full recovery of the vessel. The channels are new small capillaries through organized thrombus, so flow is only partly restored.

Where it is taught: Section C01, lines of Zahn paragraph, the fates table, and the stop and predict box of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C02-BD1

Brain dump 1 · foundational retrieval: The brakes on thrombin and the inherited thrombophilias

A competent response contains

· Thrombin converts fibrinogen to fibrin and activates platelets and factors V, VIII, and XI.

· Brake one: antithrombin inactivates thrombin and factor Xa and other serine protease factors, and heparan sulfate on the endothelium greatly speeds the reaction.

· Brake two: thrombin binding thrombomodulin on endothelium activates protein C, and activated protein C with protein S as cofactor cleaves and inactivates factors Va and VIIIa.

· Brake three: plasmin dissolves fibrin.

· Factor V Leiden: point mutation in factor V (arginine 506 to glutamine), autosomal dominant; factor Va resists cleavage by activated protein C, so thrombin generation continues. Most common inherited cause, heterozygous in about 2 to 15% of people of European descent, risk about 5-fold heterozygous and about 50-fold homozygous.

· Prothrombin G20210A: mutation in the 3' untranslated region of the prothrombin gene raises plasma prothrombin; second most common, about 1 to 2% of the population.

· Antithrombin deficiency: reduced or defective antithrombin, so thrombin and factor Xa are not inactivated; heparin works by enhancing antithrombin, so its effect is blunted.

· Protein C or S deficiency: factors Va and VIIIa are not inactivated; homozygous protein C deficiency causes neonatal purpura fulminans.

· All four mainly cause venous thrombosis.

· Suspect an inherited thrombophilia when thrombosis occurs before about age 50, recurs, appears in unusual sites such as cerebral sinuses or mesenteric or hepatic veins, or runs in the family.

· Mild hyperhomocysteinemia is a further inherited or acquired risk; homocysteine injures endothelium and promotes arterial and venous thrombosis.

Distinctions that matter

· Factor V Leiden changes the substrate so the brake cannot act, while protein C or S deficiency removes the brake itself; the end result on factor Va is the same.

· Antithrombin acts on thrombin and factor Xa, while activated protein C acts on factors Va and VIIIa, so the two brakes work at different points of the cascade.

Common misconceptions

· Saying factor V Leiden means too much factor V. The amount is normal; the mutated factor Va simply resists cleavage by activated protein C.

· Expecting these disorders to cause arterial events. They mainly cause venous thrombosis, while antiphospholipid syndrome and hyperhomocysteinemia cause both.

Where it is taught: Section C02, opening paragraph and the inherited hypercoagulable states table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C02-BD2

Brain dump 2 · integration and application: Three clotting puzzles explained from the mechanism

A competent response contains

· Case one is antiphospholipid syndrome: autoantibodies bind phospholipid-binding proteins, mainly beta2-glycoprotein I, and activate platelets, endothelium, and complement, causing arterial and venous thrombosis and recurrent pregnancy loss.

· The lupus anticoagulant prolongs the partial thromboplastin time in the test tube because the antibodies interfere with the phospholipid in the assay, while the patient clots in vivo; anticardiolipin antibodies can give a false-positive nontreponemal syphilis test.

· Antiphospholipid syndrome may be primary or secondary to systemic lupus erythematosus.

· Case two: warfarin blocks vitamin K dependent synthesis of factors II, VII, IX, and X and of proteins C and S. Protein C has a short half-life, so protein C and factor VII fall first, before factors II, IX, and X.

· In those first days a person with low protein C is transiently more hypercoagulable, and dermal microvessels thrombose, which is warfarin-induced skin necrosis.

· Case three: pregnancy and the postpartum period raise hepatic synthesis of fibrinogen and other factors (hypercoagulability), lower protein S (hypercoagulability), and compress pelvic veins with the uterus (stasis).

· Acquired states are far more common than inherited ones and most act through more than one limb of the triad, so a first thrombosis often appears only when an acquired trigger is added to an inherited trait.

· Heparin is less effective in antithrombin deficiency because heparin works by enhancing antithrombin, and there is less of it to enhance.

Distinctions that matter

· A prolonged clotting time in vitro and thrombosis in vivo are not a contradiction: the antibody interferes with the phospholipid in the assay while activating cells in the patient.

· Warfarin skin necrosis is about half-lives, not dose: the brake disappears before the procoagulant factors do.

Common misconceptions

· Reading a long partial thromboplastin time as a bleeding risk in this syndrome. The patient clots, and the long time is an artifact of the assay.

· Assuming an inherited thrombophilia should have caused thrombosis earlier. Most carriers need an acquired trigger such as pregnancy, estrogen, surgery, or immobility.

Where it is taught: Section C02, warfarin paragraph, antiphospholipid paragraph, and the acquired hypercoagulable states table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-BD1

Brain dump 1 · foundational retrieval: Where an embolus lodges, and the nonthrombotic emboli

A competent response contains

· An embolus travels until it reaches a vessel too narrow to pass, so its source predicts its destination.

· Material entering systemic veins passes through the right heart to the pulmonary arteries; material from the left heart, aorta, or large arteries lodges in systemic arteries.

· A venous embolus reaches the systemic circulation only through a right-to-left opening such as a patent foramen ovale or atrial septal defect, which is paradoxical embolism.

· The great majority of emboli are dislodged thrombi.

· About 80% of systemic thromboemboli come from intracardiac mural thrombi: about two-thirds of those form over a left ventricular infarct and about a quarter in a dilated fibrillating left atrium; the rest come from aortic aneurysms, ulcerated plaques, valve vegetations, and paradoxical emboli.

· About 75% lodge in the lower limbs and about 10% in the brain, with the intestines, kidneys, spleen, and upper limbs making up the rest; these organs are supplied by end arteries, so infarction is the usual result.

· Fat and marrow embolism follows long-bone or pelvic fracture, less often soft tissue trauma or burns: fat globules block microvessels, free fatty acids released from them injure endothelium over the next days, and platelets adhere to fat and are consumed; fat embolism syndrome appears 1 to 3 days after injury with hypoxemia, tachypnea, neurologic change, petechial rash, and thrombocytopenia.

· Amniotic fluid embolism occurs in labor and the immediate postpartum period: fluid enters maternal veins through tears in membranes or uterine veins, triggering an anaphylactoid reaction and activating coagulation, with sudden dyspnea, cyanosis, shock, seizures, then disseminated intravascular coagulation, and fetal squames, lanugo hair, vernix fat, and mucin in maternal pulmonary vessels.

· Air embolism follows chest wall injury, obstetric procedures, or opened venous lines: bubbles coalesce and obstruct flow, and usually more than about 100 mL is needed for symptoms.

· Nitrogen embolism is decompression sickness from rapid ascent by divers or rapid decompression in flight, with joint and muscle pain (the bends) and respiratory distress (the chokes); chronic caisson disease causes ischemic necrosis of the femoral head, tibia, and humerus.

· Atheroembolism follows manipulation of an atherosclerotic aorta at catheterization or surgery: cholesterol crystals and plaque debris lodge in small arteries, giving renal failure, livedo reticularis, digital ischemia, and cholesterol clefts in arterioles.

· Septic emboli from infected vegetations and tumor emboli carry their infection or tumor to the site where they lodge.

Distinctions that matter

· The destination is decided by which side of the circulation the embolus entered, which is why a leg vein thrombus reaches the brain only through a shunt.

· Fat embolism and fat embolism syndrome are not the same: fat reaches the lungs in most patients with long-bone fracture, while the syndrome appears in a small minority after a delay.

Common misconceptions

· Expecting fat embolism syndrome at the moment of injury. The free fatty acid injury takes time, so the syndrome appears 1 to 3 days later.

· Assuming a small volume of air is fatal. Usually more than about 100 mL is needed for symptoms.

Where it is taught: Section C03, opening paragraph, systemic embolism paragraph, and the nonthrombotic emboli table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-BD2

Brain dump 2 · integration and application: Three pulmonary emboli of different size

A competent response contains

· Small peripheral embolus: usually clinically silent, and it organizes and is incorporated into the vessel wall.

· Medium embolus with left heart failure: hemorrhage would be the usual result because bronchial arteries keep supplying lung tissue, but left heart failure has impaired bronchial flow, so this patient infarcts, and the infarct is red because blood still enters from the bronchial circulation and congested veins.

· Saddle embolus at the bifurcation, obstructing more than about 60% of the pulmonary circulation: acute right heart failure (acute cor pulmonale), circulatory collapse, and sudden death.

· The embolized lung is ventilated but not perfused, so dead space rises; blood diverted to the remaining lung creates ventilation and perfusion mismatch and hypoxemia.

· Hypoxemia and stimulated receptors drive hyperventilation, so PaCO2 falls rather than rises.

· Repeated small emboli can cause pulmonary hypertension.

· More than 95% of pulmonary emboli arise from deep vein thrombi of the leg, most from veins above the knee, because thrombi in the larger proximal veins are big enough and loosely enough attached at the tail to break off and occlude a pulmonary artery branch.

Distinctions that matter

· Hemorrhage and infarction are different outcomes of the same sized embolus, and the state of the bronchial supply is what separates them.

· The lung in pulmonary embolism is hypoxemic with a low carbon dioxide, which is the opposite of what students expect from the word obstruction.

Common misconceptions

· Assuming every pulmonary embolus causes an infarct. The lung has two blood supplies, so most do not.

· Expecting carbon dioxide to rise because a vessel is blocked. Ventilation continues and is increased, so carbon dioxide falls.

Where it is taught: Section C03, paragraph on the effect of a pulmonary embolus, and the flow diagram caption of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-RT01

Retrieval exercise: Rebuild the route from a leg vein to a blocked pulmonary artery

A competent response contains

· Step 1: stasis, injury, or hypercoagulability in deep leg veins; stasis is the usual limb in a long flight or immobility, often with a hypercoagulable contribution.

· Step 2: a red thrombus propagates toward the heart.

· Step 3: a thrombus fragment detaches, usually the long tail, which is loosely attached because it was laid down in flowing blood away from the point of adhesion.

· Step 4: it passes through the right atrium and ventricle.

· Step 5: it lodges in a pulmonary artery branch.

· Step 6: perfusion is lost while ventilation of that lung continues, so ventilation is the one that is preserved and dead space rises.

· Chambers in order: deep leg vein, iliac vein, inferior vena cava, right atrium, right ventricle, pulmonary artery.

· A right-to-left shunt at the atrial level, a patent foramen ovale or atrial septal defect, would divert the fragment into the systemic arteries as a paradoxical embolus.

· More than 95% of pulmonary emboli arise from leg vein thrombi, most from veins above the knee.

Distinctions that matter

· The tail embolizes rather than the attached head, which is why venous thrombi growing with flow are the dangerous ones.

· Perfusion is lost and ventilation preserved, which is the definition of wasted ventilation rather than of shunt.

Common misconceptions

· Drawing the fragment traveling from the leg to the aorta directly. Without a right-to-left shunt it must stop in the lungs.

· Writing that ventilation stops in the embolized segment. Ventilation continues, which is why dead space rises.

Where it is taught: Section C03, the flow diagram from deep vein thrombosis to pulmonary embolism of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-BD1

Brain dump 1 · foundational retrieval: Draw an infarct and date it

A competent response contains

· An infarct is ischemic necrosis caused by loss of blood flow; nearly all follow arterial occlusion by thrombosis or embolism, with vasospasm, external compression, torsion, rupture, and venous outflow obstruction less common.

· Infarcts in solid organs are usually wedge shaped with the occluded vessel at the apex and the organ surface at the base, and a fibrinous exudate often covers the overlying serosa.

· The two questions that describe every infarct: how much blood it contains, white or red, and how the dead tissue breaks down, coagulative or liquefactive.

· Coagulative necrosis: ischemia denatures structural proteins and lytic enzymes alike, so cell outlines and tissue architecture persist for days while nuclei disappear.

· Time course: cell death is not visible by light microscopy for the first several hours; nuclear loss and eosinophilic cytoplasm appear over about 4 to 12 hours; neutrophils infiltrate the border within the first 1 to 2 days; macrophages and granulation tissue follow over the next week or two; the area is replaced by scar over weeks.

· The margin of a recent infarct is often hyperemic.

· The brain is the exception: little collagen framework and lipid-rich tissue digested by released lysosomal enzymes and cleared by microglia, so the infarct liquefies and leaves a fluid-filled cavity lined by reactive astrocytes rather than a scar.

· Color and necrosis pattern are independent: a brain infarct can be red and liquefactive, and a reperfused myocardial infarct red and coagulative.

· Septic infarcts, caused by infected emboli, become abscesses.

Distinctions that matter

· The necrosis pattern depends on the tissue, not on the color of the infarct.

· Neutrophils at the border and macrophages later are the clock that dates an infarct; the first several hours look normal under the microscope even though the cells are dead.

Common misconceptions

· Expecting a dead area to look dead immediately. Light microscopy shows nothing for several hours.

· Pairing white with coagulative and red with liquefactive. The two questions are answered separately.

Where it is taught: Section C04, opening paragraph and the paragraphs on necrosis pattern and time course of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-BD2

Brain dump 2 · integration and application: Same occlusion, four different outcomes

A competent response contains

· Patient one, renal artery branch: infarct forms, white, coagulative, because the kidney is a dense organ with end-arterial supply and no alternative inflow; weeks later it is a pale depressed scar.

· Patient two, pulmonary artery branch with a normal heart: usually hemorrhage without infarction, because bronchial arteries keep supplying the tissue; if infarction did occur it would be red, and the lung has a dual supply.

· Patient three, cerebral artery branch with lysis of the embolus and return of flow: infarct forms, red because blood enters damaged vessels when flow is restored, liquefactive because of the brain's lipid-rich tissue and scant collagen, and weeks later it is a fluid-filled cavity lined by reactive astrocytes.

· Patient four, slowly narrowed coronary artery: slow narrowing allowed collateral vessels to enlarge, so a small or no infarct forms; if one does form in the heart it is coagulative and white unless the artery is reperfused, in which case it is red, and it heals as scar.

· Factors deciding whether and how large an infarct forms: anatomy of the blood supply (dual supply in lung, liver, hand and forearm, or rich anastomoses protect, while end arteries in kidney and spleen do not), rate of occlusion (slow narrowing lets collaterals enlarge), vulnerability to hypoxia (neurons die after about 3 to 4 minutes, myocytes after about 20 to 30 minutes, fibroblasts survive many hours), and oxygen content of the blood (anemia, hypoxemia, or heart failure enlarge an infarct and can cause one after a partial occlusion).

· A hemoglobin of 7 g/dL would enlarge the infarct in patient four, because the slowly narrowed coronary artery leaves myocardium dependent on marginal flow, and reduced oxygen content converts a partial occlusion into a damaging one.

· These factors explain why the same occlusion causes no damage in one person and a large infarct in another.

Distinctions that matter

· Reperfusion changes the color of an infarct without changing its necrosis pattern.

· Rate of occlusion and anatomy of supply are separate protections: collaterals must be given time to enlarge, while a dual supply is there from the start.

Common misconceptions

· Treating low oxygen content as a cause of infarction by itself. It enlarges infarcts and can make a partial occlusion damaging, which is not the same as occluding a vessel.

· Expecting a brain infarct to leave a scar like the heart or kidney. It leaves a cavity, because the tissue liquefies.

Where it is taught: Section C04, white against red table, necrosis pattern paragraph, factors table, and the stop and predict box of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-CP01

Comparison table: White against red infarcts

A competent response contains

· Occlusion: white follows arterial occlusion; red follows venous occlusion, or arterial occlusion where blood can still enter.

· Tissue: white in solid dense organs with end-arterial supply; red in loose tissue, dual-supply organs, previously congested tissue, and reperfused tissue.

· Why: in a white infarct there is no alternative inflow and dense tissue limits seepage of blood into the necrotic area; in a red infarct blood from a second supply, from backed-up venous drainage, or from restored flow enters damaged vessels and leaks into the dead area.

· Examples: white in heart, kidney, and spleen; red in lung (pulmonary and bronchial arteries), small intestine (rich anastomoses), testis or ovary after torsion, brain after lysis of an embolus, and heart after reperfusion.

· Necrosis pattern: coagulative in almost every organ including heart, kidney, spleen, lung, and gut; liquefactive in the brain, whichever color the infarct is.

· Weeks later: a coagulative infarct is replaced by scar, which is pale, firm, and depressed; a brain infarct leaves a fluid-filled cyst lined by reactive astrocytes.

· Infarcts caused by infected emboli become abscesses.

Distinctions that matter

· Color answers where the blood came from; necrosis pattern answers what the tissue is made of. The two questions are independent.

· A red infarct is not a bleeding problem; it is dead tissue into which blood has entered from a second route.

Common misconceptions

· Filling the red infarct column with the brain only. Lung, gut, torsion, and reperfusion all belong there.

· Writing liquefactive for the lung because the lung is soft. Pattern follows the tissue, and only the brain liquefies.

Where it is taught: Section C04, white against red infarcts table and the necrosis pattern paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C05-BD1

Brain dump 1 · foundational retrieval: Patterns of bleeding into tissue

A competent response contains

· Petechiae: 1 to 2 mm pinpoint hemorrhages in skin, mucosa, or serosa that do not blanch, from thrombocytopenia, platelet dysfunction, raised intravascular pressure, or loss of vessel wall support as in scurvy.

· Purpura: 3 mm or larger up to about 1 cm, from the causes of petechiae plus vasculitis, trauma, and vascular fragility; palpable purpura suggests small-vessel vasculitis.

· Ecchymosis: a larger flat subcutaneous hemorrhage, a bruise, usually over 1 to 2 cm, from trauma and worsened by any bleeding disorder.

· Hematoma: a collected mass of blood within tissue, raised or space-occupying, from trauma, factor deficiencies such as hemophilia, or a ruptured vessel or aneurysm.

· Bleeding into cavities is named for the site: hemothorax, hemopericardium, hemoperitoneum, hemarthrosis.

· Pinpoint bleeds come from capillaries and venules normally sealed by platelets, so petechiae and purpura point to a platelet or small-vessel problem.

· Large deep bleeds into muscles, joints, and soft tissue follow minor injury when the fibrin clot that should stabilize the platelet plug cannot form, so they point to a coagulation factor deficiency.

· Bruise colors in order: red-blue from hemoglobin, then blue-green as heme becomes biliverdin and then bilirubin, then golden-brown from hemosiderin.

· A large hematoma can raise serum unconjugated bilirubin as it is resorbed.

Distinctions that matter

· Size separates petechiae, purpura, and ecchymosis; a hematoma is defined by being a mass within tissue rather than by size alone.

· Platelets seal small vessels, while fibrin stabilizes the plug in larger injuries, which is why the two kinds of defect bleed in different patterns.

Common misconceptions

· Reading all purpura as a platelet problem. Palpable purpura points to small-vessel vasculitis.

· Expecting bruise color change to come from the blood drying or oxidizing. It follows macrophage breakdown of hemoglobin through biliverdin and bilirubin to hemosiderin.

Where it is taught: Section C05, opening paragraph, the hemorrhage into tissues table, and the bruise paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C05-BD2

Brain dump 2 · integration and application: Volume, rate, and site in acute blood loss

A competent response contains

· Blood volume is about 70 mL/kg, so about 4.9 L in a 70 kg man; 1.5 L is about 30% of his blood volume.

· Rapid loss of up to about 10 to 15% is tolerated with little change; above about 20% compensations begin to fail and hypovolemic shock develops; about 40% is immediately life threatening.

· His hematocrit is normal because he lost whole blood, cells and plasma in the same proportion, so the concentration of red cells in what remains is unchanged even though red cell mass has fallen by about 30%.

· Two processes lower it over the next day: lower capillary hydrostatic pressure draws interstitial fluid into the vessels (transcapillary refill), and the kidney retains sodium and water through renin, angiotensin II, aldosterone, and ADH; this cell-free volume dilutes the remaining red cells.

· An early normal hematocrit therefore does not exclude severe acute blood loss.

· The baroreceptor reflex raises heart rate and constricts arterioles and veins, and the pulse pressure narrows.

· Site can matter more than volume: a small hemorrhage inside the rigid skull raises intracranial pressure and can cause herniation; a modest but rapid pericardial collection causes tamponade because the stiff sac cannot stretch quickly; retroperitoneal bleeding from a ruptured aortic aneurysm can hide liters of blood without external signs.

· Slow or recurrent external loss, as from a bleeding peptic ulcer or heavy menstruation, depletes iron and causes iron deficiency anemia, whereas blood shed into tissues or body cavities is broken down and its iron reused.

Distinctions that matter

· Volume lost and concentration measured are different things, which is why the hematocrit lags behind the hemorrhage.

· A rigid or non-compliant container turns a small volume into a dangerous one, so skull and pericardium behave unlike the retroperitoneum.

Common misconceptions

· Using a normal hematocrit to rule out major bleeding in the first hour.

· Expecting all blood loss to cause iron deficiency. Iron is reused when blood is shed internally and lost only when it leaves the body.

Where it is taught: Section C05, volume and rate paragraph, the worked example on hematocrit, and the final paragraph on site of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C06-BD1

Brain dump 1 · foundational retrieval: The mechanisms of edema and the fluid each produces

A competent response contains

· Edema is excess fluid in the interstitium.

· Normally capillaries filter slightly more fluid than they reabsorb and lymphatics return the excess; edema appears when filtration exceeds lymphatic capacity.

· Mechanisms: raised capillary hydrostatic pressure (heart failure, deep vein thrombosis or venous obstruction, portal hypertension with ascites, volume overload in renal failure) giving a transudate; lowered plasma oncotic pressure (nephrotic syndrome, cirrhosis, protein malnutrition or kwashiorkor, protein-losing enteropathy) giving a transudate; sodium and water retention (renal failure, renin, angiotensin, aldosterone activation in heart failure) giving a transudate; increased permeability (acute inflammation, burns, ARDS, anaphylaxis) giving an exudate; lymphatic obstruction (filariasis from Wuchereria bancrofti, tumor, lymph node dissection or radiation, congenital lymphatic hypoplasia) giving protein-rich lymph.

· Transudate: vessel wall intact, forces out of balance, water and small solutes cross while protein stays behind, so it is clear with low protein and low cell content and a specific gravity below about 1.012.

· Exudate: inflammatory mediators (histamine, bradykinin, leukotrienes) contract endothelial cells and open gaps, or the endothelium is directly injured, so protein, fibrinogen, and white cells leave, giving cloudy fluid with high protein and a specific gravity above about 1.020.

· Light criteria for pleural fluid mark an exudate: a fluid-to-serum protein ratio above 0.5, a fluid-to-serum LDH ratio above 0.6, or fluid LDH above two-thirds of the upper limit of normal for serum.

· Renal sodium and water retention adds to the hydrostatic and oncotic mechanisms by expanding plasma volume.

Distinctions that matter

· Transudate against exudate is a statement about the vessel wall, not about the volume of fluid or the organ involved.

· Lymphedema fluid is protein rich like an exudate but the wall is intact; the protein is there because lymph cannot be returned.

Common misconceptions

· Treating any high-protein fluid as inflammatory. Blocked lymphatics give protein-rich fluid without inflammation.

· Expecting one mechanism per patient. Heart failure raises hydrostatic pressure and adds renal retention at the same time.

Where it is taught: Section C06, opening paragraph, the mechanisms of edema table, and the transudate against exudate paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C06-BD2

Brain dump 2 · integration and application: Four swollen patients, four mechanisms

A competent response contains

· Nephrotic syndrome: albumin loss in the urine lowers plasma oncotic pressure, the fluid is a transudate with low protein, it pits, and it appears first in loose tissue such as the eyelids.

· Heart failure: raised venous and therefore capillary hydrostatic pressure, with renal sodium and water retention adding plasma volume; transudate, pits, and collects in the legs when standing and over the sacrum when lying.

· Axillary node dissection: lymphatic obstruction, so retained fluid is protein-rich lymph, the arm is affected, and it begins soft and pitting but becomes nonpitting.

· Lymphedema becomes hard because retained protein stimulates fibroblasts and adipose deposition, so skin and subcutaneous tissue thicken; over years this is elephantiasis, and blocked dermal lymphatics in breast cancer give the dimpled peau d'orange skin.

· Hypothyroid myxedema is nonpitting for a different reason: glycosaminoglycans bind water in the dermis, so the fluid is not free in the interstitium.

· Ordinary edema pits because the excess fluid is free in the interstitium: firm pressure pushes it aside and the dent fills slowly.

· It collects where hydrostatic pressure is highest, which is the legs when standing and the sacrum when lying.

Distinctions that matter

· Pitting depends on whether the fluid is free to move, which is why both fibrotic lymphedema and myxedema fail to pit for different reasons.

· Nephrotic edema starts in loose tissue such as the eyelids because the oncotic deficit is systemic, while hydrostatic edema starts where pressure is highest.

Common misconceptions

· Calling all nonpitting swelling lymphedema. Myxedema is nonpitting from bound glycosaminoglycans.

· Describing lymphedema as permanently soft. It starts soft and becomes hard as protein drives fibrosis.

Where it is taught: Section C06, mechanisms table and the final paragraph on pitting and lymphedema of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C06-RT01

Retrieval exercise: Rebuild the edema cycle in heart failure

A competent response contains

· Step 1: cardiac output falls.

· Step 2: renal perfusion falls and renin rises.

· Step 3: angiotensin II and aldosterone retain sodium and water; ADH contributes water retention.

· Step 4: plasma volume and venous pressure rise.

· Step 5: capillary hydrostatic pressure rises and edema forms; the changed Starling force is capillary hydrostatic pressure and the fluid is a transudate.

· Step 6: the failing heart is further overloaded, which lowers output again and closes the loop.

· Renal retention is meant to restore perfusion by raising filling, but in a failing ventricle the extra volume raises venous pressure and worsens both edema and the load on the heart.

· Lowering venous pressure interrupts the loop at step four, before capillary hydrostatic pressure rises.

Distinctions that matter

· The same volume expansion that would help a hypovolemic patient harms a failing ventricle, because the ventricle cannot translate added filling into output.

· Hydrostatic edema in heart failure is driven by two mechanisms at once, the raised venous pressure and the renal retention that feeds it.

Common misconceptions

· Reading the renal response as a malfunction. It is a normal response to a low perfusion signal, which is why it fires in a patient who is not volume depleted.

· Calling this fluid an exudate because it is caused by disease. The vessel wall is intact, so it is a transudate.

Where it is taught: Section C06, the edema in heart failure cycle of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-BD1

Brain dump 1 · foundational retrieval: Build the hemodynamic table of shock

A competent response contains

· Shock is widespread tissue hypoperfusion in which oxygen delivery falls below metabolic demand, from low cardiac output or low effective circulating volume.

· Mean arterial pressure equals cardiac output times systemic vascular resistance, and cardiac output depends on venous return, so perfusion fails when the volume, the pump, the path through the heart, or vascular tone fails.

· The three measurements are cardiac output, systemic vascular resistance, and pulmonary capillary wedge pressure, which approximates left atrial pressure and so left ventricular filling pressure.

· Hypovolemic: loss of blood or fluid (hemorrhage, burns, vomiting, diarrhea); output decreased, resistance increased as a reflex, wedge decreased, skin cold and clammy.

· Cardiogenic: pump failure (myocardial infarction, myocarditis, arrhythmia, acute valve rupture); output decreased, resistance increased as a reflex, wedge increased, skin cold and clammy.

· Obstructive: block to filling or outflow (tamponade, massive pulmonary embolism, tension pneumothorax); output decreased, resistance increased as a reflex, wedge increased in tamponade with equal diastolic pressures and normal or low in pulmonary embolism, skin cold and clammy.

· Septic (distributive): vasodilation and capillary leak from the host response to infection; output increased or normal early and may fall late, resistance decreased, wedge decreased or normal, skin warm and flushed early.

· Anaphylactic (distributive): mast cell and basophil release of histamine and other mediators, usually IgE mediated; output usually decreased by vasodilation and leak and may be normal, resistance decreased, wedge decreased, skin warm and flushed with urticaria and angioedema.

· Neurogenic (distributive): loss of sympathetic tone after spinal cord injury usually above about T6; output decreased, resistance decreased, wedge decreased, skin warm and dry with bradycardia.

· Low-output shock raises resistance as a reflex; distributive shock lowers resistance as the primary defect.

· Normal values: cardiac output about 4 to 8 L/min, systemic vascular resistance about 800 to 1,200 dyn.s/cm5, pulmonary capillary wedge pressure about 6 to 12 mm Hg.

· Central venous pressure follows the wedge pressure except in right-sided obstruction such as pulmonary embolism, where it rises while the wedge stays normal or low.

Distinctions that matter

· High resistance in low-output shock is a response, while low resistance in distributive shock is the disease; naming which is primary is the whole point of the table.

· Neurogenic shock is distributive but shares a low cardiac output with the low-output types, and bradycardia with warm dry skin is what marks it out.

Common misconceptions

· Grouping all three distributive types as high output. Anaphylactic and neurogenic shock usually have a low output, and only early septic shock is reliably hyperdynamic.

· Expecting the wedge pressure to be high in all obstructive shock. It is high in tamponade and normal or low in massive pulmonary embolism.

Where it is taught: Section C07, opening paragraph and the hemodynamic profiles table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-BD2

Brain dump 2 · integration and application: Read four hemodynamic profiles

A competent response contains

· Patient A: cardiogenic shock. Output 2.5 L/min is low and is the primary change, resistance 1,800 is a reflex, and a wedge of 26 mm Hg means the left ventricle is full but not emptying, which also causes pulmonary edema.

· Patient B: hypovolemic shock. Output 2.8 L/min is low, resistance 1,700 is a reflex, and a wedge of 4 mm Hg shows the ventricle is empty, so reduced venous return is the primary problem.

· Patient C: early septic shock. Resistance of 450 is the primary defect, and an output of 8.5 L/min has risen because low resistance and reflex tachycardia allow it; the wedge of 8 mm Hg is normal.

· Patient D: obstructive shock from massive pulmonary embolism. Output is low with reflex high resistance, the central venous pressure of 20 mm Hg is raised because the right ventricle cannot push blood through the lungs, and the wedge of 7 mm Hg is normal or low because the left ventricle is underfilled.

· Central venous pressure and wedge pressure diverge when the obstruction sits between them, which happens in massive pulmonary embolism; tension pneumothorax is the other cause of obstructive shock with raised right-sided pressure.

· In tamponade, fluid around the heart limits filling of all chambers, so diastolic pressures in the atria, ventricles, and the wedge rise and equalize.

· Patient C has a normal or high mixed venous oxygen saturation; tissues remain hypoxic because flow is maldistributed and mitochondrial oxygen use is impaired, unlike the low saturation of low-output shock.

Distinctions that matter

· A low output with a high wedge separates cardiogenic from hypovolemic shock, since both have low output and high resistance.

· Mixed venous saturation separates maldistributed flow from inadequate flow: it is high when blood passes tissues without giving up oxygen and low when there is not enough flow.

Common misconceptions

· Reading a high resistance as vasoconstrictor disease. In these patients it is the baroreceptor reflex doing its job.

· Assuming a high cardiac output rules out shock. Early septic shock is hyperdynamic and the tissues are still hypoxic.

Where it is taught: Section C07, hemodynamic profiles table, the paragraphs on low-output and distributive shock, and the worked example of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-CP01

Comparison table: Cardiogenic against hypovolemic against septic shock

A competent response contains

· Primary problem: cardiogenic is pump failure from myocardial infarction, myocarditis, arrhythmia, or acute valve rupture; hypovolemic is loss of blood or fluid from hemorrhage, burns, vomiting, or diarrhea; septic is vasodilation and capillary leak from the host response to infection.

· Cardiac output: decreased and primary in cardiogenic; decreased in hypovolemic as a consequence of reduced venous return; increased or normal early in septic shock as a response to low resistance, and it may fall late.

· Systemic vascular resistance: increased as a reflex in cardiogenic and hypovolemic shock; decreased as the primary defect in septic shock.

· Wedge pressure: increased in cardiogenic shock, showing a full ventricle that cannot empty; decreased in hypovolemic shock, showing an empty ventricle; decreased or normal in septic shock.

· Skin: cold and clammy in cardiogenic and hypovolemic shock from intense arteriolar constriction; warm and flushed early in septic shock from vasodilation.

· Mixed venous oxygen saturation: low in cardiogenic and hypovolemic shock because tissues extract more from a reduced flow; normal or high in septic shock because flow is maldistributed and mitochondrial oxygen use is impaired.

· Why tissues are underperfused: in cardiogenic shock the pump cannot deliver, in hypovolemic shock there is too little volume to deliver, and in septic shock there is flow but it is distributed away from nutrient capillaries and cannot be used.

Distinctions that matter

· Cardiogenic and hypovolemic shock look identical except at the wedge pressure, which is why that one measurement decides between them.

· Septic shock is the one column where the cardiac output can be normal or high while the patient is in shock.

Common misconceptions

· Expecting warm skin to mean a patient is not in shock. Vasodilation is the defect in septic shock.

· Writing a low mixed venous saturation for septic shock by analogy with the other two. Maldistributed flow and impaired oxygen use raise it.

Where it is taught: Section C07, hemodynamic profiles table and the paragraphs on low-output and distributive shock of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-PR01

Prediction exercise: Predict the profile after a spinal cord injury at T4

A competent response contains

· Sympathetic outflow to arterioles and veins below the injury is lost, because the cord injury interrupts the sympathetic pathways leaving the cord.

· Systemic vascular resistance falls, and the fall is the primary defect rather than a reflex, which makes this distributive shock.

· Venous return falls because venous capacitance rises and blood pools, so the wedge pressure falls.

· Cardiac output falls, because the reduced venous return lowers filling and stroke volume.

· Heart rate falls: cut sympathetic pathways cannot raise it and unopposed vagal tone causes bradycardia, unlike every other low-output shock, where the baroreceptor reflex produces tachycardia.

· The skin is warm and dry, because arteriolar constriction and sympathetic sweating are both lost; in hypovolemic shock intense reflex constriction makes the skin cold and clammy.

· Neurogenic shock follows cord injury usually above about T6.

Distinctions that matter

· Neurogenic shock has a low output and a low resistance together, which separates it from both hypovolemic and septic profiles.

· Bradycardia with hypotension is the single sign that points away from every reflex-intact form of shock.

Common misconceptions

· Expecting tachycardia because the blood pressure is low. The reflex pathway to the heart is interrupted.

· Calling the skin warm and flushed as in sepsis. It is warm and dry, since sympathetic sweating is lost as well.

Where it is taught: Section C07, hemodynamic profiles table and the distributive shock paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-BD1

Brain dump 1 · foundational retrieval: The three stages of shock and the organs that fail

A competent response contains

· Nonprogressive (compensated) stage: baroreceptor reflexes, catecholamines, the renin, angiotensin, and aldosterone system, and ADH raise heart rate, constrict arterioles in skin, gut, and kidney, and retain salt and water; autoregulation keeps coronary and cerebral flow nearly normal, so there is tachycardia, cool pale skin except in early sepsis, and reduced urine output, with a blood pressure that may be near normal.

· Progressive stage: perfusion is inadequate despite the reflexes; cells shift to anaerobic glycolysis and lactate accumulates, giving a high anion gap metabolic acidosis with serum lactate above the normal of about 2 mmol/L; acidosis blunts arteriolar responses to sympathetic stimulation while venules stay constricted, so blood pools and leaks, and it depresses contractility; patients become confused, oliguric, and hypotensive.

· Irreversible stage: restoring hemodynamics no longer saves the patient; ATP is depleted, lysosomal enzymes leak from injured cells, the myocardium fails, and ischemic bowel lets bacteria and endotoxin enter the blood, adding sepsis to the original insult.

· Kidney: acute tubular necrosis, because the proximal straight tubule and thick ascending limb in the outer medulla have high energy demand and little oxygen; oliguria, rising creatinine, muddy brown granular casts.

· Lung: ARDS or diffuse alveolar damage from neutrophil and cytokine injury to endothelium and type I pneumocytes; protein-rich edema, hyaline membranes, hypoxemia with a normal left atrial pressure.

· Blood: disseminated intravascular coagulation, where microthrombi consume platelets and factors, giving low platelets and fibrinogen, long prothrombin and partial thromboplastin times, high D-dimer, and schistocytes.

· Liver: ischemic hepatitis or shock liver, with centrilobular zone 3 necrosis and very high aminotransferases.

· Gut, brain, and heart: mucosal necrosis, encephalopathy, and subendocardial necrosis, since areas perfused last fail first, and gut injury allows bacterial translocation.

· Adrenal: hemorrhagic necrosis, with bilateral adrenal hemorrhage in meningococcal sepsis, the Waterhouse-Friderichsen syndrome.

· The rule: each organ fails where its blood supply is weakest relative to its metabolic demand.

Distinctions that matter

· The compensated stage can have a near normal blood pressure, so urine output, skin, and heart rate are what reveal it.

· Acidosis in the progressive stage is both a consequence of hypoperfusion and a cause of further hypoperfusion, which is what makes the stage progressive.

Common misconceptions

· Treating hypotension as the first sign of shock. Reflexes hold the pressure up through the whole compensated stage.

· Expecting the irreversible stage to be defined by a particular pressure. It is defined by the patient no longer recovering when hemodynamics are restored.

Where it is taught: Section C08, the three stage paragraphs and the multiorgan dysfunction table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-BD2

Brain dump 2 · integration and application: From one gram negative bacterium to failure in five organs

A competent response contains

· Septic shock is caused by the host response, not by the bacteria directly.

· The molecule is lipopolysaccharide, endotoxin of gram-negative bacteria, and lipid A is its toxic portion.

· It binds LPS-binding protein, CD14, and TLR4 on macrophages, neutrophils, and endothelium; gram-positive wall components signal through TLR2, and superantigens activate T cells nonspecifically.

· NF-kappa B is activated and drives release of TNF and IL-1, which induce IL-6 and chemokines, while microbial products activate complement to generate C3a and C5a.

· Inducible nitric oxide synthase in endothelium and smooth muscle makes large amounts of nitric oxide, which relaxes vascular smooth muscle through cGMP and lowers systemic vascular resistance; this is the warm flushed skin and the hypotension.

· TNF and IL-1 loosen endothelial junctions, which gives the capillary leak and a hypovolemic component, and they depress myocardial contraction.

· Cytokines push coagulation toward clotting: they induce tissue factor, reduce thrombomodulin so less protein C is activated, and raise plasminogen activator inhibitor-1, suppressing fibrinolysis; microthrombi and consumption of platelets and factors produce disseminated intravascular coagulation, which is why the tests show clotting and bleeding at the same time, with a falling platelet count and a rising D-dimer.

· The rising creatinine is acute tubular necrosis, which appears in the outer medulla because the proximal straight tubule and thick ascending limb there have high energy demand and little oxygen.

· The hypoxemia is ARDS: neutrophil and cytokine injury to endothelium and type I pneumocytes gives protein-rich edema and hyaline membranes, so the left atrial pressure is normal and the edema is not from a failing left ventricle.

· Her shock would keep worsening because the positive feedback cycle of the progressive stage is now self-sustaining: each fall in flow causes a further fall even if the original insult stops.

· In the irreversible stage, restoring hemodynamics no longer helps.

Distinctions that matter

· Low resistance, capillary leak, and microthrombi all come from the same cytokine-activated endothelium, which is why one infection produces vasodilation, hypovolemia, and clotting together.

· ARDS edema and cardiogenic pulmonary edema are separated by the left atrial pressure: normal in ARDS, raised in left ventricular failure.

Common misconceptions

· Blaming the organ damage on the bacteria spreading to each organ. The host cytokine response and hypoperfusion do the damage.

· Reading a long prothrombin time with a low platelet count as a primary bleeding disorder. Consumption by microthrombi is what lengthens the times.

Where it is taught: Section C08, septic shock mediators paragraph, the endotoxin flow diagram, the coagulation paragraph, and the multiorgan dysfunction table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-RT01

Retrieval exercise: Rebuild the endotoxin sequence and the vicious cycle

A competent response contains

· Step 1: endotoxin binds CD14 and TLR4 on macrophages and endothelium.

· Step 2: NF-kappa B is activated and TNF and IL-1 are released.

· Step 3: inducible nitric oxide synthase makes nitric oxide, which relaxes vascular smooth muscle through cGMP.

· Step 4: endothelial leak develops, tissue factor goes up and protein C activation goes down, so microthrombi form in small vessels.

· Step 5: low systemic vascular resistance, hypovolemia, microthrombi, and myocardial depression.

· Step 6: hypoperfusion and multiorgan dysfunction.

· Vasodilation, leak, and clotting all arise from the same cytokine-activated endothelium.

· Progressive shock cycle: cardiac output and arterial pressure fall; tissue hypoxia with lactic acidosis; arterioles dilate and blood pools in capillaries; capillary leak and venous return falls; acidosis depresses the myocardium; cardiac output falls further.

· The two points where falling flow causes a further fall are the pooling and leak step, which lowers venous return, and the acidosis step, which depresses the myocardium.

· Once each fall in flow causes a further fall, shock worsens even if the original insult stops.

Distinctions that matter

· Nitric oxide lowers resistance while TNF and IL-1 open junctions and depress the myocardium, so the three hemodynamic problems have three separate mediators.

· Loss of fibrinolysis matters as much as the gain in tissue factor, which is why the microthrombi persist.

Common misconceptions

· Treating the cycle as needing continued infection. It is self-sustaining once started.

· Expecting the venules to dilate with the arterioles. They stay constricted, which is what traps blood in the capillary bed.

Where it is taught: Section C08, the endotoxin flow diagram and the positive feedback cycle of progressive shock of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-CA01

Clinical application: Hypotension and bleeding gums three days after a fracture

A competent response contains

· Recognize: septic shock from gram-negative infection, in the progressive stage, with acute tubular necrosis in the kidney, ARDS in the lung, and disseminated intravascular coagulation in the blood.

· Explain: lipopolysaccharide with lipid A as its toxic portion binds LPS-binding protein, CD14, and TLR4; NF-kappa B activation releases TNF and IL-1, which induce IL-6 and chemokines; inducible nitric oxide synthase makes nitric oxide, which relaxes vascular smooth muscle through cGMP and lowers resistance, giving warm flushed skin and hypotension, while TNF and IL-1 loosen endothelial junctions to produce the leak.

· Differentiate: her resistance is low as the primary defect with a normal or high output and a normal or low wedge pressure, and her skin is warm, whereas cardiogenic and hypovolemic shock both have a low output with reflex high resistance and cold clammy skin; the wedge pressure separates those two from each other, being high in cardiogenic and low in hypovolemic shock, and systemic vascular resistance is the measurement that marks hers out as distributive.

· Investigate: for acute tubular necrosis, creatinine and urine output and the urine sediment, where muddy brown granular casts appear; for ARDS, arterial blood gas showing hypoxemia with a chest radiograph showing bilateral infiltrates and a left atrial or wedge pressure that is normal; for disseminated intravascular coagulation, platelet count and fibrinogen falling, prothrombin and partial thromboplastin times lengthening, D-dimer rising, and schistocytes on the smear.

· Decide: disseminated intravascular coagulation explains both, since microthrombi form while platelets and clotting factors are consumed in making them, leaving nothing to seal her gums and puncture sites; at the progressive stage her shock can still respond to restoring the circulation, but the positive feedback cycle means it will keep worsening until it is interrupted.

· Justify: cytokines induce tissue factor, reduce thrombomodulin so less protein C is activated, and raise plasminogen activator inhibitor-1 to suppress fibrinolysis, so clotting is driven and clot removal is blocked; the irreversible stage is defined by hemodynamic restoration no longer helping, with ATP depletion, lysosomal enzyme leak, myocardial failure, and bacterial translocation from ischemic bowel, and she has not reached that point.

Distinctions that matter

· Thrombosis and bleeding in the same patient are one process at two stages: microthrombi first, then consumption of what made them.

· Progressive and irreversible stages are separated by whether restoring hemodynamics still helps, not by how abnormal the numbers are.

Common misconceptions

· Reading warm flushed skin with hypotension as a reason to look away from shock.

· Taking her bilateral infiltrates as cardiogenic pulmonary edema. In ARDS the left atrial pressure is normal and the edema is protein rich from injured endothelium.

Where it is taught: Section C08, stages paragraphs, mediators paragraph, coagulation paragraph, and the multiorgan dysfunction table, with the hemodynamic profiles table in C07 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-Q01

Board-style question: Sudden collapse after a flight

Correct answer: A. Hyperventilation driven by hypoxemia and stimulated receptors, with ventilation continuing in lung that is no longer perfused

Why it is right: Hypoxemia and stimulated receptors increase total ventilation, and carbon dioxide is far more diffusible and more responsive to ventilation than oxygen, so the hyperventilation of the perfused lung lowers PaCO2 while the ventilated but unperfused segment adds dead space and does not clear carbon dioxide at all.

B is wrong: Reduced carbon dioxide production by hypoperfused tissues is far too small and too slow to account for an acute fall, and hypoperfused tissues in fact generate an acid load from anaerobic glycolysis.

C is wrong: The affected segments are still ventilated rather than collapsed, which is exactly why dead space rises, and loss of alveolar surface would raise carbon dioxide rather than lower it.

D is wrong: The disturbance here is a respiratory one driven by increased ventilation, not a metabolic alkalosis, and catecholamine release does not remove acid or add bicarbonate.

A competent response contains

· A saddle embolus obstructing more than about 60% of the pulmonary circulation causes acute right heart failure, circulatory collapse, and sudden death.

· Embolized lung is ventilated but not perfused, so dead space rises, and hypoxemia drives hyperventilation that lowers PaCO2.

Distinctions that matter

· Perfusion is lost while ventilation continues, so the gas pattern is a low oxygen with a low carbon dioxide rather than retention.

Common misconceptions

· Expecting carbon dioxide to rise whenever a pulmonary vessel is blocked.

Where it is taught: Section C03 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-Q02

Board-style question: A profile with a high filling pressure

Correct answer: C. Failure of the left ventricle to empty, with reflex arteriolar constriction

Why it is right: Pump failure lowers cardiac output, the baroreceptor reflex raises systemic vascular resistance and makes the skin cool and mottled, and blood backing up behind the ventricle that cannot empty raises the wedge pressure to 28 mm Hg and floods the lungs.

A is wrong: Loss of circulating volume also gives a low output with reflex constriction, but the ventricle would be underfilled and the wedge pressure low rather than 28 mm Hg.

B is wrong: Reduced sympathetic tone would lower systemic vascular resistance rather than raise it to 1,900, and it would give warm dry skin with bradycardia.

D is wrong: Cytokine-driven vasodilation with leak lowers resistance and leaves the wedge pressure normal or low, and the skin would be warm and flushed early.

A competent response contains

· A low cardiac output with a high systemic vascular resistance and a high wedge pressure is cardiogenic shock, here after a myocardial infarction.

· The high wedge pressure means the left ventricle is full but not emptying, which also explains the crackles from pulmonary edema.

Distinctions that matter

· The wedge pressure is the one measurement that separates cardiogenic from hypovolemic shock, since both have low output and reflex high resistance.

Common misconceptions

· Reading the high resistance as the primary problem when it is the baroreceptor reflex responding to a low output.

Where it is taught: Section C07 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C02-Q03

Board-style question: A long clotting time with thrombosis

Correct answer: B. Antibodies against beta2-glycoprotein I that activate platelets, endothelium, and complement

Why it is right: Antibodies to beta2-glycoprotein I activate platelets, endothelium, and complement, which produces arterial and venous thrombosis and recurrent pregnancy loss, while binding the phospholipid in the laboratory assay prolongs the partial thromboplastin time and anticardiolipin antibodies cross-react in nontreponemal syphilis testing.

A is wrong: Factor V Leiden causes activated protein C resistance and venous thrombosis, but it does not prolong the partial thromboplastin time and does not cause a false-positive syphilis screen.

C is wrong: Antithrombin deficiency leaves thrombin and factor Xa uninactivated and blunts the effect of heparin, but it does not produce an inhibitor pattern on mixing or a positive syphilis screen.

D is wrong: Prothrombin G20210A raises plasma prothrombin and causes venous thrombosis, with no effect on the partial thromboplastin time and no cross-reacting antibody.

A competent response contains

· Venous thrombosis with recurrent pregnancy loss, a prolonged partial thromboplastin time that does not fully correct on mixing, and a false-positive nontreponemal syphilis test is antiphospholipid syndrome.

· Antibodies bind phospholipid-binding proteins, mainly beta2-glycoprotein I, and activate platelets, endothelium, and complement, causing arterial and venous thrombosis.

Distinctions that matter

· An inhibitor in the plasma does not correct on mixing with normal plasma, whereas a factor deficiency does, which is why the long time here points to an antibody rather than a missing factor.

Common misconceptions

· Taking a long clotting time as evidence of bleeding risk. The antibody interferes with the phospholipid in the assay while the patient clots in vivo.

Where it is taught: Section C02 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-Q04

Board-style question: A cavity where an infarct had been

Correct answer: D. Brain tissue has little collagen framework and is lipid rich, so lysosomal enzymes liquefy it and microglia clear it away

Why it is right: In almost every organ ischemia denatures structural proteins and lytic enzymes alike, so architecture persists and the area is replaced by scar, while the brain's scant collagen and lipid-rich tissue allow lysosomal enzymes to liquefy it and microglia to clear the debris, leaving a cyst lined by reactive astrocytes.

A is wrong: Color records where blood came from and does not decide healing, and a white infarct in a solid organ scars while a brain infarct cavitates whether it is red or white.

B is wrong: Reperfusion after lysis of an embolus makes a brain infarct red, but the liquefaction and the cavity follow from the tissue itself, so an unreperfused brain infarct also cavitates.

C is wrong: Differing tolerance of hypoxia decides how quickly and how widely cells die, not what the dead tissue becomes afterwards.

A competent response contains

· The necrosis pattern depends on the tissue rather than on the color of the infarct.

· The brain has little collagen framework and lipid-rich tissue that released lysosomal enzymes digest and microglia clear, so the infarct liquefies and leaves a fluid-filled cavity lined by reactive astrocytes, while coagulative necrosis elsewhere is replaced by scar.

Distinctions that matter

· Color and necrosis pattern are separate questions, so a red brain infarct and a white kidney infarct still heal according to their tissue.

Common misconceptions

· Assuming blood in the dead tissue decides how an infarct heals.

Where it is taught: Section C04 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-I Vascular Conditions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-09-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild the three patterns of arteriosclerosis

A competent response contains

· Arteriosclerosis means thickening and loss of elasticity of arterial walls and is the umbrella term for three patterns.

· Atherosclerosis: intimal disease of elastic and large to medium muscular arteries, with lipid, inflammatory cells, and fibrous tissue forming plaques that bulge into the lumen and narrow it.

· Hyaline arteriolosclerosis: small arteries and arterioles; plasma proteins leak across injured endothelium and smooth muscle cells lay down excess matrix, giving a homogeneous pink, glassy wall with a narrowed lumen.

· Hyaline form accompanies aging, long-standing benign hypertension, and diabetes; in the kidney it causes nephrosclerosis with a finely granular cortical surface, and in diabetes it affects both afferent and efferent arterioles.

· Hyperplastic arteriolosclerosis: severe or malignant hypertension; smooth muscle cells and reduplicated basement membrane form concentric onion-skin layers, often with fibrinoid necrosis, and the markedly narrowed lumen produces renal ischemia with pinpoint cortical hemorrhages.

· Monckeberg medial calcific sclerosis: calcification of the media and internal elastic lamina of medium muscular arteries, typically femoral, tibial, radial, and ulnar, in people over about 50.

· In Monckeberg sclerosis the intima and lumen are spared, so flow is not reduced; the vessel is palpably hard and appears as a pipestem on radiographs.

· Atherosclerosis is only one of the three patterns, so arteriosclerosis is the broader term.

Distinctions that matter

· Layer and vessel size separate the three: intima of large arteries, wall of arterioles, media of medium limb arteries.

· Only atherosclerosis and arteriolosclerosis reduce the lumen; Monckeberg sclerosis stiffens the wall without obstructing flow.

Common misconceptions

· Using arteriosclerosis and atherosclerosis as synonyms, which loses the two non-plaque patterns.

· Assuming calcification of an artery must mean reduced flow. Medial calcification leaves the lumen open and only stiffens the vessel.

Where it is taught: Section C01, all three paragraphs and the patterns table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C01-BD2

Brain dump 2 · integration and application: Two patients with hardened arteries

A competent response contains

· First patient: Monckeberg medial calcific sclerosis, with calcium in the media and internal elastic lamina; the intima and lumen are spared so flow is not reduced.

· A stiff calcified artery resists cuff compression, so the cuff pressure needed to stop flow exceeds the true intraluminal pressure and the ankle reading is falsely high.

· An index above 1.40 therefore means noncompressible arteries and an unreliable result, common in diabetes and chronic kidney disease, and it does not exclude coexisting atherosclerotic disease.

· Second patient: hyperplastic arteriolosclerosis of severe or malignant hypertension.

· The onion-skin layers are concentric smooth muscle cells and reduplicated basement membrane, often with fibrinoid necrosis of the wall.

· The markedly narrowed lumen reduces renal perfusion, producing ischemia with pinpoint cortical hemorrhages.

· A finely granular cortical surface with glassy pink arterioles is hyaline arteriolosclerosis with benign nephrosclerosis, from aging, long-standing benign hypertension, or diabetes.

Distinctions that matter

· A high index and a low index both come from a stenosis question, but a high index reports wall stiffness while a low index reports lost pressure across a narrowing.

· Hyaline and hyperplastic arteriolosclerosis affect the same vessels; the material in the wall and the severity of the hypertension separate them.

Common misconceptions

· Reading an index of 1.52 as excellent perfusion. Any value above 1.40 is uninterpretable.

· Expecting Monckeberg sclerosis to cause claudication. It does not narrow the lumen, so symptoms in such a patient come from atherosclerosis alongside it.

Where it is taught: Section C01, third paragraph and the patterns table, with the index bands in Section C06 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C01-CP01

Comparison table: The patterns of arterial hardening side by side

A competent response contains

· Vessels: atherosclerosis in the aorta and large to medium muscular arteries; both arteriolosclerosis forms in arterioles and small arteries; Monckeberg sclerosis in medium muscular arteries of the limbs.

· Layer: atherosclerosis is intimal; arteriolosclerosis involves the arteriolar wall with subendothelial protein and matrix or hyperplastic smooth muscle; Monckeberg sclerosis is medial with the internal elastic lamina.

· Lesion material: lipid core with a fibrous cap of smooth muscle and collagen; leaked plasma protein plus excess matrix giving a glassy pink wall; concentric smooth muscle and reduplicated basement membrane with fibrinoid necrosis; calcium.

· Lumen and flow: narrowed and liable to thrombose; narrowed; markedly narrowed; not narrowed and flow preserved.

· Setting: smoking, high LDL, hypertension, diabetes, and age; aging, benign hypertension, and diabetes; severe or malignant hypertension; age over about 50 with diabetes or chronic kidney disease.

· Ankle pressure: atherosclerotic stenosis lowers it and lowers the index; arteriolosclerosis of small vessels does not change a limb pressure measurement; Monckeberg calcification makes the artery noncompressible so the reading is falsely high and the index can exceed 1.40.

Distinctions that matter

· The first three patterns narrow the lumen while the fourth stiffens it, so only the fourth produces a falsely high ankle pressure.

· Hyaline disease tracks long-standing mild to moderate hypertension and diabetes; hyperplastic disease tracks a severe pressure rise.

Common misconceptions

· Putting Monckeberg sclerosis in the intima because it calcifies. The calcium sits in the media and internal elastic lamina.

· Treating fibrinoid necrosis as a feature of all arteriolosclerosis. It belongs to the hyperplastic form.

Where it is taught: Section C01, patterns of arteriosclerosis table, with the index bands in Section C06 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-BD1

Brain dump 1 · foundational retrieval: Build an atheroma from the first injury

A competent response contains

· Order: endothelial dysfunction at branch points and curvatures; LDL enters the intima and is oxidized; monocytes adhere, migrate in, and become macrophages; scavenger receptor uptake of oxidized LDL makes foam cells; fatty streak forms as a flat intimal collection of foam cells; PDGF and other growth factors recruit smooth muscle from the media; a fibrous cap of collagen covers a necrotic lipid core; rupture, erosion, hemorrhage, or calcification gives a complicated plaque.

· Macrophages use scavenger receptors, which are not down-regulated by cell cholesterol, so uptake of oxidized LDL continues until the cell becomes a foam cell.

· Endothelial dysfunction is promoted by hypertension, smoking, hyperlipidemia, hyperglycemia, and inflammation, and the dysfunctional endothelium becomes more permeable and expresses VCAM-1.

· A mature atheroma has a fibrous cap of smooth muscle cells and collagen over a core of lipid, necrotic debris, foam cells, and cholesterol crystals.

· Smooth muscle cells recruited from the media build the cap.

· Distribution in descending order: abdominal aorta more than thoracic, coronary arteries, popliteal arteries, internal carotid arteries, circle of Willis.

· Modifiable risks: smoking, hypertension, raised LDL, diabetes. Fixed risks: age, male sex, family history.

· Plaques are patchy rather than uniform along a vessel.

Distinctions that matter

· Scavenger receptor uptake is unregulated, unlike LDL receptor uptake, which is why macrophages overload.

· The fatty streak is a flat foam cell collection and not obstructive; the mature plaque bulges into the lumen.

Common misconceptions

· Describing atherosclerosis as simple lipid deposition. It is a chronic inflammatory response of the wall to endothelial injury.

· Expecting plaque where flow is fastest. It forms where flow is disturbed rather than laminar, at branch points and curvatures.

Where it is taught: Section C02, opening paragraph and the atherogenesis flow of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-BD2

Brain dump 2 · integration and application: Why a small plaque can kill and a big one can ache

A competent response contains

· Patient A: a lipid-rich plaque with a thin, inflamed cap ruptured, exposing thrombogenic contents, and the overlying thrombus occluded the artery, so the event depended on stability rather than size.

· Patient B: a stable plaque narrowed the lumen gradually, and once stenosis reaches about 70% flow cannot rise to meet demand, so ischemia appears with exertion and settles with rest.

· Vulnerable plaque features: a thin fibrous cap, a large lipid core, and many inflammatory cells.

· Macrophage metalloproteinases digest the collagen of the cap, and T cell interferon gamma suppresses smooth muscle collagen synthesis, so the cap is not repaired.

· Cap rupture exposes tissue factor and collagen, which trigger thrombosis, and a thrombus forms on the disrupted surface.

· This is why many myocardial infarctions arise from plaques that were not severely stenotic.

· Complications and mechanisms: stenosis from slow plaque growth limiting flow reserve; rupture or erosion with thrombosis from cap disruption exposing tissue factor and collagen; embolism when plaque debris or thrombus breaks off downstream; aneurysm because intimal plaque starves and weakens the underlying media; hemorrhage into plaque when its microvessels rupture and expand it abruptly.

Distinctions that matter

· Stenosis produces chronic demand ischemia; thrombosis on a disrupted plaque produces acute events.

· Plaque size predicts exertional symptoms, while plaque composition predicts sudden occlusion.

Common misconceptions

· Assuming the tightest stenosis is the most dangerous lesion. A modest plaque with a thin cap can occlude a vessel outright.

· Reading hemorrhage into a plaque as the same event as rupture of the cap. The first expands the plaque from within, the second exposes its contents to blood.

Where it is taught: Section C02, vulnerable plaque paragraph, complications table, and the stop and predict box of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-RT01

Retrieval exercise: Rebuild the atherogenesis sequence

A competent response contains

· Step 1: endothelial dysfunction at branch points and curvatures, where flow is disturbed rather than laminar, which is why those sites and not straight segments.

· Step 2: LDL enters the intima and is oxidized.

· Step 3: monocytes adhere, migrate in, and become macrophages; the endothelium displays VCAM-1.

· Step 4: scavenger receptor uptake of oxidized LDL produces foam cells.

· Step 5: fatty streak, a flat intimal collection of foam cells; present in the aortas of virtually all children over about 10 and not obstructive.

· Step 6: PDGF and other growth factors recruit smooth muscle cells from the media.

· Step 7: a fibrous cap of collagen forms over a necrotic lipid core, and this mature plaque is the first lesion that meaningfully narrows the lumen.

· Step 8: rupture, erosion, hemorrhage, or calcification gives a complicated plaque.

Distinctions that matter

· The fatty streak is visible and early but does not obstruct; narrowing begins once smooth muscle and matrix build a raised plaque.

· VCAM-1 brings monocytes in; PDGF brings smooth muscle in. Different signals at different steps.

Common misconceptions

· Placing smooth muscle recruitment before foam cell formation. Foam cells and the fatty streak come first.

· Treating the fatty streak as a disease stage that must progress. It is near universal in children and only some streaks advance.

Where it is taught: Section C02, atherogenesis flow and the paragraph on fatty streaks of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C02-Q01

Board-style question: An infarct from a modest stenosis

Correct answer: C. Macrophage metalloproteinases degraded cap collagen while interferon gamma suppressed its replacement

Why it is right: The described plaque is vulnerable, and in such plaques macrophage metalloproteinases degrade the collagen of the cap while T cell interferon gamma suppresses smooth muscle collagen synthesis, so the cap tears, tissue factor and collagen are exposed, and the overlying thrombus occludes the artery.

A is wrong: Medial calcification, as in Monckeberg sclerosis, stiffens medium limb arteries without narrowing the lumen and plays no part in coronary plaque rupture.

B is wrong: Gradual plaque growth limiting flow reserve produces exertional ischemia once stenosis reaches about 70%, and at 40% it would not cause a resting occlusion or a fresh thrombus.

D is wrong: Plasma protein leakage with excess matrix is hyaline arteriolosclerosis of arterioles in hypertension and diabetes, not a mechanism of coronary cap rupture.

A competent response contains

· A thin fibrous cap, a large lipid core, and dense macrophage infiltration define a vulnerable plaque.

· Macrophage metalloproteinases digest cap collagen and T cell interferon gamma suppresses smooth muscle collagen synthesis, so the cap thins and is not repaired; rupture exposes tissue factor and collagen and a thrombus occludes the vessel.

Distinctions that matter

· Plaque stability rather than plaque size determines the risk of acute occlusion, which is why a 40% stenosis can cause a fatal infarct.

Common misconceptions

· Judging risk from the angiographic percentage alone.

Where it is taught: Section C02 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C03-BD1

Brain dump 1 · foundational retrieval: Draw LDL clearance and then break it

A competent response contains

· Hepatic LDL receptors bind apolipoprotein B-100 on LDL and remove it from plasma by clathrin-mediated endocytosis.

· Cholesterol released in the lysosome keeps SREBP-2 inactive in the endoplasmic reticulum, which lowers synthesis of HMG-CoA reductase and of new LDL receptors.

· LDLR: loss-of-function mutation in the receptor itself, the most common cause.

· APOB: loss-of-function mutation in the receptor binding site of the ligand.

· PCSK9: gain-of-function mutation in a protein that steers the receptor to lysosomal degradation, so fewer receptors return to the surface.

· Inheritance is autosomal dominant.

· Deposits are LDL-derived cholesterol taken up by macrophages: tendon xanthomas in the Achilles tendon and the extensor tendons of the hands; xanthelasma as yellow plaques on the eyelids; corneal arcus as a gray-white ring at the corneal margin.

· Tendon xanthomas are highly suggestive of the disease; xanthelasma and arcus are less specific because both occur in older adults with normal lipids, although arcus before about age 45 suggests it.

· Homozygotes may also have planar cutaneous xanthomas and lipid deposits in the aortic root and valve.

Distinctions that matter

· A loss-of-function mutation in LDLR or APOB and a gain-of-function mutation in PCSK9 all end in the same place: fewer working receptors clearing LDL.

· Tendon xanthoma is specific; eyelid and corneal deposits are common findings in older adults and carry less weight.

Common misconceptions

· Saying the liver overproduces cholesterol as the primary defect. The defect is clearance, and hepatic synthesis continues because the hepatocyte senses too little cholesterol inside itself.

· Treating PCSK9 as a receptor. It directs the receptor to lysosomal degradation.

Where it is taught: Section C03, first and third paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C03-BD2

Brain dump 2 · integration and application: Read the lipid panel and the tendons

A competent response contains

· The boy: LDL above 500 mg/dL with childhood xanthomas means little or no receptor activity, so he is homozygous, and coronary disease or infarction is expected in childhood or the teens.

· The mother: LDL above 190 mg/dL with a tendon xanthoma and early arcus means about half of normal receptor activity, so she is heterozygous, with coronary disease commonly in the fourth to fifth decade in men and somewhat later in women.

· Plasma LDL rises because clearance falls while the hepatocyte, sensing less intracellular cholesterol, keeps SREBP-2 active and continues synthesis; the lipid stays in plasma rather than entering the cell.

· Plasma LDL rises in proportion to the receptor defect, so gene dosage sets the level.

· Triglycerides are usually normal, which separates this disease from combined hyperlipidemias.

· Corneal arcus before about age 45 suggests the diagnosis, but in a 70-year-old it occurs with normal lipids and carries little weight.

· The same process makes all three lesions: LDL that is not cleared accumulates, macrophages take it up through scavenger receptors, and lipid-laden macrophages form the tendon nodule, the eyelid plaque, and the intimal plaque of premature atherosclerosis.

· Heterozygotes are about 1 in 250 people; homozygotes are rare.

Distinctions that matter

· Receptor activity, plasma LDL, and age at first coronary event all move together with gene dosage.

· Normal triglycerides with a very high LDL points at receptor-mediated clearance rather than at overproduction of triglyceride-rich particles.

Common misconceptions

· Expecting the liver to shut down cholesterol synthesis because plasma cholesterol is high. The hepatocyte responds to its own intracellular pool, not to plasma.

· Reading any corneal arcus as evidence of familial hypercholesterolemia regardless of age.

Where it is taught: Section C03, second and third paragraphs and the heterozygous against homozygous table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C04-BD1

Brain dump 1 · foundational retrieval: Build the aneurysm table from memory

A competent response contains

· An aneurysm is a localized permanent dilation of a vessel or the heart; a true aneurysm involves all three layers and a false aneurysm is a leak contained by adventitia or clot, often at an arterial puncture or graft anastomosis.

· An abdominal aorta of 3 cm or more is aneurysmal.

· Abdominal aortic: infrarenal aorta; atherosclerosis with inflammation and matrix metalloproteinases degrading the matrix; men over 60 who smoke, with hypertension and family history as added risks; rupture, with risk rising steeply with diameter.

· Thoracic aortic: ascending aorta and arch; cystic medial degeneration from hypertension or connective tissue disorders such as Marfan syndrome, and tertiary syphilis; aortic root dilation with regurgitation, compression of mediastinal structures, and dissection.

· Berry or saccular: circle of Willis, mostly anterior circulation branch points; thin or absent media at branch points, with ADPKD, Ehlers-Danlos syndrome, hypertension, and smoking; subarachnoid hemorrhage.

· Mycotic: any artery, often aorta or cerebral branches; septic emboli from infective endocarditis or bacteremic seeding of plaque, for example Salmonella; rapid growth and rupture.

· Wall stress is about pressure times radius divided by wall thickness, the law of Laplace.

· Once a segment dilates, the larger radius raises stress, the stretched wall thins which raises it further, and higher stress degrades the remaining matrix, so dilation tends to continue.

· The infrarenal media has few vasa vasorum, so it depends on diffusion from the lumen, and a thick plaque impairs that diffusion and starves the media.

Distinctions that matter

· A false aneurysm is a contained leak rather than a dilated wall, so it has no intact media of its own.

· An aneurysm always means the media has failed; what differs between types is why it failed.

Common misconceptions

· Expecting abdominal aneurysms above the renal arteries. The usual site is infrarenal.

· Treating a berry aneurysm as atherosclerotic. It arises from a congenitally thin or absent media at a branch point.

Where it is taught: Section C04, first paragraph, aneurysm types table, and the worked example on wall stress of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C04-BD2

Brain dump 2 · integration and application: Work the numbers on a growing aorta

A competent response contains

· Radius goes from 1.0 cm to 3.0 cm, so at the same pressure and wall thickness, stress is about three times greater.

· As the wall stretches it thins, and thickness is in the denominator, so stress rises further than the radius change alone predicts.

· Higher stress degrades the remaining matrix, allowing more dilation, which is why expansion accelerates with diameter and rupture risk climbs steeply above about 5.5 cm; risk is very low below 4 cm.

· Hypertension raises the pressure term, so it adds to every step of the feedback.

· Rupture presents with abdominal or back pain, hypotension, and a pulsatile abdominal mass.

· The young tall patient has cystic medial degeneration, which is loss of elastic fibers and smooth muscle with pools of ground substance in the media, as in Marfan syndrome.

· Dilation of the root pulls the valve commissures apart so the cusps no longer meet, giving aortic regurgitation and a diastolic murmur.

· Tertiary syphilis causes obliterative endarteritis of the vasa vasorum of the ascending aorta, so the media becomes ischemic; the intima has a wrinkled tree-bark appearance and the root dilates with regurgitation.

Distinctions that matter

· Marfan syndrome weakens the media directly through defective matrix, while syphilis weakens it by cutting off its blood supply, and both end in root dilation.

· Thoracic and abdominal aneurysms have different usual mechanisms, so the site points to the process.

Common misconceptions

· Calculating stress from diameter instead of radius, which gives the right ratio by luck but the wrong quantity.

· Assuming the aortic valve itself is diseased when the root dilates. The cusps are normal and simply fail to meet.

Where it is taught: Section C04, worked example on wall stress and the paragraphs on abdominal and thoracic aneurysms of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C05-BD1

Brain dump 1 · foundational retrieval: Rebuild a dissection and its consequences

A competent response contains

· In aortic dissection blood enters the media through an intimal tear and splits the wall along its length, creating a false lumen.

· The underlying problem is a weakened media. Hypertension is present in most patients and causes medial degeneration while raising the shear and wall stress that drive the split.

· Younger patients usually have a connective tissue or structural defect: cystic medial degeneration in Marfan syndrome with defective fibrillin-1 and excess TGF-beta signaling, Ehlers-Danlos syndrome, bicuspid aortic valve with its aortopathy, or Turner syndrome, often with bicuspid valve and coarctation.

· Acute triggers are abrupt pressure surges, as with cocaine or the hemodynamic load of pregnancy.

· Sequence: weakened media plus high wall stress; intimal tear, usually in the ascending aorta; blood splits the media into a false lumen; proximal spread to the aortic root with loss of valve support and regurgitation; rupture into the pericardium with hemopericardium and tamponade; branch compression causing coronary, carotid, subclavian, renal, or mesenteric ischemia; external rupture with hemothorax or exsanguination.

· Outward rupture, into the pericardium, pleural cavity, or peritoneum, is the usual cause of death.

· Stanford A involves the ascending aorta, with or without the descending aorta. Stanford B does not involve the ascending aorta and usually begins beyond the left subclavian artery.

· Heavy atherosclerosis or medial scarring tends to stop a dissection from spreading, so dissection is not a typical complication of atherosclerosis.

Distinctions that matter

· A dissection splits the media while an aneurysm dilates it; both start from a weakened media and can occur in the same aorta.

· The ascending aorta decides the Stanford type because its involvement threatens the valve, the pericardium, and the coronary arteries.

Common misconceptions

· Listing dissection as a complication of atherosclerosis. Plaque and scarring tend to arrest a dissection.

· Describing the false lumen as a rupture through the whole wall. The blood stays within the media until it ruptures outward.

Where it is taught: Section C05, first paragraph, the dissection flow, and the Stanford table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C05-BD2

Brain dump 2 · integration and application: Explain each sign in a dissecting aorta

A competent response contains

· Tearing pain moving to between the scapulae reflects the dissection advancing distally along the media, with pain tracking the splitting wall.

· Unequal arm pressures arise when the false lumen narrows a subclavian artery, so the pulse is weak and the pressure is lower on that side.

· A new diastolic murmur is aortic regurgitation from proximal spread into the root with loss of valve support.

· A widened mediastinum on the chest radiograph reflects the dilated, blood-filled aorta and surrounding hematoma.

· Hypotension with distended neck veins and muffled heart sounds means rupture into the pericardium; blood in the pericardial sac raises intrapericardial pressure above filling pressures, so ventricular filling falls and cardiac output falls, which is tamponade.

· Tamponade and aortic regurgitation both require the ascending aorta, so this is Stanford A, the type with the highest risk along with coronary occlusion.

· A tear beyond the left subclavian artery is Stanford B, typical of older patients with hypertension, with lower risk and complications mainly branch ischemia and rupture.

· Branch occlusion shows as neurologic deficits or stroke for the carotid, paraplegia for the spinal supply, oliguria for the renal arteries, and abdominal pain for the mesenteric arteries.

Distinctions that matter

· Unequal arm pressures appear in both dissection and Takayasu arteritis; sudden tearing pain is what points to dissection.

· Hypotension in dissection can come from tamponade or from external rupture and blood loss, and the neck veins separate them.

Common misconceptions

· Reading hypotension with distended neck veins as heart failure. In this setting it is impaired filling from pericardial blood.

· Expecting the pain to stay in one place. It moves as the dissection advances.

Where it is taught: Section C05, clinical paragraph, the dissection flow, Stanford table, and the stop and predict box of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C05-Q02

Board-style question: Tearing chest pain in a tall young man

Correct answer: A. Blood has ruptured from the aorta into the pericardial sac and is limiting ventricular filling

Why it is right: A Stanford A dissection in a patient with a connective tissue disorder can rupture back into the pericardial sac, and blood there raises intrapericardial pressure above filling pressures so ventricular filling and cardiac output fall, which produces hypotension with distended neck veins and muffled heart sounds.

B is wrong: Renal artery occlusion by the false lumen causes oliguria and renin-mediated hypertension rather than acute hypotension, and it does not distend the neck veins or muffle the heart sounds.

C is wrong: Fibrinoid necrosis of arterioles belongs to hyperplastic arteriolosclerosis in severe hypertension and causes renal ischemia, not acute tamponade.

D is wrong: An intimal plaque starving the media produces an aneurysm over years, and heavy atherosclerosis actually tends to stop a dissection from spreading.

A competent response contains

· Tall stature with long fingers and a high arched palate suggests Marfan syndrome with cystic medial degeneration, and sudden tearing chest pain radiating to the back is aortic dissection.

· Hypotension with distended jugular veins and muffled heart sounds means blood has ruptured into the pericardium, so intrapericardial pressure limits ventricular filling.

Distinctions that matter

· Hypotension in dissection can come from tamponade or from external rupture with blood loss, and distended neck veins point to tamponade.

Common misconceptions

· Reading raised jugular pressure in this setting as volume overload.

Where it is taught: Section C05 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C06-BD1

Brain dump 1 · foundational retrieval: Claudication, the ankle-brachial index, and the cholesterol shower

A competent response contains

· Peripheral arterial disease is atherosclerotic narrowing of the arteries to the legs, most strongly linked to smoking and diabetes.

· At rest a stenosed artery still delivers enough blood; during walking the calf arterioles dilate and demand rises, but the fixed stenosis prevents flow from increasing, so the muscle becomes ischemic.

· Intermittent claudication is cramping pain in the muscle group below the stenosis, reproduced by a set walking distance and relieved within minutes by rest.

· Superficial femoral disease causes calf pain; aortoiliac disease causes buttock and thigh pain.

· Index bands: above 1.40 means noncompressible calcified arteries and an unreliable result; 1.00 to 1.40 normal; 0.91 to 0.99 borderline; 0.90 or lower confirms disease because pressure is lost across stenoses; below about 0.40 severe disease, often with rest pain or tissue loss.

· Normally ankle pressure equals or slightly exceeds arm pressure because of pulse wave amplification, so a lower ankle pressure means flow has crossed a resistance.

· Chronic ischemia gives weak or absent pulses, cool skin, hair loss, thin shiny skin, thickened nails, and dependent rubor with pallor on elevation.

· Rest pain in the forefoot, typically at night, signals critical ischemia; hanging the foot down adds gravity to perfusion pressure, which raises flow and eases the pain. Nonhealing ulcers on the toes and pressure points and gangrene follow.

· Leriche syndrome is occlusion at the aortic bifurcation with buttock and thigh claudication, absent or diminished femoral pulses, and erectile dysfunction from reduced internal iliac flow.

· Atheroembolic disease: an ulcerated plaque in the aorta or a large artery showers cholesterol crystals downstream, classically after arterial catheterization, angiography, or vascular surgery, and occasionally with anticoagulation or thrombolysis.

· Crystals lodge in arterioles of about 100 to 200 micrometers and incite inflammation, giving blue or purple toes and livedo reticularis with intact pedal pulses because the large arteries remain open; the kidney shows progressive renal failure, and blood tests may show eosinophilia and low complement.

· Biopsy shows biconvex, needle-shaped clefts in arterioles where cholesterol crystals dissolved during tissue processing.

Distinctions that matter

· Claudication is a flow reserve problem and rest pain is a resting perfusion problem, which is why one comes with walking and the other at night.

· An index above 1.40 and an index of 0.90 or lower both signal arterial disease, but only the low value measures lost pressure.

Common misconceptions

· Expecting absent pulses in cholesterol embolization. The large arteries stay open, so the pulses are intact.

· Treating elevation as relief for ischemic rest pain. Elevation lowers perfusion pressure and makes arterial pain worse.

Where it is taught: Section C06, all paragraphs and the ankle-brachial index table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C06-BD2

Brain dump 2 · integration and application: Three painful feet, three different vessels

A competent response contains

· Patient 1: atherosclerotic peripheral arterial disease of the superficial femoral artery, a large to medium artery obstructed by intimal plaque, so pressure is lost across the stenosis and the pedal pulses are absent, with an index of 0.58 confirming disease.

· Patient 2: atheroembolic or cholesterol embolization disease after angiography, with cholesterol crystals in arterioles of about 100 to 200 micrometers, so the small vessels are blocked while the large arteries remain open and the pedal pulses stay normal.

· Patient 2's blue toes, livedo reticularis, rising creatinine, eosinophilia, and low complement all fit crystal-induced inflammation of small vessels.

· Patient 3: Monckeberg medial calcific sclerosis makes the arteries noncompressible, so the index of 1.46 is uninterpretable and does not exclude coexisting atherosclerotic disease, which the nonhealing toe ulcer suggests is present.

· In patient 3 use toe pressures and the toe-brachial index, pulse volume recordings or Doppler waveforms, or imaging, because the digital arteries are less often calcified.

· The level of obstruction explains the pulses: proximal large artery disease abolishes the pulse, distal arteriolar embolism does not.

· Patient 1 and patient 3 have arterial ulcers, which are punched out, dry, and pale and sit at the toes, heel, and pressure points, because perfusion pressure is too low to supply the tissue, unlike the shallow moist ulcers of venous disease above the medial malleolus.

Distinctions that matter

· Blue toes with normal pulses point to small vessel embolism; blue or ulcerated toes with absent pulses point to large artery stenosis.

· A normal or high index never rules out disease in a calcified limb, whereas a low index is reliable evidence of it.

Common misconceptions

· Reading eosinophilia in patient 2 as an allergy to contrast. It is part of the inflammatory response to cholesterol crystals.

· Trusting an index above 1.40 in a patient with diabetes or chronic kidney disease.

Where it is taught: Section C06, paragraphs on peripheral arterial disease and atheroembolic disease and the index table, with Monckeberg sclerosis in C01 and ulcer features in C09 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C06-PR01

Prediction exercise: Predict what happens as a femoral stenosis tightens

A competent response contains

· Resting calf flow stays adequate while the stenosis is moderate, because resting demand is low and a stenosed artery can still meet it.

· During walking the arterioles dilate and demand rises, but the fixed stenosis prevents flow from increasing, so the muscle becomes ischemic and he feels cramping calf pain that eases within minutes of rest.

· Ankle systolic pressure falls because pressure is lost across the stenosis, so the index falls below 0.90 and keeps falling as the narrowing tightens.

· Claudication distance shortens, because less of the flow reserve remains, so ischemia is reached sooner.

· Once resting flow is inadequate, forefoot pain appears at rest and typically at night; hanging the foot down adds a gravitational column to perfusion pressure, which raises flow and eases the pain.

· The leg loses hair, the skin becomes thin, shiny, and cool, the nails thicken, dependent rubor with pallor on elevation appears, and nonhealing ulcers form at the toes and pressure points, with gangrene if ischemia continues.

Distinctions that matter

· Flow reserve, not resting flow, is what a moderate stenosis consumes, which is why symptoms begin on exertion.

· Rest pain and claudication differ in whether resting perfusion is still sufficient, not in the kind of pain.

Common misconceptions

· Expecting resting flow to fall as soon as a stenosis appears. Resting flow is preserved until the narrowing is severe.

· Explaining relief on dangling the foot by reduced venous pressure. The mechanism is added gravitational perfusion pressure.

Where it is taught: Section C06, first and second paragraphs and the index table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-BD1

Brain dump 1 · foundational retrieval: Rebuild pulmonary embolism from clot to blood gas

A competent response contains

· More than 95% of pulmonary emboli arise from thrombi in the deep veins of the leg, especially the proximal popliteal, femoral, and iliac veins, because those veins are large enough to release sizable clots.

· Sequence: embolus obstructs pulmonary arteries; pulmonary vascular resistance rises abruptly; the right ventricle dilates and fails, which is acute cor pulmonale; blood is diverted to open lung units creating low ventilation to perfusion areas; hypoxemia with a widened alveolar-arterial gradient; hypoxemia and receptor stimulation drive hyperventilation; low PaCO2 gives respiratory alkalosis.

· In the obstructed region ventilation continues without perfusion, which is dead space; in the open region perfusion rises relative to ventilation, so blood leaves it less fully oxygenated; local loss of surfactant and atelectasis add shunt.

· Obstruction of roughly 60% or more of the pulmonary circulation can cause shock or sudden death, and a large saddle embolus at the bifurcation of the main pulmonary artery can cause sudden death.

· Large embolus: sudden dyspnea with tachypnea and tachycardia, hypotension and shock in massive cases.

· Small peripheral embolus: pleuritic chest pain and hemoptysis, from pleural irritation or infarction.

· Right ventricular strain: dilated right ventricle, raised jugular venous pressure, sinus tachycardia as the most common electrocardiographic finding, sometimes a new right bundle branch block or an S1Q3T3 pattern, and elevated troponin or BNP.

· D-dimer is a fibrin degradation product that is sensitive but not specific, so a raised value does not establish the diagnosis.

Distinctions that matter

· Calf vein thrombi are common but proximal thrombi supply the clinically important emboli.

· Dead space and the overperfused open units are two halves of the same mismatch, and both are needed to explain the hypoxemia.

Common misconceptions

· Expecting PaCO2 to rise because dead space has increased. The patient hyperventilates, so PaCO2 falls.

· Treating a raised D-dimer as diagnostic. It is sensitive and not specific.

Where it is taught: Section C07, first and third paragraphs and the pulmonary embolism flow of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-BD2

Brain dump 2 · integration and application: Calculate the gradient and explain the lung that lives

A competent response contains

· Alveolar PO2 equals 0.21 times (760 minus 47) minus PaCO2 divided by 0.8, which is 0.21 times 713, about 150, minus 28 divided by 0.8, which is 35, giving about 115 mm Hg.

· The gradient is 115 minus 58, about 57 mm Hg, far above the normal of roughly 5 to 15 mm Hg in a young adult.

· A PaCO2 of 28 mm Hg shows she is hyperventilating, and hypoventilation requires a raised PaCO2, so hypoventilation cannot explain the hypoxemia.

· The mechanism is ventilation-perfusion mismatch, with shunt from surfactant loss and atelectasis adding to it.

· Only about 10% of emboli cause infarction because the lung has a dual supply: bronchial arteries from the systemic circulation continue to perfuse tissue distal to the block, and alveolar oxygen diffuses into the tissue directly.

· Infarction occurs mainly when bronchial flow is already compromised, as in left heart failure, where pulmonary venous pressure is also high.

· The infarct is hemorrhagic, wedge-shaped, and based on the pleura.

· Repeated small emboli over time produce chronic thromboembolic pulmonary hypertension.

Distinctions that matter

· A widened gradient means a gas exchange defect, while a normal gradient with hypoxemia and a high PaCO2 means hypoventilation.

· Occlusion of a pulmonary artery is not the same as loss of blood supply to lung tissue, because of the bronchial circulation.

Common misconceptions

· Forgetting to subtract water vapour pressure of 47 mm Hg before applying the fraction of inspired oxygen.

· Assuming every pulmonary artery occlusion infarcts the lung. Most do not.

Where it is taught: Section C07, third and fourth paragraphs, the worked example on the gradient, and the stop and predict box of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-RT02

Retrieval exercise: Rebuild the pulmonary embolism sequence

A competent response contains

· Step 1: embolus obstructs pulmonary arteries.

· Step 2: pulmonary vascular resistance rises abruptly; the right ventricle fails because it is thin walled and built for a low-pressure circuit, and acute failure from a sudden pressure load is acute cor pulmonale.

· Step 3: the right ventricle dilates and fails.

· Step 4: blood is diverted to open lung units, creating low ventilation to perfusion areas; the obstructed unit is ventilated and not perfused, which is dead space, and the open unit is perfused in excess of its ventilation.

· Step 5: hypoxemia with a widened alveolar-arterial gradient, which separates a gas exchange defect from hypoventilation.

· Step 6: hypoxemia and receptor stimulation drive hyperventilation, so respiratory rate and minute ventilation rise.

· Step 7: low PaCO2 produces respiratory alkalosis.

· Obstruction of roughly 60% or more of the pulmonary circulation can cause shock or sudden death.

Distinctions that matter

· Dead space raises the work of breathing but the arterial hypoxemia comes from the overperfused open units.

· The left ventricle is not the chamber at risk here, because the pressure load lies upstream of it.

Common misconceptions

· Drawing the obstructed unit as unventilated. It is ventilated and not perfused.

· Expecting the arterial blood gas to show carbon dioxide retention.

Where it is taught: Section C07, pulmonary embolism flow and third paragraph of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C07-CA01

Clinical application: Sudden breathlessness after a long flight

A competent response contains

· Recognize: pulmonary embolism from a left proximal leg deep vein thrombosis, with stasis from a 14-hour flight, hypercoagulability from estrogen, and endothelial or venous wall involvement in the already thrombosed calf and thigh veins, so all three elements of Virchow triad are present.

· Explain: the embolus creates regions that are ventilated but not perfused while blood is diverted to open units that become overperfused relative to their ventilation, so blood leaving them is not fully oxygenated and arterial PO2 falls; surfactant loss and atelectasis add shunt; hypoxemia and receptor stimulation drive hyperventilation, which lowers PaCO2.

· Differentiate: pure hypoventilation raises PaCO2 and leaves the alveolar-arterial gradient normal, whereas here PaCO2 is low and the gradient is wide; a right-to-left shunt also widens the gradient but the hypoxemia corrects poorly with supplemental oxygen, while ventilation-perfusion mismatch corrects readily.

· Investigate: alveolar PO2 is 0.21 times (760 minus 47) minus 29 divided by 0.8, which is about 150 minus 36, or about 114 mm Hg; the gradient is 114 minus 56, about 58 mm Hg, far above the normal 5 to 15 mm Hg, confirming a gas exchange defect; compression ultrasound of the left leg would show a vein that cannot be compressed because clot fills its lumen.

· Decide: the right ventricle is under acute strain, shown by the raised jugular venous pressure, the loud pulmonary component of the second heart sound, and the sinus tachycardia, which is the most common electrocardiographic finding.

· Justify: bronchial arteries from the systemic circulation continue to perfuse tissue distal to the obstruction and alveolar oxygen diffuses into the tissue directly, so only about 10% of emboli infarct the lung; left heart failure, which reduces bronchial flow and raises pulmonary venous pressure, would make a hemorrhagic wedge-shaped pleura-based infarct likely instead.

Distinctions that matter

· Pleuritic pain and a clear chest radiograph fit a peripheral embolus without infarction, so pain does not prove tissue death.

· A normal chest radiograph with marked hypoxemia is itself informative, because parenchymal causes would usually show something.

Common misconceptions

· Reading the low PaCO2 as evidence against a serious problem. It is the expected consequence of hyperventilation.

· Assuming a swollen calf means the embolus came from the calf veins. The clinically important emboli come from the proximal veins.

Where it is taught: Section C07, all paragraphs, the flow, and the worked example, with Virchow triad in C08 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C08-BD1

Brain dump 1 · foundational retrieval: Build Virchow triad and the venous thrombus

A competent response contains

· Venous thrombi usually begin in the valve pockets of the deep calf veins, where flow is slowest, and can propagate proximally.

· They form in low-flow, low-pressure vessels and are rich in red cells and fibrin, which makes them red thrombi, unlike platelet-rich arterial thrombi.

· Stasis: slow flow lets activated factors accumulate and keeps platelets near the endothelium. Examples include immobility, long travel, paralysis, heart failure, and compression by the pregnant uterus.

· Endothelial injury: loss of the anticoagulant surface and exposure of tissue factor. Examples include surgery and trauma, especially hip and knee, central venous catheters, and prior deep vein thrombosis.

· Hypercoagulability: excess procoagulant or loss of natural anticoagulant. Examples include factor V Leiden, prothrombin G20210A, antiphospholipid antibodies, malignancy, estrogen and pregnancy, and nephrotic syndrome through urinary antithrombin loss.

· Major surgery and cancer are high risk because they combine all three elements.

· Nephrotic syndrome works by losing antithrombin in the urine.

· A thrombosis obstructs venous outflow, raising capillary hydrostatic pressure, while inflammation of the vein wall adds pain and warmth, giving one swollen, warm, tender leg, sometimes with dilated superficial veins and a larger calf circumference than the other side.

· Many are silent because collateral veins drain the limb, which is why pulmonary embolism may be the first sign.

· Calf pain on dorsiflexion, the Homans sign, is unreliable.

· On ultrasound the thrombosed vein cannot be compressed because clot fills its lumen.

Distinctions that matter

· Red venous thrombi and white arterial thrombi differ because flow conditions decide whether red cells and fibrin or platelets dominate.

· Swelling comes from obstructed outflow and raised capillary pressure, while pain and warmth come from inflammation of the vein wall.

Common misconceptions

· Expecting every deep vein thrombosis to produce a swollen leg. Collaterals keep many of them silent.

· Relying on the Homans sign, which the notes state is unreliable.

Where it is taught: Section C08, first and third paragraphs and the Virchow triad table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C08-BD2

Brain dump 2 · integration and application: A red cord, a swollen leg, and a wandering vein

A competent response contains

· Patient 1: superficial thrombophlebitis of a varicose vein, in the superficial venous system, with low embolic risk unless the clot extends into the deep system.

· Patient 2: deep vein thrombosis of the left leg, in the deep system of calf, popliteal, femoral, or iliac veins, with significant embolic risk that is highest with proximal clot.

· Patient 3: migratory thrombophlebitis, the Trousseau sign of malignancy, which is recurrent superficial and deep venous thrombosis appearing and resolving at different sites.

· Patient 2 has diffuse swelling because the deep veins drain most of the leg, so their occlusion raises capillary hydrostatic pressure throughout the limb; patient 1 has little swelling because deep drainage is intact.

· The responsible tumours are adenocarcinomas, classically pancreatic and also lung, gastric, and colon; painless jaundice with weight loss points first to the pancreas.

· The tumour releases tissue factor and mucin, which activate coagulation.

· The Trousseau sign of hypocalcemia is a different finding: carpal spasm under a blood pressure cuff.

· All three patients would have a normal platelet count, because these are thrombotic rather than thrombocytopenic conditions, so the count does not separate them.

Distinctions that matter

· A palpable cord with no swelling means a superficial vein; diffuse swelling of the whole limb means the deep system.

· Two different signs carry the Trousseau name, and only one concerns veins and cancer.

Common misconceptions

· Treating superficial thrombophlebitis as a source of pulmonary embolism. It rarely embolizes unless it extends into the deep veins.

· Reading recurrent thrombophlebitis at varied sites as bad luck rather than as a sign that prompts a search for a visceral adenocarcinoma.

Where it is taught: Section C08, third and fourth paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C08-Q03

Board-style question: Thrombosis at shifting sites

Correct answer: D. Tissue factor and mucin released by an adenocarcinoma activating coagulation

Why it is right: Visceral adenocarcinomas, classically pancreatic, release tissue factor and mucin that activate coagulation, producing recurrent superficial and deep venous thrombosis that appears and resolves at different sites, and painless jaundice with marked weight loss fits a pancreatic primary.

A is wrong: Antiphospholipid antibodies cause thrombosis but not this migratory pattern with painless jaundice and weight loss, and they typically prolong the activated partial thromboplastin time.

B is wrong: Urinary antithrombin loss in nephrotic syndrome causes thrombosis in the setting of heavy proteinuria and edema, which is not the picture here.

C is wrong: Compression of the deep veins by a pelvic mass produces persistent obstruction of the affected limb rather than cords that resolve in one site and appear in another, including the forearm.

A competent response contains

· Recurrent superficial and deep thrombosis appearing and resolving at different sites is migratory thrombophlebitis, the Trousseau sign of malignancy.

· It occurs with adenocarcinomas, classically pancreatic and also lung, gastric, and colon, which release tissue factor and mucin that activate coagulation; weight loss with painless jaundice points to the pancreas.

Distinctions that matter

· The Trousseau sign of malignancy is migratory venous thrombosis, while the Trousseau sign of hypocalcemia is carpal spasm under a blood pressure cuff.

Common misconceptions

· Expecting an abnormal platelet count or coagulation screen to mark a hypercoagulable state.

Where it is taught: Section C08 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C09-BD1

Brain dump 1 · foundational retrieval: The calf pump, the valves, and the skin that follows

A competent response contains

· Leg veins return blood against gravity using the calf muscle pump and one-way valves.

· During walking, calf contraction squeezes the deep veins, and competent valves let blood move only upward and from superficial to deep through the perforating veins, which keeps ankle venous pressure low during movement.

· When valves fail, the column of blood is no longer interrupted and pressure in the superficial veins stays high even with walking, which is ambulatory venous hypertension.

· Varicose veins are dilated, tortuous superficial veins, most often in the great saphenous system, produced by raised intraluminal pressure and weakness of the vein wall that make the valves incompetent.

· Risk rises with age, female sex, pregnancy, obesity, prolonged standing, and family history; pregnancy adds compression of the inferior vena cava by the uterus plus hormonal relaxation of the vein wall, and prolonged standing keeps the hydrostatic column uninterrupted.

· Prior deep vein thrombosis that damages deep valves causes secondary venous insufficiency, and superficial varicose veins rarely cause embolism.

· Portal hypertension raises pressure elsewhere and produces esophageal varices and hemorrhoids.

· Sequence: valve incompetence in superficial, perforating, or deep veins; sustained venous hypertension in the lower leg; raised capillary hydrostatic pressure giving edema; red cells leak and hemosiderin stains the skin brown; leukocyte trapping and pericapillary fibrin impair tissue oxygenation; stasis dermatitis and fibrosis, which is lipodermatosclerosis; minor trauma fails to heal, giving a venous ulcer.

· The brown colour is hemosiderin, iron released from leaked red cells.

Distinctions that matter

· Valves and the calf pump work together: the pump moves blood and the valves stop it coming back, so failure of either raises ankle pressure.

· Primary varicose disease starts in the vein wall and its valves, while secondary insufficiency follows damage to deep valves by thrombosis.

Common misconceptions

· Expecting standing venous pressure to be the problem. The abnormality is that pressure stays high during walking, when it should fall.

· Treating varicose veins as a source of pulmonary embolism.

Where it is taught: Section C09, first and second paragraphs and the venous ulcer flow of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C09-BD2

Brain dump 2 · integration and application: Two ulcers above two different problems

A competent response contains

· The woman has chronic venous insufficiency with a venous ulcer: location above the medial malleolus in the gaiter area, where pressure from incompetent perforators and the great saphenous system is highest; shallow, irregular, moist with a granulating base; mild to moderate pain eased by elevation; surrounding edema, brown pigment, and stasis dermatitis.

· The man has an arterial ulcer: at the toes, heel, pressure points, and distal foot; punched out, dry, with a pale base; severe pain that is worse with elevation; surrounding skin cool, shiny, and hairless with weak pulses.

· Elevation lowers venous pressure, which reduces capillary hydrostatic pressure and edema and helps the venous ulcer; elevation also lowers arterial perfusion pressure, which worsens ischemia and arterial pain.

· Evening edema: raised capillary hydrostatic pressure from sustained venous hypertension drives fluid into the tissue through the day, and overnight recumbency removes the hydrostatic column so the fluid is reabsorbed.

· Brown discoloration: red cells leak from congested capillaries and hemosiderin from their iron stains the skin.

· Itchy scaling skin: stasis dermatitis from chronic edema and inflammation.

· Indurated tissue: lipodermatosclerosis, fibrosis of dermis and subcutaneous fat after prolonged inflammation.

· The ulcer itself: leukocyte trapping and pericapillary fibrin impair oxygen delivery, so minor trauma fails to heal.

Distinctions that matter

· Both ulcers sit on the lower leg, and location, moisture, and the response to elevation separate them.

· Pain that improves with elevation points to venous pressure, and pain that worsens with elevation points to arterial inflow.

Common misconceptions

· Elevating the leg of a patient with an ischemic foot. It lowers perfusion pressure and increases pain.

· Reading brown pigmentation as bruising or hyperpigmentation of aging rather than as hemosiderin from chronic venous hypertension.

Where it is taught: Section C09, third paragraph, the venous ulcer flow, and the venous against arterial ulcer table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C09-RT03

Retrieval exercise: Rebuild the path from a failed valve to a broken skin surface

A competent response contains

· Step 1: valve incompetence in superficial, perforating, or deep veins; a previous deep vein thrombosis damages deep valves and causes secondary insufficiency.

· Step 2: sustained venous hypertension in the lower leg.

· Step 3: raised capillary hydrostatic pressure gives edema; the pressure that has risen is hydrostatic pressure in the capillary.

· Step 4: red cells leak and hemosiderin stains the skin brown; the iron from leaked red cells is the pigment.

· Step 5: leukocyte trapping and pericapillary fibrin impair tissue oxygenation, which are the two changes that reduce oxygen delivery.

· Step 6: stasis dermatitis, then fibrosis, which is lipodermatosclerosis; this is the step at which the changes become a visible skin disease.

· Step 7: minor trauma fails to heal, giving a venous ulcer, most often just above the medial malleolus.

Distinctions that matter

· Edema comes from pressure, while the pigment comes from red cells escaping, so the two findings reflect different consequences of the same hypertension.

· Stasis dermatitis is inflammatory and reversible in appearance, while lipodermatosclerosis is fibrotic and persistent.

Common misconceptions

· Placing the ulcer directly after edema and skipping the inflammatory and fibrotic steps that make the skin fragile.

· Attributing poor healing to edema alone. Leukocyte trapping and pericapillary fibrin impair oxygen delivery.

Where it is taught: Section C09, venous ulcer flow and the surrounding paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C10-BD1

Brain dump 1 · foundational retrieval: The three colours of a vasospastic attack

A competent response contains

· Raynaud phenomenon is episodic, reversible vasospasm of the small arteries and arterioles of the fingers and toes, and sometimes the nose and ears, triggered by cold or emotional stress.

· Digital vessels normally constrict to cold through sympathetic alpha-adrenergic signaling, mainly alpha-2 receptors, in order to conserve core heat.

· In Raynaud phenomenon that response is exaggerated and the lumen closes completely.

· Sequence: cold or stress causes intense digital artery spasm; white pallor from absent arterial inflow; blue cyanosis as stagnant blood is deoxygenated; spasm relaxes on rewarming; red reactive hyperemia with throbbing and tingling.

· White means no inflow, blue means oxygen has been extracted from blood trapped in the digit, and red means inflow has returned above baseline.

· The red phase is reactive hyperemia: during ischemia local metabolites including adenosine, carbon dioxide, hydrogen ions, and potassium accumulate and dilate the arterioles, so flow rebounds well above normal when the spasm lifts.

· Pallor is the most specific colour, and not every attack shows all three.

Distinctions that matter

· The blue phase is not a new episode of obstruction; it is the same trapped blood losing its oxygen.

· Normal cold-induced vasoconstriction and Raynaud phenomenon differ in degree, with complete closure of the lumen in the latter.

Common misconceptions

· Expecting all three colours in every attack, which leads to dismissing a real case.

· Explaining the red phase as inflammation. It is metabolite-driven reactive hyperemia.

Where it is taught: Section C10, first paragraph, the colour sequence flow, and the reactive hyperemia paragraph of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C10-BD2

Brain dump 2 · integration and application: Which Raynaud is in front of you

A competent response contains

· Patient 1: primary Raynaud phenomenon, a functional disorder of vascular reactivity with no underlying disease, with a structurally normal vessel between attacks and no tissue damage expected.

· Patient 2: secondary Raynaud phenomenon, with structural or inflammatory narrowing of the vessel, so spasm on an already narrow lumen can cause ischemic injury with digital pits, ulcers, and possible gangrene.

· Between attacks the primary patient's lumen is normal, so even complete spasm is brief and fully reversible; the secondary patient starts from a narrowed lumen, so the same spasm produces critical ischemia.

· Patient 2's tight shiny skin, asymmetric attacks, digital pits, and abnormal nailfold capillaries point to systemic sclerosis, in which intimal proliferation narrows the digital arteries.

· Other causes of secondary disease: lupus, thromboangiitis obliterans, atherosclerosis, cryoglobulinemia, and repetitive use of vibrating tools.

· Features that suggest an underlying disease: onset after about 30 to 40, asymmetric attacks, digital pits or ulcers, abnormal dilated or dropped-out nailfold capillaries, and positive autoantibodies.

· Primary disease is most common in young women, is usually symmetric, and has normal nailfold capillaries with a negative antinuclear antibody.

Distinctions that matter

· Tissue damage implies a structurally narrowed vessel, which points to secondary disease.

· Symmetry and age at onset are useful because a functional disorder affects both hands alike from a young age.

Common misconceptions

· Treating a positive antinuclear antibody as necessary for secondary disease. Nailfold changes and tissue loss count on their own.

· Assuming any patient with colour change has a connective tissue disease. Primary disease is common and benign.

Where it is taught: Section C10, fourth paragraph and the primary against secondary table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C11-BD1

Brain dump 1 · foundational retrieval: Thromboangiitis obliterans from histology to toes

A competent response contains

· Thromboangiitis obliterans is a segmental, thrombosing, acute and chronic inflammation of medium and small arteries, mainly the tibial and radial arteries and their digital branches, that spreads secondarily to adjacent veins and nerves.

· It occurs almost exclusively in heavy smokers, classically begins before about age 35 to 40, and is more often in men.

· The mechanism is thought to involve hypersensitivity or direct endothelial toxicity from tobacco products, and the disease rarely progresses once all tobacco use stops.

· Microscopically the involved segment shows a luminal thrombus containing microabscesses, which are small collections of neutrophils surrounded by granulomatous inflammation, with the internal elastic lamina relatively preserved.

· Over time the thrombus organizes and recanalizes, and the artery, vein, and nerve become bound together in fibrous tissue.

· Segmental means the lesions are sharply demarcated with normal vessel between them, which separates the disease from the diffuse plaques of atherosclerosis.

· Clinical order: cold sensitivity and Raynaud phenomenon, and nodular superficial phlebitis migrating along veins; then claudication of the instep of the foot or the forearm and hand, because the disease is distal; then intense rest pain from inflamed nerves, out of proportion to the visible lesion; then painful digital ulcers and gangrene, often needing amputation if smoking continues.

· Peripheral pulses at the wrist and ankle may be absent while proximal pulses are normal.

Distinctions that matter

· Preserved internal elastic lamina and neutrophil microabscesses within the thrombus separate this lesion from both atherosclerosis and most vasculitides.

· Pain here has two sources, ischemia and inflamed nerves, which is why it exceeds what the lesion looks like.

Common misconceptions

· Expecting proximal pulses to be lost. The disease is distal, so brachial and femoral pulses are normal.

· Treating the vein and nerve involvement as a separate problem. The inflammation spreads to them from the artery.

Where it is taught: Section C11, all three paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C11-BD2

Brain dump 2 · integration and application: A 33-year-old smoker with a black fingertip

A competent response contains

· Diagnosis is thromboangiitis obliterans, or Buerger disease.

· Heavy smoking at age 33 fits the near exclusive association with tobacco and the onset before about 35 to 40.

· Instep claudication fits distal disease of the tibial arteries and their branches rather than proximal large artery stenosis.

· A digital ulcer fits progressive occlusion of digital branches.

· Absent radial and posterior tibial pulses with normal brachial and femoral pulses fit segmental distal arterial occlusion.

· Cold-triggered colour change is the secondary Raynaud phenomenon that accompanies this disease.

· Migratory nodular phlebitis fits spread of the inflammation to adjacent veins.

· Upper limb involvement argues against atherosclerotic disease, which affects the large and medium arteries of the legs; atherosclerosis in a man of 33 would also be unusual.

· Further separating features: age under about 40 against over 50; segmental inflammatory thrombosis with microabscesses against lipid plaque with a fibrous cap; migratory superficial phlebitis and severe neural pain against calf or buttock claudication; risk factors in atherosclerosis extend beyond smoking.

· The vein nodules and the colour change show the process involves adjacent veins and the small digital arteries, not the arteries alone.

· Whether he keeps his digits depends on complete cessation of all tobacco use, because the disease rarely progresses once tobacco exposure stops.

Distinctions that matter

· Distal and upper limb involvement in a young smoker is the pattern that separates this disease from atherosclerotic peripheral arterial disease.

· Rest pain here is partly neural, which is why it is more severe than the ulcer size suggests.

Common misconceptions

· Calling this premature atherosclerosis because the patient smokes. The lesion, the vessels, and the vein and nerve involvement differ.

· Expecting a vascular procedure to settle the outcome. Continued smoking drives progression.

Where it is taught: Section C11, all three paragraphs and the comparison table, with Raynaud phenomenon in C10 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C12-BD1

Brain dump 1 · foundational retrieval: Sort the vasculitides by vessel size, then build giant cell arteritis

A competent response contains

· Vasculitis is inflammation of the vessel wall, and the Chapel Hill classification groups the vasculitides by the size of vessel mainly affected, which predicts the clinical picture.

· Large vessel disease narrows the aorta and its branches and causes limb or organ ischemia; medium vessel disease causes infarcts and aneurysms in organs; small vessel disease involves capillaries and venules, producing palpable purpura, glomerulonephritis, and alveolar hemorrhage.

· Large: giant cell arteritis, T cell granulomatous, age over 50 with polymyalgia rheumatica; Takayasu arteritis, T cell granulomatous, women under 40 to 50 and Asian.

· Medium: polyarteritis nodosa, immune complex in some cases, hepatitis B and ANCA negative; Kawasaki disease, T cell and antibody response to an unknown trigger, children under 5 with coronary aneurysm.

· Small ANCA: granulomatosis with polyangiitis, PR3-ANCA with granulomas, sinus, lung, kidney; microscopic polyangiitis, MPO-ANCA without granulomas, lung and kidney; eosinophilic granulomatosis with polyangiitis, eosinophils with MPO-ANCA in a minority, asthma and eosinophilia.

· Small immune complex: IgA vasculitis, IgA1 immune complexes, children after an upper respiratory infection.

· Variable: Behcet syndrome, neutrophilic inflammation, oral and genital ulcers with uveitis and HLA-B51.

· Giant cell arteritis almost never begins before age 50, affects women more often, and is most frequent in people of Northern European ancestry; it is T cell mediated granulomatous inflammation of large and medium arteries.

· Arteries involved: branches of the external carotid including the superficial temporal artery, the ophthalmic and posterior ciliary arteries, the vertebral arteries, and the aorta.

· Activated T cells and macrophages release cytokines including IL-6, which drives the acute-phase response.

· Symptoms: new headache with a tender thickened temporal artery from arteritis of that vessel; jaw claudication from masseter ischemia on chewing; visual loss from occlusion of the ophthalmic or posterior ciliary arteries causing anterior ischemic optic neuropathy, which can be sudden and permanent.

· The sedimentation rate is markedly raised, often above 50 mm/h, with high CRP.

· Biopsy shows granulomatous inflammation of the inner media with multinucleated giant cells and fragmentation of the internal elastic lamina; lesions are segmental skip lesions, so a short or unlucky biopsy can be negative.

· Polymyalgia rheumatica causes pain and morning stiffness of the shoulder and hip girdles and neck with a high sedimentation rate but normal creatine kinase and normal strength, because the target is periarticular synovium and bursae rather than muscle; about half of patients with giant cell arteritis have its symptoms, and a smaller fraction of patients with it develop giant cell arteritis.

· Aortic involvement in giant cell arteritis can later cause thoracic aneurysm.

Distinctions that matter

· Vessel size predicts the organ picture, and the immune mechanism predicts the laboratory findings, so both columns are needed.

· Polymyalgia rheumatica is inflammatory but not myopathic, which is why strength and creatine kinase are normal.

Common misconceptions

· Treating a negative temporal artery biopsy as excluding the disease. Skip lesions make false negatives common.

· Calling polymyalgia rheumatica a muscle disease because of the name. The inflamed tissue is periarticular.

Where it is taught: Section C12, all paragraphs and the vasculitis by vessel size table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C12-BD2

Brain dump 2 · integration and application: Jaw pain, a headache, and sudden loss of vision

A competent response contains

· Headache with a tender thickened temporal cord: granulomatous arteritis of the superficial temporal artery, a branch of the external carotid, thickens the wall and makes it tender and palpable.

· Jaw pain after a few minutes of chewing: jaw claudication from ischemia of the masseter, because the narrowed arteries cannot raise flow to meet the muscle's demand.

· Shoulder and hip aching with morning stiffness: polymyalgia rheumatica, inflammation of periarticular synovium and bursae.

· Sudden painless vision loss: occlusion of the ophthalmic or posterior ciliary arteries producing anterior ischemic optic neuropathy, which can be permanent.

· A sedimentation rate of 86 mm/h with high CRP: IL-6 released by activated T cells and macrophages drives the acute-phase response.

· Her girdle symptoms are not a muscle disease because strength and creatine kinase are normal; the inflamed tissue is periarticular synovium and bursae.

· A normal biopsy does not exclude the disease, because the lesions are segmental skip lesions and a short or unlucky sample can miss them; it also does not confirm an alternative diagnosis.

· Giant cell arteritis can involve the aorta, and that involvement can later cause a thoracic aneurysm.

Distinctions that matter

· Jaw claudication and headache are both ischemic, but one follows exertion of a muscle and the other follows inflammation of the vessel itself.

· A high sedimentation rate reports the acute-phase response rather than the site of disease, so it supports but does not localize.

Common misconceptions

· Expecting painful vision loss. The ischemic optic neuropathy of this disease is painless.

· Assuming a normal biopsy means the headache has another cause.

Where it is taught: Section C12, giant cell arteritis and polymyalgia rheumatica paragraphs of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C13-BD1

Brain dump 1 · foundational retrieval: Takayasu arteritis from arch to pulse

A competent response contains

· Takayasu arteritis is a granulomatous vasculitis of large arteries, principally the aortic arch and its branches, and sometimes the descending and abdominal aorta, the renal arteries, and the pulmonary arteries.

· It most often affects women under about 40 to 50 and is more common in Asia.

· Histology resembles giant cell arteritis: mononuclear infiltrates, granulomas with giant cells, and destruction of the media; age is what mainly separates the two, so a granulomatous large-vessel arteritis in a young woman is Takayasu and in a patient over 50 it is giant cell arteritis.

· Chronic inflammation is followed by fibrosis of the media and adventitia and thickening of the intima, giving an irregularly thickened aortic wall with stenosis or occlusion of the origins of the great vessels.

· The disease often begins with fever, fatigue, weight loss, and arthralgia, with a raised sedimentation rate, before ischemic signs appear.

· Narrowing of the subclavian arteries at their origins makes pulses in one or both arms weak or absent, which gave the older name pulseless disease; blood pressure in the affected arm is lower so the two arms differ, arm claudication occurs with use, and bruits are heard over the subclavian and carotid arteries.

· Carotid and vertebral narrowing causes dizziness, syncope, visual disturbances, and stroke.

· Renal artery stenosis activates the renin-angiotensin system and causes hypertension, which can be masked when the pressure is measured in a narrowed arm.

· Aortic root involvement can dilate the root and cause aortic regurgitation, and pulmonary artery involvement can cause pulmonary hypertension.

Distinctions that matter

· Both large-vessel granulomatous arteritides look the same under the microscope, so age and the vessels involved carry the diagnosis.

· The inflammatory phase and the ischemic phase are separated in time, which is why constitutional symptoms come first.

Common misconceptions

· Trusting a normal blood pressure measured in a narrowed arm. The true central pressure can be much higher.

· Expecting the temporal artery to be involved. This disease centres on the arch and its branches.

Where it is taught: Section C13, all three paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C13-BD2

Brain dump 2 · integration and application: Two women with a weak arm pulse

A competent response contains

· The woman has Takayasu arteritis: granulomatous inflammation followed by intimal thickening and fibrosis narrows the origin of the left subclavian artery, so pressure distal to the stenosis is lower in that arm, with a bruit over the narrowed segment.

· The man has an aortic dissection: a false lumen in the media compresses the left subclavian artery, so flow and pressure fall in that arm.

· History separates them: months of fever, fatigue, weight loss, and a high sedimentation rate against sudden severe tearing pain radiating between the scapulae.

· Time course is the most useful discriminator because a chronic inflammatory stenosis develops over months while a dissection produces its findings within minutes.

· A pressure measured in her narrowed left arm underestimates central pressure, so hypertension can be masked; if renal artery stenosis is also present, significant hypertension could be missed entirely.

· Renal artery involvement activates the renin-angiotensin system and raises blood pressure.

· In a 68-year-old a granulomatous large-vessel arteritis would be giant cell arteritis, and age over 50 is the feature that decides it.

Distinctions that matter

· Unequal arm pressures is a shared finding with very different mechanisms: a fixed inflammatory stenosis against acute compression by a false lumen.

· Measuring blood pressure in both arms matters in both diseases, for opposite reasons.

Common misconceptions

· Treating unequal arm pressures as specific for dissection.

· Assuming a young woman cannot have severe hypertension because the cuff reading is low. The reading may be taken beyond a stenosis.

Where it is taught: Section C13, second and third paragraphs, with the dissection findings in C05 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C13-CP02

Comparison table: The two granulomatous large-vessel arteritides

A competent response contains

· Age: giant cell arteritis over 50 and almost never before; Takayasu usually under 40 to 50.

· Typical patient: older woman of Northern European ancestry; young woman, often Asian.

· Main vessels: carotid branches including the temporal and the ophthalmic and posterior ciliary arteries, the vertebral arteries, and the aorta; the aortic arch and its branches, with the descending and abdominal aorta, renal arteries, and pulmonary arteries at times.

· Key findings: new headache with a tender thickened temporal artery, jaw claudication, visual loss, and polymyalgia rheumatica; weak or absent arm pulses, unequal arm pressures, arm claudication, bruits over the subclavian and carotid arteries, and renin-mediated hypertension with renal involvement.

· Histology: both granulomatous with multinucleated giant cells; giant cell arteritis also shows fragmentation of the internal elastic lamina with segmental skip lesions, and Takayasu progresses to fibrosis of the media and adventitia with intimal thickening.

· Laboratory: markedly raised sedimentation rate, often above 50 mm/h, with high CRP in giant cell arteritis; raised sedimentation rate with constitutional symptoms in Takayasu.

· Late aortic complication: thoracic aneurysm in giant cell arteritis; aortic root dilation with regurgitation in Takayasu, and pulmonary hypertension with pulmonary artery involvement.

Distinctions that matter

· The histology column is nearly identical, which is exactly why the age and vessel rows carry the diagnosis.

· Both can involve the aorta, and the consequence differs: aneurysm in one and root dilation with regurgitation in the other.

Common misconceptions

· Expecting one of the two to be non-granulomatous.

· Filling the vessel row for giant cell arteritis with the temporal artery alone. It also involves ophthalmic, posterior ciliary, and vertebral arteries and the aorta.

Where it is taught: Section C13, comparison table, with the giant cell arteritis paragraphs in C12 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C14-BD1

Brain dump 1 · foundational retrieval: Polyarteritis nodosa, vessel by vessel

A competent response contains

· Polyarteritis nodosa is a systemic necrotizing vasculitis of small and medium muscular arteries, typically renal and visceral arteries, that spares the pulmonary arteries and does not involve arterioles, capillaries, or venules.

· Because the smallest vessels are spared it does not cause glomerulonephritis.

· It most often affects young to middle-aged adults.

· A substantial minority, historically about a third, have chronic hepatitis B, and in them immune complexes containing hepatitis B surface antigen deposit in the vessel wall, a type III mechanism.

· ANCA is characteristically negative.

· The lesion is segmental transmural inflammation with fibrinoid necrosis of the wall, initially rich in neutrophils, eosinophils, and mononuclear cells.

· Necrosis weakens the wall, producing small aneurysms that give involved arteries a beaded or nodular appearance, which is the source of the name, and it promotes thrombosis and infarction.

· The hallmark is that acute and healed fibrotic lesions coexist in the same patient and even the same artery, showing that injury occurs in waves.

· Angiography of the renal or mesenteric arteries shows multiple microaneurysms and stenoses.

· Manifestations: renal artery involvement causes renal infarcts and renin-mediated hypertension; mesenteric arteritis causes abdominal pain worse after meals, bleeding, and bowel infarction or perforation; inflammation of the vasa nervorum causes mononeuritis multiplex such as foot drop; skin shows painful nodules, livedo reticularis, and ulcers; testicular pain can occur; fever, malaise, and weight loss are common.

· Renal involvement with hypertension is the leading cause of death.

Distinctions that matter

· Medium-vessel disease produces infarcts and aneurysms, while small-vessel disease produces capillary inflammation with purpura and glomerulonephritis.

· The absence of lung involvement and of glomerulonephritis is positive information, because it tells you which vessels are spared.

Common misconceptions

· Expecting glomerulonephritis because the kidney is involved. The disease injures renal arteries and spares the glomerular capillaries.

· Looking for ANCA to confirm the diagnosis. It is characteristically negative.

Where it is taught: Section C14, all three paragraphs and the features table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C14-BD2

Brain dump 2 · integration and application: Explain a patient with fever, foot drop, and bloody stools

A competent response contains

· Diagnosis is polyarteritis nodosa associated with chronic hepatitis B.

· Hypertension and the raised creatinine: arteritis of the renal and intrarenal arteries reduces glomerular perfusion and activates renin, and renal infarcts reduce function.

· Postprandial abdominal pain with dark stools: mesenteric arteritis causes ischemia that worsens when digestive demand rises, with mucosal bleeding.

· Foot drop: mononeuritis multiplex from infarction of nerves through involvement of the vasa nervorum.

· Painful calf nodules: inflamed, aneurysmal small and medium cutaneous arteries.

· Fever, malaise, and weight loss: systemic inflammation.

· No red cell casts and no hematuria because arterioles, capillaries, and venules are spared, so there is no glomerulonephritis; this confirms medium-vessel rather than small-vessel disease.

· Hepatitis B matters because immune complexes containing hepatitis B surface antigen deposit in the vessel wall as a type III mechanism, and those deposits would be found in the wall.

· Angiography would show multiple microaneurysms with stenoses, because fibrinoid necrosis weakens the wall at multiple points.

· Hemoptysis with red cell casts and a positive ANCA would point to an ANCA-associated small-vessel vasculitis such as microscopic polyangiitis.

· Acute and healed lesions in the same artery show that the injury occurs in waves rather than all at once.

Distinctions that matter

· A rising creatinine without casts or hematuria is renal artery disease, while casts and hematuria mean glomerular capillary disease.

· Hepatitis B supports an immune complex mechanism, which is deposit-rich, unlike the pauci-immune ANCA diseases.

Common misconceptions

· Interpreting any renal failure in a vasculitis as glomerulonephritis.

· Expecting lung findings. The pulmonary arteries are spared, so normal chest imaging fits the diagnosis.

Where it is taught: Section C14, all three paragraphs and the features table, with the ANCA diseases in C16 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C15-BD1

Brain dump 1 · foundational retrieval: Kawasaki disease and the coronary arteries

A competent response contains

· Kawasaki disease, or mucocutaneous lymph node syndrome, is an acute, self-limited vasculitis of medium arteries that mainly affects children under 5 and is most common in children of East Asian, especially Japanese, ancestry.

· It is the leading cause of acquired heart disease in children in developed countries.

· The trigger is unknown; the leading view is that an infectious agent provokes an exaggerated T cell and antibody response in genetically susceptible children, with activated macrophages and cytokines damaging the vessel wall.

· The arteritis is transmural with mixed inflammatory cells, and fibrinoid necrosis is usually less prominent than in polyarteritis nodosa.

· Diagnosis is clinical: fever lasting at least 5 days plus most of the five other principal features, with fever plus at least four of the five supporting classic disease.

· Features: high fever for 5 days or more; bilateral nonexudative conjunctival injection sparing the limbus; red cracked lips with strawberry tongue and diffuse oral redness; polymorphous rash often on trunk and perineum; red swollen hands and feet with later periungual desquamation; cervical lymphadenopathy, usually unilateral, at least 1.5 cm.

· Peeling of the fingers and toes typically follows in the second to third week, and irritability is marked.

· Laboratory: raised sedimentation rate and CRP, leukocytosis, and thrombocytosis after the first week.

· The key complication is coronary artery aneurysm in roughly a quarter of untreated children: inflammation destroys the internal elastic lamina and media of the coronary arteries, which then dilate.

· Inside an aneurysm flow is slow and turbulent and the high platelet count adds to the tendency to thrombose, so coronary thrombosis can cause myocardial infarction or sudden death in a young child; aneurysms can rupture and healing can leave stenosis.

· Myocarditis, pericarditis, and valvular regurgitation can occur in the acute phase.

Distinctions that matter

· The mucocutaneous findings are temporary while the coronary lesion is the lasting problem, so the visible illness and the dangerous one are not the same.

· Fibrinoid necrosis is prominent in polyarteritis nodosa and usually less so here, although both are medium-vessel diseases with aneurysms.

Common misconceptions

· Treating the rash and mucosal findings as the reason the disease matters. The coronary arteries decide the outcome.

· Expecting exudative conjunctivitis. The conjunctival injection is nonexudative and spares the limbus.

Where it is taught: Section C15, all three paragraphs and the principal features table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C15-BD2

Brain dump 2 · integration and application: A red-eyed, red-tongued, miserable toddler

A competent response contains

· Diagnosis is Kawasaki disease.

· Criteria counted: fever for six days, which exceeds the 5-day requirement, plus conjunctival injection, lip and oral changes with strawberry tongue, polymorphous trunk and perineal rash, red swollen extremities, and cervical lymphadenopathy of 2 cm, which is all five other features, so the criteria are met.

· The eye finding is bilateral nonexudative conjunctival injection, distinguished by the absence of discharge and by sparing of the limbus.

· In the second to third week periungual desquamation is expected, with peeling of the fingers and toes.

· The platelet count is raised because thrombocytosis develops after the first week as part of the acute inflammatory response.

· The coronary arteries are the concern: inflammation destroys the internal elastic lamina and the media, so the artery dilates into an aneurysm.

· Inside the aneurysm flow is slow and turbulent and the high platelet count adds to the tendency to clot, so coronary thrombosis can cause myocardial infarction or sudden death in a young child.

· Scarlet fever is caused by group A streptococcal infection, usually with exudative pharyngitis, and although it also gives a strawberry tongue and rash it does not cause conjunctivitis or the swollen desquamating extremities, and it does not cause coronary aneurysms.

Distinctions that matter

· Strawberry tongue and rash are shared with scarlet fever, and conjunctival injection with extremity changes are not.

· Thrombocytosis is part of the course rather than an unrelated finding, and its timing after the first week is informative.

Common misconceptions

· Dismissing the diagnosis because a single feature is missing. Fever plus four of the five supports classic disease.

· Attributing the irritability to the fever alone. Marked irritability is itself part of the illness.

Where it is taught: Section C15, second and third paragraphs and the principal features table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C16-BD1

Brain dump 1 · foundational retrieval: How ANCA injures a capillary, and the three diseases that follow

A competent response contains

· ANCA target enzymes stored in neutrophil granules: PR3-ANCA against proteinase 3 gives a cytoplasmic pattern called c-ANCA, and MPO-ANCA against myeloperoxidase gives a perinuclear pattern called p-ANCA.

· Mechanism: infection or cytokines prime the neutrophil; PR3 and MPO move to the cell surface; ANCA bind them there; the bound neutrophil degranulates and releases reactive oxygen species while adhering to the endothelium; small vessels, meaning arterioles, capillaries, and venules, are injured.

· Pauci-immune means little or no immunoglobulin or complement is deposited in the lesion, and these diseases are pauci-immune because injury depends on activated cells rather than on deposits.

· An immune complex vasculitis instead shows immunoglobulin and complement in the vessel wall on immunofluorescence.

· The kidney and lung are hit hardest because they are rich in capillaries: necrotizing crescentic glomerulonephritis causes hematuria, red cell casts, and rapidly progressive renal failure, and pulmonary capillaritis causes alveolar hemorrhage with hemoptysis.

· Combined lung hemorrhage and glomerulonephritis is a pulmonary-renal syndrome, which also occurs with anti-GBM disease, identified by linear IgG along the basement membrane on immunofluorescence.

· Granulomatosis with polyangiitis: PR3-ANCA in most; necrotizing granulomas; chronic sinusitis, nasal ulcers, septal perforation, saddle nose, and otitis; lung nodules that cavitate with hemorrhage; pauci-immune crescentic glomerulonephritis; the classic triad of upper airway, lung, and kidney.

· Microscopic polyangiitis: MPO-ANCA in most; no granulomas; upper airway usually spared; capillaritis with hemorrhage; pauci-immune crescentic glomerulonephritis; like the former without granulomas or upper airway disease.

· Eosinophilic granulomatosis with polyangiitis: MPO-ANCA in a minority, about 40%; eosinophil-rich granulomas; allergic rhinitis and nasal polyps; asthma with transient infiltrates; kidney involvement less common and milder; blood eosinophilia, mononeuritis multiplex, and cardiac involvement.

Distinctions that matter

· Granulomas, eosinophils, and the ANCA type separate the three diseases, and no single one of the three is sufficient by itself.

· Pauci-immune and immune complex disease can look similar clinically, and immunofluorescence settles which it is.

Common misconceptions

· Expecting immune deposits in a vasculitis with glomerulonephritis. These diseases are pauci-immune.

· Assuming ANCA must be positive in eosinophilic granulomatosis with polyangiitis. It is positive in fewer than half.

Where it is taught: Section C16, first and second paragraphs and the ANCA comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C16-BD2

Brain dump 2 · integration and application: Three patients who all cough blood

A competent response contains

· Patient 1: granulomatosis with polyangiitis, placed there by destructive upper airway disease with a saddle nose, cavitating lung nodules, glomerulonephritis, and PR3-ANCA, which is the classic triad of upper airway, lung, and kidney with necrotizing granulomas.

· Patient 2: microscopic polyangiitis, placed there by lung hemorrhage and glomerulonephritis with palpable purpura and MPO-ANCA but no granulomas and no upper airway destruction.

· Patient 3: eosinophilic granulomatosis with polyangiitis, placed there by long-standing asthma and nasal polyps progressing to marked eosinophilia, transient infiltrates, mononeuritis multiplex, and cardiac involvement.

· Each renal biopsy would show necrotizing crescentic glomerulonephritis with little or no immunoglobulin or complement, which is what pauci-immune means, and in the third patient renal involvement is less common and milder.

· Patient 1 has upper airway destruction because necrotizing granulomas form in the upper and lower respiratory tract; patient 2 has small-vessel injury without granulomas, so the airway is usually spared.

· In patient 3 blood and tissue eosinophilia drives injury through eosinophil granule proteins, ANCA is positive in fewer than half of such patients and MPO-ANCA in about 40%, and cardiac involvement is the main cause of death.

· Anti-GBM disease would be suspected in lung hemorrhage with glomerulonephritis when ANCA is negative, and immunofluorescence showing linear IgG along the glomerular basement membrane would identify it.

Distinctions that matter

· All three can injure lung and kidney, so granulomas, the upper airway, and eosinophils are what separate them.

· Pulmonary-renal syndrome names a pattern rather than a disease, and immunofluorescence distinguishes pauci-immune from linear IgG disease.

Common misconceptions

· Reading a saddle nose as trauma rather than as destruction of cartilage by necrotizing granulomas.

· Assuming asthma is incidental in patient 3. It is the first phase of the disease.

Where it is taught: Section C16, second and third paragraphs and the ANCA comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C16-Q04

Board-style question: Sinuses, cavities, and casts

Correct answer: B. PR3-ANCA with a cytoplasmic immunofluorescence pattern

Why it is right: Upper airway destruction with saddle nose, cavitating lung nodules, and pauci-immune necrotizing crescentic glomerulonephritis is granulomatosis with polyangiitis, in which PR3-ANCA giving a cytoplasmic immunofluorescence pattern is present in most patients.

A is wrong: Linear IgG along the glomerular basement membrane identifies anti-GBM disease, which causes a pulmonary-renal syndrome but not destructive sinus disease or cavitating nodules, and its biopsy is not pauci-immune.

C is wrong: IgA-containing complexes in the mesangium identify IgA vasculitis or IgA nephropathy, which give palpable purpura and mesangial disease rather than saddle nose and lung cavities, and the lesion is deposit-rich.

D is wrong: Hepatitis B surface antigen in circulating immune complexes is associated with polyarteritis nodosa, which spares the pulmonary arteries and the small vessels and therefore causes no glomerulonephritis.

A competent response contains

· Destructive upper airway disease with a saddle nose, cavitating lung nodules, and crescentic glomerulonephritis is the classic triad of granulomatosis with polyangiitis.

· Almost no immunoglobulin or complement on immunofluorescence makes the lesion pauci-immune, which fits ANCA disease, and PR3-ANCA with a cytoplasmic pattern is positive in most such patients.

Distinctions that matter

· Pauci-immune disease points to ANCA, while linear IgG and granular IgA deposits point to other mechanisms entirely.

Common misconceptions

· Expecting deposits to be found because there is glomerulonephritis.

Where it is taught: Section C16 of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C17-BD1

Brain dump 1 · foundational retrieval: IgA vasculitis and Behcet syndrome from mechanism to sign

A competent response contains

· IgA vasculitis, formerly Henoch-Schonlein purpura, is the most common systemic vasculitis of childhood, typically between about 3 and 15 years, and often follows an upper respiratory infection.

· Patients have an excess of galactose-deficient IgA1, which is recognized by autoantibodies to form immune complexes.

· The complexes deposit in small vessels of the skin, gut, joints, and glomerular mesangium, where they activate complement through the alternative and lectin pathways; this is a type III immune complex mechanism, and immunofluorescence shows IgA in vessel walls and the mesangium.

· Skin: dermal leukocytoclastic vasculitis gives palpable purpura over the buttocks and the extensor surfaces of the legs, where gravity and hydrostatic pressure favour deposition.

· Joints: arthralgia or arthritis, usually of the knees and ankles.

· Gut: vasculitis causes colicky abdominal pain and bleeding and predisposes to intussusception, often ileoileal.

· Kidney: mesangial glomerulonephritis identical to IgA nephropathy, with hematuria and sometimes proteinuria, more often severe in adults.

· Platelet count and coagulation tests are normal because the purpura comes from inflamed vessels, which excludes thrombocytopenic purpura.

· Behcet syndrome is a variable-vessel vasculitis affecting arteries and veins of any size, most common along the old Silk Road from the Mediterranean and Middle East to Japan, associated with HLA-B51, and driven by neutrophil hyperactivity.

· Its features: recurrent painful oral aphthous ulcers in nearly all patients, genital ulcers, uveitis that can cause blindness and sometimes shows a hypopyon, erythema nodosum and pathergy in the skin, venous thrombosis and arterial aneurysms, and neurologic involvement.

· Pathergy is an exaggerated pustular response to a needle prick.

· Palpable purpura with normal platelets in a child points to IgA vasculitis; recurrent mucosal ulcers with uveitis point to Behcet syndrome.

Distinctions that matter

· One disease is deposit-driven and confined to small vessels, the other is neutrophil-driven and affects vessels of any size.

· Purpura from vessel inflammation and purpura from low platelets look similar, and the platelet count separates them.

Common misconceptions

· Expecting a low platelet count because the child has purpura.

· Treating the renal lesion as unrelated to IgA nephropathy. It is identical.

Where it is taught: Section C17, all three paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.

NPLEX-P1-CARD-18-09-C17-BD2

Brain dump 2 · integration and application: Purpura in a child and ulcers in a young adult

A competent response contains

· The boy has IgA vasculitis following an upper respiratory infection.

· Palpable purpura on buttocks and the backs of the legs: dermal leukocytoclastic vasculitis where galactose-deficient IgA1 immune complexes deposit, favoured at those sites by gravity and hydrostatic pressure.

· Swollen painful knees and ankles: immune complex deposition in the joints giving arthralgia or arthritis.

· Colicky abdominal pain: gut vasculitis with ischemia of the bowel wall.

· Dark stools: bleeding from the inflamed gut vessels.

· Blood in the urine: mesangial glomerulonephritis from IgA deposition in the glomerular mesangium.

· A normal platelet count matters because it shows the purpura comes from inflamed vessels rather than from low platelets, which rules out thrombocytopenic purpura.

· He is at risk of intussusception, often ileoileal.

· Kidney biopsy would show mesangial IgA deposition with a mesangial glomerulonephritis identical to IgA nephropathy.

· The second patient has Behcet syndrome, associated with HLA-B51, driven by neutrophil hyperactivity; the needle finding is pathergy and the layer of pus in the anterior chamber is a hypopyon.

· Behcet syndrome is called a variable-vessel vasculitis because it affects arteries and veins of any size, which is why he has both a venous thrombosis and a risk of arterial aneurysm, whereas IgA vasculitis is confined to small vessels.

Distinctions that matter

· Both diseases give skin lesions, and purpura on dependent surfaces differs from erythema nodosum and pathergy.

· Venous thrombosis and arterial aneurysm in the same patient is what variable-vessel involvement looks like in practice.

Common misconceptions

· Ordering a platelet count to confirm the diagnosis in the boy. It is normal, and its value is in excluding another cause.

· Reading the pustule at the needle site as an infection rather than as pathergy.

Where it is taught: Section C17, first, second, and third paragraphs and the comparison table of the CV-18-I Vascular Conditions notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-J Vascular Neoplasms

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-10-C01-BD1

Brain dump 1 · foundational retrieval: Hemangioma against malformation, and the life of an infantile hemangioma

A competent response contains

· A hemangioma is a benign neoplasm of endothelial cells that forms blood-filled vascular channels; a vascular malformation is an error in vessel development, present at birth, that grows in proportion to the child and never regresses.

· Benign vascular tumors form well-formed channels lined by a single layer of bland endothelium; malignant ones form irregular channels lined by crowded atypical cells.

· Life cycle, in order: absent or faint precursor patch at birth; appears in the first weeks of life; GLUT1-positive endothelium proliferates rapidly for months; endothelial cells die by apoptosis and fibrofatty tissue replaces them; most have regressed by early childhood.

· Incidence is about 4 to 5% of infants, and it is the most common tumor of infancy.

· It is more common in girls and in preterm and low birth weight infants.

· Superficial lesions are bright red and raised; deep lesions are bluish.

· Lesional endothelium expresses GLUT1 in every phase, a marker otherwise typical of placental capillaries; vascular malformations and congenital hemangiomas, which are fully formed at birth, are GLUT1 negative.

· A capillary hemangioma is a well-circumscribed but unencapsulated lobule of closely packed thin-walled capillaries in skin, subcutaneous tissue or oral mucosa.

· A cavernous hemangioma is made of large dilated thin-walled channels filled with blood, separated by scant connective tissue, less well circumscribed, and often extending into deeper tissue.

· The infantile lesion regresses because its proliferating endothelium runs a programme that ends in apoptosis; the cavernous lesion does not regress because its vessels are mature rather than proliferating.

· Involution can leave lax skin or telangiectasias behind.

Distinctions that matter

· Proliferation against malformation is the dividing line: a hemangioma has dividing endothelial cells, a malformation has structurally abnormal vessels whose cells are not proliferating abnormally.

· Capillary and cavernous describe the calibre of the channels, not the timing of the lesion, so neither word tells you whether it will regress.

Common misconceptions

· Treating any red birthmark as a hemangioma. A lesion fully formed at birth that grows with the child is a malformation.

· Expecting cavernous lesions to involute like infantile ones. Their channels are mature, so they persist, and slow flow leads instead to thrombosis and calcification.

Where it is taught: Section C01, opening paragraphs, the life cycle flow and the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-BD2

Brain dump 2 · integration and application: Site, growth rate and the harm they cause

A competent response contains

· Harm depends on where the lesion grows and how fast, not on malignancy.

· Periorbital lesion blocking the visual axis or compressing the globe: amblyopia.

· Beard distribution lesion: involvement of the subglottic airway, so airway obstruction.

· Large segmental facial lesion: part of PHACE syndrome, which is posterior fossa malformations, hemangioma, arterial anomalies, cardiac defects such as coarctation, and eye anomalies.

· Large hepatic lesion: it expresses type 3 iodothyronine deiodinase, which inactivates thyroid hormone and can cause consumptive hypothyroidism.

· Ulceration is the most common complication, because rapid growth outstrips the blood supply of the overlying skin.

· The newborn with a patch present at birth that grows in proportion to the child has a port-wine stain, a capillary malformation: GLUT1 negative, no regression, often in a trigeminal distribution, linked to Sturge-Weber syndrome with a somatic GNAQ mutation.

· Kasabach-Merritt phenomenon, the platelet-trapping consumptive coagulopathy, arises in kaposiform hemangioendothelioma and tufted angioma, both GLUT1 negative, and not in the ordinary infantile hemangioma.

Distinctions that matter

· Site decides the complication while growth rate decides ulceration, so two lesions of the same size can carry quite different risks.

· Kasabach-Merritt belongs to two specific GLUT1-negative lesions, which is why the GLUT1 result is worth knowing in a bleeding infant.

Common misconceptions

· Attaching Kasabach-Merritt phenomenon to the ordinary strawberry hemangioma.

· Expecting a port-wine stain to fade in childhood. It persists and in fact darkens and thickens with age.

Where it is taught: Section C01, the paragraph on complications, the cavernous paragraph and the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-RT01

Retrieval exercise: Rebuild the life cycle of an infantile hemangioma

A competent response contains

· Step 1: absent or only a faint precursor patch at birth.

· Step 2: appears in the first weeks of life.

· Step 3: GLUT1-positive endothelium proliferates rapidly for months, then growth plateaus.

· Step 4: endothelial cells die by apoptosis and fibrofatty tissue replaces them.

· Step 5: most have regressed by early childhood.

· The marker is GLUT1, otherwise typical of placental capillaries.

· Regression is by apoptosis, with fibrofatty replacement, sometimes leaving lax skin or telangiectasias.

· The feature that marks this as a tumor rather than a malformation is proliferation after birth followed by programmed regression; a malformation is present at birth and never regresses.

Distinctions that matter

· Growth plateau and involution are two separate stages, and the plateau comes first.

· GLUT1 positivity is present in every phase, so a negative result argues against the infantile lesion even during proliferation.

Common misconceptions

· Writing that the lesion is present and fully formed at birth. At most there is a faint precursor patch.

· Describing involution as the vessels simply shrinking. The endothelial cells die and fibrofatty tissue takes their place.

Where it is taught: Section C01, the life cycle flow diagram of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-CP01

Comparison table: Infantile hemangioma, cavernous hemangioma and port-wine stain

A competent response contains

· At birth: infantile lesion absent or a faint precursor; cavernous variable and often found later; port-wine stain present.

· Growth: infantile rapid for months then plateaus; cavernous slow or none; port-wine stain in proportion to the child, darkening and thickening with age.

· Regression: infantile yes over years; cavernous no; port-wine stain no.

· GLUT1: infantile positive; cavernous negative; port-wine stain negative.

· Histology: infantile lobules of small packed capillaries; cavernous large dilated blood-filled spaces; port-wine stain ectatic dermal capillaries and venules.

· Sites: infantile head and neck skin; cavernous skin, liver as the most common benign liver tumor, and brain; port-wine stain face, often in a trigeminal distribution.

· Associations: infantile lesion with PHACE when large and segmental on the face; cavernous with thrombosis and dystrophic calcification; port-wine stain with Sturge-Weber syndrome from a somatic GNAQ mutation.

Distinctions that matter

· Present at birth and never regressing means malformation; appearing after birth, growing and then fading means infantile hemangioma.

· A hepatic cavernous lesion is usually a small solitary incidental finding in an adult, which is a different clinical situation from a large hepatic infantile hemangioma in a baby.

Common misconceptions

· Filling the GLUT1 row as positive for all three. Only the infantile lesion is positive.

· Putting liver under the infantile row as the classic site. The liver is the classic site for the cavernous lesion as the most common benign liver tumor.

Where it is taught: Section C01, the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-BD1

Brain dump 1 · foundational retrieval: Three red papules that are not hemangiomas of infancy

A competent response contains

· Pyogenic granuloma: any age, often after minor trauma, and on the gingiva in pregnancy; a rapidly growing exophytic red polyp enlarging over weeks on the fingers, face, gingiva or oral mucosa; histology shows lobules of capillaries divided by fibrous septa in edematous stroma with inflammatory cells, resembling exuberant granulation tissue, with a collar of epidermis at the base and a thin frequently ulcerated surface; it bleeds with slight contact.

· Its histologic name is lobular capillary hemangioma, and the name pyogenic granuloma is inaccurate twice over because the lesion is neither infectious nor granulomatous.

· The pregnancy form is granuloma gravidarum, sitting on the gingiva; estrogen and progesterone amplify the vascular response of the gingiva to local irritants such as dental plaque, and it often regresses after delivery.

· Cherry hemangioma (senile angioma): small bright red dome-shaped papules on the trunk and proximal limbs, appearing from early adulthood and multiplying with age until nearly universal in older adults; histology shows dilated capillaries and venules in the papillary dermis; they do not regress and have no malignant potential.

· Bacillary angiomatosis: an infection rather than a neoplasm, in immunocompromised hosts and especially people with AIDS and very low CD4 counts; red to purple papules and nodules of skin, sometimes with bone, lymph node or liver involvement, including peliosis hepatis with B. henselae.

· The organisms are Bartonella henselae, from cats and their fleas, and Bartonella quintana, from the human body louse; both are small gram-negative bacilli.

· They induce host cells to produce hypoxia-inducible factor 1 alpha, which raises VEGF and drives capillary proliferation.

· Histology shows capillary proliferation with plump epithelioid endothelial cells, a neutrophilic infiltrate with nuclear dust, and clumps of granular purple material that a Warthin-Starry silver stain resolves into bacilli.

· Because living bacteria sustain the proliferation, the lesions disappear when the organism is eradicated.

Distinctions that matter

· Pyogenic granuloma is a reactive capillary proliferation, while cherry hemangioma is an age-related collection of dilated vessels and bacillary angiomatosis is an infection; only the infantile hemangioma in the same table is a true neoplasm driven from within the endothelium.

· Neutrophils with nuclear dust and silver-stain-positive bacilli mark bacillary angiomatosis, whereas the inflammatory cells of pyogenic granuloma sit in edematous stroma with no organisms.

Common misconceptions

· Reading the word pyogenic as evidence of infection and treating the lesion as an abscess.

· Expecting cherry hemangiomas to carry malignant potential or to regress. They persist and are benign.

Where it is taught: Section C02, the three descriptive paragraphs and the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-BD2

Brain dump 2 · integration and application: A purple papule in a man with a CD4 count of 40

A competent response contains

· Both bacillary angiomatosis and Kaposi sarcoma fit violaceous skin papules in a man with a CD4 count of 40 cells per microliter, so the color of the lesions does not decide between them.

· Bacillary angiomatosis on biopsy: rounded capillaries, plump epithelioid endothelial cells, a neutrophilic infiltrate with nuclear dust, and silver-stain-positive bacilli on Warthin-Starry staining.

· Kaposi sarcoma on biopsy: fascicles of spindle cells around slit-like spaces with extravasated red cells, lymphocytes and plasma cells, and nuclear staining for LANA.

· Fever and systemic illness favour an infection, and the cat at home fits Bartonella henselae acquired from cats and their fleas.

· Bacillary angiomatosis resolves when the organism is eradicated, because living bacteria sustain the proliferation through hypoxia-inducible factor 1 alpha and VEGF.

· Kaposi sarcoma depends on immune recovery, because the latent viral genes keep driving the spindle cells until cytotoxic T cells and NK cells can again control the infected cells and the inflammatory cytokines that feed them.

Distinctions that matter

· Rounded capillaries with plump endothelium against spindle cells around slit-like spaces is the architectural difference, and it is more reliable than any clinical feature.

· One lesion needs the bacterium present and the other needs the immune system absent, which is why the two respond to quite different changes in the host.

Common misconceptions

· Deciding between the two from the color or distribution of the papules.

· Assuming both lesions will persist until immunity recovers. Bacillary angiomatosis disappears once the bacteria are cleared.

Where it is taught: Section C02, the bacillary angiomatosis paragraph and the worked example on a purple papule in a man with AIDS of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-CP02

Comparison table: Red and purple vascular lesions side by side

A competent response contains

· Infantile hemangioma: infant in the first weeks of life; an endothelial neoplasm; packed capillaries that are GLUT1 positive; grows then regresses.

· Pyogenic granuloma: any age, after trauma, or on the gingiva in pregnancy; a reactive capillary proliferation; lobular capillaries in edematous inflamed stroma with an ulcerated surface; bleeds easily, and the pregnancy form often regresses after delivery.

· Cherry hemangioma: adults, increasing with age; cause unknown and age related; dilated capillaries and venules in the papillary dermis; persists and is benign.

· Bacillary angiomatosis: immunocompromised hosts, especially advanced AIDS; Bartonella henselae or quintana; epithelioid endothelium with neutrophils and silver-stain-positive bacilli; resolves when the infection is eradicated.

· Kaposi sarcoma: older Mediterranean men, people with AIDS, transplant recipients, and the African endemic groups; HHV-8 with immune dysfunction; spindle cells with slit-like spaces, lymphocytes, plasma cells and hemosiderin; a low-grade malignancy whose course depends on immune status.

Distinctions that matter

· Neutrophils and bacteria point to bacillary angiomatosis; spindle cells and slit-like spaces point to Kaposi sarcoma.

· Only two of the five rows hold a true neoplasm of endothelium, and the other three are a reactive proliferation, an age-related ectasia and an infection.

Common misconceptions

· Grouping all five as tumors. The cause column is the point of the table.

· Expecting the course column to follow the appearance of the lesion rather than the cause.

Where it is taught: Section C02, the table of red and purple vascular lesions of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-BD1

Brain dump 1 · foundational retrieval: How HHV-8 drives the spindle cell

A competent response contains

· HHV-8 (Kaposi sarcoma herpesvirus) is a double-stranded DNA gammaherpesvirus, spread mainly through saliva and also by sexual contact, transplanted organs and blood.

· Infection is necessary but not sufficient: most infected people never develop the tumor, and the usual cofactor is impaired T-cell immunity.

· The tumor cell is the spindle cell, of endothelial lineage and largely lymphatic endothelial, expressing CD31 and CD34 along with podoplanin and PROX1.

· In most spindle cells the virus is latent, kept as a circular episome expressing only a few genes, each of which pushes the cell towards division or survival.

· Viral cyclin is a homolog of cyclin D: it binds CDK6, resists the CDK inhibitors p16, p21 and p27, and phosphorylates Rb, freeing E2F to drive the G1 to S transition.

· LANA tethers the viral episome to host chromosomes so daughter cells inherit it, and it binds and inhibits both p53 and Rb, removing the checkpoint that would arrest or kill a cell proliferating without a signal.

· Viral FLIP activates NF-kappaB and blocks apoptosis.

· A small fraction of cells enters lytic replication, secreting viral IL-6 and expressing a constitutively active viral G protein-coupled receptor that increases VEGF; these paracrine signals promote angiogenesis and inflammation and so recruit and stimulate neighbouring cells.

· Early lesions are polyclonal and advanced nodules may become clonal, which is why Kaposi sarcoma is described as lying between reactive hyperplasia and true neoplasia.

Distinctions that matter

· Latent genes supply proliferation and survival within the infected cell, while lytic genes supply paracrine signals that act on the neighbours.

· Viral cyclin phosphorylates Rb and LANA inhibits Rb and p53, so Rb is hit twice by two different mechanisms.

Common misconceptions

· Treating HHV-8 infection as sufficient to cause the tumor. Impaired T-cell immunity is the usual cofactor.

· Calling the spindle cell a fibroblast or smooth muscle cell. It carries endothelial and lymphatic endothelial markers.

Where it is taught: Section C03, the pathogenesis paragraphs and the flow diagram of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-BD2

Brain dump 2 · integration and application: From slit-like spaces to the findings at the bedside

A competent response contains

· Patch: a flat pink, red or purple macule; histology shows irregular dilated vascular spaces with lymphocytes, plasma cells and macrophages, resembling granulation tissue.

· Plaque: a raised violaceous plaque; histology shows dermal spindle cells around jagged channels with extravasated red cells and hemosiderin-laden macrophages.

· Nodule: a purple to brown nodule that may ulcerate; histology shows sheets of plump spindle cells with slit-like spaces holding red cells and eosinophilic hyaline globules.

· Purple color: the slit-like channels are leaky, so red cells escape into the tissue.

· Brown color of old lesions: hemoglobin from those extravasated red cells is broken down to hemosiderin.

· Swelling of the limb: involvement of lymphatics and lymph nodes causes lymphedema, which fits the lymphatic endothelial lineage of the spindle cell.

· Bleeding from a gut lesion: mucosal and visceral lesions are made of leaky vascular spaces under a thin surface.

· Nuclear staining for LANA identifies HHV-8 in the spindle cells at every stage.

· When immunosuppression is lowered the lesions often shrink, because restored cytotoxic T-cell and NK cell activity kills infected spindle cells and reduces the inflammatory cytokines feeding them, even though the latent viral genes themselves have not changed.

· HHV-8 also causes primary effusion lymphoma and multicentric Castleman disease.

Distinctions that matter

· Color tracks the age of the lesion because it tracks the fate of the extravasated red cells, from purple to brown.

· Regression on restoring immunity shows that growth depends on the host as well as on the virus, which is the argument for calling the early lesion closer to hyperplasia than to neoplasia.

Common misconceptions

· Reading regression after reduced immunosuppression as loss of the virus. The virus persists; the control over it returns.

· Expecting LANA staining only in advanced nodules. It marks the spindle cells at every stage.

Where it is taught: Section C03, the staging table, the clinical findings paragraph, the immune control paragraph and the stop and predict box of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-RT01

Retrieval exercise: Rebuild the path from infection to a nodule

A competent response contains

· Step 1: HHV-8 infects endothelial-lineage cells and becomes latent.

· Step 2: viral cyclin phosphorylates Rb and LANA inhibits p53 and Rb.

· Step 3: viral FLIP activates NF-kappaB and blocks apoptosis.

· Step 4: lytic cells release viral IL-6 and raise VEGF.

· Step 5: weak T-cell and NK control lets infected cells expand.

· Step 6: spindle cells form leaky slit-like spaces.

· Step 7: patch progresses to plaque, then nodule.

· Steps 2 and 3 are the latent programme, step 4 is the lytic programme, and step 5 is the host immune system.

· Host proteins put out of action by the latent genes: Rb, p53, and the apoptotic pathway downstream of NF-kappaB activation; the CDK inhibitors p16, p21 and p27 are resisted.

· The lytic step releases viral IL-6 and the viral G protein-coupled receptor signal, and VEGF rises.

· The immune step depends on loss of control by cytotoxic T cells and NK cells.

· Latent genes give proliferation and survival, lytic genes give paracrine growth signals, and immune suppression removes the brake, so losing any one of the three stalls the sequence.

Distinctions that matter

· The latent and lytic programmes run in different cells at the same time, so both contribute to one lesion.

· Immune suppression is permissive rather than causative: it removes a brake instead of supplying a growth signal.

Common misconceptions

· Putting the immune step first as though immunosuppression caused the infection. Infection comes first and latency is established before the immune state matters.

· Assigning viral IL-6 and VEGF to the latent programme. They come from the small lytic fraction.

Where it is taught: Section C03, the flow diagram from infection to nodule of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-PR01

Prediction exercise: Predict what happens when single viral genes are lost

A competent response contains

· Viral cyclin inactivated: Rb phosphorylation falls, E2F stays bound, and the G1 to S transition is no longer forced.

· Viral cyclin inactivated: p16, p21 and p27 regain their braking effect, because it was the viral cyclin and CDK6 complex that resisted them.

· LANA lost: the episome is no longer tethered to host chromosomes, so daughter cells fail to inherit it and the infection is lost from the dividing population.

· LANA lost: p53 and Rb activity is restored, so a cell dividing without a signal is arrested or killed at the checkpoint.

· Viral FLIP lost: NF-kappaB signalling falls and apoptosis is no longer blocked, so infected cells die more readily.

· Lytic genes silenced: viral IL-6 and VEGF fall, angiogenesis and inflammation decline, and neighbouring cells lose the paracrine stimulus that recruited and stimulated them.

· Immune function restored with all viral genes intact: the lesions often shrink, because cytotoxic T cells and NK cells again kill latently infected cells and the inflammatory milieu that fed them subsides.

Distinctions that matter

· LANA does two separable jobs, one mechanical in keeping the episome and one regulatory in inhibiting p53 and Rb, so losing it costs the virus both persistence and checkpoint evasion.

· Removing a viral gene acts inside the infected cell, while restoring immunity acts on the infected cell from outside; both reduce the lesion but by different routes.

Common misconceptions

· Expecting restored p53 and Rb to have no effect because the cell is already transformed. Early lesions are polyclonal and still depend on these drivers.

· Assuming the lytic genes matter only for making new virus. Their products are the paracrine growth signals for the lesion.

Where it is taught: Section C03, the latent and lytic gene paragraphs, the immune control paragraph and the flow diagram of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C04-BD1

Brain dump 1 · foundational retrieval: Build the four forms of Kaposi sarcoma from memory

A competent response contains

· Every form contains HHV-8 and shows the same histology, so what differs is the host.

· Classic: older men, usually over 60, of Mediterranean, Eastern European or Ashkenazi Jewish descent and HIV negative; skin of the distal lower limbs with multiple plaques and nodules; indolent over years to decades with uncommon visceral spread.

· Endemic African: HIV-negative children and adults in sub-Saharan Africa, where most people acquire HHV-8 in childhood through saliva; adults have skin of the limbs and may have bone infiltration, children have lymph node disease; adults are locally aggressive and the childhood lymphadenopathic form is rapidly progressive.

· Transplant-associated (iatrogenic): solid organ recipients on T-cell immunosuppression, months to years after transplant; skin, mucosa, lymph nodes and viscera; can be aggressive and often regresses when immunosuppression is reduced.

· AIDS-associated (epidemic): people with HIV infection, mainly men who have sex with men; skin anywhere including face and trunk, oral mucosa and hard palate, gastrointestinal tract, lungs and lymph nodes; the most aggressive and widespread form, an AIDS-defining illness that improves with immune reconstitution.

· The two host variables are how common HHV-8 infection is in the population and how deeply T-cell immunity is suppressed.

· Mild immune decline gives slow disease confined to the skin of the legs; profound suppression gives multifocal disease of skin, mucosa, lymph nodes and viscera.

Distinctions that matter

· Population prevalence of the virus explains where the classic and endemic forms occur, while depth of immunosuppression explains how fast and how widely any form spreads.

· Transplant disease arises either from reactivation of the recipient's latent virus or from virus carried in the donor organ, which is a different route from the sexual and salivary spread behind the AIDS form.

Common misconceptions

· Expecting different histology for each form. The histology is the same and only the host differs.

· Treating the endemic African form as mild because the patients are HIV negative. The childhood lymphadenopathic form progresses quickly.

Where it is taught: Section C04, the opening paragraph, the table of forms and the four descriptive paragraphs of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C04-BD2

Brain dump 2 · integration and application: Four patients, four courses

A competent response contains

· The 74-year-old Mediterranean man has classic Kaposi sarcoma; the cofactor is age-related decline in T-cell function in a population where HHV-8 infection is relatively common, and the course over a year is slow growth of plaques and nodules on the feet and lower legs.

· The 6-year-old in sub-Saharan Africa has the endemic lymphadenopathic form; the cofactor is childhood acquisition of HHV-8 through saliva in a high-prevalence population, and the course is rapid progression even without skin lesions.

· The woman 14 months after kidney transplantation has transplant-associated disease; the cofactor is pharmacological T-cell suppression, and the course can be aggressive across skin, mucosa, lymph nodes and viscera.

· The man with HIV infection has AIDS-associated disease; the cofactor is CD4 depletion, and the course is the most aggressive and widespread, with pulmonary lesions causing dyspnea, hemoptysis and pleural effusions.

· The classic patient is the one most likely to die of an unrelated cause, because the disease is indolent over years to decades and visceral spread is uncommon.

· The man with HIV infection has the AIDS-defining illness.

· The transplant recipient's disease is the one most likely to shrink without treatment aimed at the tumor, because lowering immunosuppression restores T-cell control.

· The AIDS form is more common in men who have sex with men because HHV-8 spreads sexually and through saliva within those networks, which reflects transmission of the virus rather than a property of HIV itself.

· During early immune reconstitution lesions may flare transiently, because recovering immune cells increase local inflammation before overall control improves.

Distinctions that matter

· Pace and extent follow immune status, so the same virus and the same histology can give decades of indolent leg disease or widespread visceral disease within months.

· A flare during early immune recovery is inflammatory, not a sign that the tumor has escaped control.

Common misconceptions

· Reading the distribution of the AIDS form as evidence that HIV itself makes the tumor more aggressive in one group. The difference reflects how HHV-8 spreads.

· Expecting all four patients to need the same intensity of concern. The classic form rarely determines the outcome of the patient's life.

Where it is taught: Section C04, the table of forms and the paragraphs on each form of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C05-BD1

Brain dump 1 · foundational retrieval: Myxoma and rhabdomyoma from the ground up

A competent response contains

· Primary cardiac tumors are rare and metastases to the heart are far more common.

· Myxoma is the most common primary cardiac tumor in adults, usually appearing in middle age and more often in women.

· About 90% of myxomas arise in the atria, with the left atrium affected about four times as often as the right, typically from the interatrial septum near the fossa ovalis.

· Gross appearance ranges from a firm sessile mass to a soft gelatinous villous lesion on a stalk.

· Histology shows stellate or globular myxoma cells, thought to derive from primitive multipotent mesenchymal cells, in abundant acid mucopolysaccharide ground substance, often with hemorrhage and chronic inflammation.

· About 10% of myxomas are familial as part of Carney complex, an autosomal dominant disorder caused by loss-of-function mutations in PRKAR1A, a regulatory subunit of protein kinase A; these tumors are often multiple, recur, arise in younger patients and unusual sites, and come with spotty skin pigmentation and endocrine overactivity.

· Rhabdomyoma is the most common primary cardiac tumor of infants and children, a hamartoma of cardiac muscle cells, often multiple, within the ventricular myocardium.

· Rhabdomyoma is strongly associated with tuberous sclerosis, in which mutation of TSC1 (hamartin) or TSC2 (tuberin) removes inhibition of mTOR and allows excess cell growth; inheritance is autosomal dominant.

· Its large glycogen-filled myocytes have cytoplasm stretched in strands from a central nucleus to the cell membrane, the so-called spider cells, and many lesions regress spontaneously during early childhood.

Distinctions that matter

· Myxoma is a true neoplasm of primitive mesenchymal cells, while rhabdomyoma is a hamartoma of cardiac muscle, which is why only the rhabdomyoma regresses.

· Adult plus left atrium suggests myxoma; child plus multiple ventricular masses suggests rhabdomyoma and tuberous sclerosis.

Common misconceptions

· Calling the myxoma matrix fibrous. It is abundant acid mucopolysaccharide ground substance, which is why the tumor is gelatinous and friable.

· Treating the spider cell as a tumor cell of malignant type. It is a glycogen-filled cardiac myocyte within a hamartoma.

Where it is taught: Section C05, the myxoma and rhabdomyoma paragraphs and the table of primary cardiac tumors of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C05-BD2

Brain dump 2 · integration and application: Why a benign tumor makes a patient faint

A competent response contains

· Sequence: a stalked mass attached to the atrial septum; it prolapses into the mitral orifice in diastole; ball-valve obstruction to left ventricular filling; left atrial and pulmonary venous pressure rise; dyspnea, pulmonary edema and positional syncope; friable fragments break off as systemic emboli.

· The three groups of findings are obstruction, embolisation and constitutional symptoms.

· Obstruction mimics mitral stenosis with a diastolic murmur, and as the tumor drops into the ventricle there is a low-pitched early diastolic tumor plop; repeated trauma can damage the leaflets and cause mitral regurgitation.

· Obstruction varies with tumor position, so the murmur and the symptoms change with posture and the symptoms are episodic.

· A left atrial tumor sheds friable tumor or overlying thrombus into the systemic circulation, causing stroke or limb ischemia; a right atrial myxoma embolises to the lungs.

· Many myxomas secrete IL-6, which drives the hepatic acute phase response, raising ESR and CRP and producing anemia of inflammation, with fever, weight loss, fatigue and arthralgia.

· Echocardiography shows a mobile mass attached to the septum.

· A rhabdomyoma causes trouble by obstructing a valve orifice or outflow tract and by causing arrhythmias.

· Its problems differ because it sits within the ventricular myocardium rather than hanging free in an atrium, so it blocks flow from inside the wall and disturbs conduction instead of acting as a mobile ball valve or shedding emboli.

Distinctions that matter

· Obstruction from a mobile mass varies with position, while fixed valvular stenosis does not, which is what makes the symptoms postural.

· The constitutional symptoms come from a secreted cytokine, not from spread of a benign tumor.

Common misconceptions

· Explaining fever and a raised ESR in this setting as infection by default. IL-6 from the tumor produces the same acute phase picture.

· Assuming any atrial myxoma causes stroke. A right atrial tumor embolises to the lungs.

Where it is taught: Section C05, the flow diagram and the paragraph on the three groups of findings, with the rhabdomyoma paragraph of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C05-CA01

Clinical application: Breathlessness, fainting on standing and a high ESR

A competent response contains

· Recognize: a left atrial myxoma, attached to the interatrial septum near the fossa ovalis, and the sound after the second heart sound is a low-pitched early diastolic tumor plop.

· Explain: in diastole the stalked mass prolapses into the mitral orifice and obstructs left ventricular filling as a ball valve, so left atrial and pulmonary venous pressures rise and give exertional breathlessness; because obstruction depends on where the tumor sits, a change of position can block filling abruptly, cardiac output falls, and she faints.

· Differentiate: rheumatic mitral stenosis gives fixed obstruction that does not change with posture and would not explain the fever, weight loss, high ESR and anemia, and there is no history of rheumatic fever; infective endocarditis is argued against by repeatedly negative blood cultures, and it would not produce a mobile mass on the atrial septum with a tumor plop.

· Investigate: echocardiography, which shows a mobile mass attached to the interatrial septum and its movement into the mitral orifice in diastole; the raised ESR and the raised CRP are the laboratory findings already present that support the systemic picture, alongside the anemia.

· Decide: embolisation of friable tumor fragments or overlying thrombus is the most damaging complication, and from a left atrial tumor the brain is the organ most likely to be affected, giving stroke.

· Justify: many myxomas secrete IL-6, which drives the hepatic acute phase response, so fever, weight loss, a high ESR and anemia of inflammation all follow from one secreted cytokine; negative blood cultures do not argue against the diagnosis because the systemic picture is cytokine driven rather than infective.

Distinctions that matter

· Postural change in the murmur and in the symptoms separates a mobile mass from fixed valvular stenosis.

· A benign tumor is dangerous here through mechanics and embolisation rather than through invasion or metastasis.

Common misconceptions

· Reading fever, weight loss and a high ESR with a new murmur as endocarditis without weighing the mobility of the lesion and the negative cultures.

· Expecting a benign cardiac tumor to be harmless. Obstruction, emboli and IL-6 all cause harm without any malignant behavior.

Where it is taught: Section C05, the flow diagram, the paragraph on the three groups of findings and the worked example on posture-dependent syncope of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C06-BD1

Brain dump 1 · foundational retrieval: Angiosarcoma: what it is, where it sits, what it looks like

A competent response contains

· Angiosarcoma is a malignant neoplasm of endothelial cells, occurring mainly in older adults.

· It can arise anywhere, most often in the skin and especially the scalp and face of elderly patients, and in soft tissue, breast and liver; it is the most common primary malignant tumor of the heart in adults, usually in the right atrium.

· Well-differentiated tumors form irregular anastomosing channels that dissect between collagen bundles, lined by plump atypical multilayered endothelial cells.

· Poorly differentiated tumors grow as solid sheets of anaplastic spindle or epithelioid cells with mitoses and necrosis, so endothelial markers are needed to show the lineage; the two markers named are CD31 and ERG.

· Risk factors: vinyl chloride in PVC production workers, main site liver; arsenic from pesticides, smelting or contaminated well water, main site liver with skin and lung cancers as well; Thorotrast, a radiocontrast used about 1930 to 1950, main site liver with cholangiocarcinoma and hepatocellular carcinoma; chronic lymphedema in the arm after radical mastectomy with axillary dissection, which is Stewart-Treves syndrome, arising in the lymphedematous limb; prior breast or chest wall irradiation, arising in the irradiated field; and no identified exposure in older adults, arising in the scalp and face skin.

· Behavior: it infiltrates widely beyond its visible margin, is often multifocal in the skin, recurs locally and metastasizes frequently to the lungs.

· Five-year survival is about 30%, and hepatic tumors do worse.

Distinctions that matter

· Channel formation separates well from poorly differentiated tumors, and once channels are lost only the endothelial markers establish the lineage.

· The hepatic risk factors act through the liver itself, while lymphedema and radiation act on a damaged field.

Common misconceptions

· Expecting a clear surgical margin from the visible edge. The tumor infiltrates well beyond what can be seen.

· Reading a solid sheet of anaplastic cells as excluding a vascular tumor. CD31 and ERG settle the lineage.

Where it is taught: Section C06, the opening paragraph, the risk factor table and the final paragraph on behavior of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C06-BD2

Brain dump 2 · integration and application: Same tumor, two different organs

A competent response contains

· Both patients have angiosarcoma, a malignant endothelial tumor, but the injury differs, so the organ differs.

· Sequence in the PVC worker: inhaled vinyl chloride monomer in PVC manufacturing; hepatic CYP2E1 oxidizes it to chloroethylene oxide; the epoxide forms DNA adducts in sinusoidal cells; mutations accumulate over decades of latency; malignant endothelial proliferation arises in the liver.

· The liver rather than the lung is the target because the metabolic activation step happens there, so the reactive epoxide is generated next to hepatic sinusoidal cells.

· The woman has Stewart-Treves syndrome, angiosarcoma arising about a decade after chronic lymphedema of the arm following radical mastectomy with axillary node dissection.

· The two features of a chronically lymphedematous limb thought to allow the tumor are impaired local immune surveillance and growth factor rich lymphedematous tissue.

· Angiosarcoma arising in a previously irradiated breast or chest wall field often carries MYC amplification.

· The shared property is a long latent period, often decades, between the exposure and the tumor.

· The other two hepatic exposures are arsenic, from pesticides, industrial exposure or contaminated well water, and Thorotrast, a thorium dioxide contrast agent used from about 1930 to 1950 that was taken up by hepatic and splenic macrophages and retained for life, irradiating nearby cells with alpha particles.

Distinctions that matter

· A chemical that needs metabolic activation picks the organ that activates it, whereas a field injury picks the field, which is why the same tumor appears in liver or limb.

· Thorotrast and radiation both injure by ionizing radiation, but one is retained inside hepatic macrophages for life and the other is delivered to an external field.

Common misconceptions

· Expecting vinyl chloride to cause lung tumors because it is inhaled. The target follows the site of metabolic activation.

· Treating the long gap between exposure and tumor as evidence against a causal link. Decades of latency are typical for all of these exposures.

Where it is taught: Section C06, the two risk factor paragraphs, the vinyl chloride flow diagram, the risk factor table and the stop and predict box of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-Q01

Board-style question: A red lesion that appeared after birth

Correct answer: A. Rapid growth for several months, then apoptosis of the endothelium with fibrofatty replacement

Why it is right: Infantile hemangioma proliferates rapidly for several months, then growth plateaus and the endothelial cells die by apoptosis and are replaced by fibrofatty tissue, so most lesions have regressed by early childhood; preterm birth, female sex, the faint precursor patch, the lobules of packed capillaries and GLUT1 staining all fit this lesion.

B is wrong: Growth in proportion to the child with darkening and thickening in adult life describes a port-wine stain, which is present at birth and GLUT1 negative.

C is wrong: Progressive dilatation of large thin-walled channels with thrombosis and calcification describes a cavernous hemangioma, whose mature vessels do not proliferate and which is GLUT1 negative.

D is wrong: Infiltration of deep tissue with metastasis to the lungs describes angiosarcoma, a malignant tumor of older adults with atypical multilayered endothelium rather than bland packed capillaries.

A competent response contains

· A faint precursor patch at birth, appearance in the first weeks of life, steady enlargement, lobules of packed capillaries and GLUT1-positive endothelium together identify an infantile hemangioma.

· The natural history is rapid proliferation for months, a plateau, then apoptosis of the endothelium with fibrofatty replacement over years.

Distinctions that matter

· GLUT1 positivity separates the infantile hemangioma from vascular malformations and congenital hemangiomas, which are GLUT1 negative.

Common misconceptions

· Reading rapid postnatal growth as a sign of malignancy. Proliferation followed by programmed regression is the expected behavior here.

Where it is taught: Section C01 of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-Q02

Board-style question: Purple nodules in an immunocompromised host

Correct answer: C. Bacterial induction of hypoxia-inducible factor 1 alpha with a rise in VEGF

Why it is right: Bartonella henselae, acquired here from kittens, induces host production of hypoxia-inducible factor 1 alpha, which raises VEGF and drives capillary proliferation, and the neutrophilic infiltrate with Warthin-Starry positive bacilli is the histological signature of that infection.

A is wrong: A viral cyclin homolog binding CDK6 and phosphorylating Rb in spindle cells is the mechanism of Kaposi sarcoma, which would show spindle cells around slit-like spaces and no bacilli.

B is wrong: A somatic activating GNAQ mutation in dermal endothelium belongs to the port-wine stain and Sturge-Weber syndrome, a capillary malformation present from birth.

D is wrong: Endothelial type 3 iodothyronine deiodinase expression belongs to large hepatic infantile hemangiomas and causes consumptive hypothyroidism rather than a neutrophilic vascular proliferation.

A competent response contains

· Rounded capillaries with plump epithelioid endothelium, neutrophils with nuclear dust and silver-stain-positive bacilli in an immunocompromised host identify bacillary angiomatosis.

· Bartonella induces host cells to produce hypoxia-inducible factor 1 alpha, which raises VEGF and drives the capillary proliferation, so the lesions resolve when the organism is eradicated.

Distinctions that matter

· Neutrophils and visible bacilli point to bacillary angiomatosis, while spindle cells around slit-like spaces point to Kaposi sarcoma in the same patient population.

Common misconceptions

· Assuming any purple nodule in advanced HIV infection is Kaposi sarcoma. The inflammatory cells and the vessel shape decide it.

Where it is taught: Section C02 of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-Q03

Board-style question: The gene that removes two checkpoints at once

Correct answer: B. It binds and inhibits both p53 and Rb, removing the checkpoint response to unscheduled proliferation

Why it is right: LANA is the latency-associated nuclear antigen detected by the nuclear stain; besides tethering the viral episome to host chromosomes so daughter cells inherit it, it binds and inhibits both p53 and Rb, so the checkpoint that would arrest or kill an inappropriately dividing cell is lost.

A is wrong: Activating NF-kappaB to block apoptosis is the action of viral FLIP, not of the tethering nuclear antigen.

C is wrong: A constitutively active G protein-coupled receptor that raises VEGF is expressed by the small lytic fraction of cells and acts as a paracrine signal rather than on the episome.

D is wrong: Resisting the inhibitors p16, p21 and p27 is a property of the viral cyclin and CDK6 complex.

A competent response contains

· The nuclear antigen that tethers the HHV-8 episome to host chromosomes is LANA, latency-associated nuclear antigen.

· LANA also binds and inhibits both p53 and Rb, removing the checkpoint that would arrest or kill a cell proliferating without a signal.

Distinctions that matter

· Each latent gene has its own job: viral cyclin drives the cell cycle, LANA keeps the episome and disables p53 and Rb, and viral FLIP blocks apoptosis.

Common misconceptions

· Attributing all of the latent effects to one protein. The question turns on which protein does which job.

Where it is taught: Section C03 of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C06-Q04

Board-style question: A liver mass decades after an industrial exposure

Correct answer: D. Hepatic cytochrome P450 2E1 oxidized the monomer to a reactive epoxide that formed DNA adducts in sinusoidal cells

Why it is right: Vinyl chloride monomer is oxidized by hepatic cytochrome P450 2E1 to chloroethylene oxide, a reactive epoxide that forms DNA adducts in sinusoidal cells, so mutations accumulate in hepatic endothelium over decades and the malignant proliferation arises in the liver.

A is wrong: Retention of particles by hepatic macrophages with alpha particle irradiation describes Thorotrast, a thorium dioxide contrast agent, not an inhaled monomer from PVC manufacturing.

B is wrong: Impaired lymphatic drainage in a growth factor rich tissue describes the mechanism proposed for Stewart-Treves angiosarcoma in a chronically lymphedematous limb.

C is wrong: MYC amplification is characteristic of secondary angiosarcoma arising in a previously irradiated breast or chest wall field.

A competent response contains

· Irregular anastomosing channels dissecting between collagen bundles with plump multilayered atypical CD31-positive cells in the liver of a former PVC worker identify hepatic angiosarcoma from vinyl chloride.

· Hepatic CYP2E1 oxidizes vinyl chloride to chloroethylene oxide, a reactive epoxide that forms DNA adducts in sinusoidal cells, which is why the liver is the target organ.

Distinctions that matter

· Metabolic activation in the liver explains the organ for vinyl chloride, while lymphedema and radiation explain the field for the other classic exposures.

Common misconceptions

· Expecting an inhaled carcinogen to strike the lung. The site of activation, not the route of entry, picks the organ here.

Where it is taught: Section C06 of the CV-18-J Vascular Neoplasms notes.

Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-K Infectious Vascular Diseases

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-11-C01-BD1

Brain dump 1 · foundational retrieval: Build a vegetation from a normal valve

A competent response contains

· Normal endothelium resists bacterial attachment, so most cases need two things at once: a surface bacteria can grip and bacteria in the blood.

· Step 1: a turbulent high-velocity jet (regurgitant or stenotic), a ventricular septal defect, a prosthetic ring, or repeated injection of particulate material injures valve endothelium and exposes subendothelial collagen and tissue factor.

· Step 2 and 3: platelets and fibrin deposit on the injured area and form a small sterile mass, which is nonbacterial thrombotic endocarditis.

· Step 4: organisms entering during a transient bacteremia adhere to the deposit. Viridans streptococci make extracellular dextran (a glucan) and other adhesins that stick to fibrin and platelets, so they need a damaged valve.

· Staphylococcus aureus carries surface proteins that bind fibrinogen and fibronectin and can attach to and invade intact endothelium.

· Step 5 and 6: more fibrin covers the bacteria and shields them, and the growing vegetation sheds organisms into the blood continuously.

· The vegetation is avascular, so neutrophils penetrate it poorly, bacteria reach very high densities, and bacteremia is persistent rather than intermittent, which is why blood cultures stay positive.

· Staphylococcus aureus can skip the first three steps and infect a valve that was never damaged, which is why it produces acute disease on normal valves.

Distinctions that matter

· Adhesion to a sterile platelet and fibrin deposit is what viridans streptococci need; invasion of intact endothelium is what S. aureus can add.

· A vegetation is infected; nonbacterial thrombotic endocarditis is the sterile deposit that precedes it.

Common misconceptions

· Writing that bacteria settle on any valve they reach. Normal endothelium resists attachment, which is why turbulence and a sterile deposit usually come first.

· Explaining persistent bacteremia by saying the immune system is weak. The vegetation is avascular and fibrin covered, so neutrophils cannot get in.

Where it is taught: Section C01, first three paragraphs and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C01-BD2

Brain dump 2 · integration and application: Reason from the setting to the organism

A competent response contains

· Patient 1: viridans streptococci (for example S. sanguinis, S. mutans) from mouth flora released at extraction, adhering to the deposit on a regurgitant mitral valve through dextran.

· Patient 2: Staphylococcus aureus from skin flora injected into the blood, reaching the right heart first, so the tricuspid valve is involved; Pseudomonas and Candida also occur in injection drug use.

· Patient 3: Staphylococcus epidermidis or another coagulase-negative staphylococcus, introduced at operation and growing as a biofilm on foreign material.

· Patient 4: enterococci, mainly Enterococcus faecalis, from urinary and gut flora entering the blood during manipulation.

· Patient 5: Streptococcus gallolyticus (formerly S. bovis), which is associated with colorectal cancer, so the colon is examined.

· Patient 6: culture-negative endocarditis from fastidious or intracellular organisms, the HACEK group, Coxiella burnetii, or Bartonella, which grow poorly in routine culture.

· Acute course: patients 2 and 3, from virulent organisms that destroy a valve over days. Subacute course: patients 1, 4, 5, and 6, smoldering for weeks with fever, weight loss, and fatigue.

· Virulence of the organism decides the course, not the valve, which is why the time line in a question points at the organism.

· Late prosthetic valve infection behaves more like native valve disease, with streptococci and S. aureus becoming common.

Distinctions that matter

· Early and late prosthetic valve infection have different organisms: early is skin flora from the operation, late resembles native valve disease.

· Negative cultures are a clue to the organism rather than evidence against endocarditis.

Common misconceptions

· Naming S. aureus for every case. The source of the bacteremia and the surface available decide the organism.

· Assuming injection drug use always means a left-sided valve. Injected organisms reach the right heart first and the tricuspid valve is the usual site.

Where it is taught: Section C01, organism table and the worked example of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C01-RT01

Retrieval exercise: Rebuild the vegetation sequence and mark where vegetations sit

A competent response contains

· Step 1: a turbulent jet or foreign material injures valve endothelium.

· Step 2: collagen and tissue factor are exposed and platelets and fibrin deposit.

· Step 3: a sterile deposit forms, which is nonbacterial thrombotic endocarditis.

· Step 4: during a transient bacteremia organisms adhere through dextran or fibrinogen and fibronectin binding proteins.

· Step 5: more fibrin covers the bacteria and shields them from neutrophils.

· Step 6: the vegetation grows and sheds organisms into the blood continuously.

· Vegetations sit along the line of valve closure: on the atrial surface of the mitral and tricuspid valves and the ventricular surface of the aortic and pulmonary valves.

· Those are the low-pressure sides, the surfaces that receive the regurgitant jet, which is why the deposit forms there.

· The mitral valve and then the aortic valve are involved most often because they work at higher pressures.

· In injection drug use the tricuspid valve is the most common site, because injected bacteria and particulate debris reach the right heart before anything else.

Distinctions that matter

· The vegetation forms on the downstream, lower pressure side of the valve, not on the side the pressure is generated.

· Higher working pressure explains left-sided predominance; route of entry explains the tricuspid exception.

Common misconceptions

· Placing mitral vegetations on the ventricular surface. The mitral regurgitant jet strikes the atrial surface, so the deposit forms there.

Where it is taught: Section C01, third paragraph and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C01-Q01

Board-style question: Organism in right-sided endocarditis

Correct answer: C. Staphylococcus aureus

Why it is right: Skin flora are injected directly into the venous blood, Staphylococcus aureus surface proteins bind fibrinogen and fibronectin and can invade intact endothelium, so it infects the normal tricuspid valve and destroys it over days, and fragments of the vegetation lodge in the pulmonary arteries as cavitating septic emboli.

A is wrong: Streptococcus gallolyticus bacteremia is associated with colorectal cancer and a subacute left-sided course, not with injected skin flora and lung lesions.

B is wrong: Enterococcus faecalis endocarditis follows gastrointestinal or genitourinary manipulation, usually in older adults, and runs a subacute course.

D is wrong: Staphylococcus epidermidis infects prosthetic material as a biofilm, typically early after valve surgery, and this patient has a native valve.

A competent response contains

· Injection drug use, a short febrile course, and a holosystolic murmur at the left lower sternal border that increases with inspiration indicate tricuspid regurgitation from right-sided endocarditis.

· Cavitating peripheral lung nodules are septic pulmonary emboli from a right-sided vegetation.

· Staphylococcus aureus is the expected organism in injection drug use and the most common cause of endocarditis overall.

Distinctions that matter

· Right-sided vegetations embolize to the lungs; left-sided vegetations embolize to the systemic circulation.

Common misconceptions

· Reading cavitating lung nodules as a primary pneumonia. They are the first capillary bed downstream of the tricuspid valve.

Where it is taught: Section C01, organism table and the worked example of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C02-BD1

Brain dump 1 · foundational retrieval: Build the complication table by mechanism

A competent response contains

· Valve destruction: acute regurgitation, heart failure, chordal rupture. Clue is a new murmur with pulmonary edema.

· Perivalvular abscess: AV block and fistula between chambers. Clue is an aortic or prosthetic valve with new PR prolongation.

· Left-sided emboli: stroke, brain abscess, splenic and renal infarcts or abscesses, ischemic limbs or digits, Janeway lesions, splinter hemorrhages. Clue is a neurologic deficit or flank or left upper quadrant pain.

· Right-sided emboli: septic pulmonary emboli and lung abscess. Clue is injection drug use with cavitating nodules.

· Immune complexes: glomerulonephritis, Osler nodes, Roth spots. Clue is hematuria with red cell casts and low complement.

· Arterial wall infection: mycotic aneurysm. Clue is sudden headache or hemorrhage weeks into the illness.

· Ring abscess is most common with the aortic valve and with prosthetic valves, and the aortic annulus lies next to the AV node and the His bundle.

· The immune complex lesion is type III hypersensitivity: complexes deposit in glomeruli, activate complement, and cause a proliferative glomerulonephritis with hematuria and red cell casts and low complement levels; rheumatoid factor may be positive.

· Heart failure is the leading cause of death. Bacterial enzymes and neutrophil products perforate leaflets and tear chordae, and sudden severe regurgitation gives the ventricle no time to dilate, so pressure is transmitted back to the lungs and pulmonary edema follows.

Distinctions that matter

· Osler nodes are painful and attributed to immune complex vasculitis; Janeway lesions are painless and attributed to septic microemboli.

· A new murmur points to destroyed valve tissue; a new PR prolongation points to infection spreading into the annulus.

Common misconceptions

· Memorizing the peripheral signs as a list. Each one belongs to a mechanism, and the mechanism is what a question tests.

· Treating splinter hemorrhages as specific. They are linear and nonspecific and also follow trauma.

Where it is taught: Section C02, all four mechanism paragraphs and the mechanism table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C02-BD2

Brain dump 2 · integration and application: Sort four new findings back to their mechanisms

A competent response contains

· Patient 1: left-sided emboli from a mitral or aortic vegetation. Fragments enter the systemic circulation and lodge in the first capillary bed downstream, giving splenic infarction and stroke.

· Patient 2: right-sided emboli from a tricuspid vegetation. Fragments reach the lungs as septic pulmonary emboli, producing multiple peripheral nodules that may cavitate, with cough, pleuritic pain, and hemoptysis.

· Patient 3: perivalvular (ring) abscess extending from the aortic valve. The aortic annulus lies next to the AV node and the His bundle, so a new and lengthening PR interval or a higher degree of AV block in a patient with endocarditis suggests an abscess.

· Patient 4: immune complexes. Continuous antigen release drives antibody production, complexes deposit in glomeruli and activate complement, giving hematuria with red cell casts and low complement, and the painful finger pad nodules are Osler nodes.

· Sudden severe headache in week 6 suggests rupture of a mycotic aneurysm with intracranial hemorrhage, typically in a branch of the middle cerebral artery.

· A septic embolus lodges in the vasa vasorum or lumen of an artery, infection weakens the wall, and the weakened wall dilates and can rupture.

· Sudden severe regurgitation meets a ventricle of normal size and compliance, so the regurgitant volume raises left atrial and pulmonary venous pressure sharply and pulmonary edema follows. Over years the ventricle dilates and accommodates the same volume at a lower filling pressure.

· Large, mobile vegetations embolize most often.

Distinctions that matter

· The side of the heart decides where emboli land, because emboli stop in the first capillary bed downstream.

· AV block in endocarditis is a structural complication of the annulus, separate from the valve lesion itself.

Common misconceptions

· Explaining hematuria in endocarditis as embolic every time. Red cell casts with low complement point to immune complex glomerulonephritis.

· Expecting chronic and acute regurgitation of the same severity to look alike. Time to remodel is what separates them.

Where it is taught: Section C02, paragraphs on local destruction, embolism, and immune complexes, with the mechanism table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C02-CA01

Clinical application: Six weeks of fever, blood in the urine, and a changing murmur

A competent response contains

· Recognize: subacute infective endocarditis of the mitral valve, expected organism a viridans streptococcus from the dental extraction. The findings named are Osler nodes (painful finger pad nodules), Janeway lesions (painless macules on the palms), Roth spots (retinal hemorrhages with pale centers), and immune complex glomerulonephritis.

· Explain: the extraction released mouth flora into the blood; her long-standing regurgitant jet had already injured the atrial surface of the mitral valve and laid down a sterile platelet and fibrin deposit; viridans

streptococci adhere to that deposit through dextran and other adhesins, fibrin covers them, neutrophils cannot penetrate, and the vegetation grows and sheds organisms continuously.

· Differentiate: Janeway lesions are usually attributed to septic microemboli; Osler nodes, Roth spots, and the glomerulonephritis are attributed to immune complexes. Embolic lesions are occlusive and painless at the skin, while immune complex lesions are inflammatory and tender, and the low complement with red cell casts marks complex deposition rather than occlusion.

· Investigate: dysmorphic erythrocytes and erythrocyte casts with no bacteria mean the bleeding is glomerular rather than from the urinary tract, and low C3 and C4 mean complement has been consumed. This is a hypocomplementemic proliferative glomerulonephritis from type III (immune complex) hypersensitivity, with complexes deposited in the glomerular capillaries.

· Decide: a new and lengthening PR interval or a higher degree of AV block would indicate extension into the aortic annulus as a perivalvular abscess. A new loud murmur with sudden dyspnea and pulmonary edema would indicate that a leaflet has perforated or a chord has ruptured.

· Justify: the aortic annulus lies next to the AV node and the His bundle, so an abscess there interrupts conduction and lengthens the PR interval. A ventricle facing sudden severe regurgitation has had no time to dilate, so the extra volume raises left atrial and pulmonary venous pressure abruptly and fluid moves into the alveoli.

Distinctions that matter

· Glomerular bleeding gives dysmorphic cells and red cell casts; bleeding from the urinary tract does not.

· A changing murmur reports the valve, while a changing PR interval reports the annulus.

Common misconceptions

· Reading the renal failure here as embolic infarction. Low complement with red cell casts points to immune complex glomerulonephritis.

· Treating a 6-week course as evidence against endocarditis. A low-virulence organism on a damaged valve smolders for weeks.

Where it is taught: Section C02, with vegetation formation and organism prediction in Section C01 of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C03-BD1

Brain dump 1 · foundational retrieval: Trypanosoma cruzi from the bug to the tissue

A competent response contains

· Trypanosoma cruzi is a flagellated protozoan found in Mexico, Central America, and South America.

· The vector is the triatomine (reduviid, or kissing) bug, which lives in cracks in mud and thatch houses and feeds at night, often on the face.

· The bug defecates while or after feeding, and infective trypomastigotes in the feces are rubbed into the bite wound or onto the conjunctiva or other mucosa, so the feces and not the bite itself carry the parasite in.

· Other routes: blood transfusion, organ transplantation, mother to fetus, and food or drink contaminated with bug feces.

· Trypomastigotes invade local cells, transform into intracellular amastigotes, multiply, and burst out as trypomastigotes that spread through the blood.

· Amastigotes are the intracellular nonflagellated replicating form; trypomastigotes are the form that travels in the blood.

· Acute entry findings: a chagoma, an indurated skin lesion at a skin entry site, and Romana sign, painless unilateral swelling of the eyelids when entry is through the conjunctiva.

· The acute phase lasts weeks to about two months and most infections are mild or unnoticed; trypomastigotes are numerous enough to be seen on a blood smear.

· Acute myocarditis or meningoencephalitis occurs in a small minority, mainly young children and immunocompromised people.

· In nonendemic countries most cases are in immigrants from endemic regions.

Distinctions that matter

· Amastigote and trypomastigote are different stages with different locations: inside cells and multiplying, against in the blood and spreading.

· A chagoma marks a skin entry site and Romana sign marks a conjunctival entry site.

Common misconceptions

· Saying the bite inoculates the parasite. The parasite is in the bug's feces and is rubbed into the wound or the eye.

· Expecting severe acute illness. Most acute infections are mild or unnoticed, and severe acute disease is limited to young children and immunocompromised people.

Where it is taught: Section C03, first two paragraphs and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C03-BD2

Brain dump 2 · integration and application: Explain each sign of chronic Chagas disease

A competent response contains

· Enlarged heart with low ejection fraction: decades of inflammation with patchy myocardial fibrosis and myocyte loss give a dilated cardiomyopathy.

· Right bundle branch block with left anterior fascicular block, and AV block: fibrosis has destroyed conduction tissue.

· Thin bulging apex: thinning of the left ventricular apex produces an apical aneurysm, in which mural thrombus forms and embolizes.

· Dysphagia: loss of myenteric (Auerbach) plexus neurons in the esophagus abolishes coordinated peristalsis and sphincter relaxation, giving an achalasia-like megaesophagus.

· Severe constipation: the same neuron loss in the colon gives megacolon.

· Parasites persist in cardiac muscle and enteric neurons; tissue sections show lymphocytic infiltrates and fibrosis with few parasites, and an autoimmune response contributes to the damage.

· Sudden death can be the first sign because scarred myocardium supports reentrant ventricular tachycardia.

· Parasitemia is now too low for a blood smear, so chronic disease is identified by antibodies against T. cruzi (serology).

· About 20 to 30 percent of infected people develop chronic disease, years to decades after a lifelong indeterminate phase in which the immune response controls but does not clear the parasite.

Distinctions that matter

· Acute disease is found by seeing the parasite; chronic disease is found by detecting antibody, because parasite numbers have fallen while tissue damage has grown.

· Conduction block and dilated cardiomyopathy come from the same fibrosis, but the block reflects where the fibrosis sits rather than how much muscle is lost.

Common misconceptions

· Expecting abundant parasites in chronic heart tissue. The sections show lymphocytes and fibrosis with few organisms.

· Treating the megaesophagus as a muscle problem. The neurons of the myenteric plexus are what has been lost.

Where it is taught: Section C03, third paragraph, the comparison table, and the final paragraph of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C03-CP01

Comparison table: Acute against chronic Chagas disease

A competent response contains

· Timing and share: acute is weeks after infection and affects everyone infected; chronic appears years to decades later in about 20 to 30 percent.

· Parasites: acute has abundant trypomastigotes in the blood; chronic has few, persisting in heart and gut tissue.

· Findings outside the heart: acute gives fever, chagoma, Romana sign, and lymphadenopathy, and is often without symptoms; chronic gives megaesophagus with dysphagia and megacolon with constipation.

· Heart: acute myocarditis is rare; chronic gives fibrosis, conduction block, apical aneurysm with thrombi, arrhythmia, and dilated cardiomyopathy.

· Detection: blood smear in the acute phase, serology in the chronic phase.

· Microscopy: acute tissue contains amastigotes in infected cells with inflammation; chronic tissue shows lymphocytic infiltrates and fibrosis with few parasites.

· Most infected people never progress beyond the indeterminate phase.

Distinctions that matter

· The two phases differ in parasite burden and in tissue damage in opposite directions, which is why the test changes with the phase.

· The indeterminate phase is a third state, not a mild form of either phase.

Common misconceptions

· Assuming chronic disease follows every infection. It develops in about 20 to 30 percent.

Where it is taught: Section C03, acute against chronic table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C04-BD1

Brain dump 1 · foundational retrieval: Build the three stages of Lyme disease

A competent response contains

· Borrelia burgdorferi is a spirochete transmitted by Ixodes ticks: Ixodes scapularis, the black-legged or deer tick, in the eastern United States, and Ixodes pacificus on the Pacific coast.

· White-footed mice are the main reservoir and deer sustain the adult tick population.

· Risk is concentrated in the Northeast, the mid-Atlantic states, and the upper Midwest (Wisconsin and Minnesota), with a smaller Pacific coast focus.

· Most infections are acquired in late spring and summer from nymphs, which are tiny and easily missed.

· The spirochete sits in the tick midgut and must migrate to the salivary glands after feeding begins, so transmission usually requires attachment for more than about 36 hours.

· Early localized, days to about 1 month: erythema migrans with fever, fatigue, headache, and myalgia, from the spirochete spreading through the skin around the bite.

· Early disseminated, weeks to months: multiple erythema migrans lesions, facial nerve palsy (unilateral or bilateral), lymphocytic meningitis, radiculoneuritis, and carditis with AV block, from bloodstream spread to skin, nerves, meninges, and heart.

· Late, months to years: recurrent arthritis of large joints, especially the knee, and chronic neuropathy or encephalopathy, from persistent antigen and immune inflammation in synovium and nerve.

· Erythema migrans is an expanding red patch at least about 5 cm across that may clear centrally into a target shape.

· The same tick can transmit Babesia and Anaplasma, so coinfection is possible.

· Not every patient passes through every stage, and many do not recall the tick bite.

Distinctions that matter

· The stages follow where the spirochete is: in the skin around the bite, then in the blood reaching nerves, meninges, and heart, then persisting in synovium and nerve.

· Nymphs rather than adults transmit most infections, which is why the bite is so often unnoticed.

Common misconceptions

· Expecting a recalled tick bite. Nymphs are tiny and most patients do not remember one.

· Treating the attachment time as arbitrary. It reflects the migration of the spirochete from the tick midgut to the salivary glands.

Where it is taught: Section C04, first two paragraphs and the stages table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C04-BD2

Brain dump 2 · integration and application: Lyme carditis and why the testing changes with the stage

A competent response contains

· This is early disseminated Lyme disease, which is when carditis occurs, in a small percentage of untreated patients.

· Spirochetes and a lymphocytic infiltrate inflame the myocardium and the conduction system, most often at the AV node.

· AV block of any degree can occur, from a prolonged PR interval to complete heart block, and the degree can change within hours because the tissue is inflamed rather than destroyed.

· The block usually resolves, because the node is inflamed rather than scarred. Myopericarditis can occur but heart failure is uncommon.

· Erythema migrans appears before antibodies are reliably detectable, so early localized disease is recognized clinically in an exposed patient.

· Disseminated and late disease are confirmed by serology, usually a screening immunoassay followed by a confirmatory test such as an immunoblot.

· Bilateral facial palsy is unusual and helps point to Lyme disease rather than idiopathic (Bell) palsy.

· Chagas disease also blocks conduction, but through fibrosis built over years with a dilated cardiomyopathy, apical aneurysm, and megaesophagus or megacolon, in someone exposed in Latin America.

· A perivalvular abscess from endocarditis also blocks conduction, but the patient has a fever with an infected aortic or prosthetic valve, a murmur, and a PR interval that lengthens as the abscess extends.

Distinctions that matter

· Inflammation reverses and scar does not, which is why Lyme block resolves while Chagas block persists.

· Testing follows the antibody response rather than the symptom: too early for serology in the skin stage, reliable once the infection has disseminated.

Common misconceptions

· Ordering serology to confirm erythema migrans. Antibodies are not yet reliably detectable at that stage.

· Assuming complete heart block means permanent conduction damage. In Lyme carditis the node is inflamed and the block usually resolves.

Where it is taught: Section C04, carditis and diagnosis paragraphs, with Section C02 on ring abscess and Section C03 on Chagas disease of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C04-Q02

Board-style question: Heart block after a summer rash

Correct answer: A. Lymphocytic inflammation of the atrioventricular node around a disseminated spirochete

Why it is right: Borrelia burgdorferi disseminates from the skin through the blood to the heart, and spirochetes with a lymphocytic infiltrate inflame the myocardium and the conduction system at the AV node, so AV block of any degree appears and can change within hours; the facial palsy and the preceding erythema migrans place her in the same stage.

B is wrong: Fibrosis of the conduction system from decades of Trypanosoma cruzi persistence requires exposure in Latin America and years of infection, and it comes with a dilated cardiomyopathy, apical aneurysm, and megaesophagus or megacolon.

C is wrong: A ring abscess from endocarditis occurs in a febrile patient with an infected aortic or prosthetic valve and a murmur, and the PR interval lengthens progressively rather than appearing after a summer rash.

D is wrong: Immune complex deposition in endocarditis injures glomeruli and small vessels, producing hematuria with red cell casts, Osler nodes, and Roth spots, not block of the conduction system.

A competent response contains

· A 9 cm expanding patch with central clearing in August in rural Wisconsin is erythema migrans, which marks Lyme disease in an endemic area.

· P waves at 82/min unrelated to QRS complexes at 40/min are complete AV block, and the narrow QRS places the escape pacemaker at the junction.

· A facial palsy that includes the forehead is a peripheral seventh nerve palsy, another feature of early disseminated disease.

· Carditis in early dissemination comes from spirochetes and a lymphocytic infiltrate inflaming the conduction system, most often at the AV node.

Distinctions that matter

· Rash, facial palsy, and heart block together place the illness in the early disseminated stage.

Common misconceptions

· Reading complete heart block in a young patient as degenerative conduction disease. The exposure history and the rash point to a treatable infection of the node.

Where it is taught: Section C04 of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-BD1

Brain dump 1 · foundational retrieval: Rickettsia rickettsii from the tick to the vessel wall

A competent response contains

· Rickettsia rickettsii is a small obligate intracellular gram-negative bacterium that stains poorly with Gram stain, and it must be inside a host cell to replicate.

· Vectors are Dermacentor ticks: the American dog tick in the east and the Rocky Mountain wood tick in the west, with the brown dog tick in parts of Arizona and Mexico.

· Most United States cases occur in the southeastern and south-central states, such as North Carolina, Tennessee, Arkansas, Missouri, and Oklahoma, in spring and summer.

· Symptoms begin about 2 to 14 days after the bite.

· The target cell is the vascular endothelial cell. Rickettsiae attach through outer membrane proteins, are taken up, escape the phagosome into the cytoplasm, multiply there, and polymerize host actin into tails that push them into neighboring endothelial cells, so infection spreads cell to cell along the vessel.

· The lesion is a disseminated small vessel vasculitis, reaching skin, brain, lungs, kidneys, heart, and gut.

· Rash: appears about day 2 to 5 as small blanching pink macules on the wrists and ankles, spreads centripetally to the arms, legs, and trunk, and becomes petechial as red cells leak from damaged capillaries.

· Involvement of the palms and soles is characteristic but often late.

Distinctions that matter

· Spread cell to cell by actin tails keeps the organism inside the endothelium, which is why the disease is a vasculitis rather than a bacteremic illness with abscesses.

· The name points to the Rocky Mountains while the cases are concentrated in the southeast and south-central states.

Common misconceptions

· Expecting the rash first. Fever and headache come first and the rash appears about day 2 to 5, and some patients never develop one.

· Describing the rash as spreading outward from the trunk. It starts at the wrists and ankles and spreads centripetally.

Where it is taught: Section C05, first two paragraphs, the flow diagram, and the rash paragraph of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-BD2

Brain dump 2 · integration and application: Derive every laboratory finding from the leaking vessel

A competent response contains

· Thrombocytopenia: platelets adhere to damaged endothelium and exposed subendothelium and are consumed there. The marrow is producing platelets normally; they are being used up at the vessel wall.

· Hypoalbuminemia with edema: fluid and albumin leak through damaged capillaries into the interstitium.

· Hyponatremia: the fall in effective circulating volume stimulates ADH release, so water is retained and the serum sodium falls.

· Petechial rash: red cells escape from damaged capillaries, so the blanching macules become nonblanching petechiae.

· Noncardiogenic pulmonary edema: the same leak occurs in pulmonary capillaries, so fluid enters the alveoli with normal cardiac filling pressures.

· Confusion and seizures: vasculitis in cerebral vessels with edema; coma can follow in severe disease.

· Shock and acute kidney injury: hypovolemia from plasma loss, with myocardial involvement, lowers perfusion pressure, and the kidney is injured by both hypoperfusion and its own vasculitis. Liver enzymes may rise, and disseminated intravascular coagulation and gangrene of digits can occur but are less common.

· The classic triad is fever, headache, and rash with a history of a tick bite or exposure; all three are present at first in only a minority.

· The diagnosis is made on clinical grounds because the course can be rapidly fatal and antibodies are often absent in the first week.

· Other rashes on the palms and soles: secondary syphilis (copper-colored macules or papules, generalized lymphadenopathy, condylomata lata), hand, foot, and mouth disease from coxsackie A (vesicles with oral ulcers, usually in young children), and infective endocarditis (Janeway lesions with fever and a murmur).

Distinctions that matter

· Low platelets here are consumptive, not a production problem, which is why the marrow is normal.

· Myalgia, nausea, vomiting, and abdominal pain are common and can mislead toward a gastrointestinal illness.

Common misconceptions

· Explaining the hyponatremia as sodium loss. Water is retained because hypovolemia drives ADH release.

· Waiting for serology before treating the illness as spotted fever. Antibodies are often absent in the first week and the course can be rapidly fatal.

Where it is taught: Section C05, laboratory paragraph, triad paragraph, and the palms and soles table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-PR01

Prediction exercise: Predict what happens as the vasculitis spreads

A competent response contains

· Permeability rises and interstitial fluid increases, because infected endothelial cells and the inflammation around them no longer hold the barrier.

· Serum albumin falls and effective circulating volume falls, because fluid and protein escape through the damaged wall.

· ADH release rises and the serum sodium falls, because the drop in effective circulating volume is a stimulus for ADH.

· The platelet count falls, and the vessel wall accounts for it: platelets adhere to damaged endothelium and are consumed, with the marrow still producing.

· The lesions become petechial and nonblanching as red cells escape, appearing first on the wrists and ankles.

· Oxygenation falls as fluid enters the alveoli, which is noncardiogenic pulmonary edema from leaking pulmonary capillaries.

· Blood pressure falls and urine output falls, because continued plasma loss causes hypovolemia and shock, with acute kidney injury from hypoperfusion and renal vasculitis.

Distinctions that matter

· Every item traces to one event at the vessel wall, so the prediction is the same reasoning applied to different organs.

· Pulmonary edema here is permeability edema with normal cardiac filling pressures, unlike cardiogenic edema.

Common misconceptions

· Predicting a rising hematocrit and a rising albumin together. Albumin is lost with the plasma, so it falls.

· Predicting marrow failure from the low platelet count. Platelets are consumed at the injured endothelium.

Where it is taught: Section C05, flow diagram and laboratory paragraph, with the stop and predict box of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-Q03

Board-style question: Low sodium and low platelets in a spotted fever

Correct answer: D. Infection of vascular endothelium causing plasma leak and platelet consumption at the vessel wall

Why it is right: Rickettsiae replicate inside endothelial cells and push into neighboring cells using host actin, so small vessels throughout the body become inflamed and leaky: plasma and albumin escape into the interstitium giving edema and hypoalbuminemia, the fall in effective circulating volume releases ADH

and lowers the serum sodium, red cells escape to give petechiae, and platelets adhere to the damaged wall and are consumed.

A is wrong: Immune complex deposition with complement activation is the mechanism of the glomerulonephritis, Osler nodes, and Roth spots of endocarditis, and would come with a murmur and low complement rather than tick exposure and a centripetal petechial rash.

B is wrong: Rickettsia rickettsii replicates in endothelium rather than in platelets or megakaryocytes, and the marrow continues to produce platelets while they are consumed peripherally.

C is wrong: Antibody-dependent enhancement explains severe dengue in a second infection with a different serotype, and this patient has a tick exposure with a rash that began at the wrists and ankles.

A competent response contains

· Tick exposure in North Carolina in May with fever, severe headache, and petechiae that began at the wrists and ankles identifies Rocky Mountain spotted fever.

· Rickettsia rickettsii infects vascular endothelium and spreads cell to cell, producing a disseminated small vessel vasculitis.

· Edema with hypoalbuminemia comes from plasma leak, hyponatremia from ADH released in response to the fall in effective circulating volume, and thrombocytopenia from platelet consumption at the damaged wall.

Distinctions that matter

· One lesion at the endothelium explains all three laboratory abnormalities, which is what the question is testing.

Common misconceptions

· Looking for a separate cause for each abnormal value. They share a single vascular mechanism.

Where it is taught: Section C05 of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C06-BD1

Brain dump 1 · foundational retrieval: Yellow fever from the mosquito to the liver

A competent response contains

· Yellow fever and dengue are flaviviruses, enveloped positive-sense single-stranded RNA viruses, carried mainly by Aedes aegypti, a daytime-biting mosquito that breeds in standing water near houses.

· Yellow fever occurs in tropical Africa and South America. A jungle (sylvatic) cycle between monkeys and forest mosquitoes maintains the virus, and Aedes aegypti carries it between people in towns.

· After 3 to 6 days the illness begins with fever, headache, myalgia, and back pain, and most patients recover.

· About 15 percent improve briefly and then enter a toxic phase.

· The virus damages hepatocytes especially in the middle of the lobule, zone 2, which is midzonal necrosis. This is unusual because most toxic and ischemic injury is centrilobular.

· Dying hepatocytes undergo apoptosis and appear as Councilman bodies, shrunken eosinophilic cells, with microvesicular fat and relatively little inflammation.

· Liver failure causes jaundice, which gives the disease its name, and loss of clotting factor synthesis combines with platelet consumption to cause bleeding.

· Blood swallowed or vomited after gastric bleeding is darkened by acid, which produces black vomit.

· Kidney failure and shock follow, and a large share of patients in the toxic phase die.

· Laboratory clues are very high aminotransferases and bilirubin with a long prothrombin time.

Distinctions that matter

· Midzonal necrosis separates yellow fever from the centrilobular pattern of most toxic and ischemic liver injury.

· Councilman bodies are apoptotic hepatocytes, which is why inflammation is relatively sparse.

Common misconceptions

· Expecting a heavy inflammatory infiltrate in the yellow fever liver. The injury is apoptotic with relatively little inflammation.

· Explaining the bleeding by liver failure alone. Loss of clotting factor synthesis and platelet consumption act together.

Where it is taught: Section C06, opening paragraph and the yellow fever paragraph, with the comparison table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C06-BD2

Brain dump 2 · integration and application: Why the second dengue infection is the dangerous one

A competent response contains

· Dengue has four serotypes, DENV-1 to DENV-4.

· Antibodies against the earlier serotype bind the new serotype but do not neutralize it.

· The Fc portion of the bound antibody attaches to Fc gamma receptors on monocytes and macrophages, so more virus enters these cells than would enter without antibody; viral load and cytokine release rise.

· This is antibody-dependent enhancement, and infants carrying maternal antibody are the other group at risk through the same route.

· Severe dengue usually appears around defervescence, about days 3 to 7, as the fever falls.

· At the capillary wall, endothelial activation by viral proteins and cytokines makes capillaries leak plasma.

· A hematocrit rising from 40% to 48% means plasma has left the vessels while red cells stayed behind, which is hemoconcentration.

· It warns of shock before the blood pressure falls because intravascular volume is already being lost while compensation still maintains pressure.

· As the leak continues, fluid collects as pleural effusions and ascites, and hypovolemic shock can follow, while platelets fall and bleeding may occur.

· Classic dengue causes sudden high fever, often 39 to 40 °C (102.2 to 104 °F), headache with pain behind the eyes, rash, and severe muscle, joint, and bone pain (breakbone fever), with leukopenia and thrombocytopenia.

Distinctions that matter

· The rising hematocrit measures plasma leaving the vessels, while the falling platelet count measures consumption, so the two findings report different processes.

· Severe dengue begins as the fever settles rather than at the height of the fever.

Common misconceptions

· Reading a rising hematocrit as improvement. It is a sign of plasma loss and warns of shock.

· Assuming the first infection is the dangerous one. Pre-existing non-neutralizing antibody is what makes the second infection with a different serotype worse.

Where it is taught: Section C06, dengue paragraph and the worked example on the second infection of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C06-CP01

Comparison table: Yellow fever against dengue

A competent response contains

· Where: yellow fever in tropical Africa and South America; dengue throughout the tropics and subtropics worldwide.

· Main target: hepatocytes for yellow fever; monocytes, macrophages, and dendritic cells for dengue, with endothelium affected.

· Hallmark pathology: midzonal necrosis with Councilman bodies for yellow fever; capillary plasma leak for dengue.

· Key findings: jaundice, black vomit, bleeding, and renal failure for yellow fever; severe myalgia and arthralgia, retro-orbital pain, and rash for dengue.

· Laboratory clue: very high aminotransferases and bilirubin with a long prothrombin time for yellow fever; low platelets, low white cells, and a rising hematocrit for dengue.

· Severe form linked to: the toxic phase after a brief remission for yellow fever; a second infection with a different serotype for dengue.

· Both are Aedes-borne flaviviruses with thrombocytopenia and bleeding, and the liver is what separates them.

Distinctions that matter

· Both viruses share a family and a mosquito, so the organ damaged is what identifies the disease.

· Bleeding occurs in both, but from failed clotting factor synthesis in yellow fever and from consumption with endothelial injury in dengue.

Common misconceptions

· Grouping both diseases as the same hemorrhagic illness. The hepatitis of yellow fever and the plasma leak of dengue produce different pictures.

Where it is taught: Section C06, yellow fever against dengue table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C07-BD1

Brain dump 1 · foundational retrieval: The three shared effects of hemorrhagic fever viruses

A competent response contains

· Viral hemorrhagic fever is a syndrome rather than one disease, caused by enveloped RNA viruses from several families.

· Flaviviruses (yellow fever, dengue), positive-sense single-stranded RNA; filoviruses (Ebola, Marburg), negative-sense single-stranded RNA and filamentous; arenaviruses (Lassa), ambisense segmented single-stranded RNA; bunyaviruses (hantaviruses, Crimean-Congo hemorrhagic fever), negative-sense segmented single-stranded RNA.

· First effect: the viruses infect monocytes, macrophages, and dendritic cells early and spread with them to lymph nodes, liver, and spleen. Many block type I interferon signaling, so replication outpaces the innate response, infected dendritic cells fail to start an effective adaptive response, and many lymphocytes die by bystander apoptosis.

· Second effect: infected macrophages release large amounts of TNF-alpha, IL-1, IL-6, and other mediators, a cytokine storm, and these, with direct infection of endothelium in some viruses, loosen endothelial junctions and increase permeability.

· Third effect: activated macrophages express tissue factor, which starts coagulation throughout the circulation; platelets and clotting factors are consumed faster than they are made, which is consumptive coagulopathy, also called disseminated intravascular coagulation.

· Hepatocyte injury lowers clotting factor production further.

· The patient therefore leaks plasma and bleeds at the same time.

· Despite the name, visible bleeding is often modest; most deaths come from loss of fluid out of the vessels, shock, and multiorgan failure.

Distinctions that matter

· Permeability and coagulopathy are separate effects of the same infected macrophage, which is why shock and bleeding appear together.

· Interferon blockade is about timing: it lets replication run ahead of the host response.

Common misconceptions

· Attributing the deaths to blood loss. Plasma leak with shock and multiorgan failure accounts for most of them.

· Treating all of these viruses as one family. They come from several families and share mechanisms rather than taxonomy.

Where it is taught: Section C07, first two paragraphs, the flow diagram, and the virus table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C07-BD2

Brain dump 2 · integration and application: Work out who is at risk in a filovirus outbreak

A competent response contains

· Ebola and Marburg are filoviruses: enveloped, nonsegmented, negative-sense single-stranded RNA viruses with long filamentous virions. Fruit bats are the likely reservoir.

· Outbreaks begin when a person handles an infected animal or its tissues, or, for Marburg, enters bat-inhabited caves or mines. They occur in sub-Saharan Africa.

· Spread between people is by direct contact with blood, vomit, stool, or other body fluids of a symptomatic patient or a body after death, or with contaminated objects.

· Health care workers are at high risk through contact with the body fluids of symptomatic patients, and people preparing bodies for burial are at high risk through contact with the body after death.

· Patients become infectious when symptoms start, viral load rises as the illness worsens, and virus can persist in semen and other protected sites for months after recovery.

· Ordinary airborne spread does not occur.

· Incubation is about 2 to 21 days. Illness starts with sudden fever, weakness, myalgia, and headache, followed by profuse vomiting and diarrhea; a maculopapular rash, conjunctival injection, and bleeding from gums, venipuncture sites, or the gut may appear in the second week.

· The profuse vomiting and diarrhea account for most of the fluid and electrolyte loss.

· Laboratory findings: thrombocytopenia, leukopenia, high aminotransferases, and evidence of coagulopathy and kidney injury.

· Case fatality is about 50 percent on average for Ebola, ranging widely between outbreaks, and similarly high for Marburg.

Distinctions that matter

· Infectiousness begins with symptoms, which is why contact tracing focuses on symptomatic patients and on bodies after death.

· Persistence in semen and other protected sites is a separate route from the acute illness.

Common misconceptions

· Describing these viruses as airborne. Ordinary airborne spread does not occur; transmission is by direct contact with body fluids.

· Attributing the volume loss to bleeding. Vomiting and diarrhea account for most of it.

Where it is taught: Section C07, filovirus paragraphs and the virus table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C07-RT01

Retrieval exercise: Rebuild the common pathway of the hemorrhagic fever viruses

A competent response contains

· Step 1: the virus infects macrophages, dendritic cells, and in some cases endothelium.

· Step 2: interferon signaling is blocked and replication spreads to liver, spleen, and lymph nodes.

· Step 3: cytokine storm with TNF-alpha, IL-1, and IL-6 released.

· Step 4: endothelial junctions loosen, so plasma leaks and hypovolemia develops.

· Step 5: macrophage tissue factor triggers consumptive coagulopathy.

· Step 6: low platelets and clotting factors with liver injury cause bleeding.

· Step 7: shock and multiorgan failure.

· Blocking interferon signaling buys the virus time, so replication outpaces the innate response and the adaptive response starts late and poorly.

· Tissue factor is expressed by activated macrophages, and the consumption of platelets and clotting factors is consumptive coagulopathy, also called disseminated intravascular coagulation.

· Volume is lost at step 4 through the leaking endothelium and again at step 7 as shock deepens; bleeding becomes likely at step 5 when factors and platelets are consumed and at step 6 when liver injury lowers factor synthesis.

· Shock and bleeding appear together because the same activated macrophage both releases the cytokines that loosen the junctions and expresses the tissue factor that consumes the clotting system.

Distinctions that matter

· Leak and coagulopathy run in parallel from one cell, rather than one being a late consequence of the other.

Common misconceptions

· Drawing bleeding as the first event. Plasma leak and consumption of the clotting system come first, and bleeding follows.

Where it is taught: Section C07, flow diagram and the paragraph on the three shared effects of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C08-BD1

Brain dump 1 · foundational retrieval: Build the course of viral myocarditis

A competent response contains

· Myocarditis is inflammation of the myocardium with injury to myocytes that is not caused by ischemia, so ischemia must be excluded as the cause.

· Viruses are the most common cause in developed countries. Coxsackie B virus is the classic cause: an enterovirus, a nonenveloped positive-sense RNA picornavirus spread by the fecal-oral and respiratory routes.

· Adenovirus, parvovirus B19, human herpesvirus 6, influenza, and SARS-CoV-2 are also important.

· The typical patient is a young adult, more often male, with cardiac symptoms one to a few weeks after a respiratory or gastrointestinal illness, because the virus first replicates in the pharynx or gut and then reaches the heart through the blood.

· Course: viral infection of pharynx or gut; viremia with coxsackie B entering myocytes through CAR; direct viral injury to myocytes; T-cell attack with a lymphocytic infiltrate and myocyte necrosis; most recover while some develop persistent inflammation; fibrosis and dilation give dilated cardiomyopathy.

· The receptor is the coxsackie-adenovirus receptor (CAR), a junctional protein on cardiac myocytes, used by coxsackie B viruses and adenoviruses.

· CD8 cytotoxic T cells and natural killer cells destroy infected myocytes, and cytokines depress contractility.

· In the minority with persistent inflammation, autoimmunity can develop against cardiac proteins such as myosin, and progressive myocyte loss with fibrosis leaves a dilated cardiomyopathy.

· Parvovirus B19 infects coronary microvascular endothelium rather than myocytes.

Distinctions that matter

· Direct viral injury dominates the first days; the immune response dominates the following weeks.

· Most patients recover fully, and the minority with persistent inflammation are the ones at risk of chronic dilated cardiomyopathy.

Common misconceptions

· Attributing all the myocyte loss to the virus. Cytotoxic T cells and natural killer cells do much of the killing.

· Expecting cardiac symptoms during the viral illness. They follow it by one to a few weeks.

Where it is taught: Section C08, first two paragraphs and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C08-BD2

Brain dump 2 · integration and application: Separate myocarditis from an infarction and from the other infiltrates

A competent response contains

· Troponin is raised because necrotic myocytes release it, and the myocytes are dying from inflammation rather than ischemia, which is why the coronary arteries are normal.

· Inflamed tissue disturbs conduction and repolarization, so the ECG shows ST and T changes, AV or bundle branch block, or ventricular arrhythmias.

· Myocarditis is an important cause of sudden death in young athletes.

· When the pericardium is involved (myopericarditis), chest pain is pleuritic and the ECG may show diffuse ST elevation.

· Loss of contracting myocytes lowers ejection fraction and causes heart failure with dyspnea, fatigue, and edema.

· Endomyocardial biopsy shows an inflammatory infiltrate, mainly lymphocytes, with necrosis of adjacent myocytes in a pattern that is not ischemic.

· A single biopsy can miss the disease because the inflammation is patchy.

· Cardiac MRI often shows edema and late gadolinium enhancement in the subepicardial or midwall layers, unlike the subendocardial or transmural pattern of infarction.

· Many eosinophils suggest hypersensitivity or eosinophilic myocarditis, and multinucleated giant cells suggest the rapidly progressive giant cell myocarditis.

· Because chest pain, ST elevation, and raised troponin can mimic infarction, the clue is a young patient with a recent viral illness and normal coronary arteries.

Distinctions that matter

· Raised troponin reports myocyte death without saying what killed the cell, so the coronary anatomy and the MRI pattern are what separate inflammation from ischemia.

· Diffuse ST elevation with pleuritic pain points to pericardial involvement, while infarction gives changes in the leads of one territory.

Common misconceptions

· Taking a negative biopsy as proof there is no myocarditis. Inflammation is patchy and a single sample can miss it.

· Reading raised troponin with ST elevation as infarction in every patient. A young patient with a recent viral illness and normal coronary arteries fits myocarditis.

Where it is taught: Section C08, findings paragraph and the biopsy and imaging paragraph, with the stop and predict box of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C08-Q04

Board-style question: Biopsy after a viral illness

Correct answer: B. A lymphocytic infiltrate with necrosis of adjacent myocytes

Why it is right: A gastrointestinal illness a week earlier fits coxsackie B, which reaches the heart through the blood and enters myocytes through the coxsackie-adenovirus receptor; the immune response then

dominates, with CD8 cytotoxic T cells and natural killer cells killing infected myocytes, so the biopsy shows a mainly lymphocytic infiltrate with necrosis of the myocytes beside it.

A is wrong: Multinucleated giant cells with extensive myocyte loss indicate giant cell myocarditis, which is rapidly progressive and not the expected finding after a self-limited viral illness.

C is wrong: Intracellular amastigotes with interstitial fibrosis indicate chronic Chagas disease, which requires exposure in Latin America and develops years to decades after infection.

D is wrong: An eosinophil-rich infiltrate with little myocyte necrosis suggests hypersensitivity or eosinophilic myocarditis rather than a viral cause.

A competent response contains

· A young man with pleuritic chest pain, diffuse ST elevation, raised troponin, global hypokinesis, and normal coronary arteries one week after a febrile diarrheal illness has viral myocarditis, most often from coxsackie B.

· Coxsackie B enters myocytes through the coxsackie-adenovirus receptor, and CD8 cytotoxic T cells and natural killer cells destroy the infected cells.

· Biopsy therefore shows a lymphocytic infiltrate with necrosis of adjacent myocytes in a pattern that is not ischemic.

Distinctions that matter

· The histologic pattern names the cause: lymphocytes for viral myocarditis, eosinophils for hypersensitivity, giant cells for giant cell myocarditis.

Common misconceptions

· Expecting an ischemic pattern of necrosis because the troponin is raised and the ST segments are elevated.

Where it is taught: Section C08 of the CV-18-K Infectious Vascular Diseases notes.